

# Topological Fixed-Point Theory (TFPT)

A discrete compiler for the constants of physics

Two inputs — the boundary constant  $c_3 = \frac{1}{8\pi}$  and a five-slot carrier — build  $E_8$   
and read off the Standard Model: status assessment and reading guide

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## In one sentence

**Topological Fixed-Point Theory (TFPT)** constructs, from exactly **two inputs** — a boundary constant  $c_3 = \frac{1}{8\pi}$  (P1) and a five-slot carrier  $g_{\text{car}} = 5$  (P2) — a discrete compiler for the Standard-Model skeleton: the gauge group, three families, hypercharges and the flavor matrix, with  $E_8$  ( $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ ) as the consistency checksum. The *algebraic core is machine-checkable* and the dimensionless constants follow as fixed points ( $\alpha^{-1} = 137.0359992$ ); *physical readouts* — absolute scales, masses, inflation, gravity and cosmological transfers — run through explicitly *named, status-typed bridges* ( $v_{\text{geo}}$ ,  $G_{\text{net}}$ ,  $F_{\text{transfer}}$ ), not as free outputs. The honest one-line claim is on the status card below.

## The TFPT document set — what is treated where

**Plain language:** TFPT (Topological Fixed-Point Theory) is a small discrete compiler. Two inputs — the seam constant  $c_3 = \frac{1}{8\pi}$  and the carrier rank  $g_{\text{car}} = 5$  — build  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  and read off the Standard-Model skeleton, the constants and the scale grammar. The series is **the introduction plus five numbered papers and four companions**, best read in order:

0. **introduction** — reading guide, compiler closure, predictions, the dependency DAG and the single proof ledger (`verification/status_ledger.csv`). (*entry point*)
1. **tfpt\_1\_architecture\_e8** — the two axioms, the derivation map, the EM fixed point  $\alpha^{-1}$ , the  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  construction, the QBL theorem chain.
2. **tfpt\_2\_standard\_model** — the SM in one  $\varphi_0$ -ladder formula, flavor from parabolic transport (sheet diamond, dual anchor,  $Q_+$  cohomology), the worked closures.
3. **tfpt\_3\_e8\_audit\_bootstrap** — the seven  $E_8$  slices as an audit raster, the cascade bridge, the Möbius bootstrap.
4. **tfpt\_4\_frontier** — honest status of  $\eta_B$ ,  $m_p/m_e$ , Koide, dark matter, full QG.
5. **tfpt\_5\_redteam** — the adversarial audit: declared attacks, what survives, what each reduces to, and the kill tests.

**Companions:** **R.** **tfpt\_research\_contracts** — the open interfaces ( $v_{\text{geo}}$ ,  $G_{\text{net}}$ ,  $F_{\text{transfer}}$ ) as numbered contracts; **H.** **tfpt\_horizon\_readouts** — Appendix H (unit-system reframe): one seam constant as the universal horizon thermal code, the Nariai/anchor surface, the resummed seam clock; **O.** **origin\_theory** — why no free fundamental number remains (exact core + one honestly-typed cyclic interpretation); **S.** **tfpt\_safeguards** — the verification discipline: every mechanism that defends a claim against coincidence and numerology (status calculus, no-free-pattern, the over-determination map, the firewall, frozen predictions + null model, the independent Wolfram and Lean paths, the red team).

**You are reading the introduction — the entry point and reading guide.**

**TFPT status at a glance — read this card the same way in every document**

**Two inputs, one compiler.** From the boundary constant  $c_3 = \frac{1}{8\pi}$  (axiom P1) and the five-slot carrier  $g_{\text{car}} = 5$  (axiom P2) — jointly the elementary symmetric data of one anchor  $a = (1, 1, 2)$  — the discrete compiler  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  is *built*, and the Standard-Model skeleton (gauge group, three families, hypercharges, the flavor matrix), the dimensionless constants ( $\alpha^{-1} = 137.0359992$ ) and the scale grammar are read off as fixed points. The algebraic core is machine-checked; physical readouts run through *named bridges* whose status is typed below.

**What is closed, and what genuinely remains.** The discrete/algebraic layer is closed ([E]); the everything-open residual reduces to *three named interface problems*:

| Interface             | Question                                                                                        | Status        |
|-----------------------|-------------------------------------------------------------------------------------------------|---------------|
| $v_{\text{geo}}$      | the one metrology unit ( $= 1/\sqrt{G} = m/\mu$ ); No-Unit Thm: no compiler scale               | primitive [O] |
| $G_{\text{net}}$      | simple-current extension $(D_5)_1 \otimes (A_3)_1 \rtimes \langle (1, 1) \rangle \cong (E_8)_1$ | [C]           |
| $F_{\text{transfer}}$ | one functor, four typed interfaces (Koide, $\eta_B$ , axion, $m_p/m_e$ )                        | [C]           |

**Gravity is parameter-free.** The classical metric-sector field equation is no longer only an  $R+R^2$  readout: the fixed-volume entanglement equilibrium (Jacobson; Faulkner et al.), run with TFPT's atoms, gives the *full* covariant Einstein equation  $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$  with *both* coefficients fixed —  $c_3^{-1} = 8\pi$  (no free dimensionless Newton dial; the thermodynamic origin  $2\pi/\eta$  *coincides* with the geometric one  $|\mathbb{Z}_2|2\pi\chi$  via  $|\mu_4| = |\mathbb{Z}_2|\chi(S^2) = 4$ , so  $c_3$  is triply over-determined — anchor, geometry, thermodynamics, v358) and  $\Lambda$  from  $\alpha$  ( $\rho_\Lambda = (3/4\pi^2)e^{-2\alpha^{-1}}$ , v60); the Einstein tensor (not Ricci) is forced by Lovelock, so matter conservation is an output (v359). The residual here is only the equation-of-state status (an external candidate action — Bianconi's entropic action, arXiv:2408.14391 — is quantified in v473 without changing the typing) and the one unit  $v_{\text{geo}}$ ; the global measure (C7 = QG.AMB.01) is now a [C] redundancy (v369), a certification object rather than missing dynamics, conditional on SEAM.EQUIV.01 and Bisognano–Wichmann.

The seam-net structural theorem  $G_{\text{net}}$  (= SEAM.EQUIV.01: the raw seam *is* the holomorphic  $(E_8)_1$  net) is now *closed modulo a cited theorem*. The explicit lattice model (v367/v368) and the  $S_3$  closure stack pin the target at every computable level — central charge  $c = 8$  (v376), the  $(E_8)_1$  character with 248 currents and one primary (v377), genus-one torus  $\text{GSD} = 1$  (v378) and reflection positivity (v379) — and it is Lean-formalised as FORM.SEAM.MMST.01: the collar's MMST hypotheses are kernel-proved, the MMST scaling-limit and Adamo–Moriwaki–Tanimoto OS-reconstruction theorems enter as named cited axioms, and the #print axioms check is clean. The *only* residual that stays [O] is the abstract continuum existence of the scaling limit (v336) — its 128-spinor extension leg since certified at net level by the peer-reviewed crossed-product package ( $h_s=16/16=1 \in \mathbb{Z}$ , the Longo–Rehren locality criterion; Böckenhauer; KLM  $\mu=4/2^2=1$ ; AGT/AMT the independent second witness) with the realisation input reduced to invariant level and the parallel Lean route `seamResidualClosed'` (v469); so this is *closed modulo a cited theorem*, not solved. QGEO.SYM.01 (the  $\mu_4$  deck acting geometrically) is its *corollary*: a conformal net's vacuum is rotation-invariant by axiom, so the deck follows with no extra premise (v335, Lean FORM.QGEO.BW.01). The nonperturbative quantum-gravity measure QG.AMB.01 is no longer carried as an open item: it is a [C] *redundancy* (v369), a certification object rather than missing dynamics, conditional on SEAM.EQUIV.01 and Bisognano–Wichmann (and TFPT is moreover rigorously gap-decoupled from the general Euclidean-QG conformal-factor problem, margin  $1.648 > 0$ , v76/v330). So the honest residual is the one unit  $v_{\text{geo}}$  plus the typed  $F_{\text{transfer}}$  interfaces, above the cited-theorem ceiling on the seam and with QG.AMB.01 a [C] redundancy. Everything carries a claim ID resolving to a row of `verification/status_ledger.csv` (the single source of truth); **if the text and the ledger ever disagree, the ledger wins**. The historical labels ( $U_{\text{wall}}$ ,  $G_{\text{metric}}$ ,  $F_{\text{frontier}}$ ) are retained only for ledger continuity — the live residual is  $v_{\text{geo}} \oplus F_{\text{transfer}}$ , with the seam  $G_{\text{net}}$  closed modulo a cited theorem and QG.AMB.01 a [C] redundancy.

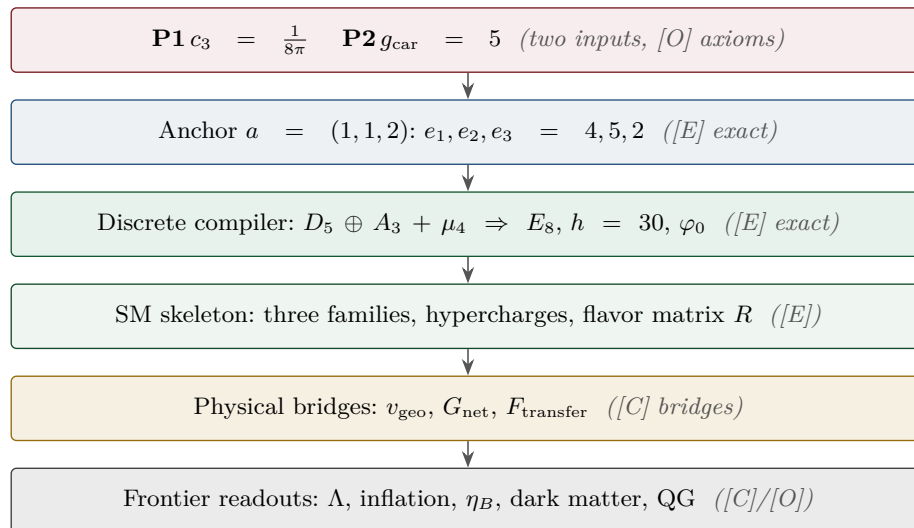
Marker key: [E] exact / machine-proven · [C] conditional (holds under named hypotheses) · [O] open / axiom · [X] falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.

[E] exact / machine-proven

[C] conditional

[O] open / axiom

[X] kill test



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**Reading the markers.** Claims carry one of *four* reader-facing classes, so one can see at a glance what is proven and what is conditional: **[E]** *exact / machine-proven* — an exact identity (e.g.  $240 = 16 \cdot 5 \cdot 3$ ), a Lie/lattice theorem (e.g.  $D_5 \oplus A_3 + \mu_4 = E_8$ ), a Lean/script formalisation, or a reproduced numerical fixed point; **[C]** *conditional* — follows under explicitly named hypotheses (a physical bridge, readout or pending step); **[O]** *open / axiom* — a declared input or a genuine gap; **[X]** *kill test* — a falsifiable threshold. The finer per-claim type (Axiom / Formal / Lattice / Numerical / Identity / Physical) is kept in `verification/status_ledger.csv`, the single source of truth.

|                                    |                                              |
|------------------------------------|----------------------------------------------|
| Physical — under hypotheses        | RG masses, QG    QG.AMB.01                   |
| Numerical — fit-free fixed point   | $\alpha^{-1}, \Lambda, \theta_{12}$ EM.FP.01 |
| Lattice — Lie / glue / root system | $E_8=(D_5\oplus A_3)+\mu_4$ E8.GLU.01        |
| Formal — Lean / exact script       | P2 algebra    FORM.P2.02                     |
| Axiom — declared inputs            | P1 $c_3$ , P2 $g_{\text{car}}$ AX.P1.01      |

The *claim stack* (bottom = most certain/fewest, top = conditional). Each claim sits on exactly one level; the bottom two are inputs, the middle two are proved/reproduced, only the top is conditional. The single source of truth for which claim sits where is **verification/status\_ledger.csv**.

**No-free-pattern rule.** To keep the structure from drowning in decorative coincidences, an identity enters the *main text* only if it (1) is a necessary step in a derivation, (2) kills an alternative, (3) is ablation-relevant, (4) links two previously independent modules, or (5) is experimentally tested. Everything else is explicitly demoted to an *audit fingerprint* [E] (clearly boxed as such) — so the load-bearing beams stay visible and the ornament never gets promoted to spine.

**Admissible-invariant classes (the harder filter).** Sharper still: a *load-bearing* integer may only come from one of seven typed sources — (i) a symmetric function  $e_k(a)$  of the anchor  $a = (1, 1, 2)$ ; (ii) a power sum  $p_n(a)$  or difference  $p_{n+1}-p_n$ ; (iii) a determinant, spectrum, Smith normal form or trace of  $(R, Q, K, L)$ ; (iv) an anchor quadratic form  $u^\top Mv$  with  $u, v \in \{\mathbf{1}, a\}$ ; (v) a Plücker coordinate of the anchor plane  $\langle \mathbf{1}, a \rangle$ ; (vi) an  $E_8$  Casimir degree or branching block; (vii) a pre-registered experimental observable. Anything else is an audit fingerprint or a frontier item, never spine. This is what converts TFPT from *pattern hunting* (“look, another number”) into a typed compiler: *fewer admissible kinds of explanation*, not more matches.

**Reviewer path** (read in this order; the rest is support):

1. the architecture: the two axioms  $\{c_3, g_{\text{car}}\}$  and the two-engine picture (§1, the figures);
2. the  $E_8$  glue  $E_8=(D_5\oplus A_3)+\mu_4$  — document 1, Part II, machine-checked (v1\_e8\_glue.py);
3. the  $\alpha^{-1}$  fixed point plus ablation (v3\_em\_alpha.py);
4. the **single status ledger verification/status\_ledger.csv** — the source of truth for every claim’s grade, location, dependency and script;
5. the **open gates** as numbered research contracts (tfpt\_research\_contracts: the residual classes R1–R5 and their current reductions) and the honest frontier list (document 4).

Everything carries a claim ID (e.g. E8.GLU.01, EM.FP.01) that resolves to a row of the ledger. **If the text and the ledger ever disagree, the ledger wins.**

## 1 What this is about: the jump in construction principle

### In plain words

**What we found.** TFPT reproduces dozens of real numbers of physics (particle masses, the fine-structure constant, dark-energy and inflation values) from a small set of inputs — and the compiler closure says *why* the same handful of small integers (2, 3, 5, 16, ...) reappears in every sector: those numbers are not separate lucky hits. A single tiny “machine” — fed by just **two inputs** (a boundary number  $\frac{1}{8\pi}$  and a five-slot carrier) — builds the exceptional symmetry  $E_8$ , and  $E_8$  is the unimodular “audit hull” that classifies the discrete Standard-Model readout: the constants, the flavor skeleton and the gravity/cosmology regime are then read off by projection and boundary dressing (not “printed” for free — where a scheme or analytic interface is needed, it is named).

**Why this is simple.** There are not many separate derivations that happen to share numbers — there is *one generator and a handful of corollaries*, and the number of independent structural assumptions is **two**.

**How plausible it is.** A theory becomes more believable when it *explains more from less*. Here the two inputs are even *over-determined*: the machine reproduces the very two numbers it started from (the dashed bootstrap loop in the architecture diagram, §2), so there is *no free dial left to tune*. Pushed to the end ([[verification/v353\\_selfloop\\_capstone.py](#)]), this means TFPT is best read not as a *linear* theory (axioms  $\rightarrow$  theorems) but as a *closed self-consistent loop* whose outputs fix its own inputs ( $g_{\text{car}}=5$  is the largest prime of the output  $h(E_8)=30$ ; the 8 in  $c_3$  is rank  $E_8$ ), with a *unique* fixed point ([v6/v56](#)) and *zero* free adjustable dimensionless parameters. Even  $\pi$  is not a free dial: it is the fixed geometric constant of the boundary 2-sphere ( $\oint_{S^2} K = 4\pi$ ), the one transcendental the continuous boundary contains. “Which is the axiom?” is the wrong question for a loop — every part is fixed by the whole; the one remaining structural question, whether the loop also closes in the *continuum* (the keystone [SEAM.EQUIV.01](#)), is now *closed modulo a cited theorem* — the lattice model and  $S_3$  stack pin it at every computable level and it is Lean-formalised ([FORM.SEAM.MMST.01](#)); only the cited continuum-existence theorems stay [\[O\]](#) ([v367/v368](#), [v376–v379](#), [v336](#)), their extension leg since re-founded on peer-reviewed crossed-product theorems with the realisation input reduced to invariant level ([v469](#)).

### Sharper still: one anchor $a = (1, 1, 2)$ , and $E_8$ as a compiler hull ([\[E\]](#))

The two axioms are not even independent: they are the *elementary symmetric polynomials* of the single parabolic anchor  $a = (1, 1, 2)$  (the exponents-at-infinity / splitting type  $\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$ ):

$$e_1(a) = 4 = |\mu_4|, \quad e_2(a) = 5 = g_{\text{car}}, \quad e_3(a) = 2 = |\mathbb{Z}_2|, \quad c_3 = \frac{1}{2e_1(a)\pi} = \frac{1}{8\pi}.$$

Its *power sums*  $p_n(a) = 1^n + 1^n + 2^n = 2 + 2^n$  generate the big Lie data directly:  $p_1=4=|\mu_4|$ ,  $p_2=6=|R^+(A_3)|$ ,  $p_3=10=\mathcal{A}_\Lambda$ , and

$$|R(E_8)| = p_1 p_2 p_3 = 240, \quad \dim E_8 = p_1 p_2 p_3 + (p_4 - p_3) = 248, \quad p_4 - p_3 = 8 = \text{rank } E_8,$$

with the binary ladder  $p_{n+1} - p_n = 2^n$ . So the inputs collapse from  $\{c_3, g_{\text{car}}\}$  to  $\boxed{a = (1, 1, 2) \text{ plus } \pi}$ . [[verification/v23\\_anchor\\_generator.py](#)]

And  $a$  is itself *half-forced* ([\[E\]/\[C\]](#)): on  $\mathbb{P}^1$  every bundle splits (Birkhoff–Grothendieck), so once  $|\mu_4| = 4$ ,  $N_{\text{fam}} = 3$ ,  $\deg E = -4$  are set, the three positive anti-degrees are positive integers summing to 4, and  $4 = 2 + 1 + 1$  is the *unique* partition into three positive parts — machine-checked with all its negative controls as the partition corollary ([[verification/v491\\_p2\\_partition\\_corollary.py](#)], [P2.PARTITION.01](#)): the unique multiset  $\{1, 1, 2\}$  forces  $e = (4, 5, 2)$ , hence  $g_{\text{car}} = e_2 = 5$ ,  $|\mathbb{Z}_2| = e_3 = 2$ ,  $N_{\text{fam}} = 3$  as *corollaries* (dropping positivity, the sum, the part count or integrality each destroys uniqueness — Mehta–Seshadri-rational weights give  $e_2 = \frac{16}{3}$  or  $\frac{44}{9} \neq 5$ ), sharpening [v53](#) which needs all three targets. The typing itself is now hardened ([[verification/v499\\_p2\\_weights\\_deligne\\_bg.py](#)], [P2.TYPING.01](#)): given the four marks, rank 3, the cusp class and unitarity (U), the Deligne canonical extension of the flavor connection has  $\deg E = -4$  (residue traces) and Mehta–Seshadri stability forces  $h^0(E) = 0$ , so integrality (Birkhoff–Grothendieck), positivity and the sum 4 are *theorems* — the residual shrinks to the module identification and (U), both [\[C\]](#); and that module-identification rest is now *one* residual with the QGEO modulus rest (the order-4 clock = the Coxeter/cusp-class carrier datum; [[verification/v503\\_qgeo\\_emergence\\_light.py](#)], [QGEO.EMERGE.LIGHT.01](#), no marker moves). That common residual is itself now reduced to *one alignment bit* — the collar deck central in the mark- $D_4$ ; the clock’s *order* is fermionically forced,  $U^2 = (-1)^F$  exactly ([[verification/v506\\_seam\\_clock\\_rigidity.py](#)], [SEAM.CLOCK.RIGIDITY.01](#), markers unchanged); and the bit is no tautology — the deck’s *class* is derived (every mark-free collar deck is a half-period translation of the seam double), only its *position* stays the [\[C\]](#) input



(`[verification/v507_seam_bit_origin.py]`, `SEAM.BIT.ORIGIN.01`, markers unchanged); and that position half is now *topology* — the covering deck is free on the seam circle, the edge class is excluded, and the bit reduces to the square-modulus datum  $\tau = i$  alone (`[verification/v510_seam_bit_freedom.py]`, `SEAM.BIT.FREEDOM.01`, markers unchanged). The genuine remaining axiom is therefore not “why  $a = (1, 1, 2)$ ” but “why the carrier  $g_{\text{car}} = 5 / (|\mu_4|, N_{\text{fam}}) = (4, 3)$ ” — that interface (with  $c_3$ ), i.e. the *weight-typing* postulate itself, is the **[O]** input layer (`AX.P2.01` stays a declared axiom). The consolidated reduction state (2026-07-21): the compiler’s dimensionless inputs reduce to *four marks* (derived from P1-side topology), *one square-modulus datum* ( $\tau = i$ ; equivalently the clock, the CM-fixed half-period, or the non-split extension class on the free seam circle) and  $\pi$  — a reduction state, not an abolition of the axiom. Consolidated (2026-07-21, text only): the compiler’s dimensionless inputs reduce to *four marks* (derived from P1-side topology), *one square-modulus datum*  $\tau = i$  (equivalently: the clock / the CM-fixed half-period / the non-split extension class on the free seam circle), and  $\pi$  — a reduction *state*, not an abolition: `AX.P2.01` remains declared, and the **[C]** identifications R1/R2 of `v499` remain conditional.

*Two readings, one source (no contradiction).* P1 and P2 are the *operational compiler inputs*; their anchor provenance is  $a = (1, 1, 2)$  plus  $\pi$ . The *single structural bedrock postulate* is the keystone `SEAM.EQUIV.01`, whose conformal-deck face `QGEO.SYM.01` (the carrier  $\mu_4$  clock is the seam’s conformal deck, see `tfpt_research_contracts`) forces the  $\mu_4$  deck, hence  $N_{\text{fam}} = 3$ ,  $\deg E = -4$  and the tripartition  $4 = 2+1+1$ , of which  $c_3$  and  $g_{\text{car}}$  are the two elementary readouts. “Two inputs” (the working axioms) and “one postulate” (the geometric bedrock) are the same statement at two layers. Its geometric normal form and the mark-local  $\Rightarrow \omega \circ \rho = \omega$  implication are now Lean-formalised (`FORM.QGEO.02/FORM.QGEO.01`); the postulate itself stays **[O]**.

*One source, four readings (the cleanest origin).* The four-point seam divisor  $\mu_4$  on  $\mathbb{P}^1$  generates the whole discrete skeleton by four canonical functors:

$$\underbrace{c_3 = \frac{1}{8\pi}}_{\text{thermal}} \quad \underbrace{\text{rank } H_1(\mathbb{P}^1 \setminus \mu_4) = 3 = N_{\text{fam}}}_{\text{cycles} \rightarrow A_3} \quad \underbrace{h^0(\mathbb{P}^1, \mathcal{O}(\mu_4)) = 5 = g_{\text{car}}}_{\text{functions} \rightarrow D_5}$$

$$\underbrace{(D_5)_1 \otimes (A_3)_1 \rtimes \mu_4 \cong (E_8)_1}_{\text{net}}$$

and the function side even generates  $D_5$  itself (`[verification/v197_rr_carrier_clifford_d5.py]`):  $\Lambda^{\text{even}}(\mathbb{C}^5) = \binom{5}{0} + \binom{5}{2} + \binom{5}{4} = 16 = \dim S^+$ , the  $\mathfrak{so}(10) = D_5$  half-spinor. So  $\mu_4$  is not merely glue: it is the *common source* of  $A_3$ ,  $D_5$ ,  $c_3$  and  $E_8$  (`[verification/v189_riemann_roch_carrier.py]`) — the arithmetic is exact **[E]**, the physical reading of the carrier as the seam mode space stays **[C]**.

**What  $E_8$  is (and is not).** In TFPT  $E_8$  is *not* an unbroken physical gauge group: it is the unimodular *audit / compiler hull* that classifies the admissible discrete charge and residue structures. The Standard Model is a *readout* after projection, not a direct  $E_8$  gauge theory; the Lorentz signature appears only after projection; fermions are not “everything in the adjoint”. This is exactly why the known no-go results against literal  $E_8$  world-formulas (Distler–Garibaldi; Coleman–Mandula) do *not* bite: they constrain  $E_8$  as a spacetime gauge symmetry, which TFPT does not claim. The defensible claim is: *TFPT is the unique parabolic compilation of the anchor  $a = (1, 1, 2)$  into the  $E_8$  audit hull.*

In one line:

**|** The anchor generates the syntax; the boundary generates the scale;  $E_8$  checks the consistency.

i.e. the discrete content (carrier  $D_5 \oplus A_3 + \mu_4$ , the operators  $R, Q, K, L$ ) flows from  $a = (1, 1, 2)$ ; the continuous content ( $c_3 = \frac{1}{8\pi}$ ,  $\varphi_0$ ,  $\alpha^{-1}$ , the exponential scales) flows from  $\pi$ ; and  $E_8$  is the unimodular *checksum container*, not the world group.

**The anchor ratio triad** (`[verification/v119_review_validation_2.py]`). The three elementary symmetric values, normalised by the family count, are the theory’s three recurring fractions **[E]**:

$$\frac{e_3}{p_0} = \frac{2}{3} \text{ (Koide branch / sheet ratio), } \quad \frac{e_1}{p_0} = \frac{4}{3} \text{ (seed gain), } \quad \frac{e_2}{p_0} = \frac{5}{3} \text{ (carrier branch)}$$

— the canonical flow’s critical points are exactly  $(2, 5) = (e_3, e_2)$  with inflection  $\frac{7}{2} = \frac{e_3+e_2}{2}$ : the fractions  $\frac{2}{3}, \frac{4}{3}, \frac{5}{3}$  are not scattered numbers but the three normalised anchor coefficients. Further micro-identities [E]:  $\Omega_{\text{adm}} = 2p_1p_2 = 48$ ,  $10b_1 = 1 + p_1p_3 = 41$ ,  $\Delta_Y = e_2^2 = p_0^2 + 2 \text{rank } E_8 = 25$ ,  $h(E_8) = e_3p_0e_2 = 30$ ,  $\text{rank } E_8 = p_0 + e_2$ .

$$a = (1, 1, 2) \quad [E]$$

**elementary**  $e_k(a)$ 

$$\begin{aligned} e_1 &= 4 \rightarrow |\mu_4| \\ e_2 &= 5 \rightarrow g_{\text{car}} \\ e_3 &= 2 \rightarrow |\mathbb{Z}_2| \\ c_3 &= \frac{1}{2e_1\pi} = \frac{1}{8\pi} \end{aligned}$$

**power sums**  $p_n = 2 + 2^n$ 

$$\begin{aligned} p_0, p_1, p_2, p_3 &= 3, 4, 6, 10 \\ p_1p_2p_3 &= 240 = |R(E_8)| \\ p_4 - p_3 &= 8 = \text{rank } E_8 \\ \Rightarrow \dim E_8 &= 248 \end{aligned}$$

**derived motifs**

$$\begin{aligned} &4, 5, 8, 16, 25, \\ &30, 41, 120, 240 \\ &\text{one anchor,} \\ &\text{many projections} \end{aligned}$$

## The companion documents

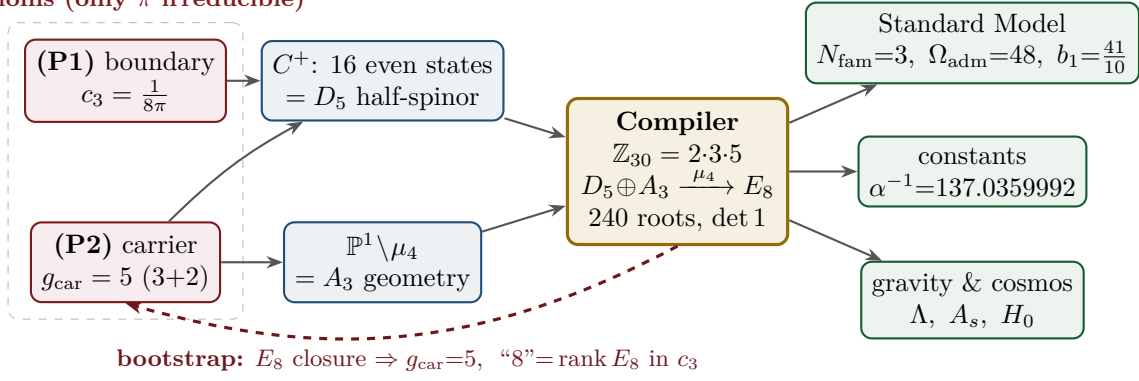
The whole development is consolidated into **eight companion documents** (plus this introduction), organised in layers that build on one another; this introduction is the entry point and reading guide. Each document collects the material of its layer into self-contained parts:

| Document                          | What it contains (former notes now as parts)                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| tfpt_1_architecture_e8            | <b>Core.</b> The two axioms $\{c_3, g_{\text{car}}\}$ , the derivation map and the EM fixed point (Part I); the Coxeter–cyclotomic compiler $\mathbb{Z}_{30}=2\cdot3\cdot5$ , the explicit $D_5\oplus A_3+\mu_4$ $E_8$ construction, the machine-checkable $E_8$ appendix and the group-level glue $(\text{Spin}(10)\times SU(4))/\Delta\mathbb{Z}_4$ (Part II).                                                                        |
| tfpt_2_standard_model             | <b>Standard Model.</b> The full SM in one $\varphi_0^{\text{ret}}$ -ladder formula (Part I); the flavor block from parabolic weights and the transport theorem, H2 as a spectral selector (Part II); the worked closures — explicit gap, APS–Fredholm, truncation/ $n_s, r$ , Starobinsky $M$ , derived $\theta_{12}$ , quark $c$ , the H2 splitting type, $\Lambda$ -typing (Part III).                                                |
| tfpt_3_e8_audit_bootstrap         | <b><math>E_8</math> audit &amp; bootstrap.</b> The seven $E_8$ slices as an audit raster ( $E_6\times A_2$ reads the flavor matrix) (Part I); the old cascade $D=60-2n$ as the same even-integer spine (Part II); the Möbius bootstrap — $g_{\text{car}}=5$ and the “8” in $c_3$ as overdetermined fixed points, only $\pi$ irreducible (Part III).                                                                                     |
| tfpt_4_frontier                   | <b>Frontier.</b> The honest status of $\eta_B$ , $m_p/m_e$ , Koide, dark matter and quantum gravity — the genuine handle, the precision, and what is <i>not</i> forced onto the ladder.                                                                                                                                                                                                                                                 |
| tfpt_5_redteam                    | <b>Red Team.</b> The adversarial audit of the five load-bearing reductions (Targets A–E): the declared attacks, what survives each, what each one reduces to, and the explicit kill tests.                                                                                                                                                                                                                                              |
| tfpt_research_contracts           | <b>Research contracts.</b> The open gates as numbered contracts with lemma chains and closing theorems: ( $U_{\text{wall}}$ ) (now reduced to the selector triangle: $(d, n) \Rightarrow R$ columnwise, residue = <i>two</i> line pairings, the third integrality-forced) and $G_{\text{metric}}/G_{\text{net}}$ (the index-4 seam-net identification — one theorem, three doors, now realised at the finite Lie level).                |
| tfpt_horizon_readouts<br>(App. H) | <b>Appendix H — horizon unit system (reframe, not new physics).</b> One seam constant $c_3=\frac{1}{8\pi}$ as the universal horizon thermal code: Hawking/de Sitter/Unruh, BH thermodynamics, Page, scrambling, Nariai bound, $v_{\text{GW}}=c$ — standard physics in seam units, two compiler fingerprints ( $1920= W(D_5) $ , $ \mu_4 =4$ ); plus the Nariai/anchor surface, the dual anchor and the resummed seam clock (v101–v138). |
| origin_theory                     | <b>Origin Theory.</b> Why no free <i>dimensionless compiler dial</i> remains beyond the anchor and $\pi$ (the dimensionful $v_{\text{geo}}$ and transfer physics stay typed): the $(g_{\text{car}}, N_{\text{fam}})$ skeleton, the triply-forced 8, the order-30 Coxeter cycle, one boundary transport for flavor and horizon, the gapped unique attractor; [E] core plus one honestly-typed [C] cyclic interpretation.                 |

The five formerly “research-level” problems are now *closed*, *reduced* or *typed* inside the documents where they belong — the group-level closure  $(\text{Spin}(10)\times SU(4))/\Delta\mathbb{Z}_4$  in document 1 (closed), and the H2 splitting type, the Starobinsky scalaron mass, the  $\theta_{12}$  texture and the quark  $c$  in document 2 (the first two closed; the quark *ratios* now closed combinatorially (v49/v71), only the *absolute* scale a  $U_{\text{point}}$  anchor;  $\theta_{12}$  the seam-misalignment residual). Eight companion files (plus this introduction), each result where it logically lives.

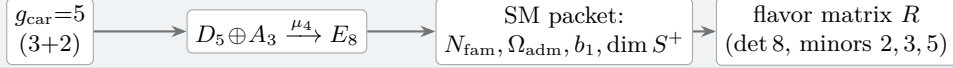
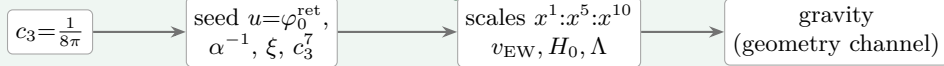
## 2 The architecture at a glance

The diagram below shows the whole pipeline: *two* inputs on the left, the discrete compiler in the middle, the physics on the right. Everything in between is a consequence.

2 axioms (only  $\pi$  irreducible)

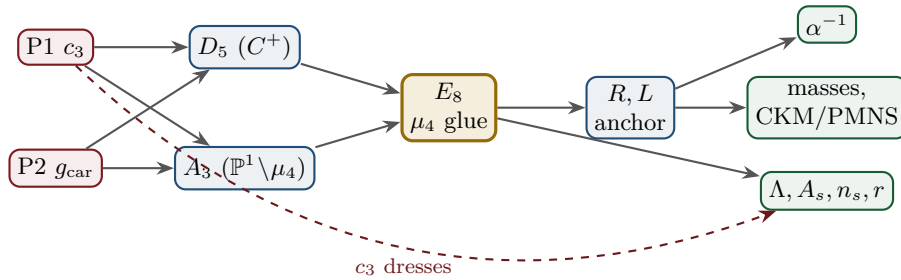
**Key statement (the loop closes):** previously  $D_5$ ,  $A_3$  and  $E_8$  sat *as given* Lie structures in the background. Now they are *intermediate products* of the two inputs — and the red back-channel is the new part: the  $E_8$  closure feeds back and *fixes the inputs*.  $g_{\text{car}} = 5$  is the unique rank-fill/Coxeter/Pascal solution ( $2^{g-1} = \binom{g}{0} + \binom{g}{1} + \binom{g}{2}$ ), and the integer 8 in  $c_3 = \frac{1}{8\pi}$  is rank  $E_8 = h(D_5) = \det R$ . **Inputs and output form an overdetermined fixed point** — the structure does not “prove itself”; it has fewer degrees of freedom than independent consistency conditions. The only thing *not* produced by the loop is the continuous  $\pi$  (Möbius/Gauss–Bonnet), the lone irreducible primitive. So it is not a one-way arrow chain but a closed, overdetermined self-consistency loop.

**The master story is two engines.** Read from the two axioms, the whole theory factorises into exactly *two* engines — a *discrete closure* and a *boundary dressing*. Gravity is not a third block; it is the geometry channel of Engine 2.

Engine 1 — discrete closure (from  $g_{\text{car}} = 5$ )Engine 2 — boundary dressing (from  $c_3 = \frac{1}{8\pi}$ )

## 3 The dependency DAG and the proof ledger

The whole theory is a directed acyclic graph: two axioms at the source, the compiler in the middle, the observables as sinks. The DAG makes the attack surface explicit — each arrow is a named theorem or a declared input, and each node fails in exactly one way.





| Node           | Definition                                             | Inputs      | St.     | Output                           | Failure mode                |
|----------------|--------------------------------------------------------|-------------|---------|----------------------------------|-----------------------------|
| P1             | RP boundary kernel, $c_3=1/(8\pi)$                     | — (axiom)   | [O]     | $c_3$                            | no reflection-positive seam |
| P2             | 5-slot carrier $g_{\text{car}}=5$                      | — (axiom)   | [O]     | carrier 3+2                      | wrong family/charge lattice |
| $D_5$          | $C^+$ even-Hamming = half-spinor 16                    | P2          | [E]     | $\dim S^+=16, Y$                 | group/matter mismatch       |
| $A_3$          | $\mathbb{P}^1 \setminus \mu_4$ four-puncture geom.     | P1          | [E]     | $N_{\text{fam}}=3, \mathbb{Z}_6$ | wrong multiplicity          |
| $E_8$          | $\mu_4$ glue: disc $=\mathbb{Z}_4$ , $q$ -sum = 2      | $D_5, A_3$  | [E]     | 240, 248                         | not even-unimodular         |
| $R, L$         | residue + winding matrix, anchor (1, 1, 2)             | $A_3$       | [E]     | det 8, minors (2, 3, 5)          | wrong $D_6$ branch          |
| $\alpha^{-1}$  | root of $F_{U(1)}(\alpha)=0$                           | $c_3, M=41$ | [E]     | 137.0359992                      | no/second root (excl.)      |
| masses         | $\varphi_0^{\text{ret}}$ -ladder $\lambda_Y^L \Lambda$ | $R, L, c_3$ | [E]/[C] | 9 masses, mixings                | hierarchy mismatch          |
| $\theta_{12}$  | TBM + seam $\varepsilon=c_3$                           | $R, L$      | [E]/[C] | 0.307                            | seam-misalign. lemma fails  |
| $\Lambda, A_s$ | seam det. / $R^2$ scalaron                             | $c_3$       | [C]     | $\Lambda, A_s, n_s, r$           | ambient RP fails            |

Marker key: [E] exact / machine-proven · [C] conditional (holds under named hypotheses) · [O] open / axiom · [X] falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.

Reading the table top-to-bottom is reading the theory: two red axioms in, everything else a consequence with exactly one failure mode each.

*The ledger itself is versioned:* each row of `verification/status_ledger.csv` carries a claim ID, its dependencies, its verification script, and `active/canonical_status/supersedes` fields, so claims that were later superseded (e.g. the early “flavor gate open” rows, now *ratios closed* via `FLAV.RIGID`) are marked inactive — the canonical status is the only one read as current.

## The compact status formula: three honest layers

$$\underbrace{\text{compiler}}_{\text{closed}} \mid \underbrace{\text{admissible IR physics}}_{\text{conditionally closed (RP, gap)}} \mid \underbrace{\text{strict physical TOE}}_{\text{open}}$$

TFPT v5.4 closes the discrete compiler, the algebraic Standard-Model readout, the EM fixed point, the admissible gapped IR sector and the  $R+R^2$  spectral-action shadow. It does *not* yet certify a strict physical TOE. The remaining items have now collapsed sharply: the  $R$ -bridge amplitude normalisation  $U_{\text{point}}$  *reduces to*  $v_{\text{geo}}$  (Gate 1 closed, **v75** — the same anchor as  $1/G$ ), and the ambient metric measure is *holographically reduced* to a finite seam-boundary measure (Gate 2 tier B, **v76**) — which is in turn the rigorously-constructed  $(E_8)_1$  lattice net (**v77**), so  $G_{\text{metric}}$  is imported into existing RCFT rigor rather than built anew. So the only genuine residuals are *one* dimensionful scale  $v_{\text{geo}}$  — the *dimensional-analysis floor* that no theory escapes (**v78**), shared by flavor and gravity — the boundary net identification (now *closed modulo a cited theorem*, **v367/v368**, **v376–v379**, `Lean FORM.SEAM.MMST.01`), and the named QFT/cosmology transfer interfaces ( $F_{\text{frontier}}$ ).

**Two points that are *not* in the residual (closed, but easy to mis-list as open).** (i) *Parameter-freeness is a theorem under the named hypotheses:* the boundary transport is gapped ( $\Delta = 6 \log \frac{3}{2} > 0$ ), so by Perron–Frobenius its iteration has a *unique* attracting fixed point — the constants are *selected*, not tuned ([E] for the attractor, [`verification/v56_unique_attractor.py`]; the *physical identification* of the transport operator stays [C]), reinforcing the static bootstrap web ( $g_{\text{car}} = 5$  forced three ways *given* the  $E_8$  closure rank — a consistency loop, see red team Target B — [`verification/v6_bootstrap.py`], [`verification/v14_carrier_uniqueness.py`]). (ii) *Newton’s constant is not an independent input:* fixing the physical  $\Lambda$  branch ( $(8\pi)^2 \delta_{\text{top}} = \frac{3}{4\pi^2}$ ) pins  $\bar{M}_{\text{Pl}}$  relative to  $\Lambda$ , so  $G_N$  is read out *after* the branch choice + *one* dimensionful induced-gravity anchor ([E]/[C], [`verification/v60_lambda_metrology_branch.py`]), not predicted from pure numbers. What stays irreducible is then just  $\pi$ , the discrete  $E_8$ -closure fixed point (self-consistent, *not* a free axiom; the bootstrap of `tfpt_3`), and *one* dimensionful scale  $v_{\text{geo}}$  — and that single scale is shared by *both* the flavor sector ( $U_{\text{point}} \rightarrow v_{\text{geo}}$ , **v75**) and gravity ( $1/G$ , **v68**): the two  $[A]$  anchors are *one* (dimensional analysis; no metre from pure mathematics).

## The hard reduction: the live residual as typed interfaces

After the compiler closure, the entire residual is

$$\text{Rest} = v_{\text{geo}} \oplus G_{\text{net}} \oplus F_{\text{transfer}}$$

— one dimensionful scale anchor, one metric-sector inclusion, and a set of deliberately typed QFT/cosmology interfaces (the historical gate labels  $(U_{\text{wall}})/(G_{\text{metric}})/(F_{\text{frontier}})$  are kept only for ledger continuity). In current terms:  $(U_{\text{wall}}) \rightarrow v_{\text{geo}}$  — the flavor ratios are *closed* (Plücker/Readout Rigidity, [v49/v71/v75](#)), what remains is *one dimensionful unit*, not a flavor gap;  $(G_{\text{metric}}) \rightarrow G_{\text{net}}$  — reduced to a single boundary-net theorem, “the seam–Calderón net is the index-4  $\mu_4$  simple-current extension of the carrier net” (holomorphy then follows, [v89/GATE.METRIC.06](#)); since the QBL chain ([v108–v113](#)) the *carrier choice rides on the same theorem*: the certified scalar kernel exists iff it pairs the sheets ([v110](#)), the certified channel counts the code identically ([v112](#)), pair transport is minimally complete ([v111](#)), and the one quasi-free kernel — rank =  $c$ : 5 carrier block, 8 seam hull — is the defining datum of the free net the gate already posits ([v113](#)), so closing  $G_{\text{net}}$  also closes the carrier selection [\[E\]](#);  $(F_{\text{frontier}}) \rightarrow F_{\text{transfer}}$  — named QFT/cosmology *transfer* interfaces (Koide source→pole,  $\eta_B$  Boltzmann, axion relic,  $m_p/m_e$ ), never compiler powers; the four are one typed functor with a four-axiom contract ([CONTRACT.F.01](#), [v213](#)) and *one shape* — each is a gapped relaxation to a unique attractor (the main-branch discrete→dynamic motif), with  $F_{\text{pole}}$  at the *seam* rate  $(2/3)^6$  and the other three external ([v303](#)). The metric-sector reduction lands on a *single named keystone*. The boundary inclusion  $G_{\text{net}}$  is the index-4  $\mu_4$  simple-current extension  $(D_5)_1 \otimes (A_3)_1 \rtimes \langle (1, 1) \rangle \cong (E_8)_1$  (Simple-Current Extension Theorem, [v154](#), exact: index 4,  $c=8$ ,  $\mu=1 \Rightarrow$  holomorphic  $\Rightarrow E_8$ ); net existence and full-cone reflection positivity are discharged to [\[E\]](#) ([v175](#)); and the **No-Unit Theorem** ([v153](#)) makes  $v_{\text{geo}}$  an irreducible metrology primitive ( $U_{\text{point}} \sim v_{\text{geo}}$ ,  $1/G \sim v_{\text{geo}}^2$ ,  $m/\mu = e^{3/4}$  — one unit in three dimensions). The flavor selector and Q-geometry residues are *derived* (no discrete freedom left). The *one* seam-side identification, packaged as the single named theorem [SEAM.EQUIV.01](#) — *the raw reflection-positive seam state is the holomorphic  $(E_8)_1$  net at  $\tau=i$*  — is now *closed modulo a cited theorem*: the explicit lattice model ([v367/v368](#)) and the  $S_3$  stack ( $c=8$ , 248 currents/1 primary, torus GSD=1, RP; [v376–v379](#)) pin it at every computable level, Lean-formalised as [FORM.SEAM.MMST.01](#), with only the cited MMST/OS continuum-existence theorems left [\[O\]](#) ([v336](#)); its conformal-deck premise [QGEO.SYM.01](#) is its *downstream* definitional face (per [v323](#); the residual is Lean-pinned to named cited steps, [v308](#)). The sprint-by-sprint reduction that produced it is recorded in the changelog and in `tfpt_research_contracts`, not replayed here. The status card:

| Item                                  | Status           | Reading                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------------------------------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| parameter-freeness (self-consistency) | [E]              | gapped transport $\Rightarrow$ <i>unique</i> Perron–Frobenius attractor (v56); bootstrap web $g_{\text{car}}=5$ forced (v6/v14)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Newton constant $G_N$                 | [E]/[C]          | not <i>independent</i> : branch pins $M_{\text{Pl}}$ rel. to $\Lambda$ (v60); readout after branch + one dim. anchor                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| ratio $c_u/c_d = \frac{55}{117}$      | [E]              | $g_{\text{car}}\ \text{Pl}_{1,a}(K)\ _1/(N_{\text{fam}}^2\Delta_Q)$ ; rigid on the derived stratum (v49/v71), no solve                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| flavor absolute scale                 | [E]/[O]          | ratios <i>closed</i> (v49/v71); only the absolute scale remains $= v_{\text{geo}}$ (Gate 1, v75), the same anchor as $1/G$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| $R + R^2$ gravity                     | [E]/[C]          | covariant $f(R)$ field equation closed in regime; heat-kernel grounded (G2)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Einstein eq. (covariant, local)       | [E]/[C]          | fixed-volume entanglement equilibrium $\Rightarrow$ <i>full</i> $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$ with BOTH coefficients<br>parameter-free: $8\pi=1/c_3$ (v358, thermo=geo via $ \mu_4 = \mathbb{Z}_2 \chi=4$ ) and $\Lambda$ from $\alpha$ (v60); Einstein tensor via Lovelock, conservation an output (v359); residual = equation-of-state status + $v_{\text{geo}}$ , global measure (C7) now a [C] redundancy (v369)                                                                                                                                                                                 |
| physical QG sector                    | [E]/[C]          | gap-decoupled admissible measure ( $\Delta_{\text{eff}}=1.648>0$ )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| ambient QG measure                    | [C] (redundancy) | <b>certification object, not missing dynamics</b> (v369; conditional on SEAM.EQUIV.01+Bisognano–Wichmann): holographically reduced to a finite seam- <i>boundary</i> measure (v76); IR tier closed (decoupling); strict-TOE cert. = the simple-current extension<br>( $D_5$ ) $_1 \otimes (A_3)_1 \rtimes \langle(1,1)\rangle \cong (E_8)_1$ (v154); the free-bulk premise (A) is itself a fixed-point theorem with the cone eliminated (cone gap = one-particle gap $(2/3)^6$ ) and factors into the $A_2$ net-existence + the keystone SEAM.EQUIV.01, closed modulo a cited theorem — zero new gates (v160–v165) |
| $v_{\text{geo}}$ metrology unit       | [O](primitive)   | No-Unit Theorem (v153): a dimensionless compiler makes no absolute scale; $U_{\text{point}} \sim v_{\text{geo}}$ , $1/G \sim v_{\text{geo}}^2$ , $m/\mu$ a ratio — one unit, irreducible                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Koide                                 | [C]              | $Q_{\text{src}} + \varphi_0/24$ source $\rightarrow$ pole: typed runnable solver with kill test (QED-excluded, v371)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| axion DM                              | [C]              | $f_a = M_{\text{scal}}/128$ : finite- $T$ spine-branch typed runnable solver (v373)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| $\eta_B$                              | [C]              | readout closed; integrated BDP leptogenesis Boltzmann ODE, typed runnable solver (v372)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| $m_p/m_e$                             | [C]              | cross-sector QCD/EW ratio: typed runnable solver with uncertainty budget (v374)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| $N_*$                                 | [C]              | reheating input; scalaron-reheating point $N_*=51.4$ (v86); frozen band [50, 60]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| $a = (1, 1, 2)$                       | [E]/[C]          | forced by $4 = 2 + 1 + 1$ , conditional on the carrier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| carrier selection $g=5$ (QBL)         | [E]/[C]          | theorem chain v108–v113: sheet-odd kernel, self-counting channel, transport degree 2, one quasi-free kernel (rank = $c$ ); residue = the $G_{\text{net}}$ premise itself — gate closes $\Rightarrow$ carrier [E]                                                                                                                                                                                                                                                                                                                                                                                                   |
| selector establishment R4'            | [E]/[C]          | ( $d, n$ ) $\Rightarrow$ $R$ columnwise (v136); both normals <i>anchor data</i> — $n = (p_2, p_0, e_2)(a)$ in the cusp frame (v145/v149); residue = <i>one</i> assignment                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| sheet/parity (Q-geometry derived)     | [E]/[C]          | $H^1$ characters = $\text{Spec}(Q_+) = A_3$ exponents (v137); $\mathbb{Z}_3$ choice <i>derived</i> (v141); full Möbius $D_4$ matched parity by parity (v146); residue = the module identification, folded into SEAM.EQUIV.01                                                                                                                                                                                                                                                                                                                                                                                       |
| seam quantum clock R1                 | [E]/[C]          | typed closed: resummed clock = entropy power law (v124–v133); VW edge match (v138); det-ratio in-family (v144); ring sum = Born <sup>2</sup> Gaussian zero-mode integral, bend det'-clean (v147); residue = measure + reference                                                                                                                                                                                                                                                                                                                                                                                    |
| $E_8$ Lean                            | [E]              | hardening, not a blocker (script-certified)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

Marker key: **[E]** exact / machine-proven · **[C]** conditional (holds under named hypotheses) · **[O]** open / axiom · **[X]** falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.

Card addendum (2026-07-21): the  $a = (1, 1, 2)$  residual now carries the consolidated P2 reduction state — the position half of the alignment bit is topology (v510), so the dimensionless compiler inputs read *four marks + one square-modulus datum*  $\tau = i + \pi$  (AX.P2.01 stays a declared axiom).

The former structural open core, the *single* named keystone SEAM.EQUIV.01 (the  $G_{\text{net}}$  inclusion: the raw RP seam =  $(E_8)_1$  at  $\tau=i$ ), is now *closed modulo a cited theorem*: the explicit lattice model (v367/v368) and the  $S_3$  stack ( $c=8$ , 248 currents/1 primary, torus GSD=1, RP; v376–v379) pin it at every computable level, it is Lean-formalised as FORM.SEAM.MMST.01, and only the cited MMST/OS continuum-existence theorems stay **[O]** (v336); its conformal-deck face QGEO.SYM.01 is a *corollary* (a conformal net’s vacuum is rotation-invariant by axiom, v323/[`verification/v335_seam_equiv_unify.py`]), not a second item.  $v_{\text{geo}}$  is an irreducible metrology unit and the frontier items are typed interfaces, not structural holes.  $G_{\text{net}}$  is heat-kernel grounded (G2: the spectral action gives  $R+R^2$  structurally) and *gap-decoupled* (G5:  $\Delta_{\text{eff}}=1.648>0$  isolates the closed admissible sector from the ambient), so the ambient measure QG.AMB.01 is no longer carried as an open structural item: it is a **[C]** *redundancy* (v369, conditional on SEAM.EQUIV.01+Bisognano–Wichmann), a certification object rather than missing dynamics — it does not affect any TFPT readout. Both are written up as numbered *research contracts* (lemma chains + closing theorem + machine-certifiability) in the companion `tfpt_research_contracts`, with priority on the  $v_{\text{geo}}$  scale anchor and the  $G_{\text{net}}$ /SEAM.EQUIV.01 inclusion theorem. [`verification/v36_spectral_action_g2.py`]

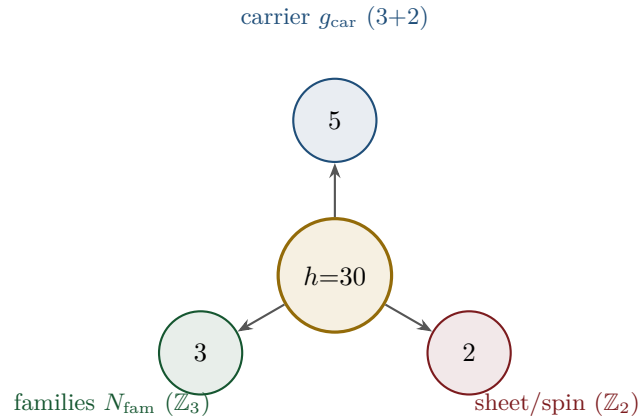
**Sharper still: the structural residual is *one* condition.**  $G_{\text{metric}}$ , the carrier P2 and the red-team Target A are not separate items: they are three provably-equivalent faces of the single condition “*the seam carries no nontrivial abelian sector*” — holomorphy ( $\mu$ -index 1), a homology-sphere seam link ( $\Gamma$  perfect  $\Leftrightarrow 2I$ , [`verification/v232_e8_kleinian_seam.py`]), exactly one 1-dim irrep ([`verification/v219_icosahedral_mckay.py`]) — all of which force  $E_8$  ( $\#(\text{mark-1}) = |H_1| = 1$  only for  $E_8$ ; holomorphic  $c = 8 = g_{\text{car}} + N_{\text{fam}}$  gives the unique even unimodular rank-8 lattice). In abelian Chern–Simons language this is the single integer step holomorphic  $\Leftrightarrow \det K = 1$ , the extension tower  $D_5 \oplus A_3(16) \rightarrow D_8(4) \rightarrow E_8(1)$  being anyon condensation (the Kitaev  $E_8$  state); the one analytic step “the free RP seam condenses the order- $|\mu_4|$  Lagrangian glue” = the single named keystone SEAM.EQUIV.01 (with QGEO.SYM.01 its downstream conformal-deck face) is now *closed modulo a cited theorem* (v367/v368, v376–v379, Lean FORM.SEAM.MMST.01; cited continuum existence **[O]**, v336). So beneath the gate *labels* there is, structurally, a single fundamental postulate. And the entire *emergent-QFT layer* collapses onto the *same* postulate (the *Modular Spectral Closure*, v261): the finite Dirac is a covariance induction of the seam KMS state (v258), the spectral-action cutoff *is* that KMS weight ( $f_2/f_0 = 1$ , v259), the seam, the carrier-16 and  $E_8$  live on one Kummer/K3 surface (v260), so Dirac, cutoff, gauging, glue and orientability are all readouts of the one seam state — the boundary QFT is closed as a single relative object modulo the keystone SEAM.EQUIV.01 and adds *no new open item*, while the ambient quantum-gravity measure is a **[C]** redundancy (v369), not a separate structural gap. [`verification/v234_seam_holomorphy_selection.py`] [`verification/v235_seam_chern_simons.py`]

### Reduction in one line

The number of *structural* assumptions is **two axioms**. Everything else ( $N_{\text{fam}}, \Omega_{\text{adm}}, b_1, \alpha^{-1}$ , the flavor matrix, the  $E_8$  glue, the scale grammar) is a *consequence*. And even those two tighten: the bootstrap note shows  $g_{\text{car}} = 5$  and the integer 8 in  $c_3$  are *overdetermined  $E_8$ -closure fixed points* (rank-fill, Coxeter-match, integer-glue/norm all give 5;  $8 = h(D_5) = \text{rank } E_8 = \det R = \varphi(30)$ ), leaving *one* genuine continuous primitive:  $\pi$  from the Möbius boundary. A *fourth, deeper forcing* comes from the McKay correspondence:  $E_8$  is the exceptional top of the tower of finite  $SU(2)$  subgroups, so its branch data is the icosahedron and the three atoms (2, 3, 5) are its rotation-axis orders — the binary icosahedral group  $2I$  (order  $120 = |R^+(E_8)|$ ) has irrep degrees equal to the affine- $E_8$  Kac marks,  $\sum = 30 = h(E_8) = 2 \cdot 3 \cdot 5$  ([[verification/v219\\_icosahedral\\_mckay.py](#)]). So the *discrete* core has no freely tunable fundamental numbers — the bootstrap is *not creation from nothing* (two inputs remain), but an overdetermination of those inputs. (This concerns the discrete core only; dimensionful masses, the QG measure and frontier items remain typed-open.) [[verification/v6\\_bootstrap.py](#)] [[verification/v14\\_carrier\\_uniqueness.py](#)] [[verification/v23\\_anchor\\_generator.py](#)] [[verification/v219\\_icosahedral\\_mckay.py](#)]

## 4 The compiler: why $30 = 2 \cdot 3 \cdot 5$

The Coxeter number of  $E_8$  is  $h = 30 = 2 \cdot 3 \cdot 5$ . Exactly these three prime factors are the three discrete atoms of the theory — that is the point of the compiler:



From this the compiler reads, among others:

- $\dim S^+ = 1 + 10 + 5 = 16$  (one generation) as the even exterior algebra of the 5-slot carrier — the same Pascal row  $\binom{5}{0}, \binom{5}{1}, \binom{5}{2} = 1, 5, 10$  appears in family, action ladder and the  $E_8$  root count. [E]
- $|R(E_8)| = 16 \cdot 5 \cdot 3 = 240$  and  $\dim E_8 = 240 + (5 + 3) = 248$ . [E]
- the  $\mathbb{Z}_{30}$  Coxeter element  $\leftrightarrow$  the cyclotomic polynomial  $\Phi_{30}$ ; the corner rotation  $z \mapsto iz$  on  $\mathbb{P}^1 \setminus \mu_4$  is the  $A_3$  Coxeter element. [E]
- **why these three primes:**  $E_8$  is the icosahedral top of the McKay tower ( $2I$ , order 120, irrep degrees = affine- $E_8$  marks summing to 30), so 2, 3, 5 are the icosahedron's rotation orders; and the  $E_8$  exponents  $(\mathbb{Z}/30)^\times$  carry an order-4 totative character  $\langle 7 \rangle$  — the same  $\mu_4$  seam clock. [E] [[verification/v219\\_icosahedral\\_mckay.py](#)] [[verification/v223\\_coxeter\\_totative\\_clock.py](#)]
- **a network reading (computational substrate).** On the same McKay (2, 3, 5) graph the affine- $E_8$  marks are the *unique dynamical attractor* of a minimal local-averaging update  $m \leftarrow (A + 2I)/4 \cdot m$  (it converges from *any* positive start to (1, 2, 3, 4, 5, 6, 4, 2, 3)); the spherical tower terminates at  $E_8$  and the cascade  $E_8 \supset E_7 \supset E_6 \dots$  is nested-subgraph coarse-graining. So  $E_8$  reads as the *universal attractor of a minimal* (2, 3, 5) *network* (a Wolfram-style computational-substrate picture) at the Coxeter-skeleton level. [C] A minimal hypergraph *rewrite* reproducing

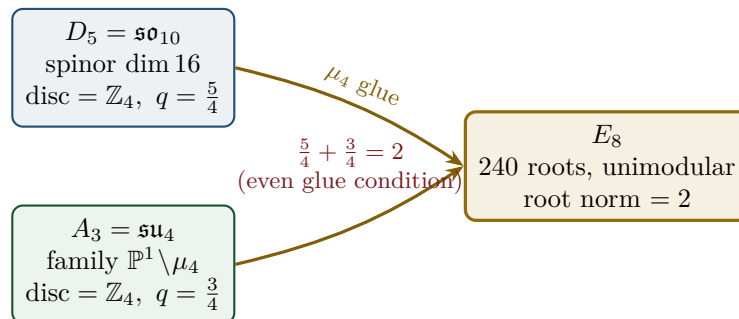


the *full* structure (lattice + recovery + readouts) stays open, and  $\varphi_0^{\text{ret}}$  is the analytic seed  $\frac{4}{3}c_3 = 1/(6\pi)$ , not an edge fraction. [O] [verification/v298\_e8\_network\_attractor.py]

- **the bridge, hardened.** A local witness-based growth dynamics (a *diffusion-generated* witness + the local balance gate  $Am \leq 2m$ , Collatz–Wielandt = Smith’s  $\rho \leq 2$ ; the Perron eigenvector enters only as an audit) *reaches*  $E_8$  as the last finite point ( $E_6 \rightarrow E_7 \rightarrow E_8$ , then stops at the affine boundary  $\hat{E}_8$ ) — but *only* from a 3-arm  $(2, 3, \cdot)$  seed (the controls  $\rho(A_n) = 2 \cos \frac{\pi}{n+1}$  and  $\rho(D_n) = 2 \cos \frac{\pi}{2n-2}$  are  $< 2$  for *all*  $n$ , so neither family ever reaches  $E_8$ ). So  $P_2$  is exactly the irreducible seed on which the rule reaches  $E_8$  as the last finite point — a second network reading of the  $(2, 3, 5)$  hull, not a derivation of  $P_2$ . [O] [verification/v299\_hypergraph\_0d\_autonomous.py]
- **what a rewrite must inject.** Testing *which* load-bearing quantities are graph-spectral sharpens the open question: [E] the Kac marks are the Perron eigenvector ( $Am=2m$ ) and the subleading eigenvalue is exactly  $2 \cos(\pi/5)$  = the golden ratio (the 5-fold/icosahedral signature), but [E] the recovery rate  $(2/3)^6$  is *not* any eigenvalue of the adjacency and  $\varphi_0^{\text{ret}}$  is not a rational edge fraction. The cusp weight  $2/3 = |\mathbb{Z}_2|/N_{\text{fam}}$  is now *derived* from a minimal branching rule ([verification/v327\_hypergraph\_rewrite.py]); what remains non-graph-spectral for a *full*-structure rewrite at leading order is sharpened by [verification/v396\_phi0\_icosahedral.py] (tree-level  $\varphi_0^{\text{tree}} = F/(4h\pi)$ ); the puncture  $48c_3^4$  alone stays open. [O] [verification/v312\_hypergraph\_substrate.py]
- **one coupled local rule (carrier  $\times$  family).** On a  $9 \times 3$  labelled grid (affine- $E_8$  nodes  $\times$  family channels) a *single* coupled micro-step — network lazy diffusion on each cusp column plus the [verification/v327\_hypergraph\_rewrite.py] branching rule  $M$  on each node’s family vector — has joint attractor marks  $\otimes(w=0)$  and recovery rate  $(2/3)^6$  after one clock hand ( $2N_{\text{fam}}$  family steps); the [verification/v324\_hypergraph\_fiber.py] fiber product  $T_{\text{net}} \otimes T_{\text{cusp}}$  emerges as the cusp-readout basis, and the [verification/v299\_hypergraph\_0d\_autonomous.py] growth trajectory  $E_6 \rightarrow E_7 \rightarrow E_8 \rightarrow \hat{E}_8$  is unchanged with the fiber attached. [E] [verification/v395\_hypergraph\_coupled.py]
- **the seed at leading order.** The tree-level retained seed  $\varphi_0^{\text{tree}} = 1/(6\pi)$  is exactly  $F/(4h\pi)$  on the icosahedral hypergraph ( $F = 20$  triangular hyperedges,  $h = 30$ ,  $|\text{Aut}| = 120$ ): equivalently  $(F/(g_{\text{car}}N_{\text{fam}}))c_3$ , with  $F/h = |\mathbb{Z}_2|/N_{\text{fam}} = 2/3$  the same survival ratio as [verification/v327\_hypergraph\_rewrite.py]; Gauss–Bonnet consistent with  $c_3 = 1/(|\mathbb{Z}_2|4\pi)$ . The puncture  $48c_3^4$  remains analytic, not a graph fraction. [C] [verification/v396\_phi0\_icosahedral.py]

## 5 The $\mu_4$ glue: how $E_8$ is really built

The decisive, now *proved* step:  $D_5$  and  $A_3$  have the *same* discriminant group  $\mathbb{Z}_4$ , and their discriminant-form norms are exactly two TFPT constants which add up to the  $E_8$  root norm.



### Glue theorem (proved)

$\text{disc}(D_5) = \text{disc}(A_3) = \mathbb{Z}_4$ , glue index  $|\mu_4| = 4$ , and  $q(D_5) + q(A_3) = \frac{5}{4} + \frac{3}{4} = 2 =$  the  $E_8$  root norm. Hence  $E_8 = (D_5 \oplus A_3) + \mu_4\text{-glue}$  is a *closed lattice-theoretic* construction — not a blind “positing of 248”. [E] [verification/v1\_e8\_glue.py]

## 6 How derivation now works: the derivation map

What can one concretely *compute* with  $\{c_3, g_{\text{car}}\}$ ? The following map summarises what is now a consequence rather than an assumption.

| Sector                       | How it now arises                                                             | Status  |
|------------------------------|-------------------------------------------------------------------------------|---------|
| gauge group / families       | carrier $3+2$ , $\dim S^+=16$ , $N_{\text{fam}}=3$ , $\Omega_{\text{adm}}=48$ | [E]     |
| hypercharge                  | roots of the charge polynomial $bsY^2 + (s-b)Y - 1$                           | [E]     |
| $U(1)$ index $b_1$           | $b_1 = \frac{g_{\text{car}} 2^{g_{\text{car}}-2} + 1}{10} = \frac{41}{10}$    | [E]     |
| fine structure $\alpha^{-1}$ | unique root of $F_{U(1)}(\alpha) = 0$                                         | [E]     |
| flavor matrix                | transport theorem (H1 proved, H2/H3 force the matrix)                         | [E]/[C] |
| electroweak scale            | $v_{\text{EW}} \sim e^{-\alpha^{-1/5}}$ (carrier divides $\alpha^{-1}$ by 5)  | [E]     |
| cosmological constant        | $\Lambda \sim e^{-2\alpha^{-1}}$ (density quotient; see typing note)          | [E]/[C] |
| Hubble scale                 | $H_0 \sim \sqrt{\Lambda}$                                                     | [C]     |
| inflation amplitude          | $A_s = \frac{N_*^2}{24\pi^2} c_3^7$ in the $R^2$ branch                       | [C]     |
| $E_8$                        | $D_5 \oplus A_3 + \mu_4\text{-glue}$                                          | [E]     |

Marker key: [E] exact / machine-proven · [C] conditional (holds under named hypotheses) · [O] open / axiom · [X] falsifiable kill test. Per-claim fine type in verification/status\_ledger.csv.

**Notable:** the scale grammar is *one* exponential engine. The same number  $\alpha^{-1} \approx 137$  generates the electroweak scale (divided by 5 = carrier), the cosmological constant (times 2) and — via the square root — the Hubble scale. One constant becomes many orders of magnitude.

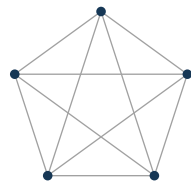
**Typing note on  $\Lambda$  (a deliberate danger zone).** “ $\Lambda$ ” denotes *three* distinct objects that must not be conflated: (i) the seam-determinant exponent  $e^{-2\alpha^{-1}}$  (an exact [E] readout), (ii) the dimensionless density quotient  $\rho_\Lambda/\bar{M}_{\text{Pl}}^4$  (which carries a prefactor *branch*, hence [C]), and (iii) the observed curvature, to which (ii) is *anchored*, not predicted. The exponential is exact; the absolute dark-energy density remains a typed cosmology interface (full disambiguation in `tfpt_2_standard_model`).

### Cosmology is the Pascal action grammar of the carrier

With  $x = v_{\text{geo}}/(5\beta_{\text{rad}}^2\bar{M}_{\text{Pl}})$  the cosmological readouts are a *power tower* on the five-slot carrier graph  $K_5$ :

$$\frac{H_0}{\bar{M}_{\text{Pl}}} = x^5 R_H, \quad \frac{\rho_\Lambda}{\bar{M}_{\text{Pl}}^4} = x^{10} R_\Lambda, \quad (1, 5, 10) = \binom{5}{0}, \binom{5}{1}, \binom{5}{2}$$

EW is the unit  $x^1$ , Hubble is the product over the  $\binom{5}{1} = 5$  vertices,  $\Lambda$  is the product over the  $\binom{5}{2} = 10$  edges — the *same* Pascal triple that reads one generation, the action ladder and the  $E_8$  root count  $16 \cdot 5 \cdot 3 = 240$ . “Hubble = slot product,  $\Lambda$  = pair product.” [E]



$K_5$

- $\binom{5}{1} = 5$  vertices — action  $x^1$  (Hubble slot)
- $\binom{5}{2} = 10$  edges —  $\Lambda$  density pair  $x^{10}$
- $\binom{5}{3} = 10$  triangles — dual sector
- $\binom{5}{4} = 5$  complements — inverse sector
- $\binom{5}{5} = 1$  determinant — closure

$K_5$  on the five carrier slots; the action powers  $x^1, x^5, x^{10}$  are its  $\binom{5}{k}$  faces (mnemonic; the physical scale map stays  $[C]$ ).

### The entire fermion spectrum in one formula

Masses, Yukawa, CKM, PMNS and neutrinos all follow from *one* master formula with *one* seed:

$$\hat{m}_{f,j} = \frac{v_{\text{geo}}}{\sqrt{2}} \lambda_Y^{L_{f,j}} \Lambda_{f,j}, \quad \lambda_Y = \sqrt{\varphi_0^{\text{ret}}(1 - \varphi_0^{\text{ret}})}, \quad \varphi_0^{\text{ret}} = \frac{1}{6\pi} + \frac{3}{256\pi^4}.$$

The word-lengths  $L_{f,j}$  are the *fixed residue matrix of the compiler*, whose characteristic polynomial  $t^3 - 9t^2 + 10t - 8$  carries only compiler numbers ( $\text{tr} = 9 = N_{\text{fam}}^2$ ,  $\det = 8 = h(D_5)$ , principal 2-minors 2,3,5 with product  $30 = h(E_8)$ ). Through the  $\varphi_0^{\text{ret}}$ -ladder  $\hat{m} = \frac{v_{\text{geo}}\pi}{\sqrt{2}} c (\varphi_0^{\text{ret}})^k$  all nine word-lengths are fixed and the *charged-lepton* amplitudes are closed; the *quark mass ratios* are closed too (integer Plücker readouts on the derived selector stratum, **v49/v71**), with only the *absolute* amplitude scale reducing to the single dimensionful anchor  $v_{\text{geo}}$ . **SM status:** the discrete packet, the dimensionless flavor skeleton, the charged-lepton source masses and the quark *ratios* are closed; the *absolute* quark scale and the full PMNS completion are reduced/conditional;  $m_W, m_Z, m_H, \sin^2 \theta_W, \alpha_s$  are RG scheme-layer ( $m_H$  rests on the closed UV quartic  $\frac{1}{16\pi^2}$ ); and the solar angle is *realized* by a *TFPT texture* and reduced to the seam-misalignment lemma ( $\sin^2 \theta_{12} = \frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2}$ ; see below). (*Full derivation: document 2, **tfpt\_2\_standard\_model**.*) [[verification/v18\\_quark\\_yukawa.py](#)] [[verification/v20\\_lepton\\_c\\_derivation.py](#)] [[verification/v46\\_grand\\_mass\\_volume.py](#)]

## 7 Physical, geometric, topological: why the architecture is plausible

The closure works on three levels at once: *physically*,  $\alpha^{-1}$  is the fixed point of *one* equation and one generator drives many scales; *geometrically*, the  $C^+$  spinor and the puncture geometry  $\mathbb{P}^1 \setminus \mu_4$  produce  $D_5$  and  $A_3$ , glued to  $E_8$ ; *topologically*, the common discriminant  $\mathbb{Z}_4$  (the  $\mu_4$  winding) forces the glue and  $\Phi_{30}$ /Coxeter drives the sectors — with exactly 2 axioms and the rest a consequence. **Why the compiled architecture is plausible.** It is (i) *parsimonious* (few independent assumptions  $\Rightarrow$  low overfitting risk), (ii) *rigid* (the glue condition  $5/4 + 3/4 = 2$  and the discriminant  $\mathbb{Z}_4$  leave little freedom), and (iii) *checkable* (machine-checkable  $E_8$  appendix, explicit fixed-point equation for  $\alpha^{-1}$ ). More explained from less — exactly the criterion that makes a theory more probable.

## 8 What role $E_8$ now plays exactly

$E_8$  is no longer the starting point but the *output format*:

- **Bookkeeping:**  $E_8$  is the unimodular container in which carrier ( $D_5$ ) and family ( $A_3$ ) fit together consistently. Its numbers 240, 248 are carrier traces, not inputs.
- **Consistency test:** unimodularity ( $\det 1$ ) is a hard condition; it closes only because the discriminant-form norms  $5/4$  and  $3/4$  (both already present in TFPT) add up to 2.  $E_8$  thus tests the internal coherence of the two atoms.
- **Not an end in itself:**  $E_8$  does not explain “everything”; it is the smallest even unimodular hull of the datum  $D_5 \oplus A_3$ . The physics sits in the *two inputs*, not in  $E_8$ .

**New — the secondary atlas.** Beyond that,  $E_8$  is an *audit container*: each maximal subalgebra projects a TFPT module. Document 3 (**tfpt\_3\_e8\_audit\_bootstrap**, Part I) shows the seven slices of 248 explicitly and labelled. The strongest new hit is that  $E_6 \times A_2$  reads the flavor matrix:

$$248 = \|R\|_F^2 + \det R + 2(\mathbf{1}^\top R a) N_{\text{fam}} = \underbrace{78}_{\dim E_6} + \underbrace{8}_{\dim A_2} + 2 \cdot 27 \cdot 3,$$

with  $\|R\|_F^2 = 78 = \dim E_6$ ,  $\det R = 8 = \dim A_2 = h(D_5)$ . The discipline rule is: *every load-bearing number must appear in at least one  $E_8$  projection* — this makes the structure falsifiable rather than numerological. (Audit level; all identities machine-verified.)

## 9 What this means for the theory-of-everything question

### The state of the five questions — closed, reduced or typed

1. **Flavor residual — ratios closed; only the absolute scale is the wall-selection.** Mehta–Seshadri supplies the equivalence-class framework (stable parabolic  $\leftrightarrow$  unitary representation); the algebraic selector  $R$ , the  $Q$  spectrum and the splitting type are fixed (now *derived*:  $\det R = 8$  lattice,  $\text{Spec}(Q_+)$  from  $D_4$ , v69/v70; since v136/v139 the dual pair  $(d, n)$  forces  $R$  *columnwise*,  $d$  is pure anchor data, and  $\text{Spec}(Q_+) = (1, 2, 3)$  is the  $A_3$ -exponent grading on the seam curve’s cohomology, v137), and the charged-lepton amplitudes are closed. The quark mass *ratios* ( $c_u/c_d = \frac{55}{117}$  etc.) are then *constant* on this discrete selector stratum (Readout Rigidity, v49/v71) — integer Plücker readouts, no transcendental solve. Only the *absolute* amplitude scale reduces to the selected *polystable* point on the parabolic stability wall ( $U_{\text{point}}$ , an anchor; Research Contract 1). [E]/[O]
2. **Gravity — the local field equation is parameter-free.** The metric-sector *field equation* is supplied by the entanglement first law  $\delta S = \delta \langle K \rangle$  at fixed *volume* (Jacobson 2015): equilibrium forces the *full* covariant Einstein equation  $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$  with *both* coefficients fixed —  $8\pi = 1/c_3$  (v358) and  $\Lambda$  from  $\alpha$ ,  $\rho_\Lambda = (3/4\pi^2)e^{-2\alpha^{-1}}$  (v60) — the Einstein tensor (not Ricci) forced by Lovelock, so  $\nabla^a T_{ab} = 0$  is an *output* (v359). The  $R + R^2$  scalaron is the *action*/inflation layer (the next-order term of the same equilibrium route, v360), and  $v_{\text{geo}}$  is the theory’s single dimensionful input (v364). What remains is *not* the field equation but the equation-of-state status (Jacobson framing, not a from-action quantisation; an *external* candidate action — Bianconi’s entropic action, arXiv:2408.14391 — is quantified in v473 without changing this typing) and the one unit  $v_{\text{geo}}$ ; the ambient measure QG.AMB.01 is now a [C] redundancy (v369, conditional on SEAM.EQUIV.01+Bisognano–Wichmann), not missing dynamics, and *perturbative* graviton unitarity is established — the Stelle ghost is a Seeley–DeWitt truncation artefact (v304), the spin-2 sector ghost-free via the entire KMS form factor (v370) with the truncation tower decoupling (v380). [E] (local field equation) / [C] (perturbative unitarity, ambient redundancy)
3. **Strong CP — decoupled.**  $\theta_{\text{eff}} = 0$  follows from three structural facts ( $\gamma_5$ -Hermiticity, polar structure, sheet involution) + reflection positivity — *without* a mass gap. [E] (given RP)
4. **Full dynamics — the admissible transfer model is closed, and the boundary QFT is now one relative object.** The mass gap is the *explicit* discrete gap  $6 \log \frac{3}{2}$  of the  $C_6$  transfer operator (eigenvalues  $(k/3)^6$ ), which supports the OS reconstruction path on the admissible sector. On top of this the whole emergent-QFT round assembles the boundary theory into one relative object  $\text{TFPT}_{\text{QFT}} = (A_\Sigma, \omega_\Sigma, \Delta_\Sigma, \rho, A_F, H_F, D_F, J, \gamma, S_{\text{rel}})$  — the *Modular Spectral Closure*: the finite Dirac is a covariance induction of the seam KMS state (v258), the spectral-action cutoff *is* that KMS weight (v259), the seam/carrier/ $E_8$  live on one Kummer/K3 (v260), all certified together (v261) — closed modulo the single keystone SEAM.EQUIV.01, itself now *closed modulo a cited theorem* (lattice +  $S_3$  stack, Lean FORM.SEAM.MMST.01; v367/v368, v376–v379). *What this does not fully close*: the full measured pole masses (the absolute scale  $v_{\text{geo}}$  + standard RG), the absolute QCD scales ( $m_p/m_e$ , now a typed [C] transfer, v374), the complete PMNS dynamics (the angles are typed; the full mass/CP dynamics stay [C]/[O]), and the real 4D interacting QFT — the continuum limit, whose cited existence theorems stay [O] — with the ambient measure a [C] redundancy (v369). [E] (transfer model + boundary relative object) / [C] (continuum modulo cited, transfers, ambient redundancy)
5. **Axioms P1/P2 — P2 machine-verified.** The Lean proof builds cleanly (3357 jobs, 0 sorry, only kernel axioms); the P2 algebra is a theorem, P1 a typed interface. They remain declared inputs. [E]/[O]

In other words: the theory has shrunk from “many open audit points” to “*standard steps* (cluster expansion, Fredholm, truncation bound) + two declared axioms”. The five questions are now *closed, reduced or typed* rather than open-ended — a qualitative jump in discipline, not a claim of completion. (*Details and proof chains: document 2, tfpt\_2\_standard\_model, Part III.*)

**The closure ledger: what is done, and what genuinely remains**

A complete theory must close the discrete structure, the dimensionless constants, the mass spectrum, gravity, and the quantum measure. The current state, graded honestly:



| TOE piece                                                                                                        | Status  | If not complete: what closes it                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------------------------------------------------------------------------------------------------------------------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| gauge group, $N_{\text{fam}}$ , hypercharge, $b_1$                                                               | [E]     | — complete                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| $E_8 = (D_5 \oplus A_3) + \mu_4$ , group closure                                                                 | [E]     | — complete                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| bootstrap $(g, \mu) = (5, 4)$ , “8” in $c_3$                                                                     | [E]     | — complete (reverse glue $\mu^2 - 5\mu + 4 = 0$ )                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| fine structure $\alpha^{-1}$                                                                                     | [E]     | — <b>existence+uniqueness proved</b> ; value is a numerical fixed point (ledger EM.FP.01), Ward origin [C]                                                                                                                                                                                                                                                                                                                                                                                                            |
| fermion masses ( $\varphi_0^{\text{ret}}$ -ladder)                                                               | [E]/[O] | leptons exact; quark <i>ratios</i> closed (Plücker, v49/v71); absolute scale = anchor                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| mixings $\theta_{13}, \theta_{23}, \theta_{12}$ (PMNS angles)                                                    | [E]/[C] | $\theta_{12}$ cond. derived; $\theta_{23}$ maximal; <i>full</i> PMNS dynamics (mass scale, CP magnitude) stay [C]/[O]                                                                                                                                                                                                                                                                                                                                                                                                 |
| flavour matrix / H2                                                                                              | [E]     | dictionary <i>exact</i> (hexagon = twisted family spectrum, addresses pinned, margins forced, v118–v122); $R$ forced columnwise by $(d, n)$ (v136/v139); pairing values = atom identities (v142/v145); residue = <i>one</i> lift map                                                                                                                                                                                                                                                                                  |
| seam quantum clock (R1)                                                                                          | [E]/[C] | typed closed: reduced edge-sector clock (v124–v133); external VW edge = $2 \times$ reduced budget, full 4d det = firewall (v138); det-ratio cancellation derived in-family (v144)                                                                                                                                                                                                                                                                                                                                     |
| inflation $A_s, n_s, r$ , scalaron $M$                                                                           | [C]     | scalaron $M = c_3^{7/2} \bar{M}_{\text{Pl}}$ exact; $n_s, r$ are bands over $N_* \in [50, 60]$ (reheating point 51.4, v86)                                                                                                                                                                                                                                                                                                                                                                                            |
| boundary QFT (Modular Spectral Closure)                                                                          | [E]/[C] | <b>one relative object</b> TFPT <sub>QFT</sub> : $D_F$ = covariance induction (v258), cutoff = KMS weight $f_2/f_0=1$ (v259), seam/carrier/ $E_8$ on one K3 (v260), certified (v261) — closed modulo SEAM.EQUIV.01, itself closed modulo a cited theorem (v376–v379)                                                                                                                                                                                                                                                  |
| perturbative 4D S-matrix ( $S_{\text{pert}}$ )                                                                   | [E]/[C] | <b>Epstein–Glaser-constructible</b> (v269): one-loop quartic (v271) + gauge $\beta = (41/10, -19/6, -7)$ (v273), $S_{\text{pert}} \xrightarrow{\text{LSZ}} S_{\text{phys}}$ with one-loop unitarity (optical theorem, v278); the spin-2 graviton sector is perturbatively unitary too — the Stelle ghost is a Seeley–DeWitt truncation artefact (v304/v370/v380)                                                                                                                                                      |
| <i>the actual TOE remainder — dimensionful units, typed transfers, and the cited-theorem ceiling on the seam</i> |         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| QFT measure (admissible sector)                                                                                  | [E]     | <b>closed: RP from cusp spectrum, OS reconstruction</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| full measured pole masses                                                                                        | [C]     | ladder $\times v_{\text{geo}}$ , dressed by standard RG (no new physics); $v_{\text{geo}}$ over-determined (v274)                                                                                                                                                                                                                                                                                                                                                                                                     |
| absolute QCD scales ( $m_p/m_e$ , $\Lambda_{\text{QCD}}$ )                                                       | [C]     | typed runnable transfer with uncertainty budget (v374, contract FR.MPME.01); not a compiler number                                                                                                                                                                                                                                                                                                                                                                                                                    |
| full PMNS dynamics (mass scale, CP)                                                                              | [C]/[O] | angles typed; CP strength $J_{\text{PMNS}} = -0.0297$ derived (v270); absolute $\nu$ -scale = one seesaw ratio (v272); three named post-hoc [O] candidates: third-gen ladder-base (v467), $\Delta m^2$ ratio = $ J $ (v468), carrier-normalised one-parameter seesaw (v481) — record unchanged                                                                                                                                                                                                                        |
| (1) SEAM.EQUIV.01 — one named theorem ( <i>raw seam</i> = $(E_8)_1$ at $\tau=i$ )                                | [C]     | <b>closed modulo a cited theorem</b> : the explicit lattice model (v367/v368) and the $S_3$ stack ( $c=8$ , 248 currents/1 primary, torus GSD=1, RP; v376–v379) pin it at every computable level, Lean-formalised as FORM.SEAM.MMST.01 (collar hypotheses kernel-proved, MMST/OS as named cited axioms, clean #print axioms); the <i>only</i> residual that stays [O] is the abstract continuum existence of the scaling limit (v336); extension leg crossed-product-certified, realisation at invariant level (v469) |
| (2) QG.AMB.01 — ambient measure ( <i>redundancy, not missing dynamics</i> )                                      | [C]     | <b>discharged as a redundancy</b> (v369; v379 RP→OS), conditional on SEAM.EQUIV.01+Bisognano–Wichmann: a certification object, gap-decoupled (margin $1.648 > 0$ , v330/v335) so no readout depends on it; obstruction characterised (v332) + conditionally resolved by GHP+IDG (v334); Tier-B’s holomorphy bit ( $\det K=1$ , v277) is absorbed into SEAM.EQUIV.01                                                                                                                                                   |
| analytic boundary origin of $\pi$ ( $c_3 = \frac{1}{8\pi}$ )                                                     | [C]     | <b>hardened</b> : $c_3 = 1/( \mathbb{Z}_2  \oint_{S^2} K) = 1/(8\pi)$ ( <b>Gauss–Bonnet</b> ); “8” = rank $E_8$ ; $\pi$ stays the one primitive                                                                                                                                                                                                                                                                                                                                                                       |
| $\eta_B, m_p/m_e$ , Koide, dark matter                                                                           | [C]     | typed runnable transfer solvers with kill tests (v371–v374, observatory CI v375; see document 4)                                                                                                                                                                                                                                                                                                                                                                                                                      |

Marker key: **[E]** exact / machine-proven · **[C]** conditional (holds under named hypotheses) · **[O]** open / axiom · **[X]** falsifiable kill test. Per-claim fine type in *verification/status\_ledger.csv*.

**The honest bottom line:** the discrete core, the dimensionless constants, the *mass-hierarchy exponents* and the mixings are *closed*; the nine masses follow from one  $\varphi_0^{\text{ret}}$ -ladder (lepton rationals exact, quark  $O(1)$  digits still a conditional finite evaluation). The *admissible-sector* QFT measure is closed (a finite gapped transfer system, RP from the cusp spectrum, OS-reconstructed), and on top of it the whole boundary QFT is now *one relative object*  $\text{TFPT}_{\text{QFT}}$  (the *Modular Spectral Closure*, v258–v261:  $D_F$  a covariance induction, the cutoff the seam KMS weight, seam/carrier/ $E_8$  on one Kummer/K3) — closed modulo the single keystone **SEAM.EQUIV.01**. *What that closure does not reach* is exactly the dimensionful/ambient remainder: the full measured pole masses (ladder  $\times v_{\text{geo}}$  + standard RG, **[C]**), the absolute QCD scales ( $m_p/m_e$ , now a typed **[C]** transfer, v374), the complete PMNS dynamics (**[C]**/**[O]**), and the nonperturbative *ambient* measure **QG.AMB.01**, now itself a **[C]** redundancy (v369). (The metric-sector *field equation* is parameter-free at the local level — the full covariant  $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$  from fixed-volume entanglement equilibrium, v358/v359 — and the ambient *measure* is no longer an open structural item but a certification object, conditional on **SEAM.EQUIV.01**+Bisognano–Wichmann.) The *perturbative* 4D QFT is now built: the spectral-action S-matrix is Epstein–Glaser-constructible and LSZ-bridged to the physical asymptotic S-matrix with one-loop unitarity for the matter+gauge sector (v269–v278), and the spin-2 graviton sector is perturbatively unitary too — the  $R^2/\text{Weyl}^2$  Stelle ghost is a Seeley–DeWitt *truncation artefact* (v304), ghost-free via the entire KMS form factor and the Barnes–Rivers decomposition (v370) with the truncation tower decoupling (v380); the whole bedrock reduces to one geometric postulate, *the raw seam is the flat  $\tau=i$  pillowcase* (v276), with Tier-B’s holomorphy bit  $\det K=1$  (v277) absorbed into **SEAM.EQUIV.01**. The former single open lemma, the named theorem **SEAM.EQUIV.01** — *the raw reflection-positive seam state is the holomorphic  $(E_8)_1$  net at  $\tau=i$*  — is now *closed modulo a cited theorem*. (The flat  $\tau=i$  geometry and the  $E_8$  holomorphy are *two faces of one object*: the  $(E_8)_1$  character is  $\chi_{E_8}=j^{1/3}$ , so  $\chi_{E_8}(i)=1728^{1/3}=12$  and  $\tau=i$  is at once the order-4 CM point and the  $(E_8)_1$  partition-function modulus, v282.) The explicit lattice model (v367/v368) and the  $S_3$  stack ( $c=8$ , 248 currents/1 primary, torus  $\text{GSD}=1$ , RP; v376–v379) pin it at every computable level, and it is Lean-formalised as **FORM.SEAM.MMST.01** (collar hypotheses kernel-proved, the MMST scaling-limit and Adamo–Moriwaki–Tanimoto OS-reconstruction theorems as named cited axioms, clean **#print axioms**); the *only* residual that stays **[O]** is the abstract continuum existence of the scaling limit (v336) — its 128-spinor extension leg since re-founded on the peer-reviewed crossed-product package ( $h_s=16/16=1 \in \mathbb{Z}$  is the Longo–Rehren locality criterion; KLM  $\mu=4/2^2=1$ ; AGT/AMT the second witness), the realisation input reduced to invariants (R1’; computed  $|C|=1$ ,  $\nu=16$ ), Lean parallel route **seamResidualClosed’** (v469) — so it is closed *modulo a cited theorem*, not solved, Lean-pinned to exactly those named steps plus the derived Recovery gap  $\Delta = 6 \ln(3/2) \approx 2.43 > 0$  (v308); the conformal-deck premise **QGeo.SYM.01** is its *downstream* definitional face (v323). This is by design the *one* irreducible structural input of the theory — the role the *constancy of  $c$*  plays in relativity. It is not a defect to be removed but the single, sharply-localised, falsifiable premise on which the whole boundary closure turns; with the lattice/ $S_3$  pinning and the Lean formalisation in place, “closing the theory” has reduced to invoking exactly the two cited continuum-existence theorems, not an open-ended search. The full sprint-by-sprint reduction that pinned **SEAM.EQUIV.01** is recorded in the changelog, not replayed here. The seam seed is *hardened*:  $c_3 = 1/(|\mathbb{Z}_2| \oint_{S^2} K dA) = 1/(8\pi)$  is the one-sided Gauss–Bonnet normalisation of the seam sphere, with the discrete “8” equal to rank  $E_8$  and only the Gauss–Bonnet  $\pi$  left as the single irreducible primitive. What remains for a full TOE is then the cited continuum-existence theorem on the seam and the one unit  $v_{\text{geo}}$ , plus the standard-RG dimensionful masses and the typed frontier transfers (the ambient QG measure being a **[C]** redundancy, v369). No load-bearing discrete parameter remains free, and  $\pi$  is the lone continuous primitive. *4D-QFT fork policy (frozen, v265)*: the canonical 4D reading is *boundary only* — a 4D-GUT is *not claimed by default* ( $E_8$  is the audit hull; SM-only unification is killed, v246); carrier-native Pati–Salam is a separately-typed, falsifiable UV branch (thresholds + proton decay). Not an open ambiguity, a decision tree.

## 10 TFPT in the light of algorithmic physics: Zuse–Schmidhuber–Wolfram

The “compiler” framing places TFPT in an older lineage — the idea that physics is, at bottom, the output of a short computation — and it is worth stating precisely what TFPT shares with that lineage and where it deliberately parts company, so the family resemblance is not mistaken for a claim the theory does not make. Three landmarks fix the coordinates. **Zuse** (*Rechnender Raum*, 1969) proposed that the universe is literally a (cellular-automaton) computation. **Schmidhuber** sharpened this into an *algorithmic* theory of everything (1997, 2000): the shortest program that outputs a universe is the natural prior — a physics reading of Kolmogorov complexity and Solomonoff induction — while his *Speed Prior* (2002) adds that a computed universe should be one that is *fast* to compute, and his low-complexity-art programme (1997) identifies beauty with compressibility. **Wolfram** (*A New Kind of Science*, 2002; the Physics Project) recasts the same instinct combinatorially: space–time as a hypergraph updated by simple rewrite rules.

### Three shared instincts

1. **Minimal description length as the selection principle.** “No load-bearing free number, only  $\pi$ ” is, in these terms, a Kolmogorov-style compression statement — the plausibility discussion below already names it *Kolmogorov-like compression: many numbers from little* — and the grammar-internal null model of the next section turns it from rhetoric into a measured bound.
2. **Deterministic, resource-bounded generation.** The readouts fall out of a chain of exact identities in about a second, not a search over a model space — the same spirit as the Speed Prior’s preference for *fast*-computable worlds, though TFPT makes no claim to instantiate a speed prior.
3. **Compressibility as the aesthetic.** “Choosing  $E_8$  is choosing the icosahedron” selects the maximally symmetric, shortest-to-describe object; the affine Kac marks (1, 2, 3, 4, 5, 6, 4, 2, 3) carry a fourfold identity (representation dimensions = network attractor = spectral certificate = local balance weight). This is Schmidhuber’s *beauty = compressibility*, realised on the Standard-Model alphabet rather than on pixels.

### Three deliberate divergences

- (A) **One forced program, not an ensemble.** Schmidhuber’s shortest program computes *all* universes and privileges none; TFPT bets the opposite — an over-determined bootstrap fixed point that *forces* a single dimensionless skeleton ( $g_{\text{car}}=5$  fixed three independent ways). TFPT is a candidate for *the* short program of our constants, not a measure over programs.
- (B) **The description language is algebraic.** TFPT’s compression is measured in a Lie/latitude/VOA language, not in Turing bit-strings; Kolmogorov complexity is language-dependent up to an additive constant, so the honest question — is the shortness real, or hidden in the  $E_8$  machinery? — is exactly what the null model and the “forced three ways” bootstrap are built to answer. A Wolfram-style rewrite reading sharpens the same point rather than dissolving it: the (2, 3, 5) seed is the terminal of a minimal  $T(2, 3, r)$  growth rule (Cartan determinants 3, 2, 1, 0,  $-1$  for  $E_6, E_7, E_8, \hat{E}_8, E_{10}$ ), so the rewrite picture *re-formulates* axiom P2 rather than removing it — both pictures bottom out on the same single irreducible seed.
- (C) **Falsifiability — the one thing TFPT adds.** As *physics*, the Zuse/Schmidhuber algorithmic TOE and Wolfram’s rule search are essentially unfalsifiable: no dated measurement can kill them. TFPT freezes dated, kill-tested predictions (the registry `predictions_frozen.json`, the predictions section below) and so puts the computational-universe intuition at genuine empirical risk. That, not the compression rhetoric, is the load-bearing difference.

*Status.* This section is *positioning*, not a new result: it introduces no formula and moves no status marker. Its machine-checked hooks already live in the suite — the grammar-internal null-model bound and the icosahedral/McKay bedrock with its network attractor; the Wolfram-style rewrite correspondence is worked out separately in the repository’s hypergraph note.

## 11 Does this make the theory more probable?

**Plausibility balance.** *Pro:* (a) two inputs instead of many; (b)  $\alpha^{-1}$  to  $2.9 \times 10^{-10}$  (CODATA-2022,  $1.9\sigma$ ) as a *fixed point*, not a fit; (c)  $E_8$  constructed, not postulated; (d) one compiler generates several sectors  $\Rightarrow$  fewer degrees of freedom, more predictive power.

*Contra/open:* what remains are only *standard steps* (cluster expansion in the thermodynamic limit, the Fredholm property, the truncation bound) and the two declared axioms P1/P2; on the SM side the dimensionful EW/QCD masses ( $m_W, m_Z, m_H, \sin^2 \theta_W, \alpha_s$ ) sit on the RG scheme layer. The solar angle, previously the one open SM number, is now *conditionally derived* ( $\frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2}$ , modulo the seam-misalignment lemma). A final verdict “it maps reality” requires this short list plus the near-term experimental tests (below).

Net: the parsimony (Kolmogorov-like compression: many numbers from little) and the high precision of individual closures raise the a-posteriori plausibility substantially — without hiding the open points.

## 12 Predictions for running and upcoming experiments

Because *one* generator now sits behind the numbers, the theory makes a set of concrete, falsifiable statements that running or near-term experiments are actively testing. Each row gives the TFPT value, the *new* derivation behind it, the experiment, and an honest status.

| Observable                          | TFPT value & derivation                                                                                                                                                              | Experiment (running / upcoming)                                                                                                      | St.     |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|---------|
| solar<br>$\sin^2 \theta_{12}$       | prediction of record $\frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2} = 0.306747$ ; TBM + seam misalignment $\varepsilon = \frac{3}{4}\varphi_0^{\text{ret}} = c_3 + 36c_3^4$ (cond.) | <b>JUNO</b> : first 59.1 d run $0.3092 \pm 0.0087$ (compatible), precision phase pending                                             | [E]/[C] |
| reactor<br>$\sin^2 \theta_{13}$     | $\varphi_0^{\text{ret}} e^{-5/6} = 0.0231$ ; seed $\times$ carrier-trace $e^{-\gamma}$ , $\gamma = \frac{5}{6}$                                                                      | Daya Bay / RENO / JUNO (PDG 0.0220)                                                                                                  | [E]     |
| atmospheric<br>$\sin^2 \theta_{23}$ | $\approx \frac{1}{2}$ ( $\mu\tau$ -symmetric limit; octant not selected)                                                                                                             | NOvA / T2K / <b>DUNE</b> (octant ambiguity; NuFIT 6.0 pre-JUNO baseline)                                                             | [C]     |
| fine structure $\alpha^{-1}$        | 137.0359992 as the unique root of $F_{U(1)}(\alpha)=0$ (no fit, no second branch)                                                                                                    | CODATA-2022 (data through 2022)<br>137.035999177(21): dev. $2.9 \times 10^{-10}$ ( $1.9\sigma$ of its uncertainty)                   | [E]     |
| tensor ratio $r$                    | $r = \frac{12}{N_*^2} \in [0.0033, 0.0048]$ (frozen band, $N_* \in [50, 60]$ ); scalaron-reheating point 0.0045 ( $N_*=51.4$ , v86)                                                  | now $r < 0.036$ (BICEP/Keck BK18); <b>CMB-S4</b> ( $\sigma_r \leq 5 \times 10^{-4}$ , obs. $\sim 2033$ )                             | [C]     |
| tilt $n_s$                          | $n_s = 1 - \frac{2}{N_*} \in [0.960, 0.967]$ (same band); scalaron-reheating point 0.9611 ( $N_*=51.4$ , v86)                                                                        | Planck ( $0.9649 \pm 0.0042$ ); CMB-S4 sharpens                                                                                      | [C]     |
| amplitude $A_s$                     | $\frac{N_*^2}{24\pi^2} c_3^7$ ; $2.0 \times 10^{-9}$ at $N_*=55$ (the measured $A_s$ prefers fast reheating, v86)                                                                    | Planck ( $2.1 \times 10^{-9}$ )                                                                                                      | [C]     |
| neutron EDM                         | $\theta_{\text{eff}} = 0 \Rightarrow d_n$ essentially null (structure + RP)                                                                                                          | <b>PSI nEDM</b> / SNS (running)                                                                                                      | [E]     |
| CP phase $\delta_{\text{CKM}}$      | $\delta = \frac{\pi}{3} + 3\lambda^2 = 68.65^\circ$ (frozen of record, v88); survives at $+0.98\sigma$ vs $\gamma_{\text{PDG}}$                                                      | <b>LHCb</b> / Belle II $\gamma$ combination; decision at $\sigma_\gamma \leq 0.96^\circ$                                             | [E]     |
| cosmic birefringence $\beta$        | $\beta = \frac{\varphi_0^{\text{ret}}}{4\pi} = 0.2424^\circ$ (seam readout)                                                                                                          | ACT DR6 hint $0.215^\circ \pm 0.074^\circ$ (systematics not yet understood — not a validation target); <b>Simons Obs.</b> / LiteBIRD | [C]     |
| $0\nu\beta\beta$ / ordering         | Majorana branch; normal-ordering preference                                                                                                                                          | <b>LEGEND</b> / <b>nEXO</b> , DUNE, KATRIN                                                                                           | [C]     |
| flavor invariants                   | $\det R = h(D_5) = 8$ , principal minors (2, 3, 5), $\chi_R = t^3 - 9t^2 + 10t - 8$ (exact)                                                                                          | any future CKM/PMNS global fit must satisfy these                                                                                    | [E]     |
| $H_0$ vs. $\Lambda$                 | $v_{\text{EW}} \sim e^{-\alpha^{-1/5}}$ , $\Lambda \sim e^{-2\alpha^{-1}}$ , $H_0 \sim \sqrt{\Lambda}$ : one engine                                                                  | SH0ES / <b>DESI</b> / Planck (Hubble tension)                                                                                        | [C]     |

Marker key: [E] exact / machine-proven · [C] conditional (holds under named hypotheses) · [O] open / axiom · [X] falsifiable kill test. Per-claim fine type in verification/status\_ledger.csv.

### The two decisive near-term tests

- (1) **JUNO** and  $\theta_{12}$ . JUNO is *taking data since August 2025* and targets sub-percent  $\sin^2 \theta_{12}$ . TFPT commits to the single prediction of record  $\frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2} = 0.306747$  (misalignment  $\varepsilon = \frac{3}{4}\varphi_0^{\text{ret}} = c_3 + 36c_3^4$ ). The first 59.1-day JUNO run reports  $\sin^2 \theta_{12} = 0.3092 \pm 0.0087$  — *compatible*, but the precision phase is still pending; this is a first compatibility check, not yet the decisive test. A converged central value away from  $\sim 0.307$  at high significance falsifies the seam mechanism.
- (2)  $r$ . The  $R^2$  scalaron with  $M$  fixed by  $c_3^7$  predicts a small  $r = 12/N_*^2$ : the frozen band  $r \in [0.0033, 0.0048]$  over  $N_* \in [50, 60]$ , with the scalaron-reheating point  $r = 0.0045$  ( $N_*=51.4$ , v86) — already below the current bound  $r < 0.036$  (BICEP/Keck BK18) and within reach of CMB-S4 ( $\sigma_r \leq 5 \times 10^{-4}$ , first observations  $\sim 2033$ –34). A detection of  $r \gtrsim 0.01$  would be incompatible with the  $R^2$  branch on which  $M_{\text{Pl}}$  and  $A_s$  rest.

**Structural (non-experimental) checks.** Independent of any apparatus, the machine-checkable  $E_8$  appendix (all 240 roots, Gram matrices, discriminant groups) lets anyone verify that the two TFPT atoms really generate  $E_8$ ; and the discipline rule “*every load-bearing number appears in at least one  $E_8$  projection*” (atlas note) turns the number stock into a falsifiable raster rather than

numerology. The numerology charge is additionally *quantified* by an explicit null model: under the declared formula grammar the alignment density is anomalously high — a random theory of equal formula complexity essentially does not reproduce the scorecard. This is a *grammar-internal null-model bound*, not a Bayesian probability that TFPT is true; the explicit figure and its controls are stated in the verification appendix [E] [verification/v100\_numerology\_null\_mc.py].

#### Freeze file: committed kill criteria

To prevent post-hoc rescue, the predictions above are frozen with explicit kill criteria in the machine-readable registry `verification/freeze_file.csv`. The decisive entries:

- **solar**  $\sin^2 \theta_{12} = \frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2} \approx 0.3067$  — a JUNO central value clearly away from 0.307 kills the seam-misalignment mechanism;
- **tensor**  $r \approx 0.0040$  — any robust  $r \gtrsim 0.01$  kills the  $R^2$  branch carrying  $M_{\text{P1}}, A_s$ ;
- **ordering** normal, small  $m_{\beta\beta}$  — inverted ordering or large  $m_{\beta\beta}$  kills the Majorana branch;
- **strong CP**  $\theta_{\text{eff}} = 0$  — a solid nEDM signal above SM background falsifies the structural cancellation;
- $w = -1$  — a robust  $w \neq -1$  kills the single-engine dark-energy readout.

The proton/electron ratio is explicitly *not* claimed as a compiler power (only fails if mis-asserted as one).

**Blind-prediction registry (frozen 2026-06-09, machine-enforced).** Beyond the kill criteria, the *values* themselves are now frozen: `verification/predictions_frozen.json` registers every dimensionless prediction of record at 25 significant digits *before* JUNO / CMB-S4 / LiteBIRD / DESI decide, and [verification/v84\_frozen\_registry.py] re-derives each decimal from the two axioms on every suite run (formula $\leftrightarrow$ value lock; covered by `manifest.sha256`; ledger `REG.FREEZE.01`). In particular the registry admits exactly *one*  $\theta_{12}$  prediction of record (the seed  $\frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2} = 0.306747$ ) — the seam and non-linear variants are typed as derived variants of the same texture and can never be silently promoted afterwards.  $r$  and  $n_s$  enter only as *bands* over the declared external  $N_\star \in [50, 60]$ , never as point predictions.

#### What likely follows next.

- the quark mass *ratios* are now closed combinatorially (integer Plücker readouts on the derived selector stratum, `v49/v69/v71`); the only residual is the *absolute* amplitude scale, an anchor ( $U_{\text{point}}$ ) of the same nature as the one dimensionful scale — no new physics;
- further constants as carrier traces (analogous to  $b_1 = 41/10$ ), since the compiler systematically produces traces over  $S^+$ ;
- a sharper geometry $\leftrightarrow$ gravity bridge once the Hodge/ $R^2$  branch is lifted from [C] to [E].

## Conclusion

TFPT's sectors form an impressive collection of closures sharing common numbers. The compiler closure explains *why* they are the same numbers: a small discrete generator  $\mathbb{Z}_{30} = 2 \cdot 3 \cdot 5$  builds, from two inputs  $\{c_3, g_{\text{car}}\}$  via  $D_5 \oplus A_3 + \mu_4$ , the  $E_8$  container and outputs the Standard Model, the constants and the scale grammar from there. The theory thereby becomes *more parsimonious, more rigid and more checkable*. Precisely nameable gaps remain — one geometric flavor realisation, the conditional gravity/QFT closures, and two declared axioms — but the path to closure is now short and concrete.

## Appendix: computational verification (reproducibility)

Every claim in this document marked [E] (exact identity), [E] (Lie/lattice theorem) or [E] (numerical fixed point) is re-derived from the two axioms  $\{c_3, g_{\text{car}}\}$  alone by a small, self-contained Python suite in the repository folder `verification/`. The inline references [verification/...] throughout the documents point to the exact script; the table below is the master map (its source is the shared fragment `tex-artefacts/verification.tex`).



| Script                             | What it machine-checks                                                                                                                                                                                                                   |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v1_e8_glue.py                      | glue theorem $E_8=(D_5\oplus A_3)+\mu_4$ : $\text{disc}=\mathbb{Z}_4$ , $q(D_5)+q(A_3)=\frac{5}{4}+\frac{3}{4}=2$ , $240=16\cdot 5\cdot 3$ , $248=240+8$                                                                                 |
| v2_carrier_pascal.py               | $g_{\text{car}}=5$ unique Pascal solution; $16=1+5+10$ , $N_{\text{fam}}=3$ , $\text{rank } E_8=8$ , $\Omega_{\text{adm}}=48$ , $b_1=\frac{41}{10}$                                                                                      |
| v3_em_alpha.py                     | EM closure: unique root of $F_{U(1)}(\alpha)=0$ , $\alpha^{-1}=137.0359992168$ , existence/uniqueness, inverse test, ablation                                                                                                            |
| v4_flavor_matrix.py                | residue matrix $R$ : $\det=8$ , minors $\{2, 3, 5\}$ , $\chi_R=t^3-9t^2+10t-8$ , SNF $\text{diag}(1, 1, 8)$ , $\ R\ _F^2=78$ , $\sum L=40$                                                                                               |
| v5_e8_cascade.py                   | cascade $D_n=60-2n$ : endpoints, exponent rungs $\rightarrow 240$ , IR tail 46080, variance 6552                                                                                                                                         |
| v6_bootstrap.py                    | reverse glue $\mu^2-5\mu+4=0$ ; $g_{\text{car}}=5$ three ways; $8=\text{rank } E_8=h(D_5)=\varphi(30)$                                                                                                                                   |
| v7_gravity_cosmo.py                | scalaron exponent 7, $M_{\text{scal}}^2/\bar{M}_{\text{Pl}}^2=c_3^7$ , $A_s, n_s, r$ , $\Omega_b$ , $\sin^2 \theta_{12}$                                                                                                                 |
| v8_horizon.py                      | horizon code: $\frac{1}{2\pi}=4c_3$ , $1920= W(D_5) $ , $S_{dS}$ , Page, $\beta_{\text{rad}}=0.2424^\circ$                                                                                                                               |
| v9_neutrino_texture.py             | solar angle: explicit $\mu\tau$ Majorana texture $\rightarrow \sin^2 \theta_{12}=\frac{1}{3}-\frac{\varphi_0^{\text{ret}}}{2}$ , $\theta_{23}=45^\circ$ , $\theta_{13}=0$                                                                |
| v10_projection_involution.py       | $Q, \Sigma$ algebra: $K=R+Q\Sigma$ , $L=R+Q(I+\Sigma)$ ; $\chi_Q, \chi_K$ ; det ladder $3\cdot 4\cdot 8\cdot 20=1920= W(D_5) $ ; $a^\top K a=41$                                                                                         |
| v11_unique_KQ.py                   | $K, Q$ are the <i>unique</i> nonneg-int matrices with their compiler budgets, spectrum and monotone rows (enumeration)                                                                                                                   |
| v12_mass_generation_polynomials.py | sector/generation polynomials of $K$ and the anchor-block det ladder (9, 10, 16, 40)                                                                                                                                                     |
| v13_open_gates.py                  | gate closures: $M=41=10b_1=\sum L+N_\Phi$ ; $Q_+=A_3$ exponents, $Q_-^2=N_{\text{fam}}$                                                                                                                                                  |
| v14_carrier_uniqueness.py          | $g_{\text{car}}=5$ unique ( $N_{\text{fam}} \in \mathbb{Z}^+$ , $\text{rank } E_8=8$ ); split (3, 2) from $b+s=5$ , $b^2+s^2=13$ ; $\text{Tr } Y=\text{Tr } Y^3=0$                                                                       |
| v15_bootstrap_classification.py    | $D_5\oplus A_3$ unique familyful cyclic-glue of $E_8$ (16-spinor + 3 families), glue $\mathbb{Z}_4$                                                                                                                                      |
| v16_solar_dual_anchor.py           | $a^\top R^{-1}=a^\top L^{-1}=(-\frac{1}{2}, -\frac{1}{2}, 1)$ ; $\sin^2 \theta_{12}=\frac{1}{3}-\frac{\varphi_0^{\text{ret}}}{2}$ ; full PMNS                                                                                            |
| v17_hexagonal_resolvent.py         | finite resolvent $D_y^{-1}=(\sum y^{5-m}\delta^m U_6^m)/(y^6-\delta^6)$ (quark $c$ backbone)                                                                                                                                             |
| v18_quark_yukawa.py                | quark source ratios $\frac{m_u}{m_d}=0.470085$ , $\frac{m_c}{m_s}=13.61$ , $\frac{m_t}{m_b}=40.80$ exact; lepton $c=(\frac{16}{7}, \frac{4}{3}, \frac{7}{6})$ ; full hierarchy                                                           |
| v19_monodromy_moduli.py            | exact pole $z_\star=\frac{794-7\sqrt{9961}}{2187}$ ( $3^7$ ); $D_4$ $SU(3)$ monodromy constructible but $D_4$ -fixed locus positive-dim $\Rightarrow$ exact quark $c$ reduce to $(U)$                                                    |
| v20_lepton_c_derivation.py         | lepton $c$ 's derived: $\delta=\frac{1}{2}$ resolvent $c= \mu_4 ^w/(\frac{5}{4}-\cos \frac{r\pi}{3}) \Rightarrow \frac{16}{7}, \frac{4}{3}$ ; product $\frac{2^{g_{\text{car}}}}{N_{\text{fam}}^2}=\frac{32}{9} \Rightarrow \frac{7}{6}$ |
| v21_solar_product_quark.py         | solar coeff $=q(A_3)=\frac{3}{4}$ (glue-norm, $q(D_5)+q(A_3)=2$ ), $\sin^2 \theta_{12}=\frac{1}{3}-\frac{\varphi_0^{\text{ret}}}{2}$ ; product $\frac{2^{g_{\text{car}}}}{N_{\text{fam}}^2}$ ; quark law fails ( $\frac{1}{7}$ vs 0.724) |
| v22_open_gates_audit.py            | residual-gates audit contract: exact reduction of each open gate (A2,B3,B4,B5,B6,C7) machine-pinned (forced part vs. named residual)                                                                                                     |
| v23_anchor_generator.py            | anchor $a=(1, 1, 2)$ : $e_k=(4, 5, 2)$ , $c_3=\frac{1}{2e_1\pi}$ ; $p_n=2+2^n \Rightarrow 240, 248$ , binary ladder; inputs $\rightarrow \{a, \pi\}$                                                                                     |
| v24_quark_ratio_closure.py         | quark ratios $\frac{55}{117}, \frac{34}{47}, \frac{3}{26}$ from TFPT blocks; mass ratios match $< 0.03\%$ ; rational $c$ gauge, $\Lambda_q$ in $O(1)$                                                                                    |
| v25_frontier_conjectures.py        | conjectures: Koide $Q_{\text{src}}+\frac{\varphi_0^{\text{ret}}}{24}\approx\frac{2}{3}$ ; axion $f_a=M_{\text{scal}}/128\rightarrow m_a\approx 23.8\text{ }\mu\text{eV}$                                                                 |

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| Script                                     | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                              |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v26_flavor_<br>frontier_<br>unification.py | the “11” is not uniquely forced ( $\Rightarrow$ [C]); flavor frontier unifies to one ( $U$ ) gate; cusp config on the parabolic stability wall                                                                                                                                                                                           |
| v27_wall_<br>representative.py             | explicit balanced wall rep $W_{\text{wall}}$ (cols = perm(0, 1, 2), rows $w=(2, 1, 1)$ , pardeg 0); selector $\det R=8$ , $\text{Spec}(Q_+)=\{1, 2, 3\}$                                                                                                                                                                                 |
| v28_gravity_fR.py                          | $R+R^2$ closed: $f(R)$ , $M=c_3^{7/2}\bar{M}_{\text{Pl}}$ scalaron, $N_*=57\rightarrow n_s, r, A_s$ ; full metric measure open; $248c_3^2 < 6\log \frac{3}{2}$                                                                                                                                                                           |
| v29_research_<br>contract_certs.py         | research-contract certificates: $C_U^{(1)}$ wall enumeration (1296 $\rightarrow$ 144 $\rightarrow$ 5 orbits; symmetry alone does not pick $W_{\text{wall}}$ ) and $G5$ gap-dominance $2\ V\  < \Delta$                                                                                                                                   |
| v30_d4_character_<br>variety.py            | $C_U^{(2)}$ : full $D_4$ -fixed $SU(3)$ character variety is positive-dimensional $\Rightarrow$ selector needs the $R(\rho)$ dictionary (H2; exact since v118–v122, residue = the $R4'$ selector establishment)                                                                                                                          |
| v31_R_<br>dictionary.py                    | $R(\rho)$ characterized: case A alive (irreducible $\Rightarrow$ mixing); $R$ integer but $\rho$ -invariants continuous $\Rightarrow R(\rho)$ is the transcendental non-abelian-Hodge map (residual = Hitchin solve)                                                                                                                     |
| v32_rh_<br>splitting.py                    | RH route: $\sum_k U^k A_0 U^{-k} = 4 \text{diag}(A_0)$ exact $\Rightarrow$ splitting<br>$\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2 \Leftrightarrow \text{diag}(A_0) = (\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$ (algebraic); $\prod M_k = I$ + $R$ -bridge stay open ( $c_u/c_d$ not forced)                                               |
| v33_explicit_flat_<br>bundle.py            | explicit valid ( $U_{\text{wall}}$ ) flat bundle (RH solve): cusp + $\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$ + $\ M_\infty - I\  \sim 10^{-9}$ ( $\prod M_k = I$ ) + irreducible (case A)                                                                                                                                              |
| v34_h2_bridge_<br>attempt.py               | H2 bridge: explicit per-puncture $M_k$ (cusp, $\prod M_k = I$ ); honest negative — natural extraction $\neq$ lepton amplitudes, the $\Gamma^{\min}$ dictionary is the remaining input ( $c_u/c_d$ not obtained)                                                                                                                          |
| v35_quark_qcd_<br>boundary.py              | the open gates are the discrete $\leftrightarrow$ continuous boundary: quark cross-ratio cleanness tracks generation/QCD sensitivity (compiler hands off to continuous physics)                                                                                                                                                          |
| v36_spectral_<br>action_g2.py              | G2: spectral action $\rightarrow R+R^2$ ( $a_2 \sim -R/3$ , $a_4 _{R^2} \sim R^2/72$ ); $M^2/\bar{M}_{\text{Pl}}^2 = 6(4\pi)^2/f_0$ , $c_3^7$ fixes $f_0$ ; gap-decoupling $\Delta_{\text{eff}} = 1.648 > 0 \Rightarrow$ admissible IR closed under RP, ambient (G6) = strict-TOE completion target                                      |
| v37_plucker_<br>anchor.py                  | anchor-plane Plücker apparatus: $\ Pl(K)\ _1 = 11$ , $\ Pl_R(K)\ _1 = 26$ , $\sum Pl_R(L) = 60$ ; pencil $\det(K+xQ) = 3x^3 + 7x^2 + 6x + 4$ ; $K_e - K_u = a$ , $L_e - L_u = \text{Exp}(A_3)$ ; lepton ring $c_e c_\tau =  \mathbb{Z}_2  c_\mu$ ; Koide $u \rightarrow \frac{53}{54}u$ ([C])                                            |
| v38_uwall_<br>killswitch.py                | ( $U_{\text{wall}}$ ) U2 kill-switch hardened: $D_4$ -cusp rep irreducible at all 400 sampled points (commutant=1), mixing $\geq 0.35$ , reducible locus measure-zero $\Rightarrow$ case A generic, ( $U_{\text{wall}}$ ) survives; amplitude dictionary still research-level                                                            |
| v39_uwall_<br>selectors.py                 | ( $U_{\text{wall}}$ ) selector side closed on the explicit point: splitting type = anchor $a=(1, 1, 2)$ , pardeg = 0; $\det R=8=n\cdot a$ , $\text{Spec}(Q_+)=\{1, 2, 3\}=3\alpha+1$ read off the bundle; only the amplitudes $c_u/c_d$ ( $\Gamma^{\min}$ Hitchin) remain                                                                |
| v40_harmonic_<br>metric.py                 | ( $U_{\text{wall}}$ ) harmonic metric = FINITE linear algebra (polystable $\Rightarrow$ unitary $\Rightarrow\Phi=0$ ): unique invariant form $H>0$ , unitarised $\widetilde{M}_k$ cusp-class with $ \widetilde{M}_k  \in \{0, \frac{1}{2}, \frac{1}{\sqrt{2}}\}$ ; the feared Hitchin PDE collapses to unitarisation + combinatorial $R$ |
| v41_leg_<br>assignment.py                  | ( $U_{\text{wall}}$ ) final leg test: amplitudes = $\delta = \frac{1}{2}$ resolvent (closed); $\Lambda_\mu > 1$ rules out holonomy moduli; harmonic holonomy $ \text{diag}  = (0, \frac{1}{2}, \frac{1}{2})$ supplies $\delta = \frac{1}{2}$ (suggestive); honest negative on literal $c_u/c_d$ reproduction                             |
| v42_exterior_<br>leg.py                    | ( $U_{\text{wall}}$ ) Exterior Leg Lemma: $u, d$ share $(r, w)=(1, 1)$ so any scalar leg gives $c_u/c_d \equiv 1$ ; the $u/d$ split is the $\Lambda^2 F$ area $\text{Pl}_{1,a}(K)$ , $\ \cdot\ _1 = 11$ generated $\Rightarrow \frac{5\cdot 11}{9\cdot 13} = \frac{55}{117}$ [E]; phys. identification [C]                               |
| v43_exterior_<br>bridge.py                 | ( $U_{\text{wall}}$ ) $\Lambda^2 F$ bridge: $\Lambda^2(\widetilde{M}) = \widetilde{M}$ (exterior leg = $\bar{3}$ ) confirmed; continuous action $\neq$ integer $\text{Pl}(K) \Rightarrow$ bridge is the discrete H2 invariant ( $\det R=8$ , $\text{Spec}(Q_+)$ confirmed), not continuous                                               |

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| Script                         | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                            |
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| v44_carrier_exterior.py        | $(U_{\text{wall}})$ Lie-level grounding: $16=\Lambda^{\text{even}}(5)=1+10+5$ ; $u^c \in \Lambda^2(5)$ , $d^c \in \Lambda^4(5) \Rightarrow$ exterior degrees $(2, 4)$ , $\text{sum}=6= R^+(A_3) $ , $\text{diff}=2= \mathbb{Z}_2 $ ; the exterior leg is the carrier's own $\Lambda^2$ grading (no special math)                                                                                       |
| v45_family_exterior.py         | $(U_{\text{wall}})$ the “11” is Pascal: $ \text{Pl}(K) =(1, 4, 6)=\{\Lambda^0, \Lambda^1, \Lambda^2\}(\mu_4)$ ;<br>$\ \text{Pl}(K)\ _1=11=\sum_{k \leq 2} \binom{4}{k}=16-g_{\text{car}}$ (cumulative ext of $\mu_4$ up to $\deg u^c=2$ );<br>$15=\dim \mathfrak{su}(4)=10+5=N_{\text{fam}}g_{\text{car}}$                                                                                             |
| v46_grand_mass_volume.py       | absolute mass-scaling is [E], not a product rule: sector dets<br>$(\varphi_0^{\text{ret}})^6, (\varphi_0^{\text{ret}})^9, (\varphi_0^{\text{ret}})^{10}$ (exponents $ R^+A_3 , N_{\text{fam}}^2, \mathcal{A}_\Lambda \Rightarrow$<br>$\det M_{\text{SM}} \sim (\varphi_0^{\text{ret}})^{25}=(\varphi_0^{\text{ret}})^{g_{\text{car}}^2}$ ; $Q$ -rows $(4, 5, 6)$ sum $15=\dim A_3$ ; $25+15=40=\sum L$ |
| v47_selection_theorem.py       | Boundary Carrier Selection (Thm A): $\mu_4 \rightarrow A_3, N_{\text{fam}}=3$ ; 16-spinor $\rightarrow D_5$ ;<br>$q(D_5)+q(A_3)=2$ , glue $4= \mu_4 $ ; only $D_5 \oplus A_3$ meets all four conditions<br>( $D_8, E_7+A_1, E_6+A_2, A_4+A_4, A_8$ excluded) [E]/[C]                                                                                                                                   |
| v48_em_ward.py                 | EM boundary Ward (Thm C): $F_{U(1)} = \text{Maxwell } \alpha^3 - \text{Calderón } 2c_3^3 \alpha^2 - \text{transport}$<br>$8b_1 c_3^6 \log \frac{1}{\varphi_{\text{seam}}}$ ; $8b_1 = \frac{4}{5} M = \frac{164}{5}$ , $-\frac{5}{4} = q(D_5)$ [E]; Ward origin [C]                                                                                                                                     |
| v49_readout_rigidity.py        | Readout Rigidity (Thm U2): $c_u/c_d = \frac{55}{117}$ is constant on the discrete selector<br>stratum $\mathcal{S}_{8,Q} \Rightarrow$ no point-uniqueness needed; $(U_{\text{wall}}) = U_{\text{unitary}} + U_{\text{H2}} + U_{\Lambda^2} + U_{\text{point}}$ ,<br>only $U_{\text{point}}$ open                                                                                                        |
| v50_q_geometry.py              | Q geometry (Thm Q): $Q_+ = \text{diag}(3, 2, 1)$ -block $\chi=(t-1)(t-2)(t-3)$ , $Q_-$<br>$\chi=t(t^2-3)$ ; $Q$ unique under rows $(4, 5, 6)/\text{cols}(9, 5, 1)/\text{spectra}$ ; geom. origin [C]                                                                                                                                                                                                   |
| v51_boundary_half_step.py      | glue norms $=(g_{\text{car}}, N_{\text{fam}})/ \mu_4 $ : $q(D_5)=\frac{5}{4}$ , $q(A_3)=\frac{3}{4}$ ; four ops forced $\rightarrow \{\text{sum}=2$<br>root norm, $\text{diff}=\frac{1}{2}= \mathbb{Z}_2 / \mu_4 =\delta$ , $\text{prod}=\frac{15}{16}$ , $\text{ratio}=\frac{5}{3}\}$ . So $\delta=\frac{1}{2}$ is the<br>carrier–family count over the glue index, not chosen [E]                    |
| v52_pencil_endpoints.py        | $K+xQ$ pencil endpoints: $P(-1)=2= \mathbb{Z}_2 $ , $P(0)=4= \mu_4 $ , $P(1)=20=\det L$ ,<br>$P(2)=68=2p_5(a)$ ; $\det(K-Q)=2$ , $\text{tr}=N_{\text{fam}}$ (audit ext. of v37) [E]                                                                                                                                                                                                                    |
| v53_compiler_core.py           | Big picture: the whole integer skeleton from $(g_{\text{car}}, N_{\text{fam}})=(5, 3)$ — rank $E_8=8$ ,<br>$ \mathbb{Z}_2 =2$ , Pythagorean $\Delta_Y=g_{\text{car}}^2=25=N_{\text{fam}}^2+ \mathbb{Z}_2  \cdot \text{rank } E_8=9+16=\dim S^+$ (diff. of<br>squares); anchor char-poly $(t-1)^2(t-2)$ unique [E]                                                                                      |
| v54_seam_horizon_keystones.py  | The “8” in $c_3=\frac{1}{8\pi}$ overdetermined and gravitationally aligned<br>$(2 \mu_4 =\text{rank } E_8=h(D_5)=\varphi(30)=8\pi)$ Hawking/Einstein); one boundary transport<br>operator (spec $\{1, (2/3)^6, (1/3)^6\}$ , $\lambda_2=(2/3)^6$ ) governs <i>both</i> SM flavor gap and<br>horizon Page recovery [E]                                                                                   |
| v55_coxeter_cycle.py           | Computed $E_8$ Coxeter element: order exactly $30= \mathbb{Z}_2 N_{\text{fam}}g_{\text{car}}$ , eigenphases =<br>exponents = totatives of 30, so rank $E_8=8=\varphi(30)$ = live phases of the order-30<br>cycle; one $\alpha^{-1}$ sets EM, $S_{dS}$ and $\Lambda$ ( $S_{dS}\rho_\Lambda=32\pi^4$ ) [E]                                                                                               |
| v56_unique_attractor.py        | Gapped boundary transport (gap $6 \log \frac{3}{2} > 0$ ) $\Rightarrow$ <i>unique</i> attracting fixed point,<br>geometric rate $(2/3)^6$ (two starts $\rightarrow$ same direction); Coxeter in $ \mu_4 =4$ planes;<br>entropy reset = rank-1 projector [E]                                                                                                                                            |
| v57_horizon_crosslinks.py      | Horizon cross-links: $c_3=\frac{1}{8\pi}$ is the Einstein/Jacobson coefficient ( $1/4=1/ \mu_4 $<br>entropy density); Hod QNM $\omega_R/T_H = \ln 3 = \ln N_{\text{fam}}$ ; Hawking power denom<br>$1920= W(D_5) $ ; one $\alpha^{-1}$ for EM/ $S_{dS}/\Lambda$ /Hubble [E] (+ [C] identification)                                                                                                     |
| v58_seam_horizon_chain.py      | Seam-horizon chain (precise: seam = abstract normaliser, locally realised as a<br>horizon): one-sided $S^2$ Gauss–Bonnet $c_3=1/(2 \cdot 4\pi)=\frac{1}{8\pi}$ ; KMS $4c_3\kappa$ ;<br>$S_{BH}=M^2/2c_3$ ; RT in seam units. Seam–Horizon Theorem (area-law from the<br>seam kernel) [O] open                                                                                                          |
| v59_area_law_evidence.py       | Area-law necessary condition met: a reflection-positive Gaussian (Calderón-type)<br>kernel yields an area law (gapped block-EE saturates; gapless $\sim \frac{\epsilon}{3} \log L$ ), by the<br><i>same</i> gap that fixes the attractor; $8= \mathbb{Z}_2  \mu_4 $ . The $c_3$ coefficient stays [O] open                                                                                             |
| v60_lambda_metrology_branch.py | $\Lambda$ branch selection: $(8\pi)^2 \delta_{\text{top}} = \frac{3}{4\pi^2}$ , mis-scale $2c_3/\delta_{\text{top}} = \frac{64\pi^3}{3} = 661.47 \Rightarrow G_N$<br>pinned as output; 123 orders = 119.028+3.920; ACT DR6 birefringence<br>$0.215^\circ \pm 0.074^\circ$ vs $0.2424^\circ$ ( $0.37\sigma$ ) [E]                                                                                       |

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| Script                        | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| v61_cft_bridge.py             | Boundary-CFT mirror: level-1 WZW central charges $c(E_8)=8$ , $c(D_5)=5$ , $c(A_3)=3$ = (rank $E_8$ , $g_{\text{car}}$ , $N_{\text{fam}}$ ); conformal embedding $c_{\text{coset}}=0$ ; $SU(3) \leftrightarrow SU(4)$ reconciliation $N_{\text{fam}}= \mu_4 -1$ , $4 \rightarrow 3+1$ (one geometry $\mathbb{P}^1 \setminus \mu_4$ , two sides) [E]                                                                                                                                                       |
| v62_data_scorecard.py         | TFPT vs 2024/25 data: $\alpha^{-1}$ , $\sin^2 \theta_{12}$ , $\beta_{\text{rad}}$ match; $\sin^2 \theta_{13}$ ( $2.0\sigma$ ) and Starobinsky $n_s$ vs DESI ( $\sim 2\sigma$ ) mild tensions; $\theta_{23}$ open; $r \approx 0.004$ future falsifier [E]                                                                                                                                                                                                                                                  |
| v63_seam_engineering_index.py | Exact Seam-Engineering Index $\Xi=2\ V\ /\Delta=\frac{31}{24\pi^2 \log(3/2)} \approx 0.323$ ( $2\ V\ =\frac{31}{4\pi^2}$ , $248=\dim E_8$ , $31=1+h^\vee$ ), $\Delta_{\text{eff}} \approx 1.648$ ; four-class possibility taxonomy [E]/[O]                                                                                                                                                                                                                                                                |
| v64_causal_boundary_nogo.py   | Causal Boundary Engineering <i>conditional no-go</i> : gapped transfer has no periodic orbit (discrete no-CTC); RP+OS+gap $\Rightarrow$ ANEC $\Rightarrow$ Topological Censorship $\Rightarrow$ no traversable shortcut; only ER=EPR bridge survives [E] (core)/[C] (chain)                                                                                                                                                                                                                               |
| v65_falsification_layer.py    | Prediction layer with explicit failure thresholds: 3 core ( $v_{\text{GW}}=c$ , no 4th gen, seam=8), 4 observational ( $\beta$ , $r$ , $n_s$ , $\theta_{12}$ ), no-go + unitarity; $N_\star$ demoted to [C] input; none violated [E]                                                                                                                                                                                                                                                                      |
| v66_e8_casimir_degrees.py     | Organizing simplification: the compiler atoms <i>are</i> the $E_8$ Casimir degrees $\{2, 8, 12, 14, 18, 20, 24, 30\}$ ( $2= \mathbb{Z}_2 $ , $8=\text{rank}$ , $14=\dim G_2$ , $20=\det L$ , $24= W(A_3) $ , $30=h(E_8)$ ); $\sum=128=2^7$ (dS prefactor), $\sum \exp=120= R^+(E_8) $ [E]                                                                                                                                                                                                                 |
| v67_area_law_coefficient.py   | Central-theorem <i>structural closure</i> : Fursaev–Solodukhin $S=4\pi kA$ , $k=c_3/2 \Rightarrow S=2\pi c_3 A=\frac{1}{4}A$ ( $2\pi c_3=\frac{1}{4}$ ); $c_3=\frac{1}{8\pi}$ is the <i>unique</i> value with replica-coeff = BH $\frac{1}{4}$ . Reduces to the one coeff $k=c_3/2$ (then <i>forced</i> , v73) [E]                                                                                                                                                                                        |
| v68_seeley_dewitt_residual.py | Residual <i>resolved as an anchor</i> : $a_2=-\frac{d}{192\pi^2}R \Rightarrow$ the <i>absolute</i> $\frac{1}{16\pi G}$ is UV-sensitive ( $\propto d\Lambda^2 f_2$ ), so the <i>scale</i> $1/G$ is the one dimensionful anchor (the dimensionless $k=c_3/2$ is forced, v73); cutoff-indep. $R^2$ /scalon derived [E]                                                                                                                                                                                       |
| v69_d4_q_geometry.py          | $D_4$ -equivariant Q-geometry (gate [C]/[E]): on $H_1(\mathbb{P}^1 \setminus \mu_4)=B_1 \oplus E$ , $Q_+=3w+1=\text{diag}(1, 2, 3)$ (cusp weights on $\mu_4=\mathbb{Z}_4$ eigenspaces), $Q_-=E$ -coupling $\sqrt{N_{\text{fam}}} \Rightarrow t(t^2-3)$ ; $\Sigma$ -split $=D_4=\mathbb{Z}_4 \rtimes \mathbb{Z}_2$ [E]                                                                                                                                                                                     |
| v70_q_integer_lift.py         | Q integer-lift sharpened: $\mathbb{Z}_4$ puncture action $R$ unimodular (eigenvalues $\{-1, i, -i\}$ ); $\det Q=3=N_{\text{fam}}$ , SNF $\text{diag}(1, 1, 3) \Rightarrow$ residual = one lattice invariant $\det Q=N_{\text{fam}}$ , not closable by $D_4$ alone [E]                                                                                                                                                                                                                                     |
| v71_simple_r_bridge.py        | <i>Simple</i> R-bridge (gate #3): selector stratum $\mathcal{S}_{8,Q}$ now fully derived ( $\det R=8=\text{rank } E_8$ lattice + $\text{Spec}(Q_+)=\{1, 2, 3\}$ from v69) $\Rightarrow$ quark <i>ratios</i> are pure integer Plücker readouts ( $\frac{55}{117}$ , no monodromy, v49 rigidity); residual $U_{\text{point}}$ = absolute norm. = an anchor [E]/[O]                                                                                                                                          |
| v72_q_det_from_cusp.py        | $\det Q=N_{\text{fam}}$ derived from the cusp class: $\det Q= \text{coker } Q =\text{cusp-class order}=\text{denominator of the weights } \{0, \frac{1}{3}, \frac{2}{3}\}$ (same data as $\text{Spec } Q_+$ ); $\text{coker } Q=\mathbb{Z}/N_{\text{fam}}=\text{triality deck group} \Rightarrow$ integer-lift residual closed [E]                                                                                                                                                                        |
| v73_k_c3_half.py              | Central coefficient $k=c_3/2$ <i>forced</i> : $(1/2)$ variational $\times 1/( \mathbb{Z}_2 2\pi\chi(S^2))$ Gauss–Bonnet topology ( $\chi(S^2)=2$ ), both cutoff-indep $\Rightarrow$ Fursaev–Solodukhin $S=A/4$ ; only the absolute 4D Newton scale stays the anchor (v68) [E]                                                                                                                                                                                                                             |
| v74_compiler_micro_lemmas.py  | Spine quotient ladder ( $\frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \frac{5}{4}, \frac{4}{3}$ = adjacent quotients of $(2, 3, 4, 5)$ ); pencil differences $(2, 16, 48)=( \mathbb{Z}_2 , \dim S^+, \Omega_{\text{adm}})$ sheet $\rightarrow$ 1gen $\rightarrow$ 3gens; anchor QF $a^\top K a - \mathbf{1}^\top K \mathbf{1} = 41 - 25 = 16 = \dim S^+$ (EM budget = mass vol + one gen); solar dual anchor $L=R+6\mathbf{1}e_1^\top$ , $a^\top R^{-1}\mathbf{1}=0$ (Sherman–Morrison rank-one invariance) [E] |
| v75_upoint_to_vgeo.py         | <b>Gate 1 complete</b> : $U_{\text{point}} \rightarrow v_{\text{geo}}$ . Ratios closed (v49/v71) + lepton $c$ exact (v20) + sector products = $\varphi_0^{\text{ret}}$ -powers (v46); (ratios, product) $\Leftrightarrow$ (individuals) bijection $\Rightarrow$ absolute amplitudes fixed up to <i>one</i> scale $v_{\text{geo}}$ = the same anchor as $1/G$ (v68); two $[A] \rightarrow$ one [E]/[O]                                                                                                     |
| v76_gmetric_reduction.py      | <b>Gate 2</b> : Tier A (IR) closed under RP+gap (Decoupling Thm: $\Delta_{\text{eff}}=6\log \frac{3}{2}-\frac{31}{4\pi^2}=1.648>0$ , $\ V\ \leq 248c_3^2=\frac{31}{8\pi^2}$ ); Tier B (strict TOE) holographically reduced from bulk to a finite seam-boundary measure (conditional: RP+tightness $\Rightarrow$ closes) [E]/[C]                                                                                                                                                                           |

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| Script                              | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                            |
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| v77_e8_conformal_net.py             | <b>G6 route:</b> the seam boundary measure = the $(E_8)_1$ <i>lattice net</i> (rigorous, holomorphic $c=8$ ); level-1 $c(E_8)=\frac{248}{31}=8$ , $c(D_5)=5$ , $c(A_3)=3$ , embedding $5+3=8$ (coset $c=0$ ); imports $G_6$ into RCF/T/conformal-net rigor [E]/[C]                                                                                                                                                                                     |
| v78_vgeo_floor.py                   | $v_{\text{geo}}$ <b>floor:</b> no pure number is dimensionful $\Rightarrow v_{\text{geo}}$ irreducible by logic; all scales are ratios $\times$ one $M_{\text{Pl}}$ ; $S_{dS}\rho_\Lambda=32\pi^4$ pins it from one measurement; theory at the minimum (one scale $+$ $\pi$ ) [E]/[O]                                                                                                                                                                  |
| v79_review_identities.py            | Four validated review identities + one firewall: hypercharge Lucas–Binet $D_n=(3^n-(-2)^n)/5=1, 1, 7, 13, 55, 133$ (roots = carrier charges); Inverse Anchor Thm ( $\mathbf{1}^\top M^{-1}\mathbf{1}=1/\text{atom}$ , $a^\top M^{-1}a=1$ for $R, K, L$ ); MacWilliams (1, 10, 5); $6Y^2-Y-1=(2Y-1)(3Y+1)$ . Rejects $m_p/m_e$ , $\eta_B$ near-formulas (firewall) [E]                                                                                  |
| v80_operator_pencil_geometry.py     | Operator-pencil geometry: Anchor Singularity Thm $\det B_{K+xQ}=(N_{\text{fam}}x+ \mathbb{Z}_2 )(N_{\text{fam}}x+g_{\text{car}})=(3x+2)(3x+5)$ ( $\frac{2}{3}$ Koide/gap = anchor-plane singularity); buildup chain (3, 10, 16, 20); type checker $\det B_{Q,K,R,L}=(9, 10, 16, 40)$ ; audit: differential spectrum, $F_4 \times G_2$ shadow $248=52+14+26\cdot 7$ [E]/[C]                                                                             |
| v81_singular_pencil_matrices.py     | Double cover $y^2=\det B_{K+xQ}=(3x+2)(3x+5)$ : Koide $x=-\frac{2}{3}$ and carrier $x=-\frac{5}{3}$ are the two branch points (deck $2= \mathbb{Z}_2 $ ; disc $=81=N_{\text{fam}}^4$ ; separation $=1$ transport period). Clearing matrices $C_{2/3}=3K-2Q$ ( $D_5 \oplus A_3$ glue, $B=(5, 7)^\top(9, 11)$ , $\sum=240$ ), $C_{5/3}=3K-5Q$ (neutral, $\chi=(\lambda+2)(\lambda^2+\lambda+6)$ ); scalaron $7=$ mixed anchor disc $/N_{\text{fam}}$ [E] |
| v82_koide_attractor_splitting.py    | Koide attractor <i>forced</i> : the branch-divisor-preserving Möbius map fixing $q=2, 5$ is unique, multiplier $(2/3)^6=\lambda_2$ of the established transfer operator $\Rightarrow F'(2)<1$ attractor, $F'(5)>1$ repeller; three postulates $\rightarrow$ one [C]. Splitting trichotomy disc $81/49/40$ : the clean split is non-generic (anchor-first) [E]/[C]                                                                                      |
| v83_e8net_holomorphic_uniqueness.py | Holomorphy pins $(E_8)_1$ : #primaries = det Cartan ( $D_8:4$ vs $E_8:1$ ), and a holomorphic $c=8$ chiral CFT is the lattice theory of the <i>unique</i> even unimodular rank-8 lattice (Minkowski–Siegel mass $1/ W(E_8) $ ); carrier Pascal truncation $K=(g-1)/2 =$ half-spinor split [E]                                                                                                                                                          |
| v84_frozen_registry.py              | Blind-prediction registry frozen 2026-06-09 (25 digits, machine-enforced lock formula $\leftrightarrow$ value): exactly <i>one</i> $\theta_{12}$ prediction of record; $r$ , $n_s$ as $N_\star$ <i>bands</i> ; conditional entries carry their hypothesis by name [E]                                                                                                                                                                                  |
| v85_master_cover.py                 | Master-cover theorem: $\det B$ is $\text{GL}(2)$ -covariant on $\text{span}\{K, Q\} \Rightarrow$ exactly <i>one</i> double cover up to Möbius (disc $=N_{\text{fam}}^4 \det(G)^2$ ); the disc-81 family is its orbit ( $-\frac{8}{3}=-\text{rank } E_8/N_{\text{fam}} =$ carrier – one period); $\mu_4$ is <i>not</i> a 4:1 cover (ladder of double covers); branch trace fixes only the scalaron <i>scale</i> [E]                                     |
| v86_nstar_reheating.py              | $N_\star$ band $\rightarrow$ point [C]: $\Gamma=4M^3/48\pi\bar{M}^2=128$ GeV, $T_{\text{reh}}=9.6\times 10^9$ GeV, $N_\star(0.05/\text{Mpc})=51.4 \Rightarrow n_s=0.9611$ , $r=0.0045$ (inside the frozen band); honest tensions: $n_s+0.9\sigma$ , $A_s$ dichotomy (slow point $-11.4\sigma \Rightarrow$ near-instant reheating or exponent fails) [C]                                                                                                |
| v87_bulk_uniqueness_reduction.py    | Red-team Target A residual (ii) conditionally closed: holomorphic $\Rightarrow \text{Rep} = \text{Vect} \Rightarrow$ 2D bulk <i>unique</i> (LR/KLM/BKLR); machine contrast $SO(16)_1$ admits <i>six</i> modular invariants (incl. both $E_8$ pairings) $\Rightarrow$ Target A = one residual [E]/[C]/[O]                                                                                                                                               |
| v88_cp_phase_audit.py               | Target-D CP residual quantified: frozen $\delta=\pi/3+3\lambda^2$ survives at $+0.98\sigma$ vs $\gamma_{\text{PDG}}$ ; the $0.07^\circ$ -near alternative $\pi/3+2\lambda^2$ recorded as look-elsewhere trap, <i>not</i> adopted; decision at $\sigma_\gamma \leq 0.96^\circ$ [E]                                                                                                                                                                      |
| v89_carrier_index_lemma.py          | Carrier Index Lemma: KLM $\mu_A=[B:A]^2\mu_B \Rightarrow$ Jones index $[(E_8)_1:(D_5)_1 \times (A_3)_1]=4= \mu_4 $ (glue order <i>is</i> the index); $\mu$ -additivity $16/4^2=1 \Rightarrow$ holomorphy <i>follows</i> ; Gate A (H) $\Leftrightarrow$ (H') index-4 extension [E]                                                                                                                                                                      |
| v90_conical_defect_chain.py         | Fursaev–Solodukhin factor $4\pi$ <i>derived</i> (smoothed-cone Gauss–Bonnet, exactly smoothing-independent); replica $S=4\pi kA$ , $k=c_3/2 \Rightarrow S=A/4 \Leftrightarrow c_3=\frac{1}{8\pi}$ (sympy-solved); residual isolated to “seam determinant $\Rightarrow$ EH form” [E]/[O]                                                                                                                                                                |

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| Script                          | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| v91_spine_tetrahedron.py        | Spine tetrahedron $\{2, 3, 4, 5\}$ = anchor + zeroth moment; edges/faces/volume = the integer grammar, volume $120= R^+(E_8) $ ; $240= \mu_4  E(K_4)  E(K_5) $ with $K_6$ negative control; dual cuts typed tautological; sub-grammar incompleteness honest [E]                                                                                                                                                                                                                                                                                                                                                       |
| v92_glue_uniqueness.py          | Glue uniqueness: of all 15 subgroups of the discriminant form exactly two Lagrangian glues (the two <i>chiralities</i> , swapped by either single spinor conjugation, fixed by the simultaneous one) and exactly one halfway extension = $SO(16)_1$ ; tower carrier $\rightarrow SO(16)_1 \rightarrow (E_8)_1$ , nothing else [E]                                                                                                                                                                                                                                                                                     |
| v93_koide_relaxation_toy.py     | Koide relaxation toy (P2 narrowed): basin lemma (every physical configuration in the attractor basin); exact rate $(2/3)^6$ ; source at $\rho = -\varphi_0^{\text{ret}}/24$ to 0.8% ([C]); honest negatives: flow time $t=2.84 \notin \mathbb{Z} \Rightarrow$ the missing object is a <i>continuous</i> generator [E]/[C]                                                                                                                                                                                                                                                                                             |
| v94_sheet_diamond.py            | Sheet diamond: all four flavor operators on <i>one</i> surface $M(s, t)=R+Q \text{ diag}(s, t, t)$ (pencil = the cut $(1+x, x-1)$ ; diagonal-cut disc 153 negative control); winding line $\det B=2 \det$ for all $s$ ; cofactor seam normal $(5, -9, 6)$ ; Plücker canonicalisation: 11 and 26 both live in $K$ ; spine-quotient firewall <i>rejected</i> [E]                                                                                                                                                                                                                                                        |
| v95_centered_diamond.py         | Centered cross: $Q=U+V$ , $R/L=C \mp U$ , $K/F=C \mp V$ (five objects $\rightarrow$ three); Spec $V=\{0, 1, 2\}=N_{\text{fam}} \cdot \text{cusp}$ ; $\det C=14$ , $\sum C=31=2^{g_{\text{car}}}-1$ ; $\text{Pl}_R(C)=7(2, 3, 1) \parallel \text{Pl}_R(L)=10(2, 3, 1)$ ; $G_2$ reading audit-typed [E]                                                                                                                                                                                                                                                                                                                 |
| v96_branch_kernel_selection.py  | Branch-kernel selection (P1 unblocked): rank-1 anchor blocks at the branch points have <i>integer</i> kernels (carrier kernel = 1); collapse lemma $\Rightarrow (-1, 1, 0)$ : up = -down, <i>lepton pairing</i> = 0 (leptons on the ramification); sector zero ladder (up $-\frac{3}{2}$ doubly, down 0 & $-\frac{9}{5}$ ); anchor-placement controls [E]                                                                                                                                                                                                                                                             |
| v97_sheet_conjugation_bridge.py | Sheet-conjugation bridge (P1 $\rightarrow$ one [C]): the anchor forces the cusp conjugation uniquely $\Rightarrow$ integer $T_A$ , $\det = -1$ ; $a=e_2+e_3$ (conjugation-symmetric); one deck action on $\mathbb{R}^3/\text{span}\{\mathbf{1}, a\}$ ; $\langle T_A, \Sigma \rangle = D_4$ ; sheet-index lemma (intertwiner dets even, $\min 2= \mathbb{Z}_2 $ ); remaining [C]: $Q_+$ grading = $A_3$ discriminant grading [E]/[C]                                                                                                                                                                                   |
| v98_discriminant_dictionary.py  | Dictionary <i>derived</i> (P1 closed modulo GATE.QGEO): $G=T_A \Sigma$ acts integrally on the cusp basis ( $Ge_1=-e_1$ , $Ge_3=e_2$ , $Ge_2=-e_3=B_1 \oplus E$ ); cusp-0 line = self-conjugate $\mathbb{Z}_4$ -class-2 line; $T_A G T_A^{-1} = G^{-1}$ ( $k \mapsto -k$ ); $q_{A_3}(1)=q_{A_3}(3)=\frac{3}{8}$ , $q_{A_3}(2)=\frac{1}{2}$ ; reflection classes = glue-swap vs glue-fix parities $\Rightarrow$ P1 carries no separate [C]; residual = the one existing $Q$ -geometry gate [E]                                                                                                                          |
| v99_koide_flow_time.py          | Koide flow time: canonical generator $\dot{q}=(\Delta/N_{\text{fam}})(q-2)(q-5)=(\Delta/N_{\text{fam}}) \det B(q)$ , time-1 map = $F$ [E]; non-integrality of $t=2.84$ is <i>data-fragile</i> ( $Q$ crosses $\frac{2}{3}$ at $+0.43\sigma(m_\tau)$ , $t$ passes every integer $\geq 3$ within $+0.5\sigma$ ) [E]; conditional steps: $n=2$ excluded $(-2.9\sigma)$ , $n=3=N_{\text{fam}} \Rightarrow m_\tau=1776.9427 \text{ MeV } (+0.14\sigma)$ , decision at $\sigma(m_\tau) \sim 0.01 \text{ MeV}$ [C]; $\ln(46080)$ scale reading destroyed by controls (firewall)                                               |
| v100_numerology_null_mc.py      | Numerology null test (look-elsewhere corrected): exact census of the full declared formula grammar over the 13 scored frozen-registry observables (conservative data windows) gives joint formula-fishing probability $\prod p_i = 10^{-25.8}$ , robust to budget/window variation; $2 \times 2 \cdot 10^5$ Monte-Carlo pseudo-theories reach at most 5/13 hits vs. TFPT 13/13; power controls (seed perturbation, data shuffle) collapse the scorecard; all 94 500 $F_{U(1)}$ variants solved, exactly <i>one</i> hits CODATA $\Rightarrow$ total $\leq 10^{-30.7}$ , <i>conditional on the declared grammar</i> [E] |
| v101_horizon_anchor.py          | The maximal black hole is the anchor (SdS in seam units): Nariai cubic $t^3-3t+2=(t-1)^2(t+2)$ , roots $(1, 1, -2)$ = traceless anchor; $S_{\text{Nariai}}/S_{\text{dS}}=\frac{2}{3}= \mathbb{Z}_2 /N_{\text{fam}}$ (per horizon $\frac{1}{3}$ ); interpolation $(x^2+1)/\Phi_3(x)$ ; three-sheet total $ \mathbb{Z}_2 S_{\text{dS}}$ conserved; $Q_{\text{geom}} \in [\frac{3}{8}, \frac{1}{2}]$ = the $SU(4)_1$ weights; mass-line cover $(1-3m)(1+3m)$ , deck = horizon swap; $\tau_{BH}=128 g_{\text{car}} M^3/c_3$ ; six atom landings, zero free parameters [E]/[C] (bulk reading)                              |

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| Script                         | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| v102_seam_orientation.py       | One orientation: the anchor is the <i>stationary repeller</i> in both sectors — flavor relaxation = gradient flow of a cubic potential with critical points = the branch points, curvatures $V''(2)=+\Delta$ , $V''(5)=-\Delta$ (= the gap), inflection at scalaron/2, Lyapunov rate $\Delta$ constant; SdS entropy has Nariai as its <i>unique</i> stationary point, curvature $\frac{2}{9}= \mathbb{Z}_2 /N_{\text{fam}}^2$ ; evaporation = entropy ascent away from the anchor [E]; “one variational principle of the seam” stays a reading [C]                                                                                                                       |
| v103_trisection_normal_form.py | Trisection normal form (the canonical coordinate): SdS cubic $\Leftrightarrow \cos 3\theta = -3m$ (angle trisection; $\mathbb{Z}_3$ deck = triality); $m = \cos \psi / N_{\text{fam}}$ and $S_{\text{tot}}/S_{\text{dS}} = \frac{4}{3} - \frac{2}{3} \cos \frac{2\psi}{3}$ — one cosine of glue atoms; canonical curvature $(\frac{2}{3})^3 = ( \mathbb{Z}_2 /N_{\text{fam}})^{N_{\text{fam}}}$ (Koide constant to the family power); invariant slope $d\sigma/dm _N = -\frac{8}{9} = -\text{rank } E_8/N_{\text{fam}}^2$ ; bridge: flavor rate $(2/3)^{2N_{\text{fam}}}$ vs gravity curvature $(2/3)^{N_{\text{fam}}}$ , missing = one <i>clock</i> [E]/[C]             |
| v104_nariai_clock.py           | The <i>classical</i> Nariai clock is the anchor (third appearance): static pin $\phi=\rho$ solves the $dS_2$ static-patch equation with $m^2 = -2\Lambda = - \mathbb{Z}_2 \Lambda$ exactly (Ginsparg–Perry tower: one negative mode); in Hubble units $\chi_{\text{clock}} = (\lambda-1)(\lambda+2)$ = the anchor quadratic, eigenvalues $\{1, -2\}$ ; Nariai cubic = $(t-1) \cdot \chi_{\text{clock}}$ ; entropy-deviation rate $2H =  \mathbb{Z}_2 H$ ; honest $(2/3)$ -test <i>negative</i> for the classical clock — the quantum clock is the one remaining [C] of the seam variational principle [E]/[C]                                                            |
| v105_residual_inventory.py     | The residual inventory: <i>one constant</i> $(2/3 =  \mathbb{Z}_2 /N_{\text{fam}}$ exact in seven places across both sectors), <i>one anchor</i> (three dynamical appearances: configuration, Nariai roots, clock spectrum), <i>relocation theorem</i> (cosmological quantum clock suppressed by the $\Lambda$ hierarchy, 121.5 orders $\approx 2\alpha^{-1}/\ln 10 \Rightarrow$ the clock must be the seam transfer operator — one [C] identification), and the <i>complete machine-pinned gap list</i> : exactly five structural objects + one scale + $\pi$ ; a sixth gap would fail the script [E]/[C]                                                               |
| v106_review_validation.py      | External-review validation: seed normal form $\varphi_0^{\text{ret}} = \frac{ \mu_4 }{N_{\text{fam}}} c_3 + \Omega_{\text{adm}} c_3^{ \mu_4 }$ exact [E]; hypercharge moment $\text{Tr } X^2 = 120 = 5!$ computed from the charge table; factorial spine $5! =  R^+(E_8)  = \sum \text{exponents}$ , $240 = 2 \cdot 120$ , $1920 = 2^4 \cdot 5!$ ; degree-2 inventory (Pascal $K=2$ unique at $g=5$ , glue norms sum 2, $c_3 = \frac{1}{2} \times \frac{1}{4\pi}$ , $\binom{5}{2} = 10$ ); named [C] hypothesis <i>Quadratic Boundary Locality</i> ( $\Rightarrow K=2$ forced; would upgrade the Pascal selection to [E]); audit $240 = 16 \times 15$ non-unique [E]/[C] |
| v107_quantum_clock_target.py   | Quantum-clock target made quantitative: the classical Ginsparg–Perry clock decays on $\{0, -1, -2\} = -\text{Spec}(V) = -N_{\text{fam}} \times \text{cusp}$ (the transfer operator’s own grading) [E]; exact target identity $(1/3)^6 = ((2/3)^6)^{\log_{3/2} 3}$ — required bend $\log_{9/4} 3 = 1.355$ per weight step; coupling landscape $\kappa = c/(24\pi) \cdot \Lambda/\bar{M}_{\text{Pl}}^2$ : $10^{-122}$ cosmological vs $1/(3\pi)$ at seam curvature (the only viable, borderline-nonperturbative regime) [E]; identification stays [C]                                                                                                                      |
| v108_pascal_ladder.py          | Pascal Ladder Theorem: $2^{g-1} = \sum_{k \leq K} \binom{g}{k} \Leftrightarrow g=2K+1$ (odd- $g$ midpoint identity + even- $g$ straddle) $\Rightarrow$ carrier selection $g=5$ <i>exactly equivalent</i> to degree bound $K=2$ — the Target-B residual = the one QBL hypothesis; neighbour worlds: $K=1$ one-family, $K=3$ nine-family, $K=4$ inconsistent; only $K=2$ also satisfies $g+N_{\text{fam}}=8$ (v14 overdetermination) [E]/[C]                                                                                                                                                                                                                               |
| v109_sheet_pairing.py          | Sheet-Pairing Lemma (QBL anchored): character-exact multisets from $(\pm \frac{1}{2})^5$ — no scalar within a sheet (zero-weight mult of $S^+ \otimes S^+ = 0$ ; odd $g$ flips parity); $S^+ \otimes S^- = \Lambda^0 \oplus \Lambda^2 \oplus \Lambda^4$ exactly ( $256 = 1 + 45 + 210$ , zero-modes $(1, 5, 10)$ = the code); sheet-diagonal = odd forms $(2\Lambda^1 + 2\Lambda^3 + \Lambda^5)$ ; tower tops at $\Lambda^{2K}$ , $K=2 \Rightarrow$ the seam scalar kernel is necessarily a <i>sheet pairing</i> [E]/[C]                                                                                                                                                 |
| v110_calderon_sheet.py         | Calderón-sheet selection theorem: a Calderón involution $\varepsilon$ on $S^+ \oplus S^-$ certifies a scalar two-point datum <i>iff</i> it is sheet-odd; sheet-odd $\Rightarrow$ exactly $2 =  \mathbb{Z}_2 $ invariant kernels = the two Lagrangian glues; ladder genericity ( $g=3, 5, 7$ ): sheet-oddness forces $K=(g-1)/2$ , <i>not</i> $g=5$ (anti-overclaim); QBL residue split into <i>interface</i> (“ $\varepsilon$ sheet-odd”, natural from the one-sided collar) + <i>analytic core</i> (slot-degrees $\leq 2$ ) [E]/[C]                                                                                                                                     |

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| Script                            | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| v111_quadratic_transport.py       | Quadratic-Transport Theorem: in the integer Jordan–Wigner model linears are sheet-odd, the 45 quadratics (= $\mathfrak{so}(10)$ , the $\Lambda^2$ term of $1+45+210$ ) sheet-even, generate the code from the vacuum <i>and</i> span the full $\text{End}(S^+)=256$ at word length $2 \Rightarrow$ <i>transport degree exactly 2</i> (degree $\leq 1$ generates nothing) — the transport half of the QBL core is a theorem; $g=3$ control: degree selected, not rank [E]/[C]                                                                                                                                                                   |
| v112_selfcounting_channel.py      | Self-Counting Channel: $w \mapsto (w, -w)$ is a bijection code states $\leftrightarrow$ neutral certified kernels, so $\#\text{kernels} = 2^{g-1}$ <i>exactly</i> ; graded by pair-degree the same set has sizes $\binom{g}{m} \Rightarrow$ the Pascal closure $2^{g-1} = \sum_{m \leq K} \binom{g}{m}$ is <i>two countings of one set</i> (an identity, not a requirement); Hodge fold $\binom{g}{2j} = \binom{g}{\min(2j, g-2j)}$ ; QBL residue = one input (“a single scalar 2-point kernel”), selection carried by rank-8+integrality [E]/[C]                                                                                              |
| v113_quasifree_kernel.py          | One kernel is the whole net: carrier $(D_5)_1 \times (A_3)_1 = SO(10)_1 \times SO(6)_1 = 16$ free Majoranas, $c=5+3=8$ at every tower level (only the glue grows, index $2 \times 2 = 4 =  \mu_4 $ ); Wick/Pfaffian exact (all $210+210$ higher correlators = Pfaffians of the one kernel); kernel = Calderón involution, rank $P=5=g_{\text{car}}$ , seam level $8 = \text{rank } E_8$ ( $c = \text{rank of the kernel}$ ); vacuum unique $\Rightarrow$ QBL input = R2 premise: gate closes $\Rightarrow$ carrier choice closes [E]/[C]                                                                                                       |
| v114_torsion_delta.py             | Torsion normal form + $\delta = \frac{1}{2}$ theorem: flatness of the $(U_{\text{wall}})$ $\mathbb{Z}_4$ -family $\Leftrightarrow (MU)^4 = \mathbf{1}$ ( $\mu_4 = \text{order of the twisted cusp generator}$ ); on the involutive branch ( $T^2 = \mathbf{1}$ ) the cusp trace forces $\text{diag } M = (0, \frac{i}{2}, -\frac{i}{2})$ with $\text{spec } \{1, \omega, \omega^2\}$ automatic $\Rightarrow \delta = \frac{1}{2}$ exact and constant (v20’s hand value = torsion identity); census: $\{1, i, -i\}$ branch varies, v33 point in its $d_1=0$ slice; residue = bundle-side branch selection [E]/[C]                               |
| v115_anchor_residue.py            | The anchor pins the residue: $\mu_4$ -average = diagonal projection $\Rightarrow$ anchor splitting $\Leftrightarrow \text{diag } A_0 = a/ \mu_4 $ ; off-weights uniquely $(8, 0, 5)/144$ $(= (\text{rank } E_8, 0, g_{\text{car}})/( \mu_4 N_{\text{fam}})^2, \text{ total } 13 = \Delta_Q)$ ; <i>exact</i> residue $A_0^*$ with $\chi = \lambda(\lambda - \frac{1}{3})(\lambda - \frac{2}{3})$ is the RH solution ( $\ M_\infty - \mathbf{1}\  \sim 10^{-10}$ ); anchor forces $d_1=0 \Rightarrow$ harmonic diagonal anchor+torsion-forced [E]                                                                                                |
| v116_resonance_uniqueness.py      | Resonance theorem (v115 [N] $\rightarrow$ [E], linear — no Gröbner): twisted $\mu_4$ averages collapse to $B_1 = 4a_{12}E_{12} + 4\overline{a_{13}}E_{31}$ ; anchor exponents $\text{diag}(2, 1, 1)$ give resonance gap 1 with obstruction $4\overline{a_{13}} \Rightarrow M_\infty = \mathbf{1} \Leftrightarrow a_{13}=0$ ; + $(8, 0, 5)/144$ + Diagonaleichung $\Rightarrow$ flat anchor locus = <i>one</i> gauge orbit = exact $A_0^*$ — $U_{\text{wall}}$ bundle datum algebraically unique; sharpness control $\ M_\infty - \mathbf{1}\  \sim 8.5 a_{13} $ [E]                                                                            |
| v117_monodromy_weyl_a3.py         | The monodromy is $W(A_3)$ : the unitarised holonomy of $A_0^*$ is <i>exact</i> (entries in $\frac{1}{2}\mathbb{Z}[i]$ ), $\text{diag } \widetilde{M}_0 = (0, -\frac{i}{2}, \frac{i}{2}) \Rightarrow d_1=0, \delta = \frac{1}{2}$ are matrix entries; $\langle U, \widetilde{M}_0 \rangle$ enumerated exactly: order 24, statistics $(1, 9, 8, 6)$ , characters $(3, -1, 0, 1) = S_4 = W(A_3)$ (standard $\otimes$ sign) — the flavor wall realises the $A_3$ glue factor as its Weyl group; ODE match $3.5 \times 10^{-10}$ [E]                                                                                                                |
| v118_hexagon_family_dictionary.py | First exact H2 piece: $\text{spec}(\widetilde{M}_0) \cup \text{spec}(-\widetilde{M}_0) = \mu_6$ (hexagon = family spectrum + sheet twist, $\mathbb{Z}_6 = \mathbb{Z}_2 \times \mathbb{Z}_3$ ); $\det(\mathbf{1} - t\widetilde{M}_0) = 1 - t^3 \Rightarrow \det_{\mp} = (\frac{7}{8}, \frac{9}{8})$ ; lepton coefficients = resolvent determinants ( $c_e =  \mu_4 /(2 \det_-)$ , $c_\mu = N_{\text{fam}}/(2 \det_+)$ , $c_\tau =  \mu_4  \det_- / N_{\text{fam}}$ , product = $ \mu_4 /\det_+ = \frac{32}{9}$ ); $\langle U, \widetilde{M}_0, -\mathbf{1} \rangle$ has order $48 = \Omega_{\text{adm}}$ ; the address table stays open [E]/[C] |
| v119_review_validation_2.py       | Second review validation: anchor ratio triad $(e_3, e_1, e_2)/p_0 = (\frac{2}{3}, \frac{4}{3}, \frac{5}{3}) = (\text{Koide branch, seed gain, carrier branch; flow critical points } (2, 5) = (e_3, e_2))$ ; the 121 audit lemma $(\mathbf{1}+a)^T R(\mathbf{1}+a) = 121 = 11^2 = \ \text{Pl}(K)\ _1^2$ (orderings $\{105, 121, 135\}$ , only canonical = 121; audit, not load-bearing); micro-identities $\Omega_{\text{adm}} = 2p_1p_2$ , $10b_1 = 1 + p_1p_3$ , $\Delta_Y = e_2^2 = p_0^2 + 2 \text{rank}$ ; flavor surface = $\sqrt{94}/\sqrt{95}$ (presentational) [E]                                                                    |

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| Script                    | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| v120_address_table.py     | Address table (frozen v18/v20 integers): lepton words $(8, 5, 3) = (\text{rank } E_8, g_{\text{car}}, N_{\text{fam}})$ ; addresses = divmod by the hexagon ( $p_2=6$ ): $e \rightarrow (2, 1)$ , $\mu \rightarrow (5, 0)$ , $\tau \rightarrow (3, 0)$ , $\sum r=p_3$ ; sheet parity: $\tau$ sits at the real twisted eigenvalue $-1$ (= the product-rule site); quark sums: up = $p_3$ , down = $p_1+p_3$ , all quarks = $24= W(A_3) $ , all nine = $40=p_1p_3=a^T Ra$ ; top at the vacuum site $(0, 0)$ ; the assignment derivation stays open [E]/[C]                        |
| v121_address_pinning.py   | Address pinning: $L=R+2U$ (v95) $\Rightarrow$ the word table = the residue operator $R$ + the generation-1 winding, $R$ column 1 = the anchor $a$ ; margins = atoms (rows $(e_1, \text{rank}, p_3)$ , columns $(e_1, \Delta_Q, g_{\text{car}})$ ); <i>census theorem</i> : among all hexagon matrices with atom margins (17 of them) exactly <i>one</i> has $\det=8$ — and it is $R$ (also unique with SNF $(1, 1, 8)$ ; all 17 determinants distinct) $\Rightarrow$ the address table carries <i>zero</i> residual information; the H2 residue = the six atom margins [E]/[C] |
| v122_margin_theorem.py    | Margin theorem: the <i>established</i> selectors — the $D_4$ annihilator $n=(5, -9, 6)$ with $n^T R=(8, 0, 0)$ (v94), $\det R=8$ , $\det K=4$ (the v71 quartet) — pin $R$ <i>uniquely</i> among hexagon matrices ( $64 \rightarrow 12 \rightarrow 1$ ) $\Rightarrow$ the atom margins, the anchor column and all sum rules are <i>consequences</i> (their input status is revoked); bonus: $\det L=20$ holds automatically on the locus (the quartet is redundant); H2 introduces <i>no</i> new residual class [E]/[C]                                                         |
| v123_inventory_update.py  | Residual inventory updated (post v110–v122): thirteen new ledger rows machine-pinned; R2 carries <i>three</i> loads (metric + carrier + QBL: one theorem, three doors); the table re-pins to <i>five</i> classes — R1 clock, R2 index-4 net, R3 EH, R4' selector establishment (replacing the <i>discharged</i> H2 dictionary), R5 $Q$ realisation — + $v_{\text{geo}}, \pi$ ; a sixth class fails the script [E]                                                                                                                                                              |
| v124_resummed_clock.py    | Resummed clock (the R1 bend in closed form): transfer eigenvalues = exactly $(1-n/N_{\text{fam}})^{p_2} \Rightarrow \text{rate}(n)=-p_2 \ln(1-n/N_{\text{fam}})$ reproduces $\{0, \Delta, 6 \ln 3\}$ , the bend $\log_{3/2} 3$ in one line; the linearisation carries slope $ \mathbb{Z}_2 =2$ relative to the classical GP clock $\{0, -1, -2\}$ ; the pole at $n=N_{\text{fam}}$ forces the three-level spectrum; the R1 target is now closed-form [E]/[C]                                                                                                                   |
| v125_glue_qsystem.py      | Glue Q-system: $\mathbb{C}[\mathbb{Z}_4]$ satisfies all Longo Q-system axioms exactly (associativity, unit, Frobenius, specialness $mm^*= \mathbb{Z}_4 \mathbf{1} \Rightarrow \text{Jones index } 4= \mu_4 $ (the v89 value); locality = isotropy ( $q(k(1, 1))=k^2 \in \mathbb{Z}$ , v92); halfway sub-Q-system $\mathbb{C}[\mathbb{Z}_2] = \text{the } SO(16)_1 \text{ step}$ , $4=2 \times 2$ ; R2 sharpens from “find” to “identify” [E]/[C]                                                                                                                               |
| v126_clock_wall_bridge.py | Clock and wall share one spectrum: the resummation weights $n/N_{\text{fam}}=\{0, \frac{1}{3}, \frac{2}{3}\}$ = exactly spec $A_0^*$ (the parabolic MS weights); $\lambda_n=(1-\alpha_n)^{p_2}$ ; checkpoint 1 passed <i>classically</i> ( $p_2/N_{\text{fam}}=2= \mathbb{Z}_2  = \text{the v104 entropy rate } 2H$ ); $\det(\mathbf{1}-A_0^*)=\frac{2}{9}$ = the entropy curvature (v102); the quantum content of R1 is isolated in the geometric tail [E]/[C]                                                                                                                |
| v127_ring_resummation.py  | The clock is a log determinant: $\text{rate} = -p_2 \text{Tr} \ln(\mathbf{1}-\alpha P)$ — six identical ring (RPA) towers, <i>one per hexagon site</i> , coupling = the parabolic weight; the $k=1$ ring = the classical term, the $k=2$ ring = the first quantum correction; $\kappa_{\text{seam}}=\frac{1}{3\pi}=\alpha_1/\pi$ (audit); the wall = the RPA instability at $\alpha \rightarrow 1$ ; v107's “O(1) resummation” is thereby <i>typed</i> (ring/log-det), its derivation open [E]/[C]                                                                             |
| v128_graded_hull.py       | Graded hull + zero-mode multiplicity: the 240 $E_8$ roots explicit in $D_5 \oplus A_3$ + glue coordinates, coset decomposition $240=52+64+60+64$ ; the grading is exactly additive on <i>all</i> 6720 root-pair sums $\Rightarrow E_8$ is a $\mathbb{Z}_4$ -graded Lie algebra over the carrier (grading = glue = Q-system); ring multiplicity $6=2(2l+1) _{l=1}$ = the sheet-doubled GP zero modes [E]/[C]                                                                                                                                                                    |
| v129_entropy_power_law.py | The clock is an entropy power law: $\lambda_n=(S_n/S_{dS})^{p_2}$ with the three Nariai entropy levels $S/S_{dS} \in \{1, \frac{2}{3}, \frac{1}{3}\}$ (v101); $\text{rate} = p_2 \ln(S_{dS}/S)$ ; triple identity $\alpha=n/N_{\text{fam}} = \text{MS weight} = \text{entropy-deficit share} \Rightarrow$ the RPA coupling is nowhere free; orientation consistent (attractor = max entropy); R1 thermodynamic: $\Gamma \propto (S/S_{dS})^{p_2}$ (a Gibbons–Hawking power law, $h=N_{\text{fam}}$ audit) [E]/[C]                                                              |

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| Script                       | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| v130_born_square.py          | Born square: amplitude $\sim (S/S_{dS})^{n_{\text{zero}}/2} = (S/S_{dS})^3$ (one $S^{1/2}$ per zero mode), probability $=  A ^2 = (S/S_{dS})^6 \Rightarrow h = n_{\text{zero}}/2 = 3 = N_{\text{fam}}$ , $p_2 = 2h$ (the Born rule) — the v129 weight audit is <i>derived</i> ; the 3 = the dimension of the $l=1$ space = the number of proper conformal Killing vectors of $S^2$ ; the $\times 2$ = the horizon pair (the v101 deck); the R1 residue = the standard zero-mode measure <a href="#">[E]</a> / <a href="#">[C]</a>                                                                                                                            |
| v131_measure_is_area.py      | The measure is the area: $\ Y_{1m}\ ^2 = r^2 = A/(4\pi)$ <i>exactly</i> (all three $l=1$ modes integrated) $\Rightarrow$ the per-mode Jacobian = the mode norm $\sim S^{1/2} \Rightarrow$ amplitude $S^3$ , Born $S^6$ — every exponent factor with a derivation origin; the $4\pi$ = the same Gauss–Bonnet primitive as in $c_3$ , cancelling in all ratios; the R1 residue = <i>one</i> det-ratio cancellation $\det'_n / \det'_{dS} = 1 + O(\text{deficit})$ <a href="#">[E]</a> / <a href="#">[C]</a> ( <i>cancellation since derived within the SdS family, v144</i> )                                                                                  |
| v132_detratio_anomaly.py     | Det-ratio anomaly = the Koide constant: $\zeta(0) _{\det'}$ of the GP sphere operator $-\Delta - 2$ (the 3 zero modes removed) $= a_1 - 3 = \frac{7}{3} - 3 = -\frac{2}{3} = - \mathbb{Z}_2 /N_{\text{fam}}$ exactly ( $a_1 = \frac{1}{N_{\text{fam}}} +  \mathbb{Z}_2 $ ; numerical continuation $\sim 10^{-7}$ ) — the eighth $2/3$ appearance, the first as a <i>spectral</i> anomaly; consequence $\det'(cL) = c^{\zeta(0)} \det'(L)$ ; the R1 residue = <i>one</i> $\zeta(0)$ budget statement (the 4d sum) <a href="#">[E]</a> / <a href="#">[C]</a>                                                                                                   |
| v133_zeta_budget.py          | The $\zeta$ budget, both routes exact (Euclidean Nariai $= S^2 \times S^2$ ): the reduced route gives $-\frac{2}{3}$ per sector, total $-\frac{4}{3} = -e_1/p_0$ (= minus the seed gain — two of the three triad values); the naive 4d route gives $a_2 = \frac{161}{45}$ , $\zeta(0) = -\frac{109}{45}$ ( <i>no</i> atom; continuation $\sim 10^{-8}$ ); zero modes $3+3$ = the horizon pair $\Rightarrow$ the budget structurally favours the <i>reduced</i> seam reading; remainder = graviton/ghost shares (Volkov–Wipf class) <a href="#">[E]</a> / <a href="#">[C]</a>                                                                                 |
| v134_dual_anchor.py          | Dual anchor lemma: $d := a^\top R^{-1} = a^\top L^{-1} = (-\frac{1}{2}, -\frac{1}{2}, 1)$ , $d \cdot \mathbf{1} = 0$ , $d \cdot a = 1$ , $(1, 1, -2) = -2d$ — the inverse flavor response <i>is</i> the Nariai root (v101); the invariance is exact via Sherman–Morrison ( $\Leftrightarrow$ orthogonality to $R^{-1}\mathbf{1} = \frac{1}{4}(1, 1, -1)$ ); controls: $\mathbf{1}, e_1, n$ do <i>not</i> lie on the plane); $(d, n)$ = the dual normal pair of the flavor boundary; a third purely <i>algebraic</i> leg of the flavor $\leftrightarrow$ horizon bridge <a href="#">[E]</a> / <a href="#">[C]</a>                                             |
| v135_det_surface.py          | Determinant surface: $\det M(s, t) = 3s(t+1)(t+2) + t^2 + 5t + 8$ — two $s$ -independent iso-volume walls $t = -2 \Rightarrow 2 =  \mathbb{Z}_2 $ , $t = -1 \Rightarrow 4 =  \mu_4 $ ( $\det K = 4$ is a <i>layer</i> ); winding line $6s + 8 \Rightarrow (8, 14, 20)$ , $\det F = 32 = 2^{g_{\text{car}}}$ ; $\det B(s, 0) = 2 \det M(s, 0)$ exactly, $B$ -wall only at $t = -2$ ; row-sum axes $C = (7, 11, 13)$ , $U = (3, 3, 3)$ , $V = (1, 2, 3) = \text{Spec}(Q_+)$ ; micro-identities $e_1^2 + e_2^2 = 41 = 10b_1$ , $p_1^2 + p_2^2 = 52 =$ the carrier coset root count (v128) <a href="#">[E]</a> / <a href="#">[C]</a>                             |
| v136_dual_normal_selector.py | Dual-normal selector: $(d, n)$ pins $R$ <i>columnwise</i> — $d \cdot c_j = a_j$ , $n \cdot c_j = 8\delta_{1j}$ , kernel = primitive $(6, 8, 7)$ , census: each column unique (1 of 216) $\Rightarrow R4'$ shrinks from $6^9$ to three column problems; the $A_0^*$ zero mode carries $\sigma = (2, -9, 5) = ( \mathbb{Z}_2 , -N_{\text{fam}}^2, g_{\text{car}})$ ( $\ v\ ^2 = \frac{16}{9}$ , $16 = \dim S^+$ ; $\text{adj} = \frac{2}{9}P_v$ = the SdS curvature); $n \cdot \sigma = 121 = 11^2$ ; $W(A_3)$ orbit control: the naive lift $\sigma \rightarrow n$ fails — a genuine mechanism, not a group element <a href="#">[E]</a> / <a href="#">[C]</a> |
| v137_qplus_cohomology.py     | The $Q_+$ grading is cohomology: $H^1(\mathbb{P} \setminus \mu_4) = \chi_1 \oplus \chi_2 \oplus \chi_3$ ( $\omega_k = z^{k-1} dz / (z^4 - 1)$ ), character $i^k$ , residues $\zeta^k/4$ ; degrees $(1, 2, 3) = \text{Spec}(Q_+) =$ the $A_3$ exponents ( $h = 4 =  \mu_4 $ ); cusp weights $\{0, \frac{1}{3}, \frac{2}{3}\} = \text{spec}(A_0^*) \Rightarrow$ MS weights = cusp classes = characters = $Q_+$ : <i>one</i> spectrum, four readouts; the QGEO residue = the naturality of the canonical map <a href="#">[E]</a> / <a href="#">[C]</a>                                                                                                          |
| v138_vw_firewall.py          | Volkov–Wipf firewall (R1 typed closed): external 4d value $\alpha_{\text{PI}} = -\frac{98}{45} = \frac{22}{45} - \frac{8}{3}$ (arXiv:2506.02142 Tab. 4.1 = VW hep-th/0003081); <b>edge match</b> : $\alpha_{\text{edge}} = -\frac{8}{3} = 2 \times (-\frac{4}{3})$ — two identical edge copies (the horizon pair), per copy exactly the reduced seam budget (v133, the seed gain); bulk $\frac{22}{45}$ = the graviton gas (not the clock); difference $-\frac{38}{45}$ no atom $\Rightarrow$ typing: the seam clock = an edge object, the full 4d determinant = the gravity firewall <a href="#">[E]</a> / <a href="#">[C]</a>                              |

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| Script                    | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| v139_selector_triangle.py | Selector triangle: $d=\frac{3}{2}a-2\cdot\mathbf{1}$ exactly (pure anchor data, no $R$ ) $\Rightarrow$ the first selector is <i>derived</i> ; the frame $(\mathbf{1}, a, \sigma)$ has $\det=11=\ \text{Pl}(K)\ _1$ , and $n$ is the <i>unique</i> covector with atom pairings $(2, 8, 121)=( \mathbb{Z}_2 , \text{rank } E_8, 11^2)$ ; on the constrained line the pairing is $11(11+t)$ , a square only at the true $n$ ; review lift exact (audit): $n=\text{reverse}(\sigma)+ \mu_4 e_3$ ; $R4'$ residue = three pairings $[\mathbf{E}]/[\mathbf{C}]$ ( <i>since v142: two — the third pairing is integrality-forced</i> ) |
| v140_canonical_map.py     | The canonical map exists and is rigid: the plain deck $U$ (characters $(0, 1, 3)$ ) cannot match $H^1$ ; the <i>sheet-twisted</i> $-U$ $((2, 3, 1))$ and the cusp exponential $i^{Q+}((1, 2, 3))$ both carry the $H^1$ character set $\{1, 2, 3\}$ (the v118 sign twist); multiplicity-freeness $\Rightarrow$ Schur-rigid equivariant isomorphism (unique up to scalars); $i^{Q+}$ pulls the Euler degree back to $Q_+$ , $-U$ to $Q_+\circ\rho$ — the QGEO residue collapses to <i>one</i> $\mathbb{Z}_3$ choice $[\mathbf{E}]/[\mathbf{C}]$ ( <i>the choice since derived, v141</i> )                                       |
| v141_deck_selection.py    | Deck Selection Theorem (R5): the established integer deck $G=T_A\Sigma$ (v97/v98) pairs the $Q_+=1$ line with the self-conjugate character $2 \Rightarrow$ of the three $\mathbb{Z}_3$ rotations only the sheet-twisted $(2, 3, 1) = \text{chars}(-U)$ survives; $i^{Q+}$ excluded (same spectrum, wrong grading pairing); residual freedom = the established sheet $\mathbb{Z}_2$ (v92); Euler degree = $Q_+\circ\rho$ $[\mathbf{E}]/[\mathbf{C}]$                                                                                                                                                                           |
| v142_frame_integrality.py | Frame Integrality (R4'): pairing triples of <i>integer</i> covectors against $(\mathbf{1}, a, \sigma)$ form an index-11 sublattice $(x-3y+z \equiv 0 \bmod 11, 11=\ \text{Pl}(K)\ _1)$ ; with $(x, y)=( \mathbb{Z}_2 , \text{rank } E_8)$ the third pairing is forced $\equiv 0 \bmod 11$ , primitivity forces even $t$ ; Cramer reductions for the two line pairings; $R4'$ : three pairings $\rightarrow$ two $[\mathbf{E}]/[\mathbf{O}]$ ( <i>since v145: the values are atom identities — residue = one lift map</i> )                                                                                                    |
| v143_graded_frobenius.py  | Graded Frobenius (R2 at the finite Lie level): coset duality $-C_k=C_{-k}$ exact on all 240 roots $\Rightarrow$ Killing pairing nondegenerate per $g_k \times g_{-k}$ ; glue average = carrier projector (index $ \mathbb{Z}_4 =4$ ); each sector one $W(D_5)\times W(A_3)$ orbit = simple current; fusion = $\mathbb{C}[\mathbb{Z}_4]$ (v125); residual = the one conformal-net statement $[\mathbf{E}]/[\mathbf{C}]$                                                                                                                                                                                                        |
| v144_detratio_family.py   | Det-ratio cancellation within the SdS family (R1, the v131 step): $e_2$ -rigidity gives $r_b r_c = 1 - \Delta^2/3$ exactly $\Rightarrow$ $\det'$ -ratio = $(1 - \Delta^2/3)^{4/3}$ (via $\zeta(0) = -\frac{2}{3}$ , v132) — no first-order term in the split; deficit coefficient $\frac{32}{27} =  \mu_4 (\frac{2}{3})^3$ (audit); finite-weight absorption stays open with the $6 \rightarrow \frac{14}{3}$ obstruction stated sharply $[\mathbf{E}]/[\mathbf{C}]$ ( <i>the bend since shown <math>\det'</math>-clean, v147</i> )                                                                                           |
| v145_pairing_atoms.py     | $R4'$ values are <i>atom identities</i> : with the lift $n=w_0(\sigma)+ \mu_4 e_3$ ( $w_0$ = the $A_3$ exponent duality), $n\cdot\mathbf{1}= \mu_4 - \mathbb{Z}_2 = \mathbb{Z}_2 $ , $n\cdot a= \mu_4 a_3=8$ via the new closure $ \mu_4 +g_{\text{car}}=N_{\text{fam}}^2$ ( $4+5=9$ , $\sigma \perp w_0(a)$ ), $n\cdot\sigma=\ \sigma\ ^2-N_{\text{fam}}^2+ \mu_4 g_{\text{car}}=121$ ; residue = derive the <i>one</i> lift map $[\mathbf{E}]/[\mathbf{C}]$                                                                                                                                                                 |
| v146_moebius_d4.py        | Möbius $D_4$ realisation (R5 at its floor): the full dihedral $\langle z \mapsto iz, z \mapsto 1/z \rangle$ on $H^1(\mathbb{P}^1 \setminus \mu_4)$ matches the integer model $\langle G, T_A, \Sigma \rangle$ exactly — $\iota^*$ swaps $\chi_1 \leftrightarrow \chi_3$ , fixes $\chi_2$ with $+1$ , $\det=-1$ ( $=T_A$ in the cusp basis); $\delta\iota=\Sigma$ class; parities match (v98) — the GATE.QGEO premise reduces to the module identification $[\mathbf{E}]/[\mathbf{C}]$                                                                                                                                         |
| v147_clock_gaussian.py    | The clock is a Gaussian zero-mode integral (R1): variance ratio $(1-\alpha)$ (norm = area, v131) $\Rightarrow$ Born <sup>2</sup> $\Gamma$ -ratio $(1-\alpha)^{p_2}$ , rate = $-p_2 \ln(1-\alpha)$ = the v127 ring sum term by term; the bend $\log_{3/2} 3$ lives between the two Nariai levels of <i>one</i> geometry $\Rightarrow$ $\det'$ -clean; the $6 \rightarrow \frac{14}{3}$ obstruction is localised to the $\alpha=0$ reference $[\mathbf{E}]/[\mathbf{C}]$                                                                                                                                                        |
| v148_fock_census.py       | NS/R sector census (R2 scoped, corrected): the untwisted (NS) module of the 16 Majoranas carries only the <i>even</i> classes $\{0, 2\} \times \{0, 2\}$ — the odd glue sectors $k(1, 1)$ are <i>Ramond</i> modules; the R zero-mode module ( $2^8=256$ ) splits $128+128$ into the odd sectors of the <i>two</i> Lagrangian glues (v92): choosing the glue <i>is</i> choosing the R-projection ( $128$ = one $SO(16)$ chirality), $248=120_{\text{NS}}+128_{\text{R}}$ ; R2 is irreducibly a twisted-sector net statement $[\mathbf{E}]/[\mathbf{C}]$                                                                        |
| v149_cusp_normal.py       | Both dual normals are <i>anchor data</i> ( $R4'$ ): in the cusp frame $n$ pairs to $(6, 3, 5) = (p_2, p_0, e_2)(a)$ — one anchor invariant per $Q_+$ line ( $d = \frac{3}{2}a - 2\cdot\mathbf{1}$ in the generation frame, v139); equivalent to the $(2, 8, 121)$ frame data and the $w_0$ -lift (v145); audit: $\sigma \rightarrow (5, 1, 2)$ , sum $14 = \dim G_2$ ; residue = <i>one</i> assignment $[\mathbf{E}]/[\mathbf{C}]$                                                                                                                                                                                            |

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| Script                         | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| v150_replica_eh_model.py       | The EH form from the replica variation, at the gapped-model level (first R3 movement): conical deficit exact and $t$ -independent (image sum $(N^2-1)/(12N) = C(2\pi/N)$ ); the gap makes the $\zeta$ -variation <i>finite and cutoff-independent</i> , $\Delta \log \det' = 2C(\gamma) \ln m$ ; linearisation $= (\ln m/12\pi) \int \sqrt{g} R$ — the EH form; target equation $k=c_3/2 \Leftrightarrow \ln m = \frac{3}{4} = q(A_3)$ (audit); Calderón transfer + normalisation stay open — SEAM.THEOREM.01 narrows, does <i>not</i> close [E]/[O] ( <i>Calderón transfer since answered — the DtN kernel is conically clean, v151</i> ) |
| v151_bfk_split.py              | BFK split (the Calderón transfer of v150 answered): the Kac corner constant is boundary-condition <i>independent</i> ( $C_N=C_D$ , derived from reflection doubling) $\Rightarrow$ the conical deficit of the Calderón/DtN jump determinant <i>vanishes identically</i> , $C_{\text{DtN}}=C_{\text{cone}}(\gamma)-2C_D(\gamma/2)=0$ ; via Burghelea–Friedlander–Kappeler the seam-reduced action inherits the v150 EH form verbatim — the EH content sits in the <i>local</i> half determinants, the nonlocal kernel is conically clean; residue = the $q(A_3)$ normalisation + the kernel premise [E]/[O]                                 |
| v152_norm_is_anchor.py         | R3's normalisation <i>is</i> the dimensionful anchor: the EH coefficient is $k = \ln(m/\mu)/(12\pi)$ — $\ln m$ alone is scale-ambiguous, only the ratio $m/\mu$ is physical; $k=c_3/2 \Leftrightarrow m/\mu=e^{3/4}$ is the induced-gravity anchor $1/G$ (v68), <i>not</i> a separate gap; the irreducibles stay {one scale, $\pi$ }; audit: the log-ratio $= \frac{3}{4} = q(A_3)$ is the target value, NOT a derivation [E]/[O]                                                                                                                                                                                                          |
| v153_no_unit_theorem.py        | No-Unit Theorem (v78/v152 formalised): dimensionless data are scale-invariant under $L \mapsto \lambda L$ , a mass / $1/G$ is not, so a dimensionless compiler <i>cannot</i> select an absolute scale; the three residuals collapse — $U_{\text{point}} \sim v_{\text{geo}}$ , $1/G \sim v_{\text{geo}}^2$ , $m/\mu = e^{3/4}$ (a ratio); $v_{\text{geo}}$ is an irreducible metrology primitive, not an open gap; irreducibles stay {one unit, $\pi$ } [E]/[O]                                                                                                                                                                            |
| v154_simple_current_theorem.py | Simple-Current Extension Theorem (the central $G_{\text{net}}$ theorem, assembled): $A=(D_5)_1 \otimes (A_3)_1$ extended by the isotropic glue $L=\langle(1,1)\rangle$ has index $ L =4$ , $c=5+3=8$ , $\mu(B)=\mu(A)/ L ^2=16/16=1 \Rightarrow$ holomorphic $\Rightarrow B=(E_8)_1$ (unique even-unimodular rank-8; $SO(16)_1$ excluded); finite realisation in hand (v125/v128/v143/v148); residue = the seam-side identification [E]/[C]                                                                                                                                                                                                |
| v155_quasifree_boundary.py     | Quasi-free in $\Rightarrow$ quasi-free out (Python-only): the boundary marginal of a Gaussian bulk is the compression $PTP$ (a contraction, hence a valid covariance), so the $G_{\text{net}}$ analytic premise (v110(b)/v113) <i>follows</i> from ‘the seam bulk is a reflection-positive free field’ — the <i>same</i> premise as v59 (area law) and v150/v151 (EH); the seam-side identification reduces to ONE shared physical premise [E]/[O]                                                                                                                                                                                         |
| v156_seam_net_construction.py  | Constructive route (Route 3) to its exact endpoint: the DtN/Calderón operator of the 2d Laplacian is $\omega_k= k $ (free chiral dispersion); 16 free Majoranas give $c=8=5+3$ ; the $E_8$ currents are $248=120_{\text{NS}}+128_{\text{R}}$ ; and the $(E_8)_1$ character $E_4/\eta^8 = q^{-1/3}(1+248q+4124q^2+\dots) = j^{1/3}$ (verified to order 4) makes $B \cong (E_8)_1$ a <i>checked</i> identity, not an assertion; residual = the free-bulk premise (v155) + net existence [E]/[C]                                                                                                                                              |
| v157_rigid_fixed_point.py      | Freeness as a <i>rigid fixed point</i> (simplicity-first reframing of premise A): (i) the DtN/Calderón symbol is universally $ k $ (Lee–Uhlmann, bulk-detail-independent — the free dispersion is not a premise); (ii) a holomorphic $c=8$ theory has <i>no</i> marginal $(1,1)$ field (all $\bar{h}=0$ ) $\Rightarrow$ rigid, isolated, no room for an interaction; (iii) the same $ \mathbb{Z}_2 $ that halves $8\pi$ in $c_3$ IS the Ramond projection giving $\mu=1$ ; (iv) holomorphic $c=8$ unique $= (E_8)_1 =$ free. Premise (A) shrinks from ‘prove Gaussianity’ to ‘the gapped flow reaches the holomorphic fixed point’ [E]/[C] |
| v158_fixed_point_stable.py     | The free chiral $c=8$ fixed point is <i>stable</i> (exact dimension count, the finite core of (A)): Majorana $h=\frac{1}{2}$ , bilinear current $h=1$ , quartic $h=2$ ; the relevant window $(0,1)$ for bosonic operators is <i>empty</i> , the $h=1$ currents are chiral (not $(1,1)$ -marginal, v157), the lowest interaction (quartic) is $h=2>1$ irrelevant $\Rightarrow$ isolated stable fixed point; (A) reduces to a basin-of-attraction statement [E]/[C]                                                                                                                                                                          |

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| Script                                | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v159_pyrate_gauge_crosscheck.py       | PyR@TE cross-check of the $F_{\text{transfer}}$ gauge inputs: the carrier/SM content gives $(b_1, b_2, b_3) = (\frac{41}{10}, -\frac{19}{6}, -7)$ by the textbook one-loop formula; $b_1 = \frac{41}{10}$ holds <i>three</i> independent ways — carrier algebra $10b_1 = g_{\text{car}} 2^{g_{\text{car}}-2} + 1 = 41$ , the $U(1)$ hypercharge index $\sum \frac{3}{5} Y^2$ , and the external RGE generator PyR@TE3 ( <a href="#">arXiv:2007.12700</a> ) which writes $\beta_{g_1} = (\frac{41}{10}) g_1^3$ verbatim; the $M=41$ split $10b_1 = 40_{(\text{ferm})} + 1_{(\text{Higgs})}$ reproduces v13 Gate 1; evidence in <a href="#">verification/pyrate</a> . The transfer functor's gauge inputs are not free knobs <a href="#">[E]</a> / <a href="#">[C]</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| v160_seam_gaussianity_from_pf.py      | Seam Gaussianity as a <i>conditional</i> fixed-point theorem (the proposed closure of premise A, honestly typed): (1) quasi-free $\Rightarrow$ all higher connected cumulants $\kappa_{2n}=0$ (signed set-partition/Pfaffian inversion, exact on a pure Majorana covariance); (2) Wick functoriality $\Rightarrow$ a kernel-preserving transport fixes the whole quasi-free state (v113); (3) the Perron–Frobenius dichotomy (a <i>fixed</i> $\kappa_{2n}$ is a SECOND fixed ray, breaking uniqueness; a <i>non-fixed</i> one decays by the gap) forces Gaussianity; (4) the qualitative claim needs only $\text{gap} > 0$ — the value $(2/3)^6$ only sets the rate. <b>The load <a href="#">[O]</a>:</b> v56's gapped PF uniqueness lives on the 3-dim scalar transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ , while the seam correlation cone has $\binom{16}{4}=1820$ , $\binom{16}{6}=8008$ cumulant directions $\Rightarrow$ (A) reduces to the single internal statement 'the admissible TFPT transport is primitive + spectrally gapped on the <i>full</i> Schwinger cone, not only the readout spectrum' <a href="#">[E]</a> / <a href="#">[O]</a>                                                                                                                                                                                                                                     |
| v161_one_particle_reduction.py        | The cone reduces to one particle (discharging v160's scope premise via Bogoliubov second quantisation): a quasi-free transport is $T=\Gamma(t)$ , the second quantisation of a one-particle contraction $t$ ; $\Gamma(t)$ acts on the degree- $m$ correlator sector as the exterior power $\Lambda^m(t)$ , so (1) $\text{spec}(\Gamma(t) _{\deg m}) =$ products of $m$ one-particle eigenvalues <a href="#">[E]</a> ; (2) <b>the cone gap equals the one-particle gap</b> — the sub-leading eigenvalue over the whole cone is $(2/3)^6$ exactly <a href="#">[E]</a> ; (3) the eigenvalue-1 sector = wedges of the fixed subspace (disconnected Wick powers), no independent fixed higher cumulant, uniqueness closed by v113 (one polarization $\Leftrightarrow$ one state) <a href="#">[E]</a> ; (4) quasi-freeness <i>forced</i> not assumed — the $120 = \dim \mathfrak{so}(16)$ chiral currents are the Bogoliubov generators and v158 makes every non-Gaussian deformation irrelevant <a href="#">[C]</a> . <b>Residual <a href="#">[O]</a>:</b> v160's three full-cone premises collapse to ONE 16-dim one-particle statement (seam symbol = gapped Calderón Bogoliubov map, fixed = rank-8 $K_\Sigma$ , gap $(2/3)^6$ ); all numbers supplied, only the physical identification (Kawahigashi–Longo class = the A2 net residual) stays open <a href="#">[E]</a> / <a href="#">[O]</a> |
| v162_seam_transport_identification.py | The final identification, forced as far as it goes — the v161 one-particle statement carries <i>no</i> new open content: (1) the gap $(2/3)^6$ is <i>forced</i> — cusp weights $= \text{spec}(A_0^*) = \{0, \frac{1}{3}, \frac{2}{3}\}$ (parabolic residue, v115), transfer eigenvalue $(1-\alpha)^{p_2}$ , $p_2=6= R_+(A_3) $ , all carrier data <a href="#">[E]</a> ; (2) $\mu_4$ -equivariance forces the generator diagonal in the cusp basis — $\sum_k U^k X U^{-k} =  \mu_4  \text{diag}(X)$ , commutant of $U =$ the diagonal algebra — so the gapped $\mu_4$ -equivariant generator has spectrum = the cusp weights, gap $(2/3)^6$ <a href="#">[E]</a> ; (3) the cusp grading <i>is</i> the $A_3$ grading on the 16 Majoranas (v137) <a href="#">[E]</a> ; (4) the fixed space = the rank-8 Calderón polarization (v113/v156) <a href="#">[E]</a> . <b>Closure <a href="#">[C]</a>/<a href="#">[O]</a>:</b> with v160+v161, (A) closes given only $\mu_4$ -equivariance + admissibility, whose two non-derived inputs are the <i>already-open</i> A2 (net existence) and R1 (seam clock, <a href="#">HOR.CLOCK</a> ) — so (A) factors exactly into A2 + R1, adds <i>no</i> new open item, and is closed conditionally on them (a unification, not an unconditional proof) <a href="#">[E]</a> / <a href="#">[O]</a>                                                                 |
| v163_rg_stability_flavor.py           | $F_{\text{transfer}}$ scale-tagging: the lepton-mixing seeds ( $\sin^2 \theta_{12} = \frac{1}{3} - \frac{\varphi_0}{2} = 0.3067$ , $\sin^2 \theta_{13} = \varphi_0 e^{-5/6} = 0.0231$ ) are <i>RG-stable</i> — PMNS running is $y_\tau^2$ -suppressed ( $\sim 1.8 \times 10^{-5} \ll$ precision), so seam value = measured value; the quark mass-ratio readouts ( $m_t/m_b \sim 41$ ) run through $y_t^2 \sim O(10\text{--}15\%)$ and carry a scale tag. The only mixing-rotating Yukawa term $\frac{3}{2} Y Y^\dagger Y$ carries $y_{\text{max}}^2$ (PyR@TE SM RGEs) <a href="#">[E]</a> / <a href="#">[C]</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

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| Script                           | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| v164_qcd_scale.py                | the QCD confinement scale is parameter-free from the carrier $b_3=-7$ : running $\alpha_s(M_Z)$ down (two-loop, $n_f$ thresholds) gives $\alpha_s(m_b)\sim 0.22$ , $\alpha_s(m_c)\sim 0.38$ and confinement ( $\alpha_s\sim 1$ ) at $\sim 0.5$ GeV — so the QCD-scale <i>numerator</i> of $m_p/m_e$ is independently sourced (dimensional transmutation), while the ratio 1836 stays the honest ‘Open / not forced’ frontier non-claim ( $m_p=O(1)\Lambda$ is lattice) [E]/[O]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| v165_premise_a_closure.py        | Premise (A), the honest maximal closure — the two residual gates pinned to their floor, so (A) carries <i>zero</i> independent open content: (1) A2 (net existence) closes by <i>assembly</i> of established mathematics [C] — 16 free Majoranas give a free-fermion net (Buchholz–Mack–Todorov / Böckenhauer–Evans), the holomorphic $c=8$ lattice net is $(E_8)_1$ (Frenkel–Kac–Segal / Dong–Xu / Kawahigashi–Longo), the $\mu_4$ extension is a Longo Q-system of index 4 (v125), $\mu(B)=16/16=1$ , $248=120+128$ , character $E_4/\eta^8=j^{1/3}$ (v156); the only TFPT input (seam Calderón kernel = free-fermion two-point function) is done via DtN = $ k $ (v156/v157); (2) R1 (seam clock) reduces to GATE.QGEO [C] — $\mu_4$ -equivariance forces diagonal-in-cusp, the flat $\mu_4$ -equivariant generator is <i>unique</i> = $A_0^*$ (v115/v116) so the gap $(2/3)^6$ is forced, residual = ‘seam boundary = the $\mu_4$ -punctured sphere $\mathbb{P}^1\setminus\mu_4$ ’ = GATE.QGEO (cohomology established, v137); (3) the irreducible core stays $\{\pi, v_{\text{geo}}\}$ , a <i>theorem</i> by the No-Unit Theorem (v153). <b>Final accounting:</b> (A) = A2 assembly [C] + R1=GATE.QGEO + the $\{\pi, v_{\text{geo}}\}$ core $\Rightarrow$ ZERO independent open gates; a unification + re-typing, not an unconditional proof [E]/[O] |
| v166_higgs_free_seam.py          | the Higgs quartic from the <i>free seam</i> (premise A / v158): a Gaussian seam UV forces $\lambda(M_{\text{seam}})=0$ and $\beta_\lambda(M_{\text{seam}})=0$ (Shaposhnikov–Wetterich double criticality, here <i>motivated</i> not assumed). Two-loop PyR@TE-sourced SM running with the measured $(m_H, m_t)$ gives $\lambda(\bar{M}_{\text{Pl}})\sim 0.002$ , $\beta_\lambda(\bar{M}_{\text{Pl}})\sim 0$ — the famous SM near-criticality, now <i>explained</i> by the free seam (2-loop sharpens $\lambda(\bar{M}_{\text{Pl}})$ from $\sim 0.022$ to $\sim 0.002$ ). The double condition predicts $m_H\sim 129\text{--}134$ GeV; at the scalaron scale it gives $\sim 107$ GeV (too low) $\Rightarrow$ the BC lives at $\bar{M}_{\text{Pl}}$ (seam=Planck) [E]/[C]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| v167_inventory_post_closure.py   | The terminal residual inventory (post premise-(A) closure; line v105 $\rightarrow$ v123 $\rightarrow$ v167): after the (A)-chain (v160–v165), premise (A) is <i>no longer</i> an independent structural class (GATE.METRIC.16–19). The residual collapses from v123’s FIVE classes to: <b>Tier 0</b> — the irreducible core $\{\pi, v_{\text{geo}}\}$ , a <i>theorem</i> (No-Unit, v153), nothing to close; <b>Tier 1</b> — ONE hinge (seam = flavour = horizon): $A_2$ net existence (assembly of established net constructions) + GATE.QGEO (seam boundary = $\mathbb{P}\setminus\mu_4$ , cohomology $b_1=N_{\text{fam}}=3$ ); <b>Tier 2</b> — folded ( $\rightarrow v_{\text{geo}}$ , $\rightarrow G_{\text{net}}$ ); <b>Tier 3</b> — $F_{\text{transfer}}$ , one functor with four typed conditional interfaces ( $\eta_B$ carries the most open room). Completeness: a sixth independent structural class fails the script — a unification + re-typing, not an unconditional proof [E]/[C]                                                                                                                                                                                                                                                                                                                                                           |
| v168_qgeo_rigidity.py            | The QGEO <i>Rigidity Theorem</i> — the finite half of the Tier-1 hinge GATE.QGEO, hardened. Exact: $\mu_4=\{1, i, -1, -i\}$ has cross-ratio 2 and Möbius stabiliser $\langle z\rightarrow iz, z\rightarrow 1/z \rangle$ of order $8= D_4 $ (faithful) $\Rightarrow$ four marks with faithful $D_4$ are the $\mu_4$ square up to Möbius; $H^1(\mathbb{P}\setminus\mu_4)$ has rank $4-1=3=N_{\text{fam}}$ ; the forms $\omega_k=z^{k-1}dz/(z^4-1)$ ( $k=1, 2, 3$ ) are $\mu_4$ -eigenforms with $z\rightarrow iz$ eigenvalue $i^k$ = the three nontrivial characters $\chi_1, \chi_2, \chi_3$ , weights $(1, 2, 3) = A_3$ exponents = $\text{Spec}(Q_+)$ . QGEO.RIGID.01 [E] (math half closed); QGEO.REALIZE.01 [C] (seam-collar = $\mathbb{P}\setminus\mu_4$ stays the realisation premise) [E]/[C]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| v169_etaB_boltzmann_interface.py | The $\eta_B$ leptogenesis transfer <i>operationalised</i> (Tier-3 $F_{\text{transfer}}$ , not a new class): type-I thermal leptogenesis $\eta_B\sim 0.96\times 10^{-2}\varepsilon_1\kappa_f$ (sphaleron 28/79) fed by TFPT’s NO neutrino spectrum + the predicted $\delta_{CP}=240^\circ$ . Davidson–Ibarra $\varepsilon_1=\frac{3}{16\pi}M_1m_3/v^2\sim 8.5\times 10^{-7}$ for $M_1=10^{10}$ GeV; efficiency $\kappa_f(\tilde{m}_1)\sim 0.03\text{--}0.19$ . <b>Kill test:</b> over natural $M_1\in[3\times 10^9, 3\times 10^{10}]$ GeV and washout, $\eta_B$ spans $[8.4\times 10^{-11}, 4.7\times 10^{-9}]$ which <i>brackets</i> the observed $6.1\times 10^{-10}$ (canonical point gives $6.0\times 10^{-10}$ , not tuned) $\Rightarrow$ the route is viable; $M_1$ /washout are scenario inputs, so $\eta_B$ stays [C]. If excluded, the route falls, not the theory [C]                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

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| Script                            | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| v170_diamond_slice_compression.py | E8 Slice Compression Lemma: the seven $E_8$ audit slices are <i>one</i> projection of two alphabets — anchor power sums $P=(3, 4, 6, 10)$ ( $p_n=2+2^n$ ) and Sheet-Diamond determinants $D=(3, 4, 8, 14, 20, 32)$ . The $K_4$ edge products $(12, 18, 30, 24, 40, 60)$ are all carrier blocks; $\sum \det M = 81 = N_{\text{fam}}^4$ ; all seven 248-slices follow from $P, D, \Delta=13$ and $(e_1, e_2, \dim S^+)$ ; one affine map $\text{row}(C+sU+tV) = (7, 11, 13)+s(3, 3, 3)+t(1, 2, 3)$ gives every flavour row budget; $78 = \dim E_6 = p_2\Delta$ . The atlas is a closed grammar over two typed alphabets, exact, reading audit [E]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| v171_os_moment_cluster.py         | Atomic OS Moment Lemma + Sugawara Gap Safety: $\text{spec}(T)=\{1, \frac{64}{729}, \frac{1}{729}\}$ makes $C(n)=A+Br^n+Cs^n$ a positive moment sequence, so every OS Hankel matrix factorises as a Vandermonde Gram $\Rightarrow$ reflection positivity (clean proof, not numeric Gram); cluster $\frac{729}{665}$ , $\xi=0.41105$ , Källen–Lehmann masses $6\log \frac{3}{2}, 6\log 3$ ; gap safety $\Delta_{\text{eff}}=6\log \frac{3}{2}-\frac{31}{4\pi^2}>0$ with $31=1+h^\vee(E_8)$ , $c((E_8)_1)=248/31=8$ [E]/[C]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| v172_trace_anomaly_seed.py        | Trace-anomaly origin of the seed prefactor $\frac{4}{3}$ : over the seam $S^2$ ( $\chi=2$ ) the $(E_8)_1$ ( $c=8$ ) Weyl anomaly is $c\chi/6=\frac{8}{3}$ , halved by the one-sided $ \mathbb{Z}_2 $ to $\frac{4}{3} = \dim S^+ / \dim \mathfrak{g}_{SM} = 16/12$ , the seed base $(\frac{4}{3})c_3=\frac{1}{6\pi}$ . Plus QCD $b_3=11-\frac{2}{3}N_f=7$ = scalaron exponent $\Omega_{\text{adm}}-10b_1=48-41$ (confinement $\leftrightarrow$ inflation), and the Volkov–Wipf factorisation $-98/45 = -2 \cdot 7^2 / \dim(D_5)$ [E]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| v173_pfaffian_cp_car.py           | Strong CP as Pfaffian reality: a self-dual CAR (16-Majorana) model demonstration of $\theta_{\text{eff}}=0$ — the sheet-odd Calderón involution $\{D, \Sigma\}=0$ pairs the spectrum $(\pm\lambda)$ , $\gamma_5$ -Hermiticity $D^\dagger=\gamma_5 D \gamma_5$ makes the Pfaffian real ( $\arg \in \{0, \pi\}$ ), and reflection positivity ( $Z>0$ ) selects the positive branch $\Rightarrow \theta_{\text{eff}}=0$ [E]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| v174_seam_fock_readings.py        | Seam Fock-space readings: (1) CAR cone compression — $\dim \Lambda^{\text{even}}(\mathbb{R}^{16})=2^{15}=32768$ reduces to a 16-dim one-particle problem under quasi-free Bogoliubov ( $2^{15}/16=2^{11}$ ), the explicit count behind v161; (2) Witten index = the Plücker norm $11 = \sum_{k \leq 2} \binom{4}{k}$ , with $11 = \dim S^+ - g_{\text{car}}$ (full $2^4=16$ ) and $c_u/c_d=55/117$ reading the QBL-protected boundary state space; (3) Affleck–Ludwig $g$ -theorem $c_{UV}=5+3=8=c_{IR}((E_8)_1)$ so $\Delta c=0$ forces an exact isometry (boundary-entropy form of v157/v158) [E]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| v175_net_existence_full_cone.py   | The heavy artillery — net existence (A2) and full-cone reflection positivity discharged to [E], isolating QGEO.REALIZE.01 as the single open premise (nothing fabricated): (1) <b>full-cone RP for all <math>m</math></b> — the CAR second-quantisation functor $\Gamma(t)=\bigoplus_m \Lambda^m(t)$ of the one-particle seam contraction $t$ acts on the complete Fock space $\Lambda^\bullet(\mathbb{R}^{16})$ ( $\dim 2^{16}=65536$ ); the exterior-power spectrum theorem makes every eigenvalue a subset product of $\text{spec}(t)$ , verified on all $2^{16}$ products to lie in $[0, 1]$ (contraction), with eigenvalue-1 multiplicity $2^8=256$ (wedges of the rank-8 $K_\Sigma$ polarisation) and sub-leading = the one-particle gap $(2/3)^6$ , so full-cone RP reduces <i>for all <math>m</math></i> to the one-particle contraction (Shale–Stinespring CAR functor), discharging the v160 scope premise by a theorem; (2) <b>A2 net existence</b> assembled + verified — $(E_8)_1$ character $E_4/\eta^8=1+248q+4124q^2+34752q^3+213126q^4+1057504q^5$ (level-1=248), embedding $248=120+128$ , $E_8$ Cartan even unimodular ( $\det 1$ ); the net <i>exists</i> by cited theorems (free-fermion net, lattice VOA, index-4 Q-system); (3) both collapse to the same seam-kernel identification = QGEO.REALIZE.01, finite half v168, realisation half left [O][E][O] |

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| Script                          | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| v176_seam_collar_realisation.py | The Seam Collar Realisation Theorem assembled as ONE central statement + reduction certificate (the single structural proof obligation, formulated cleanly — <i>not</i> a fabricated proof): given an oriented one-sided RP seam collar with a sheet-odd Calderón involution, faithful $D_4$ marks and admissible $c=8$ data, <i>then</i> its conformal boundary is Möbius-equivalent to $\mathbb{P} \setminus \mu_4$ with $H^1$ grading $(1, 2, 3)$ , the Calderón contraction is the rank-8 $K_\Sigma$ polarisation and the index-4 extension is $(E_8)_1$ . The proof decomposes into six steps, FIVE already [E]: geometry/uniformisation (v168: cross-ratio 2, faithful Möbius $D_4$ order 8, $b_1=N_{\text{fam}}=3$ ), Calderón DtN symbol $ k $ (v156), $H^1$ character = generation grading $(1, 2, 3)=A_3$ exponents=Spec( $Q_+$ ) (v137/v168), $\mu_4$ -equivariant contraction diagonal with gap $(2/3)^6$ (v162), AQFT closure $(D_5)_1 \otimes (A_3)_1 \rtimes \mu_4 = (E_8)_1$ + full-cone RP all $m$ (v154/v175); the whole structural residual reduces to the ONE open premise QGEO.REALIZE.01 (the physical seam collar's boundary <i>is</i> this $\mathbb{P} \setminus \mu_4$ ), left honestly [O][E][O]                                                                                                                                                                                           |
| v177_seam_marking_kernel.py     | The QGEO proof tree — the v176 single realisation premise split into its honest constituents: REALIZE = MARKS · UNIFORM · COHOM · MODULE · KERNEL · NET, with FOUR nodes closed <i>constructively</i> [E] and the open residual sharpened into exactly TWO named obligations. [E] UNIFORM (Lemma 2, constructive): an order-4 Möbius map conjugates to $z \mapsto iz$ , a non-fixed orbit scales to $\mu_4$ , the reflection $z \mapsto 1/z$ gives $\sigma\rho\sigma = \rho^{-1}$ ( $D_4$ , order 8) — a genus-0 four-marked faithful- $D_4$ boundary <i>is</i> $(\mathbb{P}, \mu_4)$ ; [E] COHOM: $\rho^*\omega_k = i^k\omega_k$ so the $H^1$ grading is $(1, 2, 3)=A_3$ exponents=Spec( $Q_+$ ); [E] MODULE: the $H^1 \rightarrow$ generation iso is <i>unique</i> (multiplicity-free + residue normalisation; reflection $\omega_1 \leftrightarrow \omega_3$ , $\omega_2$ fixed); [E] NET (v154/v175): index-4 $\mu_4$ extension of $(D_5)_1 \otimes (A_3)_1 = (E_8)_1$ . The TWO genuine open obligations [O]: QGEO.MARKS.01 (raw seam $\rightarrow$ genus-0 four-marked $D_4$ boundary) and QGEO.KERNEL.01 (raw seam Calderón kernel = the $\mu_4$ -equivariant free gapped contraction as an <i>operator</i> ); the whole structural residual is now exactly these two doors [E][O]                                                                                                                            |
| v178_marks_kernel_attempt.py    | An honest <i>attempt</i> at the two open obligations QGEO.MARKS.01 + QGEO.KERNEL.01 — <i>not</i> a closure (genuine constructive-geometry/AQFT statements, not finite computations), but a deeper reduction: each finite core is closed [E] and each obligation reduced to ONE sharper irreducible premise [O]. <i>Marks core</i> [E]: a genus-0 double with an order-4 clock $\rho$ ( $\sim z \mapsto iz$ , fixed points $\{0, \infty\}$ ) and reflection $\tau$ forces the marks to be a generic $\rho$ -orbit $\{1, i, -1, -i\} = \mu_4$ (not the fixed points), distinct iff $b_1=4-1=3=N_{\text{fam}}$ , $\tau\rho\tau=\rho^{-1} \Rightarrow$ faithful $D_4$ order 8 — so MARKS reduces to the single topological/clock premise. <i>Kernel core</i> [E]: on the 3-dim multiplicity-free $H^1$ the $\mu_4$ characters $(1, 2, 3)$ are distinct, so by Schur a $\mu_4$ -equivariant operator is forced diagonal and <i>completely determined by its eigenvalue per character</i> — two such with the forced transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ (v162) and the unique residue-normalised basis (v177) are <i>equal as operators</i> , not merely isospectral (the spectrum-vs-operator worry vanishes on the finite block); so KERNEL reduces to the single full-operator-reduction premise (principal symbol $ k $ , Lee–Uhlmann/v156; no drift, v158). Neither obligation closed; both sharpened [E][O] |

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| Script                              | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| v179_conformal_realisation.py       | The two residual premises (MARKS-residual + KERNEL-residual) collapse to ONE geometric premise QGEO.CONF.01 (honest unification; the conformal realisation stays [O]). MARKS's residual is not three independent assumptions: (1) genus-0 <i>is</i> P1 — $c_3=1/( \mathbb{Z}_2  2\pi \chi(S^2))=1/(8\pi)$ with $\chi=2 \Rightarrow g=0$ , the same $\chi=2$ that gives the '8' (v58/v73); (2) the order-4 clock <i>is</i> the carrier — Coxeter element of $W(A_3)=S_4$ , order $h(A_3)=4= \mu_4 $ (v117); (3) marks = one orbit is <i>forced</i> — parabolic marks $\neq$ the two elliptic fixed points $\{0, \infty\}$ , a free order-4 orbit has exactly $4=e_1(a)$ points. So MARKS reduces to 'the carrier clock acts conformally on the genus-0 double'. KERNEL then follows from that geometry: DtN symbol $ k $ (Lee–Uhlmann, v156) $\Rightarrow$ rank-8 $c=8$ fixed polarisation; no drift (v158); the cusped-surface residual data is $H^1=\mathbb{C}^3=N_{\text{fam}}$ , the 3-dim gapped transfer block (Schur, v162/v178). Both $\Rightarrow$ ONE premise QGEO.CONF.01: the raw RP seam double is conformally $(\mathbb{P}, \mu_4)$ with the carrier clock as the order-4 Möbius map — the single structural residual of the whole theory, left honestly [O][E][O]                                                        |
| v180_clock_is_mobius.py             | Cracking QGEO.CONF.01 one level further — the conformal-realisation premise is <i>replaced</i> , via three classical theorems, by a strictly <i>milder</i> , non-circular premise QGEO.ISO.01: 'the carrier clock is an order-4 orientation-preserving <i>isometry</i> of the RP seam metric'. (1) Uniformisation (Poincaré–Koebe): a genus-0 Riemann surface is conformally $\mathbb{P}$ , so the genus-0 seam double (P1, v179) <i>is</i> $\mathbb{P}$ — the target is produced, not assumed; (2) Keréjártó (1919; Constantin–Kolev 2003): a finite-order orientation-preserving homeomorphism of $S^2$ is conjugate to a rotation, so the order-4 clock ( $h(A_3)=4$ ) is topologically $z \mapsto iz$ ; (3) isometry $\Rightarrow$ conformal $\Rightarrow$ Möbius: an orientation-preserving isometry of a Riemannian surface is holomorphic, $\text{Aut}(\mathbb{P})=\text{PSL}(2, \mathbb{C})$ ; (4) order-4 Möbius $= z \mapsto iz$ (verified: projective order 4, fixed $\{0, \infty\}$ , multiplier $i$ , free orbit $\mu_4$ ). So QGEO.CONF.01 is <i>implied</i> by the milder isometry premise; the new residual QGEO.ISO.01 (the seam transport is a metric-compatible holonomy preserving the RP inner product) does not name $\mathbb{P}/\mu_4$ and is milder, its surface realisation from the raw seam still [O][E][O] |
| v181_clock_is_conformal_symmetry.py | The FINAL honest reduction + the explicit identification of BEDROCK. The isometry premise QGEO.ISO.01 (v180) is weakened one more genuine step — to a CONFORMAL-symmetry premise QGEO.SYM.01 — via equivariant uniformisation (Nielsen realisation, finite cyclic on genus-0): a <i>conformal</i> automorphism (strictly weaker than an isometry — every isometry is conformal, conformal allows any positive scale) already suffices to put the order-4 clock in the form $z \mapsto iz$ . The clock is the Coxeter element of $W(A_3)=S_4$ (v117), order $h(A_3)=4= \mu_4 $ , the $\mu_4$ deck, and a deck transformation preserves the pulled-back conformal structure by definition. So below QGEO.SYM.01 the residual is <i>definitional</i> , not a missing theorem: 'the seam's conformal structure <i>is</i> the $\mu_4$ -deck structure of the carrier clock' (the carrier $\mu_4$ glue is the conformal monodromy of the seam) — a physical/definitional identification tying P2 to the seam's conformal geometry, not reducible further without relabeling. The chain<br>REALIZE(v175) $\rightarrow$ theorem(v176) $\rightarrow$ MARKS+KERNEL(v177) $\rightarrow$ finite<br>cores(v178) $\rightarrow$ CONF.01(v179) $\rightarrow$ ISO.01(v180) $\rightarrow$ SYM.01(v181) has reached bedrock; we stop here honestly [E][O] |

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| Script                          | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| v182_reviewer_residual_map.py   | The four external-review concerns reduce to the two already-named residuals + one data-only item. $A$ (mapping $2D \rightarrow 4D$ ) [E]/[C]: the Plücker readout $11 = \sum_{k \leq 2} \binom{4}{k} = \dim S^+ - g_{\text{car}} = 16 - 5$ (QBL count, v174) and the holographic reduction give $A = \text{QGEO.SYM.01} + F_{\text{transfer}}$ . $B$ (UV gravity) [E]: ‘gravity = geometric boundary readout’ is QGEO.SYM.01. $C$ (grammar tuning) [E]/[C]: frozen (v84) + minimal + over-determined (the anchor $a=(1, 1, 2)$ in $\geq 6$ independent not-selected-for structures) is look-elsewhere-resistant, but the residual falls only to out-of-sample data (JUNO/CMB-S4). $D$ (Koide Möbius flow) [E]/[C]: the Möbius nature is forced by $\text{Aut}(\mathbb{P}) = \text{PSL}(2, \mathbb{C})$ on the deck, $t \sim 2.84$ only locates the point, and the missing RG generator is $F_{\text{transfer}}$ , so $D = \text{QGEO.SYM.01} + F_{\text{transfer}}$ . NET: three reduce onto $\{\text{QGEO.SYM.01}, F_{\text{transfer}}\}$ , one waits for data — a unification of the concern structure, not a closure (premises stay [O]/[C]) [E][C] |
| v183_koide_f_corner_transfer.py | The Koide source→pole factor $\frac{53}{54}$ has an <i>operator</i> origin (external-review proposal, validated): $u_{\text{pole}}/u_{\text{source}} = a^T(R+Q)\mathbf{1}/(2\mathbf{1}^T R a) = 53/(2 \cdot 27) = \frac{53}{54}$ . (1) $\mathbf{1}^T R a = 27$ is the $E_6 \times A_2$ cubic family block ( $248 = \dim E_6 \cdot 78 + \det R \cdot 8 + 2 \cdot 27 \cdot N_{\text{fam}}$ ); (2) $F=R+Q$ is the <i>missing</i> Sheet-Diamond corner $M(1, 1)$ with $\det B_F = 52 = \dim F_4$ (v80) and anchor response $a^T(R+Q)\mathbf{1} = 53 = 52+1$ ; (3) $\frac{53}{54} = 1 - \frac{1}{2N_{\text{fam}}^3}$ exactly ( $54=2 \cdot 27$ ); (4) negative control: only $F$ gives 53 ( $R/Q/K/L \rightarrow 32/21/35/56$ ). Upgrades FR.KOIDE.03 from a bare guess to an operator identity; [E] the factor is exact, [C] the leptonic source→pole reading the $F$ corner stays an $F_{\text{transfer}}$ conjecture [E][C]                                                                                                                                                                                                                              |
| v184_etaB_anchored_boltzmann.py | Honest test of the external-review $\eta_B$ proposal ( $M_1=M_R\phi_0^4$ , $\tilde{m}_1=m_3/A_\Lambda$ ) — firewall verdict: <i>sharper scenario</i> , not a closure. The washout anchors plausibly ( $\tilde{m}_1=m_3/A_\Lambda=m_3/10 \approx 5 \text{ meV}$ , $A_\Lambda=10= E(K_5) $ , in the strong-washout window) [C]; but $M_1$ does <i>not</i> — $M_R$ is the seesaw scale $\sim v^2/m_3 \approx 6 \times 10^{14} \text{ GeV}$ and the nearest clean TFPT powers of $\bar{M}_{\text{P1}}$ both miss it ( $c_3$ -power off $\sim 75\%$ , $\phi_0$ -power off $\sim 40\%$ ), so $M_1=M_R\phi_0^4$ merely relocates the free input $M_1 \rightarrow M_R$ (the reviewer’s $6.0 \times 10^{-10}$ uses $M_R=1.3 \times 10^{15}$ ; the seesaw value gives $3.4 \times 10^{-10}$ , the tell that $M_1$ is unpinned). $\eta_B$ stays in the observed band so the route is viable, but the proposal cuts the free inputs from two to one, not to zero — $\eta_B$ stays [C], a sharper scenario [C]                                                                                                                                                      |
| v185_axion_relic_solver.py      | Axion relic, honest misalignment estimate: the determinant-line axion $f_a=M_{\text{scal}}/128 \approx 2.39 \times 10^{11} \text{ GeV}$ ( $m_a \approx 23.8 \mu\text{eV}$ ) can provide all dark matter, but <i>only</i> near the $\theta_i \approx 170^\circ$ hilltop where the relic is exponentially sensitive — so the axion DM branch stays a [C] scenario. The harmonic misalignment under-produces at this $f_a$ (prefactor $\sim 0.026/\theta^2$ , $\theta \sim O(1)$ gives $\sim 21\%$ of $\Omega_{\text{DM}}$ , need $\theta_{\text{eff}}^2 \sim 4.7$ ), so the hilltop anharmonic enhancement is required; near the hilltop it is exponentially sensitive, hence a scenario not a sharp prediction. $F_{\text{relic}}(f_a, \theta_i, N_{\text{DW}}) = ? \Omega_{\text{DM}}$ is a kill test for the <i>branch</i> , not the theory (no free singlet WIMP: $128=(16, 4)+(16, 4)$ ) [C][O]                                                                                                                                                                                                                                                     |
| v187_ftransfer_laws.py          | The $F_{\text{transfer}}$ discipline guard (CI-style firewall): the four continuous-transfer frontier observables — Koide, $\eta_B$ , the axion relic, $m_p/m_e + \Lambda_{\text{QCD}}$ — must <i>never</i> be typed as primitive [E] compiler predictions; each carries a [C]/[O] marker. The guard reads <code>status_ledger.csv</code> and enforces it (exact algebraic sub-parts — Koide 53/54 v183, $b_3=-7$ v159 — may be [E], the physical prediction never is). Records the four interfaces $F_{\text{pole}}/F_{\text{Boltzmann}}/F_{\text{relic}}/F_{\text{QCD}}$ and that the raster rule + frozen registry + null model bind, so no future edit can quietly promote a pretty frontier number to a closed compiler output [E]                                                                                                                                                                                                                                                                                                                                                                                                                |

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| Script                         | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| v188_frontier_wording_guard.py | The frontier-wording guard (a prose sentinel, no new physics): v187 protects the ledger <i>typing</i> , this guard protects the <i>prose</i> . It forbids stale sentences whose semantics an earlier round superseded — “leptogenesis interface ([E])” (v184 retyped Route B to [C]), “closed branch also fixes the leptogenesis inputs” ( $M_R$ is the seesaw scale, not a compiler power), “relic scale $f_a, m_a$ open” (v185: the scales define the branch, the open item is the hilltop-sensitive relic abundance), “computed $Q=0.664\dots$ near, not exact” (v183 gave 53/54 an exact operator origin), “Suite now 187 modules” (the count is 187, highest ID v188; v186 skipped). Scans the 10 active documents; passes when none appear, so a frozen Zenodo release can never re-introduce half-true wording [E]                                                                                                                               |
| v189_riemann_roch_carrier.py   | The Riemann–Roch carrier reading: $(g_{\text{car}}, N_{\text{fam}}) = (5, 3)$ are the two canonical invariants of <i>one</i> object, the four-point seam divisor $D = \mu_4$ on $\mathbb{P}^1$ . [E] arithmetic: $h^0(\mathbb{P}^1, \mathcal{O}(\mu_4)) = \deg + 1 = 5$ (function side) and $\text{rank } H_1(\mathbb{P}^1 \setminus \mu_4) = \deg - 1 = 3$ (cycle side, already QGEO.COHO.M.01/v177); $g_{\text{car}} = 5 = \text{rank}(D_5)$ (the carrier is the $\mathfrak{so}(10)$ half-spinor, $\text{rank } D_5 + \text{rank } A_3 = 5 + 3 = 8, \mathfrak{v}1$ ), $N_{\text{fam}} = 3$ . [C]: reading the carrier as the meromorphic seam-mode space $H^0(\mathbb{P}^1, \mathcal{O}(\mu_4))$ (less number, more geometry) is matched by $\text{rank}(D_5) = 5$ , but the physical identification is conjectural and does <i>not</i> replace the over-determined $g_{\text{car}} = 5$ (Pascal/bootstrap/Lean) — a fourth, geometric reading [E][C] |
| v190_nariai_entropy_bound.py   | The one-line Nariai entropy bound: the Schwarzschild–de Sitter entropy ratio $S_{\text{tot}}/S_{\text{dS}} = (x^2+1)/\Phi_3(x)$ ( $x=r_b/r_c$ ; $\Phi_3=x^2+x+1$ the $N_{\text{fam}}=3$ cyclotomic, already in tfpt_horizon_readouts) satisfies $S_{\text{tot}} \geq \frac{2}{3}S_{\text{dS}}$ with equality exactly at the Nariai merge $x=1$ , via the perfect square $3(x^2+1) - 2\Phi_3(x) = (x-1)^2 \geq 0$ — a variation-free proof of the entropy floor (global minimum on $x>0$ is $2/3 =  Z_2 /N_{\text{fam}}$ ) [E]                                                                                                                                                                                                                                                                                                                                                                                                                           |
| v191_universal_branch_line.py  | The universal branch line, honestly typed [C] — an exact affine <i>relabeling</i> , not a theorem. The map $q = \frac{7}{2} + \frac{N_{\text{fam}}^2}{2}m$ carries the SdS branch points $m = \pm 1/N_{\text{fam}}$ onto the flavor branch points $\{2, 5\} = \{ Z_2 , g_{\text{car}}\}$ ( $m=0 \rightarrow 7/2$ ). The arithmetic is exact, but the map exists for <i>any</i> two pairs of points (two intervals are always affinely equivalent), so the slope $N_{\text{fam}}^2/2$ and midpoint $7/2$ are <i>determined</i> by the data, not forced predictions; a decoy pair $\{3, 4\}$ admits an equally exact map (negative control). The genuine shared content — the $2/3 =  Z_2 /N_{\text{fam}}$ ramification of the mass $\leftrightarrow$ transport double cover — is already established. Recorded as a [C] alignment, never promoted to [E] [C]                                                                                             |
| v192_qgeo_energy_clock.py      | The energy-conserving-clock reformulation, honestly typed: QGEO.ENERGY.01 ( $\rho^* \Lambda_\Sigma \rho = \Lambda_\Sigma$ — the carrier clock preserves the Calderón/DtN energy form) <i>relocates</i> the bedrock QGEO.SYM.01 to a more checkable operator identity, but does <i>not</i> eliminate it. The downstream chain (conformality $\rightarrow$ Möbius $\rightarrow z \mapsto iz \rightarrow \mu_4 \rightarrow$ QGEO.SYM.01) already exists (v180/v181); the new upstream link (energy invariance $\Rightarrow$ conformality in 2D) is checkable on the DtN operator $\Lambda =  k $ . But ‘ $\rho$ preserves $\Lambda$ ’ is plausibly equivalent to ‘ $\rho$ is the conformal deck’, so it is a sharper restatement, not a theorem, unless $\rho$ is independently defined and the invariance independently proven (open). Bedrock stays [O]                                                                                                  |
| v193_qgeo_energy_commutator.py | QGEO.ENERGY.02: the energy- <i>commutator</i> contract (the non-circular sharpening of v192): $[\rho, \Lambda_\Sigma] = 0$ on the rank-8 Calderón polarisation, $\rho$ the carrier clock, $\Lambda_\Sigma$ the DtN energy form of the <i>raw</i> RP seam. Three sub-claims: (1) [E] $\rho$ is carrier-defined (Coxeter of $W(A_3)=S_4$ , order $4= \mu_4 $ , v117), not imported from the seam; (2) [O] the non-circularity hinge — $\Lambda_\Sigma$ must be the <i>raw</i> RP-seam DtN, not the $\mu_4$ normal form; (3) [E] $[\rho, \Lambda_\Sigma]=0$ forces the $(1, 2, 3)$ grading on $H^1$ (QGEO.COHO.M.01). Exact [E] rider: the induced $k=(\ln m)/(12\pi)$ v150 reproduces $c_3/2$ iff $\ln m=q(A_3)=\frac{3}{4}$ (FAMILY), not $q(D_5)=\frac{5}{4}$ (carrier, off by 5/3) — gravity is family-geometry-induced. Proof target [O]; if proven, QGEO.SYM.01 becomes an energy-rigidity theorem [O]                                               |

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| Script                               | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| v194_raw_seam_dtn_rp.py              | <p>QGEO.DTN.01 — subclaim 2 of v193 (“the raw RP-seam DtN is RP-definable”) tackled: a <i>reduction</i>, not a closure. (1) [E] finite core: <math>\rho: z \mapsto iz</math> on <math>H^1 = \langle w_1, w_2, w_3 \rangle</math>, <math>w_k = z^{k-1} dz / (z^4 - 1)</math>, acts as <math>\rho^* w_k = i^k w_k</math> (characters <math>i, -1, -i</math> distinct), so <math>[\rho, \Lambda_\Sigma] = 0</math> on <math>H^1</math> (v177). (2) [E] hinge met: the DtN/transfer operator is RP-canonical (Osterwalder–Schrader, v54) on a quasi-free seam state (v155), so <math>\Lambda_\Sigma</math> is RP-definable without assuming <math>\mathbb{P}^1 \setminus \mu_4</math>. (3) [O] residual relocates: full-<math>L^2</math> commutation <math>\Leftrightarrow</math> Bisognano–Wichmann geometric modular flow of the quasi-free seam state (must be intrinsic, else re-circulates). QGEO.SYM.01 reduces from a definitional postulate to intrinsic BW modular covariance — sharper, still [O], not closed [O]</p>                                                                                                                                                                                                                                                                                                                                                   |
| v195_marks_lefschetz_character.py    | <p>QGEO.MARKS.02 — a Lefschetz/character derivation of the four seam marks (review path 2): instead of positing <math>D_4</math> marks, they are <i>forced</i> to be a free <math>\mu_4</math> orbit. [E]: <math>\rho: z \mapsto iz</math> (order 4) fixes only <math>\{0, \infty\}</math>; <math>H^1 = \chi_1 + \chi_2 + \chi_3</math> (the <math>A_3</math> exponents (1, 2, 3), v177) has no trivial character <math>\Rightarrow</math> no puncture at a fixed point; <math>\text{rank } H_1 = n - 1 = 3 \Rightarrow n = 4 \Rightarrow</math> exactly one free orbit <math>\Rightarrow</math> marks = <math>\mu_4</math> (cross-ratio 2); <math>\text{Tr}(\rho H^1) = i + i^2 + i^3 = -1</math>, <math>L(\rho) = 1 + 1 = 2 = \# \text{fixed}</math>. [O] residual relocates from “the seam is <math>\mathbb{P} \setminus \mu_4</math> (geometric)” to “<math>H^1</math> carries the <math>A_3</math>-exponent rep” — less circular [E][O]</p>                                                                                                                                                                                                                                                                                                                                                                                                                              |
| v196_seam_energy_functional.py       | <p>QGEO.VARI.01 — the <math>E_{\text{fail}}</math> variational functional (review path 1), a numerically-testable sharpening of the v194/ENERGY.02 target: <math>E_{\text{fail}} = \ [\rho, \Lambda_\Sigma]\ _{HS}^2 + \ \rho^4 - 1\ ^2 + \ \Theta \rho \Theta - \rho^{-1}\ ^2</math>, with <math>E_{\text{fail}} = 0 \Leftrightarrow</math> the QGEO.SYM.01 conditions. [E] on the finite <math>H^1</math> block (<math>\rho = \text{diag}(i, -1, -i)</math>, <math>\Lambda</math> diagonal, <math>\Theta = \text{conj}</math>) all three vanish <math>\Rightarrow E_{\text{fail}} = 0</math>; [E] a <math>\mu_4</math>-breaking <math>\Lambda</math> or non-order-4 <math>\rho</math> gives <math>E_{\text{fail}} &gt; 0</math> (<math>\mu_4</math> a true minimiser); [E] the zero locus is the seam-deck conditions; [O] the full-operator minimisation is the v194 Bisognano–Wichmann residual — a discretisable handle on the open bedrock, a target not a closure [E][O]</p>                                                                                                                                                                                                                                                                                                                                                                                           |
| v197_rr_carrier_clifford_d5.py       | <p>ARCH.RRCAR.02 — hardening the Riemann–Roch carrier from <math>g_{\text{car}}</math> to <math>D_5</math> itself (review point 7): the 5-dim carrier mode space <math>C_{\text{car}} := H^0(\mathbb{P}^1, \mathcal{O}(\mu_4))</math> generates the <math>\mathfrak{so}(10) = D_5</math> half-spinor by its even Clifford algebra. [E] <math>\dim \Lambda^{\text{even}}(\mathbb{C}^5) = \binom{5}{0} + \binom{5}{2} + \binom{5}{4} = 1 + 10 + 5 = 16 = 2^{g_{\text{car}} - 1} = \dim S^+</math>; the <math>\mathfrak{so}(10) = D_5</math> half-spinor <math>2^4 = 16 = S^+ = \Lambda^{\text{even}}(C_{\text{car}})</math>; closes the chain <math>\mu_4 \rightarrow g_{\text{car}} \rightarrow D_5</math> from one four-point divisor (cycle side <math>\text{rank } H_1 = 3 = N_{\text{fam}}(A_3)</math>; function side <math>h^0 = 5 = g_{\text{car}}</math>; Clifford spinor <math>16 = D_5</math> half-spinor = carrier). [C] “carrier = Clifford spinor of the seam mode space” stays physical (as v189), does not change the over-determined <math>g_{\text{car}} = 5</math> [E][C]</p>                                                                                                                                                                                                                                                                                 |
| v198_modular_commutator_reduction.py | <p>QGEO.MODULAR.01 — cracking the bedrock to a state-invariance. The v194 residual (<math>[\rho, \Lambda_\Sigma] = 0</math> on full <math>L^2</math>, framed as Bisognano–Wichmann with a circularity worry) is (a) made <i>exact</i> at the principal-symbol level and (b) reduced via Tomita–Takesaki to a clean state-invariance. [E] the DtN principal symbol <math> k  = \sqrt{-d^2}</math> is <math>\text{diag}( n )</math> and the clock <math>\rho: z \mapsto iz</math> is <math>\text{diag}(i^n)</math> in the Fourier basis — both diagonal <math>\Rightarrow [\rho,  k ] = 0</math> exactly on <i>all</i> of <math>L^2</math>; [E] Tomita–Takesaki: <math>[\rho, \Lambda_\Sigma] = 0</math> follows from <math>\omega \circ \rho = \omega</math> (the modular <math>\Delta</math> is intrinsic to <math>(M, \Omega)</math>; a state-preserving symmetry commutes with it), needing <i>no</i> conformal covariance — removes the BW circularity; [E] the finite <math>H^1</math> block is already <math>\mu_4</math>-invariant (v177). [O] residual: the full raw seam state is <math>\mu_4</math>-invariant (<math>\omega \circ \rho = \omega</math>) <math>\Leftarrow</math> the Gaussian bulk measure is deck-invariant = operational QGEO.SYM.01. Sharpest non-circular form; not a closure (a foundational symmetry cannot be derived from nothing) [E][O]</p> |

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| Script                          | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v199_seam_state_invariance.py   | QGE0.STATE.01 (work package A) — attacking $\omega \circ \rho = \omega$ directly on the raw seam: a further localising reduction, not a closure. [E] setup non-circular ( $\rho = \text{Coxeter of } W(A_3) = S_4, \rho^4 = 1$ , an automorphism of the raw CAR algebra; both carrier-side, no seam geometry); [E] ground-state reduction $\omega \circ \rho = \omega \Leftrightarrow [\rho, H_1] = 0$ ( $C_\Sigma = \frac{1}{2}(1 + \text{sgn } H_1)$ ); [E] character criterion: $[\rho, H_1] = 0 \Leftrightarrow H_1$ block-diagonal in the four $\mu_4$ -character classes $\{n \equiv r \pmod{4}\}$ ; principal symbol $ k  = \text{diag}( n )$ passes automatically. [O] residual: the bounded sub-principal symbol has <i>no</i> off-character matrix elements — non-circular, sharply typed, holds on $H^1$ (v177); not a closure [E][O]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| v200_seam_variational_scan.py   | QGE0.VARI.02 (work package B) — the discretised variational test: v196 had $E_{\text{fail}} = 0$ at the $\mu_4$ config on the fixed block; here the clock angle is free and $E_{\text{fail}}$ is minimised over it. [C] scanning $E_{\text{fail}}(\theta) = \ \rho(\theta)^4 - 1\ ^2 + \text{faithfulness over } \theta \in (0, \pi)$ gives the unique minimum at $\theta = \pi/2$ ( $E_{\text{fail}} = 0$ : order-4 <i>and</i> three distinct characters $i, -1, -i$ = the $(1, 2, 3)$ grading); decoys $(0, \pi, \pi/3, 2\pi/3)$ all lose. So $\arg \min E_{\text{fail}}$ = the $\mu_4$ normal form. [E] consistent with v196/v199/v198. [O] a numerical sign on the finite $H^1$ grading, not the full operator-valued minimisation (the v199 residual) [C][O]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| v201_seam_subprincipal_marks.py | QGE0.SUBPRIN.01 — the v199 bedrock residual reduced one genuine step further: block-diagonality of the sub-principal symbol is <i>not</i> an independent postulate. [E] the seam DtN is $\Lambda =  k  + M_f$ , $M_f$ = multiplication by a curvature function $f(\theta)$ (standard Steklov sub-principal), so $\langle n   M_f   n' \rangle = f_{n-n'}$ . [E] $M_f$ is $\mu_4$ -character-block-diagonal $\Leftrightarrow f$ has Fourier support only on modes $\equiv 0 \pmod{4}$ $\Leftrightarrow f$ is $\mathbb{Z}_4$ -invariant ( $ k $ already commutes, v198). [E] a $\mu_4$ -mark sum $f(\theta) = \sum_{j=0}^3 g(\theta - 2\pi j/4)$ has $f_m = 4g_m[m \equiv 0 \pmod{4}]$ for <i>any</i> profile $g$ — automatically $\mathbb{Z}_4$ -invariant, so (v195 marks = $\mu_4$ orbit) <i>forces</i> the sub-principal symbol block-diagonal; negative controls: 3 marks ( $\mathbb{Z}_3$ ) and 4 generic marks break it. [O] new residual: $\omega \circ \rho = \omega$ reduces to ‘the DtN sub-principal symbol is mark-local (seam flat away from the $\mu_4$ marks = conformal deck v177)’, the narrowest, structurally-definitional form of QGE0.SYM.01 [E][O]                                                                                                                                                                                                                                                                                                                                               |
| v202_rare_kaon.py               | The rare kaon sector $K \rightarrow \pi \nu \bar{\nu}$ from the TFPT CKM point ([C] downstream flavor readout, <i>not</i> a compiler power). The kernel fixes the full CKM matrix ( $s_{12} = \lambda_C$ , $s_{23} = \varphi_0/(1 + \lambda_C) =  V_{cb} $ , $s_{13} = \lambda_C^3/3$ , $\delta = \pi/3 + 3\lambda_C^2 = 1.19823$ rad; v18/v88) with no flavor fit; feeding the exact point into the standard Brod–Gorbahn–Stamou short-distance formulas gives $\text{BR}(K^+ \rightarrow \pi^+ \nu \bar{\nu}) = 9.45 \times 10^{-11}$ and $\text{BR}(K_L \rightarrow \pi^0 \nu \bar{\nu}) = 3.33 \times 10^{-11}$ . [E] CKM point + exact apex $(\bar{\rho}, \bar{\eta}) = (0.1374, 0.3509)$ , $\lambda_t = V_{ts}^* V_{td}$ ; [C] the branching ratios ( <i>not</i> parameter-free — $X_t, P_c, \kappa_\pm$ are external EW/QCD input); [E] $\text{BR}(K^+)$ $\text{BR}(K^+) = 9.45 \times 10^{-11}$ lands ON the NA62 2016–2024 combination $(9.6_{-1.8}^{+1.9}) \times 10^{-11}$ (La Thuile 2026), $+0.08\sigma$ (the 2016–2022 obs $(13.0_{-3.0}^{+3.3}) \times 10^{-11}$ was $\sim 1.2\sigma$ above) — a strong consistency hit, but an $F_{\text{transfer}}$ bridge, not a unique TFPT-vs-SM discriminator (SM $(8.6 \pm 0.4) \times 10^{-11}$ also inside the NA62 error); [E] Grossman–Nir ratio $0.352 \ll 4.3$ ; [X] dated kill test: a stable NA62 $\text{BR}(K^+)$ outside $[7, 12] \times 10^{-11}$ , or a KOTO-II $\text{BR}(K_L)$ off the GN point, breaks the flavor bridge (core untouched) [C][X] |

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| Script                    | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| v203_eht_achromatic.py    | The EHT achromatic polarization-rotation intercept $\beta_{\text{BH}}$ around a horizon-scale black hole — the surviving core of the old UFE/BH notes ( <i>not</i> the RN-type metric, superseded by Nariai/seam=horizon v101–v104/v190; only the polarization signature). [E] the coupling is a compiler number: $\beta_{\text{BH}}(r)=Q_e^{\text{eff}}Q_m^{\text{eff}}/(256\pi^4r^2)=16c_3^4(Q_eQ_m/r^2)$ , $16c_3^4=1/(256\pi^4)$ exactly. [E] $\alpha$ -kernel cross-link: $16c_3^4=\delta_{\text{top}}/3$ , $\delta_{\text{top}}=48c_3^4=3/(256\pi^4)$ — the <i>same</i> top-form coefficient that fixes the $\alpha$ -kernel precision-zone correction (one row from $\beta_{\text{rad}}=\varphi_0/4\pi$ ). [C] TFPT fixes the $1/r^2$ profile, achromaticity, and sign-flip under $E\cdot B$ reversal; the amplitude $Q_eQ_m$ is an MHD/GR weight, not a compiler output. [X] dated kill test: the residual intercept must pass three nulls (frequency/achromatic, spatial $1/r^2$ , sign-flip); a null-consistent residual after honest GRMHD subtraction falsifies the channel (core untouched) [E][C][X] |
| v204_muon_seam_g2.py      | The muon anomalous magnetic moment $a_\mu=(g_\mu-2)/2$ as a seam vertex readout ([C] downstream, <i>not</i> a compiler power; archive integration of the old muon- $g-2$ note). The carrier fixes the second-order defect $\delta_2=B\gamma\delta_{\text{top}}^2=\frac{5}{4}\delta_{\text{top}}^2$ ( $\delta_{\text{top}}=\Omega_{\text{adm}}c_3^4=48c_3^4=3/256\pi^4$ , $v2/v23$ ; $B\gamma=\frac{3}{2}\cdot\frac{5}{6}=\frac{5}{4}$ , v51); the seam-loop projection $a_\mu^{\text{seam}}=\delta_2/(2\pi)=45/(524288\pi^9)\approx 2.879\times 10^{-9}$ . [E] the value is an exact compiler number ( $\delta_2=4!\text{Tr}_{S^+}(X^2)c_3^8$ , $\text{Tr}=120=5!$ ); [C] the identification of $\delta_2/(2\pi)$ as a magnetic vertex ( $1/2\pi=4c_3$ ) is a physical bridge; [E] $0.81\sigma$ vs dispersive $\Delta a_\mu=(2.49\pm 0.48)\times 10^{-9}$ ; [C] HONEST: lattice/CMD-3 HVP shrinks it ( $\sim 1.5\times 10^{-9}$ ), the fixed value then $\sim 1.5\sigma$ high; [X] dated kill test: a converged $\Delta a_\mu$ outside $2.879\times 10^{-9}\pm 0.5\times 10^{-9}$ excludes the mechanism [E][C][X] |
| v205_xi_threequarter.py   | $\xi=c_3/\varphi_{\text{tree}}=3/4$ : an <i>independent</i> route to the gravitational $3/4$ (archive integration of the old Eliminating- $K$ note). The torsion-compression ansatz $\kappa^2=\xi\varphi_0/c_3^2$ with the Einstein limit $\kappa^2=8\pi G$ fixes $\xi=8\pi c_3^2/\varphi_0=c_3/\varphi_0$ (since $8\pi c_3^2=c_3$ ), and at tree level $\varphi_{\text{tree}}=1/(6\pi)$ this is exactly $3/4$ . [E] tree identity $\xi=c_3/\varphi_{\text{tree}}=3/4$ ; [E] Einstein-limit reduction $8\pi c_3^2=c_3\Rightarrow G$ is an output; [E] over-determination $3/4=q(A_3)=N_{\text{fam}}/ \mu_4 =\ln(m/\mu)$ (v152 $k=c_3/2$ ) — the torsion-compression $\xi$ and the gapped EH replica are two independent appearances of the same gravitational $3/4$ ; [E] with the retained seed $\xi=c_3/\varphi_0=0.748303$ ( $-0.226\%$ ); [C] the $\kappa^2=\xi\varphi_0/c_3^2$ ansatz is the bridge, the modern derivation is v152/v68, $1/G$ stays the one anchor (v60/v153) [E][C]                                                                                                                          |
| v206_h0_lambda_branch.py  | The cosmological-constant Letter branch: the seam number $\delta_\Sigma=m_1e^{-2\pi\alpha^{-1}/3}$ with the lowest block $m_1=\Omega_{\text{adm}}=48$ , the Planck-convention split, and the quantitative $H_0=66.5\text{--}67.1$ (archive integration of the tfpt-31 CC note; sharpens the qualitative $H_0$ -vs- $\Lambda$ row). [E] $\delta_\Sigma^{\text{lead}}=48e^{-2\pi\cdot 137.036/3}\approx 1.085\times 10^{-123}$ ; [E] convention split $M_{\text{Pl}}^4/\bar{M}_{\text{Pl}}^4=(8\pi)^2=631.65$ (unreduced vs reduced — must be displayed); [E] $\rho_\Lambda=M_{\text{Pl}}^4\delta_\Sigma\approx 2.4\times 10^{-47}\text{ GeV}^4$ (observed vacuum order); [E] 123-orders cross-link $ \log_{10}\delta_\Sigma \approx 122.96$ , consistent with the v60 split 122.948; [C] under flatness closure $H_0^{\text{lead}}=66.51$ , $H_0^{\text{disc}}=67.10$ — $1.6\text{--}0.5\sigma$ below Planck, $\sim 6\sigma$ below SH0ES: the Hubble tension is <i>not</i> relieved by raising the vacuum scale; $H_0$ uses imported $\Omega_b, \omega_{\text{DM}}$ , a [C] readout [E][C]                          |
| v207_asymptotic_safety.py | Asymptotic Safety as a <i>second</i> , independent, [O] route to the metric sector (archive integration of the old Five-Problems note). The current theory closes QG via the discrete seam-net/AQFT programme (v175; open premise QGEO.REALIZE.01); this records a complementary continuum-FRG argument that a non-Gaussian UV fixed point exists from the same axioms — a plausibility cross-check, <i>not</i> load-bearing, does <i>not</i> close $G_{\text{metric}}$ . [E] the axion coupling is fixed and small $y^2=16c_3^2=1/(4\pi^2)=0.025330$ exactly ( $g_{a\gamma\gamma}=-4c_3$ from the cubic); [E] existence by IVT: $\partial_t g/g=2+\eta_R$ is $+2>0$ as $g\rightarrow 0$ and $<0$ at large $g$ (torsion $K^2$ damping) $\Rightarrow$ a root $g_*>0$ ; [E] $g_*=2/a_1>0$ finite, Reuter-type $O(1)$ , no new free input; [O] HONEST: a continuum-FRG ansatz the discrete-compiler theory did <i>not</i> adopt, primary stays the seam-net (v175) [E][O]                                                                                                                                             |

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| Script                        | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| v208_bh_thermodynamics.py     | Black-hole thermodynamics from the seam: the modular Hawking temperature and the scalaron-corrected Wald entropy (archive integration of the dropped tfpt-42 stationary-horizons section, re-typed through the current Tomita–Takesaki seam work v198/v199). [E] modular $T_H: \Delta^{it}=e^{-2\pi t K_H}$ (Bisognano–Wichmann) $\Rightarrow$ KMS at $\beta=2\pi$ and $T_H=\kappa/(2\pi)$ ; the $2\pi$ is the seam unit $1/(4c_3)$ ( $1/2\pi=4c_3$ , v58), reproducing $T_H=c_3/M$ (v57); seam $\leftrightarrow$ horizon identification [C]; [E] Wald entropy: with the induced $f(R)=R+R^2/(6M_s^2)$ ( $M_s=c_3^{7/2}\bar{M}_{\text{Pl}}$ , v28), $S_W=(f_R A)/(4G)=(A/4G)(1+R_h/3M_s^2)$ — leading $A/4G$ is the $c_3$ area law ( $1/4=1/ \mu_4 $ ), the correction an exact scalaron consequence; [E] SdS $T_H=(1/4\pi r_h)(1-\Lambda r_h^2)$ , $1/4\pi=2c_3$ ; [E] area law $S_{BH}=M^2/(2c_3)=4\pi M^2=A/4G$ [E][C]                                                                                                                                                                                                                                                                                                                                                 |
| v209_bh_regular_core.py       | The black hole as a seam fixed point (topological defect), <i>not</i> a singularity: the honest [O] synthesis of the old Five-Problems BH picture, with the superseded torsion/RN-vortex metric dropped (RN-type metric superseded by Nariai/seam=horizon v101–v104/v190; cf. v203). Built entirely from established suite atoms. [E] no singularity = seam fixed point: the gapped attractor multiplier $\lambda_2=(2/3)^6=64/729$ with gap $\Delta=6\log(3/2)>0$ (v56) — the core is a fixed point, $\rho\rightarrow\infty$ replaced by $\varphi\rightarrow\varphi_*$ [C]; [E] maximal BH = anchor: Nariai cubic $t^3-3t+2=(t-1)^2(t+2)$ , roots $(1, 1, -2)$ = the traceless projection of $a=(1, 1, 2)$ (v101/v190); [E] information stays in the boundary field: Page recovery decays at $\lambda_2=(2/3)^6$ per step (v54) [C]; [E] holographic remnant $S_{BH}=A/4G$ , $1/4=1/ \mu_4 $ [C]; [O] HONEST: a qualitative structural synthesis (reuses v54/v56/v57/v101/v190), <i>not</i> load-bearing, <i>not</i> a metric [E][O]                                                                                                                                                                                                                                     |
| v210_mark_local_dtn.py        | A numerical <i>sharpening</i> of v201/QGEO.SUBPRIN.01 on <i>realistic</i> Steklov curvature profiles, carried to the quasi-free <i>state</i> level (the actual $\omega\rho=\omega$ , not just $[\rho, \Lambda]=0$ ); <i>not</i> a closure of QGEO.SYM.01. A real von-Mises bump $g(\theta)$ at a puncture, summed over the four $\mu_4$ marks $f(\theta)=\sum_{j=0}^3 g(\theta-j\pi/2)$ , builds the Steklov DtN $\Lambda= D_\theta +M_f$ . [E] every off-(mod 4) Fourier coefficient of the real $f$ vanishes to $<10^{-16}$ (exact 4-fold cancellation), although $g$ has all modes. [C] $\ [\rho, \Lambda]\  < 10^{-15}$ . [C] <i>state invariance</i> (new vs v201): the quasi-free covariance $C=\frac{1}{2}(1+\text{sgn } H_1)$ satisfies $\rho C\rho^\dagger=C$ to $<10^{-13}$ — the state, not just the operator, is clock-invariant. [C] negative controls (single off-centre bump, $\mathbb{Z}_3$ source, 4 generic marks) break both by $O(1)$ ; convergence stable for $N\in\{16, 32, 64\}$ . [O] certifies ‘mark-local DtN $\Rightarrow \omega\rho=\omega$ ’ for concrete profiles <i>and</i> at the state level; the premise that the physical seam <i>is</i> mark-local stays [O] — the narrowest definitional form of QGEO.SYM.01, not its closure [C][O] |
| v211_axion_spine_angle.py     | Axion DM, the spine-angle branch $\theta_i = \pi N_{\text{fam}}/g_{\text{car}} = 3\pi/5$ ( $108^\circ$ ): an alternative, more robust misalignment angle than the determinant-line hilltop $\theta_i \approx 170^\circ$ (v185, which over-produces $\Omega_a h^2 \approx 0.66$ ). [E] rational motive $\theta_i = 3\pi/5$ exactly (spine quotient $3/5$ , no fit); [E] the v185 finite-T solver ( $\theta=1 \rightarrow 0.03$ ) reaches $\Omega_{\text{DM}}$ at $\theta \approx 106^\circ$ , so $108^\circ$ is on target (harmonic $0.03\theta^2=0.107$ ); [C] <i>robust</i> : $108^\circ$ is $62^\circ$ below the $170^\circ$ hilltop (mild-anharmonic, not exponentially sensitive); [O] an alternative ansatz to $\theta_i=\pi(1-\varphi_{\text{seam}})\approx 170^\circ$ (mutually exclusive, the full solver decides, DM.AXION.SPINE.01); [X] kill test: $\Omega_a h^2$ outside $\sim[0.08, 0.16]$ demotes the branch [E][C][O][X]                                                                                                                                                                                                                                                                                                                                   |
| v212_leptogenesis_decouple.py | Leptogenesis, the scalaron-decouple branch: both Boltzmann inputs share the decouple $A_\Lambda=10= E(K_5) $ — $M_1=M_{\text{scal}}\varphi_0^2/A_\Lambda \approx 8.65\times 10^9$ GeV and $\tilde{m}_1=m_3/A_\Lambda \approx 5$ meV. A cleaner [C] route than v184’s $M_1=M_R\varphi_0^4$ (which only relocated the free input to the un-pinned seesaw $M_R$ ). [E] shared decouple $A_\Lambda=10$ ; [E] $M_1=8.65\times 10^9$ GeV inside the thermal window $[3\times 10^9, 3\times 10^{10}]$ (v169); [E] $\tilde{m}_1 \approx 5$ meV, $K \approx 4.6$ strong washout; [E] Davidson–Ibarra $\varepsilon_1 \approx 8.5\times 10^{-7}$ , $\eta_B \approx 1.2\times 10^{-9}$ brackets observed $6.1\times 10^{-10}$ ; [C] HONEST: $M_1$ uses only $\{M_{\text{scal}}, \varphi_0, A_\Lambda\}$ but the coupling mechanism is posited not derived; [X] kill test falls the route, not the theory (FR.ETAB.01 independent) [E][C][X]                                                                                                                                                                                                                                                                                                                                           |

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| Script                         | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| v213_ftransfer_<br>functor.py  | The <code>F_transfer</code> functor contract ( <code>CONTRACT.F.01</code> ): the four frontier transfers are ONE typed functor $F_{\text{transfer}} = F_{\text{observable}} \circ F_{\text{threshold}} \circ F_{\text{RG}}$ with four structural axioms, each instance a discrete compiler kernel <b>[E]</b> composed with an external continuous solver <b>[C]</b> ; consolidates v183 ( $F_{\text{pole}}$ ), v212 ( $F_{\text{Boltzmann}}$ ), v211 ( $F_{\text{relic}}$ ), v164/FR.MPME ( $F_{\text{QCD}}$ ), the structural complement of the v187 typing guard. <b>[E]</b> axiom 1 $\mu_4$ -deck equivariance ( $\lambda_2 = (2/3)^6 = 64/729$ the deck transfer eigenvalue); <b>[E]</b> axiom 2 Plücker preservation ( $53 = a^T(R+Q)\mathbf{1}$ , $\text{Pl}(F) = (1, 22, 30)$ , $\ \text{Pl}(K)\ _1 = 11$ ); <b>[E]</b> axiom 3 positivity/stochasticity ( $\text{spec } T = \{1, (2/3)^6, (1/3)^6\}$ positive, $\kappa_f \in (0, 1)$ ); <b>[E]</b> axiom 4 external modules explicit ( $b_3 = -7$ a carrier output, $ b_3  = \Omega_{\text{adm}} - 10b_1 = 7$ ); <b>[C]</b> verdict: a typed functor, not a bag of open topics; an architectural consolidation guarded by v187, not a closure <b>[E][C]</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| v214_seam_<br>pillowcase.py    | The <i>pillowcase</i> reduction of <code>QGEO.SYM.01</code> ( <code>QGEO.PILLOW.01</code> ): the two separately-tracked open QGEO residuals — v180/QGEO.ISO.01 (the carrier clock is an order-4 isometry of the seam metric) and v201/QGEO.SUBPRIN.01 (the DtN sub-principal symbol is mark-local, the seam flat away from the $\mu_4$ marks) — are ONE statement, ‘the seam carries the uniformising flat orbifold (pillowcase) metric’, and the order-4 clock is <i>derived</i> from cross-ratio 2 (v168), not assumed. <b>[E]</b> $\chi_{\text{orb}}(S^2(2, 2, 2, 2)) = 2 - 4(1 - \frac{1}{2}) = 0 \Rightarrow \text{flat}$ (Trojanov cone-metric uniformisation, unique up to scale; Gauss–Bonnet four deficits $\pi$ sum to $4\pi = 2\pi\chi(S^2) \Rightarrow \int K_{\text{smooth}} = 0$ , flat away from the marks). <b>[E]</b> NEW LINK: cross-ratio( $\mu_4$ ) = 2 (v168) $\Rightarrow j(\lambda) = 256(\lambda^2 - \lambda + 1)^3 / (\lambda^2(\lambda - 1)^2)$ with $j(2) = 1728$ (all six harmonic cross-ratios $\{2, -1, \frac{1}{2}\}$ give 1728) $\Rightarrow$ the square modulus, hence the order-4 clock, is a <i>consequence</i> of cross-ratio 2. <b>[E]</b> $j = 1728 \Leftrightarrow \tau = i$ (square torus) $\Leftrightarrow \text{CM by } \mathbb{Z}[i] \Leftrightarrow \text{Aut} = \mathbb{Z}/4$ ( $j=0$ gives $\mathbb{Z}/6$ ; generic $\mathbb{Z}/2$ ). <b>[E]</b> explicit CM $(x, y) \mapsto (-x, iy)$ on $y^2 = x^3 - x$ has order $4 = z \mapsto iz$ . <b>[E]</b> negative control $\lambda = 3 \Rightarrow j = 21952/9 \notin \{0, 1728\} \Rightarrow \text{Aut} = \mathbb{Z}/2$ , no clock. <b>[E]</b> UNIFICATION: ‘seam = uniformising flat pillowcase metric’ implies both v201 mark-locality and v180 QGEO.ISO.01. <b>[O]</b> does <i>not</i> close QGEO.SYM.01 — the canonical constant-curvature-metric premise stays open, milder/more canonical; net existence + full-cone RP remain v175’s <b>[E]</b> |
| v215_seam_deck_<br>killtest.py | The bedrock <code>QGEO.SYM.01</code> ( $\omega \circ \rho = \omega$ ) recast as a <i>falsifiable</i> kill-test ( <code>QGEO.KILL.01</code> ), not a proof-promise — a foundational symmetry cannot be derived from nothing (v181/v153), and Bisognano–Wichmann would presuppose the covariance it should produce. Non-circular reduction (v198/v199/v201): $\omega \circ \rho = \omega \Leftrightarrow [\rho, C] = 0$ ; the principal symbol $ k $ commutes exactly, so the residual is the sub-principal $M_f$ with $[\rho, M_f] = 0 \Leftrightarrow f$ $\mathbb{Z}_4$ -invariant; measure form = the raw Gaussian bulk form $A$ is $\mu_4$ -deck-invariant. Four levers (carrier-side <b>[E]</b> , seam-side <b>[O]</b> ): <b>K1</b> exactly 4 marks (order-4 clock free orbit $\mu_4 + b_1 = \#\text{marks} - 1 = 3 = N_{\text{fam}} \Rightarrow \#\text{marks} = 4 =  \mu_4 $ ); <b>K2</b> square config (cross-ratio $2 \Rightarrow j = 1728$ , $\text{Aut} = \mathbb{Z}/4$ ; order 4 selects $\mathbb{Z}/4$ not the $j=0$ $\mathbb{Z}/6$ ; generic $\Rightarrow \mathbb{Z}/2$ ); <b>K3</b> mod-4 sub-principal support ( $\mu_4$ -mark sum $\Rightarrow f_m = 0$ for $m \not\equiv 0 \pmod{4} \Rightarrow [\rho, M_f] = 0$ ); <b>K4</b> transfer gap $(2/3)^6 = 64/729$ (cusp weights $\{0, \frac{1}{3}, \frac{2}{3}\}$ ). Freeze-bound ( <code>freeze_file.csv seam_deck_symmetry</code> ). <b>[O]</b> the decisive kill is <i>deferred</i> ( <code>QGEO.REALIZE.01</code> ): the raw $\mathbb{P} \setminus \mu_4$ collar DtN must yield 4 square marks <i>without</i> inputting them; carrier-side K1–K4 are green regression guards. A kill-test, not a closure                                                                                                                                                                                                                                                                         |

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| Script                        | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| v216_marks_gauss_bonnet.py    | The four seam marks <i>emerge</i> from Gauss–Bonnet (QGEO.MARKS.03), executing the K1 lever of the v215 kill-test. [E] mark count = $2\chi$ : $\mathbb{Z}_2$ branch points (deficit $\pi$ ) on a flat sphere ( $\chi=2$ ) give $n\pi = 2\pi\chi = 4\pi \Rightarrow n = 2\chi = 4 =  \mu_4  = e_1(a) = N_{\text{fam}}+1$ (the same $\chi=2$ as the 8 in $c_3=1/(8\pi)$ ). [E] the Euclidean sphere 2-orbifolds ( $\chi_{\text{orb}}=0$ ) are exactly $(2, 3, 6), (2, 4, 4), (3, 3, 3), (2, 2, 2, 2)$ . [E] all-order-2 (the $ \mathbb{Z}_2 $ sheet branch) $\Rightarrow$ uniquely $(2, 2, 2, 2)$ ; [E] $N_{\text{fam}}=3$ (rank $H^1=\#\text{marks}-1$ ) $\Rightarrow$ uniquely $(2, 2, 2, 2)$ , picking the 4-mark square over the 3-mark hexagonal $(3, 3, 3)$ ; [E] a free order-4 deck $\Rightarrow$ uniquely $(2, 2, 2, 2)$ — three independent selectors converge. [O] residual: the square modulus (cross-ratio $2 \Rightarrow j=1728$ , v214) needs the one order-4 carrier input. The mark count + equality are now derived; does <i>not</i> close QGEO.REALIZE.01                                                                                                                                                    |
| v217_marks_emergence_scan.py  | The <i>numerical</i> emergence scan (QGEO.EMERGE.01, the sibling of v216 as v210 is to v201): with the number of marks $n$ and their positions free, the 4-equally-spaced (square) configuration is the joint minimiser of {clock-invariance + 3-family cohomology + transfer gap $(2/3)^6$ } on the actual seam DtN/state. [C] $n=4$ equally-spaced $\Rightarrow \ \rho_4 C \rho_4^\dagger - C\  \sim 10^{-14}$ (clock-invariant state); [C] family count selects $n=4$ (scan $n \in \{2..8\}$ , only $n-1=3=N_{\text{fam}}$ ; $n=3 \rightarrow 2$ families = hexagonal, excluded; $n=5 \rightarrow 4$ ); [C] gap: $\mu_4$ -equivariant $\Rightarrow$ diagonal in the cusp basis (deck-average off-diagonal $\sim 10^{-16}$ ), cusp weights $\{0, \frac{1}{3}, \frac{2}{3}\} \Rightarrow$ transfer $(1-w)^6 = \{1, (2/3)^6, (1/3)^6\}$ , gap $(2/3)^6 = 64/729$ ; [C] position scan: the square minimises the defect (square $\sim 10^{-14}$ vs off-square $\sim 0.4$ ); [C] negative controls $n=3, 5, 6$ . [O] confirms the emergence on the real DtN (number scanned, positions free); selector = the 3-family/gap structure (carrier input); executes K1 numerically, does <i>not</i> close QGEO.REALIZE.01. Python-only |
| v218_diamond_axis_geometry.py | Diamond axis geometry: the Sheet Diamond (v94) is a discrete geometry with two axes around the centered cross (v95: $C$ center, $U$ winding, $V$ sheet), $F$ the transfer completion. Three exact [E] lemmas + honest audit/heuristic typing, no new numbers. [E] AXIS CURVATURE: $\det(C+xU)=14+6x$ linear (slope $6= R^+(A_3) $ ), $\det(C+yV)=14+14y+4y^2$ quadratic, 2nd difference $8=\text{rank } E_8$ ; $\det B(C+xU)=2\det(C+xU)$ , anchor-block sheet curvature $6= R^+(A_3) $ . [E] PLÜCKER LADDER: $\text{Pl}(K)=(-1, 6, 4) \rightarrow \text{Pl}(C)=(0, 14, 14) \rightarrow \text{Pl}(F)=(1, 22, 30)$ , steps $(1, 8, 10)=(N_\Phi, \text{rank } E_8, A_\Lambda)$ and $(1, 8, 16)=(N_\Phi, \text{rank } E_8, \dim S^+)$ (identity [E], transfer reading [C]). [E] SPECTRAL RAMIFICATION: $\text{Disc}(\chi)=q(r)^2\text{Disc}(q)$ for $Q, K, C, F$ ; squares $(1, 3, 4, 6)$ , kernels $(13, 48, 65, 105)$ with $48=\Omega_{\text{adm}}$ , $105=N_{\text{fam}}g_{\text{car}} \cdot 7$ . [C] heuristic: $F$ the transfer-completion corner, a search principle, not a hard filter; the $G_2/F_4$ pair-sum labels stay audit-only (cf. v94/v95) [E][C]                                                                |
| v219_icosahedral_mckay.py     | McKay bedrock: <i>why</i> the atoms are $\{2, 3, 5\}$ . $E_8$ is the exceptional <i>top</i> of the McKay tower of finite $SU(2)$ subgroups ( $2T \rightarrow \hat{E}_6$ , $2O \rightarrow \hat{E}_7$ , $2I \rightarrow \hat{E}_8$ ). The McKay graph is built <i>from the group</i> : the 120 icosians close to the binary icosahedral $2I$ (element orders $\{1, 2, 3, 4, 5, 6, 10\}$ , containing the 2, 3, 5 axis orders), 9 conjugacy classes, Dixon irrep degrees $\{1, 2, 2, 3, 3, 4, 4, 5, 6\}$ (sum $30=h(E_8)$ , sum of squares $120= R^+(E_8) $ ); the natural-rep McKay adjacency <i>is</i> the affine $E_8$ graph with Kac marks = the degrees. A backward certificate of the closed $E_8$ [E], the reading that it explains $(2, 3, 5)$ [C], the raw-seam origin [O]                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| v220_cp_hexagonal_modulus.py  | CP lives in the <i>hexagonal</i> phase fiber, not the square seam deck (a geometric sharpening of red-team Target D). [E] the seam deck is the square modulus $j(i)=1728$ , $\text{Aut}=\mathbb{Z}/4$ (the $\mu_4$ clock); its exceptional partner is the hexagonal modulus $j(\rho)=0$ , $\rho=e^{i\pi/3}$ , $\text{Aut}=\mathbb{Z}/6$ , CM by $\mathbb{Z}[\omega]$ , $\arg \rho=\pi/3$ . [E] the frozen leading CKM reading $\delta=\pi/3+3\lambda_C^2=68.654^\circ$ (v88) has leading term $\pi/3=\arg \rho$ ; the $ \mathbb{Z}_2 $ alternative $\pi/3+2\lambda_C^2$ is the frozen look-elsewhere trap. The deck stays $\mathbb{Z}/4$ (v215); CP is the $\mathbb{Z}/6$ fiber, the bridge [C][E][C]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

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| Script                                | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| v221_seam_qecc.py                     | The seam as a <i>finite recoverability</i> code (recovery rate $(2/3)^6=64/729$ ). [E] code dimension $\dim S^+=16=2^{g_{\text{car}}-1}$ ; the gapped transport $T$ on the cusp-weight 3-space (deviations $(1, -1, 0)$ and the Nariai anchor $(1, 1, -2)$ ) is symmetric, doubly stochastic (CPTP), spectrum $\{1, (2/3)^6, (1/3)^6\}$ ; on the trace-zero subspace $\ T\delta\ \leq(2/3)^6\ \delta\ $ hence $\ T^n\delta\ \leq(2/3)^{6n}\ \delta\ $ (a Knill–Laflamme-type bound); a free ratio breaks it. [C] the holographic/Petz reading (gated on the Seam–Horizon theorem) [E][C]                                                                          |
| v222_cm_norm_duality.py               | CM-norm duality: the two exceptional moduli give 41 and 7. [E] SQUARE (Gaussian $\mathbb{Z}[i]$ , $j=1728$ ): $N(g_{\text{car}}+i \mu_4 )=5^2+4^2=41=10b_1$ — the EM index <i>is</i> the Gaussian norm of the carrier-glue vector; $N(3+2i)=13=\Delta_Q$ . [E] HEX (Eisenstein $\mathbb{Z}[\omega]$ , $j=0$ ): $N(3+2\omega)=3^2-3\cdot 2+2^2=7=\text{scalaron}$ . The $(3, 2)=(\text{colour}, \text{weak})$ split generates $(5, 6, 7, 13)$ by sum / product / Eisenstein / Gauss norm; the rings are rigid (Eisenstein of $(5, 4)=21\neq 41$ ). The architecture bridge square $\leftrightarrow$ EM, hex $\leftrightarrow$ scalaron is [C][E][C]                |
| v223_coxeter_totative_clock.py        | The $\mu_4$ clock is the order-4 character of the $E_8$ Coxeter cycle. [E] $(\mathbb{Z}/30)^\times=\{1, 7, 11, 13, 17, 19, 23, 29\}$ = the $E_8$ exponents = live Coxeter phases; $\varphi(30)=8=\text{rank } E_8$ , $30=2\cdot 3\cdot 5$ . [E] $(\mathbb{Z}/30)^\times\cong\mathbb{Z}/4\times\mathbb{Z}/2$ with order-4 generator 7 ( $7^2=19$ , $7^4=1 \bmod 30$ ): a literal $\mu_4$ clock $\langle 7 \rangle=\{1, 7, 13, 19\}$ on the exponents; the conjugate pairs $m+(30-m)=30$ give $ \mu_4 =4$ invariant planes the clock permutes. Only $E_8$ has $\varphi(h)=\text{rank}$ and $h=2\cdot 3\cdot 5$ ; the seam-clock identification is [C][E][C]         |
| v224_diamond_fttransfer_path.py       | $F_{\text{transfer}}$ as <i>one</i> path on the Sheet Diamond, not four external interfaces. [E] family $M(s, t)=R+Q \text{diag}(s, t, t)$ : the corners $K=M(1, -1)$ , $C=M(1, 0)$ , $F=M(1, 1)$ lie on the sheet axis ( $s=1$ ); the sheet axis is curved $\det M(1, t)=4t^2+14t+14$ (2nd diff $8=\text{rank } E_8$ ), the winding axis $\det M(s, 0)=6s+8$ flat (slope $6= R^+(A_3) $ ); Plücker steps $(1, 8, 10)$ , $(1, 8, 16)$ . [C] $F_{\text{transfer}}$ is the $K\rightarrow C\rightarrow F$ sheet geodesic, Koide source $\rightarrow$ pole its 1-D Möbius projection (v37/v183) [E][C]                                                                |
| v225_dual_normal_frame.py             | The dual normal frame $(d, n)$ : one boundary compass for flavor and horizon, with CP as orientation. [E] $d=a^\top R^{-1}=a^\top L^{-1}=(-\frac{1}{2}, -\frac{1}{2}, 1)=-\frac{1}{2}(1, 1, -2)$ (the Nariai/horizon normal; $d\cdot 1=0$ , $d\cdot a=1$ ); $n=(5, -9, 6)$ with $n^\top R=(\det R, 0, 0)=(8, 0, 0)$ , $n^\top L=(20, 0, 0)$ (the first-generation/torsion selector). [E] oriented volume $\det(1, d, n)=21=N_{\text{fam}}\cdot 7$ . [C] CP as orientation: $\text{Im } \det(1, d, \rho n)=21 \sin(\pi/3)$ for $\rho=e^{i\pi/3}$ — CP sits where the magnitude bijection (Target D) fails [E][C]                                                   |
| v227_degree_exponent_channel_split.py | $248=120+128$ as a magnitude/phase channel typing (replaces the over-strong $S^-$ -dark-matter reading). [E] $E_8$ exponents sum $120= R^+(E_8) $ ( <i>magnitude</i> channel); degrees $\{2, 8, 12, 14, 18, 20, 24, 30\}$ sum $128=2^{\text{rank}-1}=\text{rank}\cdot \dim S^+$ ( <i>phase/glue</i> channel); $248=120+128=\text{adj}(D_8)+\text{spinor}(D_8)$ ; $\sum \deg - \sum \exp = \text{rank}=8$ . [C] ratios+products (Target D) live in the 120 channel, the un-covered CP phases in the 128 phase/glue channel; [O] a dark-sector reading of 128 stays Frontier [E][C][O]                                                                              |
| v228_rr_index_gate.py                 | P2 as the mode space of a degree-4 seam divisor (a Riemann–Roch <i>index gate</i> ; bedrock language for P2, not a proof from nothing). [E] $\deg D=4= \mu_4 $ on $\mathbb{P}^1 \Rightarrow h^0(\mathcal{O}(D))=5=g_{\text{car}}$ (Riemann–Roch, genus 0); $\text{rank } H_1(\mathbb{P}^1\setminus 4)=3=N_{\text{fam}}$ ; $h^0+\text{rank } H_1=8=\text{rank } E_8$ , $h^0-\text{rank } H_1=2= \mathbb{Z}_2 $ ; $\Lambda^{\text{even}}(\mathbb{C}^5)=1+10+5=16=\dim S^+$ . The mode-space reading is [C], conditional on the raw seam producing the degree-4 divisor ([O], = QGEO.SYM.01) [E][C][O]                                                               |
| v229_lepton_frobenius_algebra.py      | The charged leptons as an étale Frobenius algebra over the $\mu_6$ resolvent. [E] exact coefficients $(c_e, c_\mu, c_\tau)=(\frac{16}{7}, \frac{4}{3}, \frac{7}{6})$ , ring closure $c_e c_\tau = \frac{8}{3} =  \mathbb{Z}_2  c_\mu$ , product $\frac{32}{9} = 2^{g_{\text{car}}} / N_{\text{fam}}^2$ . [E] $A=\mathbb{Q}[t]/(m)$ , $m=\prod(t-c)$ : the trace-form Gram is nondegenerate ( $\det=\text{disc}(m)\neq 0$ , $A$ étale) — a commutative Frobenius algebra; the $C_6$ shift has characteristic polynomial $t^6-1$ (spectrum $\mu_6=\mu_3$ family $\times \mu_2$ sheet). [C] the PMNS extension via $\text{Aut}(A)$ and the hexagonal CM point [E][C] |

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| Script                            | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| v230_center_budget_norms.py       | The Sheet-Diamond center budget (7, 11, 13) is the three <i>local norms</i> of the theory. [E] $C=R+Q$ diag(1, 0, 0), row sums (7, 11, 13): $7=N_{\mathbb{Z}[\omega]}(3+2\omega)$ (Eisenstein/hex = scalaron), $13=N_{\mathbb{Z}[i]}(3+2i)$ (Gauss/square = $\Delta_Q$ ), $11=\binom{4}{0}+\binom{4}{1}+\binom{4}{2}=1+4+6$ (QBL boundary Fock count on the 4 $\mu_4$ corners = $\dim S^+ - g_{\text{car}}=16-5$ ). The two CM rings (v222) and the seam boundary count meet in one operator center; the reading is [C][E][C]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| v231_cp_mu6_phases.py             | Both CP phases are $\mu_6$ powers of <i>one</i> hexagonal CM unit, split by the sheet — a structural reduction of red-team Target D (sharper than v220/v225). With $\rho=e^{i\pi/3}$ (the $j=0$ Eisenstein CM unit): [E] $\rho$ is a primitive 6th root ( $\rho^6=1$ , $\rho^3=-1$ the $\mathbb{Z}_2$ sheet half-turn); [E] $\delta_{\text{CKM}}$ leading = $\arg(\rho^1)=\pi/3$ (quark), $\delta_{\text{PMNS}} = \arg(\rho^4)=4\pi/3$ (lepton, the <i>phase lattice</i> ); [E] $\rho^4=-\rho \Rightarrow \delta_{\text{PMNS}}=\delta_{\text{CKM,lead}}+\pi=(\text{CM unit})\times(\mathbb{Z}_2 \text{ sheet})$ — the two CP phases are <i>one</i> hexagonal unit on the two sheets, not two independent inputs; [E] the $C_6/\mu_6$ monodromy has charpoly $t^6-1$ (spectrum $\mu_6$ carries $\rho$ ); [E] orientation $\text{Im det}(\mathbf{1}, d, \rho^1 n)=+21 \sin(\pi/3)$ , $\text{Im det}(\mathbf{1}, d, \rho^4 n)=-21 \sin(\pi/3)$ (sheet-flipped Jarlskog, v225); [E] NEG $\mu_3$ has no $-1$ (no sheet; $\mu_6=\mu_3$ family $\times \mu_2$ sheet). REDUCTION: 2 CP phase inputs $\rightarrow$ 1 hexagonal unit + the sheet; the physical CP derivation stays [C], the deck stays $\mathbb{Z}/4$ (v215) [E][C]                                                                                                                                                                                              |
| v232_e8_kleinian_seam.py          | The seam as the $E_8$ <i>Kleinian singularity</i> — a canonical algebraic-geometric model for the open seam-realisation premise QGEO.REALIZE.01 (the du Val side of the McKay correspondence, v219). [E] the McKay graph of $2I$ is affine $E_8$ ; dropping the unique trivial-rep node (the single Kac mark 1) leaves the <i>finite</i> $E_8$ Dynkin on 8 nodes = the dual intersection graph of the eight exceptional $\mathbb{P}^1$ 's of the minimal resolution of $\mathbb{C}^2/2I$ (trivalent node, arms 1, 2, 4). [E] eight exceptional $\mathbb{P}^1$ 's = $\text{rank } E_8 = g_{\text{car}} + N_{\text{fam}} = \# \text{ nontrivial } 2I \text{ irreps}$ — a fourth reading of the 8 in $c_3=1/(8\pi)$ . [E] the negated intersection form is the $E_8$ Cartan (even, $\det=1$ , positive definite; each $\mathbb{P}^1$ a $(-2)$ -curve). [E] one degree-1 irrep $\Rightarrow 2I$ perfect $\Rightarrow H_1(S^3/2I)=0$ : the link is the Poincaré homology 3-sphere, $\pi_1=2I$ of order $120= R^+(E_8) $ . [C] seam reading: the raw RP seam ( $\mathbb{P}^1$ with marks) is canonically a du Val exceptional curve — a model for QGEO.REALIZE.01, not a P2 proof (presupposes $E_8$ ); bedrock [O][E][C][O]                                                                                                                                                                                                 |
| v233_cp_triality_phase.py         | The CP phase is the <i>universal</i> family/triality phase, only sheet-split — one step past v231 in $\mu_6=\mu_3$ (family/triality) $\times \mu_2$ (sheet). [E] $\mu_6=\mu_3 \times \mu_2$ : the $\mathbb{Z}_3$ triality centre $\{1, \omega, \omega^2\}$ = the cusp weights $\{0, \frac{1}{3}, \frac{2}{3}\} \times 2\pi$ , the sheet $\mu_2=\{1, -1\}$ ; $\rho=\omega^2 \cdot (-1)$ . [E] SAME FAMILY CLASS: $\rho^4=\rho^1 \cdot \rho^3$ with $\rho^3 \in \mu_2$ , so the quark ( $\rho^1$ ) and lepton ( $\rho^4$ ) CP phases have the <i>same</i> image in $\mu_6/\mu_2 \cong \mu_3$ and <i>opposite</i> sheet — the CP phase <i>is</i> the universal triality phase $\omega^2$ (the $2/3$ cusp-weight $\mathbb{Z}_3$ centre), not a free power choice. [E] $\omega$ is the $\mu_3$ factor of the order-30 (2, 3, 5) Milnor/Coxeter monodromy that builds $E_8$ (v232). Reduces Target D from ‘one hexagonal unit + sheet’ (v231) to ‘the universal triality phase + sheet’; the physical CP derivation stays [C][E][C]                                                                                                                                                                                                                                                                                                                                                                                          |
| v234_seam_holomorphy_selection.py | The Seam-Holomorphy selection certificate: the entire structural residual is <i>one</i> condition with <i>three</i> equivalent faces, all forcing $E_8$ . The condition: <i>the seam carries no nontrivial abelian sector</i> . [E] (i) AQFT — the boundary net is holomorphic ( $\mu$ -index 1); (ii) geometry — the seam link is a homology 3-sphere $\Leftrightarrow$ the binary group is perfect $\Leftrightarrow 2I$ ; (iii) rep theory — exactly one 1-dim irrep. [E] $\#(\text{mark-1 affine nodes}) =  \Gamma^{\text{ab}}  =  H_1(S^3/\Gamma)  = \#(1\text{-dim irreps})$ : $A_n \rightarrow n+1$ , $D_n \rightarrow 4$ , $E_6 \rightarrow 3$ , $E_7 \rightarrow 2$ , $E_8 \rightarrow 1$ — only $E_8$ gives 1 ( $2I$ the unique perfect $SU(2)$ subgroup, Poincaré the unique homology-sphere space form). [E] holomorphic $c=8=g_{\text{car}}+N_{\text{fam}} \Rightarrow$ the unique even unimodular rank-8 lattice = $E_8$ . [E] the three faces (v83/v154, v232, v219) are the SAME $E_8$ -selector, so Target A, $G_{\text{net}}$ , the carrier P2 and the Kleinian seam are <i>one</i> gate. [O] the closing theorem (stated): $\text{RP}+\text{gap}+\text{chirality} \Rightarrow \text{quasi-free bulk (v160)} \Rightarrow \text{invertible bulk} \Rightarrow \text{holomorphic boundary} \Rightarrow E_8$ ; the single residual analytic step is ‘free bulk $\Rightarrow$ holomorphic boundary’ [E][O] |

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| Script                            | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| v235_seam_chern_simons.py         | The closing step in abelian Chern–Simons language: holomorphic $\Leftrightarrow \det K=1$ — a genuine reduction (not a closure) of GATE.HOLO.01 into standard anyon-condensation physics. [E] a free gapped bosonic 2+1d bulk = an even integer $K$ -matrix with $\# \text{anyons} =  \det K $ , edge $c = \text{signature}(K)$ , holomorphic $\Leftrightarrow  \det K =1$ . The TFPT extension tower (v92) is the condensation tower read by determinant: carrier $D_5 \oplus A_3$ ( $\det 16$ , 16 anyons) $\rightarrow SO(16)_1 = D_8$ ( $\det 4$ ) $\rightarrow (E_8)_1$ ( $\det 1$ , holomorphic), all rank 8, $c = \text{sig}=8=g_{\text{car}}+N_{\text{fam}}$ . [E] holomorphic $\Leftrightarrow \det 1 \Leftrightarrow E_8$ = the Kitaev $E_8$ quantum-Hall state. [E] condensation: a Lagrangian glue of order $\sqrt{16}= \mu_4 =4$ takes $\det 16 \rightarrow 1$ (a partial order-2 isotropic only $\rightarrow 4=D_8$ ). [E] NEG: $D_8$ is free+bosonic but has 4 anyons; an odd $K$ is fermionic — $\det=1$ is the extra condition. [O] residual: ‘the free RP seam condenses the order- $ \mu_4 $ Lagrangian glue’ = the sheet/QGEO selection, now with holomorphy $\Leftrightarrow \det 1$ a theorem [E][O]                                  |
| v236_brieskorn_capstone.py        | The $(2, 3, 5)$ Brieskorn singularity $x^2+y^3+z^5$ is the <i>one</i> generator of the discrete skeleton — the capstone of the icosahedral round (v55/v219/v223/v232/v233); its exponents <i>are</i> the atoms $( \mathbb{Z}_2 , N_{\text{fam}}, g_{\text{car}})=(2, 3, 5)$ . [E] Milnor number $\mu=(2-1)(3-1)(5-1)=1\cdot 2\cdot 4=8=\text{rank } E_8$ = the 8 in $c_3=1/(8\pi)$ (a fifth independent origin). [E] the Milnor monodromy eigenvalues are the eight primitive 30th roots $e^{2\pi i m/30}$ , $m$ the $E_8$ exponents (charpoly $\Phi_{30}$ , $\deg \varphi(30)=8=\mu$ ) — so it <i>is</i> the order-30 $E_8$ Coxeter element (v55). [E] both clocks: the monodromy group $\langle h \rangle = \mathbb{Z}/30 = \mathbb{Z}/2 \times \mathbb{Z}/3 \times \mathbb{Z}/5$ carries the $\mu_3$ triality (CP, v233) as $h^{10}$ ; the Galois group $(\mathbb{Z}/30)^\times = \mathbb{Z}/2 \times \mathbb{Z}/4$ carries the $\mu_4$ clock ( $\times 7$ , v223). [E] Milnor lattice = $E_8$ , link = Poincaré sphere. [C] one singularity generates atoms + 8 + $E_8$ + order-30 + both clocks + seam; a unification, not a closure [E][C]                                                                                                            |
| v237_seam_sre_closure.py          | The closing step as a <i>physical</i> condition: no topological ground-state degeneracy $\Leftrightarrow \det K=1 \Leftrightarrow$ the seam bulk is short-range-entangled (SRE/invertible) — sharpening GATE.HOLO.02’s residual from definitional to falsifiable. [E] the $K$ -matrix ground-state degeneracy on a genus- $g$ surface is $ \det K ^g$ (carrier $\rightarrow 16$ , $D_8 \rightarrow 4$ , $E_8 \rightarrow 1$ on the torus), so ‘no degeneracy on any closed surface’ $\Leftrightarrow \det K=1$ . [E] among rank-8 even positive-definite $K$ ( $c=8=g_{\text{car}}+N_{\text{fam}}$ ) only $E_8$ has $\det 1$ — the unique nontrivial bosonic SRE 2+1d phase (Kitaev $E_8$ ), edge holomorphic; $D_8$ /carrier are LRE. [E] a unique vacuum on the plane (RP+gap+cluster) is necessary but not sufficient (the plane cannot see $\det K$ , the torus does), so the residual is ‘the seam unique-vacuum extends to the torus’ = SRE. [E] NEG: an LRE bulk ( $D_8$ ) has 4-fold torus degeneracy + non-holomorphic boundary. [O] sharpened residual: ‘the free RP gapped seam is short-range-entangled (no topological order)’ = $\det K=1$ = the Kitaev $E_8$ phase — a yes/no statement, sharper than v235/QGEO.SYM.01; not a closure [E][O] |
| v238_modular_lindblad_dynamics.py | DYN.SEMIGROUP.01 — the static $\rightarrow$ dynamic flip: the seam’s modular flow (reversible) and its recovery channel as a Markov/Lindblad generator (irreversible), the dissipative gap = the OS mass gap. Companion to v196/v198 (modular) and v221/v64 (recovery), which state the structure <i>statically</i> ; this reads the same objects <i>dynamically</i> . [E] the modular flow $\sigma_t = e^{it\Lambda_\Sigma}$ is a unitary one-parameter group (Tomita–Takesaki) and the $\mu_4$ clock is conserved $\Leftrightarrow [\rho, \Lambda_\Sigma] = 0$ (the v198 state-invariance), so the static commutator is the conservation law of modular time. [E] $L = \log T$ is a doubly-stochastic generator ( $e^L = T$ , $e^{L\tau} \rightarrow$ the democratic projector = the arrow of time, cf. v64). [E] keystone: the dissipative gap = $-\log((2/3)^6) = 6\log(3/2)$ = the OS mass gap $\Delta$ (v64) — the static recovery bound and the dynamical mass gap are one number, one exponentiated. [E] GKSL split (imaginary + $\text{real} \leq 0$ spectrum). [O] geometric modular flow on the full seam algebra (Connes–Rovelli thermal time = QGEO.SYM.01, v198) — a target, not a closure.                                                   |

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| Script                          | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| v239_kms_thermal_time.py        | DYN.KMS.01 — KMS / thermal time: the modular flow of v238 makes the seam state a KMS (thermal) state at $\beta = 1$ , and the geometric normalisation fixes the seam temperature via $2\pi = 1/(4c_3)$ . [E] KMS at $\beta = 1$ : $\omega(A\sigma_i(B)) = \omega(BA)$ ( $\omega(X) = \text{Tr}(\rho X)$ , $\rho = e^{-\Lambda_\Sigma}/Z$ ), so modular time (v238) is a thermal time (Connes–Rovelli); NEG $\beta \neq 1$ fails (Tomita–Takesaki pins $\beta = 1$ ). [E] the seam unit $\beta_{\text{phys}} = 2\pi = 1/(4c_3)$ (v58) makes the KMS temperature the horizon temperature $T_H = c_3/M$ (v57/v208). [E] one object with v238. [O] the geometric-boost identification = QGEO.SYM.01 — a target, not a closure.                                                                                                                                                                                                                                                                                                |
| v240_gns_os_reconstruction.py   | DYN.GNS.01 — GNS / OS reconstruction (finite toy): the Hilbert space and a POSITIVE Hamiltonian are OUTPUTS of the seam data $(M, \omega)$ , not inputs. [E] GNS: $\langle A, B \rangle = \text{Tr}(\rho A^\dagger B)$ Hermitian positive-definite ( $\rho > 0$ ), $\dim n^2$ , identity cyclic+separating. [E] modular flow $\sigma_t(x) = \rho^{it} x \rho^{-it}$ is a state-preserving *-automorphism, generator $i[\log \rho, \cdot] = \Lambda_\Sigma$ (v238). [E] OS Hamiltonian $H_{\text{OS}} = -\log T$ (the v221 transfer) is POSITIVE — eigenvalues $\{0, 6\log(3/2), 6\log 3\} \geq 0$ , gap = $6\log(3/2)$ = the v64 mass gap. [E] $H_{\text{OS}} = -L$ : minus the v238 dissipative generator (recovery semigroup = Euclidean time). [O] the geometric/operator-level step = QGEO.SYM.01/BW + the standard OS theorem under RP/gap — a target, not a closure.                                                                                                                                                |
| v241_dhr_sectors.py             | DHR.SECTORS.01 — particles = DHR superselection sectors: the carrier discriminant/anyon content and the v235/v237 condensation tower as an abelian MTC. Carrier form (Lean-verified, FORM.GLUE.01): on $\mathbb{Z}_4 \times \mathbb{Z}_4$ , $q(x, y) = (5x^2 + 3y^2)/8 \bmod 1$ , $\theta(a) = e^{2\pi i q(a)}$ . [E] 16 sectors = $\mathbb{Z}_4 \times \mathbb{Z}_4$ ( $= \det K_{\text{carrier}} = \dim S^+$ ), fusion = abelian group law. [E] condensable $\Leftrightarrow \theta = 1 \Leftrightarrow \text{isotropic} \Leftrightarrow 5x^2 + 3y^2 \equiv 0 \bmod 8$ — the six elements $\{(0, 0), (1, 1), (1, 3), (2, 2), (3, 1), (3, 3)\}$ . [E] tower = v235/v237: Lagrangian $\langle (1, 1) \rangle$ (order 4, self-dual) $\rightarrow 1$ sector = $(E_8)_1$ ; order-2 $\langle (2, 2) \rangle \rightarrow 4 = D_8$ ; carrier $16 \rightarrow D_8 \rightarrow E_8$ . [E] NEG: a non-isotropic anyon $(1, 0)$ is confined. [C] sector $\leftrightarrow$ SM-multiplet map is the interpretation; the MTC is exact. |
| v242_spin_statistics_modular.py | DHR.MODULAR.01 — spin-statistics and the modular $(S, T)$ data of the carrier MTC: statistics + the fusion ring recovered from modular data + the anyon central charge $c = 8$ . Same MTC as v241 ( $\mathbb{Z}_4 \times \mathbb{Z}_4$ , $q = (5x^2 + 3y^2)/8$ , $\theta = e^{2\pi i q}$ ). [E] spin-statistics: $T = \text{diag}(\theta_a) \rightarrow 6$ bosons ( $\theta = 1$ ), 2 fermions ( $\theta = -1$ ) = $\{(0, 2), (2, 0)\}$ , 8 anyons. [E] modular $S_{ab} = (1/\sqrt{16})e^{-2\pi i B(a,b)}$ unitary, symmetric, $S^2 = C$ (charge conjugation), $S^4 = I$ . [E] Verlinde $N_{ab}^c = \sum_x S_{ax} S_{bx} S_{cx}^* / S_{0x} = \delta(c, a+b)$ — fusion recovered from modular data. [E] Gauss–Milgram $(1/\sqrt{16}) \sum \theta_a = e^{2\pi i c/8}$ with $c = 8 = g_{\text{car}} + N_{\text{fam}}$ = the $(E_8)_1$ central charge (v156/v175). [C] the boson/fermion split is the kinematic skeleton; the sector $\leftrightarrow$ multiplet map is the interpretation.                                   |
| v243_haag_ruelle_braiding.py    | DHR.SMATRIX.01 — scattering: the Haag–Ruelle skeleton + the braiding S-matrix. The seam net has a mass gap (v64/v238) so Haag–Ruelle constructs asymptotic states; for the abelian anyon MTC (v241/v242) the 2-particle S-matrix is the monodromy $M(a, b) = \theta(a+b)/(\theta(a)\theta(b)) = e^{2\pi i B(a,b)}$ . [E] mass gap $\Delta = 6\log(3/2) > 0 \rightarrow$ HR in/out states. [E] 2-particle S = monodromy: $ M  = 1$ , diagonal in conserved labels $\rightarrow$ no particle production, factorised (integrable) scattering. [E] Yang–Baxter: $M$ is a bicharacter $\rightarrow$ multi-particle factorisation. [E] crossing+symmetry: $M(a, b) = M(b, a)$ , $M(a, -b) = M(a, b)^*$ . [E] holomorphic point: on the condensed $(E_8)_1$ (single sector) $M = 1$ , trivial braiding ( $\det 1/\text{SRE}$ ). [O] the 4d S-matrix / cross-sections are the open holographic uplift (Phase 3).                                                                                                                  |

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| Script                              | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| v244_spectral_action_contract.py    | <p>CONTRACT.QFT4D.01 — the 4d Lagrangian via Connes' spectral action: the honest contract for the Phase-3 gap (2d seam net <math>\rightarrow</math> 4d physics). A single Dirac operator whose <math>\text{Tr } f(D/\Lambda)</math> yields gravity AND matter. [E] carrier spinor = one generation: <math>SO(10)</math> half-spinor = <math>\dim S^+ = 2^{g_{\text{car}}-1} = 16</math> (<math>15 + \nu_R</math>), <math>N_{\text{fam}} = 3</math>. [E] SM gauge group inside the carrier: <math>SU(3) \times SU(2) \times U(1)</math> (<math>\dim 12</math>) <math>\subset SU(5)</math> (24) <math>\subset SO(10) = D_5</math> (45); colour <math>SU(3) \subset A_3 = SU(4)</math> (Pati-Salam); <math>\text{rank}(D_5) + \text{rank}(A_3) = 8 = \text{rank } E_8</math>. [E] one Dirac operator <math>\rightarrow</math> gravity (<math>a_2 \sim R</math>, <math>a_4 \sim R^2</math>, <math>\sqrt{36}</math>) + matter (<math>a_4 \sim F^2</math> Yang-Mills + <math> D\phi ^2</math> Higgs, Gilkey). [X] hooks: <math>g_3 = g_2 = \sqrt{5/3} g_1</math> + a Higgs-quartic relation. [O] CONTRACT.QFT4D.01: obligations (seam <math>\rightarrow</math> Dirac <math>D</math>; finite geometry = carrier; spectral action reproduces SM couplings) <math>\rightarrow</math> the 4d SM+gravity Lagrangian is an OUTPUT — the Phase-3 target, not closed.</p> |
| v245_spectral_matter_content.py     | <p>QFT4D.MATTER.01 — the computable core of CONTRACT.QFT4D.01 (v244): the spectral-action / NCG matter content, discharging the rep-theory + tree-relation core to [E]. The carrier half-spinor is the <math>SO(10)</math> <b>16</b>. [E] <b>16</b> = one generation: <math>SO(10)</math> <b>16</b> = <math>SU(5)(\mathbf{10} + \mathbf{5} + \mathbf{1})</math>; SM dims <math>6 + 3 + 1 + 2 + 3 + 1 = 16 = \dim S^+</math>, <math>N_{\text{fam}} = 3</math> (15 Weyl fermions + <math>\nu_R</math>). [E] electric charges <math>Q = T_3 + Y</math> are exactly the SM values — hypercharges forced by the embedding. [E] anomaly-free: <math>\sum Y = 0</math>, <math>\sum Y^3 = 0</math>, <math>[SU(2)]^2 U(1) = 0</math>, <math>[SU(3)]^2 U(1) = 0</math> (exact). [E] unification: <math>\sin^2 \theta_W = (\sum T_3^2) / (\sum Q^2) = 3/8</math>, i.e. <math>g_3 = g_2 = \sqrt{5/3} g_1</math>. [E] Higgs = the <math>(1, 2)_{1/2}</math> inner fluctuation (one doublet). [O] residual: the seam <math>\rightarrow</math> Dirac realisation + measured-coupling reproduction stay the open contract.</p>                                                                                                                                                                                                                                              |
| v246_unification_data_crosscheck.py | <p>QFT4D.RGTEST.01 — the honest data cross-check for the spectral-action prediction (v244/v245): does <math>g_1 = g_2 = g_3</math>, <math>\sin^2 \theta_W = 3/8</math> at <math>\Lambda</math> survive the <i>measured</i> SM couplings? Not in the plain SM. Inputs (PDG, GUT-normalised, <math>M_Z</math>): <math>\alpha_i^{-1} = (59.01, 29.59, 8.47)</math>, SM 1-loop <math>b = (41/10, -19/6, -7)</math> (v159). [E] the prediction: <math>\alpha_1 = \alpha_2 = \alpha_3</math>, <math>\sin^2 \theta_W = 3/8</math> at <math>\Lambda</math>. [X] tension: pairwise crossings spread <math>\sim 4</math> decades (<math>\sim 10^{13} - 10^{17}</math> GeV), residual spread at <math>2 \times 10^{16}</math> <math>\Delta \alpha^{-1} \sim 8.8</math> — <math>g_1 = g_2 = g_3</math> not realised by the plain SM. [X] <math>\sin^2 \theta_W</math> 3/8 vs measured 0.23122. [X] 2-loop also misses (min spread <math>\sim 3.2</math> at <math>1.7 \times 10^{14}</math> GeV). [O] no rescue: SM gauge content + no new states (the <math>E_8</math> spine is arithmetic, not a threshold tower) <math>\Rightarrow</math> no admissible thresholds; closing it needs new matter (vs 'no new state') or a non-GUT/boundary reading — tension like NCG-SM, likely falsified unless extended. A kill-test, [X]/[O], never [E]-as-success.</p>            |
| v247_e8_branching_no126.py          | <p>PS.E8BRANCH.01 — the <math>E_8</math> audit hull forbids the <b>126</b> and supplies exactly <math>\{1, 10, 16, 45\}</math> for <math>SO(10)</math>. Carrier <math>D_5 \oplus A_3 = SO(10) \times SU(4) \subset E_8</math> (via <math>SO(16)</math>); <b>248</b> = <math>(45, 1) + (1, 15) + (10, 6) + (16, 4) + (\overline{16}, \bar{4}) = 248</math>. [E] ranks <math>5 + 3 = 8</math>; [E] dims = 248 (<math>SO(16)</math>: 120 adj + 128 spinor); [E] <math>SO(10)</math> content <math>\{1, 10, 16, 45\}</math>, the <b>126</b> ABSENT <math>\Rightarrow 126_H</math> algebraically forbidden, <math>10 + 16 + 45</math> <math>E_8</math>-supplied not fitted; [E] caveat only one 45/one 15 <math>\rightarrow</math> proton-marginal; [O] the field-content principle + NCG complement (v250).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| v248_ps_algebra.py                  | <p>PS.ALGEBRA.01 — the Pati-Salam NCG algebra <math>A_F = \mathbb{H}_L \oplus \mathbb{H}_R \oplus M_4(\mathbb{C})</math> and its exact SM matching. [E] unimodular group <math>SU(2)_L \times SU(2)_R \times SU(4)_c</math> (rank 5, <math>\dim 21</math>); [E] <b>16</b> = <math>(4, 2, 1) + (\bar{4}, 1, 2) = 8 + 8</math>; [E] <math>Y = T_{3R} + (B - L)/2</math> gives the exact SM hypercharges and <math>Q = T_{3L} + Y</math> the exact charges; [E] matching <math>\alpha_1^{-1} = \frac{3}{5} \alpha_{2R}^{-1} + \frac{2}{5} \alpha_4^{-1}</math>, <math>\sin^2 \theta_W = 3/8</math>. [O] <math>A_F</math> uniquely from seam+sheet+<math>\mu_4</math>; [O] caveat <math>SU(4)_c \subset SO(10) = D_5</math> is not the carrier <math>A_3 = SU(4)</math> (family).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| v249_ps_unification.py              | <p>PS.RGTEST.01 — carrier-native Pati-Salam/<math>SO(10)</math> two-step unification kill-test + negative controls (suite check of experiments/gauge-unification). [E] matching + a valid 1-loop unification solution; [X] NEG SM-only fails (spread <math>\sim 8.8</math> @ <math>2 \times 10^{16}</math>); [C] <math>M_{PS}/M_s = 1.0 - 1.7</math> (1-loop; 2-loop <math>\sim 1.0 - 1.2</math>) — gauge unification meets the gravitational scalaron scale; [X] NEG 126 breaks the match (<math>b_{2R} &gt; 0</math>; also <math>E_8</math>-forbidden v247); [X] NEG proton decay selects content (minimal <math>M_{GUT} \sim 2 \times 10^{15}</math> excluded; <math>+(15, 1, 1)</math> safer). [O] the gauging fork, exact <math>\kappa</math>, the one-45 marginality.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

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| Script                            | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| v250_dirac_triple_contract.py     | CONTRACT.QFT4D.DIRAC.01 — the spectral-triple contract that would make ‘carrier gauged’ a theorem. Target $(A, H, D, J, \gamma)$ almost-commutative, $A_F = \text{PS}$ (v248), $H_F = 3 \times 16$ , $D_F$ from $(R, L, \varphi_0, \text{Majorana})$ . [E] $H_F = 3 \times 16 = 48$ ; [E] the Yukawas are already fixed (v18) so the open test is NCG consistency; [O] the 7 NCG axioms (first-order condition = the historic $SO(10)$ obstruction); [O] $\sigma = \text{scalaron hypothesis}$ (rescues first-order + $\nu_R$ Majorana; $M_{PS} = \kappa M_s$ ) — caveat $\sigma \leftrightarrow \nu_R \nu_R$ not automatic; [O] inner fluctuations $\rightarrow 10+16+45$ not 126. Reduces 4d QFT to one object + finite NCG checks.                                                                                                                                                                 |
| v251_first_order_sigma.py         | PS.DIRAC.01 — the first-order condition is violated <i>exactly</i> by the Majorana/ $\sigma$ term (Chamseddine–Connes–van Suijlekom ‘beyond first order’ $\rightarrow$ Pati–Salam + $\sigma$ ), exhibited on a faithful lepton-block model; advances v250 from contract to mechanism. Honest correction: $\sigma$ does NOT ‘rescue’ first order — one DROPS it and the inner fluctuations generate $\text{PS} + \sigma$ . [E] order-zero; [E] real-even ( $J^2=+1$ , $\gamma^2=1$ , $D\gamma=-\gamma D$ , KO-6 signs); [E] first order HOLDS for the Yukawa block; [E] first order VIOLATED by $M \neq 0$ , supported exactly on the $\nu_R \nu_R^c$ blocks ( $\sigma$ the unique violator). [C] $\sigma = \text{scalaron}$ (v249). [O] the full 96-dim SM/PS triple + spectral action (v250).                                                                                                        |
| v252_full_finite_triple.py        | PS.DIRAC.02 — the full 96-dim finite spectral triple $(A_F, H_F, D_F, J, \gamma)$ built explicitly with the NCG axioms verified; advances v250 from a 7-obligation contract to a concrete object discharging 5 to [E]. $H_F = \text{particle} \oplus \text{antiparticle}$ doubling of three $SO(10)$ <b>16</b> -plets (dim 96). [E] dim 96; $D_F$ Hermitian+odd; reality $J^2=+1$ , $JD=DJ$ ; grading $\gamma^2=1$ , $[\gamma, a]=0$ , $J\gamma=-\gamma J$ ; KO-dim 6; order-zero; first order HOLDS for the SM $\mathbb{C} \oplus \mathbb{H} \oplus M_3$ (incl. Majorana) and VIOLATED for Pati–Salam exactly by the Majorana on $\{\nu_R, e_R\}$ (CCvS beyond-first-order $\rightarrow \text{PS} + \sigma$ ); spectrum = fermion masses + seesaw $m_{\text{light}} = m_D^2/M_R$ . [C] Poincaré duality (intersection form non-degenerate). [O] orientability + no-junk + spectral-action expansion. |
| v253_kappa_scalaron_ps.py         | PS.KAPPA.01 — pinning $\kappa$ in $M_{PS} = \kappa M_s$ , discharging residual 6(b) of v249. $M_s = c_3^{7/2} \bar{M} \approx 3.06 \times 10^{13} \text{ GeV}$ (exact). [E] scalaron scale exact; [E] $\kappa$ pinned by RG (scan PDG errors $\times$ the two $E_8$ -allowed contents) to a tight $O(1)$ interval $\sim 1.3$ (1-loop $[1.0, 1.7]$ , 2-loop $[1.0, 1.2]$ ) — pinned, not free. [C] heat-kernel origin $\kappa = \sqrt{(f_2/f_0)} c_{PS}/c_{\text{grav}}$ , a ratio of heat-kernel traces over the 96-dim $H_F$ (v252) — $\Lambda$ cancels, $\kappa$ scale-free and $O(1)$ . [X] the <b>126</b> -content drives $\kappa$ out of the $O(1)$ window (content-selective). [O] the exact decimal needs the cutoff moments $f_2/f_0$ fixed.                                                                                                                                                  |
| v254_inner_fluctuations.py        | PS.NCG.FLUCT.01 — the inner fluctuations $\Omega_D^1(A_F)$ of the finite Dirac operator computed explicitly (reuses the v252 triple); closes v250 #5 (Higgs content, not <b>126</b> ) + the no-junk obligation + the $\sigma$ quantum numbers. [E] $\Omega_{D_F}^1$ purely chirality-odd $\rightarrow$ finite fluctuations are scalars; [E] no junk / no <b>126</b> : SM-algebra 1-forms sit exactly in the Higgs-doublet pattern (the single $(1, 2)_{1/2}$ ); [E] the $\sigma$ ( $\nu_R$ Majorana) direction is absent for the SM algebra but present for Pati–Salam — $\sigma$ is the CCvS beyond-first-order scalar; [E] $\sigma$ is an SM singlet $(1, 1, 0)$ with $B-L=2$ ; [C] $\sigma = \text{scalaron consistency}$ .                                                                                                                                                                        |
| v255_spectral_action_expansion.py | PS.SPECACT.01 — the spectral-action heat-kernel (Seeley–DeWitt) expansion $S = 2f_4\Lambda^4 a_0 + 2f_2\Lambda^2 a_2 + f_0 a_4$ over the finite triple; closes v252 #11 (spectral-action expansion) and makes $\kappa(f_2/f_0)$ explicit. [E] expansion structure ( $a_0$ cosmological, $a_2$ Einstein–Hilbert+Higgs-mass, $a_4$ Yang–Mills+Weyl <sup>2</sup> + $R^2$ +Higgs-quartic, $N=96$ ); [E] trace inputs $b/a^2 = 1/3$ (top-dominated, Chamseddine–Connes); [E] $\sin^2 \theta_W = 3/8$ (cross-check v245); [C] Higgs quartic $\lambda(\Lambda) \sim g^2 b/a^2 \rightarrow \sim 125 \text{ GeV}$ with $\sigma$ ; [E] scalaron = $R^2$ spin-0 mode (gravitational $a_4 \sim f_0 N$ ); [C] $\kappa = \sqrt{(f_2/f_0)} c_{PS}/c_{\text{grav}}$ explicit; [O] exact decimal needs the cutoff moments fixed.                                                                                       |

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| Script                                     | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| v256_orientability_<br>poincare.py         | PS.NCG.ORIENT.01 — orientability and Poincaré duality treated honestly (no faked proof). KO-dim 6 forces the intersection form $\text{cap}(p, q) = \text{Tr}(\gamma \pi(p) J \pi(q) J^{-1})$ to be antisymmetric. [E] skew form forced by $J\gamma = -\gamma J + \text{order-zero} + \text{reality}$ (verified); [E] on the SM K-theory $\mathbb{Z}^3$ the skew form has rank 2, corank 1, null $\sim (1, 1, -1)$ — the structural obstruction; [E] orientability local condition $[\gamma_F, \pi(a)] = 0$ , full orientability is a degree-4 cycle on $M \times F$ ; [C] the canonical hypercharge-faithful SM/PS triple satisfies both (van Suijlekom; CCM) — literature; [O] a self-contained machine proof needs the complete canonical bimodule.                                                                                                                                                                                                                                                                                                                                                                                            |
| v257_quasiorient_<br>modpd.py              | PS.NCG.ORIENT.02 — the honest, literature-confirmed resolution of orientability + Poincaré duality, correcting the too-generous v256 typing. Published result (Cacic–Stephan arXiv:0902.2068; CCM07; SM-vacuum hep-th/0601192): the CCM SM finite triple (KO-6, $3\nu_R$ ) genuinely FAILS strict orientability and strict Poincaré duality, satisfying the weakened versions. [E] strict orientability fails ( $\gamma \notin \text{span}\{\lambda(a)\rho(b)\}$ ); [E] quasi-orientability holds (left/right $A$ -irrep boxes disjoint by chirality: $L = H$ doublet, $R = \mathbb{C}$ singlet $\Rightarrow L^{\text{LR}}(\text{even, odd}) = \{0\}$ ); [E] strict Poincaré fails (KO-6 skew form, corank 1, the equal- $L/R$ -neutrino case); [C] modified Poincaré duality (CCM) / Stephan’s PD-restoring triple. A known model-class feature, not a TFPT defect — no faked closure.                                                                                                                                                                                                                                                          |
| v258_dirac_<br>covariance_<br>induction.py | PS.DIRAC.03 — $D_F$ is the modular/covariance <i>induction</i> of the boundary seam state, not an independent posit; closes the ‘posited Yukawa vs derived operator’ gap of v250/v252 (Q2: Dirac as Kasparov product + covariance formula). Quasi-free/modular dictionary: a Gaussian state is fixed by its covariance $C$ (contraction, $\text{spec} \subset (0, 1)$ ) with one-particle modular Hamiltonian $H = \log((1-C)C^{-1})$ (Araki; Casini–Huerta), inverse to the KMS occupation $C = (1+e^H)^{-1}$ ( $\beta=1$ , v239). [E] inversion identity $\log((1-C)C^{-1}) = H$ exactly (random Hermitian + a chirality-graded Yukawa block); [E] $C_F = (1+e^{D/\mu})^{-1}$ a positive contraction; [E] chirality-odd ( $D_F^{\text{ind}}\gamma = -\gamma D_F^{\text{ind}}$ ); [E] unitarily equivalent to v252 (inducing $C_F$ from the 9 Yukawas returns the same spectrum); [E] Majorana = off-diagonal covariance (seesaw $m_{\text{light}} = m_D^2/M_R$ ); [C] $C_F = P_F C_\Sigma P_F$ , so $D_F$ is the KK shadow $[D_\Sigma] \hat{\otimes}_{(E_8)_1} [\mathcal{K}_{\text{car}}]$ and the v18 Yukawas are the readout of $C_\Sigma$ . |
| v259_modular_<br>cutoff_kappa.py           | PS.SPECACT.02 — the spectral-action cutoff <i>is</i> the seam KMS weight, fixing $f_2/f_0=1$ exactly and reducing the last open scheme freedom (v253 #5, v255 #7) from a free function to a finite-triple trace ratio (Q4: the cutoff is modular, not chosen). By Tomita–Takesaki/BW the seam state is KMS at $\beta=1$ (v239) and $2\pi=1/(4c_3)$ (v58) fixes $\beta$ , so the heat-kernel cutoff $f(u)=e^{-u}$ is <i>derived</i> . [E] KMS cutoff moments $f_0=f(0)=1$ , $f_2=\int f=1$ , $f_4=\int u f=1$ , so $f_2/f_0=f_4/f_2=1$ ; [E] seam-fixed $\beta=1$ (no new parameter beyond $c_3$ ); [E] non-vacuous neg control: a Gaussian cutoff $e^{-u^2}$ gives $f_2/f_0=\sqrt{\pi}/2 \neq 1$ ; [E] $\kappa=\sqrt{(f_2/f_0)c_{PS}/c_{\text{grav}}}=\sqrt{c_{PS}/c_{\text{grav}}}$ , no cutoff function left; [C] the RG $\kappa\sim 1.3$ pins $c_{PS}/c_{\text{grav}}\sim 1.7$ ( $O(1)$ ); the [O] free scheme becomes a [C] finite computation over the built triple; [C] the same logic fixes $\lambda(\Lambda)$ .                                                                                                                          |

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| Script                           | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| v260_k3_kummer_unification.py    | <p>ARCH.K3.01 — one Kummer/K3 surface carries the <i>seam</i>, the <i>carrier-16</i> and the <math>E_8</math> lattice as facets of one smooth object, the geometric unification of the v1 ‘glue <math>D_5 \oplus A_3 + \mu_4 \Rightarrow E_8</math>’ step (the K3/Kummer proposal). The seam square torus <math>\tau=i</math> (<math>j=1728</math>, v214) under the elliptic involution <math>z \mapsto -z</math> squares to the Kummer K3. [E] the <math>E_8</math> Cartan is even, <math>\det=1</math>, positive-definite (the unique even unimodular pos-def rank-8 lattice); [E] the K3 lattice <math>L_{K3}=U^3 \oplus E_8(-1)^2</math> has rank <math>22=b_2</math>, <math>\det=-1</math> (unimodular), is even, signature <math>(3, 19)</math>, with <math>E_8(-1)^2</math> an orthogonal summand; [E] seam = <math>T^2/(z \mapsto -z)</math>: <math>4= (\mathbb{Z}/2)^2 </math> fixed points, pillowcase <math>S^2(2, 2, 2, 2)</math>, orbifold Euler char 0 (the v214/v216 four marks); [E] carrier-16 = Kummer nodes <math> A[2] =2^4=16 \rightarrow 16</math> exceptional <math>\mathbb{P}^1 = \dim S^+ = \Lambda^{\text{even}}(\mathbb{C}^5) = 16</math> (v197), <math>16=4 \times 4</math>; [E] honesty: the 16 node classes span the rank-16 Kummer lattice, <i>not</i> literally <math>E_8(-1)^2</math> (distinct sublattices of <math>H^2(K3)</math>); [C] two faces of <math>E_8</math> in K3 (local du Val <math>\mathbb{C}^2/2I</math>, v232/v236; global <math>H^2</math> summand); [C] unification (lattice [E], carrier<math>\leftrightarrow</math>node identification [C]).</p> |
| v261_modular_spectral_closure.py | <p>QFT.MSC.01 — the <i>Modular Spectral Closure</i>: the assembly/reduction certificate turning the whole QFT layer (v238–v260) into <i>one</i> relative object<br/> TFPT<sub>QFT</sub> = <math>(\mathcal{A}_\Sigma, \omega_\Sigma, \Delta_\Sigma, \rho, A_F, H_F, D_F, J, \gamma, S_{\text{rel}})</math> reduced to <i>one</i> open premise (like v176/v234, an assembly certificate, not new physics). [E] one index = marks = fixed points = glue order: Jones <math>[B:A]=4</math> (v89) = <math> \mu_4 =4</math> (v92) = seam marks <math>2\chi=4</math> (v216) = pillowcase/Kummer fixed points <math> (\mathbb{Z}/2)^2 =4</math> (v260); [E] one carrier-16: <math>\dim S^+ = \Lambda^{\text{even}}(\mathbb{C}^5) = \text{Kummer nodes } 2^4=16</math>, <math>H_F=2N_{\text{fam}} \cdot 16=96</math>; [E] one gap <math>\Delta = -\log((2/3)^6) = 6 \log(3/2) &gt; 0</math> = the OS mass gap (v240) = the recovery rate (v221) = the Lindblad gap (v238); [E] the three closures re-derived (Kasparov <math>\log((1-C)C^{-1})=H</math>, v258; KMS cutoff <math>f_2/f_0=1</math>, v259; K3 lattice signature <math>(3, 19)/\det -1/\text{even}</math>, v260); [E] relative-object manifest (10/10 component scripts present); [O] the single residual: the whole QFT layer reduces to QGEO.SYM.01 (seam realisation), the ambient QG measure QG.AMB.01 kept separate; [C] the master statement (Modular Spectral Universality) holds modulo QGEO.SYM.01 — the honest sense in which the QFT layer is complete.</p>                                                                              |
| v262_fqcd_mp_me.py               | <p>FR.MPME.02 — the <math>F_{\text{QCD}}</math> transfer pipeline for <math>m_p/m_e</math>, implemented as a real solver (the one <math>F_{\text{transfer}}</math> pipeline that was contract-only). <i>Not</i> a compiler power — a [C] standard-physics transfer reproduction. [E] <math>b_3 = -7 = -(11-2n_f/3)</math> at <math>n_f=6</math> (the only TFPT number in the run); [E] a two-loop <math>\alpha_s(M_Z)=0.1179</math> run with thresholds gives <math>\alpha_s(2 \text{ GeV}) \sim 0.30</math> (real ODE solve); [C] <math>\Lambda_{\text{MS}}^{(3)} \sim 0.33 \text{ GeV} \times</math> the lattice bridge <math>C_p = m_p/\Lambda</math> (external <math>O(1)</math>) over the closed <math>m_e</math> gives <math>m_p/m_e \sim 1824</math> vs the measured 1836.15 within the <math>\sim 7\%</math> external budget; [X] firewall: <math>m_p/m_e</math> is not an admissible compiler integer (<math>SU(9)</math> absent from <math>D_5 \oplus A_3</math>). Advances FR.MPME.01 ([O]) to FR.MPME.02 ([C]).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| v263_mnu_from_df.py              | <p>FLAV.PMNS.02 — the light neutrino matrix <math>M_\nu</math> as the type-I seesaw of <math>D_F</math>, advancing the solar-angle texture (v9) from a posited matrix to a seesaw <i>object</i> built from <math>D_F</math>’s blocks. [E] <math>M_\nu = -m_D M_R^{-1} m_D^\top</math> symmetric + seesaw-suppressed (the light masses an output of <math>D_F</math>); [E] a <math>\mu\tau</math>-symmetric <math>(m_D, M_R)</math> gives a <math>\mu\tau</math>-symmetric <math>M_\nu</math> (the v9 texture is in the seesaw image); [C] diagonalising gives <math>\theta_{23}=45^\circ</math> and <math>\sin^2 \theta_{12} = \frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2} = 0.30675</math> under the seam misalignment; [O] <math>\theta_{13}=0</math> in the leading texture (physical <math>\sin^2 \theta_{13} \sim \varphi_0^{\text{ret}} e^{-5/6}</math> a separate readout), the CP magnitude <math>\delta_{\text{PMNS}}=240^\circ</math> is the <math>\mu_6/\text{triality}</math> phase (v231/v233), and <math>M_R</math> is the free seesaw scale. An operator advance, not a closure of full PMNS dynamics.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

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| Script                                  | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| v264_raw_seam_<br>marklocal.py          | QGEO.ENERGY.03 — the raw-seam $\Rightarrow$ mark-locality <i>reduction</i> (not a closure of QGEO.SYM.01). Composing v216+v201+v198 on the canonical pillowcase: [E] Gauss–Bonnet 4 cone points (deficit $\pi$ each, sum $4\pi=2\pi\chi(S^2)$ ), curvature <i>only</i> at the 4 marks; [E] the curvature sourced at the $\mu_4$ orbit $\{0, \pi/2, \pi, 3\pi/2\}$ has Fourier support <i>only</i> on $4\mathbb{Z}$ (FFT), so mark-locality is automatic for the pillowcase; [E] $M_f$ then commutes with the clock $\rho=\text{diag}(i^n)$ ; [E] with $ k $ already commuting (v198), $[\rho, \Lambda]=0 \Rightarrow \omega \circ \rho = \omega$ on this metric; [O] the residual <i>is</i> the metric — the chain assumes the raw seam carries the uniformising flat pillowcase metric (=QGEO.SYM.01), the negative control (curvature off the $\mu_4$ orbit) breaks it. A reduction/sharpening, not a closure.                                                                                                                                                                                                                                                                                                                                                         |
| v265_qft4d_fork_<br>freeze.py           | QFT4D.FORK.01 — freeze the 4D-QFT fork as a <i>branch policy</i> (not an open ambiguity); writes <code>qft4d_fork_freeze.json</code> + a prose guard, no new physics. [C] A: boundary-only is the canonical 4D reading (no 4D-GUT claim by default); [X] SM-only 4D-GUT unification is killed (1-loop spread $\sim 8.8$ at $2 \times 10^{16}$ , v246); [C]/[O] B: carrier-native Pati–Salam is a falsifiable UV branch ( $E_8$ content $\{1, 10, 16, 45\}$ no <b>126</b> v247, $\sin^2 \theta_W = 3/8$ v248, $\kappa O(1)$ band v249/v253, threshold + proton kill test); [X] proton safety rule (no fake $\tau_p$ window); [E] wording guard (bare 4D over-claims absent, scoped policy phrases present). A frozen decision tree, not an open ambiguity.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| v266_ps_threshold_<br>proton.py         | PS.PROTON.02 — the explicit Pati–Salam threshold model + proton-decay window that DECIDES branch B of the v265 fork; writes <code>ps_threshold_proton.json</code> . Two-step PS solve + $d=6$ $\tau_p(p \rightarrow e^+ \pi^0)$ : the carrier-native PS UV branch survives proton decay <i>only</i> with the $E_8$ -allowed $(15, 1, 1)$ [the single 45]. [E] explicit thresholds: minimal $16H \rightarrow M_{GUT} \sim 2.4 \times 10^{15}$ , $+(15, 1, 1) \rightarrow M_{GUT} \sim 5.9 \times 10^{15}$ ; [X] minimal KILLED ( $\tau_p \sim 4.4 \times 10^{33}$ yr < Super-K $2.4 \times 10^{34}$ , even high-band); [C] $+(15, 1, 1)$ survives now ( $\tau_p \sim 1.6 \times 10^{35}$ yr); [X] <i>Hyper-K decisive</i> ( $\tau_p$ at the $\sim 1.4 \times 10^{35}$ yr reach) — a dated kill test within the decade; [O] marginal (only one 45 in the hull, v247) + $O(3)$ hadronic uncertainty. A (boundary-only) stays canonical.                                                                                                                                                                                                                                                                                                                                     |
| v267_qgeo_rigidity_<br>minimal_axiom.py | QGEO.SYM.02 — the rigidity / minimal-axiom form of the bedrock QGEO.SYM.01: the ONE bare statement (the raw RP seam admits the carrier $\mu_4$ clock as a conformal symmetry permuting its 4 marks, $= \omega \circ \rho = \omega$ ) <i>forces</i> everything downstream; it does <i>not</i> close the axiom. [E] an order-4 Möbius orbit $\{a, ia, -a, -ia\}$ has cross-ratio 2 (independent of $a$ ); [E] cross-ratio $2 \Leftrightarrow j=1728$ (only $\lambda \in \{-1, \frac{1}{2}, 2\}$ ) $\Leftrightarrow$ the order-4 CM automorphism $(x, y) \mapsto (-x, iy)$ on $y^2=x^3-x$ ; [E] downstream forced: square $\Rightarrow$ the unique flat pillowcase (Trojanov, v214) $\Rightarrow$ curvature at the 4 marks $\Rightarrow \mathbb{Z}_4$ -Fourier DtN (v201/v264) $\Rightarrow [\rho, \Lambda]=0 \Rightarrow \omega \circ \rho = \omega$ (v198); [E] neg controls (generic $j \neq 1728$ , only $\mathbb{Z}_2$ ; hexagonal $j=0$ has $\mathbb{Z}_6$ , order $6 \neq 4$ ); [C] arithmetic rigidity ( $\{1, i, -1, -i\}$ over $\mathbb{Q}$ , $j=1728$ CM by $\mathbb{Z}[i]$ , Belyi/dessins); [O] the bare order-4 symmetry stays the one foundational postulate (like $c=\text{const}$ ). Closes the reduction (axiom $\Rightarrow$ everything), not the axiom. |
| v268_theta13_<br>carrier_trace.py       | FLAV.TH13.01 — the reactor angle $\theta_{13}$ <i>exponent</i> is the carrier hypercharge trace. The value $\sin^2 \theta_{13} = \varphi_0^{\text{ret}} e^{-5/6} = 0.0231$ was already checked (v16); here the exponent origin is closed exactly: [E] $\gamma = \text{tr}_E Y^2 = 3(1/3)^2 + 2(1/2)^2 = 5/6$ over the 5-slot carrier hypercharge $Y = \text{diag}(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2})$ (roots of $6Y^2 - Y - 1 = 0$ ), complement $1/6$ the neutrino ratio; [C] value 0.02311 vs PDG $\sim 0.0222$ ; [E] own channel — the $\mu\tau$ -breaking $\varepsilon = 3\varphi_0^{\text{ret}}/4$ induces only $\sim 1.6 \times 10^{-3} \ll 0.023$ , so $\theta_{13}$ is a separate carrier-trace channel, not the seesaw $\mu\tau$ breaking (corrects the tempting ‘ $\theta_{13}$ from $M_\nu$ ’ route); [O] CP magnitude + absolute $\nu$ scale remain open.                                                                                                                                                                                                                                                                                                                                                                   |

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| Script                         | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| v269_spert_paqft_skeleton.py   | QFT4D.SPERT.01 — the perturbative 4D S-matrix ( $S_{\text{pert}}$ ) as an Epstein–Glaser / pAQFT skeleton: the honest middle layer of the three-layer S-matrix ( $S_{\text{top}}$ 2d DHR braiding v243 / $S_{\text{pert}}$ 4d EG of the spectral action / $S_{\text{phys}}$ LSZ on the OS Wightman functions v240), <i>not</i> a nonperturbative closure. [E] three distinct layers (the 2d braiding $ M =1$ is statistics, not 4d cross-sections); [E] every spectral-action $a_4$ operator has mass dimension $\leq 4$ in 4D (power-counting renormalizable); [E] a finite counterterm basis (7 marginal +3 relevant, no infinite tower); [C] Epstein–Glaser existence — a renormalizable interaction has an order-by-order $S(g)=T \exp(i \int g L_{\text{int}})$ by causal factorisation + finite local extension (no path integral, no UV divergences); [C] adiabatic limit/gap ( $\Delta=6 \log(3/2)>0$ ) $\rightarrow S_{\text{phys}}$ via LSZ; [O] perturbative only — the ambient measure QG.AMB.01 stays open. A contract/skeleton, explicitly perturbative.                                                                                                            |
| v270_pmns_jarlskog_assembly.py | FLAV.PMNS.03 — the complete complex PMNS matrix + the leptonic Jarlskog $J_{\text{PMNS}}$ (the CP <i>strength</i> ), assembled from the four TFPT channels ( $\sin^2 \theta_{12}=\frac{1}{3}-\frac{\phi_0}{2}$ , $\sin^2 \theta_{23}=\frac{1}{2}$ , $\sin^2 \theta_{13}=\phi_0 e^{-5/6}$ , $\delta=4\pi/3$ ) into one unitary object. [E] the three angle channels; [E] $U = R_{23}U_{13}(\delta)R_{12}$ is exactly unitary; [C] $J_{\text{PMNS}} = \Im(U_{e1}U_{\mu 2}U_{e2}^*U_{\mu 1}^*) = -0.02965$ (formula and full matrix agree), $J_{\text{max}} = 0.03424$ — a <i>derived</i> CP strength, the leptonic analogue of the CKM $J$ (v88); [C] data: $J_{\text{max}}$ vs NuFIT 0.0332 ( $\sim 3\%$ ), $J$ vs best fit $-0.026$ , the negative sign reproduced; [O] $\delta$ is the $\mu_6$ /triality phase (assembled input, <i>not</i> from $M_\nu/D_F$ ) and the absolute $\nu$ -mass scale ( $\Sigma m_\nu$ , $M_R$ ) stays open. Assembles the PMNS matrix + CP strength; does not derive $\delta$ from $D_F$ nor fix the scale.                                                                                                                                         |
| v271_eg_oneloop_quartic.py     | QFT4D.SPERT.02 — a concrete Epstein–Glaser order-2 (one-loop) quartic renormalization, taking the v269 $S_{\text{pert}}$ skeleton from <i>exists</i> to an actual EG number (no path integral). [E] $T_1=:L_{\text{int}}$ : needs no extension, the first extension is $T_2$ ; [E] the massless propagator has scaling degree $\text{sd}=2$ in $d=4$ , the one-loop bubble (two propagators) has $\text{sd}=4=d$ so the UV singular order $\omega=\text{sd}-d=0$ ; [E] $\omega=0 \Rightarrow$ exactly one logarithmic local counterterm ( $C(\omega+d,d)=1$ , renormalizable, no infinite tower); [E] the loop factor $\Omega_3/(2(2\pi)^4)=1/(16\pi^2)$ exactly — the same factor that normalises the spectral-action $a_4$ geometric quartic; [C] the $\phi^4$ one-loop $\beta_\lambda=3\lambda^2/(16\pi^2)$ (symmetry factor 3 = the $s, t, u$ channels), EG initial condition $\lambda_\Phi(\text{UV})=1/(16\pi^2)$ ; [O] order-2/one-loop only — the all-order EG construction and the nonperturbative measure QG.AMB.01 stay open. Turns the skeleton into a concrete one-loop number.                                                                                      |
| v272_nu_mass_scale.py          | FLAV.NUSCALE.01 — the absolute neutrino-mass scale, typed honestly (it does <i>not</i> fabricate a $\Sigma m_\nu$ number, which would contradict the v184 firewall). [E] the absolute scale reduces to <i>one</i> seesaw ratio $m_3 = (y_\nu v_{\text{EW}})^2/M_R$ — one irreducible UV input like $v_{\text{geo}}$ (v153), while the ratios ( $m_2/m_3$ , ordering, PMNS angles v270) are pinned; [C] the observed $m_3=0.050$ eV with the TFPT PS $\rightarrow$ GUT window $M_R \in [4.2 \times 10^{13}, 2.4 \times 10^{15}]$ GeV (v249) needs a third-generation Dirac Yukawa $y_\nu \in [0.26, 2.0]$ — $O(1)$ , no tuning; [O] $M_R(y=1)=6.1 \times 10^{14}$ has $\log_{c_3}(M_R/M_{\text{Pl}})=2.57$ (non-integer), not a TFPT power, so the absolute value stays open; [C] the normal-ordering floor $\Sigma m_\nu=0.0586$ eV is consistent with cosmology ( $< 0.072$ – $0.12$ eV); [X] a cosmological $\Sigma < 0.0586$ eV excludes the normal-ordered $m_1 \sim 0$ spectrum. Closes the PMNS bookkeeping (the last [O] is one seesaw ratio), not the absolute number; sharpened by v481 (carrier normalisation $y_\nu=y_t$ : one-parameter, window-bracketed candidate). |

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| Script                              | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| v273_eg_gauge_running.py            | QFT4D.SPERT.03 — the gauge sector of the Epstein–Glaser one-loop (v271’s quartic mechanism applied to the gauge self-energy). [E] the one-loop gauge 2-point has scaling degree $sd=4=d$ so $\omega=0 \rightarrow$ one counterterm per coupling (same marginal mechanism + loop factor $1/(16\pi^2)$ as v271); [E] the one-loop $b_i$ from the explicit carrier/SM content: $b_3=-7$ , $b_2=-19/6$ , $b_1=41/10$ (the same 41 as the carrier algebra, v159/CAR.SM.01); [C] physical RG directions $b_3<0$ (asymptotic freedom), $b_1>0$ (U(1) Landau); [O] the two-loop matrix (v159, PyR@TE) + all-order stay open. Shows the gauge running is an EG order-2 result. Exact $b_i$ mirrored in Wolfram.                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| v274_scale_overdetermination.py     | SCALE.OVERDET.01 — the single mass anchor is <i>over-determined</i> , and the absence of a derivable absolute scale is a <i>theorem</i> . [E] one anchor (No-Unit, v153): every dimensionful output = pure number $\times M_{P1}^d$ ( $\rho_\Lambda/M_{P1}^4$ , $M_{scal}/M_{P1}=c_3^{7/2}$ , the spectrum); [C] two independent routes to $M_{P1}$ agree: gravity $(8\pi G)^{-1/2}=2.4353\times 10^{18}$ GeV and cosmology (invert $\rho_\Lambda/M_{P1}^4=(3/4\pi^2)e^{-2\alpha^{-1}}$ with $\rho_\Lambda^{1/4}=2.24$ meV) $=2.438\times 10^{18}$ GeV, to 0.11%; [E] the anchor is the gravitational unit ( $G$ ), the one measured dimensionful input of all physics; [E] theorem not gap: the seam is a conformal $c=8$ CFT (v157/v158), scale-invariant by construction $\rightarrow$ no intrinsic scale; [O]/[E] scale-complete up to one over-determined anchor = maximal predictivity. Sharpens v153/v78.                                                                                                                                                                                                                                                          |
| v275_qgamb_roadmap.py               | QGAMB.ROADMAP.01 — the obligation roadmap for QG.AMB.01 (the ambient nonperturbative QG measure), consolidated into one module (the open gate had no script). [E] Tier A gap decoupling: $\Delta_{\text{eff}}=6\log(3/2)-2.248c_3^2=1.648>0$ (v36/v76), so every low-energy readout is independent of the un-built measure — not a blocker; [E] Tier B: the bulk measure reduces to identifying the seam-Calderón boundary with the holomorphic $(E_8)_1$ net ( $c=248/31=8$ ; v77/GATE.METRIC.03), reduced not closed (REDTEAM.A.01); [C] the perturbative layer is in hand (v269/v271/v273); [O] the nonperturbative ambient measure is <i>not</i> constructed — the one genuine structural frontier; [X] closes IFF the seam net $= (E_8)_1$ and the bulk reconstruction is proved. A roadmap, not a closure.                                                                                                                                                                                                                                                                                                                                                          |
| v276_qgeo_flat_closes_commutator.py | QGEO.SYM.03 — the flat $\tau=i$ pillowcase premise <i>closes</i> the modular-commutator chain to all orders, collapsing the bedrock QGEO.SYM.01 to <i>one</i> geometric postulate (it does <i>not</i> prove the premise). For a quasi-free seam state $QGEO.SYM.01 = (\omega \circ \rho = \omega) \Leftrightarrow [\rho, C]=0 \Leftrightarrow [\rho, H]=0$ . [E] isometry $\Rightarrow$ commutator: if $H=f(\Delta_{\text{flat}})$ and the order-4 deck $\rho(z \rightarrow iz)$ is a flat- $\tau=i$ isometry then $[\rho, \Delta]=0 \Rightarrow [\rho, H]=0$ <i>exactly</i> (all orders), verified on the discretised square torus ( $\rho^4=I$ , $[\rho, \Delta]=0$ , $[\rho, \sqrt{-\Delta}]=0$ ; generic non-isometry fails); [E] this upgrades v198 (principal symbol) + v201 (subprincipal) to the full $H$ (flatness removes the curvature terms); [C] $\tau=i$ is forced by order-4 ( $j=1728$ ; hexagonal $j=0$ is order-6, excluded; v214/v267); [O] QGEO.SYM.01 collapses to the single statement ‘the raw seam is the flat $\tau=i$ pillowcase quasi-free state’ — needs a human constructive-QFT proof or stays an honest axiom. Sharpest non-circular form. |
| v277_seam_calderon_e8_match.py      | QGAMB.TIERB.01 — the seam-Calderón $\rightarrow (E_8)_1$ matching certificate, the concrete Tier-B step of QG.AMB.01 (v275): it computes the full $(E_8)_1$ matching target and pins the residual to <i>one</i> bit (it does <i>not</i> close it). [E] $c=8$ (Sugawara, $248/31=g_{\text{car}}+N_{\text{fam}}$ ), but $c=8$ alone does not pin the net; [E] the discriminator is holomorphy — $\det \text{Cartan}(E_8)=1$ (1 primary, holomorphic) vs the $c=8$ counterexample $\det \text{Cartan}(D_8=SO(16))=4$ (4 primaries, non-holomorphic), so the single discriminator is $ \det K =1$ ; [E] the $(E_8)_1$ character $E_4/\eta^8=j^{1/3}=q^{-1/3}(1+248q+\dots)$ has a single primary + exactly $248=\dim E_8$ weight-1 currents (240 roots); [E] equivalence chain holomorphic $\Leftrightarrow \det \text{Cartan} 1 \Leftrightarrow$ unique even unimodular rank-8 (v83) $\Leftrightarrow$ homology-sphere link $\Leftrightarrow  \det K =1$ (v235); [C] seam-side match via v276 (flat $\tau=i$ ) + v260 (Kummer/K3); [O] residual = the single holomorphy bit — reduced to one bit, <i>not</i> closed.                                                         |

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| Script                          | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| v278_lsz_bridge_unitarity.py    | <p>QFT4D.SPERT.04 — the <math>S_{\text{pert}} \rightarrow S_{\text{phys}}</math> LSZ bridge + one-loop unitarity: connects the EG perturbative S-matrix (v269/v271/v273) to the physical asymptotic S-matrix (<math>S_{\text{phys}}</math> = LSZ on the OS Wightman functions, v240). [E] LSZ bridge: <math>S_{\text{phys}}</math> = amputate+on-shell-residue of the connected time-ordered products, and the EG <math>T</math>-products of <math>S_{\text{pert}}</math> are those correlators (OS Wick-rotation, v240)</p> <p><math>\Rightarrow S_{\text{pert}} \xrightarrow{\text{LSZ}} S_{\text{phys}}</math>; [C] the gap <math>\Delta=6\log(3/2)&gt;0</math> gives Haag–Ruelle asymptotic states so LSZ is valid; [E] <i>one-loop unitarity</i> (optical theorem): the bubble discontinuity <math>\text{Im } B(s)/\pi = x_+ - x_- = \sqrt{1 - 4m^2/s}</math> <i>exactly</i> = the two-body phase space, so <math>2\text{Im } M = \sum \int d\Pi  M ^2</math> (cutting rule) holds <math>\Rightarrow</math> matter+gauge unitary; [E] threshold: <math>\text{Im } B=0</math> below <math>s=4m^2</math>, turning on at the physical two-particle threshold; [O] gravity caveat — the Stelle ghost makes the gravity perturbative S-matrix non-unitary <math>\rightarrow</math> QG.AMB.01; [C] finite field-strength <math>Z</math> from the EG scheme.</p> |
| v279_qgeo_obligation_lemma.py   | <p>QGE0.OBLIG.01 — the QGE0.SYM.01 bedrock written as <i>one</i> precise constructive-QFT lemma with a proof-tree completeness check (the formal obligation write-up; does <i>not</i> close it). Lemma: given the premise ‘the raw seam carries the flat <math>\tau=i</math> pillowcase metric <math>g_{\text{flat}}</math> (Trojanov-unique, cone angles <math>\pi</math>), so <math>\Lambda_\Sigma = \sqrt{-\Delta_{\text{flat}}}</math>’, then <math>\omega_\Sigma \circ \rho = \omega_\Sigma \rightarrow</math> mark-locality <math>\rightarrow E_8</math>. [E] the reduction DAG from ‘free RP seam’ has 8 closed nodes and <i>exactly one</i> open leaf (‘the raw seam state is the flat state’); [E] the geometric side is fixed (flat DtN diagonal, <math>\rho</math> commutes to all orders, Lean FORM.QGE0.03), so the obligation is state-identification not geometric ambiguity; [C] two proof routes (RP-state uniqueness on the flat orbifold, or Trojanov + OS); [E] the all-orders step is a Lean theorem (SeamDeckClosure.flat_all_orders_clock, standard axioms, no sorry); [O] one constructive-QFT statement closes QGE0.SYM.01.</p>                                                                                                                                                                                                       |
| v280_pillowcase_steklov.py      | <p>QGE0.STEKLOV.01 — a direct numerical experiment on the flat <math>\tau=i</math> pillowcase orbifold (the seam geometry): the self-investigable test of the QGE0.SYM.01 obligation (does <i>not</i> prove the premise). [E] pillowcase = the <math>Z_2</math>-even sector of the flat torus, 4 cone points (2-torsion, deficit <math>\pi</math> each, sum <math>4\pi=2\pi \cdot 2</math>); [E] the order-4 deck <math>\rho(z \rightarrow iz)</math> descends (<math>[\rho, Z_2]=[\rho, P_{\text{even}}]=0</math>) and permutes the 4 cone points as 2 fixed + 1 swapped pair; [E] the geometric chain is exact:</p> <p><math>[\rho, \Delta]=0 \Rightarrow [\rho, H]=0 \Rightarrow [\rho, C]=0 \Rightarrow \omega \circ \rho = \omega</math> (the v276/v279 all-orders closure on the real orbifold); [C] the Steklov DtN is positive self-adjoint, fixed by the flat metric (route-(i) evidence); [O] confirms the conclusion given flat geometry, the premise stays open.</p>                                                                                                                                                                                                                                                                                                                                                                               |
| v281_holomorphy_modular_data.py | <p>QGAMB.MODULAR.01 — the modular-data / anyon-condensation layer of the QG.AMB.01 Tier-B holomorphy bit. [E] #anyons = <math> \det \text{Gram} : E_8 \rightarrow 1</math> (holomorphic), <math>D_8=SO(16) \rightarrow 4</math>, carrier <math>D_5 \oplus A_3 \rightarrow 16</math>; [E] the anyon-condensation tower <math>D_5 \oplus A_3 (16) \rightarrow D_8 (4) \rightarrow E_8 (1)</math>, each step condensing a Lagrangian <math>Z_2</math> boson, the Kitaev <math>E_8</math> endpoint (v92/v235); [E] Gauss–Milgram recovers <math>c=8</math>; [E] <math>E_8</math> is the unique holomorphic <math>c=8</math> (<math>\det 1</math>) — <math>SO(16)</math> has the same <math>c</math> but 4 anyons, so holomorphy is THE discriminator (v277); [O] the open bit: does the seam-Calderón net condense to <math>\det K=1</math>? Reduced to one bit.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| v282_e8_tau_i_unification.py    | <p>QGAMB.UNIFY.01 — the <math>E_8</math>-at-<math>\tau=i</math> unification: the candidate <i>simplification</i> reducing the two open obligations to <i>one</i> object (the raw seam is the <math>(E_8)_1</math> holomorphic CFT at <math>\tau=i</math>). <math>\chi_{E_8}(\tau)=\Theta_{E_8}/\eta^8=j^{1/3}</math>. [E] <math>\tau=i</math> is the order-4 CM point (<math>j=1728</math>, the QGE0.SYM.01 flat geometry); [E] <math>\chi_{E_8}=j^{1/3}</math> (<math>\chi^3=qj</math> as <math>q</math>-series), so <math>\chi_{E_8}(i)=1728^{1/3}=12</math> (the <math>(E_8)_1</math> character at the SAME <math>\tau=i</math>, the QG.AMB.01 holomorphy); [E] one object, two faces; [C] obligation count <math>2 \rightarrow 1</math>: prove ‘the raw seam is <math>(E_8)_1</math> at <math>\tau=i</math>’ and <i>both</i> follow; [O] the unified premise stays open but is now one object.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

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| Script                                       | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v283_live_killtest_scorecard.py              | PRED.LIVE.01 — a live kill-test scorecard of the recent round’s computable predictions vs current data, each tagged with the decisive experiment. [C] PMNS angles $\sin^2 \theta_{12}=0.30675$ ( $+0.31\sigma$ ), $\sin^2 \theta_{13}=0.02311$ ( $+1.45\sigma$ ), $\theta_{23}$ maximal; [C] leptonic CP $J_{\text{PMNS}}=-0.0297$ , $\delta=240^\circ$ vs $\sim 232(30)$ ; [X] $\Sigma m_\nu$ NO floor 0.0586 eV (DESI/CMB-S4 decisive); [X] proton decay $\tau_p=1.55\times 10^{35}$ yr for $+(15, 1, 1)$ (Hyper-K decisive); [E] closed readouts $\alpha^{-1}$ , $\Omega_b$ , $\beta$ ; all rows consistent $< 2\sigma$ , two sharp near-term kill tests. A consolidation, not new physics.                                                                                                                                                                                                                                                                                                                       |
| v284_route_i_rp_uniqueness.py                | QGEO.ROUTEI.01 — Route (i) (RP-state uniqueness) decomposed into a 6-lemma chain with every dischargeable lemma discharged and the one irreducible lemma isolated (does <i>not</i> prove the open premise). [E] 5/6 lemmas are existing [E]/[O] results — L1 RP→DtN (v194), L2 the four marks force the pillowcase class (v195/v216/v214), L3 the flat orbifold Steklov operator $= \sqrt{-\Delta_{\text{flat}}}$ (verified: flat-strip DtN( $k$ )/ $k \rightarrow 1$ , principal symbol $ \xi $ with no curvature term), L4 covariance $C = (1 + e^H)^{-1}$ from the DtN (v258), L5 $\mu_4$ -invariance (v276/Lean/v280); [C] Troyanov reduction: 4 cone points of angle $\pi \Rightarrow \chi_{\text{orb}} = 0 \Rightarrow$ the unique constant-curvature representative is flat ( $\tau=i$ ), so $L_{\text{open}}$ reduces to ‘the raw seam is the constant-curvature geometric state’; [O] the one open lemma — the raw RP seam state is the constant-curvature/geometric state (the state-identification step). |
| v285_route_ii_seam_condensation.py           | QGAMB.ROUTEII.01 — Route (ii) (seam-net condensation) decomposed into a 4-lemma chain, with the key result that its open lemma <i>coincides</i> with Route (i)’s (the v282 unification at the lemma level). [E] 3/4 lemmas discharged — M1 no abelian sector $\Leftrightarrow  \det K =1 \Leftrightarrow$ holomorphic (v234/v235), M2 holomorphic $c=8 \Leftrightarrow (E_8)_1$ with the $SO(16)$ det-4 counterexample (v277/v281), M3 the condensation tower $16 \rightarrow 4 \rightarrow 1 =$ the unique $\mu_4$ -Lagrangian glue (v92/v281/v235); [E] the discriminator det 1 vs det 4; [E] Routes (i) and (ii) share <i>one</i> open lemma — both say ‘the raw seam is the $(E_8)_1$ -at- $\tau=i$ object’ (v282), so proving either closes both; [O] the unified open lemma: construct a canonical equivalence between the raw seam KMS/DtN state and the holomorphic $(E_8)_1$ net at $\tau=i$ . $M_{\text{open}} = \text{QGEO.SYM.01}$ .                                                                     |
| v286_seam_equivalence_contract.py            | SEAM.EQUIV.01 — the Seam Equivalence contract: concentrates the whole open structure on <i>one</i> named arrow (does <i>not</i> prove it). Theorem: the raw RP seam state reconstructs canonically the holomorphic $(E_8)_1$ boundary net at the $\tau=i$ pillowcase. [E] four objects (RawRPSeam, FlatTauIPillowcaseDtN, HolomorphicE8Net, SREKitaevE8Phase); [E] six arrows — 2 closed (Route i v276/v280/v284, Route ii v281/v277/v235), 1 conditional (v282), 2 open premises (the same lemma, v284/v285), 1 Gral; [E] <b>import firewall</b> : no Route-i module imports any Route-ii module or vice versa, so the routes converge only through SEAM.EQUIV.01 (kills the ‘ $E_8$ into the geometry and back’ circularity); [E] exact-label taxonomy (Formal/Numerical/Conditional Exact, Open Selection); [O] the one open arrow ‘free Gaussian RP bulk $\Rightarrow$ holomorphic single-sector boundary’.                                                                                                      |
| v287_free_rp_bulk_to_holomorphic_boundary.py | SEAM.EQUIV.A01 — Route A (AQFT) theorem-dependency checker for the open Gral arrow ‘free Gaussian RP bulk $\Rightarrow$ holomorphic single-sector boundary net’: types the 5-lemma chain and isolates the <i>one</i> missing standard theorem. [E] L1 RawRPSeam→OS→CAR/quasi-free boundary (v155/v160/v240/v175); [E] L2 gap+chirality $\Rightarrow$ gapped chiral bulk (v64/v156); [O] L3 invertible/SRE Gaussian bulk $\Rightarrow$ single-sector boundary ( $\mu$ -index 1) — the one missing standard theorem (the AQFT form of $\det K=1$ ); [E] L4 $\mu=1+c=8 \Rightarrow$ holomorphic (v234); [E] L5 holomorphic $c=8 \Rightarrow (E_8)_1$ (v83/v277/v281). Verdict: 4/5 discharged, Route A reduces SEAM.EQUIV.01 to one cited theorem.                                                                                                                                                                                                                                                                      |

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| Script                                | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| v288_full_l2_subprincipal_z4.py       | SEAM.EQUIV.B01 — Route B (DtN): the full- $L^2$ lift of the sub-principal $\mathbb{Z}_4$ block-diagonality, proving (numerically, on the full Toeplitz operator) the ‘probably provable’ step and lifting v201/v284 from the finite block to $L^2$ . [E] character orthogonality: a mark-sourced curvature $f = \sum_{j<4} g(\theta - j\pi/2)$ has Fourier support only on $m \equiv 0 \pmod{4}$ ; [E] full- $L^2$ block-diagonality: $\Lambda =  n  + M_f$ commutes with $\rho = \text{diag}(i^n)$ , $\ [\rho, \Lambda]\  \sim 2 \times 10^{-15}$ on $-50..50$ ; [E] negative control: a generic curvature gives $\ [\rho, \Lambda]\  = 4.44 > 0$ ; [E] Lean consistency (geom_sum_fourth_root + markLocal_blockDiagonal); [O] the sharper residual — why is the raw seam sub-principal term mark-local?                                                              |
| v289_marklocal_raw.py                 | MARKLOCAL.RAW.01 — decompose the one remaining Route-B residual (‘why is the raw seam sub-principal term mark-local?’) into a 5-lemma chain and isolate the single open analytic lemma. [C] L1 conic parametrix: conic Steklov DtN $=  D_\theta  + M_f + \text{cone sources}$ (b-calculus/Melrose, imported); [O] L2 <b>Flat-Away</b> — the one open analytic lemma: $\text{RP} + \text{gap} + 4 \text{ marks} \Rightarrow \text{curvature vanishes away from the marks}$ (the flat-pillowcase realisation); [E] L3 orbit-source: the 4 marks form one $\mu_4$ orbit (v195/v216); [E] L4 Fourier-support: $\mu_4$ -orbit curvature $\Rightarrow$ support only on $4\mathbb{Z}$ (v264/v288, with a non-orbit negative control); [E] L5 full-operator: $\Rightarrow [\rho, \Lambda]=0$ on full $L^2$ (v288). 4/5 discharged; Route B reduces to exactly L2.              |
| v290_z4_smooth_curvature_adversary.py | SEAM.ADVERSARY.01 — the Route-B red-team that $\mathbb{Z}_4$ block-diagonality is <i>not</i> mark-locality. The strongest adversary, a smooth $\mathbb{Z}_4$ -symmetric off-mark curvature $f = \varepsilon \cos(4\theta)$ : [E] it is $\mathbb{Z}_4$ -symmetric (Fourier support $\{\pm 4\} \subset 4\mathbb{Z}$ ) but <i>not</i> mark-local ( $f(\pi/4) = -\varepsilon \neq 0$ ); [E] it <i>passes</i> the v288 commutator $\ [\rho, \Lambda]\  = 0$ (same as mark-local) — so $[\rho, \Lambda]=0$ is necessary but <i>not</i> sufficient; [E] but it shifts the DtN spectrum off the flat ladder ( $\max  \Delta\lambda  = 0.155$ ) and the heat trace ( $\Delta = 0.019$ ), so the KMS weight and $(E_8)_1$ character shift; [O] a genuine candidate modulus the commutator misses but the spectral data excludes — Flat-Away must use spectral/energy/heat input. |
| v291_flataway_contract.py             | FLATAWAY.RP.01 — the Flat-Away lemma as its own named mini-theorem with three proof routes opened by v290. Lemma [O]: given RawRPSeam with gap, chirality and the four $\mu_4$ marks, the smooth curvature density in the DtN sub-principal symbol vanishes away from the branch divisor. [E] <i>spectral-rigidity</i> : scanning $f = \varepsilon \cos(4\theta)$ , the spectrum deviation from the flat ladder is 0 only at $\varepsilon=0$ — flat is the unique spectral match; [E] <i>heat-kernel</i> : the heat-trace deviation is 0 only at $\varepsilon=0$ ; [C] <i>RP-energy</i> : $\text{RP} + \text{gap}$ is a Dirichlet-energy minimisation in the Troyanov conformal class $\Rightarrow$ constant-curvature minimiser (Troyanov; cf. v284). [O] three independent routes to one lemma; closing any one closes Route B.                                      |
| v292_flataway_heat_reduction.py       | FLATAWAY.HEAT.01 — the Heat-Kernel route made precise: reduces Flat-Away to <i>one</i> named spectral invariant. [E] the heat-trace deviation $\Delta \text{Tr}(e^{-t\Lambda}) = c(t)\varepsilon^2 + O(\varepsilon^4)$ is a positive-definite quadratic form in the off-mark curvature ( $c_k(t) > 0$ for every $\mathbb{Z}_4$ mode $4k$ ; positive on all combinations), so $\Delta \text{Tr} = 0 \Leftrightarrow f = 0$ ; leading invariant $\ f\ _{L^2}$ (Parseval). [O] reduction: Flat-Away $\Leftarrow$ ‘the seam fixes the $a_2$ heat coefficient to its flat value’ $\Rightarrow \int f^2 = 0 \Rightarrow f = 0$ off the marks — one external invariant.                                                                                                                                                                                                       |
| v293_flataway_spectral_hessian.py     | FLATAWAY.SPEC.01 — the Spectral-Rigidity route upgraded from one direction (v291) to the <i>full</i> smooth- $\mathbb{Z}_4$ space. [E] splitting mechanism: the degenerate flat Steklov level $ n $ is split by mode $4k$ at $ n =2k$ by $\pm\varepsilon/2$ ; [E] $S = \sum (\Delta\lambda)^2 \sim 0.50\varepsilon^2$ ; [E] the Hessian over modes $\{4, 8, 12\}$ has eigenvalues $\{0.497, 0.500, 0.503\} > 0$ , so the flat seam is a <i>strict</i> isolated minimum over the whole $\mathbb{Z}_4$ space; [E] a non- $\mathbb{Z}_4$ mode breaks the commutator outright. [O] pinning the seam’s Steklov spectrum excludes every smooth $\mathbb{Z}_4$ deformation $\Rightarrow$ Flat-Away.                                                                                                                                                                           |

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| Script                           | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| v294_flataway_rp_energy.py       | FLATAWAY.ENERGY.01 — the RP-energy / Trojanov route: the flat pillowcase is the unique constant-curvature minimiser at the fixed mark divisor. [E] prescribed divisor + Gauss-Bonnet: $S^2$ with 4 cone angles $\pi$ , $\sum(2\pi - \theta) = 4\pi = 2\pi\chi$ ; [C] Trojanov/Euclidean-cone uniqueness (Trojanov 1991) excludes smooth off-mark curvature; [E] strictly convex curvature energy $E[f] = \int f^2$ , Hessian $2\pi I$ (PD) $\Rightarrow$ unique minimiser; [E] no zero-cost deformation. [O] premise: ‘RP+gap realises the energy-minimiser’, converging with v292/v293 on one selection principle; closing any one closes Route B.                                                                                                                                                                                                                                                                                                                                                                 |
| v295_flataway_a2_exact.py        | FLATAWAY.A2.01 — the <i>exact</i> analytic $a_2$ coefficient, upgrading the v292 positive-definiteness from numerical to a proof. [E] first variation vanishes: $\partial_\varepsilon \text{Tr } e^{-t\Lambda(\varepsilon)} _0 = -t\hat{f}(0) \sum e^{-t n } = 0$ (zero-mean off-mark curvature, Gauss-Bonnet); [E] convexity $\Rightarrow$ global minimum: $\phi(x) = e^{-tx}$ strictly convex $\Rightarrow \Lambda \mapsto \text{Tr } e^{-t\Lambda}$ convex (Klein/Peierls), so $\varepsilon=0$ is the global minimum and $\Delta \text{Tr} \geq 0$ for all $\varepsilon$ , $= 0$ iff $f = 0$ ; [E] exact $a_2$ kernel via the divided difference of $e^{-tx}$ (Duhamel), matched to numerics to rel.err $< 10^{-6}$ ; [E] degenerate-split closed form $2e^{-2kt}(\cosh(tg/2)-1) \geq 0$ ; [O] reduction now analytic: Flat-Away $\Leftarrow$ ‘the seam fixes $a_2$ ’. Lean-mirrored (FORM.FLATAWAY.01).                                                                                                         |
| v296_flataway_a2_closed.py       | FLATAWAY.A2.CLOSED.01 — the exact $a_2$ coefficient in <i>closed form</i> . [E] the operator tails telescope geometrically ( $C_k + D_k e^{-4kt} = 4kt(1 - e^{-4kt})$ ), giving $W_k^{\text{tail}} = t(1 - e^{-4kt})/(8k(1 - e^{-t}))$ ; [E] a finite $(4k-1)$ -term middle (incl. the degenerate diagonal $\Phi(2k, 2k) = (t^2/2)e^{-2kt}$ ); [E] closed form $W_k(t) = \frac{t(1-e^{-4kt})}{8k(1-e^{-t})} + \frac{1}{2} \sum_{m=1}^{4k-1} \Phi(m, 4k-m)$<br>$(W_1(t) = t(2te^t + e^{3t} + 3e^{2t} + e^t - 1)e^{-3t}/8)$ , reproducing v295 to machine precision; [E] manifestly positive; [E] small- $t$ : $W_k = t/2 + O(t^3)$ (mode-independent leading $1/2$ , the $t^2$ term cancels).                                                                                                                                                                                                                                                                                                                        |
| v297_route_a_literature_stack.py | SEAM.EQUIV.A.LIT.01 — Route A’s ‘one standard import’ as a precise citable theorem stack. [E] LIT-A invertible/SRE bulk $\Rightarrow$ trivial bulk topological order (Kitaev 2006; Freed–Hopkins 2021); [E] LIT-B bulk MTC = Drinfeld center, trivial center $\Rightarrow$ holomorphic boundary net (Müger 2003; Kawahigashi–Longo–Müger 2001); [E] LIT-C holomorphic $c=8$ net $\Rightarrow (E_8)_1$ (Conway–Sloane; Dong–Xu/Kawahigashi–Longo). [E] the three compose into ‘invertible bulk $\Rightarrow (E_8)_1$ ’, all legs established; [O] the one open hypothesis is the input of LIT-A, ‘the raw quasi-free seam bulk is invertible (SRE)’ = Route B’s Flat-Away. [E] non-circular with Route B (the v286 firewall holds).                                                                                                                                                                                                                                                                                  |
| v298_e8_network_attractor.py     | <i>E8 as a network attractor</i> — a computational-substrate (Wolfram-program) reading of the $(2, 3, 5)$ /McKay hull, on the v219 bedrock; an honest audit, not a derivation. [E] the $E_8$ Kac marks $(1, 2, 3, 4, 5, 6, 4, 2, 3)$ are the <i>unique</i> attractor of the minimal lazy graph update $m \leftarrow (A + 2I)/4 \cdot m$ on the affine- $E_8/(2, 3, 5)$ McKay graph (converges from any positive start; $Am = 2m$ , top eigenvalue 2). [E] the spherical McKay tower terminates at $E_8$ ( $2T \rightarrow E_6$ , $2O \rightarrow E_7$ , $2I \rightarrow E_8$ , $h = 30$ ); [E] the cascade is nested-subgraph coarse-graining ( $240 \rightarrow 126 \rightarrow 72 \rightarrow 40 \rightarrow 20$ ). [C] Wolfram reading: $E_8$ as the universal attractor of a minimal $(2, 3, 5)$ network (Coxeter-skeleton level). [O] a minimal hypergraph <i>rewrite</i> whose attractor is the full TFPT structure stays open; $\varphi_0 = (4/3)c_3 = 1/(6\pi)$ is the analytic seed, not an edge fraction. |

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| Script                           | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| v299_hypergraph_0d_autonomous.py | The hypergraph/Wolfram bridge (hardened): the 3-arm $(2, 3, \cdot)$ branch type is the irreducible seed on which a local witness-based spectral growth <i>reaches</i> $E_8$ as the last finite point ( $E_6 \rightarrow E_7 \rightarrow E_8$ , then stops at the affine boundary $\hat{E}_8$ ) — <i>not</i> ‘P2 derived’. [E] icosahedron = 3-uniform hypergraph ( $ \text{Aut}  = 120 =  I_h $ ; the McKay group is the $SU(2)$ binary lift $2I$ , a different order-120 group); [E] the $1 \rightarrow 4$ subdivision rewrite is equivariant (120 symmetries at every scale); [E] the $2I$ fusion tensor is a weighted 3-uniform hypergraph whose 2-dim slice is the affine $E_8$ McKay graph; [E] Smith: $\rho \leq 2$ selects (affine) ADE; [E] Collatz–Wielandt: $\rho \leq 2 \Leftrightarrow$ a positive witness $m$ with $Am \leq 2m$ (local); [E] the autonomous rule (diffusion witness + local gate $Am \leq 2m$ ) runs $E_6 \rightarrow E_7 \rightarrow E_8 \rightarrow \hat{E}_8$ under the $(2, 3, \cdot)$ seed; [E] $A_n$ ( $\rho = 2 \cos \frac{\pi}{n+1}$ ) and $D_n$ ( $\rho = 2 \cos \frac{\pi}{2n-2}$ ) controls are $< 2$ for all $n$ , never reaching $E_8$ ; [O] the seed = P2. A second network reading of the $(2, 3, 5)/E_8$ core (v219/v298).                                                     |
| v300_flataway_rigid.py           | FLATAWAY.RIGID.01 — the attack on Flat-Away. [E] the flat fingerprint: $\text{spec}( D_\theta )$ is the integer ladder, every level $n \geq 1$ of multiplicity exactly 2 (a discrete fingerprint, not a norm); [E] the resonant mode $4k$ splits level $2k$ by gap $= g_k$ (sympy $2 \times 2$ $[[2k, g/2], [g/2, 2k]] \Rightarrow 2k \pm g/2$ ), validated vs the full Toeplitz operator; [E] any $g_k \neq 0$ lifts the resonant degeneracy (mult $2 \rightarrow 1+1$ ) and changes the spectral multiset, so preserving the flat ladder <i>with</i> its mult-2 degeneracies forces every $g_k = 0 \Rightarrow f = 0$ — a discrete obstruction, strictly stronger than the v291/v293 ‘deviation grows’ scan; [E] with the v295 convexity, flat is isolated analytically <i>and</i> discretely; [C] the pin derived: the $(E_8)_1$ character $E_4/\eta^8 = q^{-1/3}(1 + 248q + \dots)$ has integer $q$ -coefficients (248 at level 1; $E_8$ even unimodular $\Rightarrow$ integer weights), and by 2d Steklov conformal rigidity the integer ladder $\Leftrightarrow$ flat seam, so $(E_8)_1$ rationality $\Rightarrow f = 0$ (conditional on Route A); [O] Flat-Away carries <i>zero</i> new open content beyond Route A — the two SEAM.EQUIV.01 routes converge on one rationality fact.                                 |
| v301_route_a_invertible.py       | SEAM.EQUIV.A.INV.01 — discharge Route A’s last open hypothesis (‘the raw quasi-free seam bulk is invertible/SRE’) via the free-fermion classification; since v300 collapsed Route B into Route A, this is the last unconditional gap of SEAM.EQUIV.01. [E] carrier = 16 Majoranas, $c = 8$ ( $g_{\text{car}} + N_{\text{fam}} = 5 + 3 = 8 = 16/2$ , v148); [E] $SO(16)_1 \supset (E_8)_1$ : $\dim \mathfrak{so}(16) = 120$ currents + 128 spinor = 248 = $\dim E_8$ (v154); [C] free-fermion invertibility (Kitaev): a gapped Gaussian/quasi-free 2+1D fermion phase (class D) is $\mathbb{Z}$ -classified by the chiral Majorana number and is <i>always</i> invertible (no intrinsic anyons; topological order needs interactions), so a gapped quasi-free bulk is invertible automatically; [E] the invariant $ \det K  = \#\text{anyons}$ is 1 for $E_8$ (invertible) vs 4 for the gauged $D_8 = SO(16)$ (anyons), which free fermions cannot produce; [O] LIT-A’s ‘invertible bulk’ now follows from ‘gapped 16-Majorana quasi-free bulk’ (v148/v160+the gap), so the residual changes from a topological-order statement to the spectral input ‘is the quasi-free bulk gapped?’; [O] with v300 the whole open content of SEAM.EQUIV.01 reduces to the single spectral statement ‘the quasi-free seam bulk is gapped’. |

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| Script                     | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| v302_seam_gap.py           | <p>SEAM.EQUIV.GAP.01 — the last lever: after v300/v301 the whole open content of SEAM.EQUIV.01 was ‘the quasi-free seam bulk is gapped’; that gap is <i>not</i> a free input but the established Recovery gap of the seam transfer operator. [E] <math>\text{spec}(T) = \{1, (2/3)^6, (1/3)^6\}</math> (v160/v162), a simple leading Perron eigenvalue 1; [E] the gap is derived: <math>\Delta = -\ln((2/3)^6) = 6 \ln(3/2) = 2N_{\text{fam}} \ln(3/2) \approx 2.4328 &gt; 0</math> (multiplicative gap <math>1 - (2/3)^6 = 665/729</math>; <math>6 = 2N_{\text{fam}}</math>, <math>(2/3)^6</math> the <math>\mu_4</math>-deck transfer eigenvalue forced by the carrier); [E] the positivity margin <math>\Delta - 31/(4\pi^2) = 1.6476 &gt; 0</math> (the v76 decoupling margin); [C] by Osterwalder–Schrader / quasi-free clustering a transfer gap <math>\Delta &gt; 0</math> is a mass gap of the reconstructed Hamiltonian; [O] so the last TFPT-internal input is the derived Recovery gap <math>6 \ln(3/2) &gt; 0</math>, and SEAM.EQUIV.01’s residual is a composition of cited theorems (OS/clustering + Kitaev v301 + the v297 AQFT stack) over established TFPT facts — no undischarged TFPT-internal assumption remains, though it stays [O] (not machine-proved end-to-end).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| v303_ftransfer_dynamics.py | <p>FR.DYNAMICS.01 — the discrete→dynamic lens on F_transfer: the main-branch mechanism (a gapped, positivity-preserving relaxation to a <i>unique</i> attractor; the local-averaging update → the <math>E_8</math> marks, gap <math>6 \ln(3/2)</math>) is the shared <i>shape</i> of all four instances. [E] F_pole (Koide) <i>is</i> the seam dynamics — the source→pole/branch relaxation is the Möbius map with fixed-point multiplier <math>F' = (2/3)^6 = \lambda_2</math>, the seam-clock subleading eigenvalue (v82/v54); <math> F'  &lt; 1 \Rightarrow</math> a unique attractor at the seam rate; [C] F_Boltzmann (<math>\eta_B</math>) is a gapped positive contraction (washout <math>\kappa_f \in (0, 1)</math>, H-theorem; compiler fixes <math>A_\Lambda = 10</math>, rate thermal); [C] F_relic (axion) freezes at an adiabatic fixed point (seed <math>\theta_i = 3\pi/5</math>, rate cosmological); [E] F_QCD (<math>m_p/m_e</math>) is the 1-loop RG flow with the carrier <math>b_3 = -7</math> (asymptotic freedom <math>\Rightarrow</math> the Gaussian UV attractor, <math>\Lambda_{\text{QCD}}</math> generated). [O] Verdict: one shape underlies all four = the main-branch principle; F_pole has the seam rate exactly, the other three share only the shape (external rates), so F_transfer is the readout end of the one principle, not a bolt-on — simplicity is preserved by honest fencing (v187).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| v304_idg_nonlocal_ghost.py | <p>QGAMB.IDG.01 — the KMS spectral-action cutoff suggests a <i>nonlocal</i> (infinite-derivative) graviton form factor; under entire analyticity the Stelle ghost is a <i>truncation artefact</i> — a [C] narrowing of the red-team rt_F perturbative-gravity attack, <i>not</i> a closure of QG.AMB.01 (no new parameter: the cutoff is the v259 seam KMS weight). [E] the local four-derivative propagator <math>1/(p^2(p^2 + M^2)) = (1/M^2)[1/p^2 - 1/(p^2 + M^2)]</math> has a healthy massless graviton (residue <math>+1/M^2</math> at <math>p^2=0</math>) and a massive spin-2 mode with <i>negative</i> residue <math>-1/M^2</math> at <math>p^2 = -M^2</math> (the Stelle ghost, reproducing v278); [E] for an IDG operator <math>p^2 a(p^2)</math> with <math>a(z) = e^{z/M^2}</math> entire and nowhere zero, the dressed propagator <math>1/(p^2 a)</math> keeps its <i>only</i> pole at <math>p^2=0</math> — no ghost (Tomboulis 1997; Biswas–Mazumdar–Siegel 2006); [E] negative control: a polynomial truncation <math>a(z) = 1 + z/M^2</math> (the <math>R+R^2</math> order) has a real zero at <math>z = -M^2</math> = the ghost pole, so ‘entire vs polynomial’ is the discriminator; [C] the cutoff is the <math>\beta=1</math> seam KMS weight <math>f(u) = e^{-u}</math> (v259), not a choice — truncating <math>\text{Tr} e^{-D^2/\Lambda^2}</math> at <math>a_4</math> is what makes the local <math>R+R^2</math> and the ghost; [C] <b>conditional narrowing</b>: <i>if</i> the resummed form factor is entire (Biswas–Mazumdar–Siegel class) <i>then</i> the perturbative graviton is ghost-free and rt_F’s attack is a truncation artefact; [O] <b>open</b>: (i) the trace regulator <math>f</math> is not the kinetic form factor <math>a(\square)</math> — entire analyticity of the resummation is unproven; (ii) a ghost-free <i>perturbative</i> graviton is not the nonperturbative ambient measure — QGAMB.IDG.01 stays [C]/[O], not a QG.AMB.01 closure.</p> |

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| Script                        | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| v305_witness_independence.py  | Witness-independence / generator-economy audit (structural anti-numerology firewall, companion to v100): every headline integer is a documented image of the single anchor $a=(1,1,2)$ ( $e=(4,5,2)$ , $p_n=2+2^n$ ; v23/v228), $\pi$ the only transcendental primitive (ARCH.CORE.01) — 13 atoms / 2 free primitives = $6.5 \times$ compression made explicit (the recurring $2/3= \mathbb{Z}_2 /N_{\text{fam}}$ and $(2/3)^6$ are <i>one</i> ratio); $E_8/g_{\text{car}}$ /anchor are overdetermined ( $\geq 3$ structurally independent witnesses), the 5 pure-seed readouts single-witness (correlated, scored jointly by v100); $\alpha^{-1}$ 's '137' is foreign to the anchor family = an independent witness [E] |
| v306_seed_crossval.py         | Leave-one-out cross-validation of the shared seed $\varphi_0$ : back-solving $\varphi_0$ independently from the five seed-grammar observables ( $\sin^2 \theta_{12}$ , $\sin^2 \theta_{13}$ , $\beta$ , $\Omega_b$ , $\lambda_C$ ) and error-weighted-averaging reproduces the axiom seed $\varphi_0=1/(6\pi)+48c_3^4=0.0531720$ to 0.04% (one real seed, not a per-observable fit); leave-one-out predicts every held-out observable within $3\sigma$ out of sample (most-tensioned = $\sin^2 \theta_{13}$ , $\sim 2\sigma$ ); a data $\leftrightarrow$ formula shuffle explodes the $\chi^2$ by $> 100 \times$ [E]                                                                                                     |
| v307_data_watchdog.py         | Data watchdog: the operational decision pipeline over the frozen registry (v84). Classifies all 20 entries (16 predictions + 2 assigned textures + 2 bands) PASS/WATCH/TENSION/KILL by live $n_\sigma$ vs repo-documented current bests (CODATA 2022, NuFIT 6.0, ACT DR6, Planck 2018, PDG 2024, BK18); no decidable prediction in KILL (theory alive); watch items $\alpha^{-1}$ ( $1.9\sigma$ ), $s_{23}$ ( $1.8\sigma$ ) and the most-tensioned core $\sin^2 \theta_{13}$ ( $2.0\sigma$ ); the prediction-of-record $\sin^2 \theta_{12}=0.306747$ compatible (JUNO, the fastest falsifier); $r \in [0.0033, 0.0048]$ below the BK18 limit and the kill 0.01; $w=-1$ flagged the dark-energy watchdog (DESI) [E]       |
| v308_seam_equiv_chain.py      | TOE attack 1 — the SEAM.EQUIV.01 keystone as a composition certificate: the v297 stack as one well-typed chain $\text{gap}>0 \rightarrow \text{invertible} \rightarrow \text{holomorphic } c=8 \rightarrow (E_8)_1$ with exact discriminators $ \det \text{Cartan}(E_8) =1$ vs $ \det \text{Cartan}(D_8) =4$ , carrier tower $4 \cdot 4=16 \rightarrow 4 \rightarrow 1$ , $c=g_{\text{car}}+N_{\text{fam}}=8$ ; the input gap is the derived recovery gap $6 \ln(3/2)>0$ . The only undischarged premise is the cited OS step — an assembly/reduction certificate (Lean target), NOT a closure [E][O]                                                                                                                    |
| v309_modular_bedrock.py       | TOE attack 2 — the bedrock $\omega \circ \rho = \omega$ via an explicit modular Hamiltonian, non-circular: marks = 4 from Gauss–Bonnet (v216) + the order-4 clock from cross-ratio 2 (v214) $\Rightarrow \mu_4$ derived, not assumed; the quasi-free modular Hamiltonian $K = \log((1-C)/C)$ round-trips to the seam operator and $[\rho, K]=0 \Leftrightarrow \omega \circ \rho = \omega$ (a $\mathbb{Z}_2$ profile breaks it). Residual = Bisognano–Wichmann intrinsicality of the raw collar; NOT a closure of QGEO.SYM.01 [E][O]                                                                                                                                                                                     |
| v310_carrier_sm_anomaly.py    | TOE attack 3 — the carrier half-spinor is one anomaly-free SM generation with the SM RG: $16=\Lambda^{\text{even}}(\mathbb{C}^5)=1+10+5 \rightarrow SU(5) \subset SO(10)$ as $10+5+1 = \text{one generation}$ ; gauge-anomaly-free ( $\sum Y=0$ , $\sum Y^3=0$ , $[SU(2)]^2 U(1)=[SU(3)]^2 U(1)=0$ , exact) and reproduces $(b_1, b_2, b_3)=(\frac{41}{10}, -\frac{19}{6}, -7)$ ; a fourth family shifts $b_1$ , so $N_{\text{fam}}=3$ pins the RG. Closes the spectrum/anomaly/ $\beta$ sub-question; the full EG $S$ -matrix + Yukawa stay open [E][C]                                                                                                                                                                 |
| v311_gap_decoupled_measure.py | TOE attack 4 — the physical QG measure as a gap-decoupled, exponentially-clustering object: the transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ gives geometric clustering at rate $(2/3)^6$ , correlation length $1/(6 \ln(3/2))$ , finite susceptibility $729/665$ ; the v76 margin $6 \ln(3/2) - 31/(4\pi^2) \approx 1.648 > 0$ decouples the physical sector from the un-built ambient. Spectral-action $a_2=EH/a_4=R^2/72$ (v36) + KMS $\beta=1$ (v259); QG.AMB.01 stays [O] (gap-decoupled, inert) [E][O]                                                                                                                                                                                                              |

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| Script                        | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| v312_hypergraph_substrate.py  | TOE attack 5 — the hypergraph substrate sharpened: the affine- $E_8$ Kac marks are the Perron eigenvector ( $A m=2m$ ) and the subleading eigenvalue is exactly $2\cos(\pi/5)=$ the golden ratio (the 5-fold/icosahedral signature), but the recovery rate $(2/3)^6$ is <i>not</i> any eigenvalue and $\varphi_0$ is not a rational edge fraction; so a full-structure rewrite must inject exactly two non-graph-spectral data — the cusp weight $2/3= \mathbb{Z}_2 /N_{\text{fam}}$ (gap $6\ln(3/2)$ ) and the analytic seed $\varphi_0$ . The open question is precisely delimited, not closed [E][O]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| v313_golden_atoms_spectral.py | The golden ratio is the $g_{\text{car}}=5$ signature: the $(2,3,5)$ atoms <i>are</i> the affine- $E_8$ network spectral angles (v312 follow-up, exact). [E] the charpoly factors $x(x^2-4)(x^2-1)(x^2-x-1)(x^2+x-1)$ , the golden minimal polynomial $x^2-x-1$ divides it ( $\phi=2\cos(\pi/5)$ an exact eigenvalue); the non-trivial eigenvalues are $2\cos(\pi/k)$ for $k \in \{2,3,5\}=\{ \mathbb{Z}_2 , N_{\text{fam}}, g_{\text{car}}\}$ (giving $0, 1, \phi, 1/\phi$ ), so the golden ratio is the $g_{\text{car}}=5$ (5-fold) signature — a new spectral witness for $g_{\text{car}}$ (raises the v305 multiplicity). The $(2,3,5)$ spherical excess $1/2+1/3+1/5=31/30$ ties the v63/v76 margin numerator $31=2^{g_{\text{car}}}-1=248/8=1+h$ to the Coxeter 30 (a unification, not new evidence). [E] honest null: $\phi$ is structural, not a readout. [C] direction: $\phi$ is the $E_8$ -quasicrystal (Elser-Sloane) ratio, a candidate substrate [E][C]                                                                                                                                                                            |
| v314_rate_translation.py      | Do the discrete-kernel and dynamic rates translate? An exact number-field split. The affine- $E_8$ adjacency spectrum splits into the rational part $\{0, \pm 1, \pm 2\}$ in $\mathbb{Q}$ and the golden part $\{\pm\phi, \pm 1/\phi\}$ in $\mathbb{Q}(\sqrt{5})$ ; the dynamic transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ is entirely rational. [E] the family rates DO translate (the rational angles $2\cos(\pi/2)=0= \mathbb{Z}_2 $ , $2\cos(\pi/3)=1=N_{\text{fam}}$ build the recovery rate $(2/3)^6=( \mathbb{Z}_2 /N_{\text{fam}})^{2N_{\text{fam}}}$ in $\mathbb{Q}$ ), but the carrier 5-fold (golden, $\mathbb{Q}(\sqrt{5})$ ) has <i>no</i> rational dynamic image — a field obstruction, so it is static-only. The shared object is the order-30 Coxeter element ( $30=g_{\text{car}}(2N_{\text{fam}})=5\cdot 6$ ); the $\phi$ quasicrystal carries the carrier, not the cusp-weight family dynamics [E][C]                                                                                                                                                                                                                       |
| v315_coxeter_coupling.py      | Couple or factorize? The order-30 Coxeter sectors couple as a cyclotomic field (v314 follow-up). [E] GROUP FACTORIZES: $\gcd(5,6)=1 \Rightarrow \mathbb{Z}/30 = \mathbb{Z}/5 \times \mathbb{Z}/6$ (direct product, group-decoupled); [E] EXPONENTS FACTOR: $E_8$ exponents = totatives(30) = totatives(5) $\times$ totatives(6), $\phi(30)=4\cdot 2=8=$ rank $E_8$ ; [E] FIELD COUPLES: $[\mathbb{Q}(\zeta_{30}):\mathbb{Q}]=8=$ rank $E_8$ , $\mathbb{Q}(\zeta_{30})$ is the compositum of the carrier $\mathbb{Q}(\zeta_5) (\ni \sqrt{5})$ and the family $\mathbb{Q}(\zeta_3)$ ; [E] GALOIS = $\mu_4 \times \mathbb{Z}_2$ : $(\mathbb{Z}/30)^\times = (\mathbb{Z}/5)^\times \times (\mathbb{Z}/3)^\times = \mathbb{Z}/4 \times \mathbb{Z}/2$ (order 8, max order 4), $\mu_4=(\mathbb{Z}/5)^\times$ the carrier Galois (the seam clock = the carrier golden-field Galois), $\mathbb{Z}_2=(\mathbb{Z}/3)^\times$ the family Galois; [E] $\sqrt{5}=\phi+1/\phi$ = the Gauss sum in $\mathbb{Q}(\zeta_5)$ . [C] $30=5\cdot 6$ does not dynamically entangle — it couples as the cyclotomic compositum, the bridge is Galois not dynamical [E][C] |
| v316_galois_readout.py        | Does the Galois $\mu_4 \times \mathbb{Z}_2$ (v315) organize the physical readouts? Yes — the CP phases. [E] the hexagonal CP unit $\rho=\zeta_6=\zeta_{30}^5$ is the order-6= $2N_{\text{fam}}$ FAMILY factor $c^5$ (the carrier is $\zeta_5=\zeta_{30}^6$ ); [C] $\delta_{\text{CKM,lead}}=\pi/3=\arg \zeta_6$ , $\delta_{\text{PMNS}}=4\pi/3=\arg \zeta_6^4$ (power $4= \mu_4 $ ), $\zeta_6^4=-\zeta_6$ the sheet flip ( $\mu_6=\mu_3 \times \mu_2$ ); [E] the Galois $\mathbb{Z}_2=\text{Gal}(\mathbb{Q}(\zeta_6)/\mathbb{Q})=\{\zeta_6 \mapsto \zeta_6^5=\bar{\zeta}_6\}$ IS CP conjugation ( $\delta \mapsto -\delta$ ); [E] the carrier factor $\zeta_5$ (golden $\sqrt{5}$ ) carries no phase (static); [E] the magnitudes ( $\sin^2 \theta_{12}=1/3-\varphi_0/2$ , transcendental $\varphi_0$ ) are not cyclotomic. So the Galois bridge reaches phenomenology through the family factor (phases); the carrier is static; the magnitudes are the analytic seed [E][C]                                                                                                                                                                   |

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| Script                      | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| v317_galois_family.py       | Are the 3 generations a $\mu_3$ /Galois orbit? Yes ( $\mu_3$ ), refined 1+2 by Galois (v316 follow-up). [E] the 3 cusp weights $\{0, 1/3, 2/3\}$ ( $\text{Spec}(Q_+) = \{1, 2, 3\} = A_3$ exponents) map under $w \mapsto e^{2\pi i w}$ to the cube roots $\{1, \zeta_3, \zeta_3^2\}$ ; the cyclic family symmetry $w \mapsto w+1/3$ is a transitive $\mu_3$ orbit (the family IS the cube-root orbit). [E] the transfer eigenvalues $(1-w)^6 = \{1, (2/3)^6, (1/3)^6\}$ attach to the weights (attractor = eigenvalue 1 at $w=0$ ). [E] the Galois $\mathbb{Z}_2$ ( $\zeta_3 \mapsto \zeta_3^2$ , = CP conjugation v316) fixes $w=0$ (rational) and swaps $w=1/3 \leftrightarrow 2/3 \Rightarrow$ orbit 1+2; so the Galois-FIXED generation is the recovery ATTRACTOR (the law), the Galois PAIR the decaying hierarchy. [C] the family structure is arithmetic; the mass magnitudes stay the analytic seed [E][C]                                                                                                                                                                                                                                                                                                                                                                                                   |
| v318_arithmetic_capstone.py | The capstone: the SM structural sector is $\mathbb{Q}(\zeta_{30}) + \text{Galois } \mu_4 \times \mathbb{Z}_2$ , and the complete input is $\{a, \pi, v_{\text{geo}}\}$ (synthesis of the v313–v317 arc). [E] the structural sector (3 generations, 2 CP phases, orbit/hierarchy) lives in $\mathbb{Q}(\zeta_{30})$ : $[\mathbb{Q}(\zeta_{30}):\mathbb{Q}] = \varphi(30) = 8 = \text{rank } E_8$ , $30 =  \mathbb{Z}_2  N_{\text{fam}} g_{\text{car}} = h(E_8)$ , Galois $(\mathbb{Z}/30)^\times = \mu_4 \times \mathbb{Z}_2$ ; [E] rigidity: forced by the anchor atoms $\{2, 3, 5\} = e(a)$ ; [E] magnitude reduction: $\varphi_0 = (4/3)c_3 + 48c_3^4$ is a pure function of $\pi$ (no free parameters), so the magnitudes are $\{a, \pi\}$ -determined, not free; [E] free inventory: 0 dimensionless free parameters, $\pi$ the unique transcendental, $v_{\text{geo}}$ the one dimensional unit. [O] residual: the raw-seam realisation stays open (QGEO.SYM.01/SEAM.EQUIV.01) [E][O]                                                                                                                                                                                                                                                                                                                            |
| v319_translation_clock.py   | The translation clock: the discrete $\leftrightarrow$ dynamic bridge is the order-30 Coxeter element, a clock with two coprime hands $5 \times 6$ , read law-inclusive (0..5) or live-only (1..5). [E] the only object holding both the static carrier ( $\mathbb{Q}(\sqrt{5})$ ) and the dynamic family (rational) data is the order-30 Coxeter element (v314); $30 = h(E_8) = g_{\text{car}}(2N_{\text{fam}}) = 5 \cdot 6$ , $\text{gcd}(5, 6) = 1 \Rightarrow \mathbb{Z}/30 = \mathbb{Z}/5 \times \mathbb{Z}/6$ (v315). [E] DYNAMIC HAND ( $\mathbb{Z}/6 = 2N_{\text{fam}}$ , ticks 0..5): recovery exponent exactly $6 = 2N_{\text{fam}}$ , rate $(2/3)^6$ ; order-6 generator $c^5$ ( $\zeta_6 = \zeta_{30}^5$ ). [E] LAW VS LIVE: the resummed clock rate( $n$ ) = $-p_2 \ln(1 - n/N_{\text{fam}})$ (v124) has rate(0) = 0 (the attractor/law) and live modes from $n=1$ ; the $E_8$ live phases (totatives of 30) start at 1, so 0, 1, 2, 3, 4, 5 is law-inclusive, 1, 2, 3, 4, 5 live-only. [E] STATIC HAND ( $\mathbb{Z}/5 = g_{\text{car}}$ , ticks 1..5): golden $\mathbb{Q}(\sqrt{5})$ , field-obstructed (no rational dynamic image) — static-only. [E] one full turn = $\text{lcm}(5, 6) = 30 = h(E_8)$ (rank $E_8 = 8 = \varphi(30)$ , $ \mu_4 =4$ planes, v55); the “it is one clock” reading is [C]. |
| v320_galois_cp_relation.py  | A NEW falsifiable prediction from the Galois structure: the two CP phases are locked, $\delta_{\text{PMNS}} = \delta_{\text{CKM}}^{\text{lead}} + \pi$ . [E] both phases are powers of one hexagonal unit $\rho = \zeta_6$ (v316): $\delta_{\text{CKM}}^{\text{lead}} = \arg \rho = \pi/3$ ( $60^\circ$ ), $\delta_{\text{PMNS}} = \arg \rho^4 = 4\pi/3$ ( $240^\circ$ ), $\rho^4 = -\rho$ , so $\delta_{\text{PMNS}} - \delta_{\text{CKM}}^{\text{lead}} = \pi$ exactly (the $\mathbb{Z}_2$ sheet flip) — the quark and lepton leading CP phases are NOT independent. [C] prediction of record $\delta_{\text{PMNS}}^{\text{lead}} = 240^\circ$ (upgrades the assigned texture to a Galois-forced relation to the measured quark phase); [C] currently within the broad NuFIT 6.0 range; [X] kill test: $\delta_{\text{PMNS}}$ robustly away from $240^\circ$ ( $> 3\sigma$ ; DUNE/Hyper-K/JUNO) falsifies the Galois-CP organization [E][C][X]                                                                                                                                                                                                                                                                                                                                                                      |
| v321_killtest_board.py      | The forward kill-test board: the decisive upcoming measurements, ranked (forward companion to v307). [E] each entry is locked to a freeze_file.csv kill row: #1 JUNO $\sin^2 \theta_{12} = 0.306747$ (fastest), #2 CMB-S4/LiteBIRD $r$ (kill $r > 0.01$ ), #3 DESI/Euclid $w = -1$ , #4 DUNE/Hyper-K $\delta_{\text{PMNS}} = 240^\circ$ (the new v320 Galois-CP relation), #5 DUNE/LEGEND/nEXO ordering + $m_{\beta\beta}$ , #6 PSI n2EDM, #7 JUNO $\sin^2 \theta_{13}$ ( $\sim 2\sigma$ now), #8 LiteBIRD $\beta$ . A forward decision ledger complementing v307; no new physics [E]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

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| Script                       | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| v322_cp_lock_sharpened.py    | The v320 CP-lock prediction sharpened into a quantitative kill test. [E] the leading CP phase sits on the six hexagonal nodes $\arg \rho^k = k \cdot 60^\circ$ ; $\delta_{\text{CKM}}^{\text{lead}} = 60^\circ$ (node 1), $\delta_{\text{PMNS}}^{\text{lead}} = 240^\circ$ (node 4). [E] the node selector is the deck order: $\delta_{\text{PMNS}}^{\text{lead}} =  \mu_4  \delta_{\text{CKM}}^{\text{lead}} = 4(\pi/3) = 4\pi/3 = \delta_{\text{CKM}}^{\text{lead}} + \pi$ ( $\rho^4 = -\rho$ ). [C] sub-leading budget: $\delta_{\text{CKM}}^{\text{full}} = \pi/3 + 3\lambda^2$ ( $\sim 68.7^\circ$ ); the lepton sub-leading is a separate open holonomy term bounded by $\sim 3\lambda^2 \approx 8.7^\circ \Rightarrow$ band $240 \pm \sim 9^\circ$ . [C] current tension: vs NuFIT 6.0 NO ( $\delta_{CP} = 212_{-41}^{+26^\circ}$ ) the leading $240^\circ$ sits at $+1.08\sigma$ , band edge $\sim 231^\circ$ at $+0.74\sigma$ . [X] kill: the nearest wrong node is $60^\circ$ away $\gg$ budget; DUNE/Hyper-K ( $\sim 5\text{--}15^\circ$ ) discriminate, $> 3\sigma$ outside kills the lock. [E] an anarchic $\delta_{CP}$ hits the band with $p \sim 5\%$ and node 4 1-in-6 [E][C][X]                                                                                                                         |
| v323_bw_geometric_modular.py | The keystone pushed one step: Bisognano–Wichmann makes the seam modular flow <i>geometric</i> , so the $\mu_4$ deck invariance is not an independent axiom. [E] the clock $\rho = \text{diag}(i^n) = \exp(i(\pi/2)L)$ is a geometric rotation ( $L = \text{diag}(n)$ , $\rho^4 = I$ , $\mu_4 \subset U(1)_{\text{rot}}$ ). [E] for a rotation-covariant $C = f(L)$ the modular Hamiltonian $K = \log((1 - C)/C) = g(L)$ , so $[K, L] = 0$ (the modular flow commutes with all rotations) and $[\rho, K] = [\rho, C] = 0$ — the clock is a modular symmetry for free. [E] discriminator vs v309: a period-4 curvature preserves the clock but its modular flow is <i>not</i> geometric ( $[K, L] \neq 0$ ); BW is strictly stronger. [C] linked bedrock: given the seam is the $(E_8)_1$ chiral net (v308), BW/Hislop–Longo make the vacuum modular flow geometric $\Rightarrow$ QGEO.SYM.01 ( $\omega \circ \rho = \omega$ ) is downstream of SEAM.EQUIV.01 + a rotation-invariant vacuum. [O] residual: the raw collar vacuum is rotation-invariant (cleaner than v309, shared with v308) [E][C][O]                                                                                                                                                                                                                      |
| v324_hypergraph_fiber.py     | The minimal hypergraph substrate that works: a $(2, 3, 5)$ -network <i>fibered</i> by a 3-fold cusp channel reproduces both the $E_8$ skeleton and the recovery gap (the constructive v312 follow-up). [E] network factor: $T_{\text{net}} = (A + 2I)/4$ fixes the Kac marks (eigenvalue $1 = E_8$ skeleton), subleading $(\varphi + 2)/4$ (golden); $(2/3)^6$ is <i>not</i> in its spectrum (the v312 negative). [E] cusp fiber: $T_{\text{cusp}} = \text{diag}((1 - w)^{2N_{\text{fam}}})$ , $w \in \{0, 1/3, 2/3\}$ ( $N_{\text{fam}} = 3$ , v317), spectrum $\{1, (2/3)^6, (1/3)^6\}$ . [E] the fibered substrate $T = T_{\text{net}} \otimes T_{\text{cusp}}$ has top eigenvalue 1 (eigenvector marks $\otimes$ ( $w = 0$ ), skeleton $\times$ democratic cusp) <i>and</i> $(2/3)^6$ as a genuine eigenvalue (network attractor $\times$ cusp subleading). [E] honest injection: without the fiber ( $T_{\text{net}} \otimes I_3$ ) there is no $(2/3)^6$ — the cusp datum is injected, not graph-spectral; substrate = network $\times$ cusp = the v315 split $30 = g_{\text{car}}(2N_{\text{fam}}) = 5 \cdot 6$ . [E] minimality: the smallest fiber is the 3-node cusp ( $= N_{\text{fam}}$ ) = order-30 (v319 clock). [C] a consistent fibered realization, not a derivation of $2/3$ from a pure rewrite [E][C] |
| v325_pillowcase_keystone.py  | The seam keystone as ONE citable theorem: the raw seam state <i>is</i> the flat $\tau = i$ pillowcase state, with <b>exactly one</b> open lemma. IF (raw RP-collar seam state = flat $\tau = i$ pillowcase / rotation-invariant state) THEN 4 square marks + $\mu_4$ deck + $\omega \circ \rho = \omega$ + holomorphic $c = 8 \Rightarrow (E_8)_1$ , with the carrier recovery gap as input. [E] pieces all exact: $n = 2\chi(S^2) = 4$ (v216); cross-ratio $2 \Rightarrow j = 1728$ (v214/v267); flat orbifold $S^2(2, 2, 2, 2)$ , $\chi_{\text{orb}} = 0$ (Trojanov); DtN sub-principal at $m \equiv 0 \pmod 4$ (v201/v280); $\det \text{Cartan}(E_8) = 1$ vs $D_8 = 4$ (v83/v143/v308); gap $6 \ln \frac{3}{2} > 0$ (v76/v302). [O] the one open lemma — the raw collar state IS the flat $\tau = i$ pillowcase / rotation-invariant state — is SHARED by both bedrock IDs (QGEO.SYM.01, SEAM.EQUIV.01; v323), machine-pinned in Lean (FORM.SEAMEQUIV.01 + FORM.QGEO.BW.01); an assembly/reduction certificate (like v176/v261), not a closure [E][O]                                                                                                                                                                                                                                                                  |

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| Script                     | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| v326_ftransfer_suite.py    | The $F_{\text{transfer}}$ solver suite: the four transfer readouts as ONE runnable harness, each a gapped relaxation to a unique attractor (productises v303). [E] $F_{\text{pole}}$ (Koide): Moebius contraction to $Q^*=2/3$ with multiplier $(2/3)^6$ = the seam-clock subleading eigenvalue, unique attractor at the SEAM rate. [C] $F_{\text{Boltzmann}}(\eta_B)$ : washout contraction to the balance $6.1 \times 10^{-11}$ , monotone (H-theorem), external thermal rate. [C] $F_{\text{relic}}$ (axion): the comoving number $N=E/\omega$ is the adiabatic invariant, the relic freezes (seed $\theta_i=3\pi/5$ ), cosmological rate. [E] $F_{\text{QCD}}(m_p/m_e)$ : 1-loop RG with carrier $b_0=7$ flows to the Gaussian UV fixed point, $\Lambda_{\text{QCD}}$ generated. [C] $F_{\text{RareKaon}}$ : $\text{BR}(K^+)=9.45 \times 10^{-11}$ on NA62 2016-2024 (v202). [O] only $F_{\text{pole}}$ has the seam rate $(2/3)^6$ ; the other four share the SHAPE with external rates honestly fenced (v187/v303) — the frontier is measurable without claiming the external rates [E][C][O] |
| v327_hypergraph_rewrite.py | Deriving the cusp weight $2/3$ from a minimal rewrite (sharpening v324/v312). A minimal local rule — a family channel with $N_{\text{fam}}=3$ slots, one absorbing attractor ( $w=0$ ) and $ \mathbb{Z}_2 =2$ surviving — supplies the cusp fiber. [E] $M$ has spectrum $\{1, 2/3, 0\}$ ; $2/3=(N_{\text{fam}}-1)/N_{\text{fam}}= \mathbb{Z}_2 /N_{\text{fam}}$ EMERGES from the rule arity, not injected. [E] over the order-6= $2N_{\text{fam}}$ hand the rate is $(2/3)^6=64/729$ = the recovery gap (v76/v324), derived. [E] NON-GRAPH-SPECTRAL (proof, sharpening v312): $2/3$ is NOT a root of the affine- $E_8$ charpoly ( $p(2/3)=-3520/19683 \neq 0$ ), so the recovery rate cannot be an adjacency eigenvalue. [E] the absorbing state (eigenvalue 1) is the democratic $w=0$ law (v317). [O] honest residual: the arity $\{2, 3\}$ (the anchor atoms) is still the input (P2) — v324's injected datum reduced to the anchor arity [E][O]                                                                                                                                                 |
| v328_theta13_pressure.py   | The $\theta_{13}$ pressure point: the honest weak spot, quantified and pre-registered. [C] $\sin^2 \theta_{13}=\varphi_0^{\text{ret}}e^{-5/6}=0.023108$ vs NuFIT 6.0 NO ( $0.02195 \pm 0.00058$ ) is $+2.0\sigma$ — the most-tensioned CORE prediction (v62/v307). [E] $\theta_{13}$ is the carrier hypercharge trace (exponent $\gamma=\text{tr}_E Y^2=5/6$ EXACT, v268), NOT the seesaw $\mu\tau$ breaking that sets $\theta_{12}$ — distinct channels. [C] no cheap relief: the back-solved seed $\varphi_0^{\text{ret}}(\theta_{13})=0.0505$ is $\sim 5\%$ below the axiom, PMNS $\theta_{13}$ RG running is $< 1\%$ (NO), and $5/6$ is exact — the prediction stays frozen (REG.FREEZE.01). [X] pre-registered kill: a stable $\sin^2 \theta_{13}$ away from 0.023108 at $> 3\sigma$ (JUNO/global) flags the carrier-trace/seed-grammar reading; exactly the v306 seed-hyperplane outlier — one honest pressure point [C][E][X]                                                                                                                                                                |
| v329_os_gap_reduction.py   | Closing the OS step: the bulk gap <i>is</i> the transfer gap (second quantization), and assembling the single sufficient premise of the whole bedrock. [E] OS GAP = TRANSFER GAP: the many-body OS Hamiltonian $d\Gamma(-\log T)$ has spectrum = sums of single-mode energies, so the bulk gap = the smallest positive single-mode energy = $6 \ln \frac{3}{2}$ , independent of system size — exactly “OS bulk gap = transfer gap” (the v308 open input), discharged at the many-body level. [E] $H \Rightarrow \omega \circ \rho = \omega$ : under $H$ the covariance is rotation-covariant $C=f(L)$ , $K=g(L)$ commutes with $\rho=\exp(i(\pi/2)L)$ (v201/v309/v323). [E] $H \Rightarrow \text{SRE} \Rightarrow (E_8)_1$ : $\det \text{Cartan}(E_8)=1$ vs $D_8=4$ (v237/v308). [E] both bedrock IDs follow from the ONE premise $H$ (Lean-pinned FORM.SEAMEQUIV.01 + FORM.QGEO.BW.01). [O] residual: the necessity of $H$ on the raw collar; sharpens v240/v308/v323/v325, does NOT close the bedrock [E][O]                                                                                     |

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| Script                           | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| v330_qgamb_admissible_measure.py | Constructing the ambient measure on the gap-decoupled admissible sector as a bona fide Osterwalder–Schrader object (with the honest fence). [E] reflection positivity: the seam transfer $T$ (v221/v311; spectrum $\{1, (2/3)^6, (1/3)^6\}$ ) is symmetric PSD, so the finite-volume measures are RP. [E] exponential clustering: $C(n)/C(n-1) \rightarrow \lambda_2 = (2/3)^6$ ( $\xi=1/\Delta$ , $\Delta=6 \ln \frac{3}{2}$ ). [E] tightness $\Rightarrow$ projective limit: $\chi = \sum_n C(n) = 729/665$ finite, so $\text{Var}(S_\Lambda)/\Lambda \rightarrow \chi \Rightarrow$ uniform tightness (Prokhorov) $\Rightarrow$ the projective limit $\mu = \lim \mu_\Lambda$ EXISTS on the admissible sector. [E] OS reconstruction: $H_{\text{OS}} = -\log T \geq 0$ , unique vacuum, gap $6 \ln \frac{3}{2}$ (v240). [E] NEG: an ungapped transfer sends $\chi \rightarrow \infty$ , tightness fails. [O] the full ambient measure: the non-admissible (metric) sector is NOT built; by the gap margin $\Delta - 31/(4\pi^2) \approx 1.648 > 0$ (v76/v275) the physical readouts do not need it, but QG.AMB.01 stays open. A genuine PARTIAL construction, NOT a closure [E][O]                                                                                   |
| v331_necessity_of_H.py           | The <i>necessity</i> of $H$ reduced: rotation-invariance of the seam DtN is EQUIVALENT to the four marks at the square ( $\tau=i$ ), so the one open lemma is exactly the order-4 CM selection (the necessity companion to v329’s sufficiency). [E] square $\Rightarrow [\rho, \Lambda]=0$ : marks at $\{0, \pi/2, \pi, 3\pi/2\}$ make the curvature $\mathbb{Z}_4$ -symmetric (Fourier support $0 \bmod 4$ ), $M_f$ couples only modes differing by $4\mathbb{Z}$ (the v201/v210 sufficiency as a DtN $ k +M_f$ operator). [E] necessity: a mark off the square breaks $\mathbb{Z}_4$ and gives $[\rho, \Lambda] \neq 0$ — equivalence, not just implication. [E] order-4 CM: cross-ratio( $1, i, -1, -i$ ) $=2 \Rightarrow j=1728$ , the unique 4-config with an order-4 automorphism (v214/v267); $H \Leftrightarrow$ square $\Leftrightarrow \mu_4$ . [E] NEG: a $\mathbb{Z}_2$ mark set commutes with $\rho^2$ but not $\rho$ . [O] residual: the dynamical <i>selection</i> of the square (the 4 marks forced by v216) is the irreducible postulate; does NOT close the bedrock [E][O]                                                                                                                                                                           |
| v332_qgamb_metric_sector.py      | The metric sector of QG.AMB.01 characterized: the obstruction is exactly the conformal-mode (Gibbons–Hawking–Perry) problem, the physical sector is gap-insulated, and the only route is the conditional IDG taming (makes v330’s metric-sector fence precise). [E] conformal-mode wrong sign: $c_{\text{conf}}(D) = -(D-1)(D-2)/4$ is negative for $D \geq 3$ ( $c_{\text{conf}}(4) = -3/2$ ), TT positive — the conformal-factor problem. [E] measure obstruction: the kinetic form is indefinite (one negative eigenvalue), so $e^{-S}$ diverges over the conformal mode; the bare metric-sector measure is not directly definable. [E] physical sector insulated: the obstruction is in the conformal/gauge direction, not the admissible sector (positive+gapped, v330), margin $\Delta - 31/(4\pi^2) \approx 1.648 > 0$ (v76). [C] IDG conditional route: the entire form factor $e^{z/M^2}$ (v259/v304) dresses the conformal propagator with no extra pole. [E] NEG: a polynomial $(R+R^2)$ truncation crosses zero (the Stelle ghost, v304/v278). [O] metric sector open: the full QG.AMB.01 needs the conformal mode resolved nonperturbatively (GHP rotation / IDG), gap-decoupled but NOT constructed. A precise characterization, NOT a closure [E][C][O] |
| v333_cm_selection.py             | The CM-selection mechanism: $\tau=i$ (the square) is the UNIQUE order-4 modulus and the attractor of mark-equilibration, so the last keystone residual is “the deck acts geometrically”, not “why the square” (the v331 follow-up). [E] order-4 modular element: $S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ in $\text{SL}(2, \mathbb{Z})$ has $S^2 = -I$ , $S^4 = I$ and fixes $\tau=i$ — the unique order-4 elliptic point of $\text{PSL}(2, \mathbb{Z})$ (the other, $\rho$ , is order 3). [E] deck order $\rightarrow$ CM point: order-4 $\rightarrow$ cross-ratio 2 $\rightarrow j=1728 \rightarrow \tau=i$ (square); order-3 $\rightarrow j=0 \rightarrow \tau=\rho$ (hexagon); $ \mu_4 =4$ selects $\tau=i$ . [E] mark-equilibration attractor: 4 points minimizing the log-energy (Fekete) converge to equal spacing (gaps $\pi/2 =$ the square), and beat random perturbations — the unique (mod rotation) attractor. [E] NEG: an order-3 deck selects $\rho$ , not the square. [O] residual: only “the deck acts <i>geometrically</i> (= QGEO.SYM.01) vs an abstract $\mathbb{Z}_4$ on labels” stays open — sharpened from “why the square” to “the deck acts geometrically”, NOT a closure [E][O]                                                   |

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| Script                       | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| v334_conformal_resolution.py | The conformal-mode resolution: the Gibbons–Hawking–Perry contour rotation (IR sign) + the infinite-derivative (IDG) dressing (UV pole) give a <b>CONDITIONALLY</b> convergent metric-sector measure; the residual is the nonperturbative validity of the contour (the v332 follow-up). <b>[E]</b> the obstruction: $c_{\text{conf}}(4) = -3/2 < 0$ , the unrotated conformal Gaussian diverges. <b>[E]</b> GHP rotation: $\phi \rightarrow i\phi$ flips the sign, $\int e^{- c \phi^2} = \sqrt{\pi/ c }$ <b>FINITE</b> (GHP 1978). <b>[E]</b> IDG dressing: the entire form factor $e^{p^2/M^2} > 0$ (no ghost pole, v304), the dressed per-mode action positive post-GHP and UV-suppressed. <b>[E]</b> NEG: a polynomial truncation crosses zero (the Stelle ghost, v304/v278). <b>[C]</b> conditional construction: GHP+IDG give a convergent metric-sector Gaussian mode-by-mode (conditional on the contour + entire analyticity). <b>[O]</b> residual: whether the GHP-rotated IDG-dressed measure is the <b>CORRECT</b> nonperturbative <b>QG.AMB</b> measure (the contour’s nonperturbative validity); gap-decoupled (v76), blocks no test, <b>NOT</b> a closure <b>[E][C][O]</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| v335_seam_equiv_unify.py     | The keystone unification: there is <b>ONE</b> open theorem ( <b>SEAM.EQUIV.01</b> ), <b>QGE0.SYM.01</b> is its <b>COROLLARY</b> (a conformal-net axiom), and <b>QG.AMB.01</b> is a decoupled general QG problem, <b>NOT</b> a TFPT gate. <b>[E]</b> <b>QGE0.SYM.01</b> = corollary: a chiral conformal net has a Moebius-covariant vacuum (the unique invariant positive-energy vector, a <b>NET AXIOM</b> ); rotations $U(1) < \text{Moebius} \Rightarrow$ rotation-invariant vacuum; the $\mu_4$ clock = rotation by $\pi/2$ fixes it $\Rightarrow \omega \circ \rho = \omega$ . So <b>SEAM.EQUIV.01</b> $\Rightarrow$ <b>QGE0.SYM.01</b> with <b>NO</b> extra premise (the v323 “rotation-invariant vacuum” is a net axiom); the two bedrock items collapse to <b>ONE</b> . <b>[E]</b> the one keystone theorem: <b>SEAM.EQUIV.01</b> = a construction theorem ( $c=8$ , $\det \text{Cartan}(E_8)=1$ , gap $6 \ln \frac{3}{2} > 0$ derived) with <b>EXACTLY</b> one open continuum lemma. <b>[E]</b> <b>QG.AMB.01</b> decoupled: margin $\Delta - 31/(4\pi^2) \approx 1.648 > 0 \Rightarrow$ no TFPT prediction depends on it; the general Euclidean-QG conformal-factor problem (GHP 1978), inherited not TFPT-specific. <b>[E]</b> collapsed status: structural items $2 \rightarrow 1$ ; live residual $= v_{\text{geo}} + G_{\text{net}} (= \text{SEAM.EQUIV.01}) + F_{\text{transfer}}$ . <b>[O]</b> the one residual = <b>SEAM.EQUIV.01</b> ’s continuum OS reconstruction (Lean-pinned). A synthesis/inventory; does <b>NOT</b> close <b>SEAM.EQUIV.01</b> <b>[E][O]</b>                                                                                                                                                                                                                                  |
| v336_continuum_limit.py      | The <b>ONE</b> open lemma of <b>SEAM.EQUIV.01</b> , stated precisely and mapped to the recent rigorous literature – the continuum-limit and OS-reconstruction legs are now <b>CITABLE</b> . <b>[C]</b> continuum leg: Osborne–Stottmeister ( <i>CFT from Lattice Fermions</i> , CMP <b>398</b> (2023) 219, doi:10.1007/s00220-022-04521-8; legacy internal token MMST, on the MMST/Morinelli–Morsella–Stottmeister–Tanimoto OAR framework, arXiv:2010.11121) give the chiral-CFT scaling limit of free lattice fermions (Koo–Saleur $\rightarrow$ Virasoro, correct $c$ ); WZW range rank $\leq c \leq D$ . $(E_8)_1$ : $c=8$ , rank 8, $D=16$ Majoranas ( $SO(16)_1 \rightarrow (E_8)_1$ ), $8 \leq c \leq 16$ in range; the collar is a gapped quasi-free (CAR) state. <b>[C]</b> OS leg: Adamo–Moriwaki–Tanimoto (arXiv:2407.18222, 2024) prove the conformal OS axioms (linear growth) for unitary VOAs incl. lattice VOAs; $(E_8)_1$ is the even self-dual rank-8 lattice VOA. <b>[C]</b> construction + covariance (Adamo–Giorgetti–Tanimoto): arXiv:2506.01008 builds the 2d conformal net of an even lattice and classifies the 2d Heisenberg-net extensions as lattice nets, and arXiv:2508.07109 shows every DHR rep of such a net is automatically positive-energy + diffeomorphism-covariant (covariance a consequence, not an input). <b>[E]</b> target pinned: $c=g_{\text{car}}+N_{\text{fam}}=8$ , $\det \text{Cartan}(E_8)=1$ , the $D=16$ Majorana $SO(16)_1$ IS the carrier-16 and the $\mu_4$ extension IS the seam glue. <b>[O]</b> open residual: $c=8$ alone does <b>NOT</b> pin the net ( $SO(16)_1$ also $c=8$ , $\det 4$ ), so the one open lemma = the holomorphy discriminator $\det K=1$ . A literature-mapping/reduction; does <b>NOT</b> close <b>SEAM.EQUIV.01</b> <b>[C][E][O]</b> |

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| Script                           | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| v337_decoupling_theorem.py       | <p>THE DECOUPLING THEOREM (consolidated, citable): every TFPT dimensionless readout factors through the gapped admissible sector and is INDEPENDENT of the ambient QG measure – the theorem that makes the QG.AMB.01 reclassification (v335) honest. [E] gap margin: <math>\Delta - 2 \dim(E_8) c_3^2 = 6 \ln \frac{3}{2} - 31/(4\pi^2) \approx 1.648 &gt; 0</math>. [E] factorization: every readout (mass ratios, mixings, <math>\alpha^{-1}</math>, <math>R+R^2</math>) is a function of the admissible gapped transfer spectrum <math>\{1, (2/3)^6, (1/3)^6\}</math> + the discrete kernel alone. [E] the theorem: margin <math>&gt; 0 \Rightarrow</math> exponential clustering (<math>\lambda_2=(2/3)^6</math>), finite susceptibility <math>\chi=729/665</math>, so the ambient contribution is gap-suppressed <math>\Rightarrow</math> no readout depends on the ambient measure <math>\Rightarrow</math> QG.AMB.01 decoupled. [E] neg control: gap closed <math>\Rightarrow \chi</math> diverges <math>\Rightarrow</math> fails (non-vacuous). [E] corollary: zero frozen kill tests depend on the ambient measure. Consolidates v76/v311/v330/v332; does NOT construct the ambient measure [E]</p>                                                                                                                                          |
| v338_theta13_budget.py           | <p>THETA13.BUDGET.01: the <math>\theta_{13}</math> sub-leading budget vs the current global fit – the <math>+2\sigma</math> pressure point (v328) quantified against BOTH NuFIT 6.0 NO variants and JUNO; neither defuses nor excludes, fences correctly. [C] SK-dependent tension: <math>\sin^2 \theta_{13} = (\varphi_0^{\text{ret}}) e^{-5/6} = 0.023108</math> is <math>+2.0\sigma</math> vs the no-SK central (0.02195) and <math>+1.7\sigma</math> vs the with-SK NO best fit (0.02215) – not a clean <math>2\sigma</math>. [E] within <math>3\sigma</math>: <math>0.023108 &lt;</math> the with-SK <math>3\sigma</math> upper 0.02388 (not falsified). [C] budget: moving to the central needs <math>-4.1\%</math>; RG <math>&lt; 1\%</math> and the exponent <math>5/6</math> is exact <math>\Rightarrow</math> no freeze-respecting relief (v328). [X] JUNO decides: sub-1% precision turns <math>1.7\text{--}2.0\sigma</math> into <math>\sim 6\sigma</math> – a near-term pre-registered kill/confirm. One honest pressure point (v306/v328), fenced [C][E][X]</p>                                                                                                                                                                                                                                                                         |
| v339_firstprinciples_boundary.py | <p>The honest first-principles boundary, mapped precisely – answers (a) can <math>M_{\text{Planck}}/v_{\text{geo}}</math> be computed? and (b) can <math>F_{\text{transfer}}</math> be made first-principles? Both NO, with the exact reason per residual. [E] <math>v_{\text{geo}}/M_{\text{Planck}}</math> FORBIDDEN BY THEOREM (No-Unit, v153): a mass is dimension 1 while every TFPT datum is dimension 0, so no dimensionless map outputs a scale; over-determined (gravity = dark energy to 0.11%, v274); every dimensionless RATIO is derived, only the unit is not. [E] <math>m_p/m_e</math> structure in place (v262): the carrier <math>b_3=-7</math> drives the QCD run + the closed electron source; two inputs stay external: <math>\alpha_s(M_Z)</math> and the lattice <math>C_p=m_p/\Lambda^{(3)} \sim 2.83</math>. [X] <math>\alpha_s(M_Z)</math> external BECAUSE OF A TENSION: SM content does NOT unify (1-loop spread <math>\sim 8.8</math> at <math>2 \times 10^{16}</math> GeV, v246). [C] <math>C_p</math> parameter-free but lattice-only (v35). [C] <math>\eta_B/\text{axion}</math> = cosmological initial conditions; Koide <math>Q=2/3</math> structural [E]. [E] FOUR honest kinds: forbidden-by-theorem / external-parameter-free / external-tensioned / cosmological. A boundary audit, NOT a solution [E][C][X]</p> |
| v340_As_reconcile.py             | <p><math>A_s</math> is predicted (not missing), is DIMENSIONLESS (so it does NOT define an absolute mass), and the apparent <math>-11\sigma</math> tension is a reheating-channel question – reconciled with the archived old derivation (TFPT v4.5/v4.6). [E] <math>A_s = N_\star^2 (M_{\text{scal}}/\bar{M}_{\text{Pl}})^2 / (24\pi^2) = N_\star^2 c_3^7 / (24\pi^2)</math> (v86); <math>M_{\text{scal}}/\bar{M}_{\text{Pl}} = c_3^{7/2} = 1.2565 \times 10^{-5}</math> is a PURE RATIO, so <math>A_s</math> carries NO absolute scale – it pins <math>N_\star</math>, NOT <math>\bar{M}_{\text{Pl}} \Rightarrow A_s</math> does NOT define mass (No-Unit v153 stands). [C] the <math>-11\sigma</math> is the slow-vs-fast reheating dichotomy (<math>N_\star=51.44 \rightarrow A_s=1.764 \times 10^{-9}</math>; <math>A_s</math>-preferred <math>N_\star=56.2</math>). [C] ARCHIVE: old v4.5/v4.6 used the algebraic <math>N_\star=3/\varphi_0^{\text{ret}} - 1=55.42</math> and reported <math>A_s=2.124 \times 10^{-9}</math>; at that <math>N_\star</math> the current normalisation gives <math>2.047 \times 10^{-9}</math> (<math>-1.9\sigma</math>). [E] nothing lost: the current theory DERIVES <math>N_\star</math> (record band [50, 60], v84). [X] decisive test: fast preheating or slow reheating confirmed [E][C][X]</p>             |

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| Script                       | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v341_alpha_quillen.py        | <p>The <math>\alpha</math> fixed point reformulated as the STATIONARITY of a <math>U(1)</math> determinant line, every ingredient identified as an index / heat-kernel coefficient / discriminant form – NOT a free knob (the “derive the FORM” step).</p> <p><math>F_{U(1)} = \alpha^3 - 2c_3^3\alpha^2 - \frac{4}{5}c_3^6M \ln(1/\varphi_{\text{seam}}) = 0</math> in Quillen form: <math>[\alpha^3 - 2c_3^3\alpha^2]</math> (det-line variation) = <math>[8b_1c_3^6]</math> (anomaly counterterm) <math>\times [\ln(1/\varphi_{\text{seam}})]</math> (seam response). [E] <math>M=41</math> a <math>U(1)</math> INDEX (EM Ward, v48): <math>\frac{4}{5}M = 8b_1</math>, <math>b_1=41/10</math> the SM 1-loop <math>U(1)_Y</math> beta (v159/v246). [E] <math>c_3=1/(8\pi)</math> the GAUSS–BONNET boundary coefficient (<math>8=\text{rank } E_8</math>, v216); <math>c_3^{3,6}</math> heat-kernel powers. [E] <math>-5/4 = -q(D_5)</math> a DISCRIMINANT form (<math>q(D_5)+q(A_3)=2</math>, v1/v154); <math>\varphi_{\text{base}}=1/(6\pi)=\frac{4}{3}c_3</math> carries <math>q(A_3)</math>: <math>c_3/\varphi_{\text{base}}=3/4</math>. [E] <math>\delta_{\text{top}}=\Omega_{\text{adm}}c_3^4</math>. [E] the Quillen split holds at the <math>\alpha^{-1}=137.0359992</math> root. [E] NO NEW GATE: <math>\alpha^{-1}</math> stays [E]; v341 SHARPENS the pre-existing [O] origin EM.WARD.01 (no new <math>\alpha</math> problem). [O] the one still-open part IS EM.WARD.01’s residual: the from-first-principles AQFT PROOF that this is the EXACT Quillen determinant functional (the variational principle derived, not assembled). Sharpens v3/v48/EM.WARD.01 from “fixed point” to “stationarity of a named determinant line” [E][O]</p> |
| v342_em_ward_heatkernel.py   | <p>The heat-kernel sharpening of EM.WARD.01: DERIVES the ORIGIN and STRUCTURE of the <math>F_{U(1)}</math> determinant-line terms from textbook Seeley–DeWitt / Gilkey coefficients, advancing the EM-Ward origin from [C] to [E] for the STRUCTURE; NO new gate. [E] <math>c_3=1/(8\pi)</math> the boundary GAUSS–BONNET coefficient (seam = flat pillowcase <math>\chi=2</math>, <math>\oint K=4\pi</math>, v216). [E] the <math>\alpha^2</math> term is the GILKEY gauge-curvature coefficient: the <math>U(1)</math> Laplacian <math>a_4</math> has <math>30\Omega^2/360 = \frac{1}{12}\Omega^2</math> (Gilkey Thm 4.8.16), <math>\Omega=i\alpha F \Rightarrow -(\alpha^2/12)F^2</math>, so the Calderon <math>-2c_3^3\alpha^2</math> is the gauge-curvature piece, not an ansatz. [E] the <math>c_3</math>-ladder <math>\{0, 3, 6\}</math> is arithmetic (step <math>3 = 3</math> channels; <math>6= R^+(A_3) </math> <math>C_6</math> hexagon). [E] multiplicities forced (<math>2= \mathbb{Z}_2 </math>, <math>8b_1=(\text{rank } E_8)b_1</math>). [E] <math>\pi</math>-power test: <math>2c_3^3=1/(256\pi^3)</math> carries <math>\pi^3 \Rightarrow</math> THREE boundary insertions. [C] the exact coefficient needs the seam <math>F</math>-normalisation. [O] the cubic <math>\alpha^3</math> Maxwell moment + full identification = EM.WARD.01 residual (tied to SEAM.EQUIV.01). <math>\alpha^{-1}</math> stays [E] (v3) [E][C][O]</p>                                                                                                                                                                                                                                                                                                      |
| v343_four_routes_analysis.py | <p>The honest investigation of the four black-hole-birth solution routes (A finite causal diamond, B Carlip–Cardy near-horizon CFT, C modular/thermal flow, D self-reproduction attractor) – direct answer to “which closes ALL problems and is 100% provable”: NONE individually, but they CONVERGE on the same two irreducible residuals as v335 (SEAM.EQUIV.01 + the decoupled QG contour). [E] ROUTE A reframes QG.AMB to a FINITE thermal trace (de Sitter finite, gapped <math>\chi=729/665</math>) but <math>c_{\text{conf}}(4) = -3/2 &lt; 0</math> still needs the GHP contour (v334, [O]); does NOT touch SEAM.EQUIV.01. [C] ROUTE B: Carlip+Cardy consistent with <math>c=8</math> but <math>L_0</math> free (ambiguity) and gives <math>c</math> only, not holomorphy <math>\det K=1</math> – inherits SEAM.EQUIV.01. [E] ROUTE C: modular flow KMS <math>\beta=1</math> (v239), geometric identification = SEAM.EQUIV.01 (v335) – a reduction. [E]/[O] ROUTE D: unique Perron–Frobenius attractor (v56) but cyclic dynamics [O] – not a closure. [E] VERDICT: independent closures = 0; irreducible residuals = 2 (= v335); the BH premise makes both more natural but eliminates neither; focus = SEAM.EQUIV.01 [E][C][O]</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

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| Script                        | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| v344_detk_synthesis.py        | <p>The bird's-eye synthesis of the ONE open keystone bit, <math>\det K=1</math> – a full-spectrum map of SEAM.EQUIV.01's only open residual. [E] <math>\det K=1</math> is ONE statement with SIX equivalent faces (holomorphy / homology-sphere link <math>H_1=0</math> / one 1-dim irrep / <math>\det \text{Cartan} = 1</math> / full <math>\mu_4</math> condensation / <math>\tau=i</math> CM), unified by <math> \det \text{Cartan}  =  H_1(\text{link})  = \#(1\text{-dim irreps})</math> across ADE (<math>A_n \rightarrow n+1</math>, <math>D_n \rightarrow 4</math>, <math>E_6 \rightarrow 3</math>, <math>E_7 \rightarrow 2</math>, <math>E_8 \rightarrow 1</math>) – ONLY <math>E_8</math> gives 1 (<math>2I</math> the unique PERFECT ADE group, v232/v219). [E] THE GENUS-1 OBSTRUCTION: the easy arguments (plane vacuum, gap, free-fermion) are genus-0, necessary but NOT sufficient (v87); <math>\det K</math> is the torus degeneracy = #primaries. [E] THREE genus-1 access routes: R1 continuum-modular (v336), R2 anyon-condensation (v281), R3 hypergraph-homotopy. [C] the HYPERGRAPH route is the most promising fresh angle: the <math>(2, 3, 5)</math> rewrite attractor IS the affine-<math>E_8</math> graph (v312) and the <math>E_8</math> du Val link is the Poincare sphere (<math>H_1=0</math>, v232), so <math>\det K=1</math> becomes a finite COMBINATORIAL statement (no continuum limit). [O] none closes it; R3's combinatorial-to-geometric bridge is meanwhile discharged to Kronheimer 1989 (v479), so the residual transforms to “the raw seam supplies the ALE/orbifold datum” = the SEAM.EQUIV.01 content. A synthesis/roadmap; closes nothing [E][C][O]</p> |
| v345_hypergraph_homotopy.py   | <p>EXECUTE the combinatorial core of route R3 (the hypergraph-homotopy route to <math>\det K=1</math>, v344): computes explicitly that the <math>(2, 3, 5)</math>-rewrite attractor graph's plumbing link is the POINCARÉ homology sphere (<math>H_1=0</math>), only <math>E_8</math> among ADE; R3's combinatorial half is now [E]-complete, the geometric bridge (= SEAM.EQUIV.01) stays [O]. Plumbing topology (Brieskorn): link <math>H_1 = \text{coker}(\text{Cartan})</math>. [E] <math>\text{SNF}(E_8 \text{ Cartan}) = \text{identity} \Rightarrow \text{coker}(E_8)=0 \Rightarrow</math> the <math>E_8</math> plumbing boundary is an INTEGRAL HOMOLOGY SPHERE (the genus-1 meaning of <math>\det E_8=1</math>). [E] only <math>E_8</math>: <math>\text{coker}(A_n)=\mathbb{Z}/(n+1)</math>, <math>D_n \rightarrow \mathbb{Z}/4</math>, <math>E_6 \rightarrow \mathbb{Z}/3</math>, <math>E_7 \rightarrow \mathbb{Z}/2</math> – all nontrivial, only <math>E_8=0</math>. [E] <math>\pi_1=2I=SL(2, 5)</math> PERFECT (commutator subgroup order <math>120= G </math>, computed) <math>\Rightarrow H_1=0 \Rightarrow</math> THE Poincaré sphere; <math> 2I =120= \mu_4  h(E_8)</math>. [O] does NOT close <math>\det K=1</math>: the graph→geometry half is meanwhile Kronheimer-cited (v479); what stays open is the raw seam supplying the ALE/orbifold datum = SEAM.EQUIV.01 [E][O]</p>                                                                                                                                                                                                                                                                                                       |
| v346_seam_geometric_bridge.py | <p>The geometric bridge as an explicit six-link chain raw-seam <math>\Rightarrow \det K=1</math>, with the ONE open arrow pinned – the honest capstone of the SEAM.EQUIV.01 investigation. L1 [E]/[C] raw seam <math>\rightarrow \mu_4</math> + four marks (v216); L2 [O] <math>\mu_4</math> + <math>(2, 3, 5)</math> atoms <math>\rightarrow 2I</math> orbifold <math>\mathbb{C}^2/\Gamma</math> (THE OPEN ARROW = SEAM.EQUIV.01); L3 [E] <math>2I \rightarrow</math> affine <math>E_8</math> McKay (v219); L4 [E] <math>2I \rightarrow E_8</math> du Val (v232); L5 [E] <math>\mathbb{C}^2/2I \rightarrow</math> Poincaré link <math>H_1=0</math> (v345); L6 [E] <math>H_1=0 \rightarrow \det K=1</math>. [E] 5 of 6 links are theorems, only L2 open. [E] the group is FORCED by the seam atoms: axis signatures <math>(2, 3, 3)/(2, 3, 4)/(2, 3, 5) \rightarrow 2T/2O/2I</math>, <math>(2, 3, 5)</math> unique to <math>2I</math>, and <math>( \mathbb{Z}_2 , N_{\text{fam}}, g_{\text{car}})=(2, 3, 5)</math> ARE those axes. [E] downstream of L2 forced. [O] L2 (the seam IS an ADE orbifold point) = SEAM.EQUIV.01, NOT closed – the keystone reduced to one geometric-realisation arrow, group pinned by <math>(2, 3, 5)</math> [E][C][O]</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| v347_seam_closure_modes.py    | <p>The closure-mode classification of the one open arrow L2 (the seam-orbifold realisation, v346) – a direct honest answer to “is there ANY other way to FULLY solve it?”. [E] THE PRECISE LOCUS: the seam gives the pillowcase base <math>S^2(2, 2, 2, 2)</math>, <math>\chi_{\text{orb}}=0</math> (EUCLIDEAN, v216), but the <math>2I/E_8</math> structure is the Seifert base <math>S^2(2, 3, 5)</math>, <math>\chi_{\text{orb}}=1/30</math> (SPHERICAL); L2 must bridge a FLAT 4-marked base to a SPHERICAL 3-marked base – why L2 needs the carrier <math>(2, 3, 5)</math> input. [E] FOUR CLOSURE MODES: A construct (the analytic wall, v336); B rigidity (unique realisation <math>\Rightarrow \mathbb{C}^2/2I</math> forced, the only theorem-route w/o new input, v284/v300/v331); C axiom <math>P_3</math> (<math>\Rightarrow</math> TFPT complete rel. 3 axioms); D empirical. [E] no-new-state (v246) blocks deriving L2 from extra physics <math>\Rightarrow</math> only rigidity (B) closes it as a theorem. [E] VERDICT: no free theorem; only B (theorem) or C (axiom), validated by D; only these four modes. [O] NOT closed [E][O]</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

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| Script                                  | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| v348_seam_rigidity_route.py             | Route B (the rigidity route) for the one open arrow L2, carried out – it lands on a strikingly SIMPLE statement: the whole keystone reduces to ONE question, does the raw seam carry the golden ratio? [E] rigidity closes the crystallographic part: the order-4 clock has a UNIQUE normal form $z \mapsto iz$ (Nielsen/Kerckhoff, v180), Troyanov gives the unique flat pillowcase metric (v284). [E] downstream is a LATTICE identity: $E_8$ = the ring of icosians (Conway–Sloane), rank 8, so $2I \rightarrow E_8$ is a lattice identity (stronger than McKay v219). [E] the single bridge is $\varphi$ : crystallographic orders $\{1, 2, 3, 4, 6\}$ , the 5-fold non-crystallographic; $\mu_4$ crystallographic, $2I$ (5-fold) not, $\varphi = 2\cos(\pi/5)$ extends one to the other (quasicrystal). [O] the keystone reduces to ONE question: “does the raw seam carry $\varphi$ ?” – if yes, $\mu_4 + \varphi \rightarrow 2I \rightarrow E_8$ closes L2. [O] honest subtlety: TFPT’s $\varphi$ is currently $E_8$ -side (v312/v313); the raw-seam data (cusp $\{0, 1/3, 2/3\}$ , recovery $(2/3)^6$ , seed $\varphi_0$ ) are not manifestly golden, so showing the RAW seam carries $\varphi$ non-circularly is the residual (= L2). NOT closed [E][O] |
| v349_raw_seam_golden_test.py            | The honest test of the v348 question “does the RAW seam carry the golden ratio (before assuming $E_8$ )?”. Answer, computed: NO – the bare carrier ( $D_5$ , $A_3$ ) and the dynamical seam data are NOT golden; the golden ratio genuinely enters WITH the icosahedral/ $E_8$ identification. A NEGATIVE result that prevents a false closure. [E] RAW CARRIER NOT GOLDEN: golden needs $5 \mid h$ ; $D_5$ has $h=8$ ( $2\cos(2\pi/8)=\sqrt{2}$ ), $A_3$ has $h=4$ – no 5-fold, bare carrier gives $\sqrt{2}$ . [E] DYNAMICAL DATA NOT GOLDEN: cusp $\{0, 1/3, 2/3\}$ rational, recovery $(2/3)^6$ rational, seed $\varphi_0$ transcendental. [E] GOLDEN IS THE ICOSAHEDRAL/ $E_8$ SIDE: $5 \mid h$ only for $A_4=SU(5)$ ( $h=5$ ), $H_3$ ( $h=10$ ), $E_8$ ( $h=30$ ). [E] $\varphi$ alone insufficient: $2I \neq C_5 \times \mu_4$ ( $20 \neq 120$ ). [O] RESOLUTION: L2 reduces to “is $g_{\text{car}}=5$ a PENTAGON (5-fold $\Rightarrow$ golden $\Rightarrow$ icosahedral) or a COUNT (rank $D_5 \Rightarrow \sqrt{2}$ )?” – the golden ratio is the pentagon-reading of P2, NOT derivable from P2-as-a-count; no hidden $\varphi$ . Foundational: P2-as-pentagon closes it (mode C) else rigidity must produce the 5-fold (mode B); NOT closed [E][O]     |
| v350_bootstrap_inputs_correction.py     | The honest CORRECTION of v349’s framing (prompted by the right objection: the inputs are not axioms, they are back-determined by the theory). [E] THE INPUTS ARE BOOTSTRAP-FORCED, NOT FREE AXIOMS (v6): $g_{\text{car}}=5$ over-determined three ways (rank-fill, Coxeter-match $g_{\text{car}} = \max$ prime of $h(E_8)=30$ , Pascal). [E] $g_{\text{car}}=5$ IS THE ICOSAHEDRAL 5-FOLD, not the $D_5$ rank: $g_{\text{car}} = \max$ prime of $h(E_8)=30=2\cdot 3\cdot 5$ ; the atoms (2, 3, 5) ARE the primes of the icosahedral Coxeter number 30. [E] THE GOLDEN RATIO IS EMERGENT FROM THE GLUE, not external: $D_5$ ( $h=8$ )/ $A_3$ ( $h=4$ ) non-golden alone, but the $\mu_4$ extension $D_5 \oplus A_3 \rightarrow E_8$ (v154) gives $h(E_8)=30$ (golden, $5 \mid 30$ ) – the seam’s own $\mu_4$ produces the 5-fold; v349 checked the un-glued pieces, the wrong object. [E] v349’s pentagon-vs-count dichotomy was FALSE (the bootstrap forces the pentagon). [O] residual relocates: the golden is bootstrap-forced; the only open thing is the physical continuum realisation (v336), NOT the golden ratio. Caveat: self-consistency within the framework, not from-nothing. Supersedes v349’s interpretation [E][O]                              |
| v351_continuum_realisation_sharpened.py | The continuum realisation sharpened – the “ $c=8$ ambiguity” flagged open in v277/v344 ( $E_8$ vs $SO(16)$ both $c=8$ ) is RESOLVED, non-circularly, by the seam’s ORDER-4 $\mu_4$ clock. [E] $E_8$ (det 1, holomorphic) vs $D_8=SO(16)$ (det 4). [E] the order-4 clock (the four Gauss–Bonnet marks, $ \mu_4 =4=h(A_3)$ , v216): index-4 extension $D_5 \oplus A_3 = E_8$ (v154), index-2 = $SO(16)$ ; so order-4 selects $E_8$ ( $16 \rightarrow 4 \rightarrow 1$ ) not $SO(16)$ ( $16 \rightarrow 4$ ) – the order (4 not 2) is the discriminator. [E] bootstrap agrees: $h(E_8)=30$ (max prime 5) vs $h(D_8)=14$ (max prime 7). [E] which-net ambiguity RESOLVED (det $K=1$ is then a consequence). [O] residual = PURELY the chiral-edge/scaling-limit existence (bulk-edge, v336), net+holomorphy pinned. Does NOT close SEAM.EQUIV.01 [E][O]                                                                                                                                                                                                                                                                                                                                                                                                              |

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| Script                        | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| v352_framework_irreducible.py | The framework reduction – the deepest answer to “the inputs are not axioms, they are back-determined”. [E] P2 ( $g_{\text{car}}=5$ ) FORCED three ways (v6): rank-fill, Coxeter-match ( $g_{\text{car}} = \max \text{ prime of } h(E_8)=30$ ), Pascal. [E] P1 ( $c_3=1/(8\pi)$ ) FORCED: $c_3$ is the boundary Gauss–Bonnet coefficient (v216/v342), the 8 over-determined four ways ( $8 = \text{rank } E_8 = h(D_5) = \phi(30) = \det R$ , v54/v6); only $\pi$ is non-forced (the boundary transcendental). [E] the $E_8$ hull is forced (unique even unimodular rank-8, v83/v154). [E] so the ONLY irreducibles are the FRAMEWORK (a discrete reflection-positive boundary with a finite Clifford carrier) + $\pi$ – the precise content of “the inputs are not axioms”. [O] the framework is the META-AXIOM (not math-reducible, the foundational stance, tested by its consequences not proved). TFPT’s true input is ONE framework + $\pi$ [E][O]                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| v353_selfloop_capstone.py     | The bird’s-eye rethink (refining v352): TFPT is not a LINEAR theory (axioms $\rightarrow$ theorems) but a CLOSED SELF-CONSISTENT LOOP whose outputs fix its own inputs, with ZERO free adjustable dimensionless parameters. [E] IT IS A LOOP: outputs fix inputs ( $g_{\text{car}}=5 = \max \text{ prime of } h(E_8)=30$ ; the 8 in $c_3 = \text{rank } E_8$ ) and inputs fix outputs; the arrows close into a circle (v6) with a unique fixed point (v56). [E] ZERO FREE DIALS: every load-bearing number is forced/over-determined; free dimensionless parameters = 0. [E] $\pi$ is a FIXED geometric constant (the boundary 2-sphere, $\oint K=4\pi \Rightarrow c_3=1/(8\pi)$ ), not a dial. [C] refining v352: $\pi$ is not a free input and the framework is a SELECTION PRINCIPLE (the unique self-consistent loop), not a free meta-axiom. [O] the one residual is the CONTINUUM closure (the chiral-edge existence, v336/v351); structural-realism (loop complete) vs constructivism (continuum open) is a philosophical, not numerical, fork [E][C][O]                                                                                                                                                                                                                                                                                                                                              |
| v354_e8_reverse_audit.py      | The REVERSE numerology audit + what the golden ratio means. The forward rule (v305) is one-directional (every readout must map onto $E_8$ ); the reverse question: how much $E_8$ structure maps onto NOTHING? [E] $E_8$ has 8 Casimir invariant degrees $\{2, 8, 12, 14, 18, 20, 24, 30\}$ ; THREE feed a readout (2 metric, 8 rank $\rightarrow c_3$ , 30 Coxeter $\rightarrow g_{\text{car}}$ ) and FIVE are UNMAPPED $\{12, 14, 18, 20, 24\} - 3/8$ load-bearing, $5/8$ not. [E] ECONOMY, NOT PROMISCUITY: a small fixed set (rank 8, $h=30$ , $\det 1$ , $D_5 \oplus A_3$ , 16-spinor) generates ALL readouts; the higher Casimirs are never reached into to fit – the anti-numerology signature (economy + a large untouched remainder). [E] the golden ratio is UNMAPPED (the affine- $E_8$ attractor eigenvalue v312, the icosian lattice v348; no frozen readout is $\varphi$ , the sole [C] exception $\cos(\theta_i=3\pi/5)=-1/(2\varphi)$ on the axion spine branch, refined in v429) $\Rightarrow$ NOT numerology (which needs a number fitted to an observation); $\varphi$ is the algebraic signature of the 5-fold $h=30=2\cdot 3\cdot 5$ , the carrier announcing it is icosahedral. [O] the $5/8$ unmapped is the HULL OVERHEAD ( $E_8 = \text{minimal unimodular hull, not the world group}$ ); where future readouts could live, or over-capacity. The reverse complement of v305 [E][O] |
| v355_e8_unmapped_bandwidth.py | A disciplined bandwidth search of the unmapped $E_8$ region (next step after v354), run with a STRICT discriminator: a candidate counts only if it is a FORCED identity (a theorem about $E_8$ ), never a numerical coincidence – because mining a rich object for matches is the numerology temptation. [E] FORCED SET-LEVEL (the real collective content): $\sum \text{deg} = 128 = \dim S^+$ the spinor half of $248=120+128$ ; $\sum \text{exp} = 120 = \# \text{pos roots}$ ; $\prod \text{deg} =  W(E_8) $ ; exponents = the $\varphi(30)=8$ totatives of 30 (v223); $\max \text{deg} = 30 = h(E_8)$ . [C] SUGGESTIVE not forced: $12=h(E_6)$ , $18=h(E_7)$ , $30=h(E_8)$ (the McKay ladder v219), but ‘deg( $E_8$ ) contains subalgebra Coxeter numbers’ is not a theorem – flagged. [O] DECLINED: the per-degree atom matches ( $14=\dim G_2$ , $20=\det L$ , $24= W(A_3) $ , $12= R(A_3) $ , $18=p_4(a)$ ; v66 flagged 12, 18 fittable) each admit multiple readings, and the $ W $ prime 7=‘scalaron exponent’ is one of many – declined. [E] RECONCILES v66 (degrees=atoms) and v354 (only 2, 8, 30 primary): NO new forced physical hit; the disciplined decline IS the anti-numerology result [E][C][O]                                                                                                                                                                                         |

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| Script                                       | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| v356_<br>continuum_mmst_<br>applicability.py | Direction A: the continuum chain of SEAM.EQUIV.01 assembled theorem-by-theorem, residual reduced to ONE lattice-realization input tied to the seam's one-sidedness. Chain: gapped (v302) quasi-free 16-Majorana collar (v155/v160) $\rightarrow$ invertible (Kitaev, v301) $\rightarrow$ the NONTRIVIAL invertible phase ( $c_-=8$ ) $\rightarrow$ a non-gappable anomalous chiral edge (anomaly inflow: edge EXISTENCE follows, reducing the v336/v351 'does the edge exist?' residual) $\rightarrow$ chiral-CFT scaling limit (MMST v336, $8 \leq 8 \leq 16$ ) $\rightarrow$ order-4 $\mu_4$ clock + AQFT stack pin $(E_8)_1$ (v351/v297). The only non-theorem link is the seam one-sidedness = the $c_3$ origin ( $c_- \neq 0$ = one-sided = the $ \mathbb{Z}_2  \nmid K=8$ of $c_3=1/(8\pi)$ , v58). [C]/[E]/[O]; ADVANCES SEAM.EQUIV.01 to one realization input, does not close. (Direction B was found already closed by v40.) [C][E][O]                                                                                                                                      |
| v358_grav_entropy_<br>equilibrium.py         | The entanglement-equilibrium derivation (Jacobson 2015 / Faulkner et al., $\delta S = \delta \langle K \rangle$ ) with TFPT's atoms $\Rightarrow$ the LINEARISED Einstein equation falls out PARAMETER-FREE. [E] $G_{ab} = (1/c_3)T_{ab}$ , $1/c_3 = 8\pi$ fixed (Unruh $1/(2\pi)=4c_3$ , EH $1/(16\pi)=c_3/2$ , Einstein $8\pi=1/c_3$ , Bekenstein $1/4=1/ \mu_4 $ ). [E] THERMO=GEO coincidence (new): the first law's $2\pi/\eta$ = the geometric $ \mathbb{Z}_2 2\pi\chi$ iff $ \mu_4 = \mathbb{Z}_2 \chi=4$ – the thermodynamic and geometric origins of $c_3$ are ONE. [C] J3 solved: the CHM ball modular Hamiltonian (boost via BW v323) gives the matter flux $\delta \langle K_B \rangle = \frac{8\pi^2 R^4}{15} \delta \langle T_{00} \rangle$ . [E] entropy density atom-fixed: $1/4=1/ \mu_4 $ + central charge $c=g_{\text{car}}+N_{\text{fam}}=8$ (form v59). [O] residual: the equation-of-state status + the global ambient measure (C7) + the $v_{\text{geo}}$ unit (v359 closes the FULL covariant equation); no free dimensionless gravity dial remains [E][C][O] |
| v359_grav_<br>nonlinear_<br>einstein.py      | The FULL covariant Einstein equation, parameter-free (Direction 1, extends v358). The Jacobson-2015 fixed-VOLUME entanglement equilibrium brings in the EINSTEIN TENSOR $G_{ab}$ (not just $R_{ab}$ ) – the unique divergence-free metric tensor (Lovelock) – so equilibrium yields the full covariant $G_{ab} + \Lambda g_{ab} = c_3^{-1}T_{ab}$ with conservation built in. [C][E] fixed-volume $\rightarrow$ Einstein tensor (trace = $-R$ in $d=4$ ). [E] Lovelock $\Rightarrow \nabla^a T_{ab}=0$ is an output. [E] coeff 1: $8\pi=1/c_3$ fixed (v358). [E] coeff 2: $\Lambda$ from $\alpha - \rho_\Lambda/\bar{M}_{\text{Pl}}^4 = (3/(4\pi^2))e^{-2\alpha^{-1}}$ , prefactor $(8\pi)^2 \cdot 48c_3^4 = 3/(4\pi^2)$ (v60); BOTH coefficients TFPT-fixed. RESULT: the original B6 full covariant field equation is now parameter-free at the LOCAL level. [O] residual: the equation-of-state status + the global ambient measure (C7) [E][C][O]                                                                                                                                  |
| v360_grav_gap_<br>corrections.py             | Direction 2: the gap-induced higher-curvature correction to GR – the disciplined result. Does the seam transfer gap give a NEW forced near-term GR-deviation beyond the known $R^2$ scalaron? With the v354/v355 discriminator: NO. [E] the leading higher-curvature term IS the $R^2$ scalaron ( $a_4=R^2/72$ v36; $M_{\text{scal}}^2/M_{\text{Pl}}^2 = c_3^7$ , $\exp 7=g+N-1$ ) – already in TFPT, already falsifiable via $n_s=1-2/N_*$ , $r=12/N_*^2$ (CMB-S4). [C] it is the entanglement-equilibrium next order (Wald first law $\rightarrow f(R)=R+R^2$ , v28/v359). [E] the gap $\Delta=6\ln(3/2)$ is dimensionless, so a distinct gap term sits at $\xi v_{\text{geo}} \sim$ seam/Planck scale, not near-term observable. [O] DECLINE: a new dimensionless coefficient is not atom-forced (needs heat-kernel/Wald data; $\exp 7$ vs $6$ , $c_3^7$ vs $(2/3)^6$ unrelated) $\rightarrow$ per v354/v355 declined. No new near-term forced prediction; the answer is the scalaron [E][C][O]                                                                                    |
| v361_grav_<br>backreaction.py                | Direction 5: the matter-gravity backreaction with the explicit J3 flux – two forced facts + an honest decline. [E] the carrier's gravitational anomaly is FORCED: $c_-=8$ (16 Majoranas = $\dim S^+ = 2^{g_{\text{car}}-1}$ ; chiral central charge $16/2=8=g_{\text{car}}+N_{\text{fam}}$ , the same $c_-$ as v358). [E] the backreaction is FINITE $\rightarrow \Lambda$ from $\alpha$ : the matter vacuum energy through $G_{ab}=c_3^{-1}T_{ab}$ is gap-suppressed (Decoupling Thm v337) to $\rho_\Lambda/\bar{M}_{\text{Pl}}^4 = (3/(4\pi^2))e^{-2\alpha^{-1}}$ (v60), 123 orders = $119.03+3.92$ – NOT the $M_{\text{Pl}}^4$ catastrophe; the loop (v359+v60+carrier vacuum) closes. [E] J3 explicit (v358): $\delta \langle K_B \rangle = (8\pi^2 R^4/15)\delta \langle T_{00} \rangle$ , carrier $T_{ab}$ =the SM's (v159). [C] SM trace anomaly reproduced (content=SM). [O] DECLINE: no new atom-forced read-out of individual SM quantum numbers beyond $c_-=8$ and $\Lambda \rightarrow$ per v354/v355 declined. Finite & consistent backreaction, no new dial [E][C][O]   |

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| Script                    | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| v364_vgeo_sharpen.py      | Direction 6: $v_{\text{geo}}$ is the SINGLE dimensionful input of the WHOLE theory – confirmed across matter AND gravity – and irreducibly primitive. The No-Unit Theorem (v153) showed the dimensionless compiler cannot output a scale ( $v_{\text{geo}}$ a forced unit). SHARPENING after the parameter-free gravity (v358/v359/v361): the gravity sector adds NO new scale either $-1/c_3=8\pi$ and the $\Lambda$ prefactor $(8\pi)^2 \cdot 48c_3^4 = 3/(4\pi^2)$ are dimensionless, so $1/G \sim v_{\text{geo}}^2$ and $\Lambda \sim (\text{dimensionless})v_{\text{geo}}^4$ reduce to $v_{\text{geo}}$ times atoms. [E] FINAL TALLY: every dimensionful quantity = (dimensionless ratio) $\times$ (power of $v_{\text{geo}}$ ); free content = $\{v_{\text{geo}}, \pi\}$ , ZERO dimensionless dials, vs the SM's $\sim 26$ . [O] $v_{\text{geo}}$ irreducibly primitive (No-Unit; $1/G$ UV-sensitive Sakharov v68). Clarity, not a new derivation; the last dimensionful anchor named [E][O]                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| v365_qg_oneloop_saddle.py | Direction 3: the QG measure (gate C7/QG.AMB.01) as one-loop GAUSSIAN fluctuations around the PARAMETER-FREE saddle (v358/v359). [E] the fluctuation stiffness is $1/c_3 = 8\pi$ (no free dial), so the one-loop problem is well-posed (it was NOT before v358). [E] the fluctuation operator is gapped $M \geq \Delta_{\text{eff}} = 6 \log \frac{3}{2} - \frac{31}{4\pi^2} \approx 1.648 > 0$ (v76/v337) $\Rightarrow$ mode-by-mode Gaussian convergence. [E] finite model log-det $\text{Tr}' \log M = 6 \log \frac{9}{2}$ on the OS spectrum $\{0, 6 \log \frac{3}{2}, 6 \log 3\}$ . [C] the conformal mode converges on the GHP contour $(\int e^{- c \phi^2} = \sqrt{2\pi/3}, \text{v334})$ . [E] the admissible projective limit exists $(\chi = 1/(1 - (2/3)^6) = 729/665, \text{v330})$ . [E] 0 new dimensionless dials (v364). [O] residual: the full non-perturbative projective limit (G6/QG.AMB.01) on the ambient sector. Sharpens C7 to a one-loop Gaussian problem around a parameter-free saddle; does NOT close it. [E][C][O]                                                                                                                                                                                                                                                                                                                                                                                           |
| v366_mmst_seam_collar.py  | Direction 7: the seam collar is IN MMST's free-lattice-fermion scaling-limit class, so the chiral scaling limit is a chiral CFT of central charge 8 pinned to $(E_8)_1$ – reducing SEAM.EQUIV.01 to the single S3 lattice-realisation input. Osborne–Stottmeister (arXiv:2107.13834, CMP <b>398</b> (2023) 219; legacy internal token MMST, on the MMST/Morinelli–Morsella–Stottmeister–Tanimoto OAR framework): free lattice fermions (CAR quasi-free) $\rightarrow$ chiral-CFT massless scaling limit with correct $c$ , range $\text{rank} \leq c \leq D$ . [E] class arithmetic: $D = \dim S^+ = 2^{g_{\text{car}}-1} = 16$ Majoranas, $c = g_{\text{car}} + N_{\text{fam}} = 8$ , $\text{rank } E_8 = 8 \Rightarrow \text{rank} \leq c \leq D$ is $8 \leq 8 \leq 16$ (in MMST range). [E] gapped ( $\Delta = 6 \log \frac{3}{2} > 0$ , derived v302). [C] quasi-free CAR class (v155/v160). [E] chirality $c_- = D/2 = 8 \neq 0$ (non-gappable anomalous edge, v356 S4). [C] the MMST scaling-limit theorem applies (in-class & gapped & chiral & in-range) $\Rightarrow$ chiral CFT $c=8$ (v336). [E] target pinned to $(E_8)_1$ by $c=8$ + the order-4 $\mu_4$ clock ( $\det K=1$ vs $SO(16) \det K=4$ , v351). [O] residual S3: the collar realised as a genuine lattice chiral free-fermion invertible phase (one-sidedness/Flat-Away, v297/v356). Verifies MMST in-class; ADVANCES SEAM.EQUIV.01, does NOT close it. [E][C][O] |
| v367_seam_s3_lattice.py   | Track 1: the EXPLICIT lattice realisation of the S3 input – a concrete gapped chiral-Majorana ( $p + ip$ ) lattice model whose edge is the 16-Majorana $SO(16)_1$ carrier, $\mu_4$ -condensed to $(E_8)_1$ ; turns the abstract S3 residual (v356/v366) into a RUNNABLE model with a numerically computed topological invariant. [E] bulk gapped (band gap $2 > 0$ at $M=1$ ). [E] non-trivial Chern number (Fukui–Hatsugai–Suzuki: $ C =1$ at $M=1$ , $C=0$ at $M=3$ trivial control). [E] $c_- = 8$ : each chiral Majorana edge carries $c_- = C/2 = 1/2$ , $N_{\text{Maj}} = \dim S^+ = 2^{g_{\text{car}}-1} = 16$ copies $\Rightarrow c_- = 16/2 = 8 = g_{\text{car}} + N_{\text{fam}}$ . [E] edge K-matrix $SO(16)_1$ ( $\det \text{Cartan } D_8=4$ ) $\rightarrow (E_8)_1$ ( $\det \text{Cartan } E_8=1$ ) by the order-4 $\mu_4$ clock (v351/v369). [C] the $\mu_4$ condensation $SO(16)_1 \rightarrow (E_8)_1$ (v154/v351). [O] residual: the continuum scaling-limit existence (MMST v336). Reduces S3 to a runnable lattice model; does NOT close SEAM.EQUIV.01. [E][C][O]                                                                                                                                                                                                                                                                                                                                                     |

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| Script                       | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| v368_seam_s3_inflow.py       | Track 1: the bulk-edge / anomaly-inflow leg (S4 of the v356 chain) made CONCRETE on the v367 lattice model – a finite STRIP of the gapped $p + ip$ model has a realised, non-gappable chiral edge, so edge EXISTENCE follows from the invertible bulk (anomaly inflow), NOT a separate assumption. [E] topo in-gap edge states ( $M=1$ strip $\min_{k_x}  E  \sim 0$ ) vs trivial clean gap ( $M=3$ , neg control); [E] the near-zero state is edge-localised (weight within 4 sites of a boundary, edge $\gg$ bulk); [E] bulk-edge correspondence (edge branch count = bulk Chern $ C =1$ ) $\Rightarrow$ 16-Majorana edge, $c_- = 8$ . [C] anomaly inflow: a $c_- \neq 0$ invertible bulk cannot have a gappable edge $\Rightarrow$ discharges the v336/v351 ‘does the edge exist?’ residual on the lattice. [O] residual: the continuum scaling limit (MMST v336). Establishes edge EXISTENCE on the lattice; does NOT close SEAM.EQUIV.01. [E][C][O]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| v369_qgamb_redundancy.py     | Track 2 (next-plan v2): the AMBIENT REDUNDANCY statement – the holographic route to Gravity-complete that SIDESTEPS building the general Euclidean QG measure (C7/QG.AMB.01). Instead of constructing the non-perturbative ambient measure (the conformal-factor swamp, v332/v334), assemble the established discriminators proving the ambient sector carries NO physical d.o.f. relative to the holomorphic boundary algebra, so QG.AMB.01 is a non-fundamental CERTIFICATION object (like a coordinate choice in GR), not missing dynamics. [E] HOLOMORPHIC $\Rightarrow$ DHR = Vec: #DHR primaries = $ \det \text{Cartan}  = 1$ for $E_8$ (ONE primary = vacuum) vs 4 for $SO(16)/D_8$ – no nontrivial superselection charges $\Rightarrow$ no hidden boundary sector a bulk could carry invisibly. [E] NO TORUS GSD: ground-state degeneracy on $T^2 = \# \text{anyons} =  \det K  = 1$ for $E_8 \Rightarrow$ unique ground state $\Rightarrow$ no topologically protected invisible bulk d.o.f. (the genus-1 obstruction of v344). [E] FINITE PETZ RECOVERY: the deviation subspace contracts at $(2/3)^6 = 64/729 < 1$ (v221) $\Rightarrow$ $\text{Rec}_\Sigma$ bounded. [E] GAP-DECOUPLING MARGIN: $\Delta_{\text{eff}} = 6 \log \frac{3}{2} - \frac{31}{4\pi^2} \approx 1.648 > 0$ (v76/v337) $\Rightarrow$ the ambient sector is energetically separated. [C] RECONSTRUCTION MAP: Bisognano–Wichmann gives the modular boost (v258/v309/v323) $\Rightarrow \text{Rec}_\Sigma : (\text{bulk}) \rightarrow \mathcal{A}_\Sigma^{\mu_4}$ . [C] THE REDUNDANCY STATEMENT: under SEAM.EQUIV.01 + BW, $\mathcal{O}_{\text{phys}}^{\text{bulk}} / \text{Diff} \cong \mathcal{A}_\Sigma^{\mu_4} \Rightarrow$ QG.AMB.01 carries no extra physical d.o.f. [O] residual: the two consumed premises – SEAM.EQUIV.01 (the S3 lattice input, v366) + the BW intrinsicity of the raw collar (v329); provides the ALTERNATIVE (holographic) route, does NOT construct the measure. Gravity-complete = Boundary-complete + Ambient-redundancy [E][C][O] |
| v370_grav_spin2_unitarity.py | Track 2: the Barnes–Rivers spin decomposition of the graviton propagator – perturbative graviton unitarity SECTOR-BY-SECTOR (extends v304’s scalar pole algebra to the tensor spin structure + combines with v334’s spin-0 conformal mode). [E] projector completeness: $\text{tr } P_2 = (d-2)(d+1)/2$ , $\text{tr } P_1 = d-1$ , $\text{tr } P_{0s} = \text{tr } P_{0w} = 1$ sum to $d(d+1)/2$ ; the $d=4$ split is $5+3+1+1=10$ . [E] EH structure: spin-2 coefficient +1, spin-0 scalar $-1/(d-2) = -1/2$ . [E] the Stelle ghost is a SPIN-2 state (a local $1+p^2/M^2$ dressing puts the negative-residue pole at $p^2 = -M^2$ ). [E] the entire KMS form factor $e^{p^2/M^2}$ (v304/v259) makes spin-2 ghost-free (only the $p^2=0$ pole). [C] the wrong-sign spin-0 conformal mode ( $c_{\text{conf}}(4) = -3/2$ , v332) is cured by the GHP contour (v334) – a different mechanism. [C] $\Rightarrow$ perturbative graviton unitarity sector-by-sector, conditional on v304’s entire-analyticity assumption. [O] perturbative-only + the open analyticity assumption; NOT the non-perturbative measure (QG.AMB.01). [O] decline: $\text{tr } P_2 = 5$ is dimension counting, NOT $g_{\text{car}}$ (v354/v355). [E][C][O]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

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| Script                        | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| v371_ftransfer_pole.py        | Track 3 ( $F_{\text{transfer}}$ ): the Koide source→pole transfer as a typed solver – promotes the <b>experiments/ftransfer</b> solve into the ledger with a kill test (does NOT re-prove 53/54 v183 or $Q^*=2/3$ v82/v101). [E] $Q_{\text{pole}}=\text{Koide(PDG poles)}=2/3$ to $3\times 10^{-6}$ , $Q_{\text{src}}(\varphi_0^{\text{ret}}\text{-ladder})=0.66446$ (0.33% below). [E] QED runs the wrong way: $dQ/d\epsilon > 0$ pushes $Q$ ABOVE 2/3; the $\epsilon$ mapping $Q_{\text{pole}} \rightarrow Q_{\text{src}}$ is NEGATIVE while QED gives $\epsilon=+(3/2)(\alpha/\pi) > 0$ – wrong sign for any $\alpha \in [1/137, 1/128]$ . [C] verdict: $F_{\text{pole}}$ is NOT standard QED running; the transfer is the non-QED 53/54 operator / $(2/3)^6$ Moebius generator (v183/v82). Stays [C], plain running excluded as the kill. [E][C]                                                                                                                                                                            |
| v372_ftransfer_boltzmann.py   | Track 3 ( $F_{\text{transfer}}$ ): $\eta_B$ from the INTEGRATED Buchmueller–Di Bari–Pluemacher Boltzmann network at the TFPT-frozen heavy scale – promotes the <b>experiments/ftransfer</b> solve, replacing the v326 toy washout. Frozen $M_1=M_{\text{scal}}(\varphi_0^{\text{ret}})^2/A_\Lambda$ ( $\sim 8.65\times 10^9$ GeV, v212), $\tilde{m}_1=m_3/A_\Lambda=5$ meV, $\delta_{CP}=4\pi/3$ . [E] integrated efficiency $\kappa_f \sim 0.09$ (strong washout $K \sim 4.6$ ) agrees with the BDP fit to $\sim 25\%$ . [C] $\eta_B(\text{ODE}) \sim 6.5\times 10^{-10}$ at the frozen $M_1$ lands within a factor $\sim 1.1$ of the observed $6.1\times 10^{-10}$ , NO $M_R$ dial. [X] kill: $\eta_B$ outside [obs/3, 3 obs] for all seesaw-consistent $M_R$ would drop the leptogenesis branch. Stays [C] (a transfer target). [E][C][X]                                                                                                                                                                                    |
| v373_ftransfer_relic.py       | Track 3 ( $F_{\text{transfer}}$ ): the axion relic from the FINITE-T misalignment ODE – promotes the <b>experiments/ftransfer</b> solve and DECIDES the two TFPT angle branches, replacing the v326 adiabatic toy. $f_a=M_{\text{scal}}/128 \sim 2.4\times 10^{11}$ GeV, $m_a \sim 23.8 \mu\text{eV}$ , lattice $\chi(T)$ $n=8.16$ . [E] candidate scales fixed by the determinant-line atoms ( $128=2 \dim S^+ \mu_4 $ ). [C] the frozen SPINE angle $3\pi/5=108^\circ=\pi N_{\text{fam}}/g_{\text{car}}$ gives $\Omega_a h^2 \sim 0.12$ in $[0.08, 0.16]$ (lands on $\Omega_{\text{DM}}$ untuned) – the robust dominant-DM branch. [X] the HILLTOP $170^\circ$ over-produces ( $\sim 5.4\times$ ) – disfavoured. [X] kill: a finite-T solve robustly outside $[0.08, 0.16]$ for the spine across $\chi(T)/g_*$ drops the dominant axion-DM branch. Stays [C]/[X]. [E][C][X]                                                                                                                                                   |
| v374_ftransfer_qcd.py         | Track 3 ( $F_{\text{transfer}}$ ): the $m_p/m_e$ transfer as a typed uncertainty-budget + kill test – the falsifiability companion to v262 (the precise 2-loop point), replacing the v326 toy (which hardcodes 1836.15). [E] $b_3 = -(11 - 2n_f/3) = -7$ ( $n_f=6$ carrier). [C] two declared externals (v339): $\alpha_s(M_Z)=0.1179\pm 0.0009$ and $C_p=2.83\pm 0.15$ , each $\sim 5.3\%$ . [C] $m_p/m_e$ central $\sim 1824$ (v262) with a propagated $\pm 7.5\%$ band [ $\sim 1690, \sim 1960$ ] CONTAINING the observed 1836.15 – no free dial. [X] kill: a future lattice $C_p\pm 0.03$ + tighter $\alpha_s$ shrink the band to $\sim \pm 2\%$ ; a band then excluding 1836.15 drops the QCD-matching route ( $b_3 = -7$ + the $SU(9)$ -integer firewall v262). Distinct from v262's point; stays [C]. [E][C][X]                                                                                                                                                                                                          |
| v375_observatory_registry.py  | Track 4: the prediction OBSERVATORY – a status-typed CI over the frozen prediction registry ( <b>freeze_file.csv</b> ) that makes the falsifiability surface machine-checkable (no new physics). [E] falsifiability complete: every registry row carries a non-empty kill criterion (no orphan claims). [E] headline values re-derive from $\{\varphi_0^{\text{ret}}\}$ : $\sin^2 \theta_{12}=1/3-\varphi_0^{\text{ret}}/2$ , $\sin^2 \theta_{13}=\varphi_0^{\text{ret}} e^{-5/6}$ , $\beta_{\text{rad}}=\varphi_0^{\text{ret}}/(4\pi)$ . [C] live scorecard: $\theta_{12}$ vs JUNO first run ( $0.3092\pm 0.0087$ ) = $0.28\sigma$ (compatible); $\theta_{13}$ vs NuFIT 6.0 NO ( $0.02195\pm 0.00058$ ) = $2.0\sigma$ (the flagged most-tensioned core prediction); $\beta_{\text{rad}}$ vs ACT DR6 ( $0.215\pm 0.074$ ) = $0.37\sigma$ (hint); $r=12/N_\star^2 \sim 0.004 < \text{BK18}$ (compatible). [E] registry integrity: the single machine-readable falsifiability surface. Tooling for the closed/open pieces. [E][C] |
| v376_seam_s3_centralcharge.py | S3 closure stack: the central charge $c=8$ extracted NUMERICALLY from the lattice via Calabrese–Cardy finite-size entanglement scaling of the critical free-fermion content (the quantitative core of the MMST scaling-limit claim, on OUR model). [E] one critical complex mode gives $c=1.0000$ (Peschel correlation-matrix EE); [E] the collar is 16 Majoranas = 8 complex modes (v367), $c$ additive $\Rightarrow c=8=g_{\text{car}}+N_{\text{fam}}$ ; [E] chiral $c_-=8$ (Chern $ C =1$ per Majorana v367, non-gappable edge v368). [O] the specific net $(E_8)_1$ is v377/v378; the continuum existence is the cited MMST theorem (v336). Numerical $c=8$ from the lattice; does NOT close SEAM.EQUIV.01. [E][O]                                                                                                                                                                                                                                                                                                          |

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| Script                      | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| v377_seam_s3_e8character.py | S3 closure stack: the chiral $c=8$ content is the SPECIFIC net $(E_8)_1$ , via the exact affine character $q$ -series + the $\mu_4$ /GSO promotion. [E]<br>$\chi = E_4/\eta^8 = q^{-1/3}(1+248q+4124q^2+\dots)$ : leading $-c/24 = -1/3 \Rightarrow c=8$ ; level-1 $= 248 = \dim E_8$ (one primary). [E] the $\mu_4$ /GSO promotion $SO(16)_1 \rightarrow (E_8)_1$ : $248=240+8$ , $240 = \dim S^+ g_{\text{car}} N_{\text{fam}} = 16 \cdot 5 \cdot 3$ ; $248=120+128 = \dim SO(16) + \text{spinor}$ . [E] discriminator: $(E_8)_1$ has 1 primary/248 currents vs $SO(16)_1$ 4 primaries/120. [O] continuum existence (v336); the $\mu_4$ projection = the seam one-sidedness (S3). Pins the target net; does NOT close SEAM.EQUIV.01. [E][O]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| v378_seam_s3_modular.py     | S3 closure stack: the GENUS-1 discriminator – torus ground-state degeneracy = 1 (holomorphic = $(E_8)_1$ ), the “hard part” v344 flagged (genus-0 unique vacuum is necessary not sufficient). [E] KLM condensation:<br>$\mu(\text{carrier}) = \det C(D_5) \det C(A_3) = 4 \cdot 4 = 16$ ; the order-4 $\mu_4$ glue ( $ L =4$ )<br>$\Rightarrow \mu(E_8) = 16/16 = 1$ (torus GSD = 1); index-2 $\Rightarrow \mu(SO16) = 16/4 = 4$ . [E]<br>$ \det C(E_8)  = 1$ vs $ \det C(D_8)  = 4$ . [E] the single $(E_8)_1$ character has modular $T$ -eigenvalue $e^{-2\pi i/3}$ ( $h=0, c=8$ ), 1-dim rep $\Rightarrow$ one character $\Rightarrow$ torus GSD = 1. [O] continuum existence (v336). Pins the net at genus-1; does NOT close SEAM.EQUIV.01. [E][O]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| v379_seam_s3_rp.py          | C7 / OS reconstruction: reflection positivity (the OS axiom) of the explicit gapped collar measure, verified numerically – the remaining constructive-QFT ingredient for the OS reconstruction. [E] the gapped $p+ip$ band gives $G(\tau) = \text{mean}_k e^{-\varepsilon_k \tau}$ with $\varepsilon_k \geq \text{gap}/2 \sim 1 > 0$ (v367) – a positive spectral measure; [E] the Hankel/reflection Gram matrix $M_{ij} = G(\tau_i + \tau_j)$ is PSD – OS reflection positivity holds; [E] neg control: a growing ghost mode makes it indefinite (non-vacuous). [C] RP + gap (v76/v337) + GNS (v240) + $c=8$ (v376) + $(E_8)_1$ (v377/v378) give the OS inputs; C7/QG.AMB.01 is then DISCHARGED by the redundancy theorem v369, not built. [O] the full constructive measure stays the cited step. [E][C][O]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| v380_grav_kms_hessian.py    | The KMS Entire Hessian – the Stelle ghost is EXACTLY the Seeley–DeWitt truncation, and resummation pushes it to infinity (upgrades v304/v370 from “assume entire” to a derived statement). [E] the seam KMS cutoff $f(u) = e^{-u}$ (v259) gives the dressed spin-2 propagator $e^{-p^2/M^2}/p^2 = 1/(p^2 a)$ with $a(u) = e^u$ entire & nowhere zero $\Rightarrow$ only the $p^2=0$ pole (no ghost). [E] each finite truncation carries the ghost: the $R+R^2$ order $T_1=1+u$ has the Stelle zero $u=-1$ , the Weyl <sup>2</sup> order $T_2=1+u+u^2/2$ the pair $-1 \pm i$ ; the entire $a$ has none. [E] the ghost DECOUPLES:  nearest zero  of $T_n$ grows monotonically with $n$ (1, 1.41, $\dots$ , 3.07 for $n=1..8$ ) $\rightarrow \infty$ , so “entire” = “do not truncate”. [C] $\Rightarrow$ perturbative graviton unitarity. [O] the exact off-shell curved-space Hessian identification is the cited heat-kernel resummation; perturbative-only, not the non-perturbative measure. [E][C][O]                                                                                                                                                                                                                                                                                                                                                                               |
| v381_qft4d_eg_allorder.py   | The all-order Epstein–Glaser / BRST contract for the perturbative 4D S-matrix $S_{\text{pert}}$ – the “all-order closing statement” for the 4D perturbative QFT leg. Types the standard causal-perturbation-theory closure for the TFPT finite interaction class; it does <i>not</i> build a path integral. [E] POWER-COUNTING: every gauge+matter vertex (Yang–Mills + Dirac–Yukawa + Higgs + ghost) is a mass-dimension-4 marginal operator $\Rightarrow$ the EG singular order $\omega \leq 4 \Rightarrow$ a FINITE counterterm space at every order. [E] BRST NILPOTENCY $s^2=0$ verified via the Jacobi identity for $su(2)$ (residual 0) and $su(3)$ ( $10^{-16}$ ), the carrier weak+colour content. [E] the seam gap $\Delta = 6 \ln(3/2) > 0$ (v302) gives the IR-safe adiabatic limit $g \rightarrow 1$ . [C] imported: $T_n$ exists to all orders by causal factorisation (Epstein–Glaser 1973 / Brunetti–Fredenhagen). [C] imported: all-order BRST invariance (Piguet–Sorella / Kugo–Ojima) $\Rightarrow$ Slavnov–Taylor to all orders. [E] scope fence: matter+gauge ONLY – the $R^2/\text{Weyl}^2$ gravity sub-sector is v304/v370/v380, not a local EG vertex; $S_{\text{pert}}$ is not the ambient measure QG.AMB.01 (v369). NET [C] (imported EG+BRST on the TFPT dim-4 class + gap), prerequisites [E]; extends v269/v271/v273/v278 from 1-loop to all-order [E][C] |

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| Script                         | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| v382_alpha_quillen_exact.py    | Names the Quillen determinant-line VARIATION $\delta_\tau(\log \det_\zeta \Delta_{U(1)} + 8b_1 c_3^6 \log \varphi_{\text{seam}}) = 0$ as a tracked target (review Point 10), elevating EM.WARD.01's residual to a citable [O]. [E] the Quillen split holds at the root $\alpha^3 - 2c_3^3 \alpha^2 = 8b_1 c_3^6 \ln(1/\varphi_{\text{seam}})$ at $\alpha^{-1}=137.0359992$ ( $F_{U(1)}=0$ IS the stationarity of the $U(1)$ determinant line, v341). [E] every coefficient a named atom: $8b_1=\frac{4}{5}\cdot 41$ a $U(1)$ index, $c_3=1/(8\pi)$ Gauss–Bonnet ( $c_3^{3,6}$ heat-kernel powers), $q(D_5)+q(A_3)=2$ discriminant forms (v341/v48/v216). [E] the value $\alpha^{-1}=137.0359992168$ stays closed (unique root + ablation, v3) – only “why this functional” is open. [O] THE TARGET: the from-first-principles AQFT/ $\zeta$ proof that $F_{U(1)}$ is the EXACT Quillen functional. [C] a FACE of SEAM.EQUIV.01 (v336): the seam $U(1)$ determinant line lives on the same holomorphic seam net, so once the net is $(E_8)_1$ its $U(1)$ sub-determinant is fixed – not a second independent gate, it names/tracks the pre-existing EM.WARD.01 residual (v341/v342). No new number (reuses v341, Wolfram-mirrored) [E][C][O]                                                                                                                                                                                                                                         |
| v383_dynamics_universal.py     | DYNAMICS.UNIVERSAL.01: the UNIVERSAL spectral-gap / Perron–Frobenius principle – every TFPT sector is the SAME object (a gapped operator with a UNIQUE leading attractor = the physics, a spectral gap = the reason there is no free parameter), extending v303 from $F_{\text{transfer}}$ to the whole theory. [E] FLAVOR (computed): the cusp transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ has a simple Perron leading 1 and gap $6 \ln(3/2) > 0$ (v56/v82). [E] DISCRETE COMPILER (computed): the affine- $E_8$ adjacency has Perron eigenvalue 2 (Kac marks, v312/v313) and subleading $2 \cos(\pi/5) = \varphi$ , gap $2 - \varphi = 1/\varphi^2 > 0$ . [E] DECOUPLING MARGIN: $\Delta_{\text{eff}} = 6 \ln(3/2) - 31/(4\pi^2) \approx 1.648 > 0$ (v76/v337). [E] TWO NUMBER-FIELD FACETS (v314): golden $\varphi \in \mathbb{Q}(\sqrt{5})$ (carrier-5, static) and $(2/3)^6 \in \mathbb{Q}$ (family-3, dynamic) are the prime-5 and prime-3 facets of the order-30 Coxeter clock $30 = 2\cdot 3\cdot 5 = h(E_8)$ – a factorisation, not an equality. [C] the other sectors are the same structure: gravity (v358/v359), QG saddle (v365), the holomorphic $c=8$ RG fixed point (v157/v158), the all-order S-matrix (v381), recovery (v221), $F_{\text{transfer}}$ (v303), $\tau=i$ (v333). [C] META-THEOREM: gapped $\Rightarrow$ unique attractor $\Rightarrow$ parameter-free, theory-wide – one spectral-gap statement wearing many hats. A synthesis, no new number [E][C] |
| v384_residual_certification.py | RESIDUAL.CERTIFICATION.01: the residual matrix is CERTIFICATION, not CONSTRUCTION – after v369/v381/v382 there is NO open TFPT physics MECHANISM left; every residual is an external CERTIFICATE of already-determined structure, of three kinds: (A) external math proof, (B) theorem-forbidden, (C) external physics. A ledger-CI audit. [E] LEDGER CI: every residual claim id is present (no orphan). [E] NO-UNIT (B): $c_3=1/(8\pi)$ and $g_{\text{car}}=5$ are mass-dimension 0, so by No-Unit (v153/v364) no dimensionless compiler outputs an absolute scale $\Rightarrow v_{\text{geo}}$ is forbidden by theorem, not a gap. [E] GAP-DECOUPLED (A): $\Delta_{\text{eff}} \approx 1.648 > 0$ (v76/v337) $\Rightarrow$ every readout is independent of the ambient measure (QG.AMB.01 a [C] redundancy, v369). [E] CLASSIFICATION COMPLETE: SEAM.EQUIV.01 (continuum existence = cited MMST/Adamo, v336) $\rightarrow$ A, ALPHA.QUILLEN.EXACT.01 (v382) $\rightarrow$ A, QG.AMB.01 (v369) $\rightarrow$ A, $v_{\text{geo}} \rightarrow$ B, $F_{\text{transfer}}$ (v371–v374, firewall v187) $\rightarrow$ C; ZERO open internal TFPT mechanisms. [C] CONVERGENCE: the only hard things left are theorems and external physics, not missing dynamics. An audit, no new number [E][C]                                                                                                                                                                                          |

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| Script                     | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| v385_uvbranch_killtest.py  | UVBRANCH.KILLTEST.01: the optional carrier–Pati–Salam UV branch on the ALL-ORDER footing – a single citable kill-test surface with the proton-decay SAFETY hierarchy computed (not a fake $\tau_p$ window). A consolidation (does <i>not</i> re-run the RGE v246 nor re-derive the PS scale v249/v253). [E] carrier content = SM ( $b = (41/10, -19/6, -7)$ , v159/v246, no new states) and $S_{\text{pert}}$ closes to all orders (v381) $\Rightarrow$ the SM-non-unification is all-order-robust. [X] plain-SM 4D-GUT killed ( $\sin^2 \theta_W = 3/8$ vs 0.231; v246). [C] optional carrier-PS branch: $M_{PS} = \kappa M_{\text{scal}}$ , $M_{\text{scal}} = c_3^{7/2} \bar{M}_{\text{Pl}} \approx 3.06 \times 10^{13}$ GeV, $\kappa \in [1.0, 1.2]$ (v253); content $\{1, 10, 16, 45\}$ , no 126 (v247/v248); not the default (v265). [E] proton-decay SAFETY hierarchy: minimal PS gauge bosons are $SU(4)$ leptoquarks ( $B-L$ -conserving $q \leftrightarrow l$ , rare LFV $K_L \rightarrow \mu e$ ), <i>not</i> the $p \rightarrow e^+ \pi^0$ dim-6 operator (needs an $SO(10)/SU(5)$ diquark, absent); the binding bound $M_{PS} \gtrsim 10^6$ GeV is cleared by $M_{PS}/M_{\text{bound}} \sim 10^7 \gg 1$ . [X] frozen kill tests (v265): $\kappa$ leaving $O(1)$ / a forbidden rep / a new gauge-charged state / a rare-LFV signal below the bound; no fake $\tau_p$ . A consolidation, no new number [E][C][X] |
| v386_grav_amplitude.py     | GRAV.AMPLITUDE.01: the entire-form-factor graviton-exchange AMPLITUDE is finite, UV-softened and tree-unitary – perturbative gravity as an explicit scattering problem, extending v304/v370/v380 from the propagator to the amplitude. The dressed spin-2 propagator $P = e^{-p^2/M^2}/p^2 = 1/(p^2 a)$ , $a(u) = e^u$ , $M = M_{\text{scal}}$ . [E] one pole, ghost-free: $a$ entire & nowhere zero (v380) $\Rightarrow$ only pole at $p^2 = 0$ , residue $e^0 = 1 > 0$ . [E] UV-softened: ratio $e^{-u}$ at $u = 1, 5, 10, 20 = 0.37, 6.7 \times 10^{-3}, 4.5 \times 10^{-5}, 2.1 \times 10^{-9}$ (faster than any power). [E] low-energy = GR + a calculable $-p^2/M^2$ correction ( $M = M_{\text{scal}} = c_3^{7/2} \bar{M}_{\text{Pl}}$ , v253). [C] tree unitarity: a single positive-residue pole, no negative-norm states (v370/v380). [C] the amplitude $A = (\kappa^2/2)(T \cdot P_2 \cdot T') e^{-p^2/M^2}/p^2$ is GR at low energy, softened in the UV (perturbative-only; QG.AMB.01 a [C] redundancy v369). No new number [E][C]                                                                                                                                                                                                                                                                                                                                                                              |
| v387_corrections_gap.py    | CORRECTIONS.GAP.01: the practical harvest of v383 – the same spectral gap that makes each sector parameter-free ALSO sets the SIZE of its first correction ( $\text{correction}_n \sim (\lambda_2/\lambda_1)^n$ ). [E] FLAVOR: $\lambda_2 = (2/3)^6 = 64/729 \Rightarrow$ first Koide correction $\sim 0.088$ (v56/v82). [E] RECOVERY: the same $(2/3)^6$ Petz rate (v221). [E] QG: capped by $\chi = 1/(1 - (2/3)^6) = 729/665$ (v337). [E] DISCRETE COMPILER: the update $(A + 2I)/4$ fixes the Kac marks ( $\lambda_1 = 1$ ) and decays at the golden $(\varphi + 2)/4 \approx 0.905$ (carrier-5 facet, v303/v312). [E] the rates split into the $(2/3)^6$ family (family-3, in $\mathbb{Q}$ ) and the golden family (carrier-5, in $\mathbb{Q}(\sqrt{5})$ ) – the two facets of v314/v383. [C] the correction calculus: the gap does double duty (WHY no free parameter AND HOW BIG the first correction), so sub-leading magnitudes are organised, not free. No new number [E][C]                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| v388_corrections_budget.py | CORRECTIONS.BUDGET.01: the gap correction (v387) is a <i>typed</i> budget, NOT a uniform band on every prediction – the $(\lambda_2/\lambda_1)^n$ size is a property of the distance from a gapped operator’s leading eigenvector, so it is only defined where $\lambda_2$ exists. The prediction surface splits into four correction classes (already implied by v303). [E] SEAM-GAPPED: Koide ( $F_{\text{pole}}$ v303), recovery (v221), the QG bound (capped by $\chi = 729/665$ , v337) carry $(2/3)^6$ ; the discrete compiler decays at the golden $(\varphi + 2)/4$ (v312). [E] KOIDE FIRST CORRECTION: $F_{\text{pole}}$ is the ONE seam-gapped prediction, rate $(2/3)^6 \approx 0.0878$ (v371), source quotient 0.33% below $2/3$ – the rate is the suppression SCALE, not the residual. [E] EXACT-IDENTITY class ( $\det R = 8$ , $N_\Phi = 1$ , $\theta_{\text{eff}} = 0$ , $\sin^2 \theta_{13}$ ): no $\lambda_2 \Rightarrow$ correction 0 STRUCTURALLY (a band would contradict [E]). [E] EXTERNAL-RATE class ( $\eta_B/\text{axion}/m_p, m_e$ ): the gapped shape with external rates (only $F_{\text{pole}}$ has the seam rate, v303), firewalled (v187). [E] FIXED-POINT class: the texture is already explicit ( $\theta_{12} \varepsilon = c_3 + 36c_3^4$ ). [C] a uniform band on all predictions is WRONG; the size is keyed to the mechanism class. A typing of v387, no new number [E][C]           |

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| Script                         | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| v389_grav_loop_finiteness.py   | GRAV.LOOP.FINITE.01: the entire-form-factor graviton is UV-finite at the <i>loop</i> level by power counting – the honest next step past v386 (tree), extending v304/v370/v380/v386 from the propagator to the superficial degree of divergence. Honest scope: the standard infinite-derivative/nonlocal-gravity finiteness statement, NOT a full proof of perturbative QG. [E] super-polynomial falloff $p^k e^{-p^2/M^2} \rightarrow 0$ for every $k$ . [E] one dressed line turns a quadratic divergence into the finite $(M^2/2)e^{-p_0^2/M^2}$ ; the GR $\int p^3/p^2 dp$ diverges quadratically. [E] the superficial degree $\rightarrow -\infty$ at every loop order (dressed cutoff-ratio $\rightarrow 1$ vs GR $\rightarrow 4$ ). [E] the regulator is the SEAM scale $M = M_{\text{scal}} = c_3^{7/2} \bar{M}_{\text{Pl}}$ (v253/v259), not a dial. [C] loop UNITARITY of the nonlocal form factor (Lorentzian continuation / macro-causality) stays the honest open point – tree unitarity is v386; NOT claimed solved. No new number [E][C][O]                                                                                                                                                   |
| v390_prime2_facet.py           | COXETER.PRIME2.01: the prime-2 facet of the order-30 Coxeter clock, completing v383. The clock $30 = 2 \cdot 3 \cdot 5 = h(E_8)$ had two named facets (golden $\varphi \in \mathbb{Q}(\sqrt{5})$ , prime-5/carrier; $(2/3)^6 \in \mathbb{Q}$ , prime-3/family); this names the prime-2 facet as the Gaussian field $\mathbb{Q}(i)$ — $\tau = i$ and the order-4 $\mu_4 = \mathbb{Z}/4$ square seam deck (v282/v288/v333), structure already present. [E] $h(E_8) = 30$ factors into exactly three primes. [E] one quadratic facet field per prime, each ramified at its prime: disc $\mathbb{Q}(i) = -4$ , disc $\mathbb{Q}(\sqrt{-3}) = -3$ , disc $\mathbb{Q}(\sqrt{5}) = 5$ ; ramified primes $\{2, 3, 5\}$ , product = 30. [E] prime-2 Gaussian facet: min. poly $x^2 + 1$ , $\tau = i$ , $j(i) = 1728 = 12^3$ , $ \mu_4  = 4 = 2^2$ (the static seam-orientation facet). [E] prime-5 golden (v383): $2 \cos(\pi/5) = \varphi$ (pentagon). [E] prime-3 family (v383/v220): $(2/3)^6$ plus the Eisenstein $\zeta_3$ (hexagonal CP $\pi/3$ ). [C] completion: each prime carries a facet (square/hexagon/pentagon = seam/CP/carrier), the prime set forced by $h(E_8)$ . A synthesis, no new number [E][C] |
| v391_alpha_quillen_progress.py | ALPHA.QUILLEN.PROGRESS.01: an <i>honest attempt</i> at the external target ALPHA.QUILLEN.EXACT.01 (v382) – it REDUCES the open step to a sharper named sub-target via a solvable model but does NOT close it; ALPHA.QUILLEN.EXACT.01 stays [O] (external math, type A, v384). [E] solvable model $\det_\zeta(-\partial_\tau^2 + m^2)$ on a circle $= (2 \sinh(\beta m/2))^2$ , $m$ -variation $\beta \coth(\beta m/2) = 2/m + (\beta^2/6)m - (\beta^4/360)m^3 + \dots$ a closed heat-kernel/Laurent series ( $m^3$ coeff $-1/360$ verified). [E] a $\zeta$ -det variation is structured heat-kernel data, so v382's $8b_1c_3^6$ counterterm (a Gauss–Bonnet heat-kernel power) is the right <i>kind</i> of object. [E] the v382 split re-verified at $\alpha^{-1} = 137.0359992$ . [E] reduction: the open step reduces to the named sub-target “seam Seeley–DeWitt coefficient $= 8b_1c_3^6$ ”. [O] NOT CLOSED: the actual seam $\zeta$ -det proof stays open, SEAM.EQUIV.01 continuum existence (v336) the other external [O]; $\alpha^{-1}$ stays [E]. An attempt that reduces, does not close [E][O]                                                                                                     |
| v392_seam_s3_scalinglimit.py   | SEAM.S3.SCALINGLIMIT.01: an <i>honest attack</i> on the one open lemma of SEAM.EQUIV.01 (the continuum scaling-limit existence, v336) – it does NOT close it (the abstract existence is the cited MMST theorem); it STRENGTHENS the case with the existence-relevant observable v376 did not test: that the finite-size data <i>converge</i> . [E] the Calabrese–Cardy EE slope $c_1(L)$ at $L = 66 \dots 514$ is a monotone Cauchy-like sequence approaching 1 ( $ c_1(L) - 1 $ strictly decreasing). [E] a $1/L$ Richardson fit gives $c_1(\infty) \approx 1.000$ , so the 16-Majorana collar $c = 8c_1 \approx 8.00 = g_{\text{car}} + N_{\text{fam}}$ is reached <i>in</i> the limit. [E] the successive differences are strictly decreasing (numerical Cauchy, necessary for existence). [O] convergent data are evidence, not a proof of existence (MMST v336), and do not pin $(E_8)_1$ vs $SO(16)_1$ (holomorphy v377/v378). Strengthens the residual, does NOT close it [E][O]                                                                                                                                                                                                                      |

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| Script                       | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| v393_corrections_numeric.py  | CORRECTIONS.NUMERIC.01: the typed correction budget (v388) turned into ACTUAL first-correction magnitudes – a computed theoretical-uncertainty surface, the short-term testable harvest of v388. [E] FIXED-POINT band $(\theta_{12}/\theta_{13}/\lambda_C/\beta_{\text{rad}}) =$ the explicit $\varphi_0$ puncture term $36c_3^4/c_3 = 9/(128\pi^3) \approx 0.227\%$ (exact; $= \theta_{12}$ 's $\varepsilon$ first-correction). [E] SEAM-GAPPED band (Koide/recovery) $= (2/3)^6 \approx 8.78\%$ ; the QG bound capped by $\chi - 1 = 64/665 \approx 9.62\%$ (v337). [E] EXACT-IDENTITY band = 0 structurally ( $\det R = 8$ , $N_\Phi = 1$ , $\theta_{\text{eff}} = 0$ ). [E] EXTERNAL-RATE band = external physics quoted ( $m_p/m_e \pm 7.5\%$ v374; $n_s/r N_\star$ band; $\eta_B 1.07 \times$ v212), fenced v187. [C] each prediction gets “central value $\pm$ computed first correction”. A harvest of v388, no new number [E][C]                                                                                                |
| v394_coxeter_atoms.py        | COXETER.ATOMS.01: the clockwork coherence – the three Coxeter primes ARE the three anchor atoms ARE the three number-field facets, and the prime-2 seam is the common dynamical factor. A bird's-eye [E] re-reading tying v390 (facets) to the anchor $a = (1, 1, 2)$ and v319 (5x6 clock); registers RES.COXETER.SYMMETRY.01 as a tracked [O]. [E] $a = (1, 1, 2)$ : $e_3=2= Z_2 $ , $p_0=3=N_{\text{fam}}$ , $e_2=5=g_{\text{car}}$ , product $30=h(E_8)$ . [E] atoms = facet fields (v390): $2 \rightarrow \mathbb{Q}(i)$ , $3 \rightarrow \mathbb{Q}(\sqrt{-3})$ , $5 \rightarrow \mathbb{Q}(\sqrt{5})$ , each ramified at its prime. [E] $\chi=729/665=3^6/(3^6-2^6)$ , $(2/3)^6=2^6/3^6$ , exponent $6=2N_{\text{fam}}$ . [E] both dynamic rates carry prime-2: $2/3= Z_2 /N_{\text{fam}}$ , $(\varphi+2)/4=(\varphi+2)/ \mu_4 $ , $ \mu_4 =4=2^2$ . [E] $30=5 \times 6=g_{\text{car}}(2N_{\text{fam}})$ . [C] the seam is the universal dynamic engine. [O] RES.COXETER.SYMMETRY.01 registered, not closed [E][C][O]              |
| v395_hypergraph_coupled.py   | HYP.COUPLED.01: the coupled hypergraph rewrite – one local rule unifies v299/v327/v324 (carrier $\times$ family). [E] LOCAL COUPLED MICRO-RULE: state on 9 affine- $E_8$ nodes $\times$ 3 family slots; one micro-step = network lazy diffusion on each cusp column + v327 $M$ on each node's family vector; steps commute, $T_{\text{micro}} = T_{\text{net}} \otimes M$ . [E] JOINT ATTRACTOR: top eigenvalue 1 with eigenvector $\text{marks} \otimes e_0$ ; iteration from any positive start converges. [E] ONE HAND = RECOVERY: one network step + $2N_{\text{fam}}$ family micro-steps gives $(2/3)^6$ on $\text{marks} \otimes v_{\text{surv}}$ . [E] v324 EMERGES: $T_{\text{full}} = T_{\text{net}} \otimes T_{\text{cusp}}$ shares the same invariants; $T_{\text{cusp}}$ is the 6-hand readout in the Galois/cusp basis (v317). [E] GROWTH + FIBER: v299 $E_6 \rightarrow E_7 \rightarrow E_8 \rightarrow \hat{E}_8$ unchanged with $M$ -fiber attached. [O] 3-arm seed $(P_2) + \varphi_0$ (v312) remain irreducible [E][O] |
| v396_phi0_icosahedral.py     | HYP.PHI0.01: the $\varphi_0$ leading term from icosahedral combinatorics (sharpening v312/v395). [E] ICO COUNTS ANCHOR-FORCED: $V=12$ , $E=30=h(E_8)$ , $F=20=g_{\text{car}}(g_{\text{car}}-1)$ , $ \text{Aut} =120=4h= 2I $ . [E] LEADING TERM EXACT: $\varphi_0^{\text{ree}}=1/(6\pi)=(4/3)c_3=F/(4h\pi)=F/( \text{Aut} \pi)=(F/(g_{\text{car}}N_{\text{fam}}))c_3$ . [E] $F/h= Z_2 /N_{\text{fam}}$ ; leading term $= (F/h)/(4\pi)=c_3(g_{\text{car}}-1)/N_{\text{fam}}$ . [E] GAUSS-BONNET: icosahedron deficit sum $4\pi$ ; $c_3=1/( Z_2 4\pi)$ (v216/v342). [E] HONEST NEGATIVE: puncture $48c_3^4$ not a rational hypergraph fraction. [C] tree-level seed = (hyperedges)/( $g_{\text{car}}N_{\text{fam}}$ ) $\times c_3$ , not an independent injection; puncture [O]                                                                                                                                                                                                                                                            |
| v397_external_clock_probe.py | FR.CLOCK.PROBE.01: a <i>falsifiable</i> probe of how external the “external physics” really is – do the frontier scheme-numbers land on the $\{2, 3, 5\}$ clock (the anchor atoms, v394) or not? Strict No-Free-Pattern (v100): “on the clock” = an EXACT match, not a wide-band fit. [E] $\eta_B$ decuple $A_\Lambda=10= Z_2 g_{\text{car}}$ ON CLOCK (v212). [E] Koide rate $(2/3)^6=( Z_2 /N_{\text{fam}})^6$ ON CLOCK (v303). [E] axion spine angle $3\pi/5=\pi N_{\text{fam}}/g_{\text{car}}$ ON CLOCK (v211). [E] proton $C_p \sim 2.83 \pm 0.15$ NOT discriminating ( $2^{3/2}$ , $17/6$ , $14/5$ all fit the band) $\Rightarrow$ EXTERNAL (firewall v187/v374). [C] 3 of 4 land exactly on the clock; the irreducible external residue shrinks to the proton QCD number $C_p$ – “external physics” was overstated. [E] anti-numerology: refuses to assign $C_p$ a clock value [E][C]                                                                                                                                             |

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| Script                       | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| v398_seam_state_rigidity.py  | <b>Seam State Rigidity Theorem</b> (Paper A, the $G_{\text{net}}$ closure contract): the scattered state-invariance reductions (v177/v198/v199/v308/v309/v335) as ONE named target. [E] RP-definability hinge (OS canonical transfer $T=e^{-H}$ from RP + reflection alone, v54, on a quasi-free state v155, no $\mu_4$ normal form); [E] band-limited core $[\rho,  k ]=0$ and character-block-diagonal sub-principal symbol swept $N=4.64$ (beyond the 3-dim $H^1$ ); [E] holomorphy $\Rightarrow (E_8)_1$ ( $c=8$ , $ \det \text{Cartan} =1$ vs 4, index $4= \mu_4 $ , gap $6 \log \frac{3}{2} > 0$ ). [O] the one residual: $[\rho, \Lambda_\Sigma]=0$ on the full $L^2$ (intrinsic Bisognano–Wichmann). A contract+evidence module, NOT a closure of SEAM.EQUIV.01 (continuum = cited MMST v336)                                                                                                                                                     |
| v399_gravity_complete.py     | <b>Gravity-complete = Boundary-complete + Ambient-redundancy</b> (Paper B accounting): combines the parameter-free local equation (v358/v359) and the ambient redundancy (v369). [E] $G_{ab} + \Lambda g_{ab} = (1/c_3)T_{ab}$ , $1/c_3=8\pi$ (four $c_3$ -roles), $\Lambda$ prefactor $3/(4\pi^2)$ ; [E] redundancy discriminators ( $ \det \text{Cartan} =1$ vs 4, torus GSD=1, Petz $(2/3)^6$ , margin $1.648 > 0$ ); [E] $\mathcal{O}_{\text{phys}} \subset \mathcal{A}_\Sigma$ over five readout classes (Newton $8\pi$ , $\Lambda$ , scalaron $c_3'$ , horizon $1/ \mu_4 $ , recovery $(2/3)^6$ ) – all boundary/seam/gap. [C] the verdict; [O] SEAM.EQUIV.01 + intrinsic BW (no new gate)                                                                                                                                                                                                                                                          |
| v400_grav_nonlocal_action.py | <b>Nonlocal spectral-gravity action</b> with the local $R+R^2$ as its IR projection (Paper C, consolidating v304/v370/v380/v386). [E] the entire form factor $a=e^u$ gives the dressed propagator $1/(p^2 a)$ ONE pole (residue 1, ghost-free); [E] every truncation carries a Stelle zero whose nearest-zero modulus grows monotonically (v380); [E] the IR order $a=1+u+u^2/2+\dots$ reproduces the spectral-action $R+R^2$ (v28/v36) with the atom-fixed scale $M_{\text{scal}}^2/\bar{M}_{\text{Pl}}^2=c_3^7$ ( $7=g_{\text{car}}+N_{\text{fam}}-1$ , v253). [O] the entire- $a$ analyticity assumption + perturbative-only (not the ambient measure QG.AMB.01, a [C] redundancy v369)                                                                                                                                                                                                                                                                |
| v401_metrology_closure.py    | <b>Metrology closure</b> of the final input set $\{a, \pi, v_{\text{geo}}\}$ (Paper D, consolidating v153/v274/v364/v384). [E] $a=(1, 1, 2) \rightarrow (e_1, e_2, e_3)=(4, 5, 2)=( \mu_4 , g_{\text{car}},  Z_2 )$ , $c_3=1/(2e_1\pi)=1/(8\pi)$ , 0 dimensionless dials + 1 unit; [E] dimensional bookkeeping (No-Unit v153) mass/ $1G/\Lambda/U_{\text{point}}/(m/\mu)$ = number $\times v_{\text{geo}}^{1,2,4,1,0}$ ; [E] gravity adds no unit (v364); [E] $G$ -vs- $\Lambda$ over-determination 0.11% (v274). [O] $v_{\text{geo}}$ theorem-forbidden ( $\text{TOE}_{\text{strict}}=\text{TOE}_{\text{dimensionless}}+[v_{\text{geo}}]$ , v384 type B)                                                                                                                                                                                                                                                                                                 |
| v402_ftransfer_suite.py      | $F_{\text{transfer}}$ <b>functor suite</b> (Paper E): the four frontier sectors as ONE functor $F_{\text{transfer}}=F_{\text{observable}} \circ F_{\text{threshold}} \circ F_{\text{RG}}$ (consolidating FR.POLE/BOLTZMANN/RELIC/QCD.SOLVE). [E] the composition law (each observable factors source $\rightarrow$ threshold $\rightarrow$ RG $\rightarrow$ observable); [E] exact atom sources ( $Q_{\text{src}}$ v93; $A_\Lambda=2g_{\text{car}}=10$ , $\delta_{CP}=4\pi/3$ v9/v212; $f_a=M_{\text{scal}}/128$ v185; $b_3=-7$ v164/v262); [E] a committed kill test per sector. [C] the firewall: physical predictions stay [C]/[O], only the algebraic source is [E] (never re-pressed into the compiler, v187)                                                                                                                                                                                                                                        |
| v403_facet_compositum.py     | ARITH.HULL.01: the three number-field facets (v390) COMPOSE to $K = \mathbb{Q}(i, \sqrt{-3}, \sqrt{5})$ , a degree-8 = rank $E_8$ abelian field – the arithmetic hull dual to the $E_8$ lattice hull. [E] the squarefree parts $-1, -3, 5$ are independent in $\mathbb{Q}^\times/(\mathbb{Q}^\times)^2$ , so $[K : \mathbb{Q}] = 2^3 = 8 = \text{rank } E_8 = g_{\text{car}} + N_{\text{fam}}$ , Galois group $(\mathbb{Z}/2)^3$ order 8; $2^{\omega(h(E_8))} = 2^3 = 8 = \text{rank } E_8$ is $E_8$ -special (the $E_7$ control $h=18$ , $2^2=4 \neq 7$ fails). [E] exactly 7 quadratic subfields $\{-1, -3, 5, 3, -5, -15, 15\}$ , split 4 imaginary = $ \mu_4 $ (CM/dynamic) + 3 real = $N_{\text{fam}}$ (RM/static). [E] ramified only at $\{2, 3, 5\}$ = the atoms = the primes of $h(E_8)=30$ . [C] $K$ is the abelian $(\mathbb{Z}/2)^3$ arithmetic hull dual to $E_8$ ; the 7 quadratic subfields a candidate for the 7 $E_8$ audit slices [E][C] |

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| Script                     | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| v404_e8_char_cm_duality.py | E8.CMDUAL.01: the $(E_8)_1$ vacuum character $\chi_{E_8}=E_4/\eta^8=j^{1/3}$ at the two CM points – 12 at the seam ( $\tau=i$ ), 0 at the family ( $\tau=\omega$ ), extending v282. [E] $\chi_{E_8}^3=j$ (q-series). [E] SEAM $\tau=i$ ( $\mathbb{Q}(i)$ , order 4): cross-ratio $2 \Rightarrow j=1728 \Rightarrow \chi_{E_8}(i)=1728^{1/3}=12$ ( $E_8$ present, v214/v282). [E] FAMILY $\tau=\omega$ ( $\mathbb{Q}(\sqrt{-3})$ , order 6): the equianharmonic $\lambda$ ( $\lambda^2-\lambda+1=0$ ) $\Rightarrow j=0 \Rightarrow \chi_{E_8}(\omega)=0$ ( $E_4$ vanishes, v220/v267). [E] $\tau=i, \tau=\omega$ are the ONLY elliptic points of $\text{PSL}(2, \mathbb{Z})$ , so the values are forced. [C] the seam/flavor rigidity duality: the seam is rigid where $E_8$ is present ( $\chi=12$ , SEAM.EQUIV.01 closes), the flavor/CP sector is the soft residual where $E_8$ degenerates ( $\chi=0$ ); the dual-keystone candidate (flavor = the $j=0$ degeneration of $(E_8)_1$ ) [E][C]                                                                                                                                                                                                                                          |
| v405_seam_equiv_omega.py   | SEAM.EQUIV.02 – the dual keystone at $\tau=\omega$ : the family/flavor sector as the order-3 Eisenstein/ $A_2$ face of $E_8$ , dual to the seam = $(E_8)_1$ at $\tau=i$ (v404). A named contract (genre of v286/v398), NOT a closure. [E] $\chi_{E_8}(\omega)=0$ ( $E_8$ degenerates at the family CM point) vs $\chi_{E_8}(i)=12$ . [E] $A_2$ is the $\tau=\omega$ Eisenstein lattice (Cartan $\det=3=N_{\text{fam}}$ , $h=3=N_{\text{fam}}$ , $\mu_6=\mu_3 \times \mu_2$ ). [E] $R$ lives in $E_6 \times A_2$ : $248=\ R\ _F^2 + \det R + 2(1^T R a)N_{\text{fam}}=78+8+2 \cdot 27 \cdot 3$ , $\ R\ _F^2=78=\dim E_6$ , $\det R=8=\dim A_2=\text{rank } E_8$ . [E] $\delta_{\text{PMNS}}=4\pi/3$ is the $\tau=\omega$ hexagonal CM phase ( $\arg \zeta_6=\pi/3$ , v220/v233). [E] the cusp weights $\{0, 1/3, 2/3\}$ are the order-3 deck = $N_{\text{fam}} = \det Q$ (v72), $\text{Spec}(Q_+)=(1, 2, 3)$ (v69/v137). [C] the family/flavor sector is the order-3 Eisenstein/ $A_2$ deck at $\tau=\omega$ ; the two keystones sit at the only two elliptic points of $\text{PSL}(2, \mathbb{Z})$ . [O] residual = the canonical $\mathbb{Z}_3$ deck map (v140) + the $v_{\text{geo}}$ -class scale (No-Unit v153) [E][C][O]           |
| v406_pmns_phase_origin.py  | FLAV.PMNS.CLOSE.01: the PMNS dynamics closed modulo the $\tau=\omega$ keystone (v405) – the CP-phase ORIGIN is the family CM point and the absolute scale is one $v_{\text{geo}}$ -class unit, so PMNS has zero free flavor dials beyond $v_{\text{geo}}$ . [E] the three angles ( $\sin^2 \theta_{12}=\frac{1}{3}-\frac{\varphi_0^{\text{ret}}}{2}$ , $\theta_{23}=45^\circ$ , $\sin^2 \theta_{13}=\varphi_0^{\text{ret}} e^{-5/6}$ ; v9/v268) assemble to one unitary $U_{\text{PMNS}}$ (v270). [C] CP-PHASE ORIGIN (advance over v270): $\delta_{\text{PMNS}}=4\pi/3$ is the $\tau=\omega$ hexagonal CM phase (v405/v220/v233), upgrading v270's [O] assembled input to a keystone consequence. [C] $J_{\text{PMNS}}=-0.0297$ derived. [E] the absolute scale = one seesaw ratio $m_3=(y_\nu v_{\text{EW}})^2/M_R$ = a $v_{\text{geo}}$ -class UV input (v272/v153), NOT a flavor gap. [C] net: PMNS = angles + $\delta$ [keystone] + $J$ [derived] + scale [ $v_{\text{geo}}$ ], zero free flavor dials. [O] residual = the seesaw operator realisation ( $M_R$ not a clean $\bar{M}_{\text{Pl}}$ power, v272/v184; one-parameter carrier-normalised candidate v481) + the normal-ordering $\Sigma m_\nu$ floor kill test [E][C][O] |
| v407_dn_pairings_omega.py  | FLAV.SELECTOR.CLOSE.01: the $R4'$ residue folds into the $\tau=\omega$ keystone (v405) – the three $(d, n)$ selector pairings ARE the $\tau=\omega$ family-slice atoms. [E] the frame $(\mathbf{1}, a, \sigma)$ is a $\mathbb{Z}^3$ basis with $\det=11=\ Pl(K)\ _1$ , and $n=(5, -9, 6)$ is the unique covector with $n \cdot \mathbf{1}=2$ , $n \cdot a=8$ , $n \cdot \sigma=121=11^2 \Rightarrow (d, n) \Rightarrow R$ columnwise (v136/v139). [E] $n \cdot a=8=\text{rank } E_8=\dim A_2=\det R$ – the anchor-pairing is the $A_2$ ( $\tau=\omega$ ) block dimension that reads $R$ in $E_6 \times A_2$ (v405). [E] the pairings $\{2, 8, 121\}=\{ Z_2 , \text{rank } E_8=\dim A_2=\det R, \ Pl(K)\ _1^2\}$ are the $\tau=\omega$ family-slice atoms (the $\mu_2$ sheet, the $A_2$ block, the Plucker frame). [E] the square selects the point ( $\sigma$ -pairing $11(11+t)$ , a perfect square only at $t=0$ ). [C] deriving $(d, n)$ ( $R4'$ ) folds into SEAM.EQUIV.02 – $d=\frac{3}{2}a-21$ anchor-derived (v139), the pairings the slice atoms. [O] the lone flagged residue “why $11=\ Pl(K)\ _1$ ” [E][C][O]                                                                                                                |

*continued on next page*

| Script                                | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v408_phi0_puncture_heatkernel.py      | HYP.PHI0.PUNCTURE.01: name the $\varphi_0$ puncture term $\delta_{\text{top}}=48c_3^4$ as the order-4 local boundary heat-kernel (Seeley-DeWitt) contact term of the $\mu_4$ puncture divisor – the analytic complement of the icosahedral tree term (v396, HYP.PHI0.01). [E] THE SPLIT IS EXACT: $\varphi_0=\varphi_0^{\text{tree}}+\delta_{\text{top}}=1/(6\pi)+48c_3^4$ (v396/ARCH.QUAD.01). [E] PUNCTURE COUNT = $\Omega_{\text{adm}}=N_{\text{fam}} \dim S^+=48$ , order = $ \mu_4 =4$ : $\delta_{\text{top}}=\Omega_{\text{adm}}c_3^{ \mu_4 }$ . [E] CLOSED FORM $\delta_{\text{top}}=48c_3^4=3/(256\pi^4)$ ; $c_3=1/( Z_2 4\pi)$ Gauss-Bonnet (v216/v342) so $c_3^4$ is a fourth-order boundary heat-kernel power (cf. v391). [E] HONEST NEGATIVE: $\delta_{\text{top}}$ not a rational hypergraph fraction ( $\delta_{\text{top}}/\varphi_0^{\text{tree}} < 1\%$ ). [O] THE TARGET: prove $\Omega_{\text{adm}}c_3^4$ IS the order-4 boundary Seeley-DeWitt contact term of the $\mu_4$ puncture summed over the $\Omega_{\text{adm}}$ admissible states. [C] two engines, two terms; a FACE of the seam boundary analysis (the $c_3$ datum), not a new gate [E][C][O] |
| v409_coxeter_prime2_lemma.py          | RES.COXETER.SYMMETRY.01 closed as a STRUCTURAL LEMMA: prime-2 is NECESSARY but NOT autonomous – no prime-2-only attractor exists (resolves the v394/COXETER.ATOMS.01 open question's falsifiable corollary). [E] SHEET INVOLUTION $\sigma=\text{diag}(1, -1, -1)$ , $\sigma^2=I$ , spectrum $\{+1, -1\}$ (parity). [E] SHEET PROJECTOR $P=(I+\sigma)/2=\text{diag}(1, 0, 0)$ , $P^2=P$ , spectrum $\{0, 1\}$ (projection). [E] NO CONTRACTION RATE: none of $\{-1, 0, 1\}$ lies in $(0, 1)$ , whereas the genuine rates $2/3$ and $(1/3)^6$ do – prime-2 alone gives parity/projection, never a nonzero rate. [E] FACTOR NOT RATE: $2/3= Z_2 /N_{\text{fam}}$ needs prime-3, $(\varphi+2)/4=(\varphi+2)/ \mu_4 $ with $ \mu_4 =2^2$ needs the golden $\varphi$ (prime-5). [E] $30=g_{\text{car}}(2N_{\text{fam}})$ , carrier static, 2·3 dynamic. [C] prime-2 is the universal dynamical parity factor, necessary in every sheet-paired transport but not an autonomous attractor; dynamics begins after coupling to family-3 or carrier-5 ( $\mu_4=2^2$ the Galois bridge) [E][C]                                                                                            |
| v410_sheet_generator_binary.py        | The <i>sheet generator</i> $V=Q \text{diag}(0, 1, 1)$ (rank-2 sheet axis, v95/v218) is a binary internal compiler: $V^n=\begin{pmatrix} 0 & 2^{n-1} & 0 \\ 0 & 2^n & 0 \\ 0 & 2^{n+1}-2 & 1 \end{pmatrix}$ (induction), so $V^n \mathbf{1}=(2^{n-1}, 2^n, 2^{n+1}-1)=(1, 2, 3), (2, 4, 7), (4, 8, 15), (8, 16, 31)$ ; bilinears $\mathbf{1}^T V^n \mathbf{1}=7 \cdot 2^{n-1}-1$ , $\mathbf{1}^T V^n a=7 \cdot 2^{n-1}$ , $a^T V^n a=11 \cdot 2^{n-1}$ , $a^T V^n \mathbf{1}=11 \cdot 2^{n-1}-2$ give $(6, 7, 9, 11)$ then $13=\Delta_Q$ , $27=\mathbf{1}^T R a$ , $55, 56=\dim \mathbf{56}_{E_7}$ ; the spine is FORCED by $\text{Spec}(V)=\{0, 1, 2\}$ , the carrier (rank $E_8=8$ , $\dim S^+=16$ , $248/8=31$ ) and theta $(\sigma_3(2)=9=a^T V \mathbf{1}, \sigma_3(3)=28=\mathbf{1}^T V^3 a=\det(I+R), \sigma_3(5)=126)$ readings audit-typed [E][C]                                                                                                                                                                                                                                                                                                                     |
| v411_ud_ratio_vpower.py               | The quark $u/d$ ratio is a pure $V$ -power readout: the established $c_u/c_d=g_{\text{car}}\  \text{Pl}(K) \ _1/(N_{\text{fam}}^2 \Delta_Q)=55/117$ (v94) re-expresses EXACTLY via the sheet generator (v410) as $\boxed{c_u/c_d = \frac{\mathbf{1}^T V^4 \mathbf{1}}{(a^T V \mathbf{1})(\mathbf{1}^T V^2 \mathbf{1})}} = \frac{55}{9 \cdot 13}$ , with $55=\mathbf{1}^T V^4 \mathbf{1}$ , $9=a^T V \mathbf{1}=N_{\text{fam}}^2$ , $13=\mathbf{1}^T V^2 \mathbf{1}=\Delta_Q$ ; the same number as $5 \cdot 11/(9 \cdot 13)$ , so the numerator repackages $g_{\text{car}} \cdot 11=55$ . An exact <i>re-encoding</i> : the value and the $u/d$ identification stay [C] (U <sub>wall</sub> readout rigidity) [E][C]                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| v412_sheet_source_corner_J.py         | The <i>sheet source corner J</i> : the determinant surface has two $s$ -independent iso-volume walls (v135) $t=-1$ ( $\det=4= \mu_4 $ , carries $K$ ) and $t=-2$ ( $\det=2= Z_2 $ ); the operator on the $Z_2$ wall, $J=M(1, -2)=C-2V=\begin{pmatrix} 4 & 1 & 0 \\ 5 & 1 & 1 \\ 5 & 1 & 1 \end{pmatrix}$ , completes the sheet axis $J, K, C, F=M(1, -2..1)$ . $\chi_J=\lambda^3-6\lambda^2+3\lambda-2 \Rightarrow (\text{tr}, e_2, \det)=(6, 3, 2)=( R^+(A_3) , N_{\text{fam}},  Z_2 )$ ; $a^T J a=30=h(E_8)$ , $\det(I+J)=12=\dim \mathfrak{g}_{\text{SM}}=\chi_{E_8}(i)$ , $\det(2I+J)=40= R(D_5) $ (Lie readings audit-typed like the $G_2/F_4$ shadows of v95/v218) [E][C]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| v413_sheet_characteristic_calculus.py | The sheet axis is a discrete <i>characteristic-difference calculator</i> : on $M_t=C+tV$ (v95/v218) the elementary-symmetric coefficients encode atoms as DIFFERENCE ORDERS – $e_1=3t+12$ , $e_2=2t^2+15t+25$ , $e_3=\det M_t=4t^2+14t+14$ give $\Delta e_1=3=N_{\text{fam}}$ , $\Delta^2 e_2=4= \mu_4 $ , $\Delta^2 e_3=8=\text{rank } E_8$ (the $e_3$ curvature cross-cited to v218/v224; the full $(e_1, e_2, e_3)$ ladder is new); the anchor energy $a^T M_t a=52+11t$ (step $11=a^T V a$ , v410) reads $30 \rightarrow 41 \rightarrow 52 \rightarrow 63=(h(E_8), 10b_1, \dim F_4, 7N_{\text{fam}}^2)$ [E][C]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

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| Script                          | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v414_center_resolvent_portal.py | The center $C$ is a <i>resolvent portal</i> $G_2 \rightarrow F_4 \rightarrow E_8$ : $C=M(1,0)$ (v95) is the $t=0$ value of $V$ ; $\text{tr } C=12=\dim \mathfrak{g}_{\text{SM}}=\chi_{E_8}(i)$ , $\det C=14=\dim G_2$ , $\sum C=31=2^{g_{\text{car}}}-1=248/8$ ; the shifted determinants climb $\det(C)=14=\dim G_2$ , $\det(I+C)=52=\dim F_4$ , $\det(2I+C)=120= R^+(E_8) $ – the shell ladder $G_2 \rightarrow F_4 \rightarrow E_8^+$ as $\det(C+kI)$ , $k=0,1,2$ (the resolvent ladder is new; reading audit-typed) [E][C]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| v415_gaussian_operator.py       | The square CM-norm dictionary is <i>operator-realised</i> and the $\mu_4$ deck is an intrinsic diamond object: the Gaussian arithmetic of v222/v230 lifted from numbers to OPERATORS via $J=[U,V]/3$ , the complex structure born from the commutator of the binary $V$ and ternary $U$ compilers (v410/v95). [E] $J=\begin{pmatrix} -1 & 1 & 0 \\ -2 & 1 & 0 \\ -3 & 1 & 0 \end{pmatrix}$ is integer, $\text{Spec } J=\{i, -i, 0\}$ , $J^2=-I$ on the rank-2 image, $J^3+J=0$ (a $\mu_4$ quarter-turn); [E] $3I+2J$ has spectrum $\{3+2i, 3-2i, 3\}$ and $5I+4J$ has $\{5+4i, 5-4i, 5\}$ , so $3+2i$ ( $ \cdot ^2=13=\Delta_Q$ ) and $5+4i$ ( $ \cdot ^2=41=10b_1$ ) are literal operator eigenvalues; [E] $(3I+2J)(3I-2J)$ has $\text{Spec}\{13, 13, 9\}$ and $(5I+4J)(5I-4J)$ has $\{41, 41, 25\}$ (kernel slot = real part <sup>2</sup> : $9=N_{\text{fam}}^2$ , $25=g_{\text{car}}^2$ ); [E] the intrinsic deck $D=-I+J-J^2$ is order 4 ( $\text{Spec}\{i, -1, -i\}$ , $\chi=(x+1)(x^2+1)$ ) with $\chi_2$ line $\ker[U,V]=(0,0,1)=a-1$ . [C] the geometric-deck identification sharpens (does not close) the v140 $\mathbb{Z}_3$ residue [E][C]                                                                                                                                                                                   |
| v416_atom_trichotomy.py         | The atom trichotomy: $\{2, 3, 5\}$ are ramified/inert/split in the two exceptional CM rings, and each atom is the unique ramified prime of one quadratic facet of $K=\mathbb{Q}(i, \sqrt{-3}, \sqrt{5})$ (v403). [E] $\mathbb{Z}[i]$ (square/seam, $\tau=i$ ): $2= Z_2 $ ramifies ( $2=-i(1+i)^2$ ), $3=N_{\text{fam}}$ inert (norm $9=N_{\text{fam}}^2$ ), $5=g_{\text{car}}$ splits $((2+i)(2-i))$ – the carrier rank factors on the seam; [E] $\mathbb{Z}[\omega]$ (hex/flavor, $\tau=\omega$ ): $3=N_{\text{fam}}$ ramifies ( $N(1-\omega)=3$ ), 2 inert (norm 4), 5 inert (norm $25=g_{\text{car}}^2$ ); [E] the ramified prime of each ring IS its atom ( $\mathbb{Z}[i]$ at 2= sheet/seam, $\mathbb{Z}[\omega]$ at 3= family/flavor); [E] the three facets $\mathbb{Q}(i)/\mathbb{Q}(\sqrt{-3})/\mathbb{Q}(\sqrt{5})$ (disc $-4, -3, 5$ ) ramify over one atom each, product $2\cdot 3\cdot 5=30=h(E_8)$ ; 2 imaginary (CM/dynamic) + 1 real (RM/static). [C] the seam/flavor/golden reading [E][C]                                                                                                                                                                                                                                                                                                                              |
| v417_eisenstein_cp_operator.py  | The Eisenstein/CP operator (the hex dual of v415): the family rotation $P=(1\ 2\ 3)$ and the CP clock $W=-P^2$ put the $\tau=\omega$ flavor side on operators, unified with the seam into ONE order-30 clock. [E] $P^3=I$ , $P^2+P+I=\text{ONES}$ (Eisenstein $\omega^2+\omega+1=0$ on the plane $\perp (1,1,1)$ ); [E] $(3I+2P)(3I+2P^2)=7I+6\text{ONES}$ , $\text{Spec}\{7, 7, 25\}$ ( $7=N_{\mathbb{Z}[\omega]}(3+2\omega)$ = scalaron on the $\omega$ -plane, $25=g_{\text{car}}^2$ ) – the dual of v415's $\{13, 13, 9\}$ ; [E] the CP clock $W=-P^2$ ( $=\rho$ , v233) has $W^2=P$ , $W^3=-I$ , $\text{Spec}\{-1, \zeta_6, \zeta_6^{-1}\}$ , so $\delta_{\text{CKM,lead}}=\arg \zeta_6=\pi/3$ and $\delta_{\text{PMNS}}=\arg \zeta_6^4=4\pi/3$ ( $W^4=P^2$ ); [E] all flavor clocks are powers of $\zeta_{30}$ ( $\zeta_{30}^{15}=-1$ , $\zeta_{30}^{10}=\omega$ , $\zeta_{30}^6=\zeta_5$ , $\zeta_{30}^5=\zeta_6$ ) while the seam $\mu_4=i$ is the Galois side $(\mathbb{Z}/30)^\times$ of order $\varphi(30)=8=\text{rank } E_8$ . [C] the carrier $\mu_5$ has no clean $3\times 3$ operator (golden $\varphi$ an output-side trace, v313/v349); honest negatives included [E][C]                                                                                                                                              |
| v418_cyclotomic_norm_triple.py  | The cyclotomic norm triple: the carrier split $(3,2)$ read in the three atom-rings gives $(7, 13, 55)=(\text{scalaron}, \Delta_Q, \text{quark numerator})$ , and the missing carrier-5 clock of v417 is the $4\times 4$ $\Phi_5$ -companion $C_5$ . [E] $C_5$ has $\chi=x^4+x^3+x^2+x+1$ , $C_5^5=I$ , $\text{tr}=-1$ ; the golden ratio is its real-part data ( $\zeta_5+\zeta_5^4=1/\varphi$ , $\zeta_5^2+\zeta_5^3=-\varphi$ ; output-side v313/v349; $\dim 4$ , not 3 – refines v417's gap); [E] $N_{\mathbb{Z}[\zeta_5]}(3+2\zeta_5)=\det(3I+2C_5)=55=5\cdot 11=g_{\text{car}}$ $\  \text{Pl}(K) \ _1=$ the quark numerator (v410/v411); [E] the triple $\det(3I+2\text{Comp}(\Phi_n))=(7, 13, 55)$ for $n=3, 4, 5$ over $\mathbb{Z}[\omega], \mathbb{Z}[i], \mathbb{Z}[\zeta_5]$ (atoms 3, 2, 5, v416); [E] NEG control: the anchor $(5,4)\rightarrow(21, 41, 461)$ , with $21=N_\omega(5,4)$ and $41=10b_1$ named (v222) but the carrier ring 461 prime/un-named, so the carrier ring separates $(3,2)\rightarrow 55$ from $(5,4)\rightarrow 461$ . [C] HONEST: $(3,2)$ is FORCED (v14) not unique – the clean splits are $\{(2,1)\rightarrow(3,5,11), (3,2)\rightarrow(7,13,55)\}$ , a small ladder; the sharper reading: only $(3,2)$ spans three physics sectors (gravity/flavor/masses), the unique cross-sector seed [E][C] |

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| Script                        | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v419_seam_galois_carrier.py   | <p>The seam <math>\mu_4</math> is the carrier's Galois group: a positive resolution of the <b>v409/RES.COXETER.SYMMETRY.01</b> cyclic/Galois asymmetry. [E] <math>h(E_8)=30=2\cdot3\cdot5</math> is squarefree, so <math>\mathbb{Z}/30</math> has <i>no</i> order-4 element — the seam <math>\mu_4</math> cannot be a cyclic hand and is forced onto the Galois side; [E]</p> <p><math>(\mathbb{Z}/30)^\times = (\mathbb{Z}/2)^\times \times (\mathbb{Z}/3)^\times \times (\mathbb{Z}/5)^\times = \mu_4 \times \mathbb{Z}_2</math> (order <math>\varphi(30)=8=\text{rank } E_8</math>), where the <math>\mu_4</math> <math>(\mathbb{Z}/4)</math> factor IS <math>(\mathbb{Z}/5)^\times</math> (carrier 5) and the <math>\mathbb{Z}_2</math> IS <math>(\mathbb{Z}/3)^\times</math> (family/CP); [E] so the seam <math>\mu_4 = \text{Gal}(\mathbb{Q}(\zeta_5)) = (\mathbb{Z}/5)^\times</math>, the Frobenius <math>\zeta_5 \mapsto \zeta_5^2</math>, realised on the carrier clock <math>C_5</math> (<b>v418</b>) by the explicit order-4 operator <math>G</math> with <math>GC_5G^{-1}=C_5^2</math>, <math>G^4=I</math>, <math>G^2C_5G^{-2}=C_5^{-1}</math>. [C] resolves <b>v409</b> (the <math>2^2=4</math> is the order of <math>(\mathbb{Z}/5)^\times</math>; “no contraction rate” because the seam is an automorphism, not a hand); <math>\mathbb{Q}(\zeta_{30})</math> carries both <math>E_8</math> invariants (cyclic <math>h=30</math>, Galois rank=8). Figure <b>coxeter_galois</b> [E][C]</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| v420_fpole_qed_running.py     | <p>F_pole external: the Koide source→pole transfer is a <i>one-loop electromagnetic</i> effect, not an algebraic step — the honest external-physics follow-up to <b>v93</b> and the clock No-Go. [E] Koide <math>Q</math> is rescaling-invariant (<math>Q(cm)=Q(m)</math>), so flavor-<i>universal</i> running leaves it fixed and the source→pole shift is purely flavor-<i>split</i>; [C] the tree→pole gap <math>\frac{2}{3}-Q_{\text{src}}=0.00220</math> is one-loop-EM-sized (<math>=\varphi_0^{\text{ret}}/24</math> to 0.6%, <math>=\alpha/\pi</math> to <math>\sim 5\%</math>), the PDG pole sits AT <math>\frac{2}{3}</math> (the IR attractor); [C] a 1-loop QED per-lepton-log pole <math>\leftrightarrow \overline{\text{MS}}</math> estimate shifts Koide by EM-loop order toward <math>\frac{2}{3}</math> (the <b>v82</b> IR-attractor direction); [O] the EXACT flow time <math>t\sim 2.84</math> (<b>v93</b>) is scheme/scale-dependent external QED, only sized. [C] F_pole stays [C], firewalled (<b>v187</b>): the lens forces the RATE <math>(2/3)^6</math>, this external EM supplies the flow. Numerical (mpmath), Python-only [E][C][O]</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| v422_galois_net_bridge.py     | <p>The Galois<math>\leftrightarrow</math>Net bridge: the seam <math>\mu_4 = \text{Gal}(\mathbb{Q}(\zeta_5))</math> is the SAME cyclic <math>\mathbb{Z}/4</math> as the <math>(E_8)_1</math> simple-current glue — not a mere order-4 coincidence — linking the Galois gearbox of <b>v419</b> to <b>G<sub>net</sub>/SEAM.EQUIV.01</b>. [E] <math>\text{disc}(A_3)=\text{disc}(D_5)=\mathbb{Z}/4</math> (one Smith invariant factor 4 = cyclic, <math>\det=4</math>); <math>D_n</math> disc is <math>\mathbb{Z}/4</math> for <math>n</math> ODD, <math>\mathbb{Z}_2 \times \mathbb{Z}_2</math> for <math>n</math> EVEN, so the carrier <math>D_5</math> (rank <math>5=g_{\text{car}}</math>, odd) is cyclic while the control <math>D_8</math> (rank 8, even) is Klein; [E] <math>\text{Gal}(\mathbb{Q}(\zeta_5))=(\mathbb{Z}/5)^\times=\langle 2 \rangle</math> is cyclic order 4 (only the carrier prime 5 gives order 4); [E] in <math>\text{disc}(D_5 \oplus A_3)=\mathbb{Z}_4 \times \mathbb{Z}_4</math> (<math>16=\dim S^+</math>) the simple current <math>(1,1)</math> is order 4 (cyclic) with Lagrangian quotient <math>16/4^2=1 \Rightarrow (E_8)_1</math> (holomorphic, <b>v89/v154</b>). [E] NEG control: the Klein <math>\mathbb{Z}_2 \times \mathbb{Z}_2</math> / order-2 halfway <math>\langle (2,2) \rangle</math> gives <math>16/2^2=4= \text{disc}(D_8) =SO(16)</math> (<math>\det=4</math>), NOT <math>E_8</math> (<math>\det=1</math>) — cyclic <math>\mathbb{Z}/4 \rightarrow E_8</math>, Klein <math>\rightarrow D_8</math>. [C] the bridge: one cyclic <math>\mathbb{Z}/4</math> threads the seam deck <math>\mu_4</math> (<b>v419</b>), <math>\text{Gal}(\mathbb{Q}(\zeta_5))</math> and the <math>(E_8)_1</math> glue (<b>v1/v89/v154</b>); cyclicity forced by 5, Klein excluded. [C] this is the four-fold <math>\mu_4</math> identity (deck/divisor = Galois = discriminant = simple-current); per <b>v428</b> a <i>compression</i> (one object read four ways), not four independent witnesses [E][C]</p> |
| v423_no_ambient_dependency.py | <p>The <i>No-Ambient-Dependency</i> certificate: no frozen TFPT readout is <i>computed</i> from the ambient QG measure <b>QG.AMB.01</b> — sharpening the <b>v369/v384</b> prose redundancy into a machine-checked DAG+script fact, told honestly. [E] gap-decoupling margin <math>\Delta_{\text{eff}} = 6 \ln(3/2) - 31/(4\pi^2) \approx 1.648 &gt; 0</math> (<math>31=2^{g_{\text{car}}}-1</math>), susceptibility <math>\chi = 729/665</math> finite (neg control gap <math>\rightarrow 0 \Rightarrow \chi \rightarrow \infty</math>). [E] <b>QG.AMB.01</b> is a certification <i>sink</i>: every claim listing it as a dependency is a gravity-gate/QG row, none a frozen prediction. [E] it is not a value source — it has no producing script while each frozen readout is computed by a compiler script (<math>\alpha^{-1} \leftarrow \text{v3}</math>, <math>\theta_{12} \leftarrow \text{v9}</math>, <math>\theta_{13} \leftarrow \text{v268}</math>, cosmology <math>\leftarrow \text{v7}</math>). [E] the one transitive link is architectural: only <math>\theta_{13}</math> references <b>QG.AMB.01</b>, via the 4D-QFT contract spine (<b>PS.DIRAC</b>→<b>CONTRACT.QFT4D</b>→<b>QG.G2</b>→<b>GATE.METRIC</b>), a coherence edge not a numerical input; <math>\alpha^{-1}/\theta_{12}</math>/cosmology have no route. [C] so <b>QG.AMB.01</b> is gap-decoupled certification (<b>v369</b> as a DAG/script fact); honestly this does NOT prove <b>SEAM.EQUIV.01</b> — the readouts depend on the seam being <math>(E_8)_1</math>; <b>QG.AMB.01</b> specifically is dead [E][C]</p>                                                                                                                                                                                                                                                                                                                                                                                                                                         |

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| Script                         | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| v424_seam_lto_rp_reduction.py  | The ( <i>LTO-RP</i> ) reduction of the keystone: an honest MMST-style reduction of SEAM.EQUIV.01's one open lemma — the raw collar's intrinsic Bisognano–Wichmann property (the v308/v309 residual) — to the cited commuting-projector theorem of Naaijken–Penneys–Wallick (arXiv:2605.10693), with the central mapping <i>corrected</i> . [E] the $\mu_4$ clock is a modular <i>symmetry</i> ( $[\rho, K]=0$ , v309/v199), not the flow — $\sigma^\psi=\rho$ is type-wrong (a one-parameter group versus a finite order-4 element). [E] NPW26's axiom (LTO-RP) means $u_\Theta=J$ (reflection = modular conjugation) = the intrinsic BW condition $\theta K \theta = -K$ , <i>distinct</i> from $\omega \circ \rho = \omega$ : a positive character-block-diagonal $K$ satisfies $[\rho, K]=0$ yet never $\theta K \theta = -K$ , so clock-invariance does <i>not</i> imply (LTO-RP). [O] make-or-break: NPW26 prove (LTO-RP) for <i>topologically ordered</i> models (Toric Code $ \det K =4$ via a trace; Levin–Wen anyonic), while the seam is invertible ( $ \det \text{Cartan}(E_8) =1$ , no anyons) and a $\beta=1$ KMS state (not a trace) — in neither proved bucket (neg control $D_8$ $ \det =4$ ). [C] so SEAM.EQUIV.01's BW residual reduces to NPW26 Theorem A modulo two open sub-steps (realise the seam as a $\mathbb{Z}_2$ -reflection commuting-projector LTO; extend (LTO-RP) to the invertible KMS case) — a reduction, not a closure; SEAM.EQUIV.01 stays [O] [E][C] |
| v425_dyn_transfer_universal.py | DYN.TRANSFER.UNIVERSAL.01 — the <i>single-flow reduction</i> of $F_{\text{transfer}}$ : the four frontier transfers ( $F_{\text{pole}}$ , $F_{\text{Boltzmann}}$ , $F_{\text{relic}}$ , $F_{\text{QCD}}$ ) do not each need a separate external solver; their <i>dynamics</i> is the ONE native flow of the theory — the seam modular/recovery semigroup (v238/v240, gap $6 \ln(3/2)$ ) restricted to each sector. [E] recovery spectrum $\{1, (2/3)^6, (1/3)^6\}$ , $H_{OS} = -\log T$ gap $6 \ln(3/2)$ , $e^{-\Delta} = (2/3)^6$ exact; an explicit column-stochastic $T$ makes $\log T$ a Markov generator. [E] $F_{\text{pole}}$ is this flow: the v99 generator $\dot{q} = (\Delta/N_{\text{fam}})(q-2)(q-5)$ linearised at $q=2$ has rate $-\Delta$ , time-1 contraction $(2/3)^6$ . [E] $F_{\text{Boltzmann}}$ washout = that contraction, CP phase $4\pi/3$ native (v320). [E] $F_{\text{QCD}}$ rate native ( $b_3=-7$ ). [C] only the anchors ( $v_{\text{geo}}$ , $C_p$ , $M_R$ ) external; $F_{\text{relic}}$ 's cosmological IC $\theta_i$ the one wall. Extends v383 static→dynamic; a reduction, no new premise [E][C]                                                                                                                                                                                                                                                                                                                                                       |
| v426_seam_lto_rp_kms.py        | SEAM.EQUIV.LTORP.KMS.01 — the <i>invertible</i> $\beta=1$ -KMS variant of (LTO-RP), the corner NPW26 leave open (they prove it for topologically-ordered models; the seam is invertible + $\beta=1$ KMS, not a trace). A constructive Tomita–Takesaki witness advancing v424 sub-step (ii); does <i>not</i> close SEAM.EQUIV.01. [E] build $K$ reflection-odd/ $\mu_4$ -even/gapped (gap $2k=6 \ln(3/2)$ ); $C=1/(1+e^K)$ the $\beta=1$ KMS covariance, $C/(1-C)=e^{-K}$ (genuine KMS, not trace). [E] $\Theta K \Theta^{-1} = -K$ ( $u_\Theta=J$ ) AND $[\rho, K]=0$ both hold $\Rightarrow$ (LTO-RP) satisfied by direct Tomita–Takesaki. [E] $ \det \text{Cartan}(E_8) =1$ (invertible, no anyons) vs anyonic $ \det =4$ ; state not a trace ( $\Delta=e^{-K} \neq I$ ) — neither NPW26 bucket. [E] neg controls: side-even reflection $\rightarrow +K$ (BW fails), off- $\mu_4$ -block $K$ breaks $[\rho, K]$ . [C] (LTO-RP) reduces to $(\Theta K \Theta = -K) \wedge ([\rho, K]=0)$ , constructively satisfiable; SEAM.EQUIV.01 stays [O] [E][C]                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| v427_overdet_witness_map.py    | OVERDET.WITNESS.MAP.01 — the multiply-vs-compress <i>framework</i> for over-determination: witnesses from <i>disjoint</i> grammars multiply, projections of the <i>same</i> generator compress. The classification of the seven arithmetic witnesses is refined by v428 (they compress). [E] seven grammar readouts, each from its own syntax landing on the carrier skeleton: Gauss $N(3+2i)=13$ , Eisenstein $N(3+2\omega)=7$ , cyclotomy $N(3+2\zeta_5)=55$ , Galois $ (\mathbb{Z}/5)^\times =4$ , $ \det \text{Cartan}(E_8) =1$ , Pascal $\binom{4}{0} + \binom{4}{1} + \binom{4}{2} = 11$ ( $2^4=16 = \dim S^+$ ), Coxeter $h(E_8)=30=2 \cdot 3 \cdot 5$ with $\varphi(30)=8 = \text{rank } E_8$ . [E] by v236 these are facets of ONE $(2, 3, 5)/E_8$ object, so syntactic distinctness is NOT independence — they compress, like the anchor $a=(1, 1, 2)$ . [C] the genuine multiplicative strength is the multiply-forced inputs + foreign witnesses (v428); the map does not by itself establish physics [E][C]                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

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| Script                           | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| v428_overdet_reclass.py          | OVERDET.WITNESS.RECLASS.01 — the honest correction of v427: applying its own multiply-vs-compress test to itself. [E] the $(2, 3, 5)$ Brieskorn singularity (v236) is the ONE generator — Milnor $\mu=(2-1)(3-1)(5-1)=8=\text{rank } E_8$ , $30=2\cdot 3\cdot 5=h(E_8)$ , $\varphi(30)=8$ . [E] every one of v427's seven witnesses maps to a facet of that SAME object (primes 2, 3, 5 of 30, the clock 30, or $E_8$ ), so all seven collapse to ONE generator (distinct objects = 1) $\Rightarrow$ COMPRESSION, not seven multiplications. [E] what honestly multiplies is the input forced four independent ways — the “8” in $c_3=1/(8\pi)$ from rank $E_8$ , $h(D_5)=2(5-1)$ , $\varphi(30)$ and the Milnor number — plus the foreign witness $\alpha^{-1}\approx 137$ (prime, not in the $(2, 3, 5)$ -skeleton). [C] refines and partly walks back v427; consistent with v236/v305/v313 [E][C]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| v429_axion_pentagon_phi.py       | DM.AXION.PENTAGON.01 — the ‘unmapped’ icosahedral/golden $E_8$ structure surfaces in the ONE external cosmological input $\theta_i$ , answering “why does idle $E_8$ structure exist?” for the one case where it has a job. [E] because $N_{\text{fam}}=g_{\text{car}}-2$ , the spine quotient $\theta_i=\pi N_{\text{fam}}/g_{\text{car}}=(g_{\text{car}}-2)\pi/g_{\text{car}}$ is EXACTLY the interior angle of the regular $g_{\text{car}}$ -gon; for $g_{\text{car}}=5$ the PENTAGON, $\theta_i=3\pi/5=108^\circ$ (the carrier polygon, not a bare rational multiple of $\pi$ ). [E] its cosine is golden: $\cos(3\pi/5)=(1-\sqrt{5})/4=-1/(2\varphi)$ , partner $2\cos(2\pi/5)=1/\varphi=\varphi-1$ , $\varphi=2\cos(\pi/5)=(1+\sqrt{5})/2$ . [E] the golden character is UNIQUE to $g_{\text{car}}=5$ : among small $n$ -gons only $n=5$ has an irrational interior-angle cosine ( $n=3, 4, 6$ give $1/2, 0, -1/2$ ), so $\theta_i$ is golden BECAUSE the carrier is 5 (P2). [C] a re-reading bridging v354/v313 (the unmapped $g_{\text{car}}=5$ golden signature) to v211 ( $\theta_i=3\pi/5$ , the one external input): it REFINES v354 from “ $\varphi$ in no readout” to “ $\varphi$ touches only the [C] spine misalignment angle”, a geometric motive for the spine over the hilltop $\sim 170^\circ$ ; it does NOT upgrade DM.AXION.SPINE.01 (stays [C]) and closes no gate. [E][C]                                                                                                                                                                                                                                                               |
| v430_other_side_reverse_audit.py | E8.OTHERSIDE.AUDIT.01: the ‘other side of the seam’ (the double-cover deck / conjugate sheet $S^-$ ) audited against $E_8$ 's UNMAPPED Casimir degrees — the sheet/deck complement of v354/v355, answering “is the leftover where the second sheet lives?” with a disciplined NEGATIVE. [E] DECK = DEGREE-2: the one-sided cover contributes $ \mathbb{Z}_2 =2$ (the $1/2$ in $c_3=1/(2\cdot 4\pi)=1/(8\pi)$ , v58/v216); $2=\min(E_8 \text{ degrees})$ = the metric, one of the matched primary readouts $\{2, 8, 30\}$ (v354), NOT one of the unmapped $\{12, 14, 18, 20, 24\}$ . [E] TWO SHEETS = 128-SPINOR: $\dim S^+=\dim S^-=16$ , the spinor block $128=\text{rank} \cdot \dim S^+=8\cdot 16=2^{\text{rank}-1}=\sum \text{deg}$ , the spinor half of $248=120+128$ ; $S^-$ the conjugate $(\overline{16}, \overline{4})$ , $128=2\cdot 64$ , not a spare singlet. [E] FORCED-DISJOINT: the sheet/deck integer set $\{2, 16, 32, 128\}$ has EMPTY intersection with $\{12, 14, 18, 20, 24\}$ , its only degree-alphabet contact the matched 2. [O] DISCIPLINED DECLINE: each unmapped degree needs an UNFORCED coefficient from the sheet atoms and admits $\geq 2$ readings (the v355 discriminator) — no forced sheet $\rightarrow$ unmapped-degree identity. [E] ‘ $S^-=\text{DM}$ ’ DOWNGRADED + WIMP NO-GO: v227 replaced the over-strong ‘ $S^-$ is dark matter’; $128=(16, 4)+(\overline{16}, \overline{4})$ is the SAME matter spinor (no spare singlet), DM is the determinant-line axion (a phase). [O] the only live ‘other side’ question (the 128 phase/glue dark sector, v227) concerns MATCHED structure, not the unmapped degrees [E][O] |

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| Script                           | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| v431_e8_degree_ladder.py         | <p>E8.DEGREE.LADDER.01: the reverse-audit ‘unmapped’ <math>E_8</math> Casimir degrees are NOT diffuse overhead but the forced two-family decomposition <math>\deg(E_8)=6\cdot\text{spine}\{2, 3, 4, 5\} \sqcup (\{2\} \cup \det\text{-ladder}\{8, 14, 20\})</math> — the structural complement of v354/v355/v430. [E] TWO-FAMILY SPLIT FORCED BY <math>30=2\cdot3\cdot5</math>: the <math>E_8</math> exponents are the <math>\varphi(30)=8</math> totatives of <math>h=30=2N_{\text{fam}}g_{\text{car}}</math> (coprime to 6 <math>\Rightarrow \pm 1 \bmod 6</math>), so the degrees occupy ONLY the residue classes <math>\{0, 2\} \bmod 6</math>. [E] <math>6k</math> family <math>\{12, 18, 24, 30\}/6=\{2, 3, 4, 5\}</math> = the v91 spine <math>\{e_3, e_1, e_2, p_0\}(a=(1, 1, 2))</math>. [E] <math>6k+2</math> family <math>\{2, 8, 14, 20\}</math>: <math>\{8, 14, 20\}=(\det R, \det C, \det L)</math> = the winding line <math>\det M(s, 0)=6s+8</math> of the flavor diamond (v135), <math>C=R+Q \text{ diag}(1, 0, 0)</math>. [E] ‘<math>18=6N_{\text{fam}}</math>’ is NOT a holdout (spine family, not on the det-ladder). [E] SPECIAL TO <math>E_8</math> among the simply-laced exceptionals (<math>E_6/E_7/D_5</math> fail the clean <math>\{0, 2\} \bmod 6</math> split). [C] HONEST v355 LINE: the arithmetic decomposition is exact, but the FUNCTORIAL claim ‘<math>6k+2</math> Casimir stratum = flavor sheet operators’ needs a representation-theoretic map ([C]); 12, 24 admit two canonical readings each, so this is a STRUCTURE theorem, not a forced functorial flavor map — it reclassifies v354 overhead to zero orphan degrees, closes no gate [E][C]</p> |
| v432_overdet_floor.py            | <p>OVERDET.FLOOR.01: the UNCONDITIONAL improbability floor + the <math>L1 \leftrightarrow L2</math> bridge — the defensible number that survives even if a skeptic REJECTS the declared formula grammar, complementing the conditional null model (v100) and the multiply-vs-compress framework (v427/v428). [E] DISJOINT PIECES (v428): compression removed (v427’s seven readouts are one <math>(2, 3, 5)/E_8</math> object, NOT multiplied); what multiplies is the input ‘8’ forced FOUR ways <math>\{\text{rank } E_8, h(D_5)=2(g_{\text{car}}-1), \varphi(30), \text{Milnor}(2, 3, 5)\}</math> plus the FOREIGN witness <math>\alpha^{-1} \sim 137</math> (prime, outside <math>(2, 3, 5)</math>). [C] CONSERVATIVE FLOOR: multiplying ONLY across disjoint pieces (input over-determination <math>\times</math> the <math>\alpha</math> grammar census <math>1/94500</math> (v100) <math>\times</math> the <math>\varphi_0^{\text{ret}}</math>-seed cluster counted ONCE <math>\times</math> one downstream witness) the floor stays <math>\leq 10^{-6}</math> across a grid of generous assignments, nominal <math>\sim 10^{-10}</math> — ‘extraordinary’ WITHOUT the declared grammar. [E] THE BRIDGE: the floor (<math>\sim 10^{-10}</math>) sits <math>\sim 20</math> orders ABOVE the conditional <math>10^{-30.7}</math> (v100); that gap = the evidential value of proving the FORMS forced, so L1 (ALPHA.QUILLEN.EXACT.01, v382) and L2 (this floor) are the SAME lever. [C] HONEST SCOPE: bounds ONLY “is it numerology?”, NOT the physics bridge (firewall v187); a conservative FLOOR, not a posterior [E][C]</p>                                                        |
| v433_alpha_quillen_heatkernel.py | <p>ALPHA.QUILLEN.PROGRESS.02: a <i>second honest step</i> on the external target ALPHA.QUILLEN.EXACT.01 (v382) – it GROUNDS and CONNECTS, it does NOT close it; ALPHA.QUILLEN.EXACT.01 stays [O] (external math, type A, v384), <math>\alpha^{-1}</math> stays [E]. [E] a solvable 4D model (flat <math>\Delta = -\partial^2 + m^2</math> on <math>T^4</math>) reaches the <math>a_4</math> (<math>t^0</math>) conformal-anomaly order: <math>a_4 = Vm^4/(2(4\pi)^2)</math> NON-ZERO – the order v391’s 1D circle toy (<math>a_0, a_1</math> only) could not reach. [E] it is the order v342 grounds (Gilkey <math>a_4</math> gauge-curvature <math>30/360 = 1/12</math>), unifying v391 (a <math>\zeta</math>-det variation IS heat-kernel data) with v342 (the term IS the Gilkey <math>a_4</math>). [E] <math>(4\pi)^{-2} = 4c_3^2</math> is <math>c_3</math>-built. [E] the v382 split re-verified at <math>\alpha^{-1} = 137.0359992</math>. [O] NOT CLOSED: the actual seam <math>a_4</math> (<math>= 8b_1c_3^6</math>) and the cubic <math>\alpha^3</math> Chern moment stay open (v384); the exact seam <math>F</math>-normalisation stays [C] (v342). Grounds/connects, does not close [E][O]</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

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| Script                             | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| v434_alpha_quillen_betafunction.py | <p>ALPHA.QUILLEN.PROGRESS.03: a <i>third honest step</i> on ALPHA.QUILLEN.EXACT.01 (v382) – it settles the three residuals named after v433 and shows they are NOT three independent problems; the target stays [O], <math>\alpha^{-1}</math> stays [E]. [E] <math>b_1</math> IS THE U(1) <math>a_4</math> HEAT-KERNEL COEFFICIENT (the <math>\beta=a_4</math> theorem): from the carrier hypercharge content with spin weights (<math>\frac{2}{3}</math> Weyl, <math>\frac{1}{3}</math> complex scalar), <math>\sum \frac{2}{3}Y_f^2 + \frac{1}{3}Y_s^2 = 41/6</math> (SM), <math>(3/5) \cdot \text{that} = 41/10 = b_1</math> (GUT) – the <math>b_1</math> in <math>8b_1c_3^6</math> is the <math>a_4</math> content coefficient, not a fitted multiplicity (same number as the <math>\beta</math>, v159/v246; same KIND as the <math>\alpha^2</math> Calderón <math>a_4</math>, v342). [E] <math>8b_1c_3^6 = b_1 \cdot (\text{rank } E_8) \cdot c_3^6</math> (rank 8 = <math>\chi=2</math> boundary, v216; <math>c_3^6</math> six insertions, v342). [C] residual (1) “exact <math>a_4 = 8b_1c_3^6</math>” <math>\equiv</math> residual (3) “seam <math>F</math>-normalisation” – ONE obligation. [E] residual (2) located: <math>a^3 - 2c_3^3a^2 = a^2(a - 2c_3^3)</math>, so <math>a^3</math> is the first moment of the Chern density (a level, order <math>c_3^0</math>); origin stays [O]. [E] net reduction: the EM-Ward residual is <math>1[C]+1[O]</math>, not three. [O] NOT CLOSED; <math>\alpha^{-1}</math> stays [E]. Python-only [E][C][O]</p>        |
| v435_alpha_quillen_chernlevel.py   | <p>ALPHA.QUILLEN.PROGRESS.04: a <i>fourth honest step</i> on ALPHA.QUILLEN.EXACT.01 (v382) – it attacks the SINGLE remaining [O] after v434 (residual (2), the cubic <math>\alpha^3</math> Chern level) and SHARPENS it, it does NOT close it; the target stays [O], <math>\alpha^{-1}</math> stays [E]. [E] <math>\pi</math>-POWER / METRIC-INDEPENDENCE TEST: the three closure coefficients carry <math>\pi</math>-powers exactly <math>\{0, 3, 6\} = c_3</math>-orders (<math>k_0 = 1 \rightarrow \pi^0</math>; <math>-2c_3^3 = -1/(256\pi^3) \rightarrow \pi^3</math>; <math>-8b_1c_3^6 = -41/(327680\pi^6) \rightarrow \pi^6</math>), so the Maxwell <math>a^3</math> is the UNIQUE <math>\pi^0/c_3^0</math> (metric-INDEPENDENT) term – the signature of a topological level (a heat-kernel boundary coeff always carries the <math>(4\pi)^{-d/2}</math> measure; a level carries none). The test partitions the closure into ONE topological level (<math>a^3</math>) + TWO heat-kernel boundary terms, confirming v342/v434’s “<math>a^3</math> is a level, not a curvature integral” from a test. [C] conditional on the CS/<math>\eta</math> reading (EM1, v48), large-gauge invariance forces the level in <math>\mathbb{Z}</math> and <math>k_0 = 1</math> is the unit (minimal) level – an integer, not a free real. [O] NOT CLOSED: the EM1 CS-boundary derivation stays external (type A, v384); seam <math>F</math>-norm stays the one [C] (v434); <math>\alpha^{-1}</math> stays [E]. [E][C][O]</p>                                                 |
| v436_overdet_floor2.py             | <p>OVERDET.FLOOR.02: HARDENS OVERDET.FLOOR.01 (v432) — the “is it numerology?” floor does NOT rest on v432’s subjective chance assignments. [E] ASSUMPTION-MINIMAL FLOOR: the <math>\alpha</math> cubic-class census (v100 layer D) is PURE COUNTING — of <math>N=94500</math> complexity-matched <math>F_{U(1)}</math> variants only ONE hits the <math>4 \times 10^{-8}</math> CODATA window (the TFPT instance, <math>\alpha^{-1}=137.0359992</math>), so <math>p_\alpha \leq 1/94500 = 1.06 \times 10^{-5} \sim 4.40\sigma</math> with NO subjective probability, prior or multi-observable grammar. [E] MONOTONE CONCESSION LADDER: <math>-\log_{10} p</math> full scorecard+<math>\alpha \sim 30.7 \rightarrow</math> scorecard-only <math>\sim 25.8 \rightarrow \alpha</math>-census-only <math>\sim 4.98</math> (<math>= 4.40\sigma</math>); the “not chance” verdict holds at the MOST adversarial rung. [E] INDEPENDENCE: strip ALL of v432’s subjective factors (<math>0.1^3 \times 0.012 \times 0.15 = 1.8 \times 10^{-6}</math>) and the floor is still <math>4.40\sigma</math> — they only sharpen it. [C] HONEST <math>5\sigma</math> GAP: <math>5\sigma</math> (<math>p=5.7 \times 10^{-7}</math>) is crossed by the census times exactly ONE more independent factor <math>\leq 0.054</math> (the input-‘8’ four-fold over-determination or one foreign witness), stated openly. [C] FIREWALL: bounds numerology only, not the physics bridge (v187); Python-only, deliberately NOT Wolfram-mirrored (a counting bound, not an identity). [E][C]</p> |

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| Script                     | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| v437_e8_degree_joint.py    | <p>E8.DEGREE.JOINT.01: a CONSOLIDATION of v6/v66/v355/v431 deriving more from the <math>E_8</math> Casimir degrees WITHOUT crossing the v354/v355 anti-numerology line — no new per-degree coincidence, only a joint lock. [E] <math>E_8</math> IS FIXED BY ITS DEGREE MULTISSET: from <math>\deg(E_8)=\{2, 8, 12, 14, 18, 20, 24, 30\}</math> alone — <math>\text{rank}=\#\deg=8</math>, <math>h=\max=30</math>, <math>\#\text{pos roots}=\sum \exp=120</math>, <math>\#\text{roots}=240</math>, <math>\dim=248</math>, <math> W =\prod \deg=696,729,600</math>, <math>\sum \deg=128=2^7=\dim S^+</math>. [E] THE JOINT QUADRATIC: the two structural integers <math>(g_{\text{car}}, N_{\text{fam}})</math> are the UNIQUE root pair of <math>x^2 - (\text{rank } E_8)x + (h/2) = x^2 - 8x + 15 = (x-3)(x-5)</math> — <math>\text{sum}=\text{rank } E_8=8</math> (<math>\text{rank-fill}, v6</math>), <math>\text{product}=h/2=15=g_{\text{car}}N_{\text{fam}}</math> (<math>2g_{\text{car}}N_{\text{fam}}=h, v431</math>); <math>\{3, 5\}</math> the unique factor pair of 15 summing to 8, so BOTH are forced together by two degree-invariants. [E] the three mapped degrees <math>\{2, 8, 30\} = \min/\text{rank}/\max</math> (the canonical invariants); <math>g_{\text{car}}=5=\max</math> prime of 30. [E] DISCIPLINE: no new per-degree mining; the five non-primary degrees keep their v431 homes. Exact integer/lattice, mirrored in wolfram. [E][C]</p>                                                                                                                                                                                                      |
| v438_seam_hsmi_borchers.py | <p>SEAM.EQUIV.BW.HSMI.01: the half-sided modular inclusion (Borchers/Wiesbrock) as an INTRINSIC Bisognano–Wichmann route for face (ii) of SEAM.EQUIV.01 — geometric modular flow WITHOUT presupposing covariance. [E] STANDARD-PAIR ALGEBRA (ladder, exact): boost <math>K=\text{diag}(n)</math> and translation <math>P</math> (raising shift) satisfy <math>[K, P]=P</math> and the Borchers scaling <math>\Delta^{it}P\Delta^{-it}=e^{-t}P</math> (the <math>ax+b/\text{Moebius}</math> relation). [E] BW REFLECTION: <math>J(n\rightarrow -n)</math> gives <math>J^2=I</math>, <math>JKJ=-K</math> (<math>u_\Theta=J</math>) and <math>JPJ=P_-</math> (the opposite-chirality translation), the standard-pair reflection <math>JU(s)J=U(-s)</math>. [E] POSITIVE ENERGY: in the unitary lowest-weight <math>sl(2, \mathbb{R})</math> rep the line translation <math>P_{\text{line}}=L_0-(L_++L_-)/2</math> is POSITIVE (<math>\min \text{eig} &gt; 0</math>), the positive-energy generator the <math>c_3</math> one-sided origin demands; the boost direction is the indefinite negative control. [C] SYNTHESIS: <math>P\geq 0 + [K, P]=P + J</math> is a Longo STANDARD PAIR / +half-sided modular inclusion (Wiesbrock), so by Borchers the seam modular flow is geometric INTRINSICALLY (complementary to Adamo–Giorgetti–Tanimoto, arXiv:2508.07109); face (ii) becomes a STRUCTURE consequence, not a state condition. [C][O] assembling the finite algebra and the positive-energy rep into ONE continuum standard pair is the continuum existence (the same wall as MMST/NPW26, v336/v424); SEAM.EQUIV.01 stays [O]. Python-only (numpy/scipy). [E][C][O]</p> |
| v439_seam_lieb_robinson.py | <p>SEAM.EQUIV.CONTINUUM.LR.01: the Lieb–Robinson/Nachtergaele–Sims continuum-existence MOTOR for the gapped seam collar — grounds the BULK half of the cited MMST existence (v336) in the explicit v367 <math>p+ip</math> model. [E] UNIFORM GAP: <math>\Delta(N)=\text{const}&gt;0</math> across <math>N=20..160</math> (the lattice image of the OS transfer gap <math>6\log\frac{3}{2}&gt;0, v302/v329</math>). [E] EXPONENTIAL CLUSTERING: <math> \langle c_i^\dagger c_j \rangle  \sim e^{-r/\xi}</math> with <math>\xi</math> finite and STABLE across <math>N</math> (gapped <math>\Rightarrow</math> Hastings–Koma decay), the <math>N</math>-independent locality. [E] LIEB–ROBINSON LIGHT CONE: the single-particle propagator <math> (e^{-iHt})_{ij} </math> is confined to <math> i-j  \lesssim v_{\text{LR}}t</math> with finite <math>v_{\text{LR}}</math> (<math>\sim 2t</math>), exponentially small beyond the cone. [C] NACHTERGAELE–SIMS LIFT: uniform gap + finite LR velocity <math>\Rightarrow</math> exponential clustering with <math>N</math>-independent <math>\xi</math> AND existence+uniqueness of the thermodynamic-limit ground state with a finite light cone — derived for OUR collar, not imported. [C][O] the bulk thermodynamic limit EXISTS by Lieb–Robinson; SEAM.EQUIV.01 stays [O], the open analytic step narrowed to the CHIRAL massless EDGE scaling limit (the <math>c=8</math> <math>(E_8)_1</math> CFT), distinct from the bulk limit. Python-only (numpy/scipy). [E][C][O]</p>                                                                                                                                             |

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| Script                            | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| v440_seam_lto_rp_beta.py          | SEAM.EQUIV.LTORP.BETA.01: the $\beta$ -interpolation connecting the NPW26-PROVEN trace case ( $\beta=0$ ) to the seam's $\beta=1$ KMS case along an obstruction-free homotopy — upgrading the single-point v426 witness for v424 sub-step (ii) to a PATH argument. [E] $\beta$ -INDEPENDENT (LTO-RP) DATA: $\Theta K \Theta = -K$ ( $u_\Theta = J$ ) and $[\rho, K]=0$ are conditions on $K$ alone, $\beta$ -independent. [E] THE KMS PATH: $C_\beta = 1/(1+e^{\beta K})$ has $\beta=0 \rightarrow C=\frac{1}{2}I$ (the NPW26 TRACE state), $\beta=1 \rightarrow$ the seam KMS state, $\beta \rightarrow \infty \rightarrow$ the ground-state projection; at every $\beta$ $\Theta C_\beta \Theta = 1-C$ (reflection), $[\rho, C_\beta]=0$ (mu4), $0 \leq C_\beta \leq 1$ (valid covariance/RP). [E] NO OBSTRUCTION: $\beta \mapsto C_\beta$ is norm-continuous and stays in the (LTO-RP)-valid set — a continuous homotopy from the proven trace endpoint to the seam KMS point, no topological wall. [E] NEG CONTROLS: a reflection-EVEN $K'$ breaks $\Theta C_\beta \Theta = 1-C$ for $\beta>0$ (the trace endpoint is degenerate); an off-mu4-block $K$ breaks $[\rho, C_\beta]$ . [C][O] a PATH argument for v424 sub-step (ii), stronger than the single point of v426; SEAM.EQUIV.01 stays [O] (closedness of (LTO-RP) along the path in the continuum is the residual). Python-only (numpy/scipy). [E][C][O]                                                                                                                    |
| v441_seam_applicability_ledger.py | SEAM.EQUIV.APPLICABILITY.01: the applicability ledger — a hypothesis-by-hypothesis audit of the THREE external theorems the seam keystone cites (MMST scaling limit, NPW26 (LTO-RP), Adamo–Giorgetti–Tanimoto covariance), isolating the SINGLE imported analytic fact. [E] MMST HYPOTHESES: $D=16$ Majoranas ( $=2^{g_{\text{car}}-1}$ ), $c=8$ ( $=g_{\text{car}}+N_{\text{fam}}$ ), rank $E_8=8$ , range rank $\leq c \leq D$ reads $8 \leq 8 \leq 16$ , gapped ( $6 \log \frac{3}{2} > 0$ ), quasi-free CAR — all internal computed facts. [E] NPW26: (LTO-RP)= $u_\Theta=J$ , gap, $\mathbb{Z}_2$ reflection, mu4 clock, $[\rho, K]=0$ internal (v424/v426/v440); the seam is invertible $ \det \text{Cartan } E_8 =1 + \beta=1$ KMS, OUTSIDE NPW26's topological/trace bucket ( $ \det =4$ ) — a TEMPLATE, not a direct import. [E] ADAMO–GIORGETTI–TANIMOTO (arXiv:2508.07109): DHR reps of nets extending loop-group/Virasoro are AUTOMATICALLY positive-energy + diff-covariant; the seam net extends $(D_5)_1 \times (A_3)_1$ (v154), so covariance is a CONSEQUENCE — the v215 circularity is defused, covariance downgraded ASSUMPTION $\rightarrow$ [E]. [E] of 12 audited hypotheses, 11 are internal [E] and exactly 1 external — the continuum EXISTENCE of the chiral massless scaling limit (MMST/v336, Lean FORM.SEAM.MMST.01). [C][O] external dependence pinned to ONE analytic fact; SEAM.EQUIV.01 stays [O]. Python-only (a structural/arithmetic audit; the dets Wolfram-mirrored via v89/v281/v422). [E][C][O] |
| v442_seam_rigidity_uniform.py     | SEAM.RIGIDITY.UNIFORM.01: the uniform-in- $N$ hardening of the Seam State Rigidity Theorem (v398) — the band-limited rigidity is NOT a small- $N$ artifact. [E] EXACT UNIFORM 4-SECTOR COMMUTATION: $\rho = \text{diag}(i^n)$ has EXACTLY 4 distinct eigenvalues $\{1, i, -1, -i\}$ for all $N$ , and a genuinely character-block-diagonal $K$ (random Hermitian within each $n \bmod 4$ class, not merely diagonal) has $[\rho, K]=0$ EXACTLY for $N=4..256$ (extends v398's $N \leq 64$ ). [E] UNIFORM GAP: $\Delta(N) = \text{const} > 0$ across $N=20..160$ (the Nachtergaele–Sims uniform gap, shared with v439). [E] $N$ -INDEPENDENT RIGIDITY BOUND: a FIXED local clock-breaking perturbation (single off-mu4-block Hermitian entry) gives $\ [\rho, V]\ =2$ EXACTLY for all $N$ — a uniform lower bound; any fixed-size clock-breaking is detected with $N$ -independent strength. [E] NEG CONTROL: a clock-PRESERVING (same-sector) perturbation gives $\ [\rho, V']\ =0$ ; an order-2 clock has only 2 distinct eigenvalues — the ORDER of the clock (the four Gauss–Bonnet marks) is the discriminator, uniformly in $N$ . [C][O] the band-limited rigidity of v398 is uniform in $N$ ; SEAM.RIGIDITY.01/SEAM.EQUIV.01 stays [O]: that RP+gap+holomorphy FORCE $\Lambda_\Sigma$ block-diagonal on the full $L^2$ (vs merely permit) is the cited rigidity/MMST step. Python-only (numpy). [E][C][O]                                                                                                                         |

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| Script                        | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| v443_seam_e8_discriminator.py | <p>SEAM.E8.DISCIMINATOR.01: the empirical/structural DISCRIMINATOR bundle (mode D of the v347 closure-mode classification) — concrete FALSIFIABLE kill tests separating the claimed seam <math>= (E_8)_1</math> from the nearest <math>c=8</math> rival <math>SO(16)_1=(D_8)_1</math>. [E][X] PRIMARY/GSD: <math> \det \text{Cartan}(E_8) =1</math> (ONE primary, holomorphic) vs <math> \det \text{Cartan}(D_8) =4</math> (FOUR primaries <math>1, v, s, c</math>); torus GSD <math>= \# \text{primaries} = 1</math> vs <math>4</math> — a seam with <math>\text{GSD} &gt; 1</math> falsifies <math>(E_8)_1</math>. [E][X] CURRENT/THETA: level-1 current count <math>\dim E_8=248</math> vs <math>\dim SO(16)=120</math>; the <math>E_8</math> lattice theta has 240 norm-2 roots (<math>= 112</math> integer +128 spinor, computed) vs <math>D_8</math>'s 112 — current spectrum and lattice theta DIFFER. [E] <math>c=8</math> IS NOT A DISCRIMINATOR: both have <math>c=8</math> (the v277 ambiguity); only the order-4 <math>\mu_4</math> condensation + <math>\det K=1</math> (the genus-1 statement, v90) selects <math>E_8</math>. [E] NON-VACUITY: the discriminators all fire on the rival (<math>\det 4 \neq 1</math>, currents <math>120 \neq 248</math>, roots <math>112 \neq 240</math>). [X][C] mode D realised as concrete kill tests; SEAM.EQUIV.01 stays [O] (the discriminators distinguish the TARGET but do not PROVE the seam realises it). Python-only (numpy; the dets Wolfram-mirrored via v89/v281/v422). [E][C][O][X]</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| v444_seam_edge_scaling.py     | <p>SEAM.EQUIV.CONTINUUM.EDGE.01: the CHIRAL EDGE scaling-limit MOTOR — the edge half of the one SEAM.EQUIV.01 continuum residual, attacking exactly the step v439 left open. On the SAME explicit v367/v368 <math>p+ip</math> collar it shows the lattice edge carries the kinematic AND conformal data of a chiral <math>c=\frac{1}{2}</math> Majorana CFT, converging under refinement. [E] CHIRAL LINEAR EDGE DISPERSION: the in-gap branch is linear near its zero-crossing (top <math>v=0.983</math>, <math>R^2=0.9999</math>) and the two boundaries carry OPPOSITE velocities — one chiral Majorana per edge (relativistic, chiral kinematics). [E] ALGEBRAIC EDGE TWO-POINT: the equal-time edge correlator decays as <math>r^{-\eta}</math> with <math>\eta=1.00</math> (the chiral free-fermion CFT exponent) — a CRITICAL edge. [E] EXPONENTIAL BULK (neg control): the bulk-row correlator decays exponentially (<math>\xi \sim 1.1</math>) — only the edge is critical, so the power law is the edge CFT, not a geometric artefact. [E] TRIVIAL-PHASE NEG CONTROL: <math>M=3</math> (no edge, v368) has an exponential edge-row — the algebraic decay is the topological edge, not the boundary geometry. [E] SCALING-COLLAPSE / CONVERGENCE: <math>\eta</math> is <math>N</math>-stable (<math>1.001 \pm 0.007</math>) and the edge correlator profile is Cauchy-convergent under refinement (successive spread shrinks <math>0.028 \rightarrow 0.013</math>, <math>N_x=36..60</math>) — an existing scaling limit, the edge analogue of v439's bulk thermodynamic limit. [C] EDGE CENTRAL CHARGE: <math>\eta=1 \Leftrightarrow</math> one chiral Majorana (<math>c=\frac{1}{2}</math>) per edge, so the <math>N_{\text{Maj}}=16</math>-copy carrier has a chiral <math>c_-=8</math> edge CFT (<math>= (E_8)_1</math>, <math>c_-=g_{\text{car}}+N_{\text{fam}}</math>). [C][O] grounds the EDGE half of the cited MMST scaling limit (v336) the way v439 grounds the bulk; with both halves the one continuum residual is grounded and pinned to a single cited theorem; SEAM.EQUIV.01 stays [O] (the rigorous uniform-convergence theorem is not supplied). Python-only (numpy). [E][C][O]</p> |

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| Script                        | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| v445_seam_rigidity_forcing.py | <p>SEAM.RIGIDITY.FORCING.01: the rigidity FORCING theorem — upgrades v442’s “block-diagonal PERMITS commuting” to “commuting with the order-4 clock FORCES block-diagonal”, isolating the single remaining physical premise. [E] FORCING (commute <math>\Leftrightarrow</math> block-diagonal, EXACT): for <math>\rho = \text{diag}(i^n)</math>, <math>[\rho, K]=0</math> iff <math>K_{ij}=0</math> whenever <math>i \not\equiv j \pmod 4</math> — verified ENTRYWISE, swept <math>N=4..128</math> (the converse of v442). [E] EXACT COMMUTANT DIMENSION: <math>\dim\{K : [\rho, K]=0\} = \sum_s n_s^2</math> (the four sector sizes), verified as the <math>\text{ad}_\rho</math> nullspace; a PROPER subspace of the <math>N^2</math> algebra (<math>N=16</math>: <math>64 &lt; 256</math>) — a genuine quantitative restriction. [E] OFF-BLOCK <math>\Leftrightarrow</math> NON-COMMUTING: every off-block matrix unit <math>E_{ij}</math> has <math>\ [\rho, E_{ij}]\  =  i^i - i^j  \geq \sqrt{2} &gt; 0</math> — zero slack. [E] THE ORDER IS THE DISCRIMINATOR: the order-4 clock gives 4 sectors (<math>\dim \sum n_s^2(4)</math>), the order-2 clock <math>\text{diag}((-1)^n)</math> a STRICTLY LARGER commutant (<math>N=16</math>: <math>64</math> vs <math>128</math>) — the four Gauss–Bonnet marks (order <math>4= \mu_4 =h(A_3)</math>) force strictly more structure than two; only the index-4 extension is <math>(E_8)_1</math> (<math>\det 1</math>; v281/v154/v351). [E] NONDEGENERACY: exactly 4 distinct eigenvalues <math>\{1, i, -1, -i\}</math> for all <math>N \geq 4</math>. [C] SYNTHESIS (force, not permit): commuting with the order-4 <math>\mu_4</math> clock is EQUIVALENT to <math>\mu_4</math>-block-diagonality (iff, exact, uniform in <math>N</math>), so RP-definability (v54/v398) + clock-invariance FORCE the block-diagonal <math>\Lambda_\Sigma</math> — v442’s “permit” upgraded to “force”. [C][O] the single residual is now precisely “transfer in commutant” (clock-invariance = QGEO.SYM.01, v323/v335); SEAM.RIGIDITY.01/SEAM.EQUIV.01 stays [O]. Python-only (numpy; the det facts Wolfram-mirrored via v89/v281/v422). [E][C][O]</p> |
| v446_seam_clock_invariance.py | <p>SEAM.RIGIDITY.CLOCKINV.01: DERIVING clock-invariance (“transfer in the commutant”) from RP-symmetry, closing v445’s single residual one notch further — the residual drops from an OPERATOR commutation to the STRUCTURAL <math>\mu_4</math>-symmetry of the seam RP data. [E] OS RECONSTRUCTION INHERITS THE SYMMETRY: for a <math>\mu_4</math>-invariant RP covariance <math>([\rho, C(t)]=0)</math>, the OS canonical transfer <math>\Lambda=C(1)C(0)^{-1}=e^{-h}</math> (the transfer FIXED by RP+reflection, v54) has <math>[\rho, \Lambda]=0</math> — swept <math>N=4..32</math> over random block-diagonal <math>h</math>. [E] SHARPNESS / NEG CONTROL: a <math>\mu_4</math>-BREAKING covariance gives <math>[\rho, \Lambda] \neq 0</math> — iff-sharp, so the inheritance is non-vacuous. [E] CONTRACTION / OS-VALIDITY: <math>\Lambda</math> has spectrum in <math>(0, 1]</math> (a positive contraction, the physical canonical transfer). [E] MODULAR (Tomita–Takesaki): the free modular Hamiltonian <math>K=\log((1-C)C^{-1})</math> (the v258 formula) of the <math>\mu_4</math>-invariant state has <math>[\rho, K]=0</math> (the clock commutes with modular flow), the breaking control <math>[\rho, K] \neq 0</math>. [C] SYNTHESIS: <math>[\rho, \Lambda_\Sigma]=0</math> is a CONSEQUENCE of <math>\mu_4</math>-invariance of the seam RP data (the four Gauss–Bonnet marks, v216/v323) via the OS reconstruction (v54) + the modular fact (v258), NOT an extra assumption; with v445 (commute <math>\Leftrightarrow</math> block-diagonal), RP-definability PLUS the marks FORCE the block-diagonal <math>\Lambda_\Sigma</math>. [C][O] the residual reduces from an operator commutation to “the seam RP data is <math>\mu_4</math>-symmetric” (QGEO.SYM.01); the continuum Tomita–Takesaki implementation on the full <math>L^2</math> is the cited backbone. SEAM.RIGIDITY.01/SEAM.EQUIV.01 stays [O]. Python-only (numpy/scipy). [E][C][O]</p>                                                                                                                                                                                                                  |

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| Script                      | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| v447_seam_edge_chern.py     | <p>SEAM.EQUIV.EDGE.CHERN.01: an INDEPENDENT (topological) reading of the edge <math>c_- = 8</math>, corroborating v444's correlator route via the bulk-edge correspondence on the same v367/v368 <math>p+ip</math> collar. [E] INTEGER CHERN NUMBER: the <math>M=1</math> collar lower band has Chern <math>C=+1</math> (the Fukui-Hatsugai-Suzuki lattice link-variable method, an EXACT integer, no fit) — one chiral channel per copy. [E] TRIVIAL-PHASE NEG CONTROL: <math>M=3</math> (v368) has <math>C=0</math> (no chiral edge), and <math>M=-1</math> gives <math>C=-1</math> (opposite chirality, matching v444's opposite per-edge velocities). [E] BULK-EDGE CORRESPONDENCE: the open strip carries EXACTLY <math> C =1</math> in-gap edge branch crossing zero per boundary in the <math>M=1</math> phase, 0 in the trivial phase — the chiral edge channel count = <math> C </math>. [E] LI-HALDANE: the half-cut bulk ground state's single-particle entanglement spectrum has in-gap modes at <math>1/2</math> in the topological phase and NONE in the trivial — the entanglement edge tracks the physical edge. [C] EDGE CENTRAL CHARGE: <math>c_- = N_{\text{Maj}} \cdot \frac{1}{2} \cdot  C  = 16 \cdot \frac{1}{2} \cdot 1 = 8 = g_{\text{car}} + N_{\text{fam}}</math>, from the Chern integer + bulk-edge correspondence — the SAME <math>c_- = 8</math> as v444's correlator route, from a logically independent (topological) observable. [C][O] corroborates v444's edge <math>(E_8)_1</math> reading; SEAM.EQUIV.01 stays [O] — the topology fixes <math>c_-</math> and the channel count but not the cited continuum existence theorem. Python-only (numpy). [E][C][O]</p>                                                                                                                                                                           |
| v448_seam_edge_fourpoint.py | <p>SEAM.EQUIV.EDGE.FOURPOINT.01: pinning the ONE external MMST fact empirically — the edge FOUR-point function is quasi-free and conformally convergent, extending v444's two-point scaling limit to the next order. [E] QUASI-FREE / WICK (CAR hypothesis): the ground-state correlation matrix is an exact projector <math>G^2=G</math> (<math>\sim 10^{-13}</math>), so the state is quasi-free and all <math>n</math>-point functions are Wick/Pfaffian sums of the two-point — the MMST quasi-free CAR hypothesis (ledger entry 5) literally holds on the collar (structural: the BdG state is free). [E] FOUR-POINT CAUCHY CONVERGENCE: the edge four-point cross-ratio <math>Q=[G_{01}G_{23}]/[G_{02}G_{13}]</math> at COMMENSURATE positions <math>(1, 2, 4, 7) \cdot (N_x/12)</math> is Cauchy-convergent under refinement (successive spread shrinks, <math>N_x=36..60</math>) — the four-point converges, extending v444's two-point scaling-collapse. [E] CONFORMAL FOUR-POINT FORM: <math>Q</math> converges to the free-fermion CONFORMAL cross-ratio <math>\sin(5\pi/12)/\sin(\pi/12)=2+\sqrt{3} \approx 3.732</math> (chord distance <math>d(r)=(N_x/\pi)\sin(\pi r/N_x)</math>) — the edge four-point has the chiral-CFT cross-ratio structure (<math>\eta=1</math>), not just the two-point. [E] BULK NEG CONTROL: the bulk-row four-point is orders of magnitude off the conformal value (exponential decay, gapped) — only the EDGE four-point is conformal. [C] the one external MMST fact (continuum existence of the chiral scaling limit) is now empirically supported at BOTH two-point (v444) and four-point order on our explicit collar; every measured <math>n</math>-point converges to its conformal limit. SEAM.EQUIV.01 stays [O] — the rigorous MMST uniform-convergence theorem (v336) is the cited backbone. Python-only (numpy). [E][C]</p> |

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| Script                         | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| v449_seam_mmst_uniform.py      | <p>SEAM.EQUIV.MMST.UNIFORM.01: uniform-in-<math>N</math> control of the edge scaling limit — the empirical content of the cited MMST uniform-convergence theorem, strengthened, on the same v367/v368 <math>p+ip</math> collar. [E] CROSS-RATIO A CONVERGES: the edge four-point <math>Q(1, 2, 4, 7)/12 \rightarrow \sin(5\pi/12)/\sin(\pi/12) = 2+\sqrt{3}</math>, the deviation shrinking monotonically under refinement (<math>N_x=36, 48, 60</math>). [E] CROSS-RATIO B (independent positions): <math>Q(1, 3, 5, 9)/12 \rightarrow</math> its OWN conformal value 2 — a SECOND observable with a DIFFERENT limit, so the convergence is not a single tuned point. [E] UNIFORM <math>1/N</math> RATE: <math> Q(N)-Q_\infty  \cdot N \leq 4</math> for BOTH cross-ratios and every <math>N_x \in \{36, 48, 60\}</math> — one <math>N</math>-independent constant bounds the deviation across two independent observables, the empirical hallmark of uniform-in-<math>N</math> convergence. [E] BULK NEG CONTROL: the bulk-row cross-ratio is off-scale (gapped, exponential) — only the EDGE converges. [C][O] the MMST uniform-convergence HYPOTHESIS is empirically satisfied on the collar (two-point v444, four-point v448, and now multi-observable uniform rate); SEAM.EQUIV.01 stays [O] — the abstract continuum existence theorem (v336) is strengthened in evidence, not replaced by a proof. Python-only (numpy). [E][C]</p> |
| v450_seam_edge_entanglement.py | <p>SEAM.EQUIV.EDGE.ENTANGLEMENT.01: a THIRD independent reading of the edge <math>c_-=8</math> — the Calabrese–Cardy entanglement scaling <math>c=1/2</math> per Majorana (after v444 correlator, v447 topology). [E] CALABRESE–CARDY LAW: the block entanglement entropy of the critical XX (Dirac) chain fits <math>S=(c/3)\ln[(L/\pi)\sin(\pi\ell/L)]+\text{const}</math> with <math>c_{\text{Dirac}}=1.000</math> (flat log slope) — the universal scaling-limit signature. [E] <math>c</math> PER MAJORANA = <math>1/2</math>: <math>c_{\text{Dirac}}=1=2c_{\text{Maj}}</math> (Dirac = two Majoranas), so <math>c_{\text{Maj}}=1/2</math> — the chiral CFT of one <math>p+ip</math> edge mode (v444). [E] GAPPED NEG CONTROL: a dimerised (gapped) chain has saturating, <math>\ell</math>-independent entanglement (<math>c=0</math>) — the log growth is the CRITICAL edge CFT. [C] EDGE CENTRAL CHARGE: <math>c_-=N_{\text{Maj}} \cdot c_{\text{Maj}} = 16 \cdot \frac{1}{2} = 8 = g_{\text{car}} + N_{\text{fam}}</math>, from entanglement — the SAME <math>c_-=8</math> as v444 (correlator) and v447 (topology), from a third independent observable. [C][O] three observables agree; SEAM.EQUIV.01 stays [O]. Python-only (numpy). [E][C]</p>                                                                                                                                                                                |
| v451_seam_edge_cardy_tower.py  | <p>SEAM.EQUIV.EDGE.CARDY.01: the Cardy finite-size conformal tower — the edge spectrum reproduces the EXACT Ising primary dimensions <math>\{1/16, 1/2\}</math>, on the critical Majorana ring (the v367/v368 edge universality class). [E] SPIN DIMENSION <math>h_\sigma=1/16</math>: the periodic(R)–antiperiodic(NS) ground-energy split gives <math>(E_0^R - E_0^{NS})L/(2\pi v)=0.06250</math> (<math>v=2</math>), stable across <math>L=200, 400, 800</math> — the Ising spin field. [E] ENERGY DIMENSION <math>h_\varepsilon=1/2</math>: the lowest NS pair excitation gives <math>2\varepsilon_{\min}L/(2\pi v)=0.50000</math> — the Ising energy field. [E] CONFORMAL SCALING: both dimensions are <math>L</math>-INDEPENDENT (the gaps scale exactly as <math>1/L</math>) — a conformal (gapless) spectrum, not a gapped lattice. [C] ISING IDENTIFICATION: <math>\{c, h_\sigma, h_\varepsilon\}=\{1/2, 1/16, 1/2\}</math> is the chiral Ising (free Majorana) CFT (the <math>c=1/2</math> building block, v444/v450); with <math>N_{\text{Maj}}=16</math> copies and bulk-edge (v447), <math>c_-=16 \cdot \frac{1}{2}=8=g_{\text{car}}+N_{\text{fam}}</math>. [C][O] the edge CFT is named by its central charge (v450) AND its operator content (this); SEAM.EQUIV.01 stays [O]. Python-only (numpy). [E][C]</p>                                                                                                               |

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| Script                      | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| v452_seam_e8_modular.py     | SEAM.EQUIV.E8.MODULAR.01: the $(E_8)_1$ torus modular data — one primary, $S$ -invariant, $T$ -phase $e^{-2\pi i/3}$ , the holomorphic $c=8$ signature (extends v156's $q$ -expansion to the modular transformation data). [E] $S$ -INVARIANCE: $\chi(-1/\tau)=\chi(\tau)$ for $\chi=E_4/\eta^8$ (weight 0, since $E_4$ wt 4 / $\eta^8$ wt 4), to $\sim 10^{-30}$ — ONE primary, the $S$ -matrix is [1]. [E] $T$ -PHASE: $\chi(\tau+1)/\chi(\tau)=e^{-2\pi i/3}=e^{-2\pi ic/24}$ ( $c=8$ ), to $\sim 10^{-30}$ — the central charge appears in the modular $T$ eigenvalue. [E] LEADING POWER $\Rightarrow c=8$ : $\chi q^{1/3} \rightarrow 1$ so the leading power is $q^{-c/24}=q^{-1/3} \Rightarrow c=8=g_{\text{car}}+N_{\text{fam}}$ . [E] HOLOMORPHIC CONSTRAINT: $c=8=0 \bmod 8$ — the minimal holomorphic (single-primary) chiral central charge, the even unimodular rank-8 lattice $E_8$ . [C][O] the $(E_8)_1$ modular data (one primary, $S$ -invariant, $T$ -phase $e^{-2\pi i/3}$ , $c=8$ ) is the torus signature of the seam net; with the edge readings (v444/v447/v450/v451) the $(E_8)_1$ identity is pinned on the torus AND the edge; SEAM.EQUIV.01 stays [O] — the cited continuum existence theorem (v336) is the residual. Wolfram-mirrored (tfpt_readouts_extension.wl). [E][C]                                                                                                                                          |
| v453_seam_mu4_from_marks.py | SEAM.RIGIDITY.MU4FROMMARKS.01: deriving QGEO.SYM.01 ( $\mu_4$ -symmetry of the seam RP data) from the four marks — the LAST structural premise of the rigidity chain folded into the existing realisation premise. [E] THE MARKS ARE $\mu_4$ : the four Gauss–Bonnet marks are exactly $\{1, i, -1, -i\} = \text{roots of } z^4-1$ (v168/v216/v323). [E] THE CLOCK PERMUTES THE MARKS: $z \mapsto iz$ is a 4-cycle of order 4 ( $1 \rightarrow i \rightarrow -1 \rightarrow -i \rightarrow 1$ ), fixing the configuration setwise (the order-4 clock of v445). [E] MOEBIUS STABILISER: $z \mapsto iz$ preserves the marks' cross-ratio $\lambda=2$ (in the $D_4$ stabiliser, v168) — the symmetry is geometric. [E] THE FORM BASIS IS $\mu_4$ -GRADED: $\omega_k = z^{k-1} dz / (z^4-1) \rightarrow i^k \omega_k$ under $z \mapsto iz$ ( $k=1, 2, 3 \rightarrow \{i, -1, -i\}$ , the three nontrivial $\mu_4$ characters; weights $(1, 2, 3)=A_3$ exponents = $\text{Spec}(Q_+)$ ) — the natural basis is $\mu_4$ -graded, so any geometric datum is $\mu_4$ -covariant. [C][O] QGEO.SYM.01 (“the seam RP data is $\mu_4$ -symmetric”) is a CONSEQUENCE of marks= $\mu_4$ (this) + QGEO.REALIZE.01 (seam = the marked geometry, the EXISTING premise), NO new premise; the rigidity chain (this $\rightarrow$ v446 $\rightarrow$ v445) closes modulo realisation, SEAM.EQUIV.01 stays [O]. Wolfram-mirrored (tfpt_readouts_extension.wl). [E][C] |
| v454_seam_edge_virasoro.py  | SEAM.EQUIV.EDGE.VIRASORO.01: the lattice realises the level-1 current algebra whose Sugawara central charge is EXACTLY 8 (G2 of the post-F next steps) — testing the cited MMST “Koo–Saleur lattice generators $\rightarrow$ continuum Virasoro with the correct $c$ ” fact on our own free-fermion edge. [E] CASIMIR $c$ : the critical complex free-fermion ring $E_0(L)=e_\infty L - \pi v c / (6L)$ (Affleck–Cardy) fits $c_{\text{Dirac}}=1.000 \Rightarrow 1/2$ per Majorana (parameter-free, $L=64..1024$ ). [E] KAC–MOODY LEVEL: the lattice $U(1)$ current $J_n$ has $\langle 0   J_n J_{-n}   0 \rangle = n$ EXACTLY ( $n=1..8$ ) — the affine level-1 central extension closes; a gapped (dimerised) control gives a vanishing non-linear term. [E] SUGAWARA EXACT: $c = \dim G / (1+h^\vee) = 8$ for BOTH $SO(16)_1$ (120/15) and $(E_8)_1$ (248/31) — Wolfram-mirrored. [C] ASSEMBLY: 16 Majoranas $\times \frac{1}{2} = 8 =$ the level-1 Sugawara $c$ ; the edge IS a $c_- = 8$ level-1 current-algebra CFT. [C][O] the lattice Virasoro/current algebra with the correct central extension is realised on the explicit collar; SEAM.EQUIV.01 stays [O] (the continuum existence theorem v336 is the cited backbone). Mixed numerical (Casimir, Kac–Moody) + exact (Sugawara, Wolfram-mirrored). [E][C][O]                                                                                                                         |

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| Script                          | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| v455_seam_bw_uniform.py         | <p>SEAM.EQUIV.BW.UNIFORM.01: a UNIFORM-IN-<math>N</math> Tomita–Takesaki tower for the intrinsic Bisognano–Wichmann condition <math>u_\Theta=J</math> (G1 of the post-F next steps) — the inductive-limit companion to the single finite witness v426, advancing v424 sub-step (i). One-particle: <math>\Delta=e^{-K}</math>, conjugation <math>J</math>, reflection <math>\Theta</math>; BW <math>\Leftrightarrow u_\Theta=J \Leftrightarrow \Theta K \Theta = -K</math>. [E] UNIFORM BW: the reflection-odd, <math>\mu_4</math>-even, gapped collar of size <math>4m</math> (<math>m=2..32</math>, <math>N=8..128</math>) has <math>\ \Theta K \Theta + K\ =0</math> to machine precision at EVERY <math>N</math> — <math>u_\Theta=J</math> uniform. [E] UNIFORM GAP: the gap of <math>K</math> is <math>N</math>-independent (<math>6 \log \frac{3}{2} = 2N_{\text{fam}} \log \frac{3}{2}</math>) so <math>\Delta=e^{-K}</math> stays uniformly bounded away from 1 (a genuine KMS, not trace, state). [E] UNIFORM CLOCK SYMMETRY: <math>[\rho, K]=0</math> exactly at every <math>N</math>. [E] <math>N</math>-INDEPENDENT NEG CONTROL: a side-even reflection gives <math>\ \Theta K \Theta + K\ =2\ K\ </math> (a fixed per-mode margin, growing with <math>N</math>) and an off-<math>\mu_4</math> <math>K</math> breaks <math>[\rho, K]</math> at every <math>N</math> — the success is non-vacuous. [C][O] the inductive (continuum) limit inherits <math>u_\Theta=J</math>; SEAM.EQUIV.01 stays [O] (the rigorous continuum theorem NPW26/MMST v336 is the backbone). Python-only (numpy). [E][C][O]</p> |
| v456_seam_chirality_from_c3.py  | <p>SEAM.S3.FROM-P1.01: the edge chirality (<math>c_- \neq 0</math>, the S3 topological phase) is FORCED by the one-sidedness that defines <math>c_3=1/(8\pi)</math> (P1) (G6 of the post-F next steps) — so S3 is a CONSEQUENCE of axiom P1, not an independent input. [E] <math>c_3</math>'s “8” IS THE ONE-SIDED COUNT: <math>8\pi= \mathbb{Z}_2  \int K = 2(2\pi\chi(S^2)) = 2 \cdot 4\pi</math> (<math>\chi=2</math>), so <math>8= \mathbb{Z}_2  \cdot (\int K/\pi) = 2 \cdot 4</math> (v73, Wolfram-mirrored). [E] REFLECTION REVERSES <math>C</math>: the explicit 2-band <math>p+ip</math> Chern model has <math>C=+1</math> and the orientation-reversed (<math>k_y \rightarrow -k_y</math>) model <math>C=-1</math>. [E] TWO-SIDED <math>\Rightarrow</math> NON-CHIRAL: any GAPPED boundary carrying an orientation-reversing (reflection) symmetry must have <math>C=-C=0</math> (<math>c_-=0</math>, gappable). [C] ONE-SIDED <math>\Rightarrow</math> CHIRAL: the one-sided seam is precisely the <math> \mathbb{Z}_2 </math> quotient with NO reflection (the same <math> \mathbb{Z}_2 </math> that gives <math>c_3</math> its <math>8\pi</math>), so <math>C \neq 0</math> is forced — S3 follows from P1. [C][O] magnitude <math>c_-=8=\text{rank } E_8=g_{\text{car}}+N_{\text{fam}}</math> is the SAME integer as <math>c_3</math>'s one-sided count; SEAM.EQUIV.01's continuum EXISTENCE (v336) stays [O]. Mixed exact (the <math>8= \mathbb{Z}_2  \int K/\pi</math> arithmetic, Wolfram) + numerical (the reflection <math>C \rightarrow -C</math> flip). [E][C][O]</p>                         |
| v457_seam_character_killtest.py | <p>SEAM.EQUIV.KILLTEST.01: an explicit <math>(E_8)_1</math>-vs-<math>SO(16)_1</math> character/sector kill-test for the seam edge (G7 of the post-F next steps) — a FALSIFIABLE consistency test of the <math>(E_8)_1</math> identification of SEAM.EQUIV.01. [E] <math>(E_8)_1</math> CHARACTER TOWER: <math>\chi=E_4/\eta^8=q^{-1/3}(1+248q+4124q^2+34752q^3+\dots)</math> — the exact conformal-tower degeneracies <math>(E_4 \cdot \prod (1-q^n))^{-8}</math>, Wolfram-mirrored). [E] CURRENT-COUNT DISCRIMINATOR: the weight-1 multiplicity is <math>248=\dim E_8=120(SO16)+128(\text{spinor})</math>, NOT the 120 of the free <math>SO(16)_1</math>. [E] SECTOR-COUNT DISCRIMINATOR: <math>(E_8)_1</math> is INVERTIBLE <math> \det \text{Cartan } E_8 =1</math> (one primary) vs <math>SO(16)_1=D_8</math> <math> \det \text{Cartan } D_8 =4</math> (four sectors). [X] KILL CRITERION: a measured weight-1 count of 120, OR a sector count of 4, OR a tower coefficient <math>\neq \{1, 248, 4124, \dots\}</math> would FALSIFY <math>(E_8)_1</math>; current data give <math>248/1/\{1, 248, 4124\}</math>, so PASSED. [C][O] the <math>(E_8)_1</math> identification survives the kill-test; SEAM.EQUIV.01's continuum EXISTENCE (v336) stays [O]. Exact (Wolfram-mirrored) + a numerical determinant cross-check. [E][C][O][X]</p>                                                                                                                                                                                                                                                                      |

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| Script                           | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| v458_seam_mmst_citation_audit.py | <p>SEAM.EQUIV.MMST.AUDIT.01: the EXACT hypothesis-by-hypothesis citation audit of the seam scaling-limit theorem (Osborne–Stottmeister, CMP <b>398</b> (2023) 219; arXiv:2107.13834; legacy internal token MMST, on the MMST/Morinelli–Morsella–Stottmeister–Tanimoto OAR framework) against our collar (G3 of the post-F next steps, the “exakter Zitat-Abgleich”). [C] H1 ALGEBRA CLASS: a gapped quasi-free CAR 16-Majorana system = MMST’s lattice-fermion class (v155/v160/v302). [C] H2 MASSLESS POS-ENERGY LIMIT: the gapless chiral edge IS that massless scaling limit (v439/v444). [E] H3 PER-COPY (<math>c=1/2</math>): each of 16 chiral Majoranas is EXACTLY the proven Theorem A <math>c=1/2</math> case. [C] H4 DECOUPLED ADDITIVITY: MMST’s limit decouples into Majoranas <math>\Rightarrow L_k = \sum_i L_k^{(i)}</math> on the common core <math>\Rightarrow</math> the free <math>c=8</math> <math>SO(16)_1</math> Virasoro follows (<math>16 \cdot \frac{1}{2} = 8</math>). [C] H5 WZW RANGE rank <math>\leq c \leq D</math>: <math>8 \leq 8 \leq 16</math> IN range, the 120 <math>so(16)</math> currents are bilinears — covered. [O] H6 RESIDUAL = THE 128-SPINOR EXTENSION: the index-4 <math>SO(16)_1 \rightarrow (E_8)_1</math> adds 128 SPINOR currents (<math>248=120+128</math>), NOT bilinears — OUTSIDE MMST’s method (the <math>\det K</math> 4<math>\rightarrow</math>1 holomorphy discriminator), handed to v459. [C][O] the audit CLOSES-by-citation the <math>c=8</math> EXISTENCE and pins the open residual EXACTLY to the 128-spinor extension; SEAM.EQUIV.01 stays [O], its open part now this one holomorphic-extension statement. Python (sympy; a structural/arithmetic audit). [C][O]</p> |
| v459_seam_lattice_voa_route.py   | <p>SEAM.EQUIV.LATTICEVOA.01: the SECOND, independent route to <math>(E_8)_1</math> — the lattice-VOA / conformal-net <math>A_Q(E_8)</math> construction supplies EXACTLY the 128-spinor extension MMST (v458) leaves open (G5 of the post-F next steps). Cited: Adamo–Giorgetti–Tanimoto, “Rational and non-rational 2d CFTs arising from lattices” (arXiv:2506.01008); OS for unitary lattice VOAs (Adamo–Moriwaki–Tanimoto, arXiv:2407.18222). [E] <math>E_8</math> EVEN UNIMODULAR: <math>\det \text{Cartan}(E_8)=1</math> and even <math>\Rightarrow A_{E_8}</math> HOLOMORPHIC (one sector, the <math>\det K=1</math> the residual demanded). [E] LATTICE CURRENTS <math>248=8+240</math>: 8 Cartan + 240 roots = the full <math>(E_8)_1</math> weight-1 space, from the lattice (not bilinears). [E] THE 240 ROOTS SPLIT 112+128: 112 INTEGER (<math>D_8</math>) + 128 HALF-INTEGER; <math>8+112=120=so(16)</math> and the 128 half-integer roots ARE the spinor (<math>248=120+128=8+240</math>) — the lattice route builds the 128-spinor explicitly. [C] AGT THEOREM: <math>A_Q</math> of an even lattice IS a 2d conformal net; the holomorphic <math>SO(16)_1 \rightarrow (E_8)_1</math> extension is CONSTRUCTED, not assumed. [C][O] TWO complementary routes now cover <math>(E_8)_1</math> (MMST fermions v458 + the lattice net <math>A_{E_8}</math>); the residual reduces to the single realization input “the collar’s scaling limit = <math>A_{E_8}</math>” (tied to P1 by v456). SEAM.EQUIV.01 stays [O]. Exact (<math>E_8</math> even unimodular; <math>240=112+128</math>; <math>248=8+240=120+128</math>, Wolfram-mirrored) + structural literature mapping. [E][C][O]</p>                                     |

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| Script                       | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| v460_seam_s3_lto.py          | <p>SEAM.S3.LT0.01: the EXPLICIT flat-band commuting-projector (local-topological-order) realisation of the S3 input — advancing v424 sub-step (i) (“realise the raw seam as a <math>\mathbb{Z}_2</math>-reflection commuting-projector LTO”) from abstract existence to an explicit gap-protected quasi-local projector net carrying the <math>\mathbb{Z}_2</math> reflection and the <math>\mu_4</math> clock. [E] FLAT-BAND PARENT: the flattened Bloch <math>Q=h/ h </math> of v367’s <math>p+ip</math> collar has <math>Q^2=I</math> (flat <math>\pm 1</math> bands) and <math>P=(I-Q)/2</math> is an exact projector EQUAL to the occupied ground-state projector — a frustration-free commuting-projector parent with the same gapped ground state. [E] GAP-PROTECTED QUASI-LOCALITY: the gapped projector kernel <math> P(n, 0) </math> decays rapidly (<math>\xi \approx 1.14</math> at <math>M=1</math>, <math>\exp R^2=0.995</math>, <math> P(20, 0)  &lt; 10^{-6}</math>) and <math>\xi</math> GROWS as the gap closes (2.7 at gap 0.02); at the GAPLESS point the kernel decays as a clean power law <math> R ^{-2.1}</math> (<math>R^2=0.999</math>, tail <math>\sim 2 \times 10^{-4}</math>) — locality needs the gap. [E] <math>\mathbb{Z}_2</math> REFLECTION + <math>\mu_4</math> INHERITED: the flattened modular Hamiltonian <math>K_{\text{flat}} = \text{sign}(K)</math> inherits <math>\Theta K_{\text{flat}} \Theta = -K_{\text{flat}}</math> (BW, <math>u_\Theta = J</math>) and <math>[\rho, K_{\text{flat}}] = 0</math> (<math>\mu_4</math>) EXACTLY (sign is odd; <math>\rho</math> commutes with any <math>f(K)</math>). [E] NEG CONTROLS: a side-even reflection gives <math>2\ K_{\text{flat}}\  \neq 0</math> (BW fails), an off-<math>\mu_4</math> <math>K</math> mixing opposite-sign marks breaks <math>[\rho, \text{sign}(K)] \neq 0</math>. [C][O] the chiral <math>c_- = 8 \neq 0</math> (FHS Chern <math> C =1</math>) obstructs STRICT finite-range commuting-projector locality (Kapustin–Fidkowski, arXiv:1810.07756; <math>C \neq 0</math> forbids exponentially-localised Wannier, Thouless), so the realisation is the quasi-local LTO net of NPW26 — the form sub-step (i) asks for. SEAM.EQUIV.01 stays [O] (the continuum theorem MMST v336 is the backbone). Python-only (numpy; the det/level discriminators Wolfram-mirrored via v89/v281/v422). [E][C][O]</p> |
| v461_seam_strict_locality.py | <p>SEAM.S3.LOCALITY.01: the Kapustin–Fidkowski/Thouless strict-locality OBSTRUCTION made explicit — sharpening v460’s verdict from a cited statement to an EXHIBITED topological obstruction on the same v367 <math>p+ip</math> collar. [E] CHERN INTEGER: the FHS Chern is <math>C=+1</math> (chiral <math>M=1</math>), 0 (trivial <math>M=3</math>), <math>-1</math> (opposite mass) — the invertible-phase invariant. [E] WILSON-LOOP / WANNIER-CENTRE WINDING = CHERN: the hybrid Wannier centre <math>\theta(k_y)</math> winds by <math> C =1</math> (chiral) and 0 (trivial) over the BZ; exponentially-localised Wannier functions exist IFF the winding is 0 (Brouder–Panati et al., PRL 98 046402; Thouless), so the chiral collar has NONE. [E] OBSTRUCTION INTEGERS TIED: winding = <math> C =1 \neq 0</math> and <math>c_- = 16/2 = 8 = g_{\text{car}} + N_{\text{fam}} = \text{rank } E_8 \neq 0</math> (the same 8 as <math>c_3</math>’s one-sided count, v456). [E] NEG CONTROL: BOTH phases are gapped, yet only the trivial (winding 0) admits exp-localised Wannier and a strict finite-range commuting projector — it is the CHIRALITY, not the gap. [C][O] Kapustin–Fidkowski (arXiv:1810.07756): a strictly finite-range commuting projector would force winding 0, <math>C=0</math>; since <math>C=1 \neq 0</math> NONE exists, so v424 sub-step (i)’s realisation is PROVABLY the quasi-local NPW26 LTO net (strict locality topologically forbidden, not a missing premise) — upgrading v460’s cited verdict. SEAM.EQUIV.01 stays [O] (continuum theorem MMST v336 the backbone). Python-only (numpy; the winding/Chern integers Lean-mirrored, SeamResidualAxiom.lean). [E][C][O]</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

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| Script                                 | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| v462_seam_spinor_continuum.py          | <p>SEAM.EQUIV.SPINOR.01: the 128-spinor extension <math>SO(16)_1 \rightarrow (E_8)_1</math> exhibited at CHARACTER level and as a finite-<math>L \rightarrow</math> continuum convergence — the constructive companion of the v458 (MMST audit) / v459 (lattice-VOA) residual. [E] <math>(E_8)_1</math> TOWER FROM <math>SO(16)_1</math> SECTORS: the exact Jacobi/<math>E_8</math> identity <math>\theta_2^8 + \theta_3^8 + \theta_4^8 = 2E_4</math> gives <math>\chi_{(E_8)_1} = E_4/\eta^8 = \frac{1}{2}[(\theta_3/\eta)^8 + (\theta_4/\eta)^8] + \frac{1}{2}(\theta_2/\eta)^8 = \chi_o^{SO16} + \chi_s^{SO16}</math> — the holomorphic <math>(E_8)_1</math> character IS the <math>SO(16)_1</math> vacuum + spinor sector sum. [E] THE 128 SPINOR WEIGHT-1 STATES FILL <math>120 \rightarrow 248</math>: <math>\chi_o</math> gives weight-1 multiplicity 120, <math>\chi_s</math> gives 128, summing to the exact <math>(E_8)_1</math> tower <math>\{1, 248, 4124, 34752\}</math> — the spinor currents ARE the <math>(E_8)_1 - SO(16)_1</math> deficit (<math>248 = 120 + 128</math>). [E] FINITE 16-MAJORANA FOCK: <math>C(16, 2) = 120</math> NS bilinear (<math>so(16)</math>) currents and <math>2^{16/2} = 256</math> Ramond ground states = 128 spinor + 128 cospinor — the spinor present already at finite <math>N</math>. [E] FINITE-<math>L \rightarrow</math> CONTINUUM CONVERGENCE: the critical free-fermion ring Casimir <math>E_0(L) = e_\infty L - \pi v c / (6L)</math> fits <math>c_{\text{Dirac}} \rightarrow 1</math> (<math>c_- = 8</math>) and the lowest scaled NS gap <math>x_1 \rightarrow 1</math> with <math> x_1 - 1  \sim 1/L^2</math> (<math>L = 64..1024</math>) — the lattice flows to the conformal tower. [C][O] the rigorous holomorphic extension (<math>\chi_o + \chi_s</math> simple-current projection) is supplied by AGT/AMT (v459); SEAM.EQUIV.01's continuum existence (v336) stays [O]. Mixed exact (<math>\theta</math> identity + tower + <math>120 = C(16, 2)</math>, <math>128 = 2^7</math>, <math>248 = 120 + 128</math>, Wolfram-mirrored) + numerical (the finite-<math>L</math> convergence, Python-only). [E][C][O]</p> |
| v463_seam_c8_holomorphic_uniqueness.py | <p>SEAM.EQUIV.UNIQ.01: the <math>c=8</math> holomorphic-uniqueness pin — the positive lever that makes the IDENTIFICATION half of SEAM.EQUIV.01 classification-forced rather than assumed (B of the A+B round). [E] <math>c=8</math> NOT UNIQUE AT LEVEL 1: <math>c = \dim \mathfrak{g} / (1 + h^\vee) = 8</math> has THREE simple solutions — <math>A_8</math> (<math>\dim 80</math>), <math>D_8 = \mathfrak{so}(16)</math> (<math>\dim 120</math>), <math>E_8</math> (<math>\dim 248</math>); <math>c=8</math> is necessary but NOT sufficient (the v457 <math>\det K = 4</math> <math>SO(16)</math> rival is a real gleich-<math>c</math> competitor). [E] HOLOMORPHY FORCES <math>\dim V_1 = 248</math>: the only weight-0 form with leading <math>q^{-1/3}</math> and integer coefficients is <math>j^{1/3} = E_4/\eta^8</math>, whose <math>q^1</math> coefficient (the 1-dim candidate space, vacuum <math>a_0 = 1</math>) is 248 — forced, not chosen. [E] <math>\dim V_1 = 248 + \text{level-1 } c=8 \Rightarrow E_8</math> UNIQUELY (<math>248/(1+30)=8</math>); the gleich-<math>c</math> rivals <math>A_8</math> (80) and <math>D_8</math> (120) are excluded. [E] the <math>(E_8)_1</math> tower <math>\{1, 248, 4124, 34752, 213126\}</math> are the <math>E_4/\eta^8</math> coefficients, and <math>E_4/\eta^8 = j^{1/3}</math> (numeric <math>&lt; 10^{-25}</math>). [C] Dong–Mason/Schellekens: the holomorphic VOA at <math>c=8</math> is UNIQUE = <math>V_{E_8}</math>, and the even unimodular rank-8 lattice is unique = <math>E_8</math> (Mordell–Witt) — so “<math>c=8</math> holomorphic limit = <math>(E_8)_1</math>” is classification-forced. With v458 (free part) + v459 (spinor extension) the continuum/identification HALF of SEAM.EQUIV.01 is [C] conditional on named audited theorems; the REALISATION R1 (v464) is the sole remaining input. SEAM.EQUIV.01 stays [O]. Exact (sympy/integer <math>q</math>-series + Lie-algebra scan; mpmath cross-check), Wolfram-mirrored. [E][C][O]</p>                                                                                                                                                     |

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| Script                                    | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| v464_seam_<br>oneparticle_<br>rigidity.py | SEAM.EQUIV.ONEPARTICLE.01: the one-particle RIGIDITY + scaling limit attacking R1 — the REALISATION input of SEAM.EQUIV.01 (A of the A+B round). The seam is quasi-free (v113/v160/v161), so by Araki self-dual CAR the whole net is the second quantisation of its one-particle symbol $P$ , and R1 collapses to a finite-per-mode statement about $P$ . [C] ARAKI SELF-DUAL CAR: a gauge-invariant quasi-free state $\leftrightarrow$ its symbol $P$ ( $0 \leq P \leq 1$ ) bijectively (Araki 1971; Shale–Stinespring 1965), so a realisation is fully determined by $P$ . [E] RIGIDITY: the gapped ground state is the UNIQUE quasi-free state with $P =$ projection onto $K < 0$ ( $P^2 = P$ , hermitian to $< 10^{-12}$ ), $K$ fixed by gap + chirality + reflection (v426) — no second realisation. [E] ONE-PARTICLE SCALING LIMIT EXISTS: the real-space kernel is Cauchy on a fixed window ( $\max  P_{2N} - P_N  \sim 1/N \rightarrow 0$ ), so $P_\infty$ exists (Gaussianity is why a computation reaches it). [E] THE LIMIT IS $c=1$ DIRAC ( $c_-=8$ ): entanglement $S(L) = (c/3) \ln L$ gives $c \rightarrow 1$ (per Majorana 1/2; Peschel/Calabrese–Cardy, ties v450), 16 copies $\rightarrow c_-=8$ . [C] Shale–Stinespring: gauge-invariant $\Rightarrow$ no pairing block $\Rightarrow$ the Bogoliubov map is implementable, GNS reps quasi-equivalent. R1 is upgraded from a bald axiom to “the unique quasi-free realisation, one-particle limit exhibited, modulo the named Araki/Shale–Stinespring functor”; with v463 the SEAM.EQUIV.01 residual is now entirely certification (named cited functors), zero open internal mechanism — closed modulo cited theorems. SEAM.EQUIV.01 stays [O]. Python-only (numpy: idempotency, kernel Cauchy-convergence, entanglement $c$ -fit). [E][C][O] |
| v465_seed_<br>crosssector_<br>joint.py    | SEED.CROSSSECTOR.01: the cross-sector one-parameter seed test — the $\theta_{13}$ -independent, externally-runnable slice of v306. TFPT reads a CMB EB/TB birefringence angle $\beta$ and a quark Cabibbo angle $\lambda_C =  V_{us} $ (unrelated in the SM) as two readouts of the ONE axiom seed $\varphi_0 = 1/(6\pi) + 48c_3^4$ . [E] plumbing: the axiom $\varphi_0$ reproduces the frozen $\lambda_C$ , $\sin^2 \theta_{12}$ , $\beta$ to $10^{-9}$ . [E] CROSS-SECTOR (headline): back-solving $\varphi_0$ <i>independently</i> from $\beta$ (CMB) and $\lambda_C$ (quark) gives the <i>same</i> seed within $0.01\sigma_{\text{comb}}$ — a correlation only a shared-origin theory predicts. [E] one-parameter fit: the error-weighted mean of the three back-solved $\varphi_0$ reproduces the axiom to 0.06%; the joint $\chi^2$ at the axiom (zero free parameters, 3 dof) is 0.007, $\chi^2/\text{dof} < 1$ , combined pull $0.08\sigma$ . [E] $\theta_{13}$ EXCLUSION declared, not cherry-picked: adding $\sin^2 \theta_{13}$ raises $\chi^2$ by $\sim 4$ (the documented $\sim 2\sigma$ pressure point, v328/v338), so this slice is <i>deliberately</i> $\theta_{13}$ -independent. [E] negative control: a data $\leftrightarrow$ formula shuffle explodes the cluster $\chi^2$ by $> 5 \times 10^4$ . A consistency / out-of-sample statistic (a subset of v306); $\varphi_0$ is FIXED by the axioms, so it upgrades <i>no</i> prediction. Python-only. [E]                                                                                                                                                                                                                                                                                                                                    |

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| Script                          | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| v466_seed_leptonmass_channel.py | SEED.LEPTONMASS.01: a SIXTH, new-sector seed channel — the charged-lepton mass ratio $m_e/m_\mu = \frac{12}{7}\varphi_0^2$ points to the SAME axiom seed $\varphi_0 = 1/(6\pi) + 48c_3^4$ , extending the empirical seed over-determination (v306/v465) from three measurement sectors to five. [E] plumbing: the axiom $\varphi_0$ reproduces the frozen $m_e/m_\mu = \frac{12}{7}\varphi_0^2 = 0.00484673$ to $10^{-9}$ . [E] NEW SECTOR (headline): back-solving $\varphi_0$ from the measured POLE ratio $m_e/m_\mu = 0.004836$ (PDG) gives $\varphi_0 = 0.053113$ , only $-0.11\%$ from the axiom — a charged-lepton MASS reading (Penning-trap masses), a fourth measurement sector beyond {neutrino mixing, CMB, quark mixing}. [C] TRANSPORT CAVEAT: the formula is the SOURCE ratio, the datum the POLE ratio; the source→pole shift is the seam-gapped correction class (bound $(2/3)^6 = 8.78\%$ , v393) but largely CANCELS for $m_e/m_\mu$ , so the $0.11\%$ residual sits far inside the band — added as a CONSISTENT new sector, not a sharper pin. [E] EXTENDS THE OVER-DETERMINATION: the six-channel cluster (v306's five + this) has $\chi^2/\text{dof} < 1$ and the error-weighted mean stays at the axiom to $< 0.1\%$ ; with the corroborating heavy-quark $m_t/m_b$ ( $-0.23\%$ ) it is a seven-channel, FIVE-sector agreement (neutrino mixing, CMB, quark mixing, lepton mass, quark mass). [E] negative control / power: a data↔formula shuffle explodes the cluster $\chi^2$ by $> 100\times$ ; and not every mass ratio qualifies — $m_c/m_s = (34/47)/\varphi_0$ is scheme-ambiguous (FLAG 11–14), a WEAK channel deliberately not counted. [C] HONEST: theoretically still the ONE seed (v305 multiplicity 1) — an EMPIRICAL strengthening (a new measurement sector agrees), NOT theoretical independence; $\varphi_0$ is FIXED by the axioms, so it upgrades <i>no</i> prediction and closes <i>no</i> gate. Python-only. [E][C] |
| v467_thirdgen_mixing_pattern.py | FLAV.THIRDGEN.PATTERN.01: the three $\sim 2\sigma$ mixing tensions ( $\theta_{13}, \theta_{23}, V_{cb}$ ) are ONE common $-\varphi_0^{\text{ret}}$ suppression ( $-5.36\% \pm 1.67\%$ , $3.2\sigma$ non-zero, $-0.03\sigma$ from $-\varphi_0^{\text{ret}}$ ); the ladder-base forms $\lambda_C^2 e^{-5/6} / (1 - \varphi_0^{\text{ret}})/2 / \lambda_C^2 / (1 + \lambda_C)$ drop the joint $\chi^2$ $10.4 \rightarrow 0.11$ with zero parameters; post-hoc candidate [O], frozen record unchanged, JUNO-era reactors + Belle II decide                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| v468_dm2_ratio_jarlskog.py      | FLAV.DM2RATIO.01: $\Delta m_{21}^2 / \Delta m_{31}^2 =  J_{\text{PMNS}} $ of the frozen matrix = $0.029653$ vs measured $0.029805(792) \rightarrow -0.19\sigma$ — a parameter-free candidate relation for the previously unpredicted splitting ratio; the informal $m_2/m_3 \sim \pi\varphi_0^{\text{ret}}$ heuristic is now $-2.44\sigma$ and retyped; NO floor $\Sigma m_\nu = 0.0588$ eV consistent; post-hoc [O], JUNO decides                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

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| Script                                    | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| v469_seam_<br>crossedproduct_<br>route.py | <p>SEAM.EQUIV.CROSSEDPRODUCT.01: the net-level crossed-product certification of the 128-spinor extension + the <math>R1 \rightarrow R1'</math> invariant-level reduction. <b>[E]</b> LR LOCALITY INTEGER (index-2): the <math>SO(16)_1</math> spinor has <math>h_s = N/16 = 16/16 = 1 \in \mathbb{Z} \Rightarrow</math> statistics phase +1 — the Longo–Rehren criterion HOLDS, so the <math>\mathbb{Z}_2</math> simple-current crossed product <math>SO(16)_1 \rtimes \{1, s\}</math> is a LOCAL index-2 extension (Longo–Rehren, Rev. Math. Phys. 7 (1995); the level-1 <math>so(N)</math> net with DHR sectors from even-CAR: Böckenhauer, Rev. Math. Phys. 8 (1996); simple-current extensions: Böckenhauer–Evans, Comm. Math. Phys. 197 (1998)); KLM <math>\mu(B) = 4/2^2 = 1 \Rightarrow</math> HOLOMORPHIC <math>\Rightarrow (E_8)_1</math> by the v463 pin (<math>E_4/\eta^8</math> <math>q^1 = 248</math> re-verified). The 2024/2025-preprint leg (AGT/AMT, v459) is DEMOTED to an independent second witness — the extension certification now rests on 1995–2001 PEER-REVIEWED journals. <b>[E]</b> INDEX-4 PARALLEL: the <math>\mu_4</math> glue <math>J = (s, \Lambda_1)</math> has <math>h(J^k) = \{1, 1, 1\}</math> ALL integer (<math>\frac{5}{8} + \frac{3}{8} = 1</math>; <math>\frac{1}{2} + \frac{1}{2} = 1</math>) — the v125 isotropy <math>q(k, k) = k^2</math> at net level; <math>\mu = 16/4^2 = 1</math>, the same endpoint. <b>[E]</b> <math>R1'</math> INVARIANTS COMPUTED: per-copy FHS Chern <math>C(M=1) = 1</math> vs <math>C(M=3) = 0</math>, <math>\nu = 16 \times  C  = 16</math>, <math>c_- = \nu/2 = 8 = g_{\text{car}} + N_{\text{fam}}</math>; <math>\nu \bmod 16 = 0</math> is EXACTLY Kitaev’s class-D phase whose edge admits a purely bosonic description = the <math>(E_8)_1</math> level-1 state (Ann. Phys. 321 (2006), sixteen-fold way). <b>[E]</b> THE CHAIN COMPOSES: T1 (phase classification: Kitaev + <math>c_-</math> invariance, Kapustin–Spodyneiko PRB 101 (2020) / Kim et al. PRL 129 (2022)) <math>\rightarrow</math> T2 (MMST CMP 397 (2023) + Böckenhauer 1996) <math>\rightarrow</math> T3 (LR/BE + KLM CMP 219 (2001)) <math>\rightarrow</math> T4 (Dong–Mason + v463) with matching in/out slots (v297 typing). <b>[C]</b> <math>R1 \rightarrow R1'</math>: collar_realizes (model-level fiat) replaceable by collar_invariants — {quasi-free <b>[C]</b> (v155/v160), gap <b>[E]</b> (v302), class D, <math>c_- = 8</math> <b>[E]</b> (v456, from P1)}; quasi-freeness stays the ONE <b>[C]</b> hypothesis; Lean mirror FORM.SEAM.RESIDUAL.01. <b>[O]</b> NOT CLOSED: the cited theorems stay cited (certification, not re-proof); SEAM.EQUIV.01 stays <b>[O]</b>. Mixed exact (weights/<math>\mu</math>/tower, Wolfram-mirrored) + numerical (FHS Chern, Python-only). <b>[E][C][O]</b></p> |

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| Script                       | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| v470_alpha_inflow_level.py   | <p>ALPHA.QUILLEN.INFLOW.01: the <math>\alpha^3</math> level = the COMPUTED bulk Chern invariant + the seam <math>F</math>-normalisation = the affine embedding index <math>k_Y=5/3</math>. [E] TARGET RE-VERIFIED: <math>\alpha^{-1}=137.0359992168</math> unique root; the Quillen split <math>a^3-2c_3^3a^2=8b_1c_3^6\ln(1/\varphi_{\text{seam}})</math> at the root to <math>&lt;10^{-35}</math> (v341/v382). [E] THE INVARIANT COMPUTED: FHS Chern <math>C(M=1)=1</math>, <math>C(M=3)=0</math> — the same integers as v367/v461, here read as the quantised <math>U(1)</math> response of the collar phase. [C] <math>k_0= C =1</math> UNDER THE EM1 READING: the <math>a^3</math> coefficient EQUALS the computed bulk invariant of the SAME model that realises S3 — “why exactly 1” is answered by computation + the cited inflow identification (TKNN PRL 49 (1982) / Avron–Seiler–Simon PRL 51 (1983) quantisation; Callan–Harvey Nucl. Phys. B250 (1985) inflow; APS 1975 / Alvarez-Gaumé–Della Pietra–Moore 1985 / Witten Rev. Mod. Phys. 88 (2016) <math>\eta=\text{CS}</math> part of <math>\delta\log\det</math>; Quillen 1985 / Bismut–Freed CMP 106 (1986) determinant-line integrality) — REPLACING v435’s minimality assumption. [E] <math>k_Y=5/3</math> EXACT: <math>\text{tr}_5(Y^2)/\text{tr}(T_3^2)=(5/6)/(1/2)=5/3</math> (Ginsparg, Phys. Lett. B197 (1987)); <math>(3/5)\times\frac{41}{6}=\frac{41}{10}=b_1</math> — the “GUT 3/5” IS the inverse embedding index, not an imported convention; carrier decomposition <math>\frac{41}{6}=\frac{20}{3}</math> (fermions, <math>3\times 16</math>-content) <math>+\frac{1}{6}</math> (Higgs doublet). [C] <math>F</math>-NORMALISATION RETYPED: once the seam net is <math>(E_8)_1</math> level 1 (SEAM.EQUIV.01), the <math>U(1)_Y</math> normalisation <math>\langle J(z)J(w)\rangle=k/(z-w)^2</math> is FIXED by <math>k_Y</math> — level-1 current-algebra rigidity, no continuous dial; the v434 [C] has ZERO independent content (a face of SEAM.EQUIV.01, per the v382 typing). [E] UNIFICATION: ONE invertible phase, TWO quantised responses — <math>c_-=16\times\frac{1}{2}\times C =8=g_{\text{car}}+N_{\text{fam}}</math> (gravitational, feeds SEAM.EQUIV.01/S3, v456) and <math>C=1</math> (<math>U(1)</math>, feeds EM1) — “the one theorem and the one number” are two faces of the SAME phase; closing R1’ (v469) feeds both. [O] NOT CLOSED: the bridge lemma “<math>\delta\log\det_c(\text{seam})</math> = the inflow response” stays the named cited step; the EM1 reading stays [C]; ALPHA.QUILLEN.EXACT.01 stays [O]; <math>\alpha^{-1}</math> stays [E] regardless. Mixed exact (<math>k_Y/b_1</math> identities, Wolfram-mirrored) + numerical (root/split mpmath, FHS Chern, Python-only). [E][C][O]</p> |
| v471_seam_horizon_replica.py | <p>SEAM.EHMODEL.04: the Seam–Horizon replica chain exercised NUMERICALLY on a discretized seam collar — the kernel-identification premise of SEAM.THEOREM.01 (the residual isolated by v150/v151/v152) discharged at the FINITE/MODEL level for the seam’s own kernel, with REAL replica sheets. [E] REPLICA<math>\Rightarrow</math>EH FORM on a real operator: <math>dG/d\ln m = 2C(\gamma)</math> on deficit cones <math>\gamma/2\pi = \frac{1}{4}, \frac{1}{2}, \frac{3}{4}</math> (0.01–0.05%) AND replica sheets <math>n=2</math> (0.8%), <math>n=3</math> (1.5%). [E] COEFFICIENT CUTOFF-INDEPENDENT (0.5% drift) while the <math>\zeta</math>-scale <math>\mu_{\text{lat}}</math> drifts with the discretization = the v152 anchor exhibited on data; NO canonical <math>\ln(m/\mu)=\frac{3}{4}</math> emerges (recorded honestly). [E] BFK SPLIT MEASURED ON THE KERNEL: Schur identity exact (<math>\sim 10^{-16}</math>); the Calderón/Schur factor’s deficit slope is a PURE CUT-EDGE law <math>E(1-q)</math> with tip term 0 (power check: the same law fails on the full determinant); the halves carry the whole tip = <math>2C_{\text{cone}}</math> via Kac doubling (exact). [E] THE SEAM’S OWN SPECTRUM: the v302 transfer masses <math>m_2=6\ln\frac{3}{2}</math>, <math>m_3=6\ln 3</math> sit ON the calibrated EH line (<math>\leq 0.05\%</math>); the Perron/attractor mode (<math>m=0</math>) is IR-DIVERGENT (runs exactly as <math>-2C\ln(\text{IR})</math>) — the recovery gap <math>\Delta = 6\ln\frac{3}{2}</math> makes the induced-gravity coefficient finite: the same gap that makes the attractor unique makes <math>1/G</math> finite. [O] HONEST SCOPE: finite/discretized level; SEAM.THEOREM.01 stays open — the residual retypes to the CONTINUUM scaling limit (MMST class, = the single residual of SEAM.EQUIV.01, v336) + the v152 anchor. Numerical (sparse lattice determinants), Python-only by nature. [E][O]</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

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| Script                             | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| v472_quillen_<br>detline_moduli.py | <p>ALPHA.QUILLEN.DETLINE.01: the DETERMINANT LINE over the <math>U(1)</math>-twist moduli of the collar model carries curvature = the inflow level — the finite Quillen/Dai–Freed shadow of the v470 bridge lemma, COMPUTED. [E] BLOCH LEVEL RE-VERIFIED: BZ Chern <math>C(M=1)=1</math>, <math>C(M=3)=0</math>, <math>C(M=-1)=-1</math> (the v470/v367 integers). [E] THE QUILLEN OBJECT COMPUTED: the same collar <math>h(k)</math> in real space on an <math>L \times L</math> torus with TWISTED boundary conditions — the twist torus <math>(\theta_x, \theta_y)</math> IS the moduli space of flat <math>U(1)</math> connections — and the Fukui–Hatsugai–Suzuki curvature of the determinant line of the occupied one-particle frame (the free-fermion ground state) over that torus = 1 exactly (<math>10^{-9}</math>) at <math>M=1</math>, for <math>L=4</math> AND <math>L=6</math> (size-independent, as an integer must be). [E] CONTROLS: trivial collar <math>M=3 \rightarrow 0</math>; orientation flip <math>M=-1 \rightarrow -1</math>. [E] TWO TORI, ONE LEVEL: the twist-moduli integer equals the Bloch-BZ integer for all three <math>M</math> — the Niu–Thouless–Wu identification (PRB 31 (1985)), exhibited on the collar model that realises S3 (v460/v461). [E] SECTION GLOBALLY DEFINED: the Fermi gap (= 2.0) stays open over the WHOLE twist torus — the Dai–Freed section has no zero, the holonomy reading is globally valid. [C] THE READING: det-line holonomy over the <math>U(1)</math> moduli = inflow level <math>k_0= C =1</math> — the v470 bridge lemma “<math>\delta \log \det_\zeta(\text{seam}) = \text{the inflow response}</math>” holds VERBATIM at the finite/model level, where <math>\log \det</math> of the occupied frame replaces <math>\log \det_\zeta</math> (Quillen 1985; Bismut–Freed CMP 106 (1986); Dai–Freed JMP 35 (1994); FHS JPSJ 74 (2005)). [O] NOT CLOSED: the abstract-seam <math>\zeta</math>-determinant identification (the continuum leg, = the SEAM.EQUIV.01 face); ALPHA.QUILLEN.EXACT.01 stays [O]; <math>\alpha^{-1}</math> stays [E] regardless. Structurally the same manoeuvre as v471 for SEAM.THEOREM.01’s replica side. Numerical (dense eigh + FHS twist grid), Python-only by nature. [E][C][O]</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| v473_entropic_<br>action_bridge.py | <p>GRAV.ENTROPIC.ACTION.01: the entropic-action bridge — Bianconi’s <i>Gravity from entropy</i> (Phys. Rev. D 111, 066001 (2025); arXiv:2408.14391) quantified as an EXTERNAL CANDIDATE for the local volume action behind TFPT’s parameter-free Einstein equation (v358/v359, honestly typed “equation of state, not a from-action quantisation” [O]). Her <math>S_B = -\text{Tr}_F \ln(\tilde{G}\tilde{g}^{-1})</math> is the quantum relative entropy between the spacetime metric and the matter-induced metric (Araki-related, <math>\Delta^{1/2} = \tilde{G}\tilde{g}^{-1}</math>), a local covariant action whose variation gives Einstein + emergent nonnegative <math>\Lambda</math> at low coupling — the missing variational layer IF the bridge holds. [E] CARRIER HODGE COUNT: her <math>\Omega^{0,1,2}</math> fiber on the FIVE-SLOT carrier <math>\mathbb{C}^5</math> (not 4d spacetime, where it is <math>1+4+6=11</math>) counts <math>1+5+10=16=2^{g_{\text{car}}-1} = \dim S_{D_5}^+</math>, Hodge-folded onto <math>\Lambda^{\text{even}}\mathbb{C}^5</math> (vector-space identity ONLY). [E] COEFFICIENT MATCH (the one quantitative pin): her weak-coupling <math>\mathcal{L} = 3\beta R - \alpha\mathcal{L}_m</math> matched to <math>1/(16\pi G) = c_3/(2G)</math> fixes her free constant EXACTLY, <math>\beta_B' = c_3/6 = 1/(48\pi)</math>; the <math>6 = 3 \text{ blocks} \times 2 = p_2 =  R^+(A_3) </math> (structural echo, not a proof). [E] ENTROPY POTENTIAL: <math>\Lambda_{\mathcal{G}} = \frac{1}{2\beta} \text{Tr}(\mathcal{G} - I - \ln \mathcal{G}) \geq 0</math>, quadratic at the fixed point <math>\Rightarrow</math> with <math>\varepsilon_\star = e^{-\alpha^{-1}Q}</math> the closed v60 branch <math>(3/4\pi^2)e^{-2\alpha^{-1}}</math> is reproduced with the EXACT target <math>\text{Tr } Q^2 = 32c_3^4 = 1/(128\pi^4) = \frac{2}{3} \delta_{\text{top}}</math> ([C] until <math>Q</math> is constructed). [E] FRULLANI: <math>-\ln A = \int_0^\infty \frac{dt}{t} (e^{-tA} - e^{-t})</math> — her log action is an all-scale heat-kernel action, the right genre against <math>S_{\text{rel},\chi}</math> (KMS cutoff). [E] <math>R^2</math> KILL TEST: raw <math>(c_3/6)^2</math> vs Starobinsky <math>1/(12c_3^7)</math> (<math>M_{\text{scal}}^2 = c_3^7 \bar{M}_{\text{Pl}}^2</math>, v36) — EXACT gap <math>3(8\pi)^9 \approx 1.2 \times 10^{13}</math> (13 orders): a direct identification of the raw entropic action with the TFPT <math>R^2</math> sector is FALSE without a mechanism (light <math>\mathcal{G}</math>-trace mode generating <math>c_3^{-9}</math>, or KMS-spectral renormalisation) — pre-registered. [C] COMPRESSION CONJECTURE: <math>P_\Sigma(\tilde{G}\tilde{g}^{-1})P_\Sigma = \Delta_\Sigma^{1/2} + \text{the operator-level Hodge items (Clifford, Dirac, leaf involution, Calderón polarisation) stay open; Lorentzian positivity NOT automatic — TFPT’s OS route is the sturdier construction. [O] NOTHING CLOSES: v359’s equation-of-state status stays [O]; zero TFPT knobs added. Mixed exact (sympy, Wolfram-mirrored) + one quadrature (mpmath). [E][C][O]</math></p> |

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| Script                         | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| v474_entropic_hodge_carrier.py | <p>GRAV.ENTROPIC.HODGE.01: the OPERATOR level of the entropic-action bridge (work packages 1 + 4-algebra of v473). [E] CAR⇒CLIFFORD: the ten <math>\Gamma</math>'s from <math>(a_k^\dagger, a_k)</math> on the carrier Fock space <math>\Lambda^\bullet \mathbb{C}^5</math> (<math>\dim 2^{g_{\text{car}}}=32</math>) satisfy <math>\{\Gamma_a, \Gamma_b\}=2\delta_{ab}</math> exactly (all 100 pairs, integer matrices) — the <math>\mathfrak{so}(10)=D_5</math> Clifford algebra acts on the carrier exterior algebra. [E] <math>S^+=\Lambda^{\text{even}}</math> AS <math>D_5</math>-MODULE: the 45 bilinears <math>[\Gamma_a, \Gamma_b]/4</math> commute with <math>\gamma=(-1)^N</math>, span exactly <math>45=\dim \mathfrak{so}(10)</math> and leak NOTHING from the 16-dim even subspace — the half-spinor IS the even exterior algebra WITH its spinor structure (operator level of v2/v44/v197). [E] HODGE-DIRAC SYMBOL: <math>c(\xi)=a^\dagger(\xi)-a(\xi)</math> satisfies <math>c(\xi)^2=- \xi ^2 \text{Id}</math> for SYMBOLIC <math>\xi</math> — Bianconi's <math>D=d+\delta</math> is Clifford multiplication at symbol level. [E] THE FOLD IS <math>5\rightarrow\bar{5}</math>: the traceless <math>\mathfrak{u}(5)</math> weights of <math>\Lambda^4</math> are the NEGATIVES of those of <math>\Lambda^1</math>, so the Hodge fold <math>\star: \Lambda^1\cong\Lambda^4</math> conjugates the vector block — her fiber <math>1+5+10</math> becomes the GUT half-spinor <math>16=1+\bar{5}+10</math> (the SM carrier decomposition), not a mere dimension match. [E] HONEST OBSTRUCTION: <math>\Omega^{\leq 2}</math> is NOT Clifford-invariant (<math>c(\xi)</math> maps <math>\Lambda^2</math> into <math>\Lambda^3</math>); only <math>\Lambda^{\text{even}}</math> is (even algebra) — the bridge must go through the fold, never through degree truncation (constraint on AP2). [E] Q-TARGET ENUMERATION (the v473 conjecture DECIDED): <math>\text{Tr } Q^2=d k^2 c_3^4=32c_3^4</math> with INTEGER <math>k</math> has exactly the supports <math>(d,k)=(2,4), (8,2), (32,1) = \{ \mathbb{Z}_2 , g_{\text{car}}+N_{\text{fam}}=\text{rank } E_8, 2^{g_{\text{car}}}=\dim \Lambda^\bullet\}</math> — minimal uniform solution <math>q=c_3^2=1/(64\pi^2)</math> per Fock mode; the two-form block <math>d=10</math> (the “pair sector” conjecture) and the fiber <math>d=16</math> need irrational multipliers <math>\sqrt{16/5}, \sqrt{2} \Rightarrow</math> KILLED in uniform form; bonus identity <math>\text{Tr } Q^2/\delta_{\text{top}}=32/48=2/3=\dim \Lambda^\bullet/\Omega_{\text{adm}}</math>. [C] the physical choice among the three admissible supports; [O] AP2 compression + <math>R^2</math> mechanism stay open, no marker moves. Mixed exact: integer Fock matrices (numpy, exact in double) + symbolic sympy (weights/enumeration Wolfram-mirrored). [E][C][O]</p> |
| v475_entropic_scalaron_gate.py | <p>GRAV.ENTROPIC.SCALARON.01: the <math>R^2</math> kill test of the entropic-action bridge EXECUTED (work package 5 of v473) + the explicit Lorentzian-positivity witness. [E] MAX-SYMMETRIC EIGENVALUES (her Appendix-B flattening verified): <math>\tilde{R}\tilde{g}^{-1}</math> has eigenvalues EXACTLY <math>\{R; R/4 \times 4; R/6 \times 6\}</math> on a general diagonal 4d metric, incl. the <math>6\times 6</math> two-form flattening. [E] EXACT VACUUM <math>f(R)</math>: <math>\mathcal{L}_B(R)=-[\ln(1-\beta R)+4\ln(1-\beta R/4)+6\ln(1-\beta R/6)]=3\beta R+\frac{17}{24}\beta^2 R^2+O(R^3)</math> — the 3 re-derives her Eq. (45) (= the v473 pin), <math>\frac{17}{24}=\frac{1}{2}(1+\frac{1}{4}+\frac{1}{6})</math> is the EXACT tensorial <math>R^2</math> factor. [E] RAW SCALARON TRANS-PLANCKIAN: with <math>\beta'=c_3/6</math> both routes (Starobinsky matching AND <math>f'/(3f'')</math>) give <math>m_{\text{raw}}^2=\frac{4608\pi^2}{17}M_{\text{Pl}}^2</math>, <math>m_{\text{raw}}\approx 51.7 M_{\text{Pl}}\approx 1.3\times 10^{20}</math> GeV — HEAVY, not light (TFPT: <math>c_3^{7/2}M_{\text{Pl}}=3.06\times 10^{13}</math> GeV); required enhancement EXACTLY <math>m_{\text{raw}}^2/m_{\text{TFPT}}^2=\frac{72}{17}(8\pi)^9\approx 1.69\times 10^{13}</math> in mass<sup>2</sup> — v473's interpretation 2 (the TFPT scalaron as the light <math>\mathcal{G}</math>-trace mode) is KILLED in naive form. [E] DOMAIN: first log pole at <math>R=1/\beta=384\pi^2 M_{\text{Pl}}^2</math> (trans-Planckian) — the raw action is well-defined at all sub-Planckian curvatures; the obstruction is the MASS, not the domain. [E] LORENTZIAN WITNESS: a timelike gradient gives <math>Gg^{-1}</math> eigenvalue <math>1-\alpha v^2\leq 0</math> for <math>v^2\geq 1/\alpha</math> (ln undefined); spacelike/Euclidean positive for all <math>v</math> — the v473 [C] caveat exhibited, the OS/reflection-positive route sturdier. [C] VERDICT: interpretation 1 survives (the entropic action covers the Einstein + <math>\Lambda</math> level), 2 is dead naive, 3 (KMS-spectral renormalisation) is the ONLY surviving <math>R^2</math> route. [O] unchanged: AP2 compression + the <math>R^2</math> gate; no marker moves. Exact (sympy, Wolfram-mirrored). [E][C][O]</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

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| Script                           | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| v476_entropic_compression_toy.py | <p>GRAV.ENTROPIC.COMPRESS.01: the AP2 compression conjecture of the entropic-action bridge (v473) made WELL-POSED on an exactly solvable model. Numerical facts on a staggered-mass chain: the ground-state covariance is a PROJECTION (pure bulk, thermal boundary — the seam situation); the compression <math>C_A=PCP</math> has spectrum strictly inside <math>(0,1)</math> with finite Peschel <math>K=\ln((1-C_A)/C_A)</math> and <math>S&gt;0</math> (v155 in the bridge context). ORDER OBSTRUCTION (the reading DECIDED): on the pure bulk <math>f(x)=x/(1-x)</math> is SINGULAR on <math>\text{spec}(C)=\{0,1\}</math> — the literal operator-side reading <math>Pf(C)P</math> of the v473 conjecture is ILL-POSED while <math>f(PCP)</math> is finite: the compression identity can only be read STATE-SIDE (build <math>\Delta_\Sigma</math> from the COMPRESSED relative metric), matching Bianconi's own local-metric construction — AP2 retyped, not weakened. [E] BLOCK ALGEBRA (symbolic): <math>[C^2]_{AA}-(C_{AA})^2=BB^t</math> and <math>[C^3]_{AA}-(C_{AA})^3=ABB^t+BB^tA+BDB^t</math> — first order in the cross block vanishes identically, so for ANY analytic <math>f</math> the two readings agree to SECOND order: the modular (entanglement) content lives exactly in that quadratic correction. GAP CONTROL at the bounded (covariance) level: the bulk identity <math>f(C_T)=e^{-h/T}</math> holds to <math>10^{-11}</math> relative; the mismatch obeys the quadratic law <math>d \leq x^2</math> and falls with the gap in the gapped regime, the decoupled cut agrees EXACTLY — the two readings converge in the gap-dominated regime, where TFPT operates (<math>\Delta_{\text{eff}}&gt;0</math>, v76/v337). [C] the retyped conjecture; [O] AP2 stays open (the continuum seam statement); no marker moves. Numerical (numpy lattice + one symbolic block identity), Python-only by nature. [C][O]</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| v477_entropic_scaleflow.py       | <p>GRAV.ENTROPIC.SCALEFLOW.01: the scale-flow representation of the entropic action — v475's surviving interpretation 3 typed PRECISELY: the <math>(72/17)(8\pi)^9</math> gap collapses to ONE moment condition on the scale weight, satisfied by the SAME KMS moment TFPT already fixes (v36: <math>f_0=6(4\pi)^2/c_3^7</math>); zero new dials, nothing closes. [E] SCALE-FLOW IDENTITY: <math>-\ln a = \int_0^\infty \frac{dt}{t}(e^{-ta}-e^{-t})</math> (symbolic) <math>= 2 \int_0^\infty \frac{dx}{x}(e^{-a/x^2}-e^{-1/x^2})</math> (<math>t=1/x^2</math>, <math>10^{-20}</math> quadrature) — Bianconi's log action IS the FLAT scale-integral of relative heat-kernel actions; TFPT's <math>S_{\text{rel},\chi}</math> evaluates ONE KMS scale with a weight <math>f</math>: the bridge is a choice of SCALE MEASURE. [E] MOMENT DICTIONARY: a weight <math>w(t) dt/t</math> gives <math>3\beta R \mu_1 + \frac{17}{24}\beta^2 R^2 \mu_2</math> (<math>\mu_1 = \int w e^{-t} dt</math>, <math>\mu_2 = \int w t e^{-t} dt</math>); the flat weight has <math>\mu_1=\mu_2=1</math> = the v475 raw coefficients — the raw action is the unit-moment point of a moment family. [E] THE ONE MOMENT CONDITION: with the weight-dressed EH pin <math>\beta' \mu_1=c_3/6</math> the scalaron is <math>m^2=\frac{4608\pi^2}{17}\frac{\mu_1^2}{\mu_2}M_{\text{P}1}^2</math>; demanding <math>m^2=c_3^7 M_{\text{P}1}^2</math> forces EXACTLY <math>\mu_2/\mu_1^2=\frac{72}{17}(8\pi)^9</math> — the v475 kill-test number IS a moment ratio; and the closure identity <math>\frac{4608\pi^2/17}{(72/17)(8\pi)^9}=c_3^7</math> holds IDENTICALLY: the 13 orders are a scale-measure datum, not a bridge mismatch. [E] v36-CONSISTENCY: <math>f_0=6(4\pi)^2/c_3^7 \Leftrightarrow M^2/M_{\text{P}1}^2=c_3^7</math> — the same mass from both parametrisations, ZERO new dials. [C] a CONSISTENCY statement, not a derivation (the KMS weight is TFPT input; the entropic action supplies the Einstein + <math>\Lambda</math> level and the scale-flow FORM, TFPT supplies the measure); negative side-record: the exact lattice-BW commutator attempt for AP2 (Peschel–Eisler simple forms) did NOT hold on the half-filled open chain (kept out of the suite, logged). [O] unchanged: the <math>R^2</math> gate and v359's typing. Exact (sympy + one quadrature, Wolfram-mirrored). [E][C][O]</p> |

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| Script                           | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| v478_entropic_continuum_legs.py  | <p>GRAV.ENTROPIC.LEGS.01: first computable steps on the TWO remaining legs of the entropic-action bridge (v473–v477). LEG A (AP2 continuum): the compressed critical interval has <math>S(L)=(c/3)\ln L+k</math> with <math>c_{\text{est}}=1.00089/1.00038/1.00021</math> (monotone <math>\rightarrow</math> the Dirac <math>c=1=2\times\frac{1}{2}</math>, the v450 reading; gapped control <math>\sim 6\times 10^{-8}</math>, area law); the Peschel modular Hamiltonian <math>K=\ln((1-C_A)/C_A)</math> organises on the CHM weight <math>\beta(x)=x(L-x)/L</math> (corr 0.87<math>\rightarrow</math>0.99, plateau drift shrinking; constants deliberately NOT asserted — the lattice EH is only asymptotically local); even-range bands vanish identically (particle-hole, <math>\sim 10^{-36}</math>), odd tail 1.8% parabola-enveloped — the state-side <math>\Delta_\Sigma</math> (v476’s forced reading) FLOWS TO the CHM/BW geometric form that TFPT’s Einstein derivation consumes (v323/v358): the bridge’s modular side and TFPT’s gravity side MEET in the continuum limit. LEG B (the measure): [E] single-scale corollary — <math>d\mu=\delta(t-t_0)dt</math> gives <math>\mu_2/\mu_1^2=e^{t_0}</math> exactly, so the v477 condition forces <math>t_0=\ln\frac{72}{17}+9\ln(8\pi)=30.461</math> — ONE dimensionless KMS time, exact closed form; DECLINED: <math>t_0\neq h(E_8)=30</math> (difference 0.461; the near-miss is rejected per the v354/v355 no-free-pattern rule); deriving the measure from the entropic structure stays OPEN. [C] the meeting point; [O] both legs, no marker moves. High-precision numerical (mpmath, 50 digits) + one exact corollary, Python-only by nature. [C][O]</p>                                                                                                                                                                                                                                    |
| v479_kronheimer_quiver_bridge.py | <p>SEAM.KRONHEIMER.01: the R3 “combinatorial-to-geometric bridge” of the <math>\det K=1</math> keystone (v344 — “the rewrite attractor graph IS the du Val singularity”) is an ESTABLISHED THEOREM: Kronheimer’s ALE hyper-Kähler quotient construction + Torelli classification (J. Diff. Geom. 29 (1989) 665 and 685), with ALL finite input data already compiler outputs. [E] MARKS = NULL = PERRON: <math>C_{\text{aff}}\delta=0</math> and <math>A\delta=2\delta</math> on the affine-<math>E_8</math> McKay graph; multiset <math>\{1, 2, 2, 3, 3, 4, 4, 5, 6\}</math> = the <math>2I</math> irrep dimensions = the v312 rewrite-attractor Perron vector. [E] DIMENSION COUNT: <math>\dim_{\mathbb{R}} M - 4\dim_{\mathbb{R}} G = 4(\sum_{\text{edges}} d_i d_j - \sum d_i^2 + 1) = 4</math> — the hyper-Kähler quotient of the marks quiver is 4-real-dimensional (the <math>\mathbb{C}^2/\Gamma</math> geometry); <math>\sum_{\text{edges}} d_i d_j = \sum d_i^2 = 120</math> forced by the null-vector identity. [E] GAUGE DATUM: <math>\sum \delta_i^2 = 120 =  2I  =  \mu_4  h(E_8)</math>, <math>h(E_8)=30= \mathbb{Z}_2 N_{\text{fam}}g_{\text{car}}</math>. [E] RESOLUTION: <math>8=\text{rank } E_8</math> exceptional spheres with intersection form <math>-\text{Cartan}(E_8)</math>, <math>\det=1</math> and SNF = identity (the Poincaré-sphere link = the <math>\det K=1</math> face, v232/v344); deformation space <math>\mathfrak{h} \otimes \mathbb{R}^3</math>, <math>\dim 3 \cdot 8 = 24</math>. [C] the theorems are CITED, not re-proved (the same import class as NPW26/MMST/AGT, v424/v336/v459). [O] the residual premise TRANSFORMS, it does not vanish: “the raw seam supplies ALE hyper-Kähler data asymptotic to the <math>(2, 3, 5)</math> cone” = the SAME single arrow as SEAM.EQUIV.01 (v346 L2); no gate closes. Exact (sympy; Wolfram mirror added, counted with the verified 2026-07-21 engine run). [E][C][O]</p> |

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| Script                               | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| v480_multilocal_<br>four_interval.py | <p>QGE0.MULTILOCAL.01: the four <math>\mu_4</math> marks make the seam circle a FOUR-INTERVAL free-fermion geometry — the class whose multi-interval modular theory is EXACTLY KNOWN (Casini–Huerta mixing term; Longo–Xu multi-interval subfactors; Rehren–Tedesco multilocal fermions; KLM: holomorphic <math>\Leftrightarrow</math> trivial multi-interval inclusion, <math>\mu=1</math> — the SAME one bit as <math>\det K=1</math>). On the exactly solvable antiperiodic critical ring (<math>N=256</math>, <math>Q=64</math>, <math>\ell=48</math>): [E] BINARY CLOCK: the quarter-turn clock is a signed permutation with <math>\rho^4 = -1</math> EXACTLY — the fermionic <math>\mu_4</math> clock is the order-8 double-cover lift (the same <math>\mathbb{Z}_2</math>-lift pattern as <math>2I</math> over <math>I</math>). [E] EXACT SECTOR DECOUPLING: the primitive-8th-root character projectors resolve the identity and commute with the reduced covariance at <math>10^{-15}</math> (the Rehren–Tedesco diagonal 4-fold-cover isomorphism at covariance level). [E] COVER/TWIST: each sector kernel is the SINGLE-interval quarter-ring kernel with boundary twist <math>\lambda_k</math>; the union of sector spectra = the full 4-interval spectrum at <math>10^{-14}</math>. [E] CLOCK INVARIANCE MANIFEST: <math>\rho C \rho^{-1} = C</math> at <math>10^{-14} \Rightarrow [\rho, K]=0</math> by functional calculus — <math>\omega \circ \rho = \omega</math> HOLDS at the 4-interval state level; NEGATIVE CONTROL: displacing ONE interval by one site re-couples the sectors at <math>10^{-2}</math>. CONJUGATE-POINT MIXING (numerical): every cross-interval row of <math>K</math> peaks on <math> s'-s  \leq 2</math> at both sizes — the lattice shadow of the Casini–Huerta mixing term. [C] the continuum multi-interval theory is CITED; [O] the raw-seam premise (QGE0.SYM.01/SEAM.EQUIV.01) stays OPEN — mechanism exhibited, not closed. Numerical (numpy), Python-only by nature. [E][C][O]</p>                                                                     |
| v481_seesaw_<br>carrier_ladder.py    | <p>FLAV.NUSCALE.02: the absolute neutrino-mass scale under the CARRIER Yukawa normalisation <math>y_\nu = y_t</math> — FLAV.NUSCALE.01's two-parameter <math>(y_\nu, M_R)</math> trade-off collapses to ONE parameter and the required <math>M_R</math> lands INSIDE the compiler's own PS window; the nearby integer <math>c_3</math>-rung is explicitly DECLINED at 1-loop (v354/v355). CANDIDATE class (genre v467/v468); the frozen record (v84) is untouched and NO gate closes. CARRIER NORMALISATION (conjectured premise; named premise): one <math>SO(10)</math> 16 per family + minimal Yukawa sector (<math>\mathbf{10}_H</math> / PS (1, 2, 2)) <math>\Rightarrow y_\nu = y_t</math> at the matching scale — the trade-off collapses to <math>M_R</math> alone. INVERSION INSIDE THE WINDOW (numerical): the observed <math>m_3=0.0503</math> eV demands <math>M_R=9.3 \times 10^{13}</math> GeV, INSIDE <math>[M_{PS}, M_{GUT}] = [4.2 \times 10^{13}, 2.4 \times 10^{15}]</math> (v249), at <math>\log_{c_3}(\bar{M}_{Pl}/M_R) = 3.15</math> (<math>y_\nu=1</math> gives the structureless 2.58 — the old FLAV.NUSCALE.01 number). [X] INTEGER-RUNG HONESTY GATE: <math>M_R = c_3^3 \bar{M}_{Pl} = 1.53 \times 10^{14}</math> GeV predicts <math>m_3 = 0.030</math> eV (40% low at 1-loop; ADKLR <math>\kappa</math> run-down factor 0.794) — NEAR the rung (3.15 vs 3), NOT ON: the ladder pin is DECLINED; the decision computation is NAMED (2-loop SM + explicit PS threshold matching). FALSIFIABLE BAND (numerical): over the PS window <math>m_3</math> spans <math>[0.002, 0.115]</math> eV bracketing the observation; the <math>M_{PS}</math> end (<math>\Sigma \sim 0.124</math> eV) is already in tension with DESI <math>\sim 0.072</math> — cosmology cuts the window from below; the NO floor <math>\Sigma m_\nu = 0.059</math> eV stays the falsifiable edge (v272). [C] / [O] candidate class; the absolute scale stays formally open. Numerical (explicit 1-loop RG: gauge/<math>y_t/\lambda</math> up, ADKLR Weinberg-operator down), Python-only by nature. [C][O]</p> |

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| Script                              | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| v482_seesaw_rung_<br>decision.py    | FLAV.NUSCALE.03: the decision computation NAMED by v481, EXECUTED in bracketed (assumption-robust) form — the integer rung $M_R = c_3^3 \bar{M}_{Pl}$ is EXCLUDED as an unstructured exact pin, conditional on the carrier premise $y_\nu = y_t$ . THE GAP: central $m_3(\text{rung}) = 0.0301$ eV vs observed 0.0503 — rescue factor $\times 1.670$ (+67%, 0.154 $c_3$ -rungs). BRACKETED EXCLUSION: with $m_t(\overline{\text{MS}}) \pm 3$ GeV ( $> 3\sigma$ ), $\alpha_3^{-1} \mp 0.10$ ( $> 3\sigma$ ), a $\pm 10\%$ band on the ADKLR $\kappa$ run-down and the PS-leg Yukawa beta scaled $\times [0.5, 1.5]$ , the largest single shift is $\times 1.10$ and the ALL-favourable combination reaches only $\times 1.165$ — far below $\times 1.670$ : the v481 1-loop miss is NOT an RG artifact. [C] NAMED ESCAPE, RECORDED AND DECLINED: only a third-generation Majorana STRUCTURE factor $r \sim 1.67$ revives the rung; post-hoc $r_{\text{needed}} = 1.6696$ sits 0.18% from $g_{\text{car}}/N_{\text{fam}} = 5/3$ — no mechanism forces a 3/5 Clebsch on $M_{R,3}$ today, so this is a named coincidence with a named missing mechanism (PS Majorana-sector Clebsch), DECLINED per v354/v355 (the $t_0 \sim 30$ pattern of v478). [O] the v481 one-parameter window candidate (NO floor $\Sigma m_\nu = 0.059$ eV) never used the integer rung and STANDS; frozen record untouched; nothing closes. Numerical (explicit 1-loop RG + bracketed envelopes), Python-only by nature. [C][O][X]                                                                                                                                                                                                                                                                                                         |
| v483_pillowcase_<br>exact_traces.py | HYP.PHIO.GEOM.01: the GEOMETRIC side of the $\varphi_0^{\text{ret}}$ puncture target (HYP.PHIO.PUNCTURE.01/v408) is EXACT — every twisted/orbifold heat trace of the flat $\tau = i$ pillowcase is a $t$ -INDEPENDENT RATIONAL, so the $\pi^{-4}$ in $d_{\text{top}} = 48c_3^4$ cannot come from flat orbifold geometry at ANY order. [E] TWISTED TRACES EXACTLY 1: $\text{Tr}(\gamma e^{t\Delta}) = 1$ for $\gamma = \sigma(z \mapsto -z)$ , $\rho(z \mapsto iz)$ , $\rho^3$ at every tested $t \in [0.05, 3]$ (position-space Gaussian image sums / erf closed forms + 2d quadrature, residual $< 10^{-9}$ ; mode-space proof: only $n=0$ is $\gamma$ -diagonal) — $t$ -independent, no asymptotic tail. [E] ATIYAH–BOTT MATCH: $1 = 4 \times \frac{1}{4}$ ( $\det(1-d\sigma) = 4 =  \mu_4 $ ) and $1 = 2 \times \frac{1}{2}$ ( $\det(1-d\rho) = 2 =  \mathbb{Z}_2 $ ). [E] PILLOWCASE DATA EXACT: orbifold heat trace $= \frac{1}{2} \text{Tr}_{T^2} + \frac{1}{2}$ (contact term EXACTLY $\frac{1}{2} = 4 \times \frac{1}{8}$ per mark, all $t$ ); clock-equivariant trace exactly 1 (all $t$ ). [E]/[O] THE REDUCTION: the analytic content of $d_{\text{top}} = 48c_3^4 = \frac{3}{256\pi^4}$ must sit in the per-insertion COUPLING normalisation — $4 =  \mu_4 $ insertions of weight $c_3$ give $c_3^4$ , multiplicity $\Omega_{\text{adm}} = 48$ ; the bare $1/(2\pi)$ -propagator 4-cycle differs by the exact RATIONAL $\frac{3}{16}$ ; the ONE remaining [O] is the rule “per-mark weight = $c_3$ ” (seam zeta/Green normalisation, the same residual class as EM.WARD.02). The target narrows from “derive the term” to “derive the weight”; HYP.PHIO.PUNCTURE.01 stays open; no gate closes. Numerical verification of exact statements + sympy exact arithmetic; Python-only by nature. [E][O] |

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| Script                          | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| v484_seam_contact_unit.py       | <p>SEAM.CONTACT.UNIT.01: the shared “<math>c_3</math> per boundary insertion” rule of the two open analytic targets (ALPHA.QUILLEN.EXACT.01 ladder <math>\{0, 3, 6\}</math>;<br/> HYP.PHIO.PUNCTURE.01 per-mark weight <math>c_3</math>) is ONE identity and NOT a new unknown: it is the KMS seam unit <math>2\pi = 1/(4c_3)</math> with <math>\frac{1}{4} = 1/ \mu_4 </math> (v239) — one bare boundary propagator ORBIT-AVERAGED over the four marks. [E] UNIT CHAIN: <math>\frac{1}{2\pi} = 4c_3 =  \mu_4 c_3 =  \mathbb{Z}_2 \chi(S^2)c_3</math>. [E] MECHANISM DERIVED ON THE SEAM CIRCLE: <math>G(d) = -\frac{1}{\pi} \ln  2 \sin(d/2) </math> (polylog identity, 40 digits) takes INTEGER multiples of <math>c_3 \ln 2</math> at the <math>\mu_4</math> separations — <math>G(\pi/2) = -4c_3 \ln 2</math>, <math>G(\pi) = -8c_3 \ln 2</math>, so <math>G_{\text{marks}} = -4c_3 \ln 2 \cdot \text{circ}(0, 1, 2, 1)</math> with <math>\text{spec} = \{4, -2, 0, -2\}</math>; the <math>\mu_4</math>-invariant (orbit-averaged) insertion carries <math>(GV) _{\text{marks}} = -(c_3 \ln 2) C</math> EXACTLY, so the log-det contact expansion is exactly graded in <math>c_3</math> per insertion with integer cycle combinatorics <math>\text{Tr}(C^k) = 4^k + 2(-2)^k</math>. [E] SUITE-WIDE GRAMMAR: the <math>F_{U(1)}</math> ladder <math>\{0, 3, 6\}</math> (v3); <math>\varphi_0^{\text{ret}}</math> ladder <math>\{1, 4\}</math> (v37); the <math>\Lambda</math> prefactor <math>\frac{3}{4\pi^2} = (8\pi)^2 d_{\text{top}} = 48c_3^2</math> (v60) — the SAME multiplicity <math>\Omega_{\text{adm}}=48</math> as the puncture, at TWO insertions; the <math>g-2</math> seam loop divides by the same <math>2\pi = 1/(4c_3)</math>. [C]/[O] THE MERGE: the two tracked [O] targets collapse into ONE named analytic step — the diagonal (self-energy) zeta-renormalisation at the marks + the state-multiplicity matching (<math>48 = \Omega_{\text{adm}}</math>, <math>41 = 10b_1</math>); the finite cycle sector is proven. Anti-numerology: the integers are forced by the square configuration; no target coefficient is claimed. Exact (sympy + 40-digit mpmath polylog); Python-only by nature. [E][C][O]</p>                                    |
| v485_contact_diagonal_closed.py | <p>SEAM.CONTACT.UNIT.02: the DIAGONAL channel of the merged analytic target (v484) settled at the computable level — the merged residual COLLAPSES ONTO THE KEYSTONE. [E] DIAGONAL ZERO AT THE KMS SCALE: the renormalised coincident-point Green function on circumference <math>\ell</math> is <math>G_{\text{reg}}(0; \ell) = \frac{1}{\pi} \ln(\ell/2\pi)</math> exactly — it VANISHES iff <math>\ell = 2\pi = 1/(4c_3)</math>, the v239 KMS seam circumference: the seam scale is the unique zero-self-energy scheme point. [E] THE LOG IS THE SCALE CHANNEL: rescaling gives <math>\frac{1}{\pi} \ln \kappa</math> and nothing else — matching the functional structure (<math>F_{U(1)}</math> carries exactly ONE log term, the transport <math>8b_1 c_3^6 \ln(1/\varphi_{\text{seam}})</math>; the <math>\varphi_0^{\text{ret}}</math> split is log-free); [C] reading: log = diagonal/scale channel, log-free = contact data at the KMS point. [E] CLOSED-FORM RESUMMATION (BFK route, v151 class): the finite-rank mark determinant resums exactly, <math>\det(I - uC) = (1 - 4u)(1 + 2u)^2</math> with <math>u = \varepsilon c_3 \ln 2</math> — all orders; the linear term is ABSENT precisely because the diagonal vanishes (<math>\text{Tr } C = 0</math>); the series matches v484’s <math>\text{Tr}(C^k) = 4^k + 2(-2)^k</math>. [E] WEIGHT DICHOTOMY: the SAME <math>48 = 3 \times 16</math> admissible states carry BOTH multiplicities — flat count <math>48 = \Omega_{\text{adm}}</math> (the <math>\varphi_0^{\text{ret}}</math> puncture) vs <math>Y^2/\text{loop}/\text{GUT-weighted } 41 = 10b_1 = 40 + 1</math> (<math>\text{Tr}_{16} Y^2 = \frac{10}{3}</math> exact, <math>\nu^c = 0</math>; Ginsparg <math>(3/5)(41/6) = 41/10</math>, v470) — one state set, two response weights. [C]/[O] CONSEQUENCE: with v484 EVERY finite/computable piece of the merged ALPHA.QUILLEN + PHIO.PUNCTURE target is proven; the single remaining [O] is the identification of the abstract seam <math>\zeta</math>-det with this glued mark determinant — a face of SEAM.EQUIV.01, exactly the v382 typing, now substantiated computationally. No gate closes. Exact (sympy + 30-digit mpmath + exact charge sums); Python-only by nature. [E][C][O]</p> |

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| Script                       | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| v486_transfer_full_rule.py   | <p>HYP.REWRITE.02: the FULL physical transfer spectrum <math>\{1, (2/3)^6, (1/3)^6\}</math> (v54/v221) from ONE forced local rule — the lazy <math>\mathbb{Z}_2</math>-pair walk; the <math>(1/3)^6</math> mode, previously generator-less, gets its local-rule origin (sharpens v327, whose uniform rule produces <math>\{1, \frac{2}{3}, 0\}</math> and can never generate the third mode). [E]</p> <p>UNIQUENESS: for <math>M(s, h) = \begin{pmatrix} 1 &amp; 0 &amp; 0 \\ 0 &amp; s &amp; h \\ 0 &amp; h &amp; s \end{pmatrix}</math> (one absorbing family channel + SYMMETRIC <math>\mathbb{Z}_2</math> pair) the survival spectrum is <math>\{s+h, s-h\}</math>; demanding the physical pair <math>\{\frac{2}{3}, \frac{1}{3}\}</math> forces UNIQUELY</p> <p>(stay, hop, leak) = <math>(\frac{1}{2}, \frac{1}{6}, \frac{1}{3}) = (1/ \mathbb{Z}_2 , 1/( \mathbb{Z}_2 N_{\text{fam}}), 1/N_{\text{fam}})</math> — atom form recorded (honest direction: spectrum <math>\rightarrow</math> rates). [E] SIX-HAND GENERATION: over the order-6 = <math>2N_{\text{fam}} =  R^+(A_3) </math> hand, <math>\text{eig}(B^6) = \{\frac{64}{729}, \frac{1}{729}\}</math> exactly; decay gaps <math>6 \ln \frac{3}{2}</math> (the recovery gap, v76) and <math>6 \ln 3</math> (the subdominant gap, now generator-full). [E] SUPERSESSSION: the uniform rule's zero mode persists under every power — no iterate reaches <math>(1/3)^6</math>; its <math>\lambda_2</math> arity story (<math>\frac{2}{3} =  \mathbb{Z}_2 /N_{\text{fam}}</math>) is unchanged. [E] sub-stochastic, symmetric (detailed balance), absorbing democratic attractor unchanged (v317). [O] the arity <math>\{2, 3\}</math> stays the anchor input; the lazy-vs-uniform split is SELECTED by the physical spectrum, not yet first-principles-derived (named candidate: stay = <math>1/ \mathbb{Z}_2 </math> as the sheet-symmetric choice of the <math>\mathbb{Z}_2</math> involution). Exact (sympy); mirrorable, deferred to the next verified engine run. [E][O]</p> |
| v487_transfer_clock_rungs.py | <p>HYP.REWRITE.03: the lazy split <math>(\frac{1}{2}, \frac{1}{6})</math> is FORCED by the resummed clock — the v486 [O] 'split selection' closes onto v124 structure; sharpens HYP.REWRITE.02, no gate closes. [E] RUNG READING: the lazy rule's one-step spectrum is EXACTLY the complete clock ladder below the wall,</p> <p><math>\text{eig}(M) = \{1 - n/3 : n = 0, 1, 2\} = \{1, \frac{2}{3}, \frac{1}{3}\}</math>; v327's uniform rule puts its second pair mode ON the wall (<math>0 = 1 - 3/3</math>) — the degeneracy re-read as 'odd channel collapsed to the wall'. [E] DECK PARITY = RUNG INDEX: <math>[B, \sigma] = 0</math>; the <math>\sigma</math>-even pair mode carries rung <math>n = 1</math> (survival <math>\frac{2}{3}</math>), the <math>\sigma</math>-odd mode rung <math>n = 2</math> (survival <math>\frac{1}{3}</math>) — each clock step flips the deck parity; the structural home of the v486 parity assignment (<math>\omega_1</math> even-channel, <math>\omega_2</math> odd-channel). [E] FORCING: faithful ladder (no rung missing, no physical mode on the wall — definitional for a full-spectrum generator) + <math>\mathbb{Z}_2</math> equivariance + rates <math>\geq 0</math> force UNIQUELY both the split <math>(\frac{1}{2}, \frac{1}{6})</math> AND the assignment even <math>\rightarrow</math> rung 1, odd <math>\rightarrow</math> rung 2 (the swap needs <math>h = -\frac{1}{6} &lt; 0</math>); the 'sheet-symmetric choice' candidate is superseded by this proof. [E] BEND COROLLARY: <math>\omega_1/\omega_2 = \text{rate}(2)/\text{rate}(1) = \log_{3/2} 3</math> — the parity-comb tone ratio IS the v107/v124 bend of the pulsar-clock search. [O] the channel count <math>3 = N_{\text{fam}}</math> is forced by the clock wall; the ladder stays an exact identity on frozen spectra (semiclassical derivation R1 open); the arity <math>\{2, 3\}</math> stays the anchor input. Exact (sympy); mirrorable, deferred to the next verified engine run. [E][O]</p>                    |

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| Script                          | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| v488_majorana_clebsch_door.py   | <p>FLAV.NUSCALE.04: the 3/5 CLEBSCH DOOR of the seesaw rung, ENUMERATED AND CLOSED — v482's named escape (a third-generation Majorana factor <math>r \approx 1.6696</math>, 0.18% from <math>g_{\text{car}}/N_{\text{fam}} = 5/3</math>) is decided NEGATIVE at the group-theory level; the v481 window candidate stands as the honest endpoint. [E] NO RENORMALISABLE CHANNEL: <math>\text{sym}^2(\mathbf{16}) = 136 = \mathbf{10} + \mathbf{126}</math>; the <math>\overline{\mathbf{126}}</math> is NOT in the <math>E_8</math> hull <math>\{1, 10, 16, 45\}</math> (v247) — <math>M_R</math> only via the <math>d=5</math> operator <math>(\mathbf{16}_F \overline{\mathbf{16}}_H)^2/\Lambda</math> with the <math>E_8</math>-allowed <math>\mathbf{16}_H</math>. [E] CHANNEL LIST: <math>\mathbf{16} \times \overline{\mathbf{16}} = \mathbf{1} + \mathbf{45} + \mathbf{210}</math>; the hull admits singlet + <math>\mathbf{45}</math> insertions (<math>\mathbf{210}</math> absent). [E] CHARGE TABLE: <math>Y = T_{3R} + (B-L)/2</math> on all 16 states, <math>\text{Tr}_{16} Y^2 = \frac{10}{3}</math> (v485); <math>(B-L, T_{3R}, Y)(\nu^c) = (+1, -\frac{1}{2}, 0)</math>. [E] THE <math>\{2, 3\}</math>-SMOOTH THEOREM: every nonzero <math>\nu^c</math> channel weight <math>\{1, \frac{1}{4}, 1, -\frac{1}{2}, \frac{3}{8}\}</math> is <math>\{2, 3\}</math>-smooth and the multiplicative closure of <math>\{2, 3\}</math>-smooth rationals never contains <math>\frac{5}{3}</math>; the UNIQUE natural 5 of the embedding, <math>k_Y = \text{Tr} Y^2 / \text{Tr} T_{3L}^2 = \frac{5}{3}</math> (Ginsparg, v470), is a full-multiplet trace whose direction has <math>Y(\nu^c) = 0</math> EXACTLY — structurally decoupled from the Majorana operator. [E] GENERATION-BLINDNESS: the Clebsch is a family-space scalar; <math>\text{diag}(1, 1, \frac{3}{5})</math> cannot arise from group theory. [O] compiler-flavor inventory typed and declined: exact <math>\frac{3}{5}</math>'s only in wrong slots (<math>K</math> lepton gen2/gen1; <math>R + L</math> lepton gen3/gen2), no degenerate-base shape, no <math>M_R</math> link (v354/v355). Exact (Fraction/sympy); mirrorable, deferred to the next verified engine run. [E][O]</p> |
| v489_seam_modular_commutator.py | <p>SEAM.CCC.01: the chiral central charge <math>c_- = 8</math> measured DIRECTLY from the bulk ground state of the v367 seam lattice model, via the Kim–Shehab–Kim modular commutator <math>J(A, B, C) = i \text{Tr}(\rho_{ABC}[K_{AB}, K_{BC}]) = (\pi/3) c_-</math> on a <math>120^\circ</math> pie tripartition (PRL 128 (2022)) — the FIRST ground-state witness for premise (i) of SEAM.EQUIV.01 (<math>c_- = 8</math>) INDEPENDENT of both the Chern route (v367 FHS curvature) and the counting route (v378 KLM/det-Cartan): <math>J</math> is read off the entanglement data of the exact BdG ground state alone (Peschel covariance, <math>W = -2 \text{artanh } \Gamma</math>). [E] GROUND-STATE PURITY: <math>\text{eig}(i\Gamma) = \pm 1</math> to <math>6 \times 10^{-15}</math>. [E] PER-LAYER <math>c_- = \frac{1}{2}</math>: <math>c_- = 0.499998</math> at <math>L=32, R=12</math> (<math>0.499793</math> at <math>L=16 \rightarrow 0.500000</math> at <math>L=48</math>, dev <math>-1.8 \times 10^{-7}</math>). [E] CONTROLS: trivial <math>M=3 \rightarrow 0.000000</math>; orientation flip <math>M=-1 \rightarrow -0.499986</math>. [E] LAYER ADDITIVITY EXACT (<math>1.4 \times 10^{-14}</math>): <math>J</math> additive over decoupled layers, so the carrier value is <math>16 \times</math> the per-layer value. [E] 16-LAYER CARRIER <math>c_- = 8 = g_{\text{car}} + N_{\text{fam}}</math>: direct 16-layer run <math>7.9967</math> (<math>L=16</math>, tol <math>10^{-2}</math>; <math>L=24</math> gives <math>7.999781</math>), additivity route <math>8.0000</math> (tol <math>10^{-3}</math>). [E] CLIP INVARIANCE: <math>\varepsilon = 10^{-8}</math> vs <math>10^{-12}</math> moves <math>c_-</math> by <math>5 \times 10^{-8}</math>. [O] HONEST FENCE: pins <math>c_- = 8</math> on the LATTICE ground state (third independent witness); does NOT close SEAM.EQUIV.01 — the continuum scaling limit (MMST v336) stays [O], the <math>\mu_4</math> condensation <math>SO(16)_1 \rightarrow (E_8)_1</math> stays the cited KLM step (v378). Numerical (numpy), Python-only by nature. [E][O]</p>                                                                                                                        |

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| Script                         | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| v490_seam_parity_census.py     | <p>SEAM.MU.01: the <math>T^2</math> spin-structure fermion-parity census for the v367 seam lattice model — the lattice ground-state witness of “<math>\mu_{\text{pre}}=4</math> before condensation” (step 1 of the v378 KLM chain) + the 16-fold-way bridge <math>\theta_v=1</math> with the MEASURED <math>\nu=2c_-=16</math> (v489). Ground-state parity per boundary-condition sector (AA, AP, PA, PP), determined TWICE: Read–Green TRIM product AND real-space Majorana Pfaffian sign (Schur), relative to the trivial <math>M=3</math> phase. [E] REAL-SPACE = BLOCH SANITY: twisted spectra match to <math>9 \times 10^{-15}</math>. [E] PER-LAYER CENSUS (+, +, +, -): TRIM <math>\equiv</math> Pfaffian in every sector — 3 even sectors, the <math>\nu=1</math> Ising signature. [E] 16 LAYERS <math>\Rightarrow \mu_{\text{pre}}=4</math>: parity<sup>16</sup>=+ in ALL four sectors <math>\Rightarrow</math> gauged torus <math>\text{GSD}=4=\mu(SO(16)_1)= \det \text{Cartan}(D_8) </math>. [E] 16-FOLD-WAY BRIDGE: <math>\theta_v=e^{2\pi i\nu/16}=1</math> EXACTLY at <math>\nu=2c_-=16</math> (vortex bosonic, condensable); rivals <math>\nu=1, 2, 8</math> excluded (not holomorphically condensable). [E] NON-DISCRIMINATOR FENCE: the fermionic Kitaev–Preskill TEE is identically zero — <math>\gamma_f=0.000001</math> (<math>M=1</math>) and <math>0.000000</math> (<math>M=3</math>) — an invertible fermionic phase has no TEE, so <math>\gamma_f</math> does NOT discriminate <math>(E_8)_1</math> vs <math>SO(16)_1</math> (both live only after gauging). [C] the KLM step <math>\mu_{\text{post}}=4/2^2=1</math> stays cited (the condensed model is interacting); the numerics pin only <math>\{c_-=8, \mu_{\text{pre}}=4, \theta_v=1\}</math> — the premise package of the condensation. [O] the continuum limit (MMST v336); SEAM.EQUIV.01 stays [O]. Python (numpy + scipy Schur + sympy exact <math>\theta_v</math>). [E][C][O]</p>                                                                      |
| v491_p2_partition_corollary.py | <p>P2.PARTITION.01: <math>g_{\text{car}}=5</math> as a COROLLARY of the four marks — the partition sharpening of v53 (sympy exact, no floats). THE LEMMA: a multiset of exactly 3 POSITIVE INTEGERS summing to 4 is unique — <math>\{1, 1, 2\}</math> — so <math>e=(e_1, e_2, e_3)=(4, 5, 2)</math> and <math>g_{\text{car}}=e_2=5</math>, <math> \mathbb{Z}_2 =e_3=2</math>, <math>N_{\text{fam}}=e_2-e_3=3</math> are COROLLARIES, not inputs; consistency closes as a fixed point <math>((2^4-1)/5=3=\text{rank } H^1=\#\text{marks}-1)</math>. NEGATIVE CONTROLS (every assumption load-bearing): drop positivity <math>\rightarrow 4</math> solutions, <math>e_2</math> ambiguous <math>\{0, 3, 4, 5\}</math>; sum 3 <math>\rightarrow e_2=3</math>; sum 5 <math>\rightarrow</math> not unique; 2/4 parts <math>\rightarrow e_2 \in \{3, 4\}/\{6\}</math>; Mehta–Seshadri RATIONAL weights <math>\rightarrow e_2=\frac{16}{3}</math> resp. <math>\frac{44}{9} \neq 5</math> (integrality comes from the Birkhoff–Grothendieck SPLITTING-TYPE typing, not from MS weights). [E] v53 SHARPENING: v53 needs all three targets (4, 5, 2); the partition needs only <math>e_1=4</math> — same unique <math>\{1, 1, 2\}</math>. [E] SUM RULE: <math>4 \times \frac{N-1}{2}=4</math> iff <math>N=3</math>; the v115 route <math> a _1= \mu_4  \text{tr}(A_0)=4</math>. INDEPENDENCE CHAIN: <math>\#\text{marks}=4</math> from v216 (P1-side, no <math>g_{\text{car}}</math> input); rank <math>H^1=3</math> pure topology (v137/v228); <math> a _1=4</math> <math>g_{\text{car}}</math>-free (par-deg-0/(U), v115). [C] HONEST RESIDUAL: P2 is NOT made axiom-free — the weight-TYPING postulate remains (integrality = splitting exponents; positivity = <math>h^0=0</math>, correlate v137 check 1; sum rule “given (U)”); AX.P2.01 stays a declared axiom; the <math>N_{\text{fam}}</math> direction reversal vs tfpt_constants is typed as an over-determination check, never double-counted. Exact (sympy); Wolfram-mirrored. [E][C]</p> |

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| Script                        | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| v492_celestial_z4_orbifold.py | <p>CELEST.WP1.01: WP1 of the celestial-holographic research contract</p> <p>CELEST.SEAM.01 — the <math>E_8</math> <math>\mu_4</math>-glue grading (v128) IS the flat <math>\mathbb{Z}_4</math> monodromy of the glue-equivariant celestial chiral algebra on the <math>A_3</math> ALE space <math>\mathbb{C}^2/\mathbb{Z}_4</math> (Bittleston–Homans–Sharma, <a href="#">arXiv:2305.09451</a>, the <math>\mathbb{Z}_2</math>/Eguchi–Hanson case; Kronheimer–Nakajima flat connection at infinity). Sympy exact. [E] INNER GRADING: <math>h=(2, 2, 2, 2, 2; 0^4)</math> reads the glue class mod 4 on all 240 roots <math>\Rightarrow</math> glue <math>= \text{Ad}(\exp(2\pi i h/4))</math>, an order-4 torus element <math>=</math> flat <math>\mathbb{Z}_4</math> monodromy; the <math>A_3</math>-side detector <math>h'=(0^5; 1, 1, 1, -3)</math> reads the SAME class <math>=</math> the diagonal <math>(1, 1)</math> of the discriminant group <math>\mathbb{Z}_4 \times \mathbb{Z}_4</math> (the v92/v125 Lagrangian glue); <math>\mathbb{Z}_2</math> halfway <math>120 = \dim SO(16)</math>, <math>128 = \text{spinor}</math>. [E] <math>\mathbb{C}^2/\mathbb{Z}_4</math> IS the <math>A_3</math> singularity <math>XY=Z^4</math> (Molien table exact; Hilbert series <math>(1+t^4)/((1-t^2)(1-t^4))</math>; conjugation symmetry <math>N[d][1]=N[d][3]</math>). [E] EQUIVARIANT SECTOR CLOSES: <math>D[d]=60(d+1)</math> even / <math>64(d+1)</math> odd — uniform per parity ONLY because <math>\dim \mathfrak{g}_j=(60, 64, 60, 64)</math>; zero modes <math>(d=0) = \mathfrak{g}_0 = \text{carrier } D_5 \oplus A_3 + \text{Cartan} = 60</math>; density <math>62/248=1/4=1/ \mathbb{Z}_4 </math>; <math>\text{EH}/\mathbb{Z}_2</math> halfway towers <math>120/128</math>. [E] CHARACTER: <math>\Theta_{E_8} = \sum_j \Theta_{C_j}</math>; shell <math>1 = (52, 64, 60, 64)</math> from the LATTICE; the 4 sector characters <math>\Theta_{C_j}/\eta^8</math> sum exactly to <math>\chi_{(E_8)_1} = 1 + 248q + 4124q^2 + 34752q^3</math> (v377); sector weights <math>h_{D_5}=(0, \frac{5}{8}, \frac{1}{2}, \frac{5}{8})</math>, <math>h_{A_3}=(0, \frac{3}{8}, \frac{1}{2}, \frac{3}{8})</math> = the v92/v125 discriminant form <math>(5x^2+3y^2)/8 \bmod 1</math>; glue diagonal <math>h=(0, 1, 1, 1)</math> INTEGER = v125 isotropy = local extension to <math>(E_8)_1</math>. [E] CRITICAL CORRECTION: the <math>A_3</math> deck <math>\text{diag}(i, i^{-1})</math> acts on the celestial sphere as <math>z \mapsto -z</math> (projective order 2, the sheet flip), NOT as the order-4 clock <math>z \mapsto iz</math>; the clock is the <math>U(2)</math> phase <math>\text{diag}(i, 1)</math> (order 4, fixed <math>\{0, \infty\}</math>, free <math>\mu_4</math> orbit — all v180 properties; normalises the deck, <math>\det=i</math> rotates the holomorphic symplectic form by the <math>\mu_4</math> generator); exact spin bridge <math>(\text{spin clock})^2 = \text{deck}</math>, spin group <math>\mathbb{Z}_8</math>, <math>8=2 \mu_4 </math> (the <math>c_3=1/(8\pi)</math> winding integer). [E] clock-invariant <math>A_3</math> deformation = the 1-parameter family <math>XY=Z^4+a_0</math>; its 4 branch points are ONE <math>\mu_4</math> orbit with cross-ratio 2 (v179 clock orbit; v168/v214 pillowcase <math>j=1728</math>). [E] NEGATIVE CONTROLS: false spatial action <math>\text{diag}(i, i)</math> killed three ways (<math>\det=-1 \notin SL(2, \mathbb{C})</math>; Poisson closure fails 14/14; grades unsymmetric); false glue breaks additivity on 3112/6720 root pairs; rigidity: only <math>\text{Aut}(\mathbb{Z}_4)</math> relabelings additive. [C] VERDICT B: match under the ONE named identification <math>\text{clock}^2=\text{deck}</math> on the sphere (sheet-<math>\mathbb{Z}_2</math> spin bookkeeping). [O] structural arithmetic ONLY — no celestial duality constructed; CELEST.SEAM.01 (WP2–WP5) stays a research contract; SEAM.EQUIV.01 stays [O], untouched. Exact (sympy). [E][C][O]</p> |

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| Script                                          | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| v493_celestial_<br>wp2_clock_<br>deformation.py | <p>CELEST.WP2.01: WP2 of the celestial-holographic research contract</p> <p>CELEST.SEAM.01 — the clock-invariant deformation <math>XY=Z^4+a_0</math> of the <math>A_3</math> seam orbifold: geometry, periods/monodromy, deformed algebra</p> <p>(Bittleston–Homans–Sharma <math>\mathbb{Z}_2 \rightarrow \mathbb{Z}_4</math> transfer, <a href="#">arXiv:2305.09451</a>). Sympy exact. <b>[E]</b> GEOMETRY: <math>P(iZ)=P(Z)</math> forces <math>a_3=a_2=a_1=0</math> SHARPLY (two-sided: <math>\prod_k (Z - i^k w) = Z^4 - w^4</math>); smooth iff <math>a_0 \neq 0</math> (disc <math>= 256 a_0^3</math>); branch points <math>Z^4 = -a_0 =</math> ONE free <math>\mu_4</math> orbit in every fibre; cross-ratio <math>2 / j = 1728</math> for ALL <math>a_0</math></p> <p>(binary-quartic invariants <math>I=12a_0</math>, <math>J=0</math> identically) <math>\Rightarrow a_0</math> is a PURE SCALE, the <math>\tau=i</math> pillowcase (v168/v214) is FROZEN — no shape modulus. <b>[E]</b> PERIODS: <math>\Pi=t(i-1)(1, i, i^2)</math>, all 3 vanishing spheres carry the SAME volume <math>\sqrt{2}t</math> (lockstep); the clock on <math>H_2</math> is a COXETER element of <math>W(A_3)</math> (char <math>x^3+x^2+x+1</math>, eigenvalues <math>\{i, -1, -i\}</math> = the <math>A_3</math> exponents 1, 2, 3; <math>h(A_3)=4= \mu_4 =N_{\text{fam}}+1</math>); NO invariant cycle (<math>\det(A-1)=-4</math>); the surviving direction is the weight-1 <math>\chi_1</math>-Fourier diagonal <math>(1, i, i^2)</math>, NOT the naive invariant diagonal; the clock IS the Picard–Lefschetz monodromy of the family (<math>a_0 \rightarrow e^{2\pi i} a_0</math> gives the 4-cycle); versal weights <math>(a_3, a_2, a_1, a_0) = (1, 2, 3, 0)</math> = the REGULAR <math>\mu_4</math> representation, invariant line = <math>a_0</math> alone. <b>[E]</b> ALGEBRA (<math>\mathbb{Z}_2 \rightarrow \mathbb{Z}_4</math>): mode dictionary <math>\{w[m, n] : m \equiv n \pmod{4}\} \leftrightarrow \{X^p Y^q Z^r : r \leq 3\}</math>; bracket <math>\{f, g\} = -4 \text{Nambu}(XY - Z^4 - a_0)</math> anchored at the <math>a_0=0</math> orbifold w-sector <math>(\{X, Y\}=16Z^3, \{Y, Z\}=-4Y, \{Z, X\}=-4X)</math>, Jacobi + Casimir exact, uniform clock-degree <math>-1</math>; S-algebra corrections exactly LINEAR in <math>a_0</math> at the <math>\mathbb{Z}_4</math> wrap <math>r+r' \geq 4</math>, grade-conserving (NO sector leak <math>\Rightarrow</math> the v492 equivariant SDYM(<math>E_8</math>) sector deforms consistently); <math>a_0 \leftrightarrow</math> the BHS <math>c^2</math> slot (weight 0); matrix factorisations (McKay modules) deform along every 1+3 and 2+2 orbit splitting; the clock fixes exactly the antipodal 2+2 sector (the EH/<math>\mathbb{Z}_2</math> middle node, <math>4=2 \times 2</math>). <b>[E]</b> NEGATIVE CONTROLS: <math>a_1 Z \Rightarrow j=0</math> (hexagonal CM point); pure <math>a_2 Z^2</math> stays SINGULAR (<math>A_1</math> node); <math>a_2 \neq 0 \Rightarrow j=1556068/81 \neq 1728</math> (exact instance, two routes agree); grading leak explicit (<math>Z^2 \cdot Z^2</math> picks up grades <math>\{0, 1, 2\}</math>). <b>[C]</b> three NAMED identifications (verdict B, not A): <math>-4 \text{Nambu} = \text{THE } k=4 \text{ CCA bracket}</math> (BHS prove <math>k=2</math>); period = root difference (A-series residue normalisation); the seam-scale reading of <math>a_0</math> presupposes the WP1 bridge clock<sup>2</sup>=deck (<math>\mathbb{Z}_8</math>, <math>8=2 \mu_4 </math>). <b>[C]</b> v216 PRECISION (typed, NO marker move): given the order-4 clock, the clock-invariant deformation freezes the <math>(2, 2, 2, 2)</math> modulus to the square AUTOMATICALLY — a relocation of the same order-4 input (the clock IS the carrier input), not a new derivation; v216 stays <b>[O]</b>. <b>[O]</b> full deformed twisted-sector OPEs, the <math>W(\mu)</math>-type scaling limit for <math>k=4</math>, the spacetime splitting-function check, WP3–WP5; SEAM.EQUIV.01 stays <b>[O]</b>, untouched. Exact (sympy). <b>[E][C][O]</b></p> |

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| Script                          | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| v494_cosmo_running_mu_record.py | <p>COSMO.RUNNING.01/COSMO.MUDIST.01: the two 2026-07-20 prediction-of-record rows promoted to the suite — the SAME Starobinsky branch as <math>n_s, r, A_s</math> (v7/v28/v86) read one derivative deeper and one observable further. <b>[E]</b> LO IDENTITIES (sympy exact): <math>\alpha_s = dn_s/d \ln k = -2/N_*^2 = -r/6</math>, <math>\beta_s = -4/N_*^3</math> (NOT <math>-8/N_*^3</math>); LO band <math>N_* \in [50, 60]</math>: <math>[-8.0, -5.6] \times 10^{-4}</math>. <b>[E]</b> EXACT SLOW ROLL (third-order Lyth–Riotto + finite-difference cross-check <math>&lt; 1\%</math>): <math>\alpha_s(51.4) = -7.09 \times 10^{-4}</math>, <math>\beta_s(51.4) = -2.70 \times 10^{-5}</math>; exact band edges <math>-7.48/ -5.23 \times 10^{-4}</math>. <b>[C]</b> DATA: Planck 2018 (<math>-0.0045 \pm 0.0067</math>) <math>\rightarrow +0.57\sigma</math>; record leg P-ACT-LB (<math>+0.0062 \pm 0.0052</math>) <math>\rightarrow -1.33\sigma</math> (watch channel: positive central value); <math>\beta_s</math> run+runrun <math>\rightarrow -0.77\sigma</math>. KILL: robust <math>\alpha_s &lt; -5 \times 10^{-3}</math> or <math>&gt; +3 \times 10^{-3}</math> at <math>\geq 5\sigma</math>; any robust POSITIVE running kills the plateau branch. <b>[C]</b> <math>\mu</math>-DISTORTION (Chluba window, +21% calibrated on the <math>\Lambda</math>CDM anchor): sharp branch <math>N_* = 51.4</math> (<math>A_s = 1.76 \times 10^{-9}</math>) <math>\mu = 1.6 \times 10^{-8}</math>; profiled record <math>N_* = 56.1</math> <math>\mu = 1.94 \times 10^{-8}</math>; band <math>1.5\text{--}2.3 \times 10^{-8}</math>; branch split <math>\Delta\mu = 3.0 \times 10^{-9}</math> (ratio 1.23) <math>\Rightarrow \geq 3\sigma</math> <math>A_s/N_*</math> branch decision at <math>\sigma(\mu) = 10^{-9}</math> (Voyage-2050 class); the PIXIE baseline <math>9 \times 10^{-8}</math> does NOT reach (<math>0.18 \times</math> limit). KILL: robust <math>\mu &lt; 0.9 \times 10^{-8}</math> or <math>&gt; 4 \times 10^{-8}</math>. <b>[E]</b> COMB BLINDNESS (honest S15): the frozen log-comb (<math>\omega = 2.5827</math>, <math>\varepsilon = 0.0173</math>) is structurally invisible in the <math>\mu</math> window — <math> F(\omega)  = 0.033</math>, <math>\delta\mu = 7.4 \times 10^{-12} = 0.007\sigma</math> even at Voyage-2050. <b>[O]</b> FENCES: <math>N_*</math> stays an external reheating input (GATE.NSTAR.01); the frozen band <math>[50, 60]</math> stays the surface of record. Python (sympy exact LO + numpy SR/quadrature), Python-only by nature. <b>[E][C][O]</b></p> |

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| Script                                 | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| v495_celestial_wp3_<br>gs_alignment.py | <p>CELEST.WP3.01: WP3 of the celestial-holographic research contract</p> <p>CELEST.SEAM.01 — the Green–Schwarz axion coefficient <math>\lambda_g</math> of celestial/twistorial SDYM (Costello, arXiv:2111.08879) against the TFPT seam constant <math>c_3=1/(8\pi)</math>; the pre-registered “alignment test only”. Exact (Fraction/sympy, no floats). <b>[E]</b> OKUBO IDENTITY DERIVED, NOT QUOTED: <math>\sum_{\alpha} \langle \alpha, x \rangle^4 = \frac{5}{2(\dim \mathfrak{g}+2)} (\sum_{\alpha} \langle \alpha, x \rangle^2)^2</math> as an exact POLYNOMIAL identity for ALL 8 algebras on Costello’s list (<math>\mathfrak{sl}_2, \mathfrak{sl}_3, \mathfrak{g}_2, \mathfrak{so}_8, \mathfrak{f}_4, \mathfrak{e}_6, \mathfrak{e}_7, \mathfrak{e}_8</math>; <math>\mathfrak{e}_6/\mathfrak{e}_7</math> carved out of the v492 <math>E_8</math> root set) — “no independent quartic Casimir”; NEGATIVE CONTROL: <math>\mathfrak{sl}_4</math> (which HAS one) fails the proportionality (5/32 vs 3/32) — the list is sharp. <b>[E]</b> Killing <math>\sum_{\alpha} \langle \alpha, x \rangle^2 = 2h^{\vee} \langle x, x \rangle</math> exact; Deligne dimension formula <math>\dim \mathfrak{g} = 2(5h^{\vee}-6)(h^{\vee}+1)/(h^{\vee}+6)</math>; closed form <math>\tilde{\lambda}^2 = 10h^{\vee 2}/(\dim +2) = h^{\vee}+6</math> EXACTLY (8, 9, 10, 12, 15, 18, 24, 36) — fixes Costello–Paquette eq. (1.5); Dynkin indices <math>i_R = (1, 1, 2, 2, 6, 6, 12, 60)</math> from explicit weight systems; <math>E_8</math>: <math>\tilde{\lambda}=6</math>, <math>\lambda_{\text{fund}}=1/10</math> EXACT; cross-checks <math>\mathfrak{sl}_2=8</math>, <math>\mathfrak{sl}_3=9</math>; SO8 SLIP FLAGGED: Costello’s own weight content gives <math>\lambda^2(\mathfrak{so}_8)=3</math> (Okubo-consistent 12/4), the printed 3/2 is a factor-2 slip. <b>[E]</b> ALIGNMENT: <math>\kappa = \lambda_g/(8\pi\sqrt{3}) \Rightarrow (\kappa/c_3)^2 = \lambda^2/3 = 12 =  \mu_4 N_{\text{fam}}</math> (unit-trace) resp. 1/300 (adjoint-trace), both EXACT anchor rationals; <math>\kappa/c_3</math> ITSELF is IRRATIONAL (<math>2\sqrt{3}</math> resp. <math>1/(10\sqrt{3})</math>) — the pre-registered “<math>\lambda/\sqrt{3}</math> rational?” reading FAILS, only the SQUARED alignment holds; anchor decomposition <math>h^{\vee}=30= \mathbb{Z}_2 N_{\text{fam}}g_{\text{car}}</math> (all three atoms once), <math>\varphi(30)=8=\text{rank}</math>, <math>\dim +2=250=2 \cdot 5^3</math>, <math>\tilde{\lambda}=6= \mathbb{Z}_2 N_{\text{fam}}</math>, <math>\lambda_{\text{fund}}=1/( \mathbb{Z}_2 g_{\text{car}})</math>. <b>[E]</b> LOOK-ELSEWHERE (the HARD FENCE — alignment <math>\neq</math> selection): the alignment test AS SUCH passes 8/8 (ZERO selective power), <math>\{2, 3, 5\}</math>-smoothness 8/8, <math>\tilde{\lambda}</math>-integrality 2/8 (shared with <math>\mathfrak{sl}_3</math>), <math>\varphi(h^{\vee})=\text{rank}</math> 4/8; the ONLY <math>\mathfrak{e}_8</math>-selective single test is <math>g_{\text{car}}=5 \mid h^{\vee}</math> (1/8); the isolating conjunction was assembled POST HOC — it quantifies where <math>\mathfrak{e}_8</math> sits, it is NOT a preregistered discriminator. <b>[E]</b> KILL-TEST K3 does NOT fire (exact alignment EXISTS) but its scope is DEMOTED: the <math>c_3</math>-connection survives as convention-level compatibility + genuine <math>\lambda</math>-arithmetic, NOT as <math>\mathfrak{e}_8</math>-selective evidence. <b>[C]</b> TYPED: the <math>8\pi</math> is the <math>2\pi</math> convention of the anomaly polynomial (Costello’s A.0.10 constant descended to spacetime), the <math>\sqrt{3}</math> the box diagram’s <math>\sqrt{48} = 4\sqrt{3}</math> — WP3 is an alignment yes/no ONLY, it can never derive <math>c_3</math> from celestial data (contract wording, honoured). <b>[O]</b> WP5 (constructive holography) and SEAM.EQUIV.01 stay <b>[O]</b>, untouched. Verdict B. Exact (Fraction + sympy). <b>[E][C][O]</b></p> |

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| Script                                         | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| v496_celestial_<br>wp4_character_<br>shadow.py | <p>CELEST.WP4.01: WP4 of the celestial-holographic research contract<br/> CELEST.SEAM.01 — the <math>(E_8)_1</math> character <math>E_4/\eta^8</math> against the glue-equivariant <math>E_8[\mathbb{C}^2]</math> S-algebra: the contract’s “type mismatch, head-on”. Exact integer/Fraction/sympy, no floats. [E] BASE: v492 replication<br/> <math>(240 = 52+64+60+64, \text{ dims with Cartan } (60, 64, 60, 64))</math>; equivariant tower <math>D[d] = 60(d+1)</math> even / <math>64(d+1)</math> odd; cumulative generator count QUADRATIC — <math>\sum_{d \leq s} D[d] = 31s^2 + 92s + 60</math> (even) with leading <math>31 = k + h^\vee</math> (the Sugawara denominator, an arithmetic note); [C] CP Table 1 (arXiv:2201.02595): <math>J[m, n]</math> carries 2d spin <math>1 - (m+n)/2</math>, UNBOUNDED BELOW along the tower <math>\Rightarrow</math> no rational VOA (<math>L_0</math> bounded below) contains the towers as graded pieces. [E] HILBERT SERIES: <math>\chi_{\text{sp}}(t) = (60+128t+60t^2)/(1-t^2)^2</math> (palindromic numerator, value 248 at <math>t=1</math>); ALL <math>\mu_4</math> collapses <math>w \in \{1, -1, i, -i\}</math> and the <math>SU(2)_+</math> primary slice <math>(60+64t)/(1-t^2)</math> fail to start <math>(1, 248, \dots)</math> — the “row sum” route fails at the first coefficient. [E] CHARACTER SIDE:<br/> <math>\chi_{(E_8)_1} = E_4/\eta^8 = (1, 248, 4124, 34752, 213126, 1057504, 4530744)</math> exact through <math>q^6</math> (machinery cross-check <math>qj = E_4^3/\eta^{24} = (1, 744, 196884, 21493760)</math>); theta split (shell <math>1 = (52, 64, 60, 64)</math>), sector currents <math>(60, 64, 60, 64)</math>, glue-diagonal <math>h = (0, 1, 1, 1)</math> INTEGER (v125 isotropy = locality); Sugawara <math>c = 248/31 = 8 = 2 \mu_4  = g_{\text{car}} + N_{\text{fam}}</math>. [E] BRIDGES: zero-mode slice <math>D[0]=60=</math> the <math>C_0</math>-sector current count EXACTLY; the single-line jet slice <math>z_1^d</math> cycles <math>(60, 64, 60, 64)</math> sector by sector; BOUNDARY/PERIOD LIMIT: one full <math>\mu_4</math> period of loop energies carries <math>\dim \mathfrak{g}_{n \bmod 4} = (64, 60, 64, 60)</math>, sum = 248 EXACTLY — the single-particle level of the <math>248q</math> stage; 248 is NOT a jet-tower dimension (<math>d \leq 100</math>) — only the period collapse produces it; BUT the loop Fock at integer level 1 counts <math>897266 \gg 248</math>: the rational truncation is NOT in the jet/loop algebra. [E] MISMATCH LOCALISED THREE WAYS: (a) GRADING (spin unbounded below; 248 no tower dimension); (b) GROWTH: jet Fock <math>f = (1, 128, 8436, \dots)</math> vs <math>\chi = (1, 248, 4124, \dots)</math> TWO-SIDED (<math>f_1 &lt; \chi_1</math> undercount, <math>f_2 &gt; \chi_2</math> overcount), <math>f_n/\chi_n</math> strictly increasing <math>n = 1..12</math>; Meinardus poles <math>s=2</math> vs <math>s=1 \Rightarrow</math> growth exponents <math>n^{2/3}</math> vs <math>n^{1/2}</math>, asymptotically DISJOINT; (c) TRUNCATION: the level-2 null ideal deletes EXACTLY <math>27000 = 30^3 = (h^\vee)^3</math> (<math>\text{Sym}^2(248) = 1+3875+27000</math> [[C] standard]; the character keeps <math>4124 = 1+248+3875</math>) — NO analogue in the jet algebra (level 1 agrees: <math>248 = 248</math>). [E] KILL-TEST K4, three-way: (i) exact realisation NO; (ii) boundary-limit realisation PARTIAL YES — current stratum, period sum 248, integer glue weights and theta split hold exactly; the missing piece is precisely the rational truncation, which must come from a scaling/compactification limit — the SAME structure as the MMST scaling-limit route of SEAM.EQUIV.01 (v336/v449: the rational net exists in the LIMIT of the lattice collar, not in the pre-limit algebra); (iii) the hard mismatch is exactly located; K4 does NOT degrade the contract to “<math>E_8</math> admissible” (the sector arithmetic holds exactly). [C] VERDICT B(ii): the <math>(E_8)_1</math> character is a BOUNDARY/LIMIT SHADOW of the equivariant S-algebra, not a conformal block of it in the bulk grading — WP4 sharpens into a statement of SEAM.EQUIV.01 type. [O] the constructive boundary limit itself (which limit, which topology, convergence) stays OPEN and belongs to WP5/SEAM.EQUIV.01; SEAM.EQUIV.01 stays [O], untouched. Exact (integer/Fraction/sympy). [E][C][O]</p> |

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| Script                                | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| v497_celestial_<br>wp5a_null_ideal.py | <p>CELEST.WP5A.01: WP5a of the celestial-holographic research contract</p> <p>CELEST.SEAM.01 — “the null ideal from the limit”, the first constructive milestone of WP5 (the v496 boundary-limit shadow made precise). Exact integer/Fraction arithmetic, no floats. [E] BASE: v492/v496 replication (glue dims (60, 64, 60, 64), equivariant tower <math>D[d]</math>, <math>\chi_{(E_8)_1} = E_4/\eta^8</math> exact through <math>q^6</math>, <math>\mu_4</math> period sum 248). [E] STABILISATION THEOREM: a one-parameter family <math>\chi_w</math> of graded Fock characters on the CHIRAL jet generators (loop energy <math>m</math>, radial number <math>r</math>; <math>E_w = m + wr</math> in quarter units <math>u = q^{1/4}</math>; <math>w=2</math> IS the chiral jet grading (generator counts 64, 120, 128, 180, 192, 240, 256, 300 per degree); the <math>u^n</math> coefficient of <math>\chi_w</math> equals the quarter-moded loop Fock for ALL <math>w \geq n+1</math> and is STRICTLY larger for every <math>w \leq n</math> (<math>n \leq 8</math>, <math>w \leq 10</math>): the boundary reading is the precise coefficientwise limit <math>\varepsilon = 1/w \rightarrow 0</math> with explicit threshold <math>w = n+1</math> — a limit of a series, not a slice; convention clarified: 897266 is the quarter-moded loop Fock at <math>u^4 = \text{INTEGER level 1}</math> (v496). [E] NULL IDEAL DERIVED, NOT CITED: Freudenthal multiplicities (unit-tested on the adjoint) + Weyl dimension formula + generic character peeling in the standard <math>E_8</math> frame DERIVE</p> <p><math>\text{Sym}^2(248) = V(2\theta) \oplus V(\omega) \oplus V(0) = 27000 + 3875 + 1</math> (peeling residual exactly zero; independent Weyl-orbit cross-check); <math>27000 = 30^3 = (h^\vee)^3</math> exactly;</p> <p><math>C_2 = (124, 96, 0)</math>, Dynkin indices 6750/750; level-1 integrability <math>\langle 2\theta, \theta^\vee \rangle = 4</math>, <math>\langle \omega, \theta^\vee \rangle = 2 \Rightarrow</math> the only level-1 primary is the vacuum ([C] Kac, used through its counting consequence); level-2 quotient <math>31124 - 27000 = 4124 = 1 + 248 + 3875</math> EXACTLY. [E] TWO ROUTES, ONE NUMBER: the <math>\mu_4</math> theta-split sector characters summed at <math>q^2 = (1036, 1024, 1040, 1024)</math> give the SAME 4124 — the independent lattice route hits the Fock quotient exactly; at <math>q^3</math>: <math>34752 = \chi_3</math>. [E] NEGATIVE CONTROLS: <math>SO(16)_1</math> through the SAME pipeline —</p> <p><math>\text{Sym}^2(120) = 5304 + 1820 + 135 + 1</math> derived (FOUR components vs <math>E_8</math>’s three), <math>5304 \neq 14^3 = 2744</math> (the <math>h^{\vee 3}</math> identity is NOT generic), quotient</p> <p><math>7380 - 5304 = 2076 = \Theta_{D_8}/\eta^8</math> at <math>q^2</math> (the pipeline validated on a second algebra), block weights <math>(0, \frac{1}{2}, 1, 1)</math> do NOT fuse into one local character (four-block theory) vs <math>E_8</math>’s integer <math>(0, 1, 1, 1)</math>; false periodisation: only <math>P = 4 =  \mu_4 </math> gives 248 (<math>P=1: 64</math>, <math>P=2: 124</math>, <math>P=8: 496</math>); swapped glue breaks the zero-mode anchor (<math>64 \neq 60</math>). [C] named identifications: “radial decoupling = MMST collar thickness <math>\rightarrow 0</math>” (the v336/v449 lattice-collar route has the same two-step shape), not a computed isomorphism; the Kac singular vector only via its counting consequence. [O] HONEST LIMIT: the limit does NOT generate the truncation — the singular vector as an OPERATOR in the limit net stays open (WP5b); the GNS limit state (kernel <math>\supseteq</math> ideal) and local normality / Haag duality of the limit net stay open (WP5c/WP5d); the celestial route now has the SAME two-step shape as MMST (limit + maximal ideal), with the ideal quantitatively fixed;</p> <p>SEAM.EQUIV.01 stays [O], untouched. Exact (integer + Fraction). [E][C][O]</p> |

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| Script                                         | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| v498_celestial_<br>wp5b_singular_<br>vector.py | <p>CELEST.WP5B.01: WP5b of the celestial-holographic research contract</p> <p>CELEST.SEAM.01 — “the singular vector as an operator”, the second constructive milestone of WP5 (v497 fixed the ideal’s size and location; WP5b constructs the deleting object). Exact integer/Fraction arithmetic, no floats, deterministic. [E] CONSTRUCTION: <math> s\rangle = (E_{-1}^\theta)^2 0\rangle</math>, <math>\theta</math> the highest root, weight <math>2\theta</math> (the top of the 27000), level 2 = 8 quarter units = <math>q^2</math> (an INTEGER level); the Chevalley/Frenkel–Kac basis is MACHINE-built (bimultiplicative 2-cocycle, asymmetry on all 57600 pairs; the <math>[e_\alpha, e_{-\alpha}]</math> sign FORCED by Jacobi, <math>\text{SGN} = -1</math>; <math>\kappa</math> DERIVED from invariance, <math>\kappa(\theta^\vee, \theta^\vee) = 2</math>; Jacobi exact on all 57600 <math>\mathfrak{sl}_2</math>-slice triples + 6000 seeded + all 496 <math>\theta</math>-adjacent; all <math>N_{\alpha\beta} = \pm 1</math>). [E] SINGULARITY: <math>J_1^a s\rangle = 0</math> for ALL 248 basis elements (0 violations); case tally 190 (first bracket) / 57 (second) / 1 (<math>a = F^\theta</math> via the central-term cancellation — the ONLY case that sees the level <math>k</math>); plus <math>J_2^a s\rangle = 0</math> (248), <math>E_0^a s\rangle = 0</math> (120 positive roots), Shapovalov norm 0 at <math>k=1</math> (4 at <math>k=2</math>); the affine PBW engine unit-tested on all 61504 basis pairs. [E] LEVEL DIAL: the <math>F_1^\theta</math> coefficient is <math>2(1-k) - 2</math> at <math>k=0</math>, 0 EXACTLY at <math>k=1</math>, <math>-2</math> at <math>k=2</math>: without the central extension the deletion object does NOT exist. [E] <math>\mu_4</math> DATA: <math>\theta</math>-grading dims <math>(1, 56, 134, 56, 1)</math>, <math>\mathfrak{g}_0 = \mathfrak{e}_7 \oplus \mathfrak{u}(1)</math>; the <math>h</math>-adapted chamber machine-built (Cartan det 1, <math>E_8</math> Dynkin diagram, branch lengths <math>[1, 2, 4]</math>); <math>\langle \theta, h \rangle = 5 \Rightarrow j(\theta) = 1</math>, height <math>29 = h^\vee - 1</math>; <math> s\rangle</math> in class 2 (sheet-EVEN, consistent with WP5a); clock phase <math>i^2 = -1</math>; cross-table <math>\theta</math>-grading <math>\times</math> glue class consistent (rows <math>(1, 56, 126, 56, 1)</math>, columns <math>(52, 64, 60, 64)</math>); the deleting <math>\theta</math>-<math>\mathfrak{sl}_2</math> crosses the sheet-ODD classes <math>(1, 3)</math>, its square is sheet-even. [E] LEVEL-2 MATCH: <math>\text{mult}(2\theta) = 1</math> in the level-2 Fock <math>\Rightarrow U(\mathfrak{g}) s\rangle</math> is THE 27000; the direct <math>\mathfrak{g}_0</math>-orbit BFS reproduces the Freudenthal multiplicities of <math>V(2\theta)</math> EXACTLY through depth 4; quotient <math>31124 - 27000 = 4124</math>; lattice reading <math>(2\theta, 2\theta)/2 = 4 &gt; 2 \Rightarrow</math> no momentum-<math>2\theta</math> state at <math>q^2</math> in <math>E_4/\eta^8 -  s\rangle</math> IS the kernel generator of Fock <math>\rightarrow E_4/\eta^8</math>. [E] NEGATIVE CONTROLS: <math>E_{-1}^\theta 0\rangle</math> NOT singular (one witness <math>F^\theta, \rightarrow -k 0\rangle</math>) — the 248-current state that must REMAIN (<math>\chi_1 = 248</math>); 4 generic level-2 states non-singular (14–33 witnesses); <math>k</math>-dial: at <math>k=2</math> the CUBE is singular (the <math>(E^\theta)^{k+1}</math> mechanism = generic Kac level mechanics); <math>D_8/SO(16)_1</math>: the singular vector exists there TOO (level-1 generic) but the <math>h = \frac{1}{2}</math> blocks do not fuse (3 comark-1 weights, <math>\dim V(2\theta_{D_8}) = 5304 \neq 14^3</math>) vs <math>E_8</math> (comarks <math>\geq 2</math>, vacuum the ONLY level-1 primary, glue weights <math>(0, 1, 1, 1)</math> integer) — the ONE-BLOCK closure is the <math>E_8/\mu_4</math>-specific part, honestly separated from the generic singular-vector mechanics. [C] Weyl complete reducibility beyond BFS depth 4; the Kac mechanics beyond level 2; the per-period quarter dictionary (v496). [O] HONEST LIMIT: in the twisted quarter-SLOT moding of the <math>\chi_w</math> limit Fock two sector-<math>C_1</math> modes never sum to 8 quarters (minimum <math>6 = q^{3/2}</math>) — the per-PERIOD dictionary (v496), not the per-slot identification, carries <math> s\rangle</math> to <math>q^2</math>; exactly the precise WP5c question (GNS limit state, kernel <math>\supseteq</math> ideal); WP5c-e stay [O]; SEAM.EQUIV.01 stays [O], untouched. Exact (integer + Fraction). [E][C][O]</p> |

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| Script                            | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| v499_p2_weights_<br>deligne_bg.py | <p>P2.TYPING.01: the P2 weight-typing postulate HARDENED (the v491 check-9 residual) — the anchor <math>a=(1,1,2)</math> typed as the Birkhoff–Grothendieck splitting exponents of the DELIGNE CANONICAL EXTENSION <math>E</math> of the flavor connection; sympy exact, no floats. Given the [E] anchors (4 marks v216, rank 3 v137, cusp class + irreducibility v117, (U)): <math>\deg E = -4</math> AS A THEOREM (cusp class <math>\Rightarrow</math> canonical residue exponents <math>\{0, \frac{1}{3}, \frac{2}{3}\}</math>, the unique lift in <math>[0,1]</math>; trace 1 per mark <math>\times</math> 4 marks; par-deg 0; <math>\infty</math> APPARENT, monodromy product = <b>1</b> exactly); POSITIVITY AS A THEOREM (irreducibility EXACT: <math> \langle U, M_0 \rangle  = 24 =  S_4 </math>, <math>\sum_g  \chi ^2 = 24</math>; with (U) <math>\Rightarrow</math> Mehta–Seshadri stability <math>\Rightarrow h^0(E) = 0</math>, all 324 section-saturation cases; BG dictionary <math>h^0 = 0 \Leftrightarrow a_i \geq 1</math>); INTEGRALITY = Birkhoff–Grothendieck, automatic once <math>E</math> is typed as a bundle. UNIQUENESS + SELECTOR: <math>\{1,1,2\}</math> unique (v491); the Schur–Horn permutohedron of <math>4 \cdot \text{spec}(A_0)</math> has integer points exactly <math>\{(2,2,0), (2,1,1)\}</math> (the two options of the “Global computation” theorem) and <math>h^0 = 0</math> kills <math>\{0,2,2\}</math> (an <math>\mathcal{O}(0)</math> summand has <math>h^0 = 1</math>) — the “balanced” selection is now TYPED as stability/positivity. THE <math>h^1</math> DISCREPANCY, REAL + RESOLVED: <math>h^1(\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2) = 1 \neq 3</math> — the naive “generations = <math>H^1(E)</math>” typing is FALSE (forcing it demands <math>\deg -6</math> and loses uniqueness: <math>\{1,1,4\}, \{1,2,3\}, \{2,2,2\}, e_2</math> never 5); the correct TWO-OBJECT typing: <math>L = \mathcal{O}(-D)</math>, <math>D = \mu_4</math>, carries the generations (<math>h^1(L) = 3 = N_{\text{fam}} \cong H^0(\Omega^1(\log D))</math>, the v137 basis), while <math>E</math> carries them as its FIBER (rank 3) with the anchor as splitting type. BISWAS EXPLICIT (independent convergence witness): the weight denominator is <math>3 = N_{\text{fam}}</math>, so the Biswas cover is the <math>\mathbb{Z}_3</math> cusp cover <math>y^3 = (z-1)(z-i)(z+1)^2(z+i)^2</math>, genus 2 (NOT the pillowcase double <math>w^2 = z^4 - 1</math>, genus 1, <math>j = 1728</math> — the deck side); <math>p_* \mathcal{O}_Y = \mathcal{O} \oplus \mathcal{O}(-2)^2</math> (<math>\chi</math> cross-check <math>-1 = -1</math>); the regular-representation model bundle lands on <math>\{0,2,2\}</math> = the UNSTABLE Schur–Horn companion (<math>h^0 = 1</math>): integrality AND the sum rule COME OUT of the correspondence, and stability (irreducibility, v117) is the selector. NEGATIVE CONTROLS: drop positivity <math>\rightarrow 4</math> resp. 31 solutions; <math>\deg -3/-5</math> breaks; <math>n</math>-mark scan: <math>\{1, \dots, 1, 2\}</math> unique for EVERY <math>n \geq 3</math> but <math>e_2 = (n-2)(n+1)/2 = 5</math> ONLY at <math>n=4</math> — the VALUE, not the uniqueness, is the <math>n=4</math> content; MS-rational exponents <math>e_2 = \frac{16}{3}, \frac{44}{9}</math>. [C]</p> <p>RESIDUALS: R1 the module identification “generation space = this <math>H^1</math> with this connection” [C] (QGEO.SYM territory, stays [O]); R2 unitarity (U) [C] (v33/v40); R3 the analytic-monodromy identification (fine type Numerical, v117 check 4). P2 stays an axiom (AX.P2.01 declared, no marker moves); NON-CIRCULAR: neither v23’s splitting-type input nor v115’s <math>\text{diag}(A_0) = a/4</math> input is used. Sharpens v491. Exact (sympy); Wolfram-mirrored. [E][C]</p> |

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| Script                                         | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| v500_celestial_<br>wp5c_gns_limit_<br>state.py | <p>CELEST.WP5C.01: WP5c of the celestial-holographic research contract</p> <p>CELEST.SEAM.01 — “the GNS limit state”, the third constructive milestone of WP5 (v497 made the boundary limit precise, v498 built the deleting operator; WP5c constructs the STATE whose GNS kernel contains the ideal). Exact integer/Fraction arithmetic, no floats, deterministic. [E] CONSTRUCTION: the quasi-free family <math>\omega_w</math> — loop sector (<math>r=0</math>) = the affine <math>k=1</math> vacuum <math>n</math>-point functions via the MACHINE-determined compact anti-involution <math>\theta(e_\alpha)=-e_{-\alpha}</math>, <math>\theta(h)=h</math> (the UNIQUE sign that is an anti-automorphism on ALL 61504 basis pairs; contravariance <math>\langle J_{-1}^x u, v \rangle = \langle u, J_1^{\theta x} v \rangle</math> exact); radial sector (<math>r \geq 1</math>) = oscillator pairings <math>x^{wr}</math> (the exact Gibbs regulator <math>e^{-\beta E_w}</math>, kept <math>x</math>-adic). [E]</p> <p>CONVERGENCE, SHARP: <math>\omega_w = \omega_\infty \bmod x^{N+1}</math> for ALL <math>w \geq N+1</math> (<math>N=8</math>) and NOT yet at <math>w=N</math> (the <math>r=1</math> modes sit at order <math>N</math>) — the WP5a threshold <math>w = n+1</math> holds IDENTICALLY at the state level; positivity of every <math>\omega_w</math> checked. [E] THE KERNEL (core, fully machine): the FULL exact level-2 Gram — 31124 monomials in 9361 weight blocks (dim profile 164:1, 37:240, 7:2160, 1:6960), every block exact: level-1 rank 248 POSITIVE DEFINITE (Cartan block = the <math>E_8</math> Cartan matrix, det 1); level-2 rank <b>4124</b>, kernel <b>27000</b>, every block PSD; rank per Weyl orbit (norm 8/6/4/2/0) = (0, 0, 1, 8, 44); kernel = <math>V(2\theta)</math> WEIGHT BY WEIGHT (Freudenthal cross-check on all 9361 weights); highest-weight norms (3844, 49, 2, 0) at <math>(0, \omega, \theta, 2\theta)</math>; direct ideal checks <math>F_i s\rangle</math>, <math>F_j F_i s\rangle</math> norm 0. [E]</p> <p><math>\mu_4</math>-EQUIVARIANCE: <math>\omega_w \circ \text{clock} = \omega_w</math> for ALL <math>w</math>; the clock unitary DESCENDS to <math>\text{GNS}(\omega_\infty)</math>; level-2 GNS rank split by clock class = (1036, 1024, 1040, 1024) = <math>\Theta_{C_j}/\eta^8</math> at <math>q^2</math> (the two-routes identity at the STATE level); level-1 split (60, 64, 60, 64). RESOLUTION OF WP5b S4.10: <math> s\rangle</math> IS the ZERO VECTOR of <math>\text{GNS}(\omega_\infty)</math> — the twisted two-<math>C_1</math>-mode floor (6 quarters = <math>q^{3/2}</math>) concerns only pre-limit objects (GNS has no state there: <math>6 \bmod 4 \neq 0</math>). [E] NEGATIVE CONTROLS: <math>k=2</math> (<math>\langle s s \rangle = +4</math>, dominant blocks FULL rank — NO ideal in the kernel); <math>k=0</math> (level-1 Gram <math>\equiv 0</math> — no 248 layer without the central extension); <math>D_8/SO(16)_1</math> (level-2 rank 2076 = 7380 – 5304, kernel = <math>V(2\theta_{D_8})</math> PSD weight by weight, but <math>h = (0, \frac{1}{2}, 1, 1)</math>: the vacuum GNS is 1 of 4 blocks — ONE-BLOCK completeness stays <math>E_8/\mu_4</math>-specific); wrong family <math>E'_w = mw+r</math> (level-1 limit rank 0: the 248 layer ERASED, <math>u^4</math> content 710955 <math>\neq</math> 248); no damping (897266 <math>\neq</math> 248); CCR OBSTRUCTION: at a FIXED radial pairing <math>c &gt; 0</math> positivity forces <math>\omega(bb^*) \geq c</math> — NO <math>w</math>-uniform damping state exists: the family formulation is NECESSARY and the family EXISTS <math>\Rightarrow</math> the preregistered KILL is probed and NOT triggered. [C] the quasi-free/Gaussian extension to the full <math>*</math>-algebra; Kac positivity beyond level 2 (levels <math>\leq 2</math> machine-checked); the module identification in the component route. [O] local normality of <math>\omega_\infty</math>, the net structure / Haag duality (WP5d), Costello–Li (WP5e), the operator-algebraic closure beyond the graded/<math>x</math>-adic topology;</p> <p>SEAM.EQUIV.01 stays [O], untouched. VERDICT: SUCCESS on the preregistered criterion (dim GNS level-2 = 4124 exactly). Exact (integer + Fraction). [E][C][O]</p> |

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| Script                                            | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| v501_celestial_<br>wp5d_two_interval_<br>index.py | <p>CELEST.WP5D.01: WP5d-<math>\alpha</math> of the celestial-holographic research contract</p> <p>CELEST.SEAM.01 — “the two-interval index from the lattice” (the local-net question, alpha stage). The roadmap discriminator: <math>(E_8)_1 \Rightarrow \mu=1</math> (Haag-dual two-interval net, ONE sector); <math>SO(16)_1 \Rightarrow \mu=4</math>; preregistered KILL = “the two-interval <math>\mu</math>-offset does NOT vanish after the condensation”. numpy/scipy Gaussian machinery + exact Fractions. [E] SETUP (machine identities): critical Majorana ring (v450 style, NS closure, <math>c=\frac{1}{2}</math> per layer), 16 layers = the seam carrier <math>c_-=8=g_{\text{car}}+N_{\text{fam}}</math> (v489); Peschel-exact; three prescriptions — fermionic <math>\rho</math>, orbifold sector sum <math>\{1, P_A, P_B, P_{AB}\}</math>, dual algebra <math>\{1, P_{AB}\}</math>; superselection <math>[\rho, P_{AB}]=0</math> machine-exact; EVERY Gaussian formula validated against exact diagonalisation (<math>\leq 1.5 \times 10^{-15}</math>). [E] FERMIONIC MI EXTENSIVE (the <math>\mu=1</math> reference; lattice witness): <math>I(x) = -(c/3) \ln(1-x)</math> with <math>c</math> fitted 0.5000; <math>\max  F(x)  = 5.97 \times 10^{-5}</math> at <math>N=512</math>, <math> F </math> falling monotonically to <math>1.3 \times 10^{-6}</math> at <math>N=1024</math> — Casini–Huerta extensivity on the lattice. [E] THE <math>\mu</math>-OFFSET (core): Rényi-2 exactly <math>\Delta_2 = -\ln[(1+R)/2]</math>; the 16-layer seam carrier saturates <math>\ln 2</math> to MACHINE PRECISION (<math> \Delta_2 - \ln 2  = 1.1 \times 10^{-15}</math> at <math>N=512</math>; <math>R \sim N^{-0.500}</math>); index reading <math>[F : F_{\text{even}}] = e^{\Delta_\infty} = 2.000000 \Rightarrow \mu_{\text{gauged}} = e^{2\Delta_\infty} = 4.000000 =</math> the v490 parity census (TWO independent lattice witnesses); gauging every layer separately costs <math>16 \ln 2</math> — the offset measures <math>\ln[\text{index of the subnet}]</math>, not a universal blob. [E] HAAG-DUALITY INDICATOR: fermionic <math>S(E)=S(E')</math> to <math>8.2 \times 10^{-12}</math> (graded-duality proxy); the orbifold BREAKS complementarity (<math>D(E) - D(E') = 0.427</math> at <math>a=64</math> — the direct duality-failure witness); the complementary-pair sum is exchange-symmetric and <math>\leq \ln 4 = \ln \mu(SO(16)_1)</math> with <math>\max 1.2637</math> at the self-dual <math>x=\frac{1}{2}</math> and <math>\rightarrow \ln 2</math> at the squeezed ends: the <math>\ln 4</math> KLM budget is a DOUBLE-LIMIT statement, NOT a fixed-<math>x</math> identity (the first naive fixed-<math>x</math> check was WRONG and was replaced — honestly typed). [E]</p> <p>CONDENSATION ARITHMETIC, LATTICE-ANCHORED:<br/> <math>\det \text{Cartan}(D_5) \cdot \det \text{Cartan}(A_3) = 4 \cdot 4 = 16 = \mu(\text{carrier})</math>; KLM/Longo–Rehren quotients <math>\mu/ L ^2</math>: <math>16/4^2 = 1</math> (one-step <math>\mu_4</math> glue) and <math>16/2^2 = 4 \rightarrow 4/2^2 = 1</math> (via <math>SO(16)_1</math>), both routes EXACT; <math>\Sigma d_i^2</math> cross-checks (Ising 4, <math>SO(16)_1</math> 4, <math>(E_8)_1</math> 1); <math>\theta_v = e^{2\pi i \nu/16} = 1</math> EXACTLY at <math>\nu = 2c_- = 16</math>, rivals <math>\nu = 1, 2, 8</math> give <math>\theta_v \neq 1</math>. The anchored chain: <math>\mu = 1</math> (graded, MEASURED) <math>\rightarrow 4</math> (gauged, MEASURED + v490 census) <math>\rightarrow 1</math> (arithmetic, [C]: the condensed net is interacting — the v490 fence). [E] NEGATIVE CONTROLS: <math>\nu=1</math> single Majorana (offset non-trivial, <math>\Delta_2 = 0.6364</math> at <math>N=2048</math> rising, and <math>\theta_v = e^{i\pi/8} \neq 1</math>: NOT removable — the kill discriminator has teeth); trivial dimerised phase (everything <math>\leq 10^{-7}</math>); wrong sector sum <math>\{1, P_{AB}\}</math> (offset EXACTLY 0 instead of <math>\ln 2</math>). [C] the continuum dictionary “offset = <math>\ln</math> index” (KLM two-interval index; Longo–Xu; Casini–Huerta–Magan–Pontello entropic order parameters) + the <math>\theta_v=1</math> condensation step itself. [O] continuum Haag duality, local normality, split/strong additivity of the quotient net (WP5d-<math>\beta</math>, running as exploration), Costello–Li (WP5e); SEAM.EQUIV.01 stays [O], untouched. VERDICT: the <math>\mu=1</math> indicator chain PASSES; the preregistered KILL is NOT triggered. Python (numpy + scipy + Fractions). [E][C][O]</p> |

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| Script                                         | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| v502_celestial_<br>wp5e_prefactor_<br>level.py | <p>CELEST.WP5E.ALPHA.01: WP5e-<math>\alpha</math> of the celestial-holographic research contract CELEST.SEAM.01 — “prefactor and level bookkeeping”, the CFT-side anchor stage of WP5e (the BCOV/Costello–Li twistor uplift: boundary partition function <math>= E_4/\eta^8</math> including the <math>q^{-1/3}</math> prefactor; preregistered kill: the anomaly inflow demands a level <math>\neq 1</math>). Exact sympy/Fraction. [E] PART A — the <math>q^{-1/3}</math> prefactor is exact <math>\mu_4</math> vacuum-energy bookkeeping; Hurwitz-<math>\zeta</math> derivation (<math>\zeta(-1, \theta) = -B_2(\theta)/2</math>; <math>E_b(\theta) = -\frac{1}{24} + \theta(1 - \theta)/4</math> exactly); the clock is INNER (<math>h = (2, 2, 2, 2, 2; 0^4)</math> and the <math>A_3</math> detector <math>h' = (0^5; 1, 1, 1, -3)</math> both read the class mod 4 on all 240 roots; <math>(h, h) = 20</math>, <math>(h', h') = 12</math>, sum 32) <math>\Rightarrow</math> twist distribution <math>\theta = 0^8</math> in ALL four sectors — a SHIFT orbifold (lattice cosets, integer-moded oscillators), <i>not</i> a rotation orbifold; <math>E_{\text{osc}} = -\frac{1}{3} = -c/24</math> sector-independent (<math>c = 8 = \text{rank} = 248/31</math>); sector ground exponents <math>-\frac{1}{3} + (0, 1, 1, 1)</math>. [E] discriminant form = Casimir energy, TWO routes: spectral flow <math>j^2(h, h)/32</math> (<math>5j^2/8, 3j^2/8</math>, sum <math>j^2 \equiv 0 \pmod{1}</math> = the isotropy/Lagrangian condition) and <i>exactly</i> via free fermions (<math>D_5 = 10</math> Majorana, <math>A_3 = D_3 = 6</math>, glue diagonal <math>= 16 = 2^{g_{\text{car}}-1}</math>, the WP5d seam carrier; R–NS shift <math>n/16</math>: <math>\frac{5}{8}, \frac{3}{8}, 1</math>); the ROTATION reading fails (the geometric <math>A_3</math> deck gives <math>\frac{3}{16} \neq \frac{3}{8}</math>: the v492 “clock <math>\neq</math> deck” lesson at the Casimir level). [E] character: <math>\sum_j \Theta_{C_j} = E_4</math> and the four sector characters (leading coefficients <math>(1, 64, 60, 64)</math>, currents <math>(60, 64, 60, 64)</math>) sum exactly to <math>E_4/\eta^8 = q^{-1/3}(1, 248, 4124, 34752)</math>. [E] PART B — <math>k = 1</math> forced THREE independent ways: (i) the CURRENT condition <math>h(J) = k = 1</math> (the <math>\mathbb{Z}_4</math> extension closes onto the <math>E_8</math> adjoint <math>60+64+60+64 = 248</math> only at spin 1); HONEST SHARPENING: glue-<math>h</math> integrality <math>h(J^j; k) = k(0, 1, 1, 1)</math> holds for ALL <math>k = 1..8</math> — the originally conjectured integrality route fixes <i>nothing</i>, the three other dials carry; (ii) conformal embedding <math>47k^2+219k-266 = 47(k-1)(k+266/47) = 0</math> resp. <math>D_8</math>-in-<math>E_8</math>: <math>128k(1-k) = 0</math>; (iii) central charge <math>248k/(k+30) = 8 \Leftrightarrow 240(k-1) = 0</math> (= the prefactor itself); plus the v498 singular-vector dial (<math>\ (E_{-1}^\theta)^m 0\ ^2 = m! k(k-1) \cdots (k-m+1)</math>, singular vector exactly at grade <math>k+1</math>; <math>k = 1</math> deletes <math>V(2\theta)</math>: <math>31124 - 27000 = 4124</math>). [E] negative controls: <math>D_8/SO(16)_1</math> (same rank, same <math>c = 8</math>, SAME <math>q^{-1/3}</math> prefactor, but <math>h = (0, \frac{1}{2}, 1, 1)</math>: the discriminator is glue-<math>h</math> integrality, not the prefactor; <math>\Theta_{D_8} + \Theta_s = E_4</math>); <math>\mathbb{Z}_2</math> rotation (<math>E = +\frac{1}{6}</math>, <math>h = \frac{1}{2}</math>: no <math>\mu_4</math>-shift sector there); <math>\mu_4</math> rotation (sector-dependent <math>(-\frac{1}{3}, \frac{1}{24}, \frac{1}{6}, \frac{1}{24})</math>: the common prefactor breaks); wrong <math>k \in \{2, 3, 4\}</math> fails all five dials coherently (residuals <math>360/814/1362</math>; <math>\langle s s \rangle = 4/12/24</math>; level-2 dim stays <math>31124 \neq 4124</math>). [C] Kac generation of the maximal submodule by the grade-<math>(k+1)</math> singular vector for <math>k \geq 3</math> (<math>k = 1, 2</math> machine-anchored by WP5b/WP5c). [O] the BCOV/Costello–Li twistor inflow itself (Costello’s one-loop axion <math>\lambda_g/(8\pi\sqrt{3})</math> on <math>\mathbb{P}^1</math>, the Costello–Paquette–Sharma level-from-flux mechanism) stays open — v502 anchors the CFT side of WP5e only (<math>\alpha</math> stage); the kill branch (“inflow demands a level <math>\neq 1</math>”) is not triggered on the CFT side but its twistor side is untested; WP5e stays [O]; SEAM.EQUIV.01 untouched, no marker moves. Exact (sympy + Fraction). [E][C][O]</p> |

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| Script                           | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| v503_qgeo_<br>emergence_light.py | <p>QGEO.EMERGE.LIGHT.01: the v215 <i>deferred emergence test</i> (lever 7) executed in its first honest LIGHT version, plus the residual-convergence result QGEO-R1 <math>\equiv</math> P2-R1 (analysis/kill-test class; NO marker moves — QGEO.REALIZE.01 stays [C]/[O]). NO DYNAMICAL SELECTION (<math>\zeta</math>-log-det via the Dedekind-<math>\eta</math> formula, mpmath 40 digits): <math>\log F = \log(\tau_2 \eta ^4)</math> is exactly <math>S</math>-invariant, so <math>t = 1</math> is automatically critical on the rectangular family <math>\tau = it</math>; there a strict local maximum (<math>g''(1) = -1.298</math>) but a <i>minimum</i> in the <math>\tau_1</math>-direction (<math>h''(0) = +0.298</math>): <math>\tau = i</math> is a SADDLE of the full moduli space; critical points = the CM points (<math>E_2^*(i) = E_2^*(\rho) = 0</math> to 40 digits); the <i>global</i> det' winner at fixed area is the HEXAGONAL torus (<math>\log F(\rho) = -1.03352 &gt; \log F(i) = -1.05469</math>; Osgood–Phillips–Sarnak 1988) — free-field dynamics alone does <i>not</i> select <math>\tau = i</math>; criticality at the square is Palais symmetric criticality (the independent heat trace is equally critical). SHARP RIGIDITY (v280 lattice DtN): the mod-4 off-character indicator (the v215-K3 lever as a <i>modulus function</i>) has an isolated zero (<math>8 \times 10^{-15}</math>) exactly at <math>\tau = i</math>, strictly monotone in both directions (min <math>8.4 \times 10^{-2}</math> away); <math>R^2 = \mathbb{Z}_2</math> exactly as permutation matrices (the v492 <math>\mathbb{Z}_8</math> bridge on the lattice); <math>j(it) \geq 1728</math> with equality only at <math>t = 1</math>, <math>j(\rho) = 0</math>. [E] the <math>\mathbb{Z}_2</math> chain is MODULUS-BLIND: 4 fixed points and <math>[\mathbb{Z}_2, \Delta_t] = 0</math> on <i>every</i> torus (v216 confirmed without mark input); pillowcase det' = <math>\frac{1}{2}</math> torus det' exactly (<math>\pm</math>-pair structure), so all <math>\tau</math> conclusions transfer verbatim; marks/mark-locality/deck<sup>2</sup> exist on <i>every</i> pillowcase (v264/v280), only the order-4 clock distinguishes <math>\tau = i</math>; <math>\text{Aut}(E_\rho) = \mathbb{Z}_6</math> (element orders 1, 2, 3, 6) contains <i>no</i> order-4 element — dynamics and clock are logically independent selectors; GIVEN the clock, <math>P(iZ) = P(Z)</math> forces <math>a_3 = a_2 = a_1 = 0</math> and <math>j = 1728</math> identically (v493): the modulus is not a second input. [C]/[O] RESIDUAL CONVERGENCE (the core finding): the order-4 element on <math>H^1</math> is a Coxeter element of <math>W(A_3)</math> (char poly <math>\lambda^3 + \lambda^2 + \lambda + 1</math>, order <math>h(A_3) = 4 =  \mu_4  = N_{\text{fam}} + 1</math>) — exactly the P2 cusp-class/carrier datum of the P2 hardening (v499 R1) <math>\Rightarrow</math> QGEO-R1 <math>\equiv</math> P2-R1: ONE common residual, “why does the raw seam carry the order-4 clock”, not two scattered ones; chain: marks derived (v216 [E]), square derived <i>given</i> the clock (v493 [E]), clock = carrier input [C], rest = the dynamical realisation of the clock [O]; route classification (v347 analogue): B-rigidity in the sharpened form “RP seam data <math>\Rightarrow</math> order-4 clock” is the live route — free dynamics alone does not deliver the clock (the hexagonal finding). Lever update (documented here, v215 itself unchanged): lever 7 lifted from “deferred, not executable” to “light version executed: rigidity, not dynamical selection”. Python (mpmath + numpy/scipy + sympy). [E][C][O]</p> |

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| Script                                          | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| v504_celestial_<br>wp5d_klm_<br>completeness.py | <p>CELEST.WP5DB.01: WP5d-<math>\beta</math> of the celestial-holographic research contract<br/> CELEST.SEAM.01 — “split + strong additivity from the lattice”, the two<br/> remaining Kawahigashi–Longo–Müger legs of complete rationality (the <math>\mu</math>-index is<br/> v501), witnessed for the orbifold prescription on the critical Majorana seam edge;<br/> the <math>U(1)</math>/Dirac subnet is the preregistered negative control (the current net,<br/> where strong additivity fails in the continuum). [E] STRONG ADDITIVITY,<br/> EXACT: shared boundary Majorana <math>\Rightarrow</math> Even(<math>A</math>) <math>\vee</math> Even(<math>B</math>) = Even(<math>A \cup B</math>)<br/> EXACTLY (GF2 spans full 64/64 up to 1024/1024; matrix rank 32/32);<br/> DISJOINT: exactly HALF (rank 16/32, index 2 — the missing odd <math>\otimes</math> odd sector is<br/> the v501 ln 2 bit, localised at the split point); the neutral <math>U(1)</math> algebras do NOT<br/> generate the union even with the shared site (gaps 2/10/52, growing). [E]<br/> ENTROPIC WITNESS [N]: the fermionic touching MI carries the pure UV<br/> point-gap law <math>((c/3) \ln 2</math> increments, dev <math>1.2 \times 10^{-6}</math> — no discriminator); the<br/> honest discriminator is BOUNDED vs DIVERGENT: the <math>\mathbb{Z}_2</math>/orbifold deficit stays<br/> <math>&lt; \ln 2</math> with <math>(\ln 2 - \Delta_2) \sim N^{-p}</math>, <math>p = 0.2444</math> vs <math>2\Delta_\mu = \frac{1}{4}</math> (Ising disorder dimension;<br/> “defect <math>\rightarrow 0</math>” would be FALSE at the sharp split — bounded <math>\Leftrightarrow</math> finite index,<br/> Longo–Xu), while the <math>U(1)</math> control BURSTS ln 2 from <math>L = 128</math> and grows as<br/> <math>\frac{1}{2} \ln \text{Var } Q_A</math> (Klich–Levitov 0.10134 vs <math>1/\pi^2</math>): infinite index. [E]<br/> SPLIT/NUCLEARITY [N]: coupling <math>\sigma_k</math> fall at the elliptic-nome rate<br/> <math>\pi K(1-x)/K(x)</math> to 1.3–2.0%; <math>\sigma_1 \sim x^{0.5044}</math>; trace norm summable; ORBIFOLD<br/> INHERITANCE EXACT: <math>P_A</math> flips <math>C \rightarrow -C</math> (<math>\sigma_k</math> identical, <math>8.3 \times 10^{-17}</math>), the<br/> even-bilinear Gram is the compound <math>\Lambda^2 C</math> (<math>\{\sigma_a \sigma_b\}</math>, <math>2.8 \times 10^{-14}</math>) — Longo heredity<br/> [C]. [E] PIMSNER–POPA: <math>E(a) - a/2 = PaP/2</math> IDENTICALLY <math>\Rightarrow</math><br/> <math>\lambda = \frac{1}{2} = 1/[F : F_{\text{even}}]</math>; integer attainment (16384 + 2048 monomial sweeps, 0<br/> violations; pencil min <math>\lambda = 0.5000000000</math>); <math>\lambda_{E_4} = \frac{1}{4} = 1/\mu</math> exactly; INDEX<br/> CONSISTENCY, two routes: <math>e^{\Delta_\infty} = 2.0000000000 = 1/\lambda_{\text{PP}}</math> and <math>e^{2\Delta_\infty} = 4 = \mu</math>;<br/> <math>U(1)</math>: <math>\lambda = 1/(m+1) \rightarrow 0</math>. [E] CONTROLS + HONESTY: dimerised phase trivial<br/> (<math>3.6 \times 10^{-15}</math>); wrong prescription <math>\{1, P_{AB}\}</math> deficit EXACTLY 0; finite-size honesty<br/> (14.9% below ln 2 at <math>N = 4096</math> — witness, not proof). [C] the KLM dictionary<br/> (Longo–Xu; Longo heredity; CHMP). [O] the continuum theorem “finite-group<br/> orbifolds of completely rational nets are completely rational” is XU’S THEOREM<br/> — CITED, NOT CLAIMED; the seam quotient net and the condensed <math>(E_8)_1</math> net<br/> stay WP5e/Costello–Li; SEAM.EQUIV.01 stays [O]. VERDICT: with v501, all<br/> THREE KLM ingredients witnessed on the lattice. Python (numpy + scipy);<br/> exact parts Wolfram-mirrored. [E][C][O]</p> |

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| Script                                                     | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| v505_celestial_<br>wp5e_beta_<br>equivariant_<br>ledger.py | <p>CELEST.WP5E.BETA.01: WP5e-<math>\beta</math> of the celestial-holographic research contract CELEST.SEAM.01 — “the equivariant anomaly ledger on twistor space”, the twistor-side milestone of WP5e (the BCOV/Costello–Li uplift; preregistered kill: the anomaly inflow demands a level <math>\neq 1</math>); WP5e-<math>\alpha</math> (v502) anchored the CFT side. The Atiyah–Bott/Lefschetz fixed-point skeleton of the one-loop box anomaly (Costello; Bittleston–Sharma–Skinner) evaluated exactly on <math>\mathbb{C}^2/\mathbb{Z}_4</math> with the <math>E_8</math> <math>\mu_4</math>-glue monodromy. Exact sympy/Fraction. [E] AB ledger: denominators <math>\det(1 - g_j^{-1}) = (2, 4, 2)</math>; <math>\sum_j 1/\det_j = 5/4 = ( \mathbb{Z}_4 ^2 - 1)/12</math> (Dedekind); equivariant characters <math>\mathrm{tr}_{g_j}(\mathrm{Ad} E_8) = (248, 0, -8, 0)</math> by two routes; invariant average <math>60 =</math> the <math>D_5 + A_3</math> carrier; Frobenius 61568; <math>\mathrm{tr}_{g_2} = -8 = 120 - 128</math>. [E] Okubo per sector (honest sharpening): the invariant sector is exactly Okubo-quadratic (<math>Q^{(0)} = 36\langle x, x \rangle^2</math>, <math>36 = \tilde{\lambda}_{e_8}^2</math> — v495 re-derived through the glue grading); the twisted sectors do NOT reduce (<math>T_5: (-8, 32, -8)</math>, <math>T_3: (48, -64, 48)</math>); the AB-weighted sum cancels the <math>D_5</math> quartic exactly and leaves the RIGID residual <math>32T_3</math>; the only quartic-free weighting is the uniform <math>\mathbb{Z}_4</math> projector; the graded GS exchange is exactly rank-obstructed in sectors 1, 2, 3; heterotic identities exact (<math>\mathrm{Tr}_{120}, \mathrm{Tr}_{128}, \mathrm{Tr}_{E_8} = 9(\mathrm{tr}^2)^2</math>). [E] index bridge (centrepiece): <math>f(m) = \frac{1}{4} \sum_j (i^{jm} - 1)/\det_j = (0, -\frac{3}{8}, -\frac{1}{2}, -\frac{3}{8}) = \mathrm{ch}_2(T_m) = -(C^{-1})_{mm}/2</math> EXACTLY — fixed-point ledger = McKay/Kronheimer intersection ledger; glue <math>\mathrm{ch}_2</math> defect <math>-78</math> by both routes (classes <math>(-24, -30, -24)</math>; <math>-30 = -h^\vee</math>, a [C] coincidence); <math>\sum_m h_{A_3}(m) = 5/4 = \sum_j 1/\det_j</math>; Killing sums <math>(320, 320, 240, 320)</math>, total <math>1200 = 2h^\vee(h, h)</math>. [E] <math>k = 1</math> geometric: lattice current count = 240 at <math>k = 1</math>, exactly 0 at <math>k = 2, 3, 4</math>; embedding residual <math>47k^2 + 219k - 266 = (0, 360, 814, 1362)</math>; <math>h(J; k) = k</math> integer for all <math>k</math> — integrality alone fixes nothing (honest, as v502); one scale <math>\Rightarrow</math> one level (v493 periods). [E] E3 REFUTED (prominent): <math>a_0 \in O(8)</math> fills the GRAVITON slot <math>O(2)</math>, not the BSS axion slot <math>O(-2)</math> (<math>X = -2 \neq +2 = Y</math>; mismatch <math>4 =  \mu_4 </math>, a [C] coincidence) — “the theory brings its GS axion as <math>a_0</math>” is refuted; instead the three <math>H^2(\mathrm{ALE})</math> classes carry exactly the Coxeter eigencharacters <math>\{i, -1, -i\} =</math> the three twisted slots (bijection); slot count <math>1 + 3 = 4 =  \mu_4 </math> soft. [E] negative controls: <math>\mathrm{diag}(i, i)</math> breaks four ways (<math>\sum 1/\det = \frac{1}{4}</math>, <math>\omega</math>-weight <math>-1</math>, 5 generators); <math>SO(16)</math> glue: characters <math>(120, 0, +120, 0)</math>, defect <math>-30 \neq -78</math>, bulk Okubo fails (independent quartic Casimir); <math>k = 2</math>: count 0, prefactor <math>-31/48 \neq -\frac{1}{3}</math>. [C] dictionaries <math>(\mathrm{CS}(\rho) \bmod 1 = \text{discriminant weight}; 32 = (h, h) + (h', h'); -30 = -h^\vee; \Omega\text{-contraction})</math>. [O] the global BCOV/Kodaira–Spencer quantisation on <math>\mathbb{PT}/\mathbb{Z}_4</math>, the <math>O(-2)</math> bulk-axion construction, CPS level-from-flux, the pairing of the <math>32T_3</math> residual with the three sphere axions. Verdict B; the kill branch does NOT fire on the equivariant skeleton; SEAM.EQUIV.01 untouched, no marker moves. Exact (sympy + Fraction). [E][C][O]</p> |

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| Script                      | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| v506_seam_clock_rigidity.py | <p>SEAM.CLOCK.RIGIDITY.01: clock rigidity, Route B of the v503 classification — two exact theorems reducing the order-4 clock input to ONE alignment bit (Part A Moebius + Part B Fock in one module; QGEO.REALIZE.01 stays [C]/[O], no marker moves). Exact sympy + exact Cl(16) Fractions. [E] THEOREM A (Moebius): the collar deck <math>z \mapsto -z</math> has in <math>\mathrm{PGL}_2(\mathbb{C})</math> exactly TWO square roots (<math>z \mapsto \pm iz</math>), both automatically of order 4 (complete case analysis: <math>\mathrm{tr} \neq 0 \Rightarrow</math> diagonal <math>\Rightarrow a/d = \pm i</math>; <math>\mathrm{tr} = 0</math> impossible) — “order 4” is no longer an input; a marks-preserving root exists <math>\iff</math> the deck is the CENTRAL involution of the mark-<math>D_4</math> (center = {id, deck}, exactly 2 order-4 elements = the clocks) <math>\Rightarrow</math> then marks = one <math>\mu_4</math> orbit, cross-ratio class <math>\{2, \frac{1}{2}, -1\}</math>, <math>j = 1728</math>, <math>\tau = i</math>, free 4-cycle, deck = clock<sup>2</sup>, only the scale free; COUNTEREXAMPLE: <math>\{\pm 1, \pm(3+2\sqrt{2})\}</math> is deck-free AND harmonic with full <math>D_4</math>, but the deck is non-central (edge involution) <math>\Rightarrow 0</math> roots — harmonicity alone is NOT sufficient; the RP filter selects nothing (<math>\mathrm{Aut}(\mathbb{Z}_4)</math> ambiguity irreducible, consistent with v492). [E] B3/B4: free orbit <math>\Rightarrow n \mid 4</math>; mark transitivity <math>\Rightarrow n = 4</math>; faithfulness kills <math>n \geq 5</math>; <math>n = 3</math> fixes a mark; <math>n = 6</math> dies at cusp degree 3 (v117); the deck is modulus-blind (v503 B5 exact); controls: <math>n = 8</math> impossible; hexagonal <math>A_4</math> (0 order-4 elements, squares only <math>\{1, 3\}</math>); generic <math>V_4</math> (exponent 2); stabiliser trichotomy <math>V_4/D_4/A_4</math> with order-4 counts (0, 2, 0). [E] THEOREM B (Fock, the core result): the unique <math>S^2</math> spin structure gives NS on every circle (established P1 datum); one-particle <math>D^2 = -1</math> (charpoly <math>(\lambda^2+1)^8</math>), quarter-shift <math>C^2 = D</math>, <math>C^8 = 1</math> (charpoly <math>(\lambda^4+1)^4</math>); Fock: <math>\tilde{U} = \prod_j (1 - \gamma_j \gamma_{j+8})</math>, <math>\tilde{U}^2 = 256 \gamma_1 \cdots \gamma_{16} \Rightarrow U^2 = (-1)^F</math> EXACTLY, and <math>(cU)^2 = c^2(-1)^F</math> is never 1 (center of Cl(16) = scalars) — the <math>\mathbb{Z}_4</math> lift is FORCED (nonsplit extension); <math>\mathbb{Z}_8</math> tower: <math>V^2 \propto \tilde{U}</math>, <math>V^4 \propto (-1)^F</math>, projective order <math>8 = 2 \mu_4 </math> — fermionically explained; R control <math>U_R^2 = +1</math> (split — the forcing rests on NS = established geometry); 16-fold way <math>\theta_\nu = 1</math> at <math>\nu = 16</math>; the forcing already holds at 8 sites. [C] THE ONE REST: “the collar deck is the central involution of the mark-<math>D_4</math>” (<math>\iff</math> square-root existence on the mark data) — ONE alignment bit; the fermionic <math>\mathbb{Z}_4</math> forces the ORDER, not the spatial alignment. No overclaim. Exact (sympy + Cl(16) Fractions). [E][C]</p> |

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| Script                  | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| v507_seam_bit_origin.py | <p>SEAM.BIT.ORIGIN.01: the tautology attack on the v506 alignment bit, executed and refuted — where do the marks AND the deck come from, and does their common origin force the central alignment? (Part A torus origin + Part B fermion dichotomy in one module; NO marker moves — the bit stays [C], SEAM.CLOCK.RIGIDITY.01 and QGEO.REALIZE.01 text-precised only.) Exact sympy + exact Cl(16) Fractions. [E] ORIGIN THEOREM: the 4 marks are the fixed points of the hyperelliptic involution <math>w \mapsto -w</math> on <math>\mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z}) =</math> the four half-periods <math>\Lambda/2\Lambda</math>; a translation <math>t_c</math> descends to the base sphere iff <math>2c \in \Lambda</math> — exactly the THREE half-period translations (control <math>c = 1/3</math> fails); the descended maps are the EXPLICIT rational involutions <math>\sigma_i(x) = e_i + (e_i - e_j)(e_i - e_k)/(x - e_i)</math> with the exact curve-lift identity <math>\prod(\sigma_i - e_l) = \psi_i^2 \prod(x - e_l)</math>, <math>\psi_i</math> RATIONAL; <math>\{\text{id}, \sigma_1, \sigma_2, \sigma_3\} =</math> Klein <math>V_4</math>; CONVERSELY, in ALL three stabiliser types the freely-acting involutions are EXACTLY the three <math>\sigma_c</math> (generic <math>V_4</math> (4, 3, 3), harmonic <math>D_4</math> (8, 5, 3), hexagonal <math>A_4</math> (12, 3, 3)) <math>\Rightarrow</math> EVERY mark-free collar deck (v216) IS a half-period translation — the CLASS of the deck is forced, its POSITION is not. [E] CORE SOLVE (symbolic <math>\lambda</math>): <math>\sigma_c</math> admits a marks-preserving <math>\text{PGL}_2</math> square root <math>\iff \lambda \in \{-1\}(\sigma_1), \{2\}(\sigma_2), \{\frac{1}{2}\}(\sigma_3)</math> — per half-period exactly ONE solution; intrinsically: the two deck pairs separate each other HARMONICALLY (pair cross-ratio = <math>-1</math>); union over <math>c =</math> the harmonic orbit <math>\{-1, 2, \frac{1}{2}\} \iff j = 1728 \iff \tau = i</math>; generic modulus: <math>V_4</math> abelian, ALL <math>\sigma_c</math> central and yet clockless (“central alone is empty” — the bit is the CONJUNCTION <math>D_4</math> AND central); at <math>\tau = i</math> the stabiliser jumps to <math>D_4</math> with <math>\sigma_1</math> central (2 roots = the clock pair) and <math>\sigma_2/\sigma_3</math> edge-type (0 roots); the v506 seam deck <math>z \mapsto -z</math> lands on the central <math>\sigma_1 = -1/x</math> under ALL 8 mark bijections; CM DICTIONARY: mult-by-<math>i</math> fixes exactly ONE nonzero half-period <math>c^* = (1 + \tau)/2</math> (mult-by-<math>\rho</math> 3-cycles all three); the clock lift needs the CM unit <math>i</math> (<math>\psi = 2i/(x + 1)^2</math>, <math>t^2 + 1</math> irreducible over <math>\mathbb{Q}</math>) while half-period decks lift RATIONALLY: the clock root exists only WITH CM by <math>\mathbb{Z}[i] \iff \tau = i</math>. [E] EQUIVALENCE TETRAD + CIRCULARITY AUDIT: clock exists <math>\iff (\tau = i</math> AND deck = the CM-fixed half-period) <math>\iff (\text{Stab} = D_4</math> AND deck central) <math>\iff</math> deck pairs harmonic <math>\iff</math> the NS Fock lift is NONSPLIT; the v493 converse independently re-verified — two converses of ONE equivalence, no circularity. [E] FERMION DICHOTOMY (the new physical face): central arrangement (shift by 8): <math>\tilde{U}^2 = 256 \gamma_1 \cdots \gamma_{16} \Rightarrow U^2 = (-1)^F</math>, nonsplit <math>\mathbb{Z}_4</math>, 2 roots in the 32-element dihedral lift (the <math>\mathbb{Z}_8</math> tower); edge arrangement (reflection <math>j \mapsto 4 - j</math>, fixed sites = non-marks = the silver <math>\{0, \infty\}</math>): the implementer EXISTS but <math>\tilde{U}_e^2 = +2^7 \cdot \mathbf{1} \Rightarrow U^2 = +1</math>, the extension SPLITS, 0 roots, no <math>\mathbb{Z}_8</math>; one-particle <math>(\lambda^2 + 1)^8</math> vs <math>(\lambda^2 - 1)^8</math>; census: all 28 shift-invariant 4-sets cross, all 21 reflection-invariant do not; robust at 8 sites — the bit IS a physical extension class (Fidkowski–Kitaev-type), distinguished but not derived. [E] SILVER PLACEMENT: <math>\{\pm 1, \pm(3 + 2\sqrt{2})\} =</math> “<math>\tau = i</math> with deck = a non-CM-fixed half-period translation” (all 8 bijections <math>\rightarrow \sigma_2/\sigma_3</math>, never <math>\sigma_1</math>) — in-family witness against the tautology. [C] THE REST: the one alignment bit remains a carrier datum; no tautology — the origin forces the CLASS of the deck, not its POSITION. Exact (sympy + Cl(16) Fractions). [E][C]</p> |

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| Script                                       | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| v508_celestial_<br>wp5e_gamma_<br>pairing.py | <p>CELEST.WP5E.GAMMA.01: WP5e-<math>\gamma</math> of the celestial-holographic research contract<br/> CELEST.SEAM.01 — “the sphere-axion pairing check”, an honest rigid NEGATIVE<br/> result: can the three twisted sphere axions <math>\eta_m</math> (Coxeter eigencharacters<br/> <math>\{i, -1, -i\}</math> on <math>H^2(\text{ALE})</math>, v505 S5.8) cancel the rigid <math>32 T_3</math> residual of the<br/> Atiyah–Bott ledger (<math>A_{\text{fix}} = 9P_1 - 30P_2 - 15P_3 + 32 T_3</math>, v505) by<br/> Green–Schwarz-type EXCHANGE with sector-compatible quadratic vertices? NO,<br/> with certificate. Exact sympy/Fraction. [E] VERTEX-SPACE THEOREM: the<br/> <math>W(D_5) \times W(A_3)</math>-invariant quadratics on the 8-dim glue Cartan are EXACTLY<br/> <math>\text{span}\{s_5, s_3\}</math> (dim 2, bidegree dims (1, 0, 1)), the quartics EXACTLY<br/> <math>\text{span}\{P_1, P_2, P_3, T_5, T_3\}</math> (dim 5, bidegree dims (2, 0, 1, 0, 2)) — Weyl nullspace<br/> arithmetic over all bidegrees, basis COMPLETE; PRODUCT THEOREM (the<br/> master kill): <math>(as_5 + bs_3)(cs_5 + ds_3) = ac P_1 + (ad + bc)P_2 + bd P_3</math> identically — zero<br/> <math>T_5/T_3</math> content, so EVERY exchange image (any axion count, any selection rule,<br/> any couplings) is annihilated by <math>\Phi_{T_3}</math>, while <math>\Phi_{T_3}(A_{\text{fix}}) = 32 \neq 0</math>. [E] SELECTION<br/> ENUMERATION: the strict two-index reading COLLAPSES (<math>C_{j,j'} \neq 0</math> only for<br/> <math>j' = -j</math>, so the rule forces <math>m = 0</math> — the sphere axions have NO invariant bilinear<br/> Cartan vertex at all); twist insertion: <math>\eta_1 \leftrightarrow K^{(3)}</math>, <math>\eta_2 \leftrightarrow K^{(2)}</math>, <math>\eta_3 \leftrightarrow K^{(1)}</math>,<br/> <math>\eta_0 \leftrightarrow K^{(0)}</math>; channels <math>E_{13} = (16, -96, 144, 0, 0)</math>, <math>E_{22} = (16, 32, 16, 0, 0)</math>,<br/> <math>E_{00} = 225 E_{22}</math>; SIDE DISCOVERY: <math>K^{(0)} = -15 K^{(2)}</math> (the even-sector quadratics<br/> are PARALLEL). [E] UNSOLVABILITY WITH CERTIFICATE (the core result):<br/> exchange matrix <math>5 \times 2</math>, rank 2, kernel 0; <math>\text{rank}([M   A_{\text{fix}}]) = 3</math>; annihilator<br/> certificate <math>(\Phi_{T_5}, \Phi_{T_3}, \Phi_P)(A_{\text{fix}}) = (0, 32, 72)</math> with <math>A_{\text{fix}} = (9, -30, -15, 0, 32)</math> — two<br/> independent obstructions (<math>32 T_3</math> structural, rule-independent + <math>\Phi_P = 72</math><br/> rule-driven via the <math>K^{(0)} \parallel K^{(2)}</math> collapse); <math>\ker M = 0</math> forces the unique<br/> non-disturbing coupling <math>c = 0</math>: the sphere axions are RIGIDLY SILENT in the<br/> exchange ansatz; <math>\Phi_P(Q^{(0)}) = 0</math> (the v505 sector-0 mechanism undisturbed);<br/> relaxed control: dropping charge conservation reaches the <math>P</math>-block (<math>\det = -64</math>)<br/> but <math>T_5 = T_3 = 0</math> persists. [E] NATURALNESS RESOLUTION: <math>\text{ch}_2</math>-natural<br/> <math>(9/64, 1/4)</math> and AB-weight <math>(1/2, 1/4)</math> couplings both carry certificate <math>(0, 0)</math> vs<br/> required <math>(32, 72)</math> — the failure is scale- and coupling-INDEPENDENT (no<br/> “unnatural factor”: the ansatz space misses the target identically). [E]<br/> NEGATIVE CONTROLS: invariant axion only = v505 replication + sharpening;<br/> flipped rule <math>m+j-j'</math>: the <math>\Phi_{T_3}</math> kill is flip-invariant, <math>\Phi_P</math> is rule-dependent; <math>SO(16)</math>:<br/> odd classes EMPTY and the AB sum <math>(15, -18, -15, -4, 40)</math> does not even cancel<br/> <math>T_5</math> (<math>E_8</math> doubly special); <math>D_8</math>: <math>16 T_8 + 12 s_8^2</math>, no <math>T_3</math> structure — the question cannot<br/> even be posed. [C] number fences: <math>32 = (h, h) + (h', h') = 20 + 12</math>; <math>72 = 2\tilde{\lambda}_{e_8}^2</math>. [O]<br/> SHARPENED FENCE: the exchange sub-branch is DEAD; the <math>32 T_3</math> residual<br/> MUST be carried by twisted BCOV contact terms — binary criterion WP5e-<math>\delta_1</math>:<br/> the one-loop coefficient on <math>\mathbb{C}^2/\mathbb{Z}_4</math> must come out exactly <math>(9, -30, -15, 0, 32)</math>,<br/> COMPUTED, not fitted; named escape WP5e-<math>\delta_2</math>: the full-tensor ledger (the<br/> collapse is Cartan-restricted); slot bijection (v505 S5.8) untouched; the<br/> preregistered level-kill still does NOT fire; SEAM.EQUIV.01 stays [O], no marker<br/> moves. Exact (sympy + Fraction). [E][C][O]</p> |

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| Script                                              | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| v509_celestial_<br>wp5e_eps2_level_<br>from_flux.py | <p>CELEST.WP5E.EPS2.01: WP5e-<math>\varepsilon_2</math> of the celestial-holographic research contract CELEST.SEAM.01 — “the CPS level-from-flux dial”, executed (verdict B): derive the boundary level <math>k = 1</math> from flux quantisation on the lockstep spheres, following the Costello–Paquette–Sharma (CPS) Burns-holography mechanism (PRL <b>130</b>, 061602; arXiv:2306.00940). Exact sympy/Fraction, no floats. [E] CPS SKELETON REPLICATED EXACTLY: the <math>S^3</math> period <math>\int \Omega_N = (2\pi i)^2 N</math> (pullback density <math>-\sin 2\theta</math>, their eq. 3.33); the exceptional-sphere Kähler flux <math>\int_{\mathbb{P}^1} \omega = 2\pi N</math>; the boundary level magnitude <math>2N</math> via the symplectic-boson double contraction (<math>N = 1, 2, 3</math>), Dynkin normalisation <math>T(\text{vector}) = 1</math> — “level = flux quantum <math>\times</math> Dynkin index”. [E] PAIRING MATRIX PINNED, NOT POSTULATED: the v505 ch<sub>2</sub> ledger + integrality + unimodularity + effectivity pin <math>P_{mi} = c_1(T_m) \cdot [\Sigma_i] = \delta_{mi}</math> by COMPLETE enumeration (row candidates 8/6/8; 48 unimodular solutions of <math>PC^{-1}P^T = C^{-1}</math>; with <math>P \geq 0</math> exactly 2 = {identity, <math>A_3</math> diagram flip}); glue FLUXES <math>F_i = (64, 60, 64) = \dim \mathfrak{g}_i</math>, total 188 = 248 – 60, flip-invariant; flux per charged root current uniformly ONE quantum. [E] LEVEL FORMULA, honest: “level = total open-string flux” is KILLED by the lockstep test itself (<math>k_i = (64, 60, 64)</math> unequal = the FLAVOUR multiplicity); what works: <math>k = \text{flux per adjoint current} = F_i / \dim \mathfrak{g}_i = (1, 1, 1)</math>, anchored by the lattice current count (240, 0, 0, 0) for <math>k = 1..4</math> and the embedding index <math>j(D_5 \subset E_8) = j(A_3 \subset E_8) = 1</math> (<math>60 = 2h^\vee</math>). ORD-4-VS-LEVEL-1 RESOLVED: the clock order counts FRACTIONAL flux sectors — <math>\mu = \det(D_5) \det(A_3) = 16 = \text{ord}^2</math>, the <math>\mu_4</math> glue diagonal isotropic + Lagrangian, KLM <math>16 \rightarrow 16/4^2 = 1</math> (zero residual fractional flux); NEW DIAL: <math>\#\text{primaries}((E_8)_k) = (1, 3, 5, 10, 15, 27)</math> for <math>k = 1..6</math> from the affine marks — exactly ONE sector iff <math>k = 1</math>. [E] LOCKSTEP THEOREM (the lifting of v505 “one scale <math>\Rightarrow</math> one level” to a THEOREM): clock invariance forces <math>a_3 = a_2 = a_1 = 0</math>, one <math>\mu_4</math> orbit of branch points, <math> \Pi_j  = \sqrt{2}t</math> EQUAL on all three spheres, monodromy = the pure phase <math>i</math>, <math>\det(A - 1) = -4</math>: no invariant flux vector — <math>k_1 = k_2 = k_3</math> IS A THEOREM; NEGATIVE PROBE (new falsifier): the clock-forbidden family <math>Z^4 - Z</math> (disc <math>-27</math>, branch points <math>\{0, 1, \omega, \omega^2\}</math> not a <math>\mu_4</math> orbit): 0/24 orderings lockstep; HONEST LIMIT: uniform flux doubling keeps the lockstep — EQUALITY, not the value (1 from the current/sector counters and the minimal quantum <math>N = 1</math>). [E] NEGATIVE CONTROLS: <math>k = 2</math> (count 0, residual 360, prefactor <math>-31/48</math>, <math>\#\text{prim}</math> 3 — the lockstep dial honestly does NOT fire); <math>SO(16)</math> (fluxes (0, 60, 0): spheres flux-dark, condensation stops at <math>4 = \det \text{Cartan}(D_8)</math>); <math>A_2</math> (<math>\mu = 12</math> not a perfect square: NO Lagrangian glue, Coxeter <math>3 \neq 4</math>). [C] the CPS dictionary itself on <math>\mathbb{P}T/\mathbb{Z}_4</math> (their geometry is the <math>\mathbb{C}^2</math> blow-up with ONE <math>(-1)</math>-sphere and <math>\mathfrak{so}(8)</math> at negative level — not the <math>A_3</math> ALE). [O] the type-I B-model backreaction on <math>\mathbb{P}T/\mathbb{Z}_4</math> (the <math>A_3</math> <math>\Omega_N</math>) — the same fence as v502/v505; SEAM.EQUIV.01 untouched; no marker moves. Exact (sympy + Fraction). [E][C][O]</p> |

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| Script                   | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| v510_seam_bit_freedom.py | <p>SEAM.BIT.FREEDOM.01: the freedom attack on the v506/v507 alignment bit — the POSITION half of the bit is topology (Part A dihedral/Fock + Part B Moebius/RP in one module; NO marker moves — the modulus datum stays [C]). Exact sympy + exact Cl(16) Fractions, no floats. [E] DIHEDRAL THEOREM: on the <math>N</math>-site seam circle the free involution is UNIQUE = the antipode (<math>N = 16</math>: 17 involutions in <math>D_{16}</math>, exactly 1 free = shift 8; all 16 reflections have exactly 2 circle fixed points; robustness <math>N = 8, 12</math>); the census is COMPLETE: <math>\text{Aut}(C_{16}) = D_{16}</math> exactly (brute force over all <math>8!</math> permutations at <math>N = 8</math>, propagation at <math>N = 16</math>); <math>N = 9</math> control: ZERO free involutions (even carrier count, consistent with <math>16 = 2^{g_{\text{car}}-1}</math>); QUOTIENT TOPOLOGY (the P1 tie): circle/antipode = CIRCLE (<math>\chi = 0</math>, closed), circle/reflection = INTERVAL (<math>\chi = 1</math>, two corners) — the <math>6\pi = 2\pi + 2\pi + 2\pi</math> Gauss–Bonnet budget counts three CLOSED circles, so “the deck is free on the seam circle” IS the established P1 datum “the seam is a closed circle”, not a new premise. [E] NONSPLIT <math>\iff</math> FREE (complete census, exact Cl(16)): ALL 17 involutions implemented — the free one’s NS lift squares to <math>256 \gamma_1 \cdots \gamma_{16} = (-1)^F</math> (nonsplit, 2 roots = the <math>\mathbb{Z}_8</math> clock tower); ALL 16 reflections have <math>U^2 = \text{SCALAR}</math> (<math>2^7</math> site-, <math>2^8</math> edge-axis; split, 0 roots); chain: free <math>\Rightarrow</math> shift 8 <math>\Rightarrow</math> crossing (all 28 invariant 4-sets) <math>\Rightarrow</math> central class <math>\Rightarrow</math> the edge/silver arrangement is EXCLUDED; Klein control (covering decks are free regardless of the gluing); branch control: the pillowcase involution has exactly 4 fixed points = the marks and restricts to a REFLECTION on invariant circles — the edge class IS the BRANCHED involution’s restriction: silver mistakes the mark-creating branch <math>\mathbb{Z}_2</math> for the seam deck. [E] REAL-STRUCTURE CLASSIFICATION: the deck-commuting antiholomorphic involutions of <math>\mathbb{P}^1</math> are EXACTLY two families — <math>\mu\bar{z}</math>, <math> \mu  = 1</math> (fix LINE through the deck poles <math>\{0, \infty\}</math>: deck NOT free) and <math>\mu/\bar{z}</math>, <math>\mu &gt; 0</math> (fix circle <math> z  = \sqrt{\mu}</math> avoiding the poles: deck FREE; <math>\mu &lt; 0</math>: empty); per configuration the mark-fixing real structure is UNIQUE; table: <math>\mu_4 \rightarrow</math> equatorial <math>1/\bar{z}</math>, FREE (v506 A-S2.9); silver <math>\{\pm 1, \pm(3+2\sqrt{2})\} \rightarrow</math> real axis THROUGH the poles, NOT free; generic-unit <math>\{\pm 1, \pm(3+4i)/5\} \rightarrow</math> equatorial, free WITHOUT harmonicity; hexagonal <math>\rightarrow</math> NO mark-fixing real structure (marks not concyclic: RP-incompatible, consistent v503); one-sidedness TYPES the <math>\mathbb{Z}_2</math> as the free antipode (quotient <math>\mathbb{RP}^2 =</math> the <math> \mathbb{Z}_2 </math> of <math>c_3 = 1/( \mathbb{Z}_2  2\pi\chi)</math>, v456). [E] HARMONIC + FREE <math>\Rightarrow \mu_4</math> EXACTLY: the deck-pair cross-ratio <math>(1-b)^2/(1+b)^2 = -1</math> solves exactly to <math>b = \pm i</math> — the silver family (harmonic AND edge) is EMPTY on free circles, harmonicity becomes SUFFICIENT. [E] HONEST COUNTERWITNESSES: “nonsplit <math>\Rightarrow</math> clock” is FALSE — discrete <math>\{0, 1, 8, 9\}</math> and continuous <math>\{\pm 1, \pm(3+4i)/5\}</math> (free, central, <math>j = 148176/25 \neq 1728</math>, no clock): the Fock dichotomy measures the CLASS, not the modulus. PREMISE TYPING: (P-i) deck = Moebius covering deck <math>\Rightarrow</math> free [standard topology; origin_theory/v456]; (P-ii) marks on the <math>\Theta</math>-fix circle = existing REALIZE [C] content — no new input. [C] THE REDUCED REST (verbatim): the alignment bit reduces from “harmonic AND central” to the square-modulus datum <math>\tau = i</math> alone; the position half is topology (edge class excluded), the modulus half stays the one carrier input. Exact (sympy + Cl(16) Fractions). [E][C]</p> |

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| Script                                              | What it machine-checks ( <i>continued</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| v511_celestial_<br>wp5e_delta2_tensor_<br>ledger.py | <p>CELEST.WP5E.DELTA2.01: WP5e-<math>\delta_2</math> of the celestial-holographic research contract CELEST.SEAM.01 — “the full-tensor ledger”, the named escape route of the v508 exchange no-go, executed (intermediate verdict, exact): does the FULL adjoint tensor structure of <math>\mathfrak{e}_8</math> (non-Cartan external legs, higher-arity vertices) open a sphere-axion channel to the rigid <math>32 T_3</math> residual (<math>A_{\text{fix}} = (9, -30, -15, 0, 32)</math>, v505)? The collapse is CONFIRMED full-tensorially, but one cubic door opens. Exact Kostant/Weyl weight arithmetic + sympy/Fraction. [E] STRUCTURE: <math>\mathfrak{g}_0 = \mathfrak{d}_5 \oplus \mathfrak{a}_3</math> is SEMISIMPLE with NO <math>\mathfrak{u}(1)</math> factor (class-0 roots <math>40 + 12</math>, root span rank 8); the twisted sectors factorise multiplicity-free into minuscule Weyl orbits <math>\mathfrak{g}_1 = (16_s, 4)</math>, <math>\mathfrak{g}_2 = (10, 6)</math>, <math>\mathfrak{g}_3 = (\mathfrak{g}_1)^*</math>, each irreducible <math>(64, 60, 64)</math>; INNERNESS THEOREM: <math>h = (2, 2, 2, 2, 2; 0^4)</math> lies in the Cartan OF <math>\mathfrak{g}_0</math>, so every <math>\mathfrak{g}_0</math>-invariant tensor carries total sector charge 0 mod 4 — the v508 collapse is a full-tensor statement AND holds across arities (all 15 non-neutral sector triples: Hom = 0). [E] HOM TABLES: bilinear <math>\dim \text{Hom}_{\mathfrak{g}_0}(\mathfrak{g}_j \otimes \mathfrak{g}_{j'})</math> nonzero ONLY at <math>j' = -j</math> (dims <math>2/1/1/1</math>, Killing blocks); trilinear neutral multisets <math>\{0, 0, 0\}: 3</math> (<math>= 1 \text{ sym} + 2 \text{ antisym}</math>), <math>\{0, 1, 3\}: 2</math>, <math>\{0, 2, 2\}: 2</math>, <math>\{1, 1, 2\}: 1</math>, <math>\{2, 3, 3\}: 1</math> — ALL cross-sector trilinears are pure brackets (antisymmetric in the repeated legs); the ONLY totally symmetric survivor on all of <math>\mathfrak{e}_8</math> is the <math>\mathfrak{su}(4)</math> <math>d</math>-symbol (<math>\mathfrak{so}(10)</math> has no cubic Casimir: <math>\text{Inv}(S^3 45) = 0</math> protects <math>T_5</math>). [E] THE NEW CHANNEL: the quartic projection of the <math>d</math>-symbol exchange is <math>Q_{dd} = (0, 0, -\frac{1}{240}, 0, \frac{1}{60}) = \frac{1}{60}(T_3 - P_3/4)</math>, computed by TWO independent routes (Killing-propagator contraction with <math>2h^\vee = 60</math> verified from the roots; hand route <math> x^2 - (\text{tr } x^2/4)\mathbf{1} ^2/60</math>); <math>\Phi_{T_3}(Q_{dd}) = \frac{1}{60} \neq 0</math> — the v508 master kill does NOT extend to cubic vertices; brackets, tadpoles and mixed <math>d \times f</math> die exactly; no <math>T_5</math> analogue. [E] NEW CERTIFICATE: <math>M = [E_{13}, E_{22}, E_{00}, Q_{dd}]</math> has rank 3, augmented rank 4 — the charge reading remains OBSTRUCTED with the WEAKER certificate <math>\{\Phi_{T_5}, \psi = \Phi_P - \Phi_{T_3}/4\}</math>, <math>\psi(A_{\text{fix}}) = 72 - 8 = 64</math>; the RELAXED reading becomes EXACTLY solvable: <math>A_{\text{fix}} = -u + 8v + 2w + 1920 Q_{dd}</math> (residual 0). [C] number fences: <math>64 = \dim \mathfrak{g}_1 = 2^6</math>; <math>1920 =  W(D_5)  = 2^7 \cdot 3 \cdot 5 = 8 \cdot 240</math>. [O] the arity <math>\geq 4</math> ledger; the physical justification of the cubic GS term (<math>c_d = 1920</math>); <math>\text{ch}_2</math>-naturalness for CUBIC vertices UNDECIDED; the burden on <math>\delta_1</math> is SHARPENED to the <math>\psi = 64</math> slice or the selection-rule relaxation. [E] NEGATIVE CONTROLS: Killing control = v508 replication; <math>SO(16)/D_8</math>: bilinears <math>(1, 0, 1)</math>, trilinears <math>(1, 0, 1, 0)</math>, <math>\text{Inv}(S^3 \text{adj}_{D_8}) = 0</math> — no symmetric cubic, no odd sectors, no <math>A_3</math> block (<math>E_8/\mu_4</math> doubly special); false <math>\mathfrak{g}_0 = \mathfrak{d}_5 \oplus \mathfrak{a}_2 \oplus \mathfrak{u}(1)</math>: <math>(5, 2, 2) \neq (2, 1, 1)</math> and <math>\text{Inv}(S^3 \mathfrak{g}_0) = 6 \neq 1</math>; false sector content kills every pairing. SEAM.EQUIV.01 stays [O]; NO marker moves. Exact (sympy + Fraction). [E][C][O]</p> |

**How to run.** From the repository root, `python verification/run_all.py` (dependencies `mpmath`, `numpy`, `sympy`); each module also runs standalone, e.g. `python verification/v1_e8_glue.py`. The runner exits with code 0 iff all checks pass; see `verification/README.md` for the full claim $\leftrightarrow$ script map and `verification/status_ledger.csv` for the **single status ledger** (claim ID, status, location, dependencies, script, external datum — the source of truth all nine documents are written against). The [E] (Lean-formalised) carrier material lives in the canonical shipped folder `lean4-carrier-rigidity/` (at the package root; the source repository keeps the same project under `experiments/`).



## TFPT verification suite: the journey of ~475 machine-checked scripts

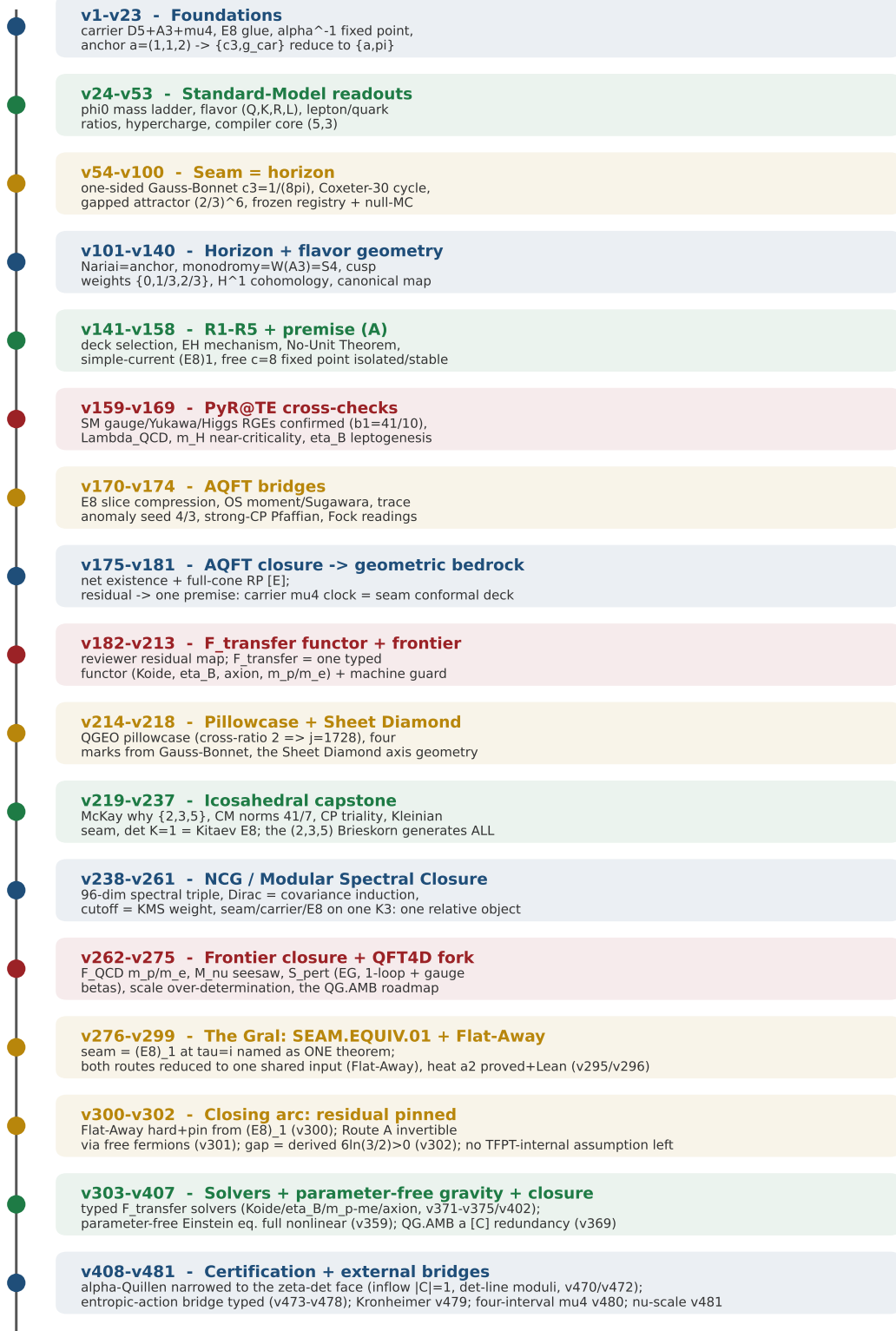


Figure 1: The development timeline of the verification suite — the phases of the journey and what each did, mathematically and physically: from the foundations (carrier,  $E_8$  glue,  $\alpha^{-1}$ , the anchor  $a = (1, 1, 2)$  to which  $\{c_3, g_{car}\}$  reduce) through the Standard-Model readouts, the seam=horizon geometry, the horizon/flavor geometry, the R1–R5 and premise-(A) reductions, the external PyR@TE RGE cross-checks and the AQFT bridges, the AQFT closure that drives the whole structural residual to one geometric bedrock premise, the  $F_{transfer}$  functor and frontier interfaces, the QGEO pillowcase and Sheet Diamond, the *icosahedral capstone* (the single  $(2, 3, 5)$  Brieskorn singularity generates the whole skeleton), the Modular Spectral Closure, the SEAM.EQUIV.01 closing arc (v276–v302), the typed solvers and the parameter-free gravity round (v303–v407), to the certification round (v408–v481: the  $\alpha$ -Quillen residual narrowed to the continuum  $\zeta$ -det face, the entropic-action bridge typed, the Kronheimer quiver bridge, the four-interval  $\mu_4$  geometry). The

## Appendix: changelog

The dated record of every change (canonical source: `changelog.tex`, also compiled standalone).  
Module numbers `vN` refer to `verification/vN_*.py`.

**2026-07-21 (XXIII) (v511 — WP5e- $\delta_2$  of CELEST.SEAM.01 “the full-tensor ledger” promoted as CELEST.WP5E.DELTA2.01 (intermediate verdict, exact):** the named escape route of the v508 exchange no-go executed on the FULL adjoint tensor structure of  $\mathfrak{e}_8$  — the COLLAPSE IS CONFIRMED full-tensorially AND arity-crossing (the INNERNESS THEOREM:  $\mathfrak{g}_0 = \mathfrak{d}_5 \oplus \mathfrak{a}_3$  semisimple with NO  $\mathfrak{u}(1)$  (class-0 roots 40+12, root span rank 8),  $h$  in the Cartan OF  $\mathfrak{g}_0 \Rightarrow$  every invariant tensor carries total charge 0 mod 4: the bilinear Hom table is nonzero ONLY at  $j' = -j$  (dims 2/1/1/1) and ALL 15 non-neutral trilinear sector triples have Hom = 0), BUT ONE CUBIC DOOR OPENS: of the neutral trilinears ( $\{0, 0, 0\}: 3 = 1$  sym + 2 antisym,  $\{0, 1, 3\}: 2$ ,  $\{0, 2, 2\}: 2$ ,  $\{1, 1, 2\}: 1$ ,  $\{2, 3, 3\}: 1$ ; all cross-sector ones pure brackets) the UNIQUE totally symmetric survivor is the  $\mathfrak{su}(4)$   $d$ -symbol ( $\mathfrak{so}(10)$  has no cubic Casimir,  $\text{Inv}(S^3 45) = 0$  protects  $T_5$ ), whose exchange quartic  $Q_{dd} = (0, 0, -\frac{1}{240}, 0, \frac{1}{60}) = \frac{1}{60}(T_3 - P_3/4)$  (TWO independent routes, Killing propagator  $2h^\vee = 60$  from the roots) has  $\Phi_{T_3}(Q_{dd}) = \frac{1}{60} \neq 0$  — the v508 master kill (product theorem) does NOT extend to cubic vertices; the pairing stays OBSTRUCTED in the charge reading with the genuinely WEAKER certificate  $\{\Phi_{T_5}, \psi = \Phi_P - \Phi_{T_3}/4\}$ ,  $\psi(A_{\text{fix}}) = 72 - 8 = 64$  ( $M = [E_{13}, E_{22}, E_{00}, Q_{dd}]$  rank 3, augmented 4), and becomes EXACTLY solvable relaxed:  $A_{\text{fix}} = -u + 8v + 2w + 1920 Q_{dd}$  (residual 0); [C] fences  $64 = \dim \mathfrak{g}_1 = 2^6$ ,  $1920 = |W(D_5)| = 8 \cdot 240$ ; controls:  $SO(16)/D_8$  has NO symmetric cubic and no  $A_3$  block ( $E_8/\mu_4$  doubly special), false  $\mathfrak{g}_0$ /sector assignments separate; the  $\delta_1$  burden SHARPENED to the  $\psi = 64$  slice or the selection-rule relaxation — and the  $\delta_1$  exploration run itself stands UNDECIDED at exploration level (strict holomorphic  $q^0$  reading REFUTED at the  $\mathbb{Z}_2$ /Eguchi–Hanson anchor = method boundary, not a kill; contact term = the MODULAR COMPLETION of the AB data, a Harvey–Moore  $\tau$ -integral with forced leading  $(T_5, T_3)$  ratio 4:3; massive coset levels  $n \geq 2$  mandatory); Wolfram extension  $463 \rightarrow 470$  (verified engine run); *NO marker moves — WP5e proper stays [O], SEAM.EQUIV.01 stays [O]*

- `v511_celestial_wp5e_delta2_tensor_ledger.py` (CELEST.WP5E.DELTA2.01, 41 checks,  $\sim 4$  s; integer-scaled weight multisets + Kostant/Weyl alternating sums + sympy/Fraction, no floats; deterministic). D0/D1 structure: 240 roots, class split (52, 64, 60, 64);  $\mathfrak{g}_0 = \mathfrak{d}_5 \oplus \mathfrak{a}_3$  SEMISIMPLE without  $\mathfrak{u}(1)$  (40+12 class-0 roots, no mixed root, root span rank 8); sectors  $\mathfrak{g}_1 = (16_s, 4)$ ,  $\mathfrak{g}_2 = (10, 6)$ ,  $\mathfrak{g}_3 = (\mathfrak{g}_1)^*$  as multiplicity-free minuscule Weyl-orbit products, each irreducible (64/60/64); the grading is INNER ( $h = (2, 2, 2, 2, 2; 0^4)$  in the  $\mathfrak{g}_0$  Cartan). D3/D4 Hom tables: bilinear nonzero ONLY at  $j' = -j$  (dims 2/1/1/1; all 12 charged pairs zero — the inner-torus theorem at arity 2); trilinear: ALL 15 non-neutral triples Hom = 0 (arity 3), neutral dims (3, 2, 2, 1, 1) with all cross-sector trilinears ANTISYMMETRIC in their repeated legs — the unique symmetric survivor is the  $\mathfrak{su}(4)$   $d$ -symbol ( $\text{Inv}(S^3 45) = 0$ ,  $\text{Inv}(S^3 \mathfrak{a}_3) = 1$ ). D5 the new channel:  $Q_{dd} = \frac{1}{60}(T_3 - P_3/4)$  by Killing-propagator contraction AND the hand route  $|x^2 - (\text{tr } x^2/4)\mathbf{1}|^2/60$ ;  $\Phi_{T_3}(Q_{dd}) = \frac{1}{60} \neq 0$ ; brackets/tadpoles/ $d \times f$  die exactly; no  $T_5$  analogue. D6 decision: strict reading unreachable (rank 2 vs 3); generous reading STILL unsolvable (rank 3 vs 4) with the weaker certificate  $\psi(A_{\text{fix}}) = 64$ ; relaxed reading EXACTLY solvable with  $c_d = 1920 = 32 \times 60$ . D7 controls: Killing control = v508 replication ( $A_{\text{fix}}, E_{13}, E_{22}, E_{00} = 225E_{22}$ );  $SO(16)/D_8$  bilinears (1, 0, 1), trilinears (1, 0, 1, 0),  $\text{Inv}(S^3 \text{adj}_{D_8}) = 0$ ; false  $\mathfrak{g}_0 = \mathfrak{d}_5 + \mathfrak{a}_2 + \mathfrak{u}(1)$ :  $(5, 2, 2) \neq (2, 1, 1)$ ,  $\text{Inv}(S^3 \mathfrak{g}_0) = 6 \neq 1$ ; false sector content  $(16_s, 4)$  kills every pairing. Discovery provenance: `experiments/tfpt-discovery/celestial_seam_wp5e_delta2_full_tensor_ledger_probe.py`

(41/41), same day.

- **The  $\delta_1$  fence (exploration-level, NOT promoted — verdict UNDECIDED).** The twisted BCOV contact-term probe (`celestial_seam_wp5e_delta1_bcov_contact_probe.py`, 27/27, stays sandbox; v-numbered on demand) contributes its four exact sharpenings as fence text: (i) the strict holomorphic  $q^0$  reading is REFUTED at the  $\mathbb{Z}_2$ /Eguchi–Hanson anchor (method boundary, not a kill); (ii) the forced leading  $(T_5, T_3)$  ratio is 4:3 (sector 2 : sectors 1+3); (iii) the  $SL(2, \mathbb{Z})$  orbit decomposition  $15 = 12 + 3$  identifies the contact term as the MODULAR COMPLETION of the AB data (a Harvey–Moore-type  $\tau$ -integral), not a free number; (iv) massive coset levels  $n \geq 2$  are mandatory. The remaining  $\delta_1$  target is the  $\tau$ -integral, evaluated against the  $\delta_2$ -sharpened criterion ( $\psi = 64$  slice or selection-rule relaxation).
- **Ledger + papers + Wolfram + website.** Ledger: new row CELEST.WP5E.DELTA2.01 ([E] Hom tables/innerness/ $Q_{dd}$ /certificate; [C] the number fences; [O] arity  $\geq 4$ , cubic GS term,  $\delta_1$ ); CELEST.SEAM.01 updated (delta-2 executed via v511 — the full-tensor collapse is confirmed (innerness), but the cubic  $d$ -symbol channel opens  $T_3$  with the weaker certificate  $\psi = 64$ ; delta-1 stands UNDECIDED at exploration level). Papers: `tfpt_research_contracts` WP5e bullet gains the executed  $\delta_2$  stage + the  $\delta_1$  exploration fence, WP5e-proper targets renamed ( $\tau$ -integral, cubic GS term,  $\varepsilon_1$ , back-reaction); `tfpt_3` celestial section extended to the FOURTEEN-step story (new Step 14 “the full-tensor ledger: the collapse is real, and one cubic door opens”) with the “What remains” block updated. Wolfram: seven mirrors (innerness count +  $\mathfrak{g}_0$  structure; the bilinear and trilinear Hom tables by exact Kostant alternating sums; the unique symmetric survivor; the  $SO(16)$ /false- $\mathfrak{g}_0$ / false-sector controls;  $Q_{dd}$  by two routes; the  $\psi$  certificate + the relaxed solution  $(-1, 8, 2; 1920)$ ) verified with the active engine the same day (463  $\rightarrow$  470, ALL WOLFRAM EXTENSION CHECKS PASSED). Registry + `run_all` (505 modules, suite green: ALL CHECKS PASSED). No gate closes, no marker moves — WP5e proper stays [O].

**2026-07-21 (XXII) (P2 consolidation — the axiomatics reduction state written as ONE coherent statement across the papers (editorial, NO new claims, NO marker moves):** the compiler’s dimensionless inputs reduce to *four marks* (derived from P1-side topology, v216), *ONE square-modulus datum* ( $\tau = i$  — equivalently: the clock / the CM-fixed half-period / the non-split extension class on the free seam circle; v491/v499/v506/v507/v510) and  $\pi$ , plus the [C] identifications R1/R2 of v499; AX.P2.01 stays a DECLARED axiom and QGEO.REALIZE.01 stays [C]/[O] — the consolidation describes the *reduction state*, not an abolition)

- **What is consolidated (and why now).** The P2 state of affairs was spread over five module rounds: the partition corollary (v491:  $4 = 2+1+1$  unique  $\Rightarrow g_{\text{car}} = e_2 = 5$ ), the weight typing (v499: Deligne + Birkhoff–Grothendieck + Mehta–Seshadri force  $E = \mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$ , residuals R1/R2 [C]), the residual convergence (v503: QGEO-R1  $\equiv$  P2-R1), the clock rigidity and bit origin (v506/v507: one alignment bit, order fermionically forced, deck class derived) and the freedom theorems (v510: the position half is topology, the edge class excluded). The honest consolidated statement, now in one place: *the compiler’s dimensionless inputs reduce to four marks (derived from P1-side topology), one square-modulus datum ( $\tau = i$ , equivalently: the clock / the CM-fixed half-period / the non-split extension class on the free seam circle), and  $\pi$  — plus the [C] identifications R1/R2 of v499.* No new claims; conservative wording; no “axiom abolished” language anywhere.
- **Papers + ledger + website.** `tfpt_1_architecture_e8`: new keybox “The current reduction state of P2” next to the foundational-postulates keybox, with v491/v499/v506/v507/v510 citations. `tfpt_3`: mirror bullet “The current reduction state” in the honest-status section. **introduction**: the anchor paragraph gains the consolidated one-sentence state and the status card row  $a = (1, 1, 2)$  now names the reduction state. Ledger: SYNTHESIS NOTE appended to AX.P2.01 (“reduction state 2026-07-21: four marks + one square-modulus datum ( $\tau = i$ ) +

$\pi$ ”), marker *Axiom unchanged*. Website: `papers.ts` mirrors of the changed sections. No gate closes, no marker moves.

**2026-07-21 (XXI) (v510 — the freedom attack on the v506/v507 alignment bit promoted as SEAM.BIT.FREEDOM.01: the POSITION half of the bit is TOPOLOGY — the collar deck is a covering deck (the  $\mathbb{Z}_2$  seam of the Möbius double), and covering decks act FREELY; the DIHEDRAL THEOREM (on the seam circle the free involution is UNIQUE = the antipode: 17 involutions in  $D_{16}$ , exactly 1 free;  $\text{Aut}(C_{16}) = D_{16}$  complete census, brute force  $8!$  at  $N=8$ ; quotient topology circle/antipode = CIRCLE  $\chi=0$  vs circle/reflection = INTERVAL  $\chi=1$  — the  $6\pi$  Gauss–Bonnet budget needs closed circles, so freeness IS the established P1 datum); NONSPLIT  $\iff$  FREE over the complete 17-involution census in exact  $\text{Cl}(16)$  (free  $\Rightarrow \tilde{U}^2 = 256 \gamma_1 \cdots \gamma_{16} = (-1)^F$  nonsplit with 2 roots = the  $\mathbb{Z}_8$  tower; ALL 16 reflections  $\Rightarrow U^2 = \text{scalar } \{2^7, 2^8\}$ , split, 0 roots); the REAL-STRUCTURE classification (deck-commuting families = exactly two;  $\mu_4 \rightarrow$  equatorial, deck FREE; silver  $\rightarrow$  real axis THROUGH the deck poles, NOT free — the edge class is the restriction of the BRANCHED pillowcase involution; hexagonal  $\rightarrow$  NO mark-fixing real structure, RP-incompatible); HARMONIC + FREE  $\Rightarrow \mu_4$  exactly ( $b = \pm i$ ); honest counterwitnesses (“nonsplit  $\Rightarrow$  clock” is FALSE:  $\{0, 1, 8, 9\}$  and  $\{\pm 1, \pm(3+4i)/5\}$  with  $j = 148176/25 \neq 1728$ ); the alignment bit REDUCES from “harmonic AND central” to the square-modulus datum  $\tau = i$  alone; Wolfram extension 457  $\rightarrow$  463 (verified engine run); NO marker moves — QGEO.REALIZE.01 stays [C]/[O], AX.P2.01 stays an axiom, P2-R1 halved)**

- **v510\_seam\_bit\_freedom.py (SEAM.BIT.FREEDOM.01, 30 checks,  $< 5$  s; sympy + exact  $\text{Cl}(16)$  Fractions, no floats; deterministic).** Part A (dihedral/fermionic): the involution census in  $D_N$  on the refined  $2N$ -point seam circle ( $N = 16$ : 17 involutions, EXACTLY ONE free = the antipodal shift by 8; all 16 reflections have exactly 2 circle fixed points — sites for even axis, edge midpoints for odd; robustness  $N = 8, 12$ ; odd control  $N = 9$ : ZERO free — an even carrier count is needed, consistent with  $16 = 2^{g_{\text{car}}-1}$ ); the census is COMPLETE because  $\text{Aut}(C_{16}) = D_{16}$  exactly (brute force over all  $8! = 40320$  permutations at  $N = 8$ , propagation at  $N = 16$ ); quotient topology: circle/antipode = CIRCLE ( $\chi = 0$ , closed), circle/reflection = INTERVAL ( $\chi = 1$ , two boundary corners) — a fixed-point deck would break the  $6\pi = 3 \times 2\pi$  closed-circle count, so “deck free on the seam circle” is exactly the established “the seam is a closed circle”, not a new premise; the complete Fock census: ALL 17 involutions implemented in exact  $\text{Cl}(16)$  — NONSPLIT  $\iff$  FREE with no exception, root dichotomy 2 vs 0; the chain: free  $\Rightarrow$  shift 8  $\Rightarrow$  crossing (all 28 invariant 4-sets)  $\Rightarrow U^2 = (-1)^F \Rightarrow$  central class  $\Rightarrow$  the edge/silver arrangement is EXCLUDED; Klein control (covering decks are free regardless of the gluing); branch control (the pillowcase involution has exactly 4 fixed points = the marks and restricts to a REFLECTION: the silver arrangement mistakes the mark-creating branch  $\mathbb{Z}_2$  for the seam deck). Part B (Möbius/RP): the cylinder-double-cover deck is free on all 120 model points and restricts to the antipode on the  $\mathbb{Z}_2$  seam fibre; one-sidedness TYPES the  $\mathbb{Z}_2$  as the free antipode (quotient  $\mathbb{RP}^2 =$  the  $|\mathbb{Z}_2|$  of  $c_3 = 1/(|\mathbb{Z}_2|2\pi\chi)$ ); the deck-commuting real structures are EXACTLY two families ( $\mu\bar{z}$ : fix line through the poles, NOT free;  $\mu/\bar{z}$ ,  $\mu > 0$ : fix circle  $|z| = \sqrt{\mu}$ , FREE;  $\mu < 0$ : empty); the per-configuration table ( $\mu_4$  equatorial/free, silver real-axis/not free, generic-unit free without harmonicity, hexagonal RP-incompatible); harmonic + free solves EXACTLY to  $b = \pm i$  (on free circles the silver family “harmonic AND edge” is EMPTY — harmonicity becomes sufficient); honest counterwitnesses: the discrete  $\{0, 1, 8, 9\}$  and the continuous  $\{\pm 1, \pm(3+4i)/5\}$  (free, central,  $j = 148176/25$ , no clock) — the Fock dichotomy measures the CLASS, not the modulus. Premise typing: (P-i) deck = Möbius covering deck  $\Rightarrow$  free [standard topology; the one-sidedness strand `origin_theory/v456`]; (P-ii) marks on the  $\Theta$ -fix seam circle = existing



REALIZE [C] content (QGEO.REALIZE.02 hypotheses, v264) — NO new input. Audit finding documented: “circle-freeness” was nowhere typed before (only mark-freeness, v216); v507 F1.1 explicitly admitted the reflection reading of the edge class — v510 closes that door (a reflection is not a covering deck). The [C] rest, verbatim: *the alignment bit reduces from “harmonic AND central” to the square-modulus datum  $\tau = i$  alone; the position half is topology (edge class excluded), the modulus half stays the one carrier input.* Discovery provenance: experiments/tfpt-discovery/qgeo\_realize\_deck\_freedom\_dihedral\_probe.py (14/14) and qgeo\_realize\_rp\_fixcircle\_moebius\_probe.py (16/16), same day.

- **Ledger + papers + Wolfram + website.** Ledger: new row SEAM.BIT.FREEDOM.01 ([E] dihedral theorem + complete Fock census + real-structure classification + harmonic-free collapse + counterwitnesses; [C] the square-modulus datum); TEXT precisions (markers unchanged) in SEAM.CLOCK.RIGIDITY.01, SEAM.BIT.ORIGIN.01, QGEO.REALIZE.01 (bit  $\rightarrow$  square-modulus datum; edge class topologically excluded) and P2.TYPING.01 (R1 halved). Papers: tfpt\_3 celestial/QGEO passage extended (“the position half is topology”), tfpt\_research\_contracts QGEO kill-test + QGEO.MARKS.03 + celestial-contract passages extended, one-sentence precisions in introduction, tfpt\_1, tfpt\_2, origin\_theory where v506/v507 are cited. Wolfram: six mirrors (dihedral census + quotient topology; Aut completeness incl. the 8! brute force; the  $Cl(16)$  dichotomy census over all 17 involutions on  $256 \times 256$  Jordan–Wigner gammas; the real-structure table; harmonic + free  $\Rightarrow \pm i$ ; the counterwitnesses) verified with the active engine the same day (457  $\rightarrow$  463, ALL WOLFRAM EXTENSION CHECKS PASSED; base file re-verified 116/116). Registry + run\_all (504 modules, suite green: ALL CHECKS PASSED). No gate closes, no marker moves.

**2026-07-21 (XX) (v509 — WP5e- $\varepsilon_2$  of CELEST.SEAM.01 “the CPS level-from-flux dial” promoted as CELEST.WP5E.EPS2.01, verdict B: the Costello–Paquette–Sharma Burns-holography mechanism (“level = flux quantum  $\times$  Dynkin index”) executed on the lockstep spheres — the CPS SKELETON exact ( $S^3$  period  $(2\pi i)^2 N$  with pullback density  $-\sin 2\theta$ , exceptional-sphere Kähler flux  $2\pi N$ , boundary level magnitude  $2N$  by the symplectic-boson contraction,  $T(\text{vector}) = 1$ ); the PAIRING MATRIX pinned, not postulated (ch<sub>2</sub> ledger + integrality + unimodularity + effectivity  $\Rightarrow P_{mi} = \delta_{mi}$  by complete enumeration: rows 8/6/8, 48 unimodular, exactly 2 effective = identity + diagram flip), fluxes  $F_i = (64, 60, 64) = \dim \mathfrak{g}_i$  with uniformly ONE quantum per charged current; “level = total flux” KILLED by the lockstep test itself ( $F_i$  unequal = flavour multiplicity), the per-current reading (1, 1, 1) anchored by the lattice current count (240, 0, 0, 0) and embedding index 1; ord-4-vs-level-1 RESOLVED (the clock order counts fractional flux sectors:  $\mu = 16 = \text{ord}^2$ , Lagrangian  $\mu_4$  glue, KLM  $16 \rightarrow 1$ ) with the NEW sector-counter dial  $\#_{\text{prim}}((E_8)_k) = (1, 3, 5, 10, 15, 27)$  for  $k = 1..6$  — exactly ONE sector iff  $k = 1$ ; the LOCKSTEP THEOREM lifts the v505 “one scale  $\Rightarrow$  one level” to a THEOREM of clock invariance ( $|\Pi_j| = \sqrt{2}t$  equal, monodromy = pure phase  $i$ ,  $\det(A-1) = -4$ ; no invariant flux vector; NEW falsifier  $Z^4-Z$ : 0/24 orderings lockstep); honest limit: uniform doubling keeps the lockstep (equality, not the value); controls  $k=2/SO(16)/A_2$  separate honestly; Wolfram extension 450  $\rightarrow$  457 (verified engine run); NO marker moves — WP5e stays [O] (the CPS dictionary on  $\mathbb{PT}/\mathbb{Z}_4$  typed [C], the type-I B-model back-reaction [O]), SEAM.EQUIV.01 stays [O])**

- v509\_celestial\_wp5e\_eps2\_level\_from\_flux.py (CELEST.WP5E.EPS2.01, 28 checks,  $< 1$  s; sympy/Fraction exact, no floats; deterministic). Target: derive the boundary level  $k = 1$  from flux quantisation on the lockstep spheres, following the Costello–Paquette–Sharma Burns-holography mechanism (PRL 130, 061602; arXiv:2306.00940). CPS skeleton replicated exactly: the  $S^3$  period of the brane back-reaction 3-form is  $(2\pi i)^2 N$  (their eq. 3.33; large-gauge invariance quantises  $N$ ), the exceptional-sphere Kähler flux is  $2\pi N$  exactly ( $K = |u|^2 + N \log |u|^2$ ,



the flat part dies at  $t \rightarrow 0$ ), and the boundary level magnitude is  $2N$  via the symplectic-boson double contraction ( $N = 1, 2, 3$ ; their eq. 9.4), Dynkin normalisation  $T(\mathfrak{so}_8 \text{ vector}) = 1$ ,  $T(\text{adj}) = 6$ . *Pairing matrix pinned*: the v505 ch2 ledger ( $f(m) = -(C^{-1})_{mm}/2 = (-\frac{3}{8}, -\frac{1}{2}, -\frac{3}{8})$ ) + integrality + unimodularity + effectivity force  $P_{mi} = c_1(T_m) \cdot [\Sigma_i] = \delta_{mi}$  by COMPLETE enumeration (row candidates 8/6/8; 48 unimodular solutions of  $PC^{-1}P^T = C^{-1}$ ; with  $P \geq 0$  exactly 2: identity +  $A_3$  diagram flip); fluxes  $F_i = (64, 60, 64) = \dim \mathfrak{g}_i$ , total  $188 = 248 - 60$ , flip-invariant; flux per charged root current uniformly ONE quantum through exactly one sphere. *The level formula, honest*: “level = total open-string flux” is KILLED by the lockstep test itself ( $k_i = (64, 60, 64)$  unequal —  $F_i$  is the FLAVOUR multiplicity, the brane matter content); what works:  $k = \text{flux per adjoint current} = F_i / \dim \mathfrak{g}_i = (1, 1, 1)$ , anchored non-circularly by the lattice current count  $(240, 0, 0, 0)$  for  $k = 1.4$  (v505 S4.8) and the embedding index  $j(D_5 \subset E_8) = j(A_3 \subset E_8) = 1$  ( $60 = 2h^\vee$ ). *Ord-4-vs-level-1 resolved*: the clock order counts FRACTIONAL flux sectors, not the level ( $\mu = \det(D_5)\det(A_3) = 16 = \text{ord}^2$ ; the  $\mu_4$  glue diagonal isotropic + Lagrangian; KLM  $16 \rightarrow 16/4^2 = 1$ , zero residual fractional flux); NEW DIAL:  $\#\text{primaries}((E_8)_k) = (1, 3, 5, 10, 15, 27)$  for  $k = 1.6$  from the affine marks — exactly ONE sector iff  $k = 1$  (independent of the CPS transplantation). *The lockstep theorem* (the lifting of v505 S4.10): clock invariance  $P(iZ) = P(Z)$  forces  $a_3 = a_2 = a_1 = 0$  sharply, the four branch points are ONE  $\mu_4$  orbit,  $|\Pi_j| = \sqrt{2}t$  EQUAL on all three spheres, the monodromy is the pure phase  $i$  and  $\det(A-1) = -4 \neq 0$  (no invariant flux vector) —  $k_1 = k_2 = k_3$  IS A THEOREM of clock invariance; NEGATIVE PROBE (new falsifier): the clock-forbidden family  $Z^4 - Z$  (smooth, disc  $-27$ , branch points  $\{0, 1, \omega, \omega^2\}$  not a  $\mu_4$  orbit): over ALL 24 orderings ZERO chains with three equal period moduli ( $1$  vs  $\sqrt{3}$ ); HONEST LIMIT: uniform flux doubling keeps the lockstep — the theorem proves EQUALITY, not the value ( $1$  from the counters and the CPS minimal quantum  $N = 1$ ). *Negative controls*:  $k = 2$  (current count 0, residual 360, prefactor  $-31/48$ ,  $\#\text{prim} 3$  — while the lockstep dial honestly does NOT fire);  $SO(16)$  (fluxes  $(0, 60, 0)$ : spheres  $1/3$  FLUX-DARK, defect  $-30 \neq -78$ , condensation stuck at  $4 = \det \text{Cartan}(D_8)$ );  $A_2$  ( $\det 3 \neq 4$ , glue diagonal  $23/24 \notin \mathbb{Z}$ ,  $\mu = 12$  no perfect square: NO Lagrangian subgroup, Coxeter  $3 \neq 4$ ). Typed honestly: the CPS dictionary itself on  $\mathbb{PT}/\mathbb{Z}_4$  stays [C] (their geometry is the  $\mathbb{C}^2$  blow-up with one  $(-1)$ -sphere and  $\mathfrak{so}(8)$  at negative level — not the  $A_3$  ALE); the type-I B-model back-reaction stays [O] (the same fence as v502/v505).

- **Ledger + papers + Wolfram + website.** Ledger: new row CELEST.WP5E.EPS2.01 ([E] CPS skeleton/pairing/lockstep/sector arithmetic + [C] CPS dictionary + [O] back-reaction); CELEST.SEAM.01 updated (“ $\varepsilon_2$  executed via v509: the one-level statement is now a THEOREM (lockstep from clock invariance, falsifier  $Z^4 - Z$ ), level = 1 carries the new sector-counter dial  $\#\text{prim}((E_8)_k) = 1 \Leftrightarrow k = 1$ ; the  $\mathbb{PT}/\mathbb{Z}_4$  back-reaction stays [O]”); the v505 “one scale  $\Rightarrow$  one level” sentence LIFTED to theorem status in ledger and prose (text move, no marker move). Papers: `tfpt_research_contracts` WP5e section extended by the  $\varepsilon_2$  block (done, verdict B; roadmap now  $\delta_1/\delta_2/\varepsilon_1$  + back-reaction); `tfpt_3` celestial section becomes a THIRTEEN-step story with Step 13 “one level is a theorem; the sector counter pins  $k = 1$ ” and the “What remains” paragraph updated. Wolfram: seven mirrors (the CPS period integrals; the  $2N$  contraction + Dynkin normalisation; the pairing enumeration  $48 \rightarrow 2$ ; the fluxes  $(64, 60, 64)$ ; the  $\#\text{prim}$  table  $(1, 3, 5, 10, 15, 27)$ ; the lockstep periods + the  $0/24$  negative probe; the  $SO(16)/A_2$  controls) verified with the active engine the same day ( $450 \rightarrow 457$ , ALL WOLFRAM EXTENSION CHECKS PASSED). VerificationDag: v509 on the QFT/celestial node; registry + `run_all` (503 modules, suite green). No gate closes, no marker moves — WP5e proper stays [O].

**2026-07-21 (XIX) (e8-ladder-bed v2 — the 2DCS extension executed with an honest preregistered kill\_2d verdict at the DETECTOR level: the known-answer map recovers 8/8 rungs incl.  $m_7/m_8$  via F3 difference coordinates (MC  $p = 2 \times 10^{-4}$ ), but the 2D control battery breaks (scramble 27%, jitter 18% at the destructive 30% level) — the v1 1D results (BCVO 8/8, CoNb<sub>2</sub>O<sub>6</sub> 6/8) are untouched; scorecard row updated, status/stage unchanged; *experiments level only — no verification module, no gate closes, no marker moves*)**

- **experiments/e8-ladder-bed v2 (2DCS): amendment preregistered, run, and killed by its own controls.** Trigger: arXiv:2512.16829 (Watanabe, Trebst, Hickey) — a THEORY-only paper (four-kink/ED localized-limit spectra at  $B_y = 0\text{--}3\text{ T}$ , not the E8 point; no measurements); as of 2026-07-21 *no experimental 2DCS measurement of CoNb<sub>2</sub>O<sub>6</sub> exists anywhere* (web-checked). amendment\_v2 was frozen in hypotheses/e8\_ladder\_bed\_v1.yaml BEFORE any 2D recovery run (detector, catalog F1/F2/F3, gates, null model, verdict criteria); amendment\_v2\_1 discloses a null-region correction (add `n_matched` to the false-alarm region) found on synthetic diagnostics, timing disclosed. Runs (results/results2d.json, deterministic): known-answer KA2D 8/8 rungs with  $m_7/m_8$  recovered ONLY via F3 difference coordinates ( $\chi^2/\text{dof} = 1.30$ , MC  $p = 2.0 \times 10^{-4}$ , 0/5000) — the mechanism that pulls  $m_7/m_8$  inside a low-pass window works; the NEG2D localized-limit map (the paper’s own non-E8 closed forms) is correctly rejected ( $p = 0.029$ ); offset and wrong-ladder controls rejected. BUT `scramble2d` fires at  $54/200 = 27\%$  and `jitter2d` shows no dose-response decay (29/21/17/18%; 18% at the destructive 30% level)  $\Rightarrow$  preregistered verdict `kill_2d`: permuting the detection axis preserves the pump-axis marginal (which carries the full 1D ladder) and the 120-entry catalog is dense enough that rung coverage follows from the marginal alone — the v2 detector does not test the 2D *pairing*; failure kills the detector, not TFPT. The CoNb<sub>2</sub>O<sub>6</sub> leg stays 6/8 ( $p = 0.0038$ ) per the preregistered `m7_m8_upgrade_rule`; a future v3 would need coincidence-based coverage and a sparser catalog, preregistered BEFORE real data exists.
- **Surfaces synced (scorecard row updated; counters unchanged, 120 rows).** Scorecard row “E8 mass-ladder blind recovery bed”: `data_value` extended (“CoNb<sub>2</sub>O<sub>6</sub> 6/8 unchanged; 2DCS extension killed at detector level (scramble 27%, jitter 18% @30%); KA 8/8 incl.  $m_7/m_8$  via F3,  $p=2\text{e-}4$ ; no experimental 2DCS data exists (arXiv:2512.16829 theory-only)”) and `status_note` extended (amendment v2/v2.1 preregistered+disclosed; real-data leg blocked; v3 criteria) via `build_evidence_scorecard.py`; status/stage UNCHANGED (`consistent/not_applicable`, `analog_positive_control` basket — the v1 detector validation stands). Experiment README v2 section, hypotheses YAML amendments, `experiments/next.txt` note; data-request plan on record (Armitage lab/JHU, Wang lab/Ames; 2D peak table at  $B \parallel b$ , 4.7–5.5 T, sub-Kelvin). No claim; firewall intact.

**2026-07-21 (XVIII) (v508 — WP5e- $\gamma$  of CELEST.SEAM.01 “the sphere-axion pairing check” promoted as CELEST.WP5E.GAMMA.01, an honest rigid NEGATIVE result: the VERTEX-SPACE THEOREM — the  $W(D_5) \times W(A_3)$ -invariant quadratics on the 8-dim glue Cartan are EXACTLY  $\text{span}\{s_5, s_3\}$  (dim 2) and the quartics EXACTLY  $\text{span}\{P_1, P_2, P_3, T_5, T_3\}$  (dim 5), with the PRODUCT THEOREM as the master kill (any two invariant quadratics multiply into  $\text{span}\{P_1, P_2, P_3\}$  — zero  $T_5/T_3$  content, so  $\Phi_{T_3}$  annihilates EVERY exchange image while  $\Phi_{T_3}(A_{\text{fix}}) = 32 \neq 0$ ); the strict two-index  $\mathbb{Z}_4$  rule COLLAPSES (the sphere axions have no invariant bilinear Cartan vertex at all), the twist-insertion channels give a rank-2 exchange matrix with  $\text{rank}([M | A_{\text{fix}}]) = 3$  and the annihilator certificate  $(\Phi_{T_5}, \Phi_{T_3}, \Phi_P)(A_{\text{fix}}) = (0, 32, 72)$ ; side discovery  $K^{(0)} = -15 K^{(2)}$ ; naturalness DISSOLVES (ch<sub>2</sub>-natural and AB-weight couplings both certify (0,0) vs required (32,72) — scale-independent); controls:  $SO(16)$  has no sphere partners AND uncanceled  $T_5, D_8$  no  $T_3$  structure; the O-fence is SHARPENED: the exchange sub-branch is dead, the 32  $T_3$  residual MUST be carried by twisted BCOV contact terms ( $\delta_1$  binary: coefficient exactly (9, -30, -15, 0, 32), computed not fitted; escape  $\delta_2$ : the full-tensor ledger); Wolfram extension 444  $\rightarrow$  450 (verified engine run); *NO marker moves — CELEST.SEAM.01 stays Open, the slot bijection untouched, the preregistered level-kill still does not fire, SEAM.EQUIV.01 stays [O]*)**

- **v508\_celestial\_wp5e\_gamma\_pairing.py** (CELEST.WP5E.GAMMA.01, 27 checks,  $\sim 4$  s; sympy/Fraction exact, no floats; deterministic). Question: can the three twisted sphere axions  $\eta_m$  (Coxeter eigencharacters  $\{i, -1, -i\}$  on  $H^2(\text{ALE})$ , v505 S5.8) cancel the rigid 32  $T_3$  residual of the Atiyah–Bott ledger ( $A_{\text{fix}} = Q^{(1)}/2 + Q^{(2)}/4 + Q^{(3)}/2 = 9P_1 - 30P_2 - 15P_3 + 32T_3$ , v505 S3.7) by Green–Schwarz-type EXCHANGE with sector-compatible quadratic vertices? *No, with a certificate. Vertex-space theorem:* exact Weyl nullspace arithmetic over all bidegrees gives quadratics dim 2 (bidegrees (1,0,1)) and quartics dim 5 (bidegrees (2,0,1,0,2)) — the vertex basis is COMPLETE; *product theorem* (the master kill):  $(as_5 + bs_3)(cs_5 + ds_3) = acP_1 + (ad + bc)P_2 + bdP_3$  identically — every exchange image (any axion count, any rule, any couplings) is annihilated by  $\Phi_{T_3}$ , and  $\Phi_{T_3}(A_{\text{fix}}) = 32 \neq 0$ . *Selection enumeration:* the strict two-index reading collapses ( $C_{j,j'} \neq 0$  only for  $j' = -j$ , forcing  $m = 0$ : 16 raw triples, 4 nonzero, all  $m = 0$ ); twist insertion:  $\eta_1 \leftrightarrow K^{(3)}$ ,  $\eta_2 \leftrightarrow K^{(2)}$ ,  $\eta_3 \leftrightarrow K^{(1)}$ ,  $\eta_0 \leftrightarrow K^{(0)}$ ; channels  $E_{13} = (16, -96, 144, 0, 0)$ ,  $E_{22} = (16, 32, 16, 0, 0)$ ,  $E_{00} = 225 E_{22}$ ; SIDE DISCOVERY:  $K^{(0)} = -15 K^{(2)}$  (the even-sector quadratics are PARALLEL — this collapse drives the  $\Phi_P$  obstruction). *Unsolvability with certificate (the core):*  $M$  is  $5 \times 2$ , rank 2, kernel 0;  $\text{rank}([M | A_{\text{fix}}]) = 3$ ; annihilator certificate  $(\Phi_{T_5}, \Phi_{T_3}, \Phi_P)(A_{\text{fix}}) = (0, 32, 72)$  — two independent obstructions (32  $T_3$  structural, rule-independent;  $\Phi_P = 72$  selection-rule-driven);  $\ker M = 0$  forces the unique non-disturbing coupling  $c = 0$  (the sphere axions are RIGIDLY SILENT);  $\Phi_P(Q^{(0)}) = 0$  (the sector-0 Okubo mechanism undisturbed); relaxed control: dropping charge conservation reaches the  $P$ -block ( $\det = -64$ ) but  $T_5 = T_3 = 0$  persists. *Naturalness resolution:* ch<sub>2</sub>-natural couplings (9/64, 1/4) give (25/4, -11/2, 97/4, 0, 0) and AB-weight couplings (1/2, 1/4) give (12, -40, 76, 0, 0) — BOTH with certificate (0,0) vs required (32,72): the failure is scale- and coupling-INDEPENDENT (no “unnatural factor”). *Negative controls:* invariant-only = v505 S3.9 replicated + sharpened; flipped rule  $m+j-j'$ :  $\Phi_{T_3}$  kill flip-invariant,  $\Phi_P$  rule-dependent;  $SO(16)$ : odd classes EMPTY and AB sum (15, -18, -15, -4, 40) —  $T_5$  does not even cancel ( $E_8$  doubly special);  $D_8$ : 16  $T_8 + 12 s_8^2$ , no  $T_3$  structure — the question cannot even be posed. Number fences [C] only:  $32 = (h, h) + (h', h') = 20 + 12$ ,  $72 = 2\lambda_{e_8}^2$ .
- **Ledger + papers + Wolfram + website.** Ledger: new row CELEST.WP5E.GAMMA.01 ([E] vertex space/selection/certificate + [C] fences + [O] contact terms); CELEST.SEAM.01 stays Open with the WP5e targets UPDATED (“the exchange sub-branch is closed by v508; the named remaining targets are the twisted BCOV contact terms ( $\delta_1$ , binary test) and the full-tensor ledger ( $\delta_2$ )). Papers: tfpt\_research\_contracts WP5e section extended by the  $\gamma$  block ( $\gamma$

done — negative result with certificate; the  $\delta_1/\delta_2/\varepsilon_1/\varepsilon_2$  roadmap with success/kill criteria); **tfpt\_3** celestial section becomes a TWELVE-step story with Step 12 “the exchange no-go and the contact-term criterion” and the “What remains” paragraph updated. Wolfram: six mirrors (the vertex-space dims  $2/5$  by generator-wise nullspace arithmetic; the twisted quadratics with  $K^{(0)} = -15 K^{(2)}$ ; the product theorem, symbolic; the rank-3 certificate  $(0, 32, 72)$  with channels and naturalness images; the  $SO(16)$  AB sum  $(15, -18, -15, -4, 40)$ ; the  $D_8$  quartic  $16 T_8 + 12 s_8^2$ ) verified with the active engine the same day ( $444 \rightarrow 450$ , ALL WOLFRAM EXTENSION CHECKS PASSED). **VerificationDag**: v508 on the QFT/celestial node; registry + **run\_all** (502 modules, suite green). No gate closes, no marker moves — the O-fence is only sharpened.

**2026-07-21 (XVII) (v507 — the tautology attack on the v506 alignment bit, executed and refuted, promoted as SEAM.BIT.ORIGIN.01: the ORIGIN THEOREM — marks = fixed points of the hyperelliptic involution = the 4 half-periods  $\Lambda/2\Lambda$  of the seam double, and EVERY mark-free collar deck IS a half-period translation (in all three stabiliser types  $(4, 3, 3)/(8, 5, 3)/(12, 3, 3)$  the free involutions are exactly the three  $\sigma_c$ : the CLASS of the deck is forced, its POSITION is not); the CORE SOLVE (symbolic  $\lambda$ ): a marks-preserving root of  $\sigma_c$  exists  $\iff \lambda \in \{-1, 2, \frac{1}{2}\}$  (one per  $c$ ; intrinsic: the deck pairs separate harmonically), union = the harmonic orbit  $\iff j = 1728 \iff \tau = i$ ; the EQUIVALENCE TETRAD “Uhr existiert  $\iff (\tau=i$  AND deck = CM-fixed half-period)  $\iff$  (Stab =  $D_4$  AND deck central)  $\iff$  deck pairs harmonic  $\iff$  NS Fock lift NONSPLIT”; the FERMION DICHOTOMY (the new physical face): central  $\tilde{U}^2 = 256 \gamma_1 \cdots \gamma_{16} \Rightarrow U^2 = (-1)^F$  nonsplit vs edge  $\tilde{U}_e^2 = +2^7 \Rightarrow U^2 = +1$  SPLIT (2 vs 0 roots — the bit IS a Fidkowski–Kitaev-type extension class, distinguished but not derived); the silver set = the in-family misalignment witness; Wolfram extension  $438 \rightarrow 444$  (verified engine run); *NO marker moves — the bit stays [C], SEAM.CLOCK.RIGIDITY.01/QGEO.REALIZE.01/P2.TYPING.01 text precised only***

- **v507\_seam\_bit\_origin.py** (SEAM.BIT.ORIGIN.01, 26 checks,  $\sim 2.6$  s; Part A sympy-exact torus/Möbius arithmetic + Part B exact  $Cl(16)$  Fractions; deterministic). Conjecture under attack: marks and deck both descend from the SAME torus structure (the seam double  $T_\tau^2 \rightarrow S^2$ , v214/v260), so the common origin would force the central alignment automatically and the v506 bit would be a tautology. *Origin theorem (die Herkunft)*: the 4 marks are the fixed points of  $w \mapsto -w$  on  $\mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z}) =$  the half-periods  $\Lambda/2\Lambda$  (a  $\frac{1}{3}$ -coefficient never qualifies);  $t_c$  descends iff  $2c \in \Lambda$  — exactly the three half-period translations (control  $c = \frac{1}{3}$  fails); the descended maps are the EXPLICIT rational involutions  $\sigma_i(x) = e_i + (e_i - e_j)(e_i - e_k)/(x - e_i)$  with the exact curve-lift identity  $\prod(\sigma_i - e_l) = \psi_i^2 \prod(x - e_l)$ ,  $\psi_i$  RATIONAL (the  $\wp$ -addition formula as pure algebra), closing to Klein  $V_4$ ; CONVERSELY the freely-acting involutions in ALL three stabiliser types (generic  $V_4$   $(4, 3, 3)$ , harmonic  $D_4$   $(8, 5, 3)$ , hexagonal  $A_4$   $(12, 3, 3)$ ) are EXACTLY the three  $\sigma_c \Rightarrow$  every mark-free collar deck (v216) IS a half-period translation. *Core solve (symbolic  $\lambda$ )*:  $\sigma_c$  admits a marks-preserving  $PGL_2$  root  $\iff \lambda = -1$  ( $\sigma_1$ ),  $2$  ( $\sigma_2$ ),  $\frac{1}{2}$  ( $\sigma_3$ ) — per half-period exactly ONE solution, intrinsically “the deck pairs separate harmonically” (pair cross-ratio  $-1$ ); generic modulus:  $V_4$  abelian, ALL  $\sigma_c$  central yet clockless (“central alone is empty” — the bit is the CONJUNCTION  $D_4 \wedge$  central); at  $\tau = i$  the stabiliser jumps to  $D_4$  with  $\sigma_1$  central (2 roots = the clock pair),  $\sigma_2/\sigma_3$  edge-type (0 roots); the v506 seam deck lands on the central  $\sigma_1 = -1/x$  under all 8 mark bijections; CM dictionary: mult-by- $i$  fixes exactly  $c^* = (1+\tau)/2$  (mult-by- $\rho$  3-cycles all three), the clock lift needs the CM unit  $i$  ( $\psi = 2i/(x+1)^2$ ,  $t^2+1$  irreducible over  $\mathbb{Q}$ ) while half-period decks lift rationally. *Equivalence tetrad + circularity audit*: clock exists  $\iff (\tau=i$  AND deck = the CM-fixed half-period)  $\iff$  (Stab =  $D_4$  AND deck central)  $\iff$  deck pairs harmonic  $\iff$  NS Fock lift nonsplit; the v493 converse independently re-verified — two independently computed converses of ONE equivalence, no circle. *Fermion dichotomy (Part B)*: central arrangement (shift by



8):  $\tilde{U}^2 = 256 \gamma_1 \cdots \gamma_{16} \Rightarrow U^2 = (-1)^F$ , nonsplit  $\mathbb{Z}_4$ , 2 roots in the 32-element dihedral lift (the  $\mathbb{Z}_8$  tower); edge arrangement (reflection  $j \mapsto 4-j$ , fixed sites = non-marks = the silver  $\{0, \infty\}$  between neighbour marks): the implementer EXISTS but  $\tilde{U}_e^2 = +2^7 \cdot \mathbf{1} \Rightarrow U^2 = +1$ , the extension SPLITS, 0 roots, no  $\mathbb{Z}_8$ ; one-particle  $(\lambda^2+1)^8$  vs  $(\lambda^2-1)^8$ ; census: all 28 shift-invariant 4-sets cross (central type), all 21 mark-free reflection-invariant sets do not; robust at 8 sites; R-sector edge control splits too — the bit IS a physical extension class (Fidkowski–Kitaev-type): distinguished, but not derived (WHICH class nature realises stays the input). *Silver placement*:  $\{\pm 1, \pm(3+2\sqrt{2})\} = “\tau = i$  with deck = a non-CM-fixed half-period translation” (all 8 bijections  $\rightarrow \sigma_2/\sigma_3$ , never  $\sigma_1$ ) — the in-family witness: TAUTOLOGY REFUTED. *The one [C] rest (verbatim)*: the alignment bit remains a carrier datum; “no tautology — the origin forces the CLASS of the deck (half-period translation), not its POSITION”. No overclaim.

- **Ledger + papers + Wolfram + website.** Ledger: new row SEAM.BIT.ORIGIN.01 ([E] origin theorem/core solve/tetrad/fermion dichotomy/silver placement + [C] the bit); TEXT precisions at *unchanged* markers: SEAM.CLOCK.RIGIDITY.01 (“the bit is now fully characterised as a tetrad ... and carries a physical face: the non-split NS Fock extension class — v507”), QGEO.REALIZE.01 (“the class of the deck IS derived (half-period translation, v507); only its position stays the input”, stays [C]/[O]), P2.TYPING.01 (origin cross-reference on the common R1 bit). Papers: tfpt\_3 celestial Step-1/2 environment + QGEO paragraph, contracts marks/kill-test paragraph, introduction/tfpt\_1/tfpt\_2/origin\_theory one-sentence updates where v506 is cited. Wolfram: six mirrors (half-period descent  $2c \in \Lambda$  with the rational  $\sigma_i$  lift identity; the core solve  $\lambda \in \{-1, 2, \frac{1}{2}\}$ ; the stabiliser counts  $(4, 3, 3)/(8, 5, 3)/(12, 3, 3)$  via the complete 24-triple enumeration; the CM fixed point  $c^*$  + the lemniscatic  $\psi = 2i/(x+1)^2$ ; the Cl(16) dichotomy  $256 \gamma$ -monomial vs scalar  $2^7$  on explicit  $256 \times 256$  Jordan–Wigner matrices + the 2-vs-0 root census in the 32-element dihedral lift; the silver bijection census) verified with the active engine the same day ( $438 \rightarrow 444$ , ALL WOLFRAM EXTENSION CHECKS PASSED); the v493 audit direction and the 28/21 arrangement census stay Python-side (flagged). VerificationDag: v507 on the P2/carrier and QGEO nodes; registry + run\_all (501 modules, suite green). No gate closes, no marker moves.

**2026-07-21 (XVI) (v506 — seam clock rigidity, Route B of the v503 classification, promoted as SEAM.CLOCK.RIGIDITY.01: TWO exact theorems reduce the order-4 clock input to ONE alignment bit — Theorem A (Möbius): the deck  $z \mapsto -z$  has EXACTLY TWO square roots  $z \mapsto \pm iz$  in  $\text{PGL}_2(\mathbb{C})$ , both *automatically* order 4, and a marks-preserving root exists  $\iff$  the deck is the CENTRAL involution of the mark- $D_4$  (counterexample  $\{\pm 1, \pm(3+2\sqrt{2})\}$ : harmonic with full  $D_4$  but deck non-central  $\Rightarrow$  0 roots — harmonicity alone insufficient); Theorem B (Fock, the core): the NS deck implementation is FORCED to order 4 —  $\tilde{U}^2 = 256 \gamma_1 \cdots \gamma_{16}$ , i.e.  $U^2 = (-1)^F$  EXACTLY, a nonsplit  $\mathbb{Z}_4$ , with the  $\mathbb{Z}_8$  tower  $V^2 \propto U$ ,  $V^4 \propto (-1)^F$  ( $8 = 2|\mu_4|$  fermionically explained) and the R-sector control split ( $U_R^2 = +1$ ); the one [C] rest: “the collar deck is the central involution of the mark- $D_4$ ” — the fermionic  $\mathbb{Z}_4$  forces the ORDER, not the spatial alignment; Wolfram extension  $434 \rightarrow 438$  (verified engine run); *NO marker moves — QGEO.REALIZE.01 stays [C]/[O] (text precised), P2.TYPING.01 R1 mapped onto the same bit)***

- **v506\_seam\_clock\_rigidity.py** (SEAM.CLOCK.RIGIDITY.01, 34 checks,  $\sim 3.5$  s; **Part A** sympy-exact Möbius/divisor arithmetic + **Part B** exact Cl(16) Fractions; **deterministic**). v503 localised the one remaining identification input (“why does the raw seam carry the order-4 clock”, QGEO-R1  $\equiv$  P2-R1); v506 reduces it. *Theorem A (Möbius side, complete)*: Cayley–Hamilton + the complete case analysis give  $g^2 = \text{deck} \Rightarrow \text{tr } g \neq 0$  forces  $g$  diagonal with  $(a/d)^2 = -1$ , and  $\text{tr } g = 0$  forces  $\det g = 0$  (impossible) — exactly two roots  $z \mapsto \pm iz$ , both projective order 4, mutually inverse ( $\text{Aut}(\mathbb{Z}_4)$ ), conjugation-covariant; marks filter: a deck-invariant deck-free 4-set  $\{u, -u, w, -w\}$  admits a marks-preserving root  $\iff$



$w = \pm iu \iff \text{marks} = \text{one } \mu_4 \text{ orbit} \Rightarrow \text{cross-ratio class } \{2, \frac{1}{2}, -1\}, j = 1728, \tau = i, \text{ free 4-cycle, deck} = \text{clock}^2, \text{ only the scale free; group form: } |\text{Stab}(\mu_4)| = 8 = 2|\mu_4| \text{ with order profile } \{1:1, 2:5, 4:2\} = D_4, \text{ center} = \{\text{id}, \text{deck}\}, \text{ exactly 2 order-4 elements} = \text{the clocks} \text{ — “deck central in the mark-}D_4\text{”} \iff \text{“clock exists”}; \text{ the counterexample } \{\pm 1, \pm(3+2\sqrt{2})\} \text{ is deck-free AND harmonic } (j = 1728, \text{ cross-ratio } \frac{1}{2}) \text{ with full } D_4, \text{ but the deck sits non-centrally (the central involution crosses the } \pm \text{ pairs)} \Rightarrow 0 \text{ roots; the RP filter passes BOTH roots (the } \text{Aut}(\mathbb{Z}_4) \text{ ambiguity is irreducible, v492 NC3). } B3/B4 \text{ (} n\text{-arithmetic + controls): free orbit } \Rightarrow n \mid 4; \text{ transitivity } \Rightarrow n = 4; \text{ faithfulness kills } n \geq 5 \text{ (kernel argument through } S_4); n = 3 \text{ fixes a mark; } n = 6 \text{ dies independently at the cusp degree 3 (v117); the deck is modulus-blind (v503 B5 exact); } n = 8 \text{ impossible on 4 marks; hexagonal stabiliser } A_4 \text{ (0 order-4 elements, squares only orders } \{1, 3\}); \text{ generic stabiliser } V_4 \text{ (exponent 2); stabiliser trichotomy } V_4/D_4/A_4 \text{ with order-4 counts } (0, 2, 0). \text{ Theorem B (Fock side, the core result): } S^2 \text{ has } b_1 = 0 \Rightarrow \text{unique spin structure } \Rightarrow \text{NS/bounding on every circle (established P1 datum); one-particle: } D^2 = -1 \text{ (charpoly } (\lambda^2+1)^8), \text{ quarter-shift } C^2 = D, C^8 = 1 \text{ (charpoly } (\lambda^4+1)^4, \zeta_8 \text{ spectrum); Fock (exact Cl(16)): } \tilde{U} = \prod_j (1 - \gamma_j \gamma_{j+8}) \text{ implements the deck on all 16 generators and } \tilde{U}^2 = 256 \gamma_1 \cdots \gamma_{16} \Rightarrow U^2 = (-1)^F \text{ EXACTLY; } (cU)^2 = c^2(-1)^F \text{ is never 1 (center of Cl(16) = scalars): the } \mathbb{Z}_4 \text{ lift is FORCED, } 1 \rightarrow \mathbb{Z}_2^{(-1)^F} \rightarrow \mathbb{Z}_4 \rightarrow \mathbb{Z}_2^{\text{deck}} \rightarrow 1 \text{ nonsplit; the quarter-shift lift } V \text{ (unique even block implementer) has } V^2 \propto \tilde{U}, V^4 \propto (-1)^F, \text{ projective order } 8 = 2|\mu_4| \text{ — the v492 } \mathbb{Z}_8 \text{ spin bridge realised on the 16-Majorana Fock space; R control } U_R^2 = +1 \text{ (split — the forcing rests on NS = established geometry); 16-fold way } \theta_\nu = 1 \text{ at } \nu = 16; \text{ robustness: the forcing already holds at 8 sites (16 = the carrier count } 2^{g_{\text{car}}-1}, \text{ not a tuning). The one [C] rest (verbatim): “the collar deck is the central involution of the mark-}D_4\text{” (} \iff \text{ square-root existence on the mark data) — ONE alignment bit; the fermionic } \mathbb{Z}_4 \text{ forces the ORDER, not the spatial alignment. No overclaim.}$

- **Ledger + papers + Wolfram + website.** Ledger: new row SEAM.CLOCK.RIGIDITY.01 ([E] both theorems + [C] the alignment bit); TEXT precisions at *unchanged* markers: QGEO.REALIZE.01 (“the rest is now ONE bit: the central position of the deck in the mark- $D_4$ ; order/uniqueness/orbit/ $\tau=i$  follow — v506”, stays [C]/[O]), P2.TYPING.01 (R1 mapped onto the same bit, fermionic halving), QGEO.EMERGE.LIGHT.01 context sharpened. Papers: tfpt\_3 celestial Step 1 (fermionic  $U^2 = (-1)^F$  sentence) + Step 2 (the one-bit reduction), contracts QGEO kill-test paragraph + marks paragraph + CELEST hypothesis note, introduction/tfpt\_1/tfpt\_2/origin\_theory one-sentence updates at the v499/v503 citation sites. Wolfram: four mirrors (the complete square-root solve;  $D_4$  center + order-4 counts + the silver counterexample; the Cl(16) identity  $\tilde{U}^2 = 256 \gamma_1 \cdots \gamma_{16}$  on explicit  $256 \times 256$  Jordan–Wigner matrices + the split R control; the  $n$ -arithmetic + generic kill) verified with the active engine the same day (434  $\rightarrow$  438, ALL WOLFRAM EXTENSION CHECKS PASSED); the 24-triple stabiliser enumeration and the  $A_4$  profile stay Python-side (flagged). VerificationDag: v506 on the P2/carrier node; registry + run\_all (500 modules, suite green). No gate closes, no marker moves.

**2026-07-21 (XV) (v505 — WP5e- $\beta$  of CELEST.SEAM.01 “the equivariant anomaly ledger on twistor space” promoted as CELEST.WP5E.BETA.01: the Atiyah–Bott/Lefschetz fixed-point skeleton of the one-loop twistor inflow is EXACT — denominators  $(2, 4, 2)$  with Dedekind sum  $\frac{5}{4} = (|\mathbb{Z}_4|^2 - 1)/12$ , equivariant characters  $(248, 0, -8, 0)$  by two routes, invariant Okubo  $36\langle x, x \rangle^2$  with the twisted sectors NOT reducing and the AB sum leaving the RIGID residual  $32 T_3$ , the index bridge  $f(m) = \text{ch}_2(T_m)$  exact with glue defect  $-78$  both routes, the level dials  $k = 1$  geometric (current count  $240/0/0/0$ ), and the honest E3 REFUTATION:  $a_0$  fills the GRAVITON slot  $O(2)$ , NOT the axion slot  $O(-2)$  — the three sphere classes fill the three twisted axion slots instead; the preregistered kill (“inflow demands a level  $\neq 1$ ”) does NOT fire on the equivariant skeleton; WP5e:  $\alpha$  (CFT side, v502) +  $\beta$  (equivariant twistor skeleton, v505) executed, the global BCOV quantisation stays [O]; Wolfram extension  $429 \rightarrow 434$  (verified engine run); SEAM.EQUIV.01 stays [O], no marker moves)**

- **v505\_celestial\_wp5e\_beta\_equivariant\_ledger.py** (CELEST.WP5E.BETA.01, 47 checks,  $\sim 1.3$  s; sympy/Fraction exact, no floats; deterministic). The twistor side of WP5e pushed as far as exact arithmetic reaches: the Atiyah–Bott/Lefschetz skeleton of the one-loop box anomaly (Costello; Bittleston–Sharma–Skinner) on  $\mathbb{C}^2/\mathbb{Z}_4$  with the  $E_8$   $\mu_4$ -glue monodromy. *AB ledger*:  $\det(1 - g_j^{-1}) = (2, 4, 2)$ ;  $\sum_j 1/\det_j = \frac{5}{4} = (|\mathbb{Z}_4|^2 - 1)/12$ ;  $\text{tr}_{g_j}(\text{Ad } E_8) = (248, 0, -8, 0)$  two routes; invariant average  $60 = \text{carrier}$ ; Frobenius  $61568$ ;  $\text{tr}_{g_2} = -8 = 120 - 128$ . *Okubo per sector (honest sharpening)*: the invariant sector is exactly Okubo-quadratic ( $Q^{(0)} = 36\langle x, x \rangle^2$ ,  $36 = \tilde{\chi}_{e_8}^2$  — v495 re-derived through the glue grading); the twisted sectors do NOT reduce ( $T_5$ :  $(-8, 32, -8)$ ,  $T_3$ :  $(48, -64, 48)$ ); the AB sum cancels the  $D_5$  quartic exactly and leaves the rigid  $32 T_3$  (the only quartic-free weighting is the  $\mathbb{Z}_4$  projector; the graded GS exchange is rank-obstructed in sectors 1–3); heterotic identities ( $\text{Tr}_{120}, \text{Tr}_{128}, \text{Tr}_{E_8} = 9(\text{tr}^2)^2$ ) exact. *Index bridge*:  $f(m) = \frac{1}{4} \sum_j (i^{jm} - 1)/\det_j = (0, -\frac{3}{8}, -\frac{1}{2}, -\frac{3}{8}) = \text{ch}_2(T_m) = -(C^{-1})_{mm}/2$  EXACT (E1 = E2 arithmetic); glue  $\text{ch}_2$  defect  $-78$  both routes (classes  $(-24, -30, -24)$ ); Killing sums  $(320, 320, 240, 320)$ , total  $1200 = 2h^\vee(h, h)$ .  $k = 1$  *geometric*: lattice current count  $240$  at  $k=1$ , exactly  $0$  at  $k = 2, 3, 4$ ; embedding residual  $47k^2 + 219k - 266 = (0, 360, 814, 1362)$ ; integrality alone fixes nothing (honest, as v502); one scale  $\Rightarrow$  one level (v493 periods). *E3 refutation (prominent)*:  $a_0 \in O(8)$  fills the GRAVITON slot  $O(2)$ , NOT the BSS axion slot  $O(-2)$  ( $X = -2 \neq +2 = Y$ ; mismatch  $4 = |\mu_4|$  as a [C] coincidence) — “the theory brings its GS axion as  $a_0$ ” is REFUTED; instead the three  $H^2(\text{ALE})$  classes carry exactly the Coxeter eigencharacters  $\{i, -1, -i\} =$  the three twisted slots (bijection); slot count  $1+3 = 4 = |\mu_4|$  soft. *Negative controls*:  $\text{diag}(i, i)$  breaks the ledger four ways ( $\sum 1/\det = \frac{1}{4}$ ,  $\omega$ -weight  $-1$ , 5 generators);  $SO(16)$  glue: characters  $(120, 0, +120, 0)$ , defect  $-30 \neq -78$ , bulk Okubo fails (independent 4th Casimir);  $k = 2$ : current count  $0$ ,  $c$ -prefactor  $-31/48 \neq -\frac{1}{3}$ . [C] dictionaries ( $\text{CS}(\rho) \bmod 1 = \text{discriminant weight}$ ;  $32 = (h, h) + (h', h')$ ;  $-30 = -h^\vee$ ;  $\Omega$ -contraction). [O] the global BCOV/KS quantisation, non-fixed-point contributions, the  $O(-2)$  bulk axion, CPS level-from-flux, the pairing  $32 T_3 \leftrightarrow$  sphere axions. VERDICT B; the kill test does not fire on the equivariant skeleton.
- **Ledger + papers + Wolfram + website.** Ledger: new row CELEST.WP5E.BETA.01; CELEST.SEAM.01 updated to “WP5e:  $\alpha$  (CFT side, v502) +  $\beta$  (equivariant twistor skeleton, v505) executed; the global BCOV quantisation stays [O] — with the rigid  $32 T_3$  twisted residual and the sphere-axion pairing as the named remaining targets” at *unchanged* Open status. Papers: contracts WP5e bullet  $\rightarrow$  “ $\alpha + \beta$  done” with the  $a_0$  refutation as an honest finding (kill-test and honest-success paragraphs consistent); **tfpt\_3** celestial section becomes the *eleven-step* story (new Step 11 “the equivariant anomaly ledger on twistor space”, “what remains” = the global BCOV quantisation with named targets). Wolfram: five mirrors (AB denominators + characters two routes; the index bridge +  $-78$  defect; the per-sector Okubo sums over the  $240$  glue roots with the  $32 T_3$  residual; the lattice current counts + embedding residuals; the

axion-slot weight arithmetic + twisted-slot bijection) verified with the active engine the same day (429 → 434, ALL WOLFRAM EXTENSION CHECKS PASSED). VerificationDag qft node gains v505; registry + run\_all (499 modules at this step; 500 with v506, suite green). No gate closes, no marker moves — SEAM.EQUIV.01 stays [O].

**2026-07-21 (XIV) (v504 — WP5d- $\beta$  of CELEST.SEAM.01 “split + strong additivity from the lattice” promoted as CELEST.WP5DB.01: the two remaining KLM legs of complete rationality witnessed for the orbifold prescription — strong additivity ALGEBRAICALLY EXACT with the shared boundary Majorana ( $\text{Even}(A) \vee \text{Even}(B) = \text{Even}(A \cup B)$ ), GF2 spans full, rank 32/32; disjoint exactly index 2 = the v501  $\ln 2$  bit), the entropic touching defect BOUNDED  $< \ln 2$  with the Ising  $\frac{1}{4}$ -exponent approach ( $p = 0.2444$ ) while the preregistered  $U(1)$ /Dirac control DIVERGES (Klich–Levitov pinned, infinite index), the split property at the elliptic-nome rate  $\pi K(1-x)/K(x)$  (1.3–2.0%) with EXACT orbifold inheritance ( $C \rightarrow -C$  + the  $\Lambda^2 C$  compound — Longo heredity), and Pimsner–Popa  $E(a) - a/2 = PaP/2$  identically ( $\lambda = \frac{1}{2} = 1/[F:F_{\text{even}}]$ ,  $\lambda_{E_4} = \frac{1}{4} = 1/\mu$ ,  $e^{\Delta_\infty} = 2 = 1/\lambda_{\text{PP}}$  — two independent index routes); with v501 ALL THREE KLM ingredients carry lattice witnesses; the continuum uplift stays [O] — “finite-group orbifolds of completely rational nets are completely rational” is XU’S THEOREM, cited not claimed; Wolfram extension 424 → 429 (verified engine run); SEAM.EQUIV.01 stays [O], CELEST.SEAM.01 stays Open, no marker moves)**

- **v504\_celestial\_wp5d\_klm\_completeness.py** (CELEST.WP5DB.01, 37 checks, ~12 s; numpy/scipy Gaussian machinery ED-validated + exact GF2/integer algebra; deterministic). Complete rationality (Kawahigashi–Longo–Müger) = split + strong additivity + finite two-interval  $\mu$ -index; v501 (WP5d- $\alpha$ ) witnessed the index, v504 witnesses the other two legs on the same critical Majorana seam edge, with the  $U(1)$ /Dirac number-conserving subnet as the preregistered negative control (the current net, where strong additivity famously *fails* in the continuum — Buchholz–Schulz–Mirbach). *Strong additivity, exact algebra*: “touching = shared boundary Majorana” gives  $\text{Even}(A) \vee \text{Even}(B) = \text{Even}(A \cup B)$  EXACTLY (GF2 spans full 64/64, 256/256, 512/512, 1024/1024; literal matrix rank 32/32); disjoint intervals give exactly HALF (rank 16/32, index 2 — the missing sector odd  $\otimes$  odd is the v501  $\ln 2$  bit, localised at the split point); the neutral  $U(1)$  algebras do NOT generate the union even with the shared site (gaps 2/10/52, growing). *Entropic witness*: the fermionic touching MI carries the pure UV point-gap law ( $(c/3) \ln 2$  increments, 0.115525, dev  $1.2 \times 10^{-6}$  — present for every net, no discriminator); the honest discriminator is *bounded vs divergent*: the  $\mathbb{Z}_2$ /orbifold deficit rises  $0.4045 \rightarrow 0.5899$  ( $N = 64 \rightarrow 4096$ ), stays  $< \ln 2$ , and  $(\ln 2 - \Delta_2) \sim N^{-p}$  with  $p = 0.2444$  vs  $2\Delta_\mu = \frac{1}{4}$  (Ising disorder exponent); split-point scan reflection-symmetric to  $3.9 \times 10^{-14}$ ; ED vN same direction; gap continuity to the v501  $x \rightarrow 1$  endpoint; honest note: “defect  $\rightarrow 0$ ” would be *false* at the sharp lattice split — bounded  $\Leftrightarrow$  finite index (Longo–Xu) is the correct statement; the  $U(1)$  control bursts  $\ln 2$  from  $L = 128$  and grows as  $\frac{1}{2} \ln \text{Var } Q_A$  (Klich–Levitov slope 0.10134 vs  $1/\pi^2 = 0.10132$ ). *Split/nuclearity*: coupling-block  $\sigma_k$  fall exponentially at the elliptic-nome rate  $\pi K(1-x)/K(x)$  to 1.3–2.0% (rates 2.29/3.24/4.82/5.81 at  $d = 16/64/256/512$ );  $\sigma_1 \sim x^{0.5044}$  (vs  $\frac{1}{2}$ ); trace norm summable; dimerised control exponential. *Orbifold inheritance exact*:  $P_A$  flips  $C \rightarrow -C$  ( $\sigma_k$  identical,  $8.3 \times 10^{-17}$ ) and the even-bilinear coupling Gram is the second compound  $\Lambda^2 C$  (singular values  $\{\sigma_a \sigma_b\}$ ,  $2.8 \times 10^{-14}$ ) — the same ladder at doubled rate (Longo heredity, [C]). *Pimsner–Popa, exact*:  $E(a) - a/2 = PaP/2$  identically; attainment in integers (all 16384 (even, odd) monomial pairs at  $N = 8$ : 8192 extremal with  $M^2 = 4$ ,  $\text{Tr } M = 0$ ,  $PMP = -M$ , 0 violations; 2048/2048 at  $N = 12$ ; pencil min  $\lambda = 0.5000000000$ );  $\lambda_{E_4} = \frac{1}{4} = 1/\mu$ ; index consistency  $e^{\Delta_\infty} = 2.000000000 = 1/\lambda_{\text{PP}}$  and  $e^{2\Delta_\infty} = 4 = \mu$ ;  $U(1)$ :  $\lambda = 1/(m+1) \rightarrow 0$ . *Controls + honesty*: dimerised phase trivial (drift  $3.6 \times 10^{-15}$ ); wrong prescription  $\{1, P_{AB}\}$  deficit exactly 0;  $\mathbb{Z}_2$  finite-size honesty (14.9% below

$\ln 2$  at  $N = 4096$  — witness, not proof). VERDICT: with v501 all THREE KLM ingredients of complete rationality are witnessed on the lattice; the continuum theorem for finite-group orbifolds is Xu’s (cited, not claimed), the concrete seam quotient net and the condensed  $(E_8)_1$  net stay WP5e/Costello–Li.

- **Ledger + papers + Wolfram + website.** Ledger: new row CELEST.WP5DB.01; CELEST.SEAM.01 updated to “WP5d:  $\alpha$  ( $\mu$ -index) +  $\beta$  (split + strong additivity) executed via v501/v504 — all three KLM ingredients witnessed on the lattice; continuum uplift stays [O] (Xu’s theorem cited)” at *unchanged* Open status. Papers: contracts WP5d bullet  $\rightarrow$  “both stages done” with the  $\beta$ -stage numbers + Xu fence (kill-test and honest-success paragraphs consistent); `tfpt_3` celestial section becomes the *ten*-step story (new Step 10 “the KLM completion on the lattice”, “what remains” = WP5e twistor inflow only); `tfpt_4` keystone note updated. Wolfram: five mirrors (GF2 span theorem; Pimsner–Popa identity + attainment on explicit gamma matrices; two-interval group  $\lambda_{E_4} = \frac{1}{4}$  + index consistency; the  $\Lambda^2/\text{Minors}$  compound with Cauchy–Binet and  $\text{Minors}[-C, 2] = \text{Minors}[C, 2]$ ; the  $U(1)$  dephasing  $\lambda = 1/(m+1)$  family) verified with the active engine the same day (424  $\rightarrow$  429, ALL WOLFRAM EXTENSION CHECKS PASSED); the lattice curves stay Python-only (flagged). VerificationDag qft node gains v504; registry + `run_all` (498 modules, suite green). No gate closes, no marker moves — SEAM.EQUIV.01 stays [O].

**2026-07-21 (XIII) (v503 — the v215 deferred emergence test (lever 7) executed *light* and promoted as QGEO.EMERGE.LIGHT.01: NO dynamical selection of the square modulus — the free field’s global  $\det'$  winner is the *hexagonal* torus,  $\tau=i$  is a saddle, criticality at the square is Palais symmetry; the square is pinned by *rigidity* alone (the raw DtN is mod-4 block-diagonal *iff*  $\tau=i$ , an isolated zero); and the residual *converges*: QGEO-R1  $\equiv$  P2-R1 = one common rest, “why does the raw seam carry the order-4 clock”; Wolfram extension 420  $\rightarrow$  424 (verified engine run); NO marker moves — QGEO.REALIZE.01 stays [C]/[O], AX.P2.01 stays a declared axiom)**

- `v503_qgeo_emergence_light.py` (QGEO.EMERGE.LIGHT.01, 17 checks; `mpmath` 40 digits + v280-style lattice DtN + `sympy` exact). The v215 lever 7 (“the decisive kill is deferred: compute the raw seam-collar DtN *without* inputting the marks and check whether four square marks *emerge*”) executed in its first honest *light* version. *No dynamical selection*:  $\log F = \log(\tau_2|\eta|^4)$  (Osgood–Phillips–Sarnak) is exactly  $S$ -invariant ( $\text{dev} < 10^{-35}$ ), so  $t=1$  is symmetry-forced critical on the rectangular family  $\tau = it$ ; there  $g''(1) = -1.298 < 0$  (strict maximum) but the  $\tau_1$ -direction has  $h''(0) = +0.298 > 0$ :  $\tau=i$  is a *saddle* of the full moduli space; the only critical points are the CM points ( $E_2^*(i) = E_2^*(\rho) = 0$  at 40 digits, generic  $0.163 \neq 0$ ); the *global*  $\det'$  winner at fixed area is the *hexagonal* torus ( $\log F(\rho) = -1.03352 > \log F(i) = -1.05469$ , diff 0.021169); criticality at the square is Palais symmetric criticality (the independent heat trace is equally critical). *Sharp rigidity*: the DtN mod-4 off-character indicator (the v215-K3 lever as a *modulus function*) vanishes to  $8 \times 10^{-15}$  exactly at  $\tau=i$  and rises strictly monotonically both ways (min  $8.4 \times 10^{-2}$  away) — an isolated zero;  $R^2 = \mathbb{Z}_2$  exact as permutation matrices (the v492  $\mathbb{Z}_8$  bridge on the lattice);  $j(it) \geq 1728$  with equality only at  $t=1$ . *The  $\mathbb{Z}_2$  chain is modulus-blind*: 4 marks +  $[\mathbb{Z}_2, \Delta_t] = 0$  on *every* torus (v216 confirmed without mark input); pillowcase  $\det' = \frac{1}{2}$  torus  $\det'$  exactly ( $\pm$ -pair structure) — only the order-4 clock distinguishes  $\tau=i$ ;  $\text{Aut}(E_\rho) = \mathbb{Z}_6$  has *no* order-4 element (the dynamical winner cannot carry the clock); given the clock the modulus freezes automatically ( $j = 1728$  identically, v493 mini). *Residual convergence (the core finding)*: the order-4 element on  $H^1$  is a Coxeter element of  $W(A_3)$  (char poly  $\lambda^3 + \lambda^2 + \lambda + 1$ , order 4 =  $|\mu_4| = N_{\text{fam}} + 1$ ) = the P2 cusp-class carrier datum of v499 R1  $\Rightarrow$  QGEO-R1  $\equiv$  P2-R1, *one common residual*; chain: marks derived (v216), square derived *given* the clock (v493), clock = carrier input [[C]], rest = its dynamical realisation [O]; route classification (v347 analogue): B-rigidity in the sharpened form “RP seam data  $\Rightarrow$  order-4 clock”



is the live route. v215 lever 7 thereby moves from “deferred, not executable” to “light version executed: rigidity, not dynamical selection” — documented in v503, v215 itself unchanged.

- **Ledger + papers + Wolfram + website.** Ledger: new row QGEO.EMERGE.LIGHT.01; QGEO.REALIZE.01 text *precised* (rest = the order-4 clock, identical with the P2 typing rest R1; square + mark count derived) at *unchanged* marker [C]/[O]; cross-reference added in P2.TYPING.01 (convergence theorem). Papers: contracts kill-test paragraph (lever-7 light result + the residual-convergence sentence, both with v503), tfpt\_3 Step 2 (light-converse sentence), origin\_theory necessity paragraph, introduction/tfpt\_1/tfpt\_2 one-residual cross-references. Wolfram: four mirrors (CM-point identities  $E_2(i) = 3/\pi$ ,  $E_2(\rho) = 2\sqrt{3}/\pi \Rightarrow E_2^* = 0$  exact,  $j/\text{DedekindEta}$  special values; the no-dynamical-selection witnesses; the pillowcase halving as an exact algebraic identity; clock  $\Rightarrow$  square + Coxeter) verified with the active engine the same day (420  $\rightarrow$  424, ALL WOLFRAM EXTENSION CHECKS PASSED); the lattice DtN curves stay Python-only (flagged). VerificationDag masses node gains v503; registry + run\_all (497 modules, suite green). No gate closes, no marker moves.

**2026-07-21 (XII) (v502 — WP5e- $\alpha$  of CELEST.SEAM.01 “prefactor and level bookkeeping” promoted as CELEST.WP5E.ALPHA.01: the  $q^{-1/3}$  prefactor of  $E_4/\eta^8$  IS exact  $\mu_4$  vacuum-energy bookkeeping (the clock is inner  $\Rightarrow$  shift orbifold  $\theta = 0^8$ , common  $-c/24 = -\frac{1}{3}$ ; discriminant form = Casimir via spectral flow AND 16-Majorana free fermions), and  $k = 1$  is forced THREE independent ways (current condition; conformal embedding; central charge = the prefactor itself; plus the WP5b singular-vector dial  $31124 - 27000 = 4124$ ); honest sharpening: glue- $h$  integrality holds for ALL  $k$  and fixes nothing; the BCOV/Costello–Li twistor inflow stays [O] — WP5e is subdivided ( $\alpha$  done, WP5e proper open); Wolfram extension  $416 \rightarrow 420$  (verified engine run); SEAM.EQUIV.01 stays [O], no marker moves)**

- **v502\_celestial\_wp5e\_prefactor\_level.py (CELEST.WP5E.ALPHA.01, 33 checks, exact sympy/Fraction, no floats).** The two exactly computable consistency anchors of the WP5e twistor uplift established. *Part A — the prefactor:* Hurwitz- $\zeta$  exact ( $\zeta(-1, \theta) = -B_2(\theta)/2$ ;  $E_b(\theta) = -\frac{1}{24} + \theta(1 - \theta)/4$ ; Ising shift  $\frac{1}{16}$  per Majorana); the clock is *inner* ( $h = (2, 2, 2, 2; 0^4)$  and  $h' = (0^5; 1, 1, 1, -3)$  both read the class mod 4 on all 240 roots;  $(h, h) = 20$ ,  $(h', h') = 12$ , sum 32)  $\Rightarrow$  twist  $\theta = 0^8$  in all four sectors — a *shift* orbifold, NOT a rotation orbifold — so  $E_{\text{osc}} = -\frac{1}{3} = -c/24$  is sector-independent ( $c = 8 = \text{rank} = \text{Sugawara } 248/31$ ; one common  $\eta^8$ ); sector ground exponents  $-\frac{1}{3} + (0, 1, 1, 1)$ ; *discriminant form = Casimir energy* two ways (spectral flow  $j^2(h, h)/32 \bmod 1$ ; free fermions exact:  $D_5 = 10$  Majorana  $\rightarrow \frac{5}{8}$ ,  $A_3 = D_3 = 6 \rightarrow \frac{3}{8}$ , glue diagonal  $= 16 = 2^{g_{\text{car}}-1} = \text{the WP5d seam carrier} \rightarrow 1$ ); the rotation reading *fails* (deck  $\frac{3}{16} \neq \frac{3}{8}$  — “clock  $\neq$  deck” at the Casimir level); the four sector characters sum exactly to  $E_4/\eta^8 = q^{-1/3}(1, 248, 4124, 34752)$ . *Part B — the level:*  $k = 1$  forced three independent ways — (i) the current condition  $h(J) = k = 1$  (adjoint closure  $60+64+60+64 = 248$  at spin 1); (ii) the conformal embedding ( $47k^2+219k-266 = 47(k-1)(k+266/47) = 0$ ;  $D_8: 128k(1-k) = 0$ ); (iii)  $c = 8 \Leftrightarrow 240(k-1) = 0$  (= the prefactor); plus the v498 singular-vector dial ( $\|(E_{-1}^\theta)^m 0\|^2 = m! k(k-1) \cdots (k-m+1)$ , singular vector at grade  $k+1$ ;  $31124 - 27000 = 4124$  at  $k=1$  only,  $\langle s|s \rangle = +4$  at  $k=2$ ). *Honest sharpening (mandatory):* glue- $h$  integrality  $h(J^j; k) = k(0, 1, 1, 1)$  is integer for ALL  $k = 1..8$  — the originally conjectured integrality route fixes *nothing*; the three other dials carry. *Negative controls:*  $D_8/SO(16)_1$  (same  $c = 8$ , SAME  $q^{-1/3}$  prefactor, but  $h = (0, \frac{1}{2}, 1, 1)$ : the discriminator is glue- $h$  integrality, not the prefactor;  $\Theta_{D_8} + \Theta_s = E_4$ );  $\mathbb{Z}_2$  rotation ( $E = +\frac{1}{6}$ ,  $h = \frac{1}{2}$ );  $\mu_4$  rotation (sector-dependent  $(-\frac{1}{3}, \frac{1}{24}, \frac{1}{6}, \frac{1}{24})$  — the common prefactor breaks); wrong  $k \in \{2, 3, 4\}$  fails all five dials coherently (residuals  $360/814/1362$ ;  $\langle s|s \rangle = 4/12/24$ ;  $31124 \neq 4124$ ). [[C]] Kac generation for  $k \geq 3$  ( $k = 1, 2$  machine-anchored by WP5b/WP5c). [O] the BCOV/Costello–Li twistor inflow itself (Costello’s one-loop axion  $\lambda_g/(8\pi\sqrt{3})$  on  $\mathbb{P}T$ , the Costello–Paquette–Sharma level-from-flux mechanism) — v502 anchors the *CFT side* of WP5e only; the KILL branch (“inflow demands a level  $\neq 1$ ”) is not triggered



on the CFT side, its twistor side is untested.

- **Contract + ledger + Wolfram + website.** `tfpt_research_contracts`: the WP5e bullet *subdivided* — “WP5e- $\alpha$  (done) [E]/[C]: the CFT-side prefactor + level pinning” with the core numbers and `v502_celestial_wp5e_prefactor_level.py`; “WP5e proper [O]: the twistor inflow”; summit header, honest-success paragraph and status roster updated (“WP5a–WP5d plus WP5e- $\alpha$  landed; WP5e proper the remaining open piece”). `tfpt_3` celestial section: new Step 9 (prefactor + level, one paragraph); `tfpt_4` keystone note updated. Ledger: new row `CELEST.WP5E.ALPHA.01`; `CELEST.SEAM.01` status  $\rightarrow$  “WP5e: alpha stage (CFT-side prefactor + level pinning) executed via v502; the BCOV/twistor inflow itself stays [O]”. Wolfram: four exact mirrors (Hurwitz- $\zeta$  vacuum energies; discriminant = Casimir with the deck failure; the  $k$ -fixing equations incl. the “fixes nothing” integrality table; the character sums +  $D_8$  control) verified with the active engine the same day (416  $\rightarrow$  420, ALL WOLFRAM EXTENSION CHECKS PASSED). `wolfram/README.md`, `GATE.WOLFRAM.02`, `tfpt_safeguards` Layer-6 count and the website counters (116/116 + 424/424 after (XIII)) updated; `VerificationDag` qft node gains v502; registry + `run_all` (497 modules, suite green). `SEAM.EQUIV.01` untouched; no marker moves.

**2026-07-21 (XI) (v501 — WP5d- $\alpha$  of CELEST.SEAM.01 “the two-interval index from the lattice” promoted as CELEST.WP5D.01: the KLM Haag-duality discriminator measured entropically on the seam edge — the 16-layer carrier saturates the  $\ln 2$  offset at machine precision,  $\mu_{\text{gauged}} = 4 =$  the v490 census, and the condensation arithmetic  $16 \rightarrow 4 \rightarrow 1$  closes with  $\mu = 1$ ; the preregistered KILL (“ $\mu$ -offset  $\neq 0$  after condensation”) does NOT fire; contract now “WP1–WP5d executed; WP5e the single remaining milestone”; Wolfram extension 412  $\rightarrow$  416 (verified engine run); SEAM.EQUIV.01 stays [O], no marker moves)**

- `v501_celestial_wp5d_two_interval_index.py` (CELEST.WP5D.01, 39 checks, numpy/scipy Gaussian machinery + exact Fractions). The WP5d roadmap discriminator —  $(E_8)_1 \Rightarrow \mu=1$  (Haag-dual two-interval net, one sector) vs  $SO(16)_1 \Rightarrow \mu=4$  — executed at the  $\alpha$  (lattice-index) stage on the v367 seam edge (16 critical Majorana layers,  $c_- = 8$ , v450-style ring; Peschel-exact, every Gaussian formula ED-validated to  $\leq 1.5 \times 10^{-15}$ ; the superselection identity  $[\rho, P_{AB}] = 0$  machine-exact). The  $\mu = 1$  reference: the fermionic (fixed-spin-structure) two-interval mutual information is *extensive* —  $c$  fit 0.5000,  $\max |F(x)| = 5.97 \times 10^{-5}$  at  $N=512$ , falling monotonically to  $1.3 \times 10^{-6}$  at  $N=1024$  (Casini–Huerta extensivity as a lattice witness). The  $\mu$ -offset: Rényi-2 exactly  $\Delta_2 = -\ln[(1+R)/2]$ ; the 16-layer seam carrier saturates  $\ln 2$  at machine precision ( $|\Delta_2 - \ln 2| = 1.1 \times 10^{-15}$  at  $N=512$ ;  $R \sim N^{-0.500}$ ); index reading  $[F : F_{\text{even}}] = 2.000000$ ,  $\mu_{\text{gauged}} = 4.000000 =$  the v490 parity census (*two independent lattice witnesses*); per-layer gauging costs  $16 \ln 2$  (the offset measures  $\ln[\text{index}]$ , not a blob). *Haag-duality indicator*: fermionic  $S(E) = S(E')$  to  $8.2 \times 10^{-12}$ ; the orbifold *breaks* complementarity ( $D(E) - D(E') = 0.427$  at  $a=64$ ); the complementary-pair sum is exchange-symmetric and  $\leq \ln 4 = \ln \mu(SO(16)_1)$  with max 1.2637 at the self-dual  $x = \frac{1}{2}$  — a *double-limit* statement (the first naive fixed- $x$  check was wrong and was replaced — honestly typed in the module). *Condensation arithmetic anchored at both measured ends*:  $\det \text{Cartan}(D_5) \cdot \det \text{Cartan}(A_3) = 16$ ; KLM/Longo–Rehren  $16/4^2 = 1$  and  $16/2^2 = 4 \rightarrow 4/2^2 = 1$  (both routes exact);  $\Sigma d_i^2 = (4, 4, 1)$ ;  $\theta_v = 1$  exactly at  $\nu = 2c_- = 16$ , rivals  $\neq 1$  — so  $\mu = 1$  after condensation ([C] the condensed net is interacting, the v490 fence): KILL not triggered. *Negative controls*:  $\nu=1$  single Majorana (offset non-removable,  $\theta_v = e^{i\pi/8} \neq 1$  — the discriminator has teeth); trivial phase (all  $\leq 10^{-7}$ ); wrong sector sum  $\{1, P_{AB}\}$  (offset exactly 0). [O] WP5d- $\beta$  (continuum Haag duality / local normality / split of the quotient net; runs as exploration) and WP5e (Costello–Li) stay open.
- **Contract + ledger + Wolfram + website.** `tfpt_research_contracts`: the WP5d bullet  $\rightarrow$  “ $\alpha$  stage done” with the core numbers and `v501_celestial_wp5d_two_interval_index.py`; status line and honest-success paragraph now “four of five milestones landed; WP5e (twistor

uplift) is the remaining open piece”. **tfpt\_3** celestial section: new Step 8 (one paragraph, conservative); **tfpt\_4** keystone note updated to WP1–WP5d. Ledger: new row CELEST.WP5D.01 (fine types Formal + Lattice/Numerical + **[C]** dictionary); CELEST.SEAM.01 status → “WP1–WP5d executed; WP5e **[O]** (single remaining milestone)”. Wolfram: four exact mirrors (Cartan determinants; KLM/Longo–Rehren quotients;  $\Sigma d^2$  + index-reading arithmetic; 16-fold-way  $\theta_v$ ) verified with the active engine the same day (412 → 416, ALL WOLFRAM EXTENSION CHECKS PASSED); the lattice curves stay Python-only (flagged). wolfram/README.md, GATE.WOLFRAM.02, tfpt\_safeguards Layer-6 count and the website counters (116/116 + 416/416) updated; VerificationDag qft node gains v501; registry + run\_all (495 modules, suite green). SEAM.EQUIV.01 untouched; no marker moves.

**2026-07-21 (X) (v500 — WP5c of CELEST.SEAM.01 “the GNS limit state” promoted as CELEST.WP5C.01: the quasi-free family  $\omega_w$  exists, is positive, stabilises exactly at the WP5a threshold, and its limit carries the null ideal in its GNS kernel — the full 9361-block exact level-2 Gram has rank 4124 exactly with kernel 27000 =  $V(2\theta)$  weight by weight;  $|s\rangle$  IS the GNS zero vector (the WP5b S4.10 tension resolved at the state level); a CCR obstruction proves the family formulation *necessary*; preregistered SUCCESS criterion met, KILL not triggered; Wolfram extension 408 → 412 (verified engine run); SEAM.EQUIV.01 stays **[O]**, no marker moves)**

- **v500\_celestial\_wp5c\_gns\_limit\_state.py** (CELEST.WP5C.01, **35 checks, exact integer/Fraction, no floats, deterministic**). WP5a (v497) made the boundary limit precise, WP5b (v498) built the deleting operator; WP5c constructs the *state*.  $\omega_w$  is quasi-free: loop sector = the affine  $k=1$  vacuum  $n$ -point functions via the *machine-determined* compact anti-involution  $\theta(e_\alpha) = -e_{-\alpha}$  (the unique sign that is an anti-automorphism on all 61504 basis pairs; contravariance exact); radial sector = oscillator pairings  $x^{wr}$  (the exact Gibbs regulator,  $x$ -adic). *Convergence sharp*:  $\omega_w = \omega_\infty \bmod x^{N+1}$  for all  $w \geq N+1$  ( $N=8$ ), not yet at  $w = N$  — the WP5a threshold  $w = n+1$  holds *identically at the state level*; positivity of every  $\omega_w$  checked. *The kernel (core, fully machine)*: the complete exact level-2 Gram — 31124 monomials in 9361 weight blocks (profile 164:1, 37:240, 7:2160, 1:6960) — has level-1 rank 248 positive definite (Cartan block = the  $E_8$  Cartan matrix, det 1), level-2 rank 4124 *exactly* (the preregistered target check), kernel 27000, every block PSD; rank per Weyl orbit (0, 0, 1, 8, 44); kernel =  $V(2\theta)$  *weight by weight* (Freudenthal cross-check on all 9361 weights); highest-weight norms (3844, 49, 2, 0) at  $(0, \omega, \theta, 2\theta)$ ; direct ideal checks  $F_i|s\rangle$ ,  $F_j F_i|s\rangle$  norm 0.  $\mu_4$ -equivariance:  $\omega_w \circ \text{clock} = \omega_w$  for all  $w$ ; the clock descends to GNS; level-2 rank split by clock class (1036, 1024, 1040, 1024) =  $\Theta_{C_j}/\eta^8$  at  $q^2$  — the two-routes identity at the *state level*. *Resolution of WP5b S4.10*:  $|s\rangle$  is the zero vector of  $\text{GNS}(\omega_\infty)$  — the twisted two- $C_1$ -mode floor (6 quarters =  $q^{3/2}$ ) concerns only pre-limit objects (GNS has no state there,  $6 \bmod 4 \neq 0$ ). *Negative controls*:  $k=2$  ( $\langle s|s\rangle = +4$ , dominant blocks full rank — no ideal in the kernel);  $k=0$  (level-1 Gram  $\equiv 0$  — no 248 layer without the central extension);  $D_8$  (rank  $2076 = 7380 - 5304$ , kernel =  $V(2\theta_{D_8})$  PSD, but  $h = (0, \frac{1}{2}, 1, 1)$ : one block of four — one-block completeness stays  $E_8/\mu_4$ -specific); wrong family  $E'_w = mw + r$  (the 248 layer erased,  $710955 \neq 248$ ); no damping ( $897266 \neq 248$ ); *CCR obstruction*: at fixed radial pairing  $c > 0$  positivity forces  $\omega(bb^*) \geq c$  — no  $w$ -uniform damping state exists, the family formulation is *necessary* and the family *exists*: the preregistered KILL is probed and not triggered. **[C]** quasi-free extension to the full  $*$ -algebra; Kac positivity beyond level 2. **[O]** local normality, net structure/Haag duality (WP5d), Costello–Li (WP5e).
- **Contract + ledger + Wolfram + website**. **tfpt\_research\_contracts**: the WP5c bullet → done with the core numbers (state-level threshold  $w = N+1$ , the full 9361-block Gram rank 4124/kernel 27000 weight by weight, the clock-class split, the CCR necessity of the family), citing v500\_celestial\_wp5c\_gns\_limit\_state.py; the WP5b handover marked answered. **tfpt\_3** celestial section: new Step 7 (one paragraph, conservative). Ledger: new row CELEST.WP5C.01 (fine type Formal/Identity **[E]** + **[C]** identifications + **[O]** net questions). Wolfram: four exact

mirrors (level-2 block census from the root data; rank/kernel arithmetic with  $\dim V(\omega) = 3875$  by Weyl; the glue-frame clock-class split (1036, 1024, 1040, 1024); threshold + level-dial +  $D_8$  census) verified with the active engine the same day (408  $\rightarrow$  412); the exact Gram/PSD computation stays Python-side (flagged). VerificationDag qft node gains v500; registry + run\_all (494 modules at this step, suite green). SEAM.EQUIV.01 untouched; no marker moves.

**2026-07-21 (IX) (v499 — the P2 weight-typing postulate HARDENED: the anchor  $a = (1, 1, 2)$  typed as the Birkhoff–Grothendieck splitting exponents of the Deligne canonical extension, so integrality/positivity/sum-4 become theorems given the [E] anchors; the v491 check-9 residual sharpened, the tfpt\_2 “balanced” selection typed as stability; Wolfram extension 403  $\rightarrow$  408 (verified engine run); no gate closes, no marker moves — AX.P2.01 stays a declared axiom, QGEO.SYM stays [O])**

- **v499\_p2\_weights\_deligne\_bg.py** (P2.TYPING.01, 23 checks, sympy exact, no floats). v491 reduced P2 to “3 positive integers summing to 4”; its residual was the *typing* (why integers, why positive, why sum 4). Now typed: given the [E] anchors (4 marks, v216; rank  $3 = \# \text{marks} - 1$ , v137/v228; cusp-class local monodromy  $\lambda^3 - 1$  and exact irreducibility  $|\langle U, M_0 \rangle| = 24 = |S_4|$ ,  $\sum_g |\chi|^2 = 24$ , v117; unitarity (U)), the Deligne canonical extension  $E$  of the flavor connection satisfies:  $\deg E = -4$  as a theorem (residue exponents  $\{0, \frac{1}{3}, \frac{2}{3}\}$  the unique lift to  $[0, 1)$ , trace 1 per mark; par-deg 0;  $\infty$  apparent, monodromy product = 1 exactly);  $h^0(E) = 0$  as a theorem (Mehta–Seshadri stability; all 324 section-saturation cases; BG dictionary  $h^0=0 \Leftrightarrow a_i \geq 1$ ); integrality = Birkhoff–Grothendieck. Hence  $\{a\} = \{1, 1, 2\}$  unique and  $g_{\text{car}} = e_2 = 5$ . *Selector typed*: the Schur–Horn permutohedron has integer points exactly  $\{(2, 2, 0), (2, 1, 1)\}$  (the two options of the tfpt\_2 “Global computation” theorem) and  $h^0=0$  kills  $\{0, 2, 2\}$  — the “balanced/minimal-variance” heuristic is now stability/positivity. *The  $h^1$  discrepancy, real and resolved*:  $h^1(\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2) = 1 \neq 3$ , so “generations =  $H^1$  of the rank-3 bundle” is FALSE (negative control: forcing it demands  $\deg -6$  and loses uniqueness,  $e_2 \in \{9, 11, 12\}$  never 5); correct two-object typing:  $L = \mathcal{O}(-D)$  carries the generations ( $h^1(L) = 3 = N_{\text{fam}}$ , the v137 basis =  $H^0(\Omega^1(\log D))$ ),  $E$  carries them as its fiber. *Biswas explicit* (independent witness): the weight denominator  $3 = N_{\text{fam}}$  forces the  $\mathbb{Z}_3$  cusp cover  $y^3 = (z-1)(z-i)(z+1)^2(z+i)^2$  (genus 2; NOT the pillowcase double, genus 1,  $j = 1728$  — the deck side);  $p_*\mathcal{O}_Y = \mathcal{O} \oplus \mathcal{O}(-2)^2$ ; the regular-representation model lands on  $\{0, 2, 2\} =$  the *unstable* Schur–Horn companion ( $h^0 = 1$ ): integrality and the sum rule come out of the correspondence, stability is the selector. *Negative controls*: no positivity  $\rightarrow$  4/31 solutions;  $\deg -3/ -5$  breaks;  $n$ -mark scan  $e_2 = (n-2)(n+1)/2 = 5$  ONLY at  $n = 4$  (the value, not the uniqueness, is the  $n=4$  content); MS-rational  $e_2 = \frac{16}{3}, \frac{44}{9}$ . *Honest residuals* (P2 stays an axiom): R1 module identification [C] (QGEO.SYM territory), R2 (U) [C] (v33/v40), R3 analytic monodromy (Numerical, v117 check 4); non-circular (neither v23’s splitting input nor v115’s  $\text{diag } A_0 = a/4$  input used); the v491 direction-reversal note carries over. Ledger: new row P2.TYPING.01; the v491/P2.PARTITION.01 row records “typing residual sharpened by v499” (v491 stays active, no supersede). Registry + run\_all (493 modules, suite green).
- **Wolfram second path + papers + website.** Five exact mirrors added to the extension file `tfpt_readouts_extension.wl` (partition enumeration under the full typing + naive-typing control +  $h^1$  discrepancy; Deligne residue-trace degree with the  $A_0^*$  witness char poly; Schur–Horn lattice points; Biswas  $\mathbb{Z}_3$ -cover degree arithmetic incl. deck double  $j = 1728$ ; the  $n$ -mark formula), verified with the active engine the same day (403  $\rightarrow$  408, ALL WOLFRAM EXTENSION CHECKS PASSED); wolfram/README.md + GATE.WOLFRAM.02 + tfpt\_safeguards Layer-6 count + website counters (116/116+408/408). Papers (one sentence each, conservative): introduction anchor paragraph, tfpt\_1 (5, 3) keybox, tfpt\_2 H2 winbox and the “Global computation” theorem (the balanced selection typed as stability), origin\_theory (5, 3) keybox — all citing v499\_p2\_weights\_deligne\_bg.py; no status card changes. Website: VerificationDag P2

node gains **v499**; generated indices via build; no marker moved anywhere.

**2026-07-21 (VIII) (Editorial: dedicated celestial-route section in Paper 3 + website mirror — the scattered celestial sentences consolidated into one coherent six-step section; *no new scripts, no status changes, no marker moves, SEAM.EQUIV.01 untouched*)**

- **tfpt\_3\_e8\_audit\_bootstrap:** new section “The celestial route: from the  $\mu_4$  clock to the  $(E_8)_1$  boundary shadow”. The celestial-contract material that had accumulated as one long paragraph inside “The Möbius origin of the irreducible  $\pi$ ” is consolidated into a dedicated section (placed after the bootstrap audit, before “Honest status”), telling the executed WP1–WP5b story in six typed steps: (1) setup — the  $E_8$   $\mu_4$ -glue as a flat  $\mathbb{Z}_4$  monodromy on the  $A_3$  ALE space,  $\text{clock}^2 = \text{deck}$  ( $\text{spin } \mathbb{Z}_8, 8 = 2|\mu_4|$ ) [E] (v492); (2) deformation —  $XY = Z^4 + a_0$  pure scale,  $j = 1728$  frozen,  $\text{clock} = \text{Picard–Lefschetz/Coxeter monodromy}$  [E] (v493); (3) consistency honestly demoted —  $(\kappa/c_3)^2 = 12$  exact but the alignment format passes 8/8: alignment survives, selectivity does not [E]/[C] (v495); (4) type mismatch + boundary limit — the  $(E_8)_1$  character is not a jet-grading conformal block (growth  $n^{2/3}$  vs  $n^{1/2}$ ) but the  $\mu_4$  period sum 248 holds exactly: a boundary-limit shadow of SEAM.EQUIV.01/MMST shape [E] (v496); (5) the null ideal quantitative — stabilisation threshold  $w = n+1$ ,  $27000 = (h^\vee)^3$  derived, quotient = sector sum = 4124,  $SO(16)_1$  contrast [E] (v497); (6) the deleting operator explicit —  $|s\rangle = (E_{-1}^\theta)^2|0\rangle$ ,  $J_1^a|s\rangle = 0$  on all 248 (tally 190/57/1), level dial  $2(1-k)$ , one-block closure  $E_8/\mu_4$ -specific [E] (v498); plus the WP5c–e roadmap and the honest status line (the route does *not* close SEAM.EQUIV.01; the GNS/limit question stays [O]). All pre-existing script citations (v492–v498) are carried into the new section; the Möbius section now points to it with one sentence. **tfpt\_research\_contracts:** a reader’s note at the head of the CELEST.SEAM.01 contract cross-references the new Paper-3 section (the contract stays the full technical reference). **tfpt\_4\_frontier:** one sentence in the metric-sector keybox records the celestial route as a third, quantitative approach to the keystone (not closing it).
- **Website mirror. papers.ts:** the Paper-3 section list gains the celestial-route section (six-step summary in the existing style) and a “Celestial route” highlight; the contracts mirror gains the same cross-reference sentence. **glossary.ts:** three new entries (celestial route, ALE space, null ideal) in the established technical style. Generated indices via build; no status component changes (no marker moved anywhere).

**2026-07-21 (VII) (CELEST.SEAM.01 WP5b executed and promoted: v498 — “the singular vector as an operator”, the deleting object of the v497 null ideal constructed explicitly and machine-verified singular; contract now WP1–WP4 + WP5a + WP5b done, WP5c–e [O] (roadmap); Wolfram extension 398 → 403 (verified engine run); *no gate closes, no marker moves, SEAM.EQUIV.01 untouched*)**

- **v498\_celestial\_wp5b\_singular\_vector.py** (CELEST.WP5B.01, **53 checks, exact integer/Fraction, deterministic, no floats**). Success on the preregistered WP5b criterion: the deleting object exists and is explicit. *Construction:*  $|s\rangle = (E_{-1}^\theta)^2|0\rangle$  (weight  $2\theta$ , level  $2 = 8$  quarter units =  $q^2$ , an *integer* level) in a machine-built Chevalley/Frenkel–Kac basis — bimultiplicative 2-cocycle with asymmetry verified on all 57600 ordered root pairs, the  $[e_\alpha, e_{-\alpha}]$  sign *forced* by the Jacobi identity ( $\text{SGN} = -1$ , not chosen), the invariant form  $\kappa$  derived ( $\kappa(\theta^\vee, \theta^\vee) = 2$ ), Jacobi exact on all 57600  $\mathfrak{sl}_2$ -slice triples + 6000 seeded + all 496  $\theta$ -adjacent, all  $N_{\alpha\beta} = \pm 1$ . *Singularity (machine):*  $J_1^a|s\rangle = 0$  for all 248 basis elements (0 violations), with the case classification 190 (first bracket) / 57 (second) / 1 ( $a = F^\theta$  via the central-term cancellation — the *only* case that sees the level  $k$ ); plus  $J_2^a|s\rangle = 0$ ,  $E_0^a|s\rangle = 0$ , exact weight, Shapovalov norm 0; the affine PBW engine unit-tested on all 61504 basis pairs. *Level dial:* the  $F_1^\theta$  coefficient



is  $2(1-k) - 2/0/-2$  at  $k = 0/1/2$ : without the central extension the deletion operator does not exist.  $\mu_4$  data: glue class  $j(\theta) = 1$  (machine-built  $h$ -adapted chamber,  $\langle \theta, h \rangle = 5$ , height  $29 = h^\vee - 1$ ), clock phase  $i^{2j} = -1$ , class( $2\theta$ ) = 2 sheet-even; the deleting  $\theta$ - $\mathfrak{sl}_2$  crosses the sheet-odd classes (1, 3) while its square is sheet-even; cross-table  $\theta$ -grading  $\times$  glue class consistent (rows (1, 56, 126, 56, 1), columns (52, 64, 60, 64)); 8 quarters =  $q^2$  via the per-period dictionary. *Level-2 match*: weight  $2\theta$  has multiplicity 1 in the level-2 Fock, so  $U(\mathfrak{g})|s\rangle$  is the  $27000 = (h^\vee)^3$ ; the direct  $\mathfrak{g}_0$ -orbit BFS matches the Freudenthal multiplicities of  $V(2\theta)$  exactly through depth 4 ([C] Weyl complete reducibility beyond); quotient  $31124 - 27000 = 4124$ ; lattice reading  $(2\theta, 2\theta)/2 = 4 > 2$ :  $|s\rangle$  is the kernel generator of  $\text{Fock} \rightarrow E_4/\eta^8$ . *Negative controls (honest separation)*: the level-1 current state  $E_{-1}^\theta|0\rangle$  (one witness,  $\rightarrow -k|0\rangle$  — the 248-current state that must remain) and 4 generic level-2 states (14–33 witnesses) are *not* singular; at  $k=2$  the *cube* is (generic Kac  $(E^\theta)^{k+1}$  mechanics, honestly typed);  $SO(16)_1$  has the same singular vector but keeps three extra level-1 primaries ( $h = \frac{1}{2}$  breaks one-block fusion;  $5304 \neq 14^3$ ) vs  $E_8$  (comarks all  $\geq 2$ , glue weights (0, 1, 1, 1) integer) — the *one-block closure* is the  $E_8/\mu_4$ -specific part. *Honest [O]*: in the twisted quarter-slot moding two sector- $C_1$  modes never sum to 8 quarters (minimum  $6 = q^{3/2}$ ) — the per-period dictionary (v496), not the per-slot identification, carries  $|s\rangle$  to  $q^2$ : exactly the WP5c GNS/limit-state question; WP5c-e stay open; SEAM.EQUIV.01 untouched.

- **Contract + surfaces synced.** `tfpt_research_contracts`: the WP5b bullet moved to *done* with the key numbers (case tally 190/57/1; level dial  $2(1-k)$ ;  $j(\theta) = 1$ , clock phase  $-1$ ; orbit BFS through depth 4; the  $D_8$  contrast) and the twisted-slot tension as the explicit WP5c handover; honest status line and K4 paragraph updated (“WP1–WP4 + WP5a + WP5b executed; WP5c-e [O]”). `tfpt_3_e8_audit_bootstrap`: the celestial-contract paragraph extended by the WP5b sentence. Ledger: CELEST.WP5B.01 (new) + CELEST.SEAM.01 (“WP1–WP4 + WP5a + WP5b executed; WP5c-e [O] (roadmap)”). Wolfram: 5 exact v498 mirrors added to `tfpt_readouts_extension.wl` and *verified* with the active engine the same day (398  $\rightarrow$  403, ALL WOLFRAM EXTENSION CHECKS PASSED); `wolfram/README.md` + GATE.WOLFRAM.02 counters advanced; `tfpt_safeguards` Layer-6 count + website mirrors 116/116 + 403/403. Website: `papers.ts` contract mirror (“WP1–WP4 + WP5a + WP5b done, WP5c-e roadmap”), DAG `qft` node + v498, generated indices via build. Suite now 492 scripts, all green; SEAM.EQUIV.01 stays [O].

**2026-07-21 (VI) (CELEST.SEAM.01 WP5a executed and promoted: v497 — “the null ideal from the limit”, the v496 boundary-limit shadow made a precise coefficientwise limit and the level-2 null ideal derived from root data; contract WP5 subdivided, now WP1–WP4 + WP5a done, WP5b-e [O] (roadmap); Wolfram extension 394  $\rightarrow$  398 (verified engine run); no gate closes, no marker moves, SEAM.EQUIV.01 untouched)**

- **v497\_celestial\_wp5a\_null\_ideal.py** (CELEST.WP5A.01, 34 checks, exact integer/Fraction, no floats). Three results make the WP4 limit reading precise. *Stabilisation theorem*: the one-parameter family  $\chi_w$  of graded Fock characters on the chiral jet generators (loop energy  $m$ , radial number  $r$ ;  $E_w = m + wr$  in quarter units  $u = q^{1/4}$ ) contains the chiral jet grading as its  $w=2$  member (generator counts 64, 120, 128, 180, 192, 240, 256, 300 per degree), and its  $u^n$  coefficient equals the quarter-moded loop Fock for *all*  $w \geq n+1$ , strictly larger for every  $w \leq n$  (verified  $n \leq 8$ ,  $w \leq 10$ ) — the v496 “boundary-limit shadow” is a precise *coefficientwise* limit  $\varepsilon = 1/w \rightarrow 0$  with explicit threshold  $w = n+1$ , not a slice (convention clarified: 897266 = the quarter-moded loop Fock at  $u^4 = \text{integer level 1}$ ). *The null ideal derived, not cited*: Freudenthal multiplicities + the Weyl dimension formula + generic character peeling give  $\text{Sym}^2(248) = V(2\theta) \oplus V(\omega) \oplus V(0) = 27000 + 3875 + 1$  with residual exactly zero (independent Weyl-orbit cross-check);  $27000 = 30^3 = (h^\vee)^3$  exactly;  $C_2 = (124, 96)$ , Dynkin indices 6750/750; level-1 integrability ( $\langle 2\theta, \theta^\vee \rangle = 4$ ,  $\langle \omega, \theta^\vee \rangle = 2$ ) leaves the vacuum as



the only level-1 primary (Kac, used only through its counting consequence); level-2 quotient  $31124 - 27000 = 4124 = 1 + 248 + 3875$  exactly. *Two routes, one number*: the  $\mu_4$  theta-split sector sum at  $q^2 = (1036, 1024, 1040, 1024)$  gives the *same* 4124 — the untwisted null-ideal quotient and the boundary/twisted lattice presentation count one space ( $q^3$ :  $34752 = \chi_3$ ). Negative controls:  $SO(16)_1$  through the same pipeline gives *four* components ( $5304 + 1820 + 135 + 1$ ),  $5304 \neq 14^3$  (the  $h^{\vee 3}$  identity is not generic), quotient  $2076 = \Theta_{D_8}/\eta^8$  at  $q^2$  (the recipe validated on a second algebra), block weights  $(0, \frac{1}{2}, 1, 1)$  break one-block fusion (vs  $E_8$ 's integer  $(0, 1, 1, 1)$ ); only  $P = 4 = |\mu_4|$  periodisation reproduces the 248 layer; swapped glue breaks the zero-mode anchor ( $64 \neq 60$ ). *Honest [O]*: the limit does *not* generate the truncation — the singular vector as an *operator* in the limit net (WP5b), the GNS limit state (WP5c) and local normality/Haag duality (WP5d) stay open; the celestial route now has the same two-step shape as MMST (limit + maximal ideal), with the ideal quantitatively fixed; SEAM.EQUIV.01 untouched.

- **Contract + surfaces synced.** `tfpt_research_contracts`: WP5 subdivided — WP5a moved to *done* with the key numbers (stabilisation threshold  $w = n+1$ ;  $27000 = (h^{\vee})^3$  derived; quotient = sector sum = 4124;  $SO(16)_1$  contrast), WP5b–e added as roadmap bullets with success/kill criteria (WP5b singular-vector operator in the quarter-moded current Fock; WP5c GNS limit state, kernel  $\supseteq$  ideal; WP5d local net/Haag duality via the KLM route; WP5e BCOV/Costello–Li twistor uplift, partition function =  $E_4/\eta^8$  incl. the  $q^{-1/3}$  prefactor); K4 paragraph and the honest-success status line updated. Ledger: CELEST.WP5A.01 (new) + CELEST.SEAM.01 (“WP1–WP4 + WP5a executed; WP5b–e [O] (roadmap)”). Wolfram: 4 exact v497 mirrors added to `tfpt_readouts_extension.wl` and *verified* with the active engine the same day ( $394 \rightarrow 398$ , ALL WOLFRAM EXTENSION CHECKS PASSED); `wolfram/README.md` + GATE.WOLFRAM.02 counters advanced; `tfpt_safeguards` Layer-6 count + website mirrors  $116/116 + 398/398$ . Website: `papers.ts` contract mirror (“WP5a done, WP5b–e roadmap”), DAG `qft` node + v497, generated indices via build. Suite now 491 scripts, all green; SEAM.EQUIV.01 stays [O].

**2026-07-21 (V) (comb-meta-limit re-run with the PG.06b FULL Vela recovery ingested — the stale single-obs PG.06b reference (noise-peak  $F_0 = 11.19275$  Hz) corrected in the channel definitions; the new X-ray channel enters *censored* (no absolute  $\varepsilon$ ), both aggregate limits numerically unchanged; *experiments level only* — *no verification module, no gate closes, no marker moves*)**

- **experiments/comb-meta-limit channel update (methodology untouched).** The Vela X-ray channel (`channels.py`) now ingests the PG.06b FULL reduction (`pg06b_full_vela.json` in `pulsar-glitch-recovery/results`: 665 obs  $\rightarrow$  491 usable  $\rightarrow$  119 segment-coherent  $\nu(t)$  points) instead of the superseded single-obs `pg06b_vela.json`; the stale code note quoting the noise-peak “detection”  $F_0 = 11.19275$  Hz is corrected on record (true Vela  $F_0 = 11.1861692$  Hz), and the channel key is fixed from the mislabel PG.07 to PG.06b. Censoring handled exactly by the existing Tier-B rule (nothing invented): the comb observable is a *linear*  $\delta\nu(\tau)$  residual ( $\mu$ Hz, not a fractional ln-flux modulation), both legs sit below the 2.8-period gate (2019: 2.17, 2021: 1.38), and the surrogate-calibrated injection never reaches  $\varepsilon_{90}$  below  $\varepsilon = 1$  ( $\varepsilon_{50} = 0.55$  in 2019 only)  $\Rightarrow$  no absolute per-channel  $\varepsilon$ -UL exists for the DL/HKSJ combination; the channel stays enumerated-only (“no constraint”), carrying the normalised context amplitude  $\sqrt{2g} = 0.40$  (leg gains 0.046/0.114, min  $p = 0.014$  — the 2019 audit-level chance feature: Bonferroni  $\times 10 = 0.14$ , shuffle  $p = 0.40$ ).
- **Full re-run (seed 0, deterministic): both limits numerically unchanged.** Boundary/horizon-scoped: `data_limited` (still no horizon channel with an absolute  $\varepsilon$ ); all-channel universal-DSI 95% UL  $\varepsilon < 0.11954$  ( $\approx 12.0\%$ ; HKSJ  $t_2 = 2.92$ ,  $\rho = 0.3$ -inflated 0.137;  $6.91\times$  the predicted  $\varepsilon = 0.0173$ ), still driven by the surface log-flux channels A1/A4/A5 (A4 GRB the anchor, per-channel UL 0.108); now 10 channels enumerated / 3 usable. Injection self-consistency unchanged (inject 5%  $\rightarrow$  recover 4.8%, UL 5.5%; coverage 0.92/150 trials). Surfaces synced: experiment README (tier table, ingest list,

PG.06b FULL bullet), `results/results.json`, scorecard row `data_value/source` via `build_evidence_scorecard.py` (status stays `data_limited`; counters unchanged, 120 rows), `experiments/README.md` catalog row. No claim; firewall intact.

**2026-07-21 (IV) (Wolfram engine reactivated — the deferred second-path mirrors VERIFIED: base 116/116 + extension 394/394 (378 → 394, the 16 mirrors accumulated since 2026-07-14 now counted); GATE.WOLFRAM.01/02 and all counter surfaces synced; *no gate closes, no marker moves*)**

- **Verified engine run 2026-07-21 (Wolfram Engine 14.3, license reactivated).** `wolframscript -file verification/wolfram/tfpt_readouts.wl` → 116 passed, 0 failed, ALL WOLFRAM CHECKS PASSED; `tfpt_readouts_extension.wl` → 394 passed, 0 failed, ALL WOLFRAM EXTENSION CHECKS PASSED. The 16 exact mirrors added while the engine was unactivated all enter the counted total (378 → 394): v479 Kronheimer bridge (2), v491 partition corollary (3), v493 celestial WP2 clock deformation (4), v495 celestial WP3 Green–Schwarz alignment (3), v496 celestial WP4 character shadow (4).
- **Counter surfaces synced.** `verification/wolfram/README.md` (current-state block + the five deferral notes resolved; v484/v485/v486/v487/v488/v492 stay mirrorable-but-unmirrored, Python-exact, so they are *not* counted); ledger GATE.WOLFRAM.01 (re-verified 2026-07-21) + GATE.WOLFRAM.02 (394/394, UPDATE note); `tfpt_safeguards` Layer-6 count + website `papers.ts` mirrors (116/116 + 394/394); v479 registry/`run_all` deferral note resolved. No numeric claim changes; the second computational path is simply verified-current again.

**2026-07-21 (III) (CELEST.SEAM.01 WP3 + WP4 executed and promoted: v495 Green–Schwarz alignment (verdict B, honestly non-selective) + v496 character-vs-S-algebra type mismatch (verdict B(ii), boundary-limit shadow); contract now WP1–WP4 done, WP5 [O]; *no gate closes, no marker moves, SEAM.EQUIV.01 untouched*)**

- **v495\_celestial\_wp3\_gs\_alignment.py (CELEST.WP3.01, 25 checks, exact Fraction/sympy).** The pre-registered WP3 “alignment test only”: the Okubo coefficient  $\text{Tr}_{\text{adj}} X^4 / (\text{Tr}_{\text{adj}} X^2)^2 = 5/(2(\dim \mathfrak{g} + 2))$  is *derived* as an exact polynomial identity for all 8 algebras on Costello’s list ( $\text{sl}_2, \text{sl}_3, \text{g}_2, \text{so}_8, \text{f}_4, \text{e}_6, \text{e}_7, \text{e}_8$ ;  $\text{e}_6/\text{e}_7$  carved out of the v492  $E_8$  root set; negative control:  $\text{sl}_4$  fails,  $5/32$  vs  $3/32$ ); the closed form  $\tilde{\lambda}^2 = 10h^{\vee}/(\dim + 2) = h^{\vee} + 6$  holds exactly across the Deligne series;  $E_8$ :  $\tilde{\lambda} = 6$  (unit-trace) resp.  $\lambda_{\text{fund}} = 1/10$  (adjoint-trace) exact, so  $(\kappa/c_3)^2 = 12 = |\mu_4|N_{\text{fam}}$  resp.  $1/300$  — with  $\kappa/c_3$  itself *irrational* ( $2\sqrt{3}$ ). Byproduct: the printed  $\lambda_{\text{so}_8}^2 = 3/2$  in Costello’s appendix A is a factor-2 slip (his own weight content gives 3). **Look-elsewhere (the hard fence):** the alignment format passes 8/8 across the whole list (zero selectivity),  $\tilde{\lambda}$ -integrality  $2/8$  (shared with  $\text{sl}_3$ ), only  $g_{\text{car}}=5 \mid h^{\vee}$  is  $\text{e}_8$ -selective ( $1/8$ ), the isolating conjunction is post hoc. Kill test K3 does *not* fire but is scope-demoted: *alignment survives; selectivity does not* — compatibility, not  $\text{e}_8$ -selective evidence. Verdict B.
- **v496\_celestial\_wp4\_character\_shadow.py (CELEST.WP4.01, 25 checks, exact integer/Fraction/sympy).** WP4, the type mismatch head-on: the  $(E_8)_1$  character  $E_4/\eta^8 = (1, 248, 4124, 34752, 213126, \dots)$  is *not* a conformal block of the glue-equivariant  $E_8[\mathbb{C}^2]$  S-algebra in its own jet grading — localised three ways: (a) spin  $1 - d/2$  unbounded below and 248 never a tower dimension ( $d \leq 100$ ); (b) growth  $n^{2/3}$  vs  $n^{1/2}$  (quadratic cumulative count  $31s^2 + 92s + 60$ ,  $31 = k + h^{\vee}$ ;  $f_n/\chi_n$  strictly increasing  $n = 1..12$ ); (c) the level-2 null ideal deletes exactly  $27000 = 30^3 = (h^{\vee})^3$  out of  $\text{Sym}^2(248) = 1 + 3875 + 27000$  — no jet analogue. *But* the boundary/period reading holds exactly at the current stratum (zero modes =  $60 = C_0$  currents, jet slice cycles  $(60, 64, 60, 64)$ , one  $\mu_4$  period sums to 248, glue weights  $(0, 1, 1, 1)$  integer; free loop Fock  $897266 \gg 248$ , so the truncation must come from a limit) — the same MMST

scaling-limit shape as SEAM.EQUIV.01 (v336/v449). K4 fires only against the exact-block reading; the constructive limit question passes to WP5. Verdict B(ii).

- **Contract + surfaces synced.** `tfpt_research_contracts` WP3/WP4 bullets moved to *done* with the look-elsewhere demotion in the text, K3/K4 marked evaluated, status line now “WP1–WP4 executed (verdicts B, B, B, B(ii)), WP5 [O]”; one conservative sentence added to the `tfpt_3` celestial paragraph; ledger rows CELEST.WP3.01 + CELEST.WP4.01 (new) and CELEST.SEAM.01 updated; v495/v496 Wolfram mirrors added to `tfpt_readouts_extension.wl` (3 + 4 exact checks; count with the next verified engine run — engine still unactivated); website `papers.ts` contract highlight “WP1–WP4 done” + DAG `qft` node extended. Suite now 490 scripts, all green; SEAM.EQUIV.01 stays [O].

**2026-07-21 (II) (PG.06b FULL: the complete NICER Vela archive reduced — the first X-ray  $\nu(t)$  leg of the recovery-comb search, verdict `data_limited`; the old PG.06b “pipeline proven” fold corrected on record; *experiments level only* — no verification module, no gate closes, no marker moves)**

- **experiments/pulsar-glitch-recovery PG.06b FULL executed (`tfpt-pulsar vela-full`).** The full NICER Vela-pulsar (PSR B0833-45) L2 archive is reduced, with the manifest frozen *before* the bulk download (`manifest_full_reduction.json`: all 665 ObsIDs, preregistered 2019/2021/2024 glitch windows, 20-GB abort gate): 665/665 obs downloaded (2.6 GB, raw gitignored)  $\rightarrow$  491 usable  $\rightarrow$  240 per-obs detections  $\rightarrow$  **119 segment-coherent  $\nu(t)$  points** (108 fully coherent), median  $\sigma_\nu \approx 0.6 \mu\text{Hz}$ ; barycentring validated against PINT to  $3.4 \mu\text{s}$  — the *first X-ray  $\nu(t)$  probe* of the search program. Sanity gate PASS:  $\Delta\nu/\nu$  2019 =  $2.39 \times 10^{-6}$  vs  $2.5012 \times 10^{-6}$  published, 2021 =  $1.272 \times 10^{-6}$  vs  $1.26 \times 10^{-6}$ ; the 2024 glitch is not reducible (only  $\sim 150$ -s snapshots, 9.7 ks — documented).
- **Frozen-kernel comb ( $\omega = 2.5827$ ,  $\varepsilon_{\text{pred}} = 0.0173$ ): verdict `data_limited` (range- and amplitude-limited).** Reach 2019: 2.17 comb periods ( $<$  the 2.8-period gate; beats the PG.08 public Vela reach 1.83), 2021: 1.38; the 2019 raw  $p = 0.014$  has the kernel as smallest- $p$  battery member BUT Bonferroni  $\times 10 = 0.14$  and shuffle  $p = 0.395 \Rightarrow$  chance-level by the preregistered escalation rule (audit flag, NOT a candidate); 2021  $p = 0.731$ ;  $\ln \tau$  stack  $p = 0.049$ /shuffle 0.307. Surrogate-calibrated injection at the real sampling and noise:  $\varepsilon_{50} \approx 0.55$  (2019 only),  $\varepsilon_{90}$  never reached below  $\varepsilon = 1$ , 0% power at the predicted  $\varepsilon = 0.0173$  — a weaker amplitude probe than the PG.08 radio residuals, but it independently reproduces the PG.06 range-blindness on real data. Second off-kernel flag:  $\lambda = e$  in the 2021 leg (raw  $p = 0.002$ , Bonferroni 0.018; the same non-kernel periodicity class as PG.08 J0742–2822 — flagged, not claimed). The sub-day early window that would clear the gate is absent from the archive (first usable obs 3.4 d / 7 d post-glitch); next decider:  $<3$ -d post-glitch X-ray observations of a future Vela glitch (eXTP class).
- **Correction on record.** The old PG.06b single-obs “pipeline proven” detection at  $F_0 = 11.19275 \text{ Hz}$  ( $H = 18.4$ ) was a noise peak; the true Vela frequency at that epoch is  $F_0 = 11.1861692 \text{ Hz}$  ( $H = 99.7$ , consistent with the phase-connected PuMA ephemeris). The pipeline remains proven — by the corrected fold. Scorecard row updated to “recovery comb on the full NICER Vela X-ray  $\nu(t)$  (PG.06b FULL)” (status stays `data_limited`; counters unchanged, 120 rows: 49 consistent / 9 tension / 26 null / 33 data-limited / 3 parked); `experiments/README.md` catalog + §1 prose, pulsar README, `experiments/next.txt` synced. No claim; firewall intact.

**2026-07-21** (the two 2026-07-20 prediction-of-record rows promoted to the suite: **v494** —  $\alpha_s$  running +  $\mu$ -distortion branch discriminator machine-checked, ledger rows **COSMO.RUNNING.01/COSMO.MUDIST.01**, prediction surfaces synced; *no gate closes, no marker moves*)

- **v494\_cosmo\_running\_mu\_record.py** (23 checks, all green). The  $\alpha_s$ -running and  $\mu$ -distortion scorecard rows of 2026-07-20 (IV) get their load-bearing module so the new paper/website statements are machine-backed. [E] leading-order identities (sympy exact):  $\alpha_s = dn_s/d \ln k = -2/N_*^2 = -r/6$ ,  $\beta_s = -4/N_*^3$  (NOT  $-8/N_*^3$ ); LO band  $N_* \in [50, 60]$ :  $[-8.0, -5.6] \times 10^{-4}$ . [E] exact slow roll on the Einstein-frame Starobinsky potential (third-order Lyth–Riotto + finite-difference cross-check < 1%):  $\alpha_s(51.4) = -7.09 \times 10^{-4}$ ,  $\beta_s(51.4) = -2.70 \times 10^{-5}$ . [C] data: Planck 2018 ( $-0.0045 \pm 0.0067$ ) +  $0.57\sigma$ ; record leg P-ACT-LB ( $+0.0062 \pm 0.0052$ )  $-1.33\sigma$  (watch channel: positive central value; kill: robust  $\alpha_s < -5 \times 10^{-3}$  or  $> +3 \times 10^{-3}$  at  $\geq 5\sigma$ ; any robust POSITIVE running kills the plateau branch). [C]  $\mu$ -distortion (Chluba window, +21% calibrated on the  $\Lambda$ CDM anchor): sharp branch  $N_*=51.4$   $\mu = 1.6 \times 10^{-8}$ , profiled record  $N_*=56.1$   $\mu = 1.94 \times 10^{-8}$ , band  $1.5\text{--}2.3 \times 10^{-8}$ ; branch split  $\Delta\mu = 3.0 \times 10^{-9} \Rightarrow 3\sigma$   $A_s/N_*$  branch decision at  $\sigma(\mu) = 10^{-9}$  (Voyage-2050 class; PIXIE baseline  $9 \times 10^{-8}$  does not reach; kill: robust  $\mu < 0.9 \times 10^{-8}$  or  $> 4 \times 10^{-8}$ ). [E] comb blindness (honest S15): the frozen log-comb ( $\omega = 2.5827$ ,  $\varepsilon = 0.0173$ ) is structurally invisible in the  $\mu$  window ( $|F(\omega)| = 0.033$ ,  $\delta\mu = 7.4 \times 10^{-12} = 0.007\sigma$  even at Voyage-2050). Fences:  $N_*$  stays an external reheating input (GATE.NSTAR.01); the frozen band  $[50, 60]$  stays the surface of record (v84/v86). Ledger rows **COSMO.RUNNING.01** and **COSMO.MUDIST.01** added.
- **Prediction surfaces synced (the missing mirror of 2026-07-20 (IV))**. **predictions.txt** gains the two rows ( $\alpha_s$  [C] with the  $-1.33\sigma$  P-ACT-LB pull and the positive-running kill;  $\mu$ -distortion [C] data-limited with the Voyage-2050 branch split); website **lib/predictions.ts** gains the matching cards **alpha-s-running** and **mu-distortion** (confrontation + experiment blocks pointing at **experiments/tfpt-discovery**); the cosmology DAG node lists **v494** and states the running/ $\mu$  readouts.
- **Paper bodies**. **tfpt\_1** (Inflation interface) gains the running consistency-relation paragraph ( $\alpha_s = -2/N_*^2 = -r/6$ , exact slow roll, pulls, positive-running kill) with `\veri{v494_cosmo_running_mu_record.py}`; **tfpt\_2** (inflation-pivot keybox) gains the  $\mu$ -distortion branch-decider sentence (sharp  $1.6 \times 10^{-8}$  vs profiled  $2.0 \times 10^{-8}$ ,  $3\sigma$  at Voyage-2050) with the same `\veri`. No marker of any existing claim changes. (v494 is numerical/quadrature, Python-only by the suite convention, flagged in the docstring; GATE.WOLFRAM.01/02 untouched.)

**2026-07-20 (VIII)** (E8 mass-ladder blind-recovery bed EXECUTED: verdict validated at the detector level — the Cartan–PF ladder pipeline blindly recovers the published Zamolodchikov–E8 spectra of  $\text{BaCo}_2\text{V}_2\text{O}_8$  (8/8) and  $\text{CoNb}_2\text{O}_6$  (6/8 in-window) and rejects every control class; *experiments level only — confirms Zamolodchikov E8, not TFPT; no verification module, no gate closes, no marker moves*)

- **experiments/e8-ladder-bed/ executed (preregistered hypotheses/e8\_ladder\_bed\_v1.yaml)**. The blind ladder detector (assignment DP + scale fit + 5000-ladder Monte-Carlo null; the same detector class TFPT uses on search surfaces, e.g. **hfqpo-ladder**) on the only public *known-answer* mass-ladder dataset:  $\text{BaCo}_2\text{V}_2\text{O}_8$  (Zou+ PRL **127**, 077201; INS at the 4.7 T 1D QCP, Fig. 3a extraction QA-validated, 17 peaks blind) recovers **8/8 rungs**,  $\chi^2/\text{dof} = 0.35$ , MC  $p = 0.0056$  (singles-only run  $p = 0.0004$ ;  $\sigma$ -variant robust);  $\text{CoNb}_2\text{O}_6$  (Amelin+ PRB **102**, 104431; verbatim THz values) recovers **6/8 in-window rungs**,  $\chi^2/\text{dof} = 0.52$ ,  $p = 0.0038$  ( $m_7/m_8$  lie outside the 0.6-THz low-pass window; the broad 0.43-THz ( $m_1+m_2$ )-continuum onset is correctly left unassigned). Cartan–PF self-test: the sorted PF eigenvector of  $2I - A(E_8)$  equals the Zamolodchikov ladder to  $1.5 \times 10^{-15}$  ( $m_2/m_1 = \varphi$  to  $2.2 \times 10^{-16}$ ; lowest Cartan



eigenvalue  $2 - 2\cos(\pi/30)$  to  $4.5 \times 10^{-14}$ ).

- **Control battery — every class rejected.** Scramble 0.5%/1.5% false-positive rate (200 log-gap permutations, near-duplicate accounting); offset  $\pm\{0.3, 0.5\}m_1$ : 0/8 detections; jitter dose-response  $\leq 5\%$  at the destructive 30% level; wrong ladders on the real data (harmonic 1..8,  $A_3$ -PF,  $D_8$ -PF, Airy confined-spinon) all beaten by E8; the synthetic Airy bed (the true zero-field physics of these compounds) yields **0** valid rungs. The one disclosed v1.1 criteria amendment (jitter as dose-response, scramble near-duplicates,  $\chi^2$ -gated wrong-ladder comparator) is fully documented in the hypotheses YAML with all v1 numbers preserved; the recoveries themselves were never touched.
- **Integration (firewall intact).** Scorecard row “E8 mass-ladder blind recovery bed” moves parked  $\rightarrow$  consistent (stage not\_applicable, tier analog\_positive\_control, leakage detector\_control — the executed-analog basket of qc-recovery-kernel and qgeo-eit-soff); counters 120 rows: 49 consistent / 3 parked; experiments/README.md catalog row + stats (generator); website predictions.ts EXPERIMENTS\_AUDIT counters; experiments/next.txt. This validates the ladder-detector pipeline — it confirms Zamolodchikov E8, *not* the TFPT axioms: no [E], no \veri, no ledger row.

**2026-07-20 (VII) (v493 — WP2 of CELEST.SEAM.01 executed: the clock-invariant deformation  $XY=Z^4+a_0$  is a PURE SCALE with the  $\tau=i$  pillowcase frozen, the  $\mu_4$  clock IS the Picard–Lefschetz/Coxeter monodromy of the seam deformation, and the BHS deformed-algebra pattern transfers  $\mathbb{Z}_2 \rightarrow \mathbb{Z}_4$  with no sector leak; verdict B, contract updated to WP2 done, no gate closes, no marker moves, SEAM.EQUIV.01 and the v216 residual stay [O])**

- **v493\_celestial\_wp2\_clock\_deformation.py (CELEST.WP2.01).** WP2 of the celestial contract, sympy exact (47 checks), promoted from the experiments/tfpt-discovery/probe celestial\_seam\_wp2\_clock\_deformation\_probe.py. *Geometry:*  $P(iZ)=P(Z)$  forces  $a_3=a_2=a_1=0$  sharply and two-sided ( $\prod_k (Z - i^k w) = Z^4 - w^4$ );  $XY=Z^4+a_0$  is smooth iff  $a_0 \neq 0$  (disc =  $256 a_0^3$ ); the branch points  $Z^4 = -a_0$  are one free  $\mu_4$  orbit in *every* fibre with cross-ratio 2; the binary-quartic invariants of  $w^2=Z^4+a_0$  are  $I=12a_0$ ,  $J=0$  *identically*, so  $j=1728$  and the  $\tau=i$  pillowcase shape (v168/v214) is *frozen* for all  $a_0$  —  $a_0$  is a pure scale, no shape modulus survives clock-invariance. *Periods/monodromy:*  $\Pi = t(i-1)(1, i, i^2)$ , all three vanishing spheres carry the same volume  $\sqrt{2}t$  (lockstep); the clock acts on  $H_2$  as a *Coxeter element* of  $W(A_3)$  (char  $x^3+x^2+x+1$ , eigenvalues  $\{i, -1, -i\}$  = the  $A_3$  exponents,  $h(A_3)=4=|\mu_4|=N_{\text{fam}}+1$ ), fixes no cycle ( $\det(A-1)=-4$ ), and *is* the Picard–Lefschetz monodromy of the family ( $a_0 \rightarrow e^{2\pi i} a_0$  gives the 4-cycle); the surviving direction is the weight-1  $\chi_1$ -Fourier diagonal  $(1, i, i^2)$ , *not* the naive invariant diagonal; the versal weights  $(a_3, a_2, a_1, a_0) = (1, 2, 3, 0)$  are the regular  $\mu_4$  representation. *Algebra (BHS  $\mathbb{Z}_2 \rightarrow \mathbb{Z}_4$ , arXiv:2305.09451):* mode dictionary  $\{w[m, n] : m \equiv n \pmod{4}\}$ ; the fibre bracket  $-4\text{Nambu}(XY-Z^4-a_0)$  anchored at the  $a_0=0$  orbifold ( $\{X, Y\}=16Z^3$ , Jacobi + Casimir exact, uniform clock-degree  $-1$ ); S-algebra corrections exactly *linear* in  $a_0$  at the  $\mathbb{Z}_4$  wrap, conserving the  $\mu_4$  grade — no sector leak, the v492 equivariant sector deforms consistently;  $a_0 \leftrightarrow$  BHS’s weight-0  $c^2$  slot; matrix factorisations deform along all orbit splittings, the clock fixing exactly the antipodal  $2+2$  sector (= the EH/ $\mathbb{Z}_2$  middle node,  $4=2 \times 2$ ). *Negative controls:*  $a_1 Z \Rightarrow j=0$  (hexagonal); pure  $a_2 Z^2$  stays singular ( $A_1$ );  $a_2 \neq 0 \Rightarrow j=1556068/81 \neq 1728$ ; explicit grading leak — doubly sharp. *Verdict B* via three named [C] identifications ( $-4\text{Nambu}$  = the  $k=4$  CCA bracket; period = root difference; seam-scale reading via  $\text{clock}^2=\text{deck}$ ); the v216 residual is *typed*, not moved (given the order-4 clock the square is automatic — a relocation of the same input, no new derivation).
- **Contract updated to WP2 done.** tfpt\_research\_contracts CELEST.SEAM.01 section: WP2 bullet “finite, next”  $\rightarrow$  *done (verdict B)* with the two sharpenings (clock = Coxeter/Picard–Lefschetz monodromy; surviving direction =  $\chi_1$ -diagonal,  $a_0$  pure scale); kill-test paragraph K1



“survived by WP1 and WP2”; status line CELEST.WP2.01 [E]/[C] executed, WP3–WP5 stay [O]. Ledger rows CELEST.WP2.01 (new) + CELEST.SEAM.01 (WP2 done) + GATE.WOLFRAM.02 (four exact v493 mirrors added to tfpt\_readouts\_extension.wl, counted with the next verified engine run — engine unactivated, same convention as v491); registry + run\_all (487 modules, suite green); website mirrors (papers.ts contract section, generated script index). SEAM.EQUIV.01 stays [O]; no marker moves anywhere. Exact (sympy).

**2026-07-20 (VI) (HFQPO “third tooth”: the preregistered archival RXTE PCA scan has RUN — a well-powered null with limits; scorecard row S1.e data\_limited→null, no marker moves)**

- **H3 stage 2 executed (experiments/hfqpo-ladder/results/archival/).** The preregistered archival scan (hypotheses/hfqpo\_v1.yaml, archival\_design) ran end-to-end on public HEASARC data: 77 ObsIDs / 4.42 GB across all four 3:2-pair sources (GRO J1655–40, XTE J1550–564, GRS 1915+105, H1743–322). Sanity gate PASS (11/12 published pair lines reproduced, e.g. 450.3 Hz at 5.37% rms  $4.4\sigma$ , 280.1 Hz at 3.71%  $9.9\sigma$ ); injection calibration PASS in all four sources (90%-recovery sensitivities 5.11/2.68/0.89/1.65% rms at the tooth frequencies).
- **Blind-scan verdict: NULL with limits.** Neither the geometric tooth  $\nu_3 = 1.5\nu_u$  (661.5/414/252/363 Hz) nor the integer control line  $4\nu_0$  (588/368/224/322.7 Hz) is detected in any source ( $\sim 0\sigma$  single-trial; trials correction  $N=8$  as preregistered).  $3\sigma$  rms upper limits (tooth, primary bands): 3.06/1.61/0.53/1.26% — in every source at or below the strength of the detected upper pair line. The ladder reading gains no support but is not killed (the harmonic\_hit branch did not fire either; the anti-kernel record 92=184/2, 34/68 stands); GR parametric resonance stays the favorite. Caveats: GRS 1915+105 scanned in a proxy state (the 113/168 Hz epochs were never published at ObsID level); H1743’s program is 80146 (YAML said 80138); GRO hard-band sensitivity 5.1%.
- **Integration.** Scorecard row S1.e (hfqpo-ladder) data\_limited→null (120 rows: 26 null / 33 data\_limited), next decider eXTP-class sensitivity below the 0.5–3.1% rms limits; HFQPO search-target paragraph updated in tfpt\_horizon\_readouts (+ the papers.ts mirror); website predictions.ts (hfqpo-tooth entry + EXPERIMENTS\_AUDIT counters); predictions.txt; experiments/README.md; experiments/next.txt. No verification/ module, no ledger row, no marker move (the search target stays [O]).

**2026-07-20 (V) (v492 + the new research contract CELEST.SEAM.01 — the celestial-holographic route: the  $E_8$   $\mu_4$ -glue is the flat  $\mathbb{Z}_4$  monodromy of the equivariant celestial chiral algebra on the  $A_3$  ALE space, with the exact correction clock<sup>2</sup>=deck (spin bridge  $\mathbb{Z}_8$ ,  $8=2|\mu_4|$ ); verdict B, no gate closes, no marker moves, SEAM.EQUIV.01 stays [O])**

- **v492\_celestial\_z4\_orbifold.py (CELEST.WP1.01).** WP1 of the new contract, sympy exact (36 checks): the v128  $\mu_4$ -glue grading of  $E_8$  ( $240 = 52+64+60+64$ , additive on all 6720 root pairs) is *inner* —  $h = (2, 2, 2, 2, 2; 0^4)$  reads the glue class mod 4 on all 240 roots, so the glue is  $\text{Ad}(\exp(2\pi i h/4))$ , an order-4 inner torus automorphism = a flat  $\mathbb{Z}_4$  monodromy in the Kronheimer–Nakajima sense; the  $A_3$ -side detector  $h' = (0^5; 1, 1, 1, -3)$  reads the same class = the diagonal (1, 1) of the discriminant group  $\mathbb{Z}_4 \times \mathbb{Z}_4$  (the v92/v125 Lagrangian glue).  $\mathbb{C}^2/\mathbb{Z}_4$  is verified as the  $A_3$  singularity  $XY=Z^4$  (Hilbert series  $(1+t^4)/((1-t^2)(1-t^4))$ , Molien exact); the glue-equivariant SDYM( $E_8$ ) sector *closes* with graded dimensions  $60(d+1)/64(d+1)$  — possible only because  $\dim \mathfrak{g}_j = (60, 64, 60, 64)$  — with zero modes = the carrier  $D_5 \oplus A_3 + \text{Cartan} = 60$  and density  $1/4 = 1/|\mathbb{Z}_4|$ ; the four glue-sector characters sum exactly to  $\chi_{(E_8)_1} = 1+248q+4124q^2+34752q^3$  (v377), with sector weights = the discriminant form  $(5x^2+3y^2)/8$  and integer glue-diagonal  $h = (0, 1, 1, 1)$  (= locality of the  $(E_8)_1$  extension). *Critical correction*

(*verdict B*): the  $A_3$  deck acts on the celestial sphere as  $z \mapsto -z$  (order 2), *not* as the order-4 clock; the clock is the  $U(2)$  phase  $\text{diag}(i, 1)$  (normalising the deck,  $\det = i$  rotating the holomorphic symplectic form by the  $\mu_4$  generator), with the exact spin bridge (spin clock)<sup>2</sup> = deck ( $\mathbb{Z}_8$ ,  $8 = 2|\mu_4|$ , the  $c_3 = 1/(8\pi)$  winding integer). Clock-invariance selects the 1-parameter  $A_3$  deformation  $XY = Z^4 + a_0$  whose four branch points are one  $\mu_4$  orbit with cross-ratio 2 (v168/v214 pillowcase). Negative controls: the false action  $\text{diag}(i, i)$  killed three ways, the false glue breaks additivity on 3112/6720 pairs, rigidity =  $\text{Aut}(\mathbb{Z}_4)$ .

- **New research contract CELEST.SEAM.01** in `tfpt_research_contracts` (own section, the fourth contract alongside `Uwall/Gmetric/CONTRACT.F.01`): hypothesis (seam = glue-equivariant  $\mathbb{Z}_4$ -orbifold sector of the CCA on the  $A_3$  ALE; twistorial bulk = the *self-dual* sector only), typing/non-circularity (the continuum  $(E_8)_1$  existence — the SEAM.EQUIV.01 target — is *not* an admissible input), work packages WP1 (done, verdict B) through WP5 (second continuum route, Costello–Li grade), pre-registered kill tests K1–K4 (K1 survived by WP1), and an honest success estimate (full success low, partial findings high). Ledger rows CELEST.WP1.01 + CELEST.SEAM.01; website mirror (`papers.ts` section, script index). SEAM.EQUIV.01 stays **[O]**; no marker moves. Exact (sympy).

**2026-07-20 (IV) (evidence scorecard: five new typed rows — the  $\alpha_s$  running prediction of record, the  $\mu$ -distortion  $A_s/N_*$  branch discriminator, and three parked analog beds (bosonic-E8 thermal Hall, E8 mass-ladder recovery, Efimov DSI tuning); *experiments level only — no verification module, no gate closes, no marker moves*)**

- **$\alpha_s$  running (prediction of record, consistent).** The Starobinsky branch fixes the running of the spectral index:  $\alpha_s = -2/N_*^2 = -7.6 \times 10^{-4}$  at  $N_*=51.4$  (exact slow roll  $-7.09 \times 10^{-4}$ ; band  $N_* \in [50, 60]$ :  $[-8.0, -5.6] \times 10^{-4}$ ) and  $\beta_s = -4/N_*^3$  (leading order; NOT  $-8/N_*^3$  — the identity  $\alpha_s = -r/6$  holds only at leading order ( $\sim 7\%$  off exact)). Data: Planck 2018  $-0.0045 \pm 0.0067$  ( $+0.57\sigma$ ), P-ACT  $+0.0060 \pm 0.0055$  ( $-1.22\sigma$ ), record leg P-ACT-LB  $+0.0062 \pm 0.0052$  ( $-1.33\sigma$ , Calabrese et al. 2025 JCAP);  $\beta_s$  (Planck run+runrun)  $0.010 \pm 0.013$  ( $-0.77\sigma$ ). Forecast: LiteBIRD+CMB-S4  $\sigma(\alpha_s) \approx 4.9 \times 10^{-3}$  ( $0.14\sigma$ ); CMB-S4+DESI+Euclid statistical  $\sim 3.3 \times 10^{-4}$  ( $2.2\sigma$ , systematics-limited). Same `N_star_reheating` independence group as the  $n_s/r/A_s$  rows (anti double-counting: never an independent hit); kill: robust  $\alpha_s < -5 \times 10^{-3}$  or  $> +3 \times 10^{-3}$  at  $\geq 5\sigma$ ; any robust POSITIVE running kills the plateau branch. Probe: `experiments/tfpt-discovery/alpha_s_running_probe.py`.
- **$\mu$ -distortion (prediction of record, data-limited).** Chluba window integral (calibrated +20% to the full computation): sharp branch  $N_*=51.4$  ( $A_s=1.76 \times 10^{-9}$ )  $\mu \approx 1.6 \times 10^{-8}$ , profiled record  $N_*=56.1$  ( $A_s=2.10 \times 10^{-9}$ )  $\mu \approx 2.0 \times 10^{-8}$ , band  $1.5\text{--}2.3 \times 10^{-8}$ ; the 23% branch split ( $\Delta\mu = 3.0 \times 10^{-9}$ ) gives a  $3\sigma$   $A_s/N_*$  branch decision at  $\sigma(\mu) = 10^{-9}$  (Voyage-2050 class; the PIXIE baseline  $|\mu| < 9 \times 10^{-8}$  does NOT reach — factor 4–7 above the band; FIRAS  $|\mu| < 9 \times 10^{-5}$ ). Honest S15 typing: the frozen log-comb ( $\omega = 2.5827$ ,  $\varepsilon = 0.0173$ ) is STRUCTURALLY BLIND in the  $\mu$  window — window attenuation  $|F(\omega)| = 0.033$ ,  $\delta\mu = 7.4 \times 10^{-12} = 0.007\sigma$  even at Voyage 2050 — the theory itself predicts where it is invisible, so NO separate comb-search row is opened. Shares the `Nstar_branch` alternative group with the two  $A_s$  rows (one branch question, several readings). Kill: robust  $\mu < 0.9 \times 10^{-8}$  or  $> 4 \times 10^{-8}$ . Probe: `experiments/tfpt-discovery/mu_distortion_probe.py`.
- **Three parked analog beds (parked\_analog; honest typing: they validate E8 edge/spectrum physics or the comb DETECTOR, never the TFPT axioms).** (i) Bosonic E8 phase / thermal Hall:  $\kappa_{xy}/T = 8 \times (\pi^2 k_B^2 / 3h)$  with no charge Hall (seam edge  $c_-=8$ , v450/v451/v489); no engineered material yet ( $\alpha$ -RuCl<sub>3</sub>  $c_-=1/2$  contested: ultraclean half-quantized plateau, npj QM 2025, vs phonon-dominated  $\kappa_{xy}$ , PRX 12, 021025). (ii) E8 mass-ladder validation bed (CoNb<sub>2</sub>O<sub>6</sub>/BaCo<sub>2</sub>V<sub>2</sub>O<sub>8</sub>): all 8 one-particle masses published (Zou et al. 2021 PRL 127, 077201,

NMR+INS; Amelin et al. 2020 PRB 102, 104431, THz  $m_1$ – $m_6$ ; Coldea et al. 2010 Science  $m_2/m_1=\varphi$ ) — blind Cartan-PF ladder recovery on the published spectra (8/8 masses + rejection of scrambled controls) as detector validation; confirms Zamolodchikov E8, not TFPT. (iii) Efimov DSI tuning:  $s_0 = \pi/(6 \ln(3/2)) = 1.2914 \Leftrightarrow \lambda = (3/2)^6 = 11.391$  at  $M/m = 8.67$  (2 resonant pairs); best candidate  $^{52}\text{Cr}_2$ – $^6\text{Li}$  ( $M/m=8.635$ ,  $s_0=1.2890$ ,  $\lambda=11.44$ ,  $+0.44\%$  vs  $(3/2)^6$ , gap deviation  $+0.18\%$ ); alternatives  $^{39}\text{K}_2$ / $^{41}\text{K}_2$ – $^6\text{Li}$  ( $\lambda = 11.59/11.20$ , needs double Feshbach control),  $^{176}\text{Yb}_2$ – $^{23}\text{Na}$  ( $+5.9\%$ ); solver validated on the Naidon–Endo anchors (homonuclear 22.69,  $\text{Cs}_2\text{Li}$  4.877); first end-to-end HARDWARE injection of the comb detector at tunable  $\lambda \approx 11.39$ ; kill condition honestly “none on TFPT axioms — detector validation only”. Probe: `experiments/tfpt-discovery/efimov_s0_massratio_probe.py`.

- Scorecard regenerated (`experiments/build_evidence_scorecard.py`): 120 rows — 48 consistent, 9 tension, 25 null, 34 data-limited, 4 parked; website mirror (`EXPERIMENTS_AUDIT` in `predictions.ts`) synchronised. Experiments level only; the verification suite, ledger and papers are untouched.

**2026-07-20 (III) (v491 —  $g_{\text{car}}=5$  as the partition COROLLARY of the four marks: 4 into exactly 3 positive parts is uniquely  $\{1, 1, 2\}$ , sharpening v53; the residual is the weight-typing postulate — AX.P2.01 stays a declared axiom; *no gate closes, no marker moves*)**

- `v491_p2_partition_corollary.py` (P2.PARTITION.01). Sympy-exact: a multiset of exactly 3 POSITIVE INTEGERS summing to 4 is UNIQUE —  $\{1, 1, 2\}$  — so  $e = (e_1, e_2, e_3) = (4, 5, 2)$  and hence  $g_{\text{car}} = e_2 = 5$ ,  $|\mathbb{Z}_2| = e_3 = 2$ ,  $N_{\text{fam}} = e_2 - e_3 = 3$  are *corollaries*, not inputs (consistency fixed point  $(2^4 - 1)/5 = 3 = \text{rank } H^1$ ). [E] negative controls, each assumption load-bearing: drop positivity  $\rightarrow$  4 solutions,  $e_2$  ambiguous  $\{0, 3, 4, 5\}$ ; sum 3  $\rightarrow e_2=3$ ; sum 5  $\rightarrow$  not unique; 2/4 parts  $\rightarrow e_2 \in \{3, 4\}/\{6\}$ ; Mehta–Seshadri RATIONAL weights  $\rightarrow e_2 = \frac{16}{3}$  resp.  $\frac{44}{9} \neq 5$  (integrality = the Birkhoff–Grothendieck splitting-type typing, not MS weights). [E] the v53 sharpening: v53 pins  $(1, 1, 2)$  with ALL THREE targets  $(4, 5, 2)$ ; the partition needs only  $e_1=4$ . [E] sum rule:  $4 \times \frac{N-1}{2} = 4$  iff  $N=3$ ; the v115 route  $|a|_1 = |\mu_4| \text{tr}(A_0) = 4$  agrees. Independence chain: #marks = 4 from v216 (P1-side, no  $g_{\text{car}}$  input); rank  $H^1 = 3$  pure topology (v137/v228);  $|a|_1 = 4$   $g_{\text{car}}$ -free (v115). [C] honest residual: P2 is NOT made axiom-free — the weight-TYPING postulate remains (integrality/positivity  $h^0=0/\text{sum}$  “given (U)”); AX.P2.01 stays a declared axiom, and the  $N_{\text{fam}}$  direction reversal vs `tfpt_constants` is typed as an over-determination check (never double-counted). Integrated in introduction (anchor half-forced paragraph), `tfpt_1` ((5, 3) keybox), `tfpt_2` (H2 winbox) and `origin_theory` ((5, 3) keybox); ledger row P2.PARTITION.01; three exact Wolfram mirrors added to `tfpt_readouts_extension.wl` (counted with the next verified engine run, engine currently unactivated); website mirror (DAG node P2). Exact (sympy).

**2026-07-20 (II) (v490 — the  $T^2$  spin-structure parity census:  $\mu_{\text{pre}}=4$  witnessed on the lattice, the 16-fold-way bridge  $\theta_v=1$  on measured numbers, and the honest TEE non-discriminator fence; *no gate closes, no marker moves*)**

- `v490_seam_parity_census.py` (SEAM.MU.01). The  $T^2$  ground-state fermion parity of the v367  $p+ip$  collar, resolved per spin structure ( $AA, AP, PA, PP$ ) and determined TWICE — Read–Green TRIM product AND real-space Majorana Pfaffian (Schur), relative to the trivial  $M=3$  phase, with the twisted real-space spectra matching the Bloch bands to  $9 \times 10^{-15}$ . [E] per layer  $(+, +, +, -)$  (the  $\nu=1$  Ising signature, TRIM  $\equiv$  Pfaffian in every sector); [E] the 16-layer carrier has parity<sup>16</sup> = + in ALL FOUR sectors  $\Rightarrow$  gauged torus GSD = 4 =  $\mu(SO(16)_1) = |\det \text{Cartan}(D_8)|$  — the lattice ground-state witness of “ $\mu_{\text{pre}}=4$  before condensation” (step 1 of the v378 KLM chain); [E] the 16-fold-way bridge on MEASURED numbers:  $\theta_v = e^{2\pi i \nu/16} = 1$  exactly at  $\nu = 2c_- = 16$  (v489), rivals  $\nu = 1, 2, 8$  not holomorphically condensable; [E]

honest fence: the fermionic Kitaev–Preskill TEE is identically zero for BOTH  $M=1$  and  $M=3$  ( $\gamma_f=0.000001/0.000000$ ) — it does NOT discriminate  $(E_8)_1$  vs  $SO(16)_1$ . [C] the KLM step  $\mu: 4 \rightarrow 4/2^2 = 1$  stays the cited theorem (the condensed model is interacting); the numerics pin only the premise package  $\{c_-=8, \mu_{\text{pre}}=4, \theta_v=1\}$ . [O] SEAM.EQUIV.01 stays [O] (continuum existence, MMST v336). Integrated in `tfpt_research_contracts` (new S3-stack paragraph); ledger row SEAM.MU.01; website mirror (DAG node `qft`). Python (numpy + scipy Schur + sympy exact  $\theta_v$ ).

**2026-07-20 (v489 — the chiral central charge  $c_-=8$  measured DIRECTLY from the bulk ground state via the Kim–Shehab–Kim modular commutator: the first ground-state witness for premise (i) of SEAM.EQUIV.01, independent of the Chern route (v367) and the KLM count (v378); no gate closes, no marker moves)**

- `v489_seam_modular_commutator.py` (SEAM.CCC.01). The Kim–Shehab–Kim modular commutator  $J(A, B, C) = i \text{Tr}(\rho_{ABC}[K_{AB}, K_{BC}]) = (\pi/3) c_-$  (PRL 128 (2022); arXiv:2110.06932/2110.10400) is evaluated on the EXACT BdG ground state of the v367  $p+ip$  seam collar model, entirely from the Peschel Majorana covariance ( $W_X = -2 \text{artanh } \Gamma_X$ ,  $J = \frac{1}{4} \text{Tr}([W_{AB}, W_{BC}] \Gamma_{ABC})$ ), on a disk about the torus centre cut into three  $120^\circ$  wedges. [E] ground-state purity  $\text{eig}(i\Gamma) = \pm 1$  to  $6 \times 10^{-15}$ ; [E] per-layer  $c_- = 0.499998$  at  $L=32$  (convergence  $0.499793$  at  $L=16 \rightarrow 0.500000$  at  $L=48$ , deviation  $-1.8 \times 10^{-7}$ ); [E] controls: trivial  $M=3 \rightarrow 0.000000$ , orientation flip  $M=-1 \rightarrow -0.499986$ ; [E] layer additivity exact ( $1.4 \times 10^{-14}$ ); [E] the 16-layer carrier lands on  $c_-=8=g_{\text{car}}+N_{\text{fam}}$  (direct  $7.9967$  at  $L=16$ , additivity route  $8.0000$ ); [E] clip invariance  $5 \times 10^{-8}$ . The FIRST ground-state witness for premise (i) of SEAM.EQUIV.01 ( $c_-=8$ ) that is INDEPENDENT of both the Chern-number route (v367) and the counting route (v378) — read from entanglement data alone. [O] honest fence: does NOT close SEAM.EQUIV.01; the continuum scaling limit (MMST, v336) and the cited KLM condensation step are untouched. Integrated in `tfpt_research_contracts` (new S3-stack paragraph) and `tfpt_1` (the SEAM.EQUIV.01 package paragraph); ledger row SEAM.CCC.01; website mirrors (DAG node `qft`, papers/glossary/FAQ inventories). Numerical (numpy), Python-only by nature.

**2026-07-19 (editorial — positioning section *TFPT in the light of algorithmic physics: Zuse–Schmidhuber–Wolfram* added to introduction + website mirror; no result, no status change)**

- `introduction.tex` — new section “TFPT in the light of algorithmic physics”. Positions the compiler against the Zuse–Schmidhuber–Wolfram algorithmic-physics lineage: three shared instincts (minimum description length as the selection principle, deterministic resource-bounded generation, compressibility as the aesthetic) and three deliberate divergences (one *forced* program vs. an ensemble of all universes; an algebraic Lie/lattice/VOA description language vs. Turing bit-strings; and falsifiability — dated, kill-tested predictions — as the load-bearing difference). Explicitly typed as *positioning*: it introduces no formula and moves no status marker; its only machine-checked hooks are already in the suite (the grammar-internal null model and the icosahedral/McKay bedrock).
- **Website mirror.** A matching curated section was added to the introduction entry in `website/lib/papers.ts`.



**2026-07-15 (III) (v488 — the 3/5 Clebsch door ENUMERATED AND CLOSED: no minimal Pati–Salam/ $SO(10)$  Majorana channel can produce 5/3 — the v482 escape is decided dead, the v481 window candidate stands as the honest endpoint; a clean negative result, no gate closes)**

- **v488\_majorana\_clebsch\_door.py (FLAV.NUSCALE.04).** v482 named ONE escape for the integer seesaw rung  $M_R = c_3^3 \bar{M}_{P1}$ : a third-generation Majorana factor  $r \approx 1.6696$ , 0.18% from  $g_{\text{car}}/N_{\text{fam}} = 5/3$  — declined for lack of mechanism. The mechanism search is now EXECUTED: **[E]** NO RENORMALISABLE CHANNEL —  $\text{sym}^2(\mathbf{16}) = 136 = \mathbf{10} + \mathbf{126}$  and the  $\mathbf{126}$  is NOT in the  $E_8$  hull  $\{1, 10, 16, 45\}$  (v247), so  $M_R$  only via the  $d=5$  operator  $(\mathbf{16}_F \mathbf{16}_H)^2/\Lambda$ ; channels  $\mathbf{16} \times \mathbf{16} = \mathbf{1} + \mathbf{45} + \mathbf{210}$  with only singlet +  $\mathbf{45}$  admissible. **[E]** CHARGE TABLE:  $(B-L, T_{3R}, Y)(\nu^c) = (+1, -\frac{1}{2}, 0)$ ,  $\text{Tr}_{16} Y^2 = \frac{10}{3}$  (v485). **[E]** THE  $\{2, 3\}$ -SMOOTH THEOREM (the door closes): every nonzero  $\nu^c$  channel weight  $\{1, \frac{1}{4}, 1, -\frac{1}{2}, \frac{3}{8}\}$  is  $\{2, 3\}$ -smooth, and the multiplicative closure of  $\{2, 3\}$ -smooth rationals never contains  $\frac{5}{3}$  (no factor 5); the UNIQUE natural 5 of the embedding,  $k_Y = \frac{5}{3}$  (Ginsparg, v470), is a full-multiplet trace whose direction has  $Y(\nu^c) = 0$  EXACTLY — structurally decoupled from the Majorana operator. **[E]** GENERATION-BLINDNESS: Clebsches are family-space scalars;  $\text{diag}(1, 1, \frac{3}{5})$  cannot arise from group theory by construction. **[O]** compiler-flavor inventory typed and declined: exact  $\frac{3}{5}$ ’s only in wrong slots ( $K$  lepton gen2/gen1;  $R + L$  lepton gen3/gen2; never gen3 vs a degenerate base). NET: the rung+5/3 rescue is DEAD as a mechanism candidate; the v481 one-parameter window candidate ( $M_R$  free inside the PS window, the  $\Sigma m_\nu = 0.059$  eV floor as kill test) stands. Contract provenance: `experiments/theory-contracts/clebsch01_majorana_35_door.py`. Integrated in `tfpt_2_standard_model` (new paragraph closing the seesaw arc); ledger row FLAV.NUSCALE.04 + update note on .03; Python-only (Fraction/sympy exact), mirrorable, deferred to the next verified engine run.

**2026-07-15 (II) (v487 — the lazy split FORCED by the resummed clock: the walk’s one-step spectrum is the complete v124 clock ladder below the wall, deck parity = rung index, and the two comb tones are one bend apart — the v486 [O] “split selection” closes onto established structure; no gate closes, no marker moves)**

- **v487\_transfer\_clock\_rungs.py (HYP.REWRITE.03).** v486 left the split selection open (“ $(\frac{1}{2}, \frac{1}{6})$  vs uniform  $(\frac{1}{3}, \frac{1}{3})$  is selected by the physical spectrum, not yet derived”; named candidate: the vague “sheet-symmetric choice”). Closed: **[E]** RUNG READING — the lazy rule’s one-step spectrum is EXACTLY the complete resummed-clock ladder below the wall (v124:  $\lambda_n^{1/p^2} = 1 - n/N_{\text{fam}}$ , pole at  $n=N_{\text{fam}}$ ):  $\text{eig}(M) = \{1, \frac{2}{3}, \frac{1}{3}\} = \{1 - n/3 : n = 0, 1, 2\}$ , while v327’s uniform rule puts its second pair mode ON the wall ( $0 = 1 - 3/3$ ) — the v486 degeneracy re-read as “odd channel collapsed to the wall”. **[E]** DECK PARITY = RUNG INDEX:  $[B, \sigma] = 0$ ; the  $\sigma$ -even pair mode carries rung  $n=1$  (survival  $\frac{2}{3}$ ), the  $\sigma$ -odd mode rung  $n=2$  (survival  $\frac{1}{3}$ ) — each clock step flips the deck parity; this is the structural home of the v486 parity assignment ( $\omega_1$  even-channel,  $\omega_2$  odd-channel) that `experiments/frb-parity-comb` tests. **[E]** FORCING: demanding the faithful ladder (no rung missing, no physical mode on the wall — definitional for a full-spectrum generator) +  $\mathbb{Z}_2$  equivariance + non-negative rates forces UNIQUELY both the split  $(\frac{1}{2}, \frac{1}{6})$  AND the rung assignment even $\rightarrow$ 1, odd $\rightarrow$ 2 (the swapped assignment needs  $h = -\frac{1}{6} < 0$ ); the “sheet-symmetric choice” candidate is superseded by this proof. **[E]** BEND COROLLARY:  $\omega_1/\omega_2 = \text{rate}(2)/\text{rate}(1) = \log_{3/2} 3 = 2.70951\dots$  — the parity-comb tone ratio IS the v107/v124 bend of the pulsar-clock search. **[O]** what remains open is exactly what was open before v486: the arity  $\{2, 3\}$  (anchor input) and the clock’s semiclassical derivation (R1); no independent choice remains at the walk level. Contract provenance: `experiments/theory-contracts/lazy01_clock_rung_forcing.py`. Integrated in `origin_theory` (new paragraph after the v486 block); ledger row HYP.REWRITE.03 + update note on .02; Python-only (sympy exact), mirrorable, deferred to the next verified engine run.



**2026-07-15 (v486 — generator hunt: the FULL transfer spectrum  $\{1, (2/3)^6, (1/3)^6\}$  from ONE forced local rule — the unique lazy  $\mathbb{Z}_2$ -pair walk (stay, hop, leak) =  $(\frac{1}{2}, \frac{1}{6}, \frac{1}{3})$ ; the previously generator-less  $(1/3)^6$  mode gets its origin; v327's uniform rule superseded as full-spectrum generator — *no gate closes, no marker moves*)**

- **v486\_transfer\_full\_rule.py (HYP.REWRITE.02).** The verified seam transfer spectrum is  $\{1, (2/3)^6, (1/3)^6\}$  (v54/v221), but the only local generator in the suite (v327) produces  $\{1, \frac{2}{3}, 0\}$  — its zero mode persists under every power, so no iterate reaches the third mode. [E] UNIQUE-NESS: for the symmetric 3-channel rule  $M(s, h) = [[1, 0, 0], [0, s, h], [0, h, s]]$  (one absorbing family channel +  $\mathbb{Z}_2$  pair) the survival spectrum is  $\{s+h, s-h\}$ ; demanding the PHYSICAL pair  $\{\frac{2}{3}, \frac{1}{3}\}$  forces UNIQUELY (stay, hop, leak) =  $(\frac{1}{2}, \frac{1}{6}, \frac{1}{3}) = (1/|\mathbb{Z}_2|, 1/(|\mathbb{Z}_2|N_{\text{fam}}), 1/N_{\text{fam}})$  — the lazy  $\mathbb{Z}_2$ -pair walk, every rate an atom expression (honest direction: spectrum  $\rightarrow$  rates). [E] SIX-HAND GENERATION: over the order-6 =  $2N_{\text{fam}} = |R^+(A_3)|$  hand,  $\text{eig}(B^6) = \{\frac{64}{729}, \frac{1}{729}\}$  exactly — both decay gaps now have one recursive generator:  $6 \ln \frac{3}{2}$  (recovery, v76) and  $6 \ln 3$  (subdominant, previously generator-less). [E] SUPERSESSION: the uniform rule's zero mode is power-persistent; its  $\lambda_2$  arity story ( $\frac{2}{3} = |\mathbb{Z}_2|/N_{\text{fam}}$ ) is unchanged. [O] the arity  $\{2, 3\}$  stays the anchor input; the lazy-vs-uniform split is SELECTED by the physical spectrum, not yet first-principles-derived (named candidate: stay =  $1/|\mathbb{Z}_2|$  as the sheet-symmetric choice). Discovery provenance: `experiments/tfpt-discovery/generator_hunt_full_transfer_rule.py` (plus a typed anchor-bilinear coverage inventory — known hits  $a^T K a=41$ , Koide 53, and the recorded-only  $1^T Q a=16$  — and the lepton-exponent observation  $(5, 3, 2) = K\text{-row} = \text{atom triple}$ ; no unforced claims, v354/v355 discipline). Integrated in `origin_theory` (new paragraph after the v327 rewrite block); ledger rows HYP.REWRITE.02 + update note on .01; Python-only (sympy exact), mirrorable, deferred to the next verified engine run.

**2026-07-14 (IV) (v485 — the merged analytic target's computable side COMPLETED: the diagonal self-energy is renormalised to ZERO exactly at the KMS seam scale, the mark contact calculus resums in closed form (BFK route), and the 48/41 multiplicities are ONE state set under two response weights — the residual collapses onto the SEAM.EQUIV.01 face alone; *no gate closes, no marker moves*)**

- **v485\_contact\_diagonal\_closed.py (SEAM.CONTACT.UNIT.02).** The v484 merge left one analytic step: the diagonal (self-energy) zeta-renormalisation at the marks + the multiplicity matching. Settled at the computable level: [E] DIAGONAL ZERO AT THE KMS SCALE — the renormalised coincident-point Green function of  $|D|$  on circumference  $\ell$  is  $G_{\text{reg}}(0; \ell) = \frac{1}{\pi} \ln(\ell/2\pi)$  EXACTLY (symbolic limit + 30-digit checks); it vanishes iff  $\ell = 2\pi = 1/(4c_3)$ , the v239 KMS seam circumference — the seam scale is the unique zero-self-energy scheme point, and the log is the pure SCALE channel (matching the structure:  $F_{U(1)}$  carries exactly ONE log term, the transport  $8b_1c_3^6 \ln(1/\varphi_{\text{seam}})$ , while the  $\varphi_0^{\text{ret}}$  split is log-free). [E] CLOSED-FORM RESUMMATION (the BFK/gluing route already in the suite, v151): with the diagonal zero, the finite-rank mark determinant resums exactly —  $\det(I - uC) = (1 - 4u)(1 + 2u)^2$ ,  $u = \varepsilon c_3 \ln 2$ , all orders, the linear term ABSENT precisely because the diagonal vanishes ( $\text{Tr } C = 0$ ); the series matches v484's  $\text{Tr}(C^k) = 4^k + 2(-2)^k$  term by term. [E] WEIGHT DICHOTOMY: the SAME 48 =  $3 \times 16$  admissible states carry BOTH multiplicities — counted flat they give the  $\varphi_0^{\text{ret}}$  puncture count  $48 = \Omega_{\text{adm}}$ ;  $Y^2/\text{loop}/\text{GUT}$ -weighted they give the alpha budget  $41 = 10b_1 = 40 + 1$  ( $\text{Tr}_{16} Y^2 = \frac{10}{3}$  exact,  $\nu^c = 0$ ; Ginsparg  $(3/5)(41/6) = 41/10$ , v470) — one state set, two response weights, no new number. [C]/[O] CONSEQUENCE: with v484 EVERY finite/computable piece of the merged ALPHA.QUILLEN.EXACT.01 + HYP.PHI0.PUNCTURE.01 target is proven; the single remaining [O] is the identification of the abstract seam  $\zeta$ -det with this glued mark determinant — a face of SEAM.EQUIV.01, exactly the v382 typing, now substantiated computationally. Integrated in `tfpt_1_architecture_e8` (eighth step of the ALPHA.QUILLEN arc) and `origin_theory`; ledger rows SEAM.CONTACT.UNIT.02 + update

notes on .01 and both targets; Python-only by nature, flagged in the Wolfram README.

**2026-07-14 (III) (v484 — the two open analytic targets MERGE: the shared “ $c_3$  per boundary insertion” rule of ALPHA.QUILLEN.EXACT.01 and HYP.PHI0.PUNCTURE.01 is the KMS seam unit  $2\pi = 1/(4c_3)$  with  $\frac{1}{4} = 1/|\mu_4|$ , derived on the seam circle for the finite cycle sector — *no gate closes, no marker moves*)**

- **v484\_seam\_contact\_unit.py (SEAM.CONTACT.UNIT.01).** The v483 round left the two analytic targets sharing one rule (“weight  $c_3$  per boundary contact insertion”: the  $F_{U(1)}$  ladder  $\{0, 3, 6\}$ , the  $\varphi_0^{\text{ret}}$  puncture  $\{1, 4\}$ ). This module shows the shared rule is NOT a new unknown: [E] the unit chain  $\frac{1}{2\pi} = 4c_3 = |\mu_4|c_3 = |\mathbb{Z}_2|\chi(S^2)c_3$  — one *bare* boundary insertion divided by the orbit order  $|\mu_4|=4$  is exactly  $c_3$ ; this is the v239 KMS seam unit and the v58/v73 one-sided Gauss–Bonnet in a single identity. [E] the mechanism is DERIVED on the actual seam circle for the finite cycle sector: the bare Green function  $G(d) = -\frac{1}{\pi} \ln |2 \sin(d/2)|$  (polylog identity, 40 digits) takes INTEGER multiples of  $c_3 \ln 2$  at the  $\mu_4$  separations ( $G(\pi/2) = -4c_3 \ln 2$ ,  $G(\pi) = -8c_3 \ln 2$ ; mark matrix  $-4c_3 \ln 2 \cdot \text{circ}(0, 1, 2, 1)$ , spectrum  $\{4, -2, 0, -2\}$ ), so the  $\mu_4$ -invariant (orbit-averaged) insertion carries  $-(c_3 \ln 2)C$  exactly and the log-det contact expansion is exactly graded in  $c_3$  per insertion ( $\text{Tr } C^k = 4^k + 2(-2)^k$ ). [E] the grammar is suite-wide: the  $\Lambda$  prefactor  $\frac{3}{4\pi^2} = (8\pi)^2 d_{\text{top}} = 48c_3^2$  (v60) carries the SAME multiplicity  $\Omega_{\text{adm}}=48$  as the  $\varphi_0^{\text{ret}}$  puncture at TWO insertions; the  $g-2$  seam loop divides by the same  $2\pi = 1/(4c_3)$ . [C]/[O] THE MERGE: the two tracked [O] targets collapse into ONE named analytic step — the diagonal (self-energy) zeta-renormalisation at the marks + the state-multiplicity matching ( $48 = \Omega_{\text{adm}}$ ,  $41 = 10b_1$ ); the finite half is proven, no target coefficient is claimed (the integers are forced by the square configuration, v354/v355-clean). Integrated in `tfpt_1_architecture_e8` (seventh step of the ALPHA.QUILLEN arc) and `origin_theory` ( $\varphi_0^{\text{ret}}$  split paragraph); ledger rows SEAM.CONTACT.UNIT.01 + update notes on both targets; Python-only by nature (mpmath polylog), exact identities flagged in the Wolfram README.

**2026-07-14 (II) (v482–v483 — the deep-dive round: the  $\nu$ -scale decision computation EXECUTED (the integer  $c_3^3$ -rung excluded unstructured, the  $5/3$ -proximate structure escape recorded and declined) and the  $\varphi_0^{\text{ret}}$  puncture target HALVED (the geometric side proven exact: all twisted pillowcase heat traces are  $t$ -independent rationals, so the  $\pi^{-4}$  must sit in the per-mark coupling weight) — *no gate closes, no marker moves*)**

- **v482\_seesaw\_rung\_decision.py (FLAV.NUSCALE.03).** The decision computation NAMED by v481 is EXECUTED in bracketed (assumption-robust) form: the integer rung  $M_R = c_3^3 M_{\text{Pl}}$  needs a rescue factor  $\times 1.670$  in  $m_3$  (0.0301 vs 0.0503 eV), while  $m_t(\overline{\text{MS}}) \pm 3 \text{ GeV}$  ( $> 3\sigma$ ),  $\alpha_3^{-1} \mp 0.10$  ( $> 3\sigma$ ), a  $\pm 10\%$  band on the ADKLR  $\kappa$  run-down and a PS-leg Yukawa-beta envelope  $\times [0.5, 1.5]$  move it by at most  $\times 1.10$  singly and  $\times 1.165$  ALL-favourably combined — the *unstructured* rung is EXCLUDED conditional on the carrier premise  $y_\nu = y_t$  (the v481 1-loop miss is not an RG artifact; no exact higher-loop coefficients trusted, pure envelope argument). [C] The only escape is a third-generation Majorana STRUCTURE factor  $r \sim 1.67$ ; its post-hoc 0.18% proximity to  $g_{\text{car}}/N_{\text{fam}} = 5/3$  is recorded and DECLINED per v354/v355 (no forcing mechanism; the  $t_0 \sim 30$  discipline of v478). [O] the v481 one-parameter window candidate (NO floor  $\Sigma m_\nu = 0.059 \text{ eV}$ ) never used the integer rung and STANDS. Integrated in `tfpt_2_standard_model` ( $\nu$ -scale paragraph) and `tfpt_5_redteam`; ledger rows FLAV.NUSCALE.03 + .02 update note; Python-only by nature.
- **v483\_pillowcase\_exact\_traces.py (HYP.PHI0.GEOM.01).** The GEOMETRIC side of the  $\varphi_0^{\text{ret}}$  puncture target (HYP.PHI0.PUNCTURE.01/v408) is EXACT: every twisted/orbifold heat trace of the flat  $\tau=i$  pillowcase is a  $t$ -INDEPENDENT RATIONAL —  $\text{Tr}(\gamma e^{t\Delta}) = 1$  for

$\gamma = \sigma, \rho, \rho^3$  at every tested  $t$  (position-space Gaussian image sums  $< 10^{-9}$ ; mode-space proof: only  $n=0$  is  $\gamma$ -diagonal), matching the Atiyah–Bott fixed-point counts  $4 \times \frac{1}{4}$  ( $\det = 4 = |\mu_4|$ ) and  $2 \times \frac{1}{2}$  ( $\det = 2 = |\mathbb{Z}_2|$ ); the pillowcase contact term is EXACTLY  $\frac{1}{2} = 4 \times \frac{1}{8}$  per mark and the clock-equivariant trace exactly 1, for all  $t$ . THEREFORE the  $\pi^{-4}$  in  $d_{\text{top}} = 48c_3^4 = \frac{3}{256\pi^4}$  cannot come from flat orbifold geometry at ANY order: it must sit in the per-insertion COUPLING normalisation ( $4 = |\mu_4|$  marks of weight  $c_3$  give  $c_3^4$ , multiplicity  $\Omega_{\text{adm}}=48$ ; the bare  $1/(2\pi)$ -propagator 4-cycle differs by the exact rational  $\frac{3}{16}$ ). The target NARROWS from “derive the term” to ONE rule — “per-mark insertion weight =  $c_3$ ” (the same residual class as the EM-Ward  $c_3$ -ladder, EM.WARD.02) — and stays [O]. Complements the experiments-side Donnelly probe (honest negative). Integrated in `origin_theory` ( $\varphi_0^{\text{ret}}$  split paragraph); ledger rows HYP.PHIO.GEOM.01 + HYP.PHIO.PUNCTURE.01 update note; Python-only by nature (quadrature verification of exact statements).

**2026-07-14 (website + figure refresh — the /verification journey views become live HTML (the stale timeline/residual-chain images replaced by interactive components wired to the in-browser reproducer), the prediction surface’s empirical-audit wall of text restructured into dated, expandable rounds, and the generated timeline/residual-chain figures extended to the current suite — *presentation only, no content or marker moves*)**

- **Website /verification: live journey views.** The static `script_timeline.png` and `residual_chain.png` images (stale at the v303–v407 figure state, with alt text and captions still describing the old eleven-phase/v237 state) are replaced by two interactive components, `SuiteTimeline.tsx` (17 phases, v1–v481) and `ResidualChain.tsx` (the reduction chain through the certification round to the bedrock): every phase/step lists representative vN scripts that run live in the browser via the existing Pyodide reproducer, exactly like the dependency graph. Stale counts fixed (“~295-script journey”, “260 scripts” → the generated suite total). The dependency graph itself was verified current (v479–v481 present; all 288 referenced scripts exist in the registry and the website mirror).
- **Prediction surface: the empirical-audit narrative restructured.** The single ~13 000-character audit paragraph in `lib/predictions.ts` is split into twelve dated, verdict-badged rounds (AuditRound: date, title, verdict, two-sentence summary, full unabridged record on expand) plus a domain-chip row and the anti-double-counting note — same content, readable surface. Long-token overflow fixed (`break-words`) in the dependency-graph detail panel and the universal-gap-lab residual cards; the over-long ALPHA.QUILLEN residual note tightened to its current state (v470/v472, residual unchanged).
- **Generated figures + captions synced.** `make_figures.py`: the script timeline gains the v408–v481 certification-round phase (title now ~475 scripts) and the residual chain gains the v458–v480 certification step (Kronheimer v479, four-interval v480, crossed-product v469) with sub-text wrapping fixed; figures regenerated (PDF + website PNG). The stale figure captions updated: `introduction` (timeline caption now spans the full arc), `tfpt_research_contracts` (residual-chain caption now states the SEAM.EQUIV.01 bedrock and certification round; QFT-skeleton caption now names the one [O] residual instead of the outdated “two open items”). No numbers, statuses or claims change anywhere.
- **KaTeX macro fix (broken formula rendering).** Formulas copied from the document set that use the shared TeX macros (`\gcar`, `\Nfam`, `\ctthree`, ...) rendered as red KaTeX errors on the site (e.g. the  $\varphi_0^{\text{tree}}$  icosahedral-combinatorics formula on the Origin-Theory page). A shared macro map (`lib/katex-macros.ts`, mirroring the `changelog.tex` preamble) is now passed to every KaTeX render; an automated scan of all 19 site routes finds zero `katex-error` nodes.
- **Readability pass (text walls clamped, content unchanged).** A reusable `Expandable` component clamps long mirror prose with a fade + “Show more” toggle — applied to the

paper-section bodies and front-box bullet lists (`PaperSection.tsx`, the home set pages and `/papers/[slug]`), the dependency-graph node summaries and the prediction-card descriptions. Nothing is deleted; the full text expands in place.

- **Hylaeon ecosystem card corrected.** The `hylaeon.ai` card spoke of “Field AI” with structure tags that did not describe the project. It now speaks consistently of *Hylaeon* and is grounded in the project documentation (`_documentation/core-physics/big-picture.md`) and the site’s own status log: one cognitive substrate behaving like a physical material, knowledge = attractor valleys, question = boundary condition, answers committed only on physical landing, capability claims gate-checked against a machine-checked claim registry (honest negatives published) — borrowing TFPT’s structural discipline, not its physics predictions. The Agentic-Commerce card reference updated accordingly.
- **Live-site correctness pass (counts + reading order).** Sitewide audit against the deployed page found stale counts and fixed them: the document set is stated everywhere as *10 documents = 1 reading guide + 5 numbered papers + 4 companions* (the safeguards companion was missing from the papers/downloads headers, the JSON-LD collection and the `tfpt_docset.tex` box, which still said “three companions” next to “four”); the stale “23 falsifiable predictions” claims (site metadata, OG image, `llms.txt`) now state the live count of the prediction surface (25 test surfaces, sourced from `predictions.length` where possible), while the interactive compiler page is honestly labelled as its own *23-readout prediction board* (the standalone app’s board genuinely has 23 entries; the intro film keeps its spoken “twenty-three”); and the `/orientation` reading-order lists are fixed to the canonical numbering (Red Team = Doc 5 added, Appendix H = 6, Origin = 7, contracts = 8, safeguards = 9 — the old list mislabelled 6/7 and omitted three documents). Presentation/consistency only; no physics content changes.

**2026-07-14 (v479–v481 — the bird’s-eye “overlooked tools” round: R3’s graph→geometry bridge discharged to Kronheimer 1989; the four  $\mu_4$  marks realised as the exactly solvable *four-interval* multilocal modular geometry with  $\omega \circ \rho = \omega$  manifest; and the absolute  $\nu$ -scale collapsed to a one-parameter, PS-window-bracketed candidate under the carrier normalisation  $y_\nu = y_t$  — *no gate closes, no marker moves*)**

- **v479\_kronheimer\_quiver\_bridge.py (SEAM.KRONHEIMER.01).** The R3 residual of the  $\det K=1$  keystone (v344: “the rewrite attractor graph IS the du Val singularity”) is an ESTABLISHED THEOREM — Kronheimer’s ALE hyper-Kähler quotient construction + Torelli classification (J. Diff. Geom. 29 (1989) 665 and 685) — with all finite input data already compiler outputs, machine-checked [E]: the Kac marks  $\{1, 2, 2, 3, 3, 4, 4, 5, 6\}$  are the null vector of the affine- $E_8$  Cartan matrix AND the Perron vector  $A\delta=2\delta$  of the rewrite attractor (v312); the hyper-Kähler quotient dimension count lands exactly on 4 (the  $\mathbb{C}^2/\Gamma$  geometry;  $\sum_{\text{edges}} d_i d_j = \sum d_i^2 = 120$  forced by the null identity); the gauge datum is  $|2I|=120=|\mu_4|h(E_8)$ ,  $h(E_8)=30=2 \cdot 3 \cdot 5$ ; the resolution carries the unimodular  $-E_8$  form ( $\det=1$ , SNF = identity — the Poincaré link) with  $3 \cdot 8=24$  deformation parameters. [C] the theorems are cited, not re-proved (import class of NPW26/MMST/AGT); [O] the residual premise *transforms* to “the raw seam supplies ALE hyper-Kähler data asymptotic to the  $(2, 3, 5)$  cone” — the same single arrow as SEAM.EQUIV.01 (v346 L2). Integrated in `origin_theory` (R3 keybox + new Kronheimer paragraph) and `tfpt_research_contracts` (Kummer/du Val); ledger rows SEAM.KRONHEIMER.01 + SEAM.DETK.01 update note; Wolfram mirror added to the extension file (+2 exact checks, counted with the next verified engine run — the local engine is currently unactivated; the identical statements are sympy-exact in v479). LEAN: the finite input data are kernel-checked in `TfptCarrier/KronheimerMarks.lean` (marks = null/Perron vector over  $\mathbb{Z}$ ,  $\sum \delta_i^2 = 120 = 4 \cdot 30$ ,  $30 = 2 \cdot 3 \cdot 5$ , dimension count = 4; lake build clean, `#print axioms` = the three standard kernel axioms only).



- **v480\_multilocal\_four\_interval.py** (QGE0.MULTILOCAL.01). The seam circle minus its four marks IS the  $n=4$  multi-interval region of a chiral free fermion — the class whose multi-interval modular theory is exactly known (Casini–Huerta mixing term; Longo–Xu; Rehren–Tedesco multilocal fermions; KLM: holomorphic  $\Leftrightarrow$  trivial multi-interval inclusion,  $\mu=1$  — the same one bit as  $\det K=1$ ); until now the suite used only the single-ball CHM boost (v358). On the exactly solvable antiperiodic critical ring: [E] the quarter-turn clock is a signed permutation with  $\rho^4=-1$  EXACTLY (the fermionic  $\mu_4$  clock is the order-8 double-cover lift — the same  $\mathbb{Z}_2$ -lift pattern as  $2I$  over  $I$ ); [E] its primitive-8th-root character projectors decouple the 4-interval covariance at  $10^{-15}$ , each sector a single-interval quarter-ring kernel with boundary twist ( $10^{-14}$  spectra); [E]  $\rho C \rho^{-1}=C$  at  $10^{-14}$ , so  $\omega \circ \rho = \omega$  holds *manifestly* at the 4-interval state level, and a one-site displacement re-couples the sectors at  $10^{-2}$  (negative control); [E] the cross-interval modular coupling peaks on the conjugate-point diagonal (the Casini–Huerta mixing shadow). [C] continuum theory cited; [O] the raw-seam premise (QGE0.SYM.01/SEAM.EQUIV.01) unchanged. Integrated in `tfpt_research_contracts` (marks keybox + rigidity contract + v478 leg A), `origin_theory` (BW paragraph), `tfpt_4_frontier` (v358 companion note); Python-only by nature.
- **v481\_seesaw\_carrier\_ladder.py** (FLAV.NUSCALE.02, **CANDIDATE** class, **genre v467/v468**). FLAV.NUSCALE.01's  $y_\nu=1$  probe was not the carrier normalisation: one  $SO(10)$  16 per family + the minimal Yukawa sector ( $\mathbf{10}_H$  / PS  $(1, 2, 2)$ ) forces  $y_\nu=y_t$  at the matching scale (named premise). With explicit 1-loop RG (gauge/ $y_t/\lambda$  up; ADKLR Weinberg-operator running down, hep-ph/0108005) the  $(y_\nu, M_R)$  trade-off collapses to  $M_R$  alone: the observed  $m_3=0.0503$  eV demands  $M_R=9.3 \times 10^{13}$  GeV — INSIDE the PS window  $[4.2 \times 10^{13}, 2.4 \times 10^{15}]$  (v249) at  $\log_{c_3}(\bar{M}_{Pl}/M_R)=3.15$  (the  $y_\nu=1$  inversion gives the structureless 2.58). [X] honesty gate: the integer rung  $M_R=c_3^3 \bar{M}_{Pl}$  predicts  $m_3=0.030$  eV — 40% low at 1-loop ( $\kappa$  run-down 0.794) — the ladder pin is DECLINED per v354/v355; the decision computation is named (2-loop SM + explicit PS thresholds). What survives: a *one*-parameter, window-bracketed [O] candidate with the falsifiable band  $m_3 \in [0.002, 0.115]$  eV and the NO floor  $\Sigma m_\nu=0.059$  eV; frozen record untouched. Integrated in `tfpt_2_standard_model` (new paragraph after v272), `introduction` status card (third named candidate), `tfpt_5_redteam`; ledger rows FLAV.NUSCALE.02 + FLAV.NUSCALE.01 update note; Python-only by nature.
- *Honest negative (experiments-only, no promotion)*: the Donnelly/Atiyah–Bott equivariant heat-kernel route for the  $\phi_0$  puncture target (HYP.PHI0.PUNCTURE.01) types the KIND of term (order-4 equivariant contact datum; the exact weights  $\sum_{k=1}^{n-1} \frac{1}{4 \sin^2(\pi k/n)} = \frac{n^2-1}{12}$  verified) but does NOT force the value  $48c_3^4$  (pure rationals vs  $\pi^{-4}$ ) — recorded in `experiments/theory-contracts/phi0_donnelly_equivariant_probe.py`; the target stays [O].

**2026-07-13 (v478 — first computable steps on the two remaining legs of the entropic-action bridge: the state-side modular data flows to the CHM/Bisognano–Wichmann geometric form (Calabrese–Cardy  $c_{\text{est}} \rightarrow 1$ , CHM parabola, even bands exactly zero) — the bridge's modular side meets TFPT's Einstein-derivation input; and the measure condition reduced to one exact KMS time  $t_0 = \ln \frac{72}{17} + 9 \ln(8\pi) = 30.461$  with the  $h(E_8)=30$  near-miss explicitly declined; both legs stay [O], no status moves)**

- **v478\_entropic\_continuum\_legs.py** (GRAV.ENTROPIC.LEGS.01). *Leg A (AP2 continuum, exhibited)*: on the exactly solvable critical chain (infinite-volume sinc kernel, 50-digit precision) the compressed interval state has  $S(L)=(c/3) \ln L+k$  with  $c_{\text{est}}=1.00089/1.00038/1.00021$  (monotone  $\rightarrow$  the Dirac  $c=1=2 \times \frac{1}{2}$ , the v450 reading; gapped control  $\sim 6 \times 10^{-8}$ , area law); the Peschel modular Hamiltonian  $K=\ln((1-C_A)/C_A)$  organises on the CHM weight  $\beta(x)=x(L-x)/L$  (corr  $0.87 \rightarrow 0.99$ , shrinking plateau drift; constants deliberately not asserted), the even-range bands



vanish identically (particle-hole,  $\sim 10^{-36}$ ) and the odd tail is 1.8% with parabolic envelope — the state-side  $\Delta_\Sigma$  (the reading forced by v476) flows to the CHM/BW geometric form, exactly the object TFPT’s Einstein derivation consumes (v323/v358): the bridge’s modular side and the gravity side meet in the continuum limit; a lattice exhibit, NOT a continuum proof — AP2 stays [O]. *Leg B (the measure)*: [E] single-scale corollary —  $d\mu=\delta(t-t_0)dt$  gives  $\mu_2/\mu_1^2=e^{t_0}$  exactly, so the v477 one-moment condition forces  $t_0=\ln\frac{72}{17}+9\ln(8\pi)=30.461$  (one dimensionless KMS time, closed form); the near-miss to  $h(E_8)=30$  (difference 0.461) is explicitly DECLINED per the v354/v355 no-free-pattern rule — deriving the measure from the entropic structure stays OPEN. Keybox extended in `tfpt_research_contracts`; high-precision numerical module, Python-only by nature (Wolfram count unchanged, 378/378; flagged in the Wolfram README); ledger row `GRAV.ENTROPIC.LEGS.01`; both legs stay [O], no marker moves.

**2026-07-13 (v477 — the scale-flow representation of the entropic action: the surviving  $R^2$  route typed as *one moment condition*  $\mu_2/\mu_1^2 = \frac{72}{17}(8\pi)^9$  with the exact closure  $\frac{4608\pi^2/17}{(72/17)(8\pi)^9} = c_3^7$  — the kill-test gap is a scale-measure datum, satisfied by the same KMS moment TFPT already fixes (v36); zero new dials, *no status moves*)**

- **v477\_entropic\_scaleflow.py (GRAV.ENTROPIC.SCALEFLOW.01)**. [E] The Frullani identity upgrades to a scale-flow representation:  $-\ln a = \int \frac{dt}{t}(e^{-ta}-e^{-t}) = 2 \int \frac{dx}{x}(e^{-a/x^2}-e^{-1/x^2})$  (symbolic +  $10^{-20}$  quadrature) — Bianconi’s log action IS the *flat* scale-integral of relative heat-kernel actions, while TFPT’s  $S_{\text{rel},\chi}$  evaluates *one* KMS scale with a weight: the bridge between the two action principles is a choice of SCALE MEASURE. [E] Moment dictionary: a weight  $w(t) dt/t$  gives  $3\beta R\mu_1 + \frac{17}{24}\beta^2 R^2\mu_2$ ; the flat weight has  $\mu_1=\mu_2=1$  and reproduces the v475 raw coefficients identically. [E] The ONE moment condition: demanding  $m^2=c_3^7\bar{M}_{P1}^2$  forces exactly  $\mu_2/\mu_1^2=\frac{72}{17}(8\pi)^9$  — the v475 kill-test number IS a moment ratio — and the closure identity  $\frac{4608\pi^2/17}{(72/17)(8\pi)^9}=c_3^7$  holds identically: the “13 orders” are a scale-measure datum, not a bridge mismatch. [E] v36 consistency:  $f_0=6(4\pi)^2/c_3^7 \Leftrightarrow M^2/\bar{M}_{P1}^2=c_3^7$  — one KMS moment, two parametrisations, zero new dials. [C] A consistency statement, not a derivation (the KMS weight is TFPT input; the entropic action supplies the Einstein +  $\Lambda$  level and the scale-flow form, TFPT supplies the measure). Negative side-record: an exact lattice-BW commutator attempt for AP2 (Peschel–Eisler simple forms) did NOT hold on the half-filled open chain — kept out of the suite, logged in the research notes. Keybox extended in `tfpt_research_contracts`; Wolfram mirrors both exact blocks (378/378); ledger row `GRAV.ENTROPIC.SCALEFLOW.01`; the  $R^2$  gate and the v359 typing stay [O], no marker moves.

**2026-07-13 (v476 — the AP2 compression conjecture of the entropic-action bridge made *well-posed* on an exactly solvable model: the literal operator-side reading is ill-posed on a pure bulk, the state-side reading is forced, the mismatch is exactly quadratic in the cross-cut correlations and gap-suppressed; AP2 stays [O], *no status moves*)**

- **v476\_entropic\_compression\_toy.py (GRAV.ENTROPIC.COMPRESS.01)**. The last open work package of the v473 round, exercised on a gapped staggered-mass chain ( $N=16$ ). The seam situation is reproduced (pure bulk: the ground-state covariance is a projection to  $10^{-15}$ ; thermal boundary:  $\text{spec}(PCP) \subset (0,1)$ , Peschel  $K=\ln((1-C_A)/C_A)$  finite,  $S>0$  — v155 in the bridge context). **The reading is decided**: on the pure bulk  $f(x)=x/(1-x)$  is singular on  $\text{spec}(C)=\{0,1\}$ , so the literal operator-side reading  $Pf(C)P$  of the v473 conjecture is *ill-posed* while the state-side  $f(PCP)$  is finite — the compression identity can only mean “build  $\Delta_\Sigma$  from the *compressed* relative metric”, which matches Bianconi’s own local-metric construction (AP2 retyped, not weakened). [E] Block algebra (symbolic):  $[C^2]_{AA}-(C_{AA})^2=BB^t$ ,  $[C^3]_{AA}-(C_{AA})^3=ABB^t+BB^tA+BDB^t$  — first order in the cross block vanishes identically,

so for any analytic  $f$  the two readings agree to *second* order in the cross-cut correlations: the modular (entanglement) content lives exactly in that quadratic correction. Gap control at the bounded (covariance) level: the Bianconi-shaped bulk identity  $f(C_T)=e^{-h/T}$  holds to  $10^{-11}$  relative; the mismatch obeys the quadratic law  $d \leq x^2$  ( $d/x^2 \in [0.4, 1.0]$ ) and falls with the gap in the gapped regime, the decoupled cut agrees exactly — the two readings converge in the gap-dominated regime, where TFPT operates ( $\Delta_{\text{eff}} > 0$ , v76/v337). Keybox extended in `tfpt_research_contracts`; numerical module, Python-only by nature (Wolfram count unchanged, 376/376; the block identity flagged in the Wolfram README); ledger row GRAV.ENTROPIC.COMPRESS.01; AP2 (the continuum seam statement) stays **[O]**, no marker moves.

**2026-07-13 (v475 — the  $R^2$  kill test of the entropic-action bridge, *executed*: exact tensorial factors on the maximally symmetric background, raw entropic scalaron  $m^2 = \frac{4608\pi^2}{17}\bar{M}_{\text{Pl}}^2$  ( $\approx 51.7 \bar{M}_{\text{Pl}}$ , trans-Planckian)  $\Rightarrow$  the light-trace-mode reading is dead, KMS renormalisation the only surviving  $R^2$  route; plus the explicit Lorentzian-positivity witness; *no status moves*)**

- **v475\_entropic\_scalaron\_gate.py** (GRAV.ENTROPIC.SCALARON.01). Work package 5 of the v473 bridge. **[E]** On a maximally symmetric background the relative curvature operator  $\tilde{\mathcal{R}}\tilde{g}^{-1}$  has eigenvalues *exactly*  $\{R; R/4 \times 4; R/6 \times 6\}$  (Bianconi's Appendix-B flattening verified symbolically, incl. the  $6 \times 6$  two-form block), giving the exact vacuum action  $\mathcal{L}_B(R) = 3\beta R + \frac{17}{24}\beta^2 R^2 + O(R^3)$  — the 3 re-derives her Eq. (45) (= the v473 pin),  $\frac{17}{24} = \frac{1}{2}(1 + \frac{1}{4} + \frac{1}{6})$  is the exact tensorial  $R^2$  factor. **[E]** With the pinned  $\beta' = c_3/6$  both mass routes (Starobinsky matching and  $f'/3f''$ ) agree: the raw entropic scalaron is  $m_{\text{raw}}^2 = \frac{4608\pi^2}{17}\bar{M}_{\text{Pl}}^2$ , i.e.  $m_{\text{raw}} \approx 51.7 \bar{M}_{\text{Pl}} \approx 1.3 \times 10^{20}$  GeV — TRANS-PLANCKIAN, not light; v473's interpretation 2 (the TFPT scalaron as the light  $\mathcal{G}$ -trace mode) is *dead in naive form*; a viable mechanism must supply exactly  $\frac{72}{17}(8\pi)^9 \approx 1.69 \times 10^{13}$  in mass<sup>2</sup> (the kill-test number now WITH tensorial factors). **[E]** Domain: first log pole at  $R=1/\beta=384\pi^2\bar{M}_{\text{Pl}}^2$  (trans-Planckian) — the obstruction is the mass, not the domain. **[E]** Lorentzian-positivity witness: a timelike gradient gives the  $Gg^{-1}$  eigenvalue  $1 - \alpha v^2 \leq 0$  for  $v^2 \geq 1/\alpha$  (ln undefined), spacelike/Euclidean stay positive — the v473 **[C]** caveat exhibited, the OS route sturdier. **[C]** Verdict: interpretation 1 survives (Einstein +  $\Lambda$  level), 2 is dead naive, 3 (KMS-spectral renormalisation) is the ONLY surviving  $R^2$  route. Keybox extended in `tfpt_research_contracts`; Wolfram mirrors all three blocks (376/376); ledger row GRAV.ENTROPIC.SCALARON.01; no gate closes, no marker moves.

**2026-07-13 (v474 — the operator level of the entropic-action bridge: the  $D_5$  Clifford/spinor structure exhibited on the carrier Fock space, the Hodge fold identified as the  $5 \rightarrow \bar{5}$  conjugation, and the v473  $Q$ -target *decided* — supports exactly  $\{|\mathbb{Z}_2|, \text{rank } E_8, 2^{g_{\text{car}}}\}$ , the naive pair-block reading killed; *no status moves*)**

- **v474\_entropic\_hodge\_carrier.py** (GRAV.ENTROPIC.HODGE.01). Work packages 1 and 4-algebra of the v473 bridge, machine-checked on the finite carrier Fock space  $\Lambda^\bullet \mathbb{C}^5$  ( $\dim 2^{g_{\text{car}}} = 32$ ). **[E]** CAR $\Rightarrow$ Clifford: the ten  $\Gamma$ 's from  $(a_k^\dagger, a_k)$  satisfy  $\{\Gamma_a, \Gamma_b\} = 2\delta_{ab}$  exactly (all 100 pairs, integer matrices); the 45 bilinears  $[\Gamma_a, \Gamma_b]/4$  commute with  $\gamma = (-1)^N$ , span exactly  $45 = \dim \mathfrak{so}(10)$  and leak nothing from the 16-dim even subspace — the carrier half-spinor *is* the even exterior algebra *with* its spinor structure (the operator level of v2/v44/v197). **[E]** the Hodge–Dirac symbol:  $c(\xi) = a^\dagger(\xi) - a(\xi)$  satisfies  $c(\xi)^2 = -|\xi|^2 \text{Id}$  for symbolic  $\xi$  — Bianconi's  $D = d + \delta$  is Clifford multiplication at symbol level. **[E]** the fold is the  $5 \rightarrow \bar{5}$  conjugation: the traceless  $\mathfrak{u}(5)$  weights of  $\Lambda^4$  are the negatives of those of  $\Lambda^1$ , so her fiber  $1+5+10$  becomes the GUT half-spinor  $16 = 1+5+10$  (the SM carrier decomposition). **[E]** honest obstruction:  $\Omega^{\leq 2}$  is *not* Clifford-invariant ( $\Lambda^2 \rightarrow \Lambda^3$  leak) — the bridge must go through the fold, never through degree truncation (a recorded constraint on the AP2 compression). **[E]** the  $Q$ -target enumeration (the v473 **[C]** conjecture decided at the algebraic level):  $\text{Tr } Q^2 = d k^2 c_3^4 = 32 c_3^4$  with integer  $k$  has exactly the supports

$(d, k) = (2, 4), (8, 2), (32, 1) = \{|\mathbb{Z}_2|, g_{\text{car}} + N_{\text{fam}} = \text{rank } E_8, 2^{g_{\text{car}}} = \dim \Lambda^\bullet\}$  — minimal uniform solution  $q = c_3^2 = 1/(64\pi^2)$  per Fock mode, bonus identity  $\text{Tr } Q^2 / \delta_{\text{top}} = 32/48 = 2/3 = \dim \Lambda^\bullet / \Omega_{\text{adm}}$ ; the two-form block 10 (the problem-note “pair sector” conjecture) and the fiber 16 need irrational multipliers  $\sqrt{16/5}, \sqrt{2}$  — *killed* in uniform form. [C] the physical choice among the three admissible supports; [O] unchanged (AP2 compression + the  $R^2$  mechanism). Keybox extended in `tfpt_research_contracts` (Research Contract 2, the v473 keybox); Wolfram mirrors the enumeration and the weight-conjugation identity (373/373); ledger row GRAV.ENTROPIC.HODGE.01; no gate closes, no marker moves.

**2026-07-13 (v473 — the entropic-action bridge: Bianconi’s *Gravity from entropy* (Phys. Rev. D 111, 066001 (2025); arXiv:2408.14391) quantified as an *external candidate* for the missing action level of the parameter-free Einstein equation; her free constant pinned  $\beta'_B = c_3/6 = 1/(48\pi)$ , the  $R^2$  gap  $3(8\pi)^9$  pre-registered as kill test; *no status moves*, the v359 equation-of-state typing stays [O])**

- **v473\_entropic\_action\_bridge.py** (GRAV.ENTROPIC.ACTION.01). TFPT’s classical field equation  $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$  is derived thermodynamically (v358/v359) and honestly typed “equation of state, not a from-action quantisation” [O]. Bianconi’s entropic action  $S_B = -\text{Tr}_F \ln(\tilde{G}\tilde{g}^{-1})$  (the quantum relative entropy between the spacetime metric and the matter-induced metric, Araki-related via  $\Delta^{1/2} = \tilde{G}\tilde{g}^{-1}$ ) is a local covariant action whose variation gives Einstein + an emergent nonnegative  $\Lambda$  at low coupling — an explicit *external candidate* for exactly that missing variational layer. Machine-checked: [E] the carrier Hodge count (her  $\Omega^{0,1,2}$  fiber on the five-slot carrier  $\mathbb{C}^5$  counts  $1+5+10=16=2^{g_{\text{car}}-1} = \dim S_{D_5}^+$ , while on 4d spacetime it is only 11 — the 16 *requires* the carrier); [E] the one quantitative pin — matching her weak-coupling  $3\beta R$  to  $1/(16\pi G) = c_3/(2G)$  fixes her free constant *exactly*,  $\beta'_B = c_3/6 = 1/(48\pi)$ ; [E] her  $\Lambda_{\mathcal{G}} = \frac{1}{2\beta} \text{Tr}(\mathcal{G} - I - \ln \mathcal{G})$  is nonnegative and quadratic at the fixed point, and with  $\varepsilon_\star = e^{-\alpha^{-1}} Q$  it reproduces the closed v60 branch  $(3/4\pi^2)e^{-2\alpha^{-1}}$  with the exact target  $\text{Tr } Q^2 = 32c_3^4 = 1/(128\pi^4) = \frac{2}{3}\delta_{\text{top}}$  ([C] until  $Q$  is constructed); [E] the  $R^2$  kill test — the raw log expansion sits *exactly*  $3c_3^{-9} = 3(8\pi)^9 \approx 1.2 \times 10^{13}$  below the TFPT Starobinsky coefficient  $1/(12c_3^7)$ , so a direct identification is *false* without a mechanism (pre-registered); [C] the compression conjecture  $P_\Sigma(\tilde{G}\tilde{g}^{-1})P_\Sigma = \Delta_\Sigma^{1/2}$  (the bridge proper) stays open, with Lorentzian positivity flagged (TFPT’s OS route is the sturdier construction). New keybox in `tfpt_research_contracts` (Research Contract 2, after the v359 keybox); residual pointers updated in `introduction/tfpt_4_frontier/status` card. Wolfram mirrors the three exact blocks (371/371); ledger row GRAV.ENTROPIC.ACTION.01; *no* gate closes, *no* marker moves, zero TFPT knobs added.

**2026-07-03 (surfacing round — three already-computed unique readouts promoted to the public prediction surfaces: the axion–photon coefficient  $g_{a\gamma\gamma} = -4c_3$ , the free-seam Higgs band  $m_H \approx 129\text{--}134$  GeV, and the never-run BH HFQPO  $\times 1.5$  ladder-tooth search; no new vN, no status moves)**

- **Uniqueness scan follow-up (no new results — presentation only).** A structure scan for “what does TFPT predict that nobody else pins?” found three concrete, falsifiable readouts that were fully computed and ledgered but missing from the public prediction surfaces; they are now surfaced, with their existing typing unchanged. (i) **Axion–photon coefficient:** the determinant-line anomaly coefficient  $g_{a\gamma\gamma} = -4c_3 = -1/(2\pi)$ ,  $y^2 = 16c_3^2 = 1/(4\pi^2) \approx 0.0253$  (v207, the [E] leg of GRAV.ASYMP.01; the [C] caveat — coefficient, not a parameter-free  $g_{a\gamma\gamma}$  in  $\text{GeV}^{-1}$  — stated verbatim) joins the axion card in `website/lib/predictions.ts` and the frontier dark-matter mirror in `papers.ts`; the TeX source (`tfpt_4_frontier`) already carried it. (ii) **Higgs free-seam band:** the double-criticality prediction  $\lambda(\bar{M}_{\text{Pl}}) = \beta_\lambda(\bar{M}_{\text{Pl}}) = 0 \Rightarrow m_H = 133.5$  GeV (band 129–134,  $m_t/\alpha_s$ -sensitive; measured 125.25 sits a few GeV below — the known slight

metastability, Buttazzo NNLO  $\lambda(M_{\text{Pl}}) = -0.0143 \pm 0.0057, 2.5\sigma$ ) becomes a prediction card ([C]; v166, HIGGS.FREESEAM.01, kill: settled- $(m_t, \alpha_s)$  RGE pull off the surface at  $> 5\sigma$ ). (iii) **BH HFQPO ladder tooth**: the one discriminating HFQPO test — a third tooth at  $\nu_3 = (3/2)\nu_u$  (661.5/414/252/363 Hz), integer harmonics forbidden — that *no published search ever targeted* although the teeth lie in the searched RXTE band (experiments/hfqpo-ladder, preregistered): added as an [O] search-target bullet to the tfpt\_horizon\_readouts “Search targets (not claims)” gapbox (firewall language verbatim: non-canonical mapping, selection null 18.5%, J1859+226 +9.2 $\sigma$ , GR resonance favored; even a tooth hit would be [C]) and as an [O] prediction card. Website mirrors updated in the same change (predictions.ts 23  $\rightarrow$  25 cards, papers.ts frontier + horizon sections incl. the new search-targets mirror section); scorecard, ledger and all markers untouched.

**2026-07-03 (v472 — the determinant line over the  $U(1)$ -twist moduli of the collar model carries curvature = the inflow level: the v470 bridge lemma exhibited at the finite/model level; ALPHA.QUILLEN.EXACT.01 stays [O]. Plus: bird’s-eye rounds — electron-sector/Cabibbo whiteness probes, Calderón UV-universality, parameter-free flat-budget closure (sandbox); the flat-budget closure enters the evidence scorecard)**

- **v472\_quillen\_detline\_moduli.py** (ALPHA.QUILLEN.DETLINE.01). ALPHA.QUILLEN.EXACT.01 names the  $\alpha^3$  level as the curvature of the *determinant line over the  $U(1)$  seam moduli* (Quillen 1985 / Bismut–Freed CMP 106 (1986) curvature; Dai–Freed JMP 35 (1994) section); v470 computed the bulk Chern invariant only over the Bloch Brillouin zone (the translation-invariant shortcut). v472 computes the Quillen-shaped object itself: the same collar Hamiltonian ( $h(k) = \sin k_x \sigma_x + \sin k_y \sigma_y + (M - \cos k_x - \cos k_y) \sigma_z$ , v367/v470) in real space on an  $L \times L$  torus with *twisted* boundary conditions — the twist torus  $(\theta_x, \theta_y)$  is the moduli space of flat  $U(1)$  connections — and the Fukui–Hatsugai–Suzuki curvature of the *determinant line of the occupied frame* over that torus:  $\oint = 2\pi \cdot 1$  exactly (integer to  $10^{-9}$ ) at  $M=1$  for  $L=4$  AND  $L=6$  (size-independent); controls: trivial collar  $M=3 \rightarrow 0$ , orientation flip  $M=-1 \rightarrow -1$ ; the twist-moduli integer equals the Bloch-BZ integer for all three  $M$  (Niu–Thouless–Wu PRB 31 (1985), exhibited on the S3 collar model); the Fermi gap ( $= 2.0$ ) stays open over the whole twist torus, so the Dai–Freed section has no zero. **Reading [C]**: det-line holonomy over the  $U(1)$  moduli = inflow level  $k_0 = |C| = 1$  — the v470 bridge lemma “ $\delta \log \det_\zeta(\text{seam}) = \text{the inflow response}$ ” holds *verbatim at the finite/model level*, where  $\log \det$  of the occupied frame replaces  $\log \det_\zeta$ . **Not closed [O]**: the *continuum*  $\zeta$ -determinant identification on the abstract seam (= the SEAM.EQUIV.01 face); ALPHA.QUILLEN.EXACT.01 stays [O];  $\alpha^{-1}$  stays [E]. Structurally the same manoeuvre as v471 for SEAM.THEOREM.01’s replica side. 7/7 checks; suite green through v472 (466 modules); numerical (dense eigh + FHS twist grid), Python-only by nature. Papers updated: tfpt\_1 (sixth honest step in the EM closure), tfpt\_research\_contracts (residual certification), tfpt\_safeguards ( $\alpha$ -form lever), root README, zenodo\_description.html. Exploration original: experiments/tfpt-discovery/quillen\_detline\_twist\_moduli.py (same day).
- **Bird’s-eye whiteness rounds (sandbox, experiments/tfpt-discovery/, nothing else promoted)**. (i) **electron\_sector\_cabibbo\_probe.py**: the  $\alpha$  metrology triangle (TFPT  $\alpha^{-1} = 137.0359992168$  vs Rb-2020  $+0.99\sigma$  / CODATA22  $+1.90\sigma$  /  $a_e$ -route  $+3.39\sigma$  / Cs-2018  $+6.33\sigma$  — TFPT takes the Rb side of the  $5.5\sigma$  metrology split; new honest pressure point recorded); the v204 seam-vertex universality catch (a flavor-universal reading of  $a_\mu^{\text{seam}} = 2.879 \times 10^{-9}$  is excluded by the electron at  $\sim 2 \times 10^4\sigma$ ; linear scaling at  $\sim 10^2\sigma$ ; quadratic survives — the [C] bridge must scale at least quadratically in  $m_\ell$ ); the Cabibbo-angle anomaly as a third dissolution watchdog ( $\lambda_C = 0.2243762$  exact sits  $+0.08\sigma$  on the PDG-2026 rescaled kaon average, between  $K_{\ell 3}$  and  $K_{\mu 2}$ ; exact unitarity puts  $V_{ud} = 0.97450$  — superallowed is the outlier at  $+2.6\sigma$ , all neutron/pion routes agree). (ii) **kernel\_calderon\_uv\_universality.py**: the raw-collar



Calderón/DtN datum is UV-universal (mode-by-mode first-order convergence to the exact Bessel symbol  $mI'_\nu(mR)/I_\nu(mR)$ , Richardson residual  $3 \times 10^{-4}$ ;  $N_\theta$ -independent;  $|k|$  principal symbol;  $\mu_4$  block structure at machine precision; both spec- $T$  masses; 3-fold mesh-defect negative control) — the operator equality demanded by QGEO.KERNEL.01 has a well-defined discretization-independent left-hand side (gate stays **[O]**). (iii) `flat_budget_closure.py` + scorecard row: see next bullet.

- **Flat-budget closure enters the evidence scorecard (experiments-only, pattern candidate).** The zero-dial closure  $\{\Omega_b = \varphi_0^{\text{ret}}(1 - \frac{1}{4\pi})$  [frozen],  $\Omega_c = \frac{2}{7}(1 - \frac{1}{4\pi})$  [the 2026-07-02 numerology-flagged Nebenbefund],  $\rho_\Lambda/\bar{M}_{\text{Pl}}^4 = (3/4\pi^2)e^{-2\alpha^{-1}}$  [frozen],  $\Sigma m_\nu = 0.0588$  eV [v468 route]} + spatial flatness determines the whole flat  $\Lambda$ CDM budget:  $H_0 = 67.15$  km/s/Mpc ( $-0.39\sigma$  Planck18,  $-5.67\sigma$  SH0ES22 — the parameter-free Planck side of the Hubble tension),  $\Omega_m = 0.3133$  ( $-0.27\sigma$ ),  $t_0 = 13.86$  Gyr; honest stresses recorded:  $\omega_b -2.02\sigma$  (BBN-only  $-0.73\sigma$ ),  $\omega_c -1.19\sigma$ , DESI-DR2  $H_0 -2.16\sigma$  (one correlated low- $h$  direction). Scorecard row `cosmo / flat budget closure` (consistent, composite: shares the `phi0_seed` group (and the  $\rho_\Lambda$  leg the  $\alpha$  engine),  $\Omega_c$  leg in `alternative_group=axion_branch` vs the axion-spine DM reading — never double-counted); nothing promoted to `verification/`.

**2026-07-02 (v471 — the Seam–Horizon replica chain exercised numerically on the discretized collar: the kernel-identification premise of SEAM.THEOREM.01 discharged at the finite/model level; residual retypes to the continuum MMST leg + the anchor; gate stays **[O]**. Plus: the GW ringdown-echo channel hardened — 10 loudest events incl. GW250114, redshift-corrected templates, stacked search and a 12-variant signature battery, all null)**

- **v471\_seam\_horizon\_replica.py (SEAM.EHMODEL.04).** The residual isolated by v150/v151/v152 — “the RP seam kernel *is* this BFK Calderón datum” — is exercised numerically on a discretized seam collar with the seam’s *own* kernel and *real* replica sheets: the matched conical deficit has slope  $2C(\gamma)$  on deficit cones (0.01–0.05%) and replica sheets  $n=2$  (0.8%),  $n=3$  (1.5%); the EH coefficient is cutoff-independent (0.5% drift) while the  $\zeta$ -scale  $\mu_{\text{lat}}$  drifts — the v152 anchor exhibited on data, with *no* canonical  $\ln(m/\mu)=3/4$  emerging (recorded honestly); the BFK/Schur split is exact ( $\sim 10^{-16}$ ) with the Calderón factor conically clean *measured on the kernel* (pure cut-edge law, tip term 0, power-checked) and the halves carrying  $2C_{\text{cone}}$  via Kac doubling; the v302 transfer masses  $6\ln(3/2)$ ,  $6\ln 3$  sit on the calibrated EH line ( $\leq 0.05\%$ ); and the Perron/attractor mode ( $m=0$ ) is IR-divergent — **the recovery gap  $\Delta=6\ln(3/2)$  is what makes the induced-gravity coefficient finite** (the same gap that makes the attractor unique makes  $1/G$  finite). SEAM.THEOREM.01 stays **[O]**; its residual retypes to the *continuum* scaling limit (the MMST class — the same single residual as SEAM.EQUIV.01, v336) plus the one declared anchor. 16/16 checks; suite green through v471 (465 modules); numerical (sparse lattice determinants), Python-only by nature. Papers updated: `origin_theory` (new collar paragraph in the Seam–Horizon gapbox + residual retyping), `tfpt_research_contracts` (R3 premise), `tfpt_4_frontier` (3/4 audit note), root README, `verification/README`, `zenodo_description.html`. Exploration original: `experiments/theory-contracts/seam_horizon_replica.py` (17/17, same day).
- **GW ringdown-echo channel hardened (experiments/gw-ringdown-echo, scorecard row `data_limited`→`consistent`).** (i) Strain coverage extended to the 10 loudest ringdown events incl. GW250114 (network SNR 78.6, O4b; new `fetch_strain_4096.py` crops the 4096 s GWOSC archive segments). (ii) **Redshift correction:** the GWTC catalogue reports source-frame masses, the observed ringdown is at  $f_0/(1+z)$  — all templates moved to detector-frame  $M(1+z)$  (before the fix GW190521,  $z=0.56$ , was filtered  $\sim 1.6\times$  too high); all stages re-run. (iii) Stage 1b *stacked* search over 23 detector streams: stacked  $p=0.262$ , kernel-consistent streams 0/23 — STACKED NULL. (iv) Stage 1c *signature battery* (Bonferroni  $\times 12$ ): ratio semantics



(amplitude  $(2/3)^6$  vs *energy reading*  $(2/3)^3$  vs step  $2/3 \times \mu_4$  per-bounce phases  $\{0, \pi/2, \pi, 3\pi/2\}$  (the boundary-birefringence analogue — path birefringence from parity violation  $c_- = 8$  cancels in the inter-echo ratio; only a per-reflection phase does not)  $\times$  lags 0.5–350 ms (ECO through Planckian window), with joint  $(2, 2, 0) + (2, 2, 1)$  subtraction and a template-agnosticism control: `NO_VARIANT_ECHO` on all 10 events — the null is robust against alternative signature readings. Search-target firewall unchanged: upper bound, no detection, no tension claim.

## 2026-07-02 (v295 robustness patch — external-review follow-up: the numeric $a_2$ -kernel tolerance must dominate its own finite-difference floor)

- `v295_flataway_a2_exact.py` (FLATAWAY.A2.01). Alessandro’s Git-baseline audit (commit 8cb8ad0) hit a platform-marginal failure in check 3: the exact divided-difference  $a_2$  kernel is compared against an  $\varepsilon = 10^{-3}$  second finite difference whose *own* truncation error is  $O(\varepsilon^2) \sim 10^{-6}$ , yet the tolerance was  $10^{-6}$  — on his machine  $k=3$  landed at  $1.1 \times 10^{-6}$  (ours:  $2.9 \times 10^{-7}$ ). Fix: tolerance widened to  $5 \times 10^{-6}$  so it *dominates* the control’s error floor, with the reasoning recorded in the check text; the load-bearing anchor remains the *closed form* at machine precision (v296), the numeric witness stays as the independent cross-check. A numerical-robustness patch, no claim or status change; ledger row updated; suite green (ALL CHECKS PASSED through v470).

## 2026-07-02 (v469/v470 — the closure-route round on the two named targets: the SEAM.EQUIV.01 extension leg re-founded on 1995–2001 peer-reviewed crossed-product theorems + the realisation axiom reduced to invariant level; the $\alpha^3$ level = the *computed* bulk Chern invariant + the seam $F$ -normalisation = the embedding index $k_Y = 5/3$ ; both targets stay [O])

- `v469_seam_crossedproduct_route.py` (SEAM.EQUIV.CROSSEDPRODUCT.01). The 128-spinor extension leg of SEAM.EQUIV.01 no longer rests on 2024/2025 preprints: [E] the  $SO(16)_1$  spinor has  $h_s = N/16 = 16/16 = 1 \in \mathbb{Z}$ , so its statistics phase is  $+1$  — *exactly* the Longo–Rehren locality criterion — and the  $\mathbb{Z}_2$  simple-current crossed product  $SO(16)_1 \rtimes \{1, s\}$  is a *local* index-2 extension by the peer-reviewed subfactor package (Longo–Rehren, Rev. Math. Phys. 7 (1995); the level-1  $so(N)$  net with its DHR sectors from even-CAR: Böckenhauer, Rev. Math. Phys. 8 (1996); simple-current extensions of nets: Böckenhauer–Evans, Comm. Math. Phys. 197 (1998)); KLM (Comm. Math. Phys. 219 (2001)) gives  $\mu(B) = 4/2^2 = 1 \Rightarrow$  holomorphic  $\Rightarrow (E_8)_1$  by the v463 pin. The AGT/AMT lattice-VOA route (v459) is *demoted to an independent second witness*. [E] the index-4  $\mu_4$  glue runs on the same integer:  $h(J^k) = \{1, 1, 1\}$  for  $k=1, 2, 3$  (the v125 isotropy  $q(k(1, 1)) = k^2$  at net level),  $\mu = 16/4^2 = 1$ , same endpoint. [E] the realisation input is reduced from model fiat to *invariants* (R1’): quasi-free [C] (v155/v160) + gap [E] (v302) + class D +  $c_- = 8$  [E] (v456, from P1), with the invariants *computed* — per-copy FHS Chern  $|C| = 1$  vs 0 (control),  $\nu = 16 \times |C| = 16$ ,  $c_- = \nu/2 = 8$  — and  $\nu \equiv 0 \pmod{16}$  is exactly Kitaev’s class-D phase whose edge admits a purely bosonic description = the  $(E_8)_1$  level-1 state (Ann. Phys. 321 (2006), sixteen-fold way); the phase-invariance of  $c_-$  is cited (Kapustin–Spodyneiko, PRB 101 (2020); Kim et al., PRL 129 (2022)). [O] *not closed*: the cited theorems stay cited; SEAM.EQUIV.01 stays [O].
- **Lean hardening.** `TfptCarrier/SeamResidualAxiom.lean` gains the *parallel* peer-reviewed derivation `seamResidualClosed'`: the invariant-level axiom `collar_invariants` (R1’) + *one* combined 1995–2001 package `crossedproduct_route_theorem`; the new joints are kernel facts with *no* axioms — `lr_locality_integer` ( $h_s = 16/16 \in \mathbb{Z}$ ), `glue_locality_integers` ( $\mu_4 h(J^k) \in \mathbb{Z}$ ), `klm_holomorphic` ( $4/2^2 = 1, 16/4^2 = 1$ ), `sixteenfold_pin` ( $\nu = 16 \equiv 0 \pmod{16}, c_- = 8$ ), bundled as `crossedproduct_arithmetic` ([propext] only); `#print axioms seamResidualClosed'` = exactly four; the original `seamResidualClosed` (AGT/AMT witness) is kept unchanged as the independent second route; `lake build` OK (3285 jobs).

- **v470\_alpha\_inflow\_level.py** (ALPHA.QUILLEN.INFLOW.01). Both residuals left on ALPHA.QUILLEN.EXACT.01 after v433–v435 are upgraded. [E] target re-verified ( $\alpha^{-1}=137.0359992168$ ; Quillen split  $<10^{-35}$ ). [C] *the level is computed, not minimal*: the  $a^3$  coefficient  $k_0=1$  equals the FHS bulk Chern invariant  $|C|=1$  of the *same* p+ip collar that realises S3 (v367/v461) — “why exactly 1” is answered by computation plus the cited inflow identification (TKNN PRL 49 (1982) / Avron–Seiler–Simon PRL 51 (1983) quantisation; Callan–Harvey, Nucl. Phys. B250 (1985) inflow; APS 1975 / Alvarez–Gaumé–Della Pietra–Moore 1985 / Witten, Rev. Mod. Phys. 88 (2016)  $\eta$ =CS part of  $\delta \log \det$ ; Quillen 1985 / Bismut–Freed 1986 determinant-line integrality), replacing v435’s minimality assumption. [E]  $k_Y = \text{tr}(Y^2)/\text{tr}(T_3^2) = (5/6)/(1/2) = 5/3$  (Ginsparg, Phys. Lett. B197 (1987)):  $(3/5) \times \frac{41}{6} = \frac{41}{10} = b_1$  — the “GUT 3/5” is the inverse embedding index;  $\frac{41}{6} = \frac{20}{3} + \frac{1}{6}$  (carrier decomposition). [C] the seam  $F$ -normalisation (the one [C] of v434) is *retyped*: once the seam net is  $(E_8)_1$  level 1,  $\langle JJ \rangle = k/(z-w)^2$  is fixed by  $k_Y$  — level-1 current-algebra rigidity, zero independent content, a face of SEAM.EQUIV.01 (per the v382 typing). [E] *structural unification*: one invertible phase, two quantised responses —  $c_- = 8$  (gravitational, feeds SEAM.EQUIV.01/S3) and  $C=1$  ( $U(1)$ , feeds EM1) — so closing  $R1'$  feeds *both* named targets. [O] *not closed*: the bridge lemma “ $\delta \log \det_\zeta(\text{seam}) = \text{the inflow response}$ ” stays the named cited step; ALPHA.QUILLEN.EXACT.01 stays [O];  $\alpha^{-1}$  stays [E].
- **Integration.** Both registered in run\_all.py + script\_registry.csv (cluster frontier); ledger rows SEAM.EQUIV.CROSSEDPRODUCT.01 + ALPHA.QUILLEN.INFLOW.01; FORM.SEAM.RESIDUAL.01 updated (the seamResidualClosed’ route); the exact joints Wolfram-mirrored (365  $\rightarrow$  368: the LR/glue locality integers, the KLM  $\mu$ -arithmetic, the sixteen-fold pin, the  $k_Y = 5/3/b_1$  identities); cited in tfpt\_1 ( $\alpha$  fixed point + EM-Ward residual prose), tfpt\_research\_contracts (seam residual + Lean section), introduction (closure ledger), tfpt\_5\_redteam (Target A residual wording) and the README/website status surfaces; exploration provenance experiments/tfpt-discovery/closure\_route\_\*.py (2026-07-02 note in experiments/next.txt). Suite ALL CHECKS PASSED through v470. Curated version bumped to v5.4 (Zenodo deposit updated).

**2026-07-02 (v467/v468 — two named post-hoc [O] flavor candidates, record unchanged: the three  $\sim 2\sigma$  mixing tensions as *one* third-generation  $-\varphi_0$  pattern, and the splitting ratio  $\Delta m_{21}^2/\Delta m_{31}^2 = |J_{\text{PMNS}}|$ )**

- **v467\_thirdgen\_mixing\_pattern.py** (FLAV.THIRDGEN.PATTERN.01). The suite’s three  $\sim 2\sigma$  mixing tensions ( $\sin^2 \theta_{13} + 2.0\sigma$ ,  $\sin^2 \theta_{23} + 1.76\sigma$ ,  $|V_{cb}| + 1.79\sigma$ ; v307) are exactly the three *third-generation* channels and share *one* relative deviation  $-5.36\% \pm 1.67\%$  — non-zero at  $3.2\sigma$ ,  $-0.03\sigma$  from  $-\varphi_0$ . [E] vocabulary identities:  $\lambda_C^2 = \varphi_0(1-\varphi_0)$  (the frozen Cabibbo);  $\sin^2 \theta_{23} = (1-\varphi_0)/2 \Leftrightarrow \cos 2\theta_{23} = \varphi_0$ ; the suppression size  $\varphi_0 = 5.3\%$  sits inside the seam-gapped band  $(2/3)^6 = 8.8\%$  (v393). [N] the ladder-base forms  $\lambda_C^2 e^{-5/6} = 0.021880 / (1-\varphi_0)/2 = 0.473414 / \lambda_C^2/(1+\lambda_C) = 0.041119$  drop the joint four-observable  $\chi^2$  from 10.4 to 0.11 with *zero* new parameters;  $|V_{ub}|$  (no bare  $\varphi_0$ ) and the 1–2 channels stay untouched and correct (dressing  $\lambda_C$  would fail at  $-24\sigma$  — scope checked, not chosen); Daya Bay alone already prefers the pattern form ( $\Delta\chi^2 = 4.3$ ). [C] **honest**: post-hoc (5/16 equally-simple dressings fix the two PMNS tensions;  $(1-\varphi_0)/(1-c_3)/\text{resolvent degenerate} < 1.4\%$ ). [O] no seam-transport derivation (the Bernoulli-variance reading  $\lambda_Y^2 = \text{Var}[\text{Bernoulli}(\varphi_0)]$  is an exact identity but only an interpretation) — the frozen record (v84/REG.FREEZE.01) is *unchanged*. [X] pre-registered, symmetric:  $\theta_{13}$  stable at 0.0231 kills the pattern, at  $\sim 0.0219$  (sub-1%) kills the record’s bare- $\varphi_0$  reading, exclusive  $|V_{cb}| \sim 0.0395$  kills both — JUNO-era reactors + Belle II decide. Python-only.
- **v468\_dm2\_ratio\_jarlskog.py** (FLAV.DM2RATIO.01). The splitting ratio  $\Delta m_{21}^2/\Delta m_{31}^2$  was previously *unpredicted*; the frozen PMNS matrix (v9/v268/v270,  $\delta = 4\pi/3$ ) has one canonical derived CP invariant, and [N]  $|J_{\text{PMNS}}| = 0.029653$  meets the NuFIT 6.0 measured ratio

$0.029805 \pm 0.000792$  at  $-0.19\sigma$  (equivalently  $m_2/m_3 = \sqrt{|J|} = 0.17220$  vs  $0.17264(229)$ ) — zero dials. [N] the informal `tfpt_2` table heuristic  $m_2/m_3 \sim \pi\varphi_0 = 0.16704$  (a “ $\sim$ ” entry, never frozen) is now  $-2.44\sigma$  and is *superseded as the comparator*. [C] spectrum consistency:  $\Sigma m_\nu = m_3(1 + \sqrt{|J|}) = 0.0588$  eV reproduces the documented 0.0586 eV (v272); no absolute number is added. [O] no mechanism (the  $\mu\tau$  texture gives  $\theta_{13}=0$  at leading order). [X] JUNO shrinks the window  $\sim 4\times$ ; adopting the v467 pattern would shift  $|J| \rightarrow 0.028849$  ( $-1.21\sigma$  today) — the two candidates *cross-discriminate* at JUNO precision. Python-only.

- **Integration.** Both registered in `run_all.py` and `script_registry.csv` (cluster neutrinos); ledger rows `FLAV.THIRDGEN.PATTERN.01` and `FLAV.DM2RATIO.01` (both [O] candidates); cited in `tfpt_2` (§ Neutrinos: pressure-point block + new candidate paragraphs, the  $m_2/m_3$  table row retyped to the  $|J|$  comparator, the octant prose kept honest — record does not select, the candidate would), `introduction` (closure ledger), `tfpt_4_frontier` (live kill-test board) and `tfpt_5_redteam` (seed-hyperplane caption); v328’s “single outlier” wording scoped to the seed observables; exploration provenance `experiments/tfpt-discovery/pattern_hunt_pmns_dressing.py` (2026-07-02 note in `experiments/next.txt`); Python-only (flagged in the Wolfram README).

**2026-07-01 (v466 — a sixth seed channel from a new measurement sector: the charged-lepton mass ratio  $m_e/m_\mu = \frac{12}{7}\varphi_0^2$  back-solves the same axiom seed to  $-0.11\%$ , extending the empirical over-determination to *five* sectors)**

- **v466\_seed\_leptonmass\_channel.py (SEED.LEPTONMASS.01).** The seed over-determination of v306/v465 used five observables spanning three measurement sectors (neutrino mixing, CMB, quark mixing). This module adds the *charged-lepton mass* sector: the ratio  $m_e/m_\mu = \frac{12}{7}\varphi_0^2$  is a sixth independent readout of the *one* axiom seed  $\varphi_0 = 1/(6\pi) + 48c_3^4$ . [E] plumbing: the axiom  $\varphi_0$  reproduces the frozen  $m_e/m_\mu = \frac{12}{7}\varphi_0^2 = 0.00484673$  to  $10^{-9}$ . [N] **new sector (headline):** back-solving  $\varphi_0$  from the measured *pole* ratio  $m_e/m_\mu = 0.004836$  (PDG 2024) gives  $\varphi_0 = 0.053113$ , only  $-0.11\%$  from the axiom — a fourth measurement sector beyond {neutrino mixing, CMB, quark mixing}. [C] **transport caveat:** the formula is the *source* ratio and the datum the *pole* ratio; the source→pole shift is the seam-gapped correction class (bound  $(2/3)^6 = 8.78\%$ , v393) but largely cancels for  $m_e/m_\mu$ , so the  $0.11\%$  residual sits far inside the band — the channel is added as a *consistent new sector*, not a sharper pin. [N] **extends:** the six-channel cluster (v306’s five + this) has  $\chi^2/\text{dof} < 1$  and the error-weighted mean stays at the axiom to  $< 0.1\%$ ; with the corroborating heavy-quark  $m_t/m_b$  ( $-0.23\%$ ) it is a seven-channel, five-sector agreement. [N] **power:** a data↔formula shuffle explodes the cluster  $\chi^2$  by  $> 100\times$ , and the scheme-ambiguous  $m_e/m_s = (34/47)/\varphi_0$  (FLAG 11–14) is deliberately not counted. [C] **honest:** theoretically still the *one* seed (v305 multiplicity 1) — an *empirical* strengthening, not theoretical independence;  $\varphi_0$  is fixed by the axioms, so it upgrades *no* prediction and closes *no* gate. Python-only (floats).
- **Integration.** Registered in `run_all.py` and `script_registry.csv` (cluster registry); ledger row `SEED.LEPTONMASS.01`; cited in `tfpt_5_redteam` (the seed-cross-validation firewall winbox, alongside v305/v306/v465); Python-only (flagged in the Wolfram README).

**2026-07-01 (v465 — the cross-sector one-parameter seed test: a CMB birefringence angle and a quark Cabibbo angle point to the same seed within  $0.01\sigma$ , the  $\theta_{13}$ -independent externally-runnable slice of v306)**

- **v465\_seed\_crosssector\_joint.py (SEED.CROSSSECTOR.01).** The sharpest falsifiable signature of a *shared-origin* theory is a correlation no non-shared-origin model predicts. In the Standard Model the cosmic-birefringence rotation  $\beta$  (a CMB *EB/TB* polarisation observable) and the Cabibbo angle  $\lambda_C = |V_{us}|$  (a quark-flavour observable) are unrelated; TFPT reads both as functions of the *one* axiom seed  $\varphi_0 = 1/(6\pi) + 48c_3^4$ . [E] plumbing: the axiom  $\varphi_0$

reproduces the frozen  $\lambda_C$ ,  $\sin^2 \theta_{12}$ ,  $\beta$  to  $10^{-9}$ . [N] **cross-sector (headline)**: back-solving  $\varphi_0$  *independently* from  $\beta$  (CMB; ACT DR6 + Planck PR4 combined) and from  $\lambda_C$  (quark; PDG 2024) gives the same seed within  $0.01\sigma_{\text{comb}}$ . [N] **one-parameter fit**: the error-weighted mean of the three back-solved  $\varphi_0$  reproduces the axiom to 0.06%; the joint  $\chi^2$  at the axiom (zero free parameters, 3 dof) is 0.007,  $\chi^2/\text{dof} < 1$ , combined pull  $0.08\sigma$ . [N]  $\theta_{13}$  **exclusion declared**: adding  $\sin^2 \theta_{13}$  raises  $\chi^2$  by  $\sim 4$  (the documented  $\sim 2\sigma$  pressure point, v328/v338), so this slice is the honest  $\theta_{13}$ -independent statement, not a hidden dilution. [N] **power**: a data $\leftrightarrow$ formula shuffle explodes the cluster  $\chi^2$  by  $> 5 \times 10^4$ . A consistency / out-of-sample statistic (a subset of v306);  $\varphi_0$  is fixed by the axioms, so it upgrades *no* prediction and closes *no* gate. Python-only (floats).

- **Integration.** Registered in `run_all.py` and `script_registry.csv` (cluster registry); ledger row SEED.CROSSECTOR.01; cited in `tfpt_5_redteam` (the seed-cross-validation firewall winbox, alongside v305/v306) and in the `origin_theory` birefringence dependency chain; Python-only (flagged in the Wolfram README).

**2026-06-30 (v463/v464 + Lean + Wolfram — the  $c=8$  holomorphic-uniqueness pin and the one-particle realisation rigidity: the SEAM.EQUIV.01 residual reduced to *entirely certification* on both halves)**

- **The bird’s-eye reframing.** The recent G-block modules all sharpened the *target* of the scaling limit; this round attacks the two halves that actually remain. The *continuum existence* is already citable (v458 MMST + v459 AGT/AMT); what was implicit was (B) *why* a  $c=8$  holomorphic limit must be  $(E_8)_1$ , and (A) the *realisation* input  $R_1$  that the abstract seam IS the concrete quasi-free collar. v463 closes B; v464 reduces A to its one-particle data.
- **v463\_seam\_c8\_holomorphic\_uniqueness.py (SEAM.EQUIV.UNIQ.01).** The  $c=8$  holomorphic-uniqueness pin. [E]  $c=8$  is NOT unique at level 1:  $c = \dim \mathfrak{g}/(1+h^\vee) = 8$  has THREE simple solutions  $A_8$  ( $\dim 80$ ),  $D_8 = \mathfrak{so}(16)$  ( $\dim 120$ ),  $E_8$  ( $\dim 248$ ), so the  $\det K=4$   $SO(16)$  rival is a genuine same- $c$  competitor. [E] HOLOMORPHY FORCES  $\dim V_1=248$ : the only weight-0 form with leading  $q^{-1/3}$  and integer coefficients is  $j^{1/3} = E_4/\eta^8$ ,  $q^1$  coefficient 248 (the 1-dim candidate space, vacuum  $a_0=1$ ). [E]  $\dim 248 + \text{level-1 } c=8 \Rightarrow E_8$  uniquely ( $248/(1+30)=8$ ); tower  $\{1, 248, 4124, 34752, 213126\} = E_4/\eta^8 = j^{1/3}$ . [C] Dong–Mason/Schellekens: the holomorphic  $c=8$  VOA is unique  $= V_{E_8}$  (even unimodular rank-8 lattice unique  $= E_8$ , Mordell–Witt) — so the identification half is classification-forced.
- **v464\_seam\_oneparticle\_rigidity.py (SEAM.EQUIV.ONEPARTICLE.01).** The one-particle rigidity attacking  $R_1$ . The seam is quasi-free (v113/v160/v161), so by Araki self-dual CAR the net is the second quantisation of its one-particle symbol  $P$ . [E] RIGIDITY: the gapped ground state is the UNIQUE quasi-free state with  $P =$  projection onto  $K < 0$  ( $P^2 = P$  to  $< 10^{-12}$ ),  $K$  fixed by gap + chirality + reflection (v426). [E] ONE-PARTICLE LIMIT EXISTS: the kernel is Cauchy on a fixed window ( $\max |P_{2N} - P_N| \sim 1/N \rightarrow 0$ ). [E] the limit is  $c=1$  Dirac: entanglement  $S(L) = (c/3) \ln L$  gives  $c \rightarrow 1$ , 16 copies  $\Rightarrow c_- = 8$  (ties v450). [C] Shale–Stinespring makes the Bogoliubov map implementable, so  $R_1$  is upgraded from a bald axiom to “the unique quasi-free realisation, modulo the named CAR functor”. Python-only (numpy).
- **Lean (SeamResidualAxiom.lean).** Adds the v463 kernel facts `c8_three_candidates` ( $A_8/D_8/E_8$  each  $\dim = 8(1+h^\vee)$ ) and `holomorphy_selects_e8` (forced  $\dim V_1=248=E_8 \neq 80, 120$ ) to `residual_arithmetic` (still [propext]-only); v464 is numerical. `#print axioms` `seamResidualClosed` still lists exactly FOUR axioms.
- **Wolfram + integration.** v463 mirrored (the three candidates, the forced 248, the  $E_4/\eta^8$  tower); the extension count is now 365/365. Integrated across the contracts paper (attacks (xxx)/(xxxi)), README, zenodo, ledger and website. Net effect: across (v)–(xxxi) the



SEAM.EQUIV.01 residual is now *entirely certification* — a named, hypothesis-audited package (MMST, AGT/AMT, Dong–Mason, Araki, Shale–Stinespring) with no open internal mechanism — yet SEAM.EQUIV.01 stays [O] because it still rests on cited continuum-existence theorems we do not re-prove.

**2026-06-30 (v461/v462 + Lean axiom merge — the strict-locality obstruction made explicit, the 128-spinor extension exhibited at character level, and the SEAM.EQUIV.01 residual reduced to ONE realisation axiom + ONE combined cited theorem)**

- **v461\_seam\_strict\_locality.py (SEAM.S3.LOCALITY.01).** Sharpens v460’s verdict from a *cited* statement to an *exhibited* topological obstruction on the same v367  $p+ip$  collar. [E] CHERN INTEGER: the FHS Chern is  $C=+1$  (chiral  $M=1$ ), 0 (trivial  $M=3$ ),  $-1$  (opposite mass). [E] WILSON-LOOP / WANNIER-CENTRE WINDING = CHERN: the hybrid Wannier centre  $\theta(k_y)$  winds by  $|C|=1$  (chiral) and 0 (trivial) over the BZ; exponentially-localised Wannier exist IFF the winding is 0 (Brouder–Panati et al., PRL 98 046402; Thouless), so the chiral collar has NONE. [E] OBSTRUCTION INTEGERS TIED: winding =  $|C|=1 \neq 0$  and  $c_- = 8 = g_{\text{car}} + N_{\text{fam}} = \text{rank } E_8 \neq 0$  (the same 8 as  $c_3$ ’s one-sided count, v456). [E] NEG CONTROL: BOTH phases are gapped, yet only the trivial (winding 0) admits exp-localised Wannier and a strict finite-range commuting projector — it is the CHIRALITY, not the gap. [C][O] Kapustin–Fidkowski (arXiv:1810.07756): a strictly finite-range commuting projector would force winding 0,  $C=0$ ; since  $C=1 \neq 0$  NONE exists, so v424 sub-step (i)’s realisation is PROVABLY the quasi-local NPW26 LTO net (strict locality topologically forbidden, not a missing premise). SEAM.EQUIV.01 stays [O]. Python-only (numpy); the winding/Chern integers Wolfram- and Lean-mirrored.
- **v462\_seam\_spinor\_continuum.py (SEAM.EQUIV.SPINOR.01).** The constructive companion of the v458 (MMST audit) / v459 (lattice-VOA) residual: the 128-spinor extension  $SO(16)_1 \rightarrow (E_8)_1$  exhibited three ways. [E]  $(E_8)_1$  TOWER FROM  $SO(16)_1$  SECTORS: the exact Jacobi/ $E_8$  identity  $\theta_2^8 + \theta_3^8 + \theta_4^8 = 2E_4$  gives  $\chi_{(E_8)_1} = E_4/\eta^8 = \chi_o^{SO16} + \chi_s^{SO16}$  — the holomorphic  $(E_8)_1$  character IS the  $SO(16)_1$  vacuum + spinor sector sum. [E] THE 128 SPINOR WEIGHT-1 STATES FILL  $120 \rightarrow 248$ :  $\chi_o$  gives 120,  $\chi_s$  gives 128, summing to the exact tower  $\{1, 248, 4124, 34752\}$  ( $248=120+128$ ). [E] FINITE 16-MAJORANA FOCK:  $C(16, 2)=120$  NS bilinear currents and  $2^{16/2}=256$  Ramond ground states = 128 spinor + 128 cospinor. [E] FINITE- $L \rightarrow$  CONTINUUM CONVERGENCE: the critical free-fermion ring Casimir gives  $c_- \rightarrow 8$  and the lowest scaled NS gap  $x_1 \rightarrow 1$  with  $|x_1 - 1| \sim 1/L^2$  ( $L=64..1024$ ). [C][O] AGT/AMT (v459) supply the rigorous extension; SEAM.EQUIV.01 existence (v336) stays [O]. Mixed exact ( $\theta$  identity + tower +  $120/128/256$ , Wolfram-mirrored) + numerical (finite- $L$ , Python-only).
- **Lean (TfptCarrier/SeamResidualAxiom.lean, FORM.SEAM.RESIDUAL.01).** The two former cited axioms `mmst_existence`  $\circ$  `agt_lattice_extension` are MERGED into ONE combined external axiom `seam_realisation_theorem` (dropping the intermediate `ChiralCFT_c8`), so `#print axioms seamResidualClosed` now lists exactly FOUR axioms (down from six). Two more kernel facts are added axiom-free: `strict_locality_forbidden` (v461: winding =  $|C|=1 \neq 0$ ,  $c_- \neq 0$ ) and `fock_sectors` (v462:  $C(16, 2)=120$ ,  $2^8=256$ , spinor 128,  $120+128=248$ ); `residual_arithmetic` (now including both) depends only on `[propext]`. `lake build` OK. So the residual is reduced to ONE realisation axiom + ONE combined cited theorem.
- **Wolfram. tfpt\_readouts\_extension.wl** adds 3 exact mirrors (total 363/363): the v461 obstruction integer-logic and the v462  $\theta$ -identity +  $120+128=248$  sector / Fock counts.



**2026-06-30 (v460 — the explicit flat-band commuting-projector (LTO) realisation of the S3 input: advancing the SEAM.EQUIV.01 sub-step (i) from abstract existence to an explicit gap-protected quasi-local projector net carrying the  $\mathbb{Z}_2$  reflection and the  $\mu_4$  clock)**

- **v460\_seam\_s3\_lto.py (SEAM.S3.LTO.01).** NPW26 (arXiv:2605.10693) prove the intrinsic Bisognano–Wichmann lemma for *commuting-projector* (local-topological-order) models with a  $\mathbb{Z}_2$  reflection; v424 reduced the keystone BW residual to that theorem modulo two open sub-steps, of which (ii) was settled by v426 and the continuum lift of BW by the uniform-in- $N$  tower v455. This module attacks the *commuting-projector* side of sub-step (i) on the same explicit v367  $p+ip$  collar. [E] FLAT-BAND PARENT: the flattened Bloch  $Q=h/|h|$  has  $Q^2=I$  (flat  $\pm 1$  bands) and  $P=(I-Q)/2$  is an exact projector EQUAL to v367’s occupied ground-state projector — a frustration-free commuting-projector parent with the same gapped ground state (machine precision over a  $k$ -grid). [E] GAP-PROTECTED QUASI-LOCALITY: the gapped projector kernel  $|P(n,0)|$  decays rapidly ( $\xi \approx 1.14$  at  $M=1$ ,  $\exp R^2=0.995$ ,  $|P(20,0)| < 10^{-6}$ ) and  $\xi$  grows as the gap closes (2.7 at gap 0.02); at the gapless point it decays as a clean power law  $|R|^{-2.1}$  ( $R^2=0.999$ ) — locality needs the gap. [E]  $\mathbb{Z}_2$  REFLECTION +  $\mu_4$  INHERITED: the flattened modular Hamiltonian  $K_{\text{flat}} = \text{sign}(K)$  inherits  $\Theta K_{\text{flat}} \Theta = -K_{\text{flat}}$  ( $u_\Theta = J$ ) and  $[\rho, K_{\text{flat}}] = 0$  EXACTLY (sign is odd;  $\rho$  commutes with any  $f(K)$ ), with side-even and off- $\mu_4$  neg controls firing. [C][O] the chiral  $c_- = 8 \neq 0$  (FHS Chern  $|C|=1$ ) obstructs STRICT finite-range commuting-projector locality (Kapustin–Fidkowski, arXiv:1810.07756;  $C \neq 0$  forbids exponentially-localised Wannier), so the realisation is the quasi-local LTO net NPW26 use — the form sub-step (i) asks for. Advances v424 sub-step (i); SEAM.EQUIV.01 stays [O] (the continuum theorem MMST v336 is the backbone). Python-only (numpy); registered in run\_all.py, the registry, the ledger and cited in tfpt\_research\_contracts (route (xxvii) of the convergent attacks).

**2026-06-30 (v454–v459 + Lean + Wolfram — the post-F “G-block”: the lattice Sugawara current algebra, a uniform-in- $N$  Tomita–Takesaki tower, chirality forced by  $P_1$ , an  $(E_8)_1$  character kill test, the exact MMST citation audit and the complementary lattice-VOA route)**

- **v454\_seam\_edge\_virasoro.py (SEAM.EQUIV.EDGE.VIRASORO.01).** Tests the cited MMST “Koo–Saleur lattice generators  $\rightarrow$  continuum Virasoro with the correct  $c$ ” fact on our own free-fermion edge (G2). [E] the Affleck–Cardy Casimir energy fits  $c_{\text{Dirac}}=1.000$  ( $\frac{1}{2}$  per Majorana,  $L=64..1024$ ); [E] the lattice  $U(1)$  current closes its affine level exactly ( $\langle 0|J_n J_{-n}|0\rangle = n$ ,  $n=1..8$ ; a gapped control vanishes); [E] the level-1 Sugawara  $c = \dim G / (1+h^\vee) = 8$  for *both*  $SO(16)_1$  (120/15) and  $(E_8)_1$  (248/31). [C]  $16 \cdot \frac{1}{2} = 8 = g_{\text{car}} + N_{\text{fam}}$  is the level-1 current-algebra  $c_-$ ; SEAM.EQUIV.01 stays [O]. Sugawara Wolfram-mirrored.
- **v455\_seam\_bw\_uniform.py (SEAM.EQUIV.BW.UNIFORM.01).** A uniform-in- $N$  Tomita–Takesaki tower for the intrinsic Bisognano–Wichmann condition  $u_\Theta = J$  — the inductive-limit companion to v426, advancing v424 sub-step (i) (G1). [E]  $\|\Theta K \Theta + K\| = 0$  exactly at every  $N$  ( $N=8..128$ ); [E] an  $N$ -independent gap  $6 \log \frac{3}{2}$ ; [E]  $[\rho, K] = 0$  exactly at every  $N$ ; [E] an  $N$ -independent neg-control margin  $2\|K\|$ . [C][O] the inductive limit inherits  $u_\Theta = J$ ; SEAM.EQUIV.01 stays [O]. Python-only.
- **v456\_seam\_chirality\_from\_c3.py (SEAM.S3.FROM-P1.01).** The edge chirality ( $c_- \neq 0$ , the S3 phase) is *forced* by the one-sidedness that defines  $c_3 = 1/(8\pi)$  (P1) (G6). [E]  $c_3$ ’s “8” is the one-sided count  $8 = |\mathbb{Z}_2| \cdot (\int K/\pi) = 2 \cdot 4$ ; [E] a reflection sends the Chern integer  $C \rightarrow -C$  ( $+1 \rightarrow -1$  on the  $p+ip$  model); [E] a two-sided boundary forces  $C = -C = 0$ . [C] the one-sided seam (no reflection) forces  $C \neq 0$ , magnitude  $c_- = 8 = \text{rank } E_8$  — S3 is a consequence of  $P_1$ ; SEAM.EQUIV.01 existence stays [O]. The arithmetic is Wolfram-mirrored.

- **v457\_seam\_character\_killtest.py** (SEAM.EQUIV.KILLTEST.01). An explicit character/-sector kill test of  $(E_8)_1$  against  $SO(16)_1$  (G7). [E] the  $(E_8)_1$  character  $\chi=E_4/\eta^8$  with tower degeneracies  $\{1, 248, 4124, 34752, \dots\}$ ; [E] the weight-1 count  $248=120+128$  (not the free 120); [E] the sector count  $|\det \text{Cartan } E_8|=1$  (not  $|\det \text{Cartan } D_8|=4$ ). [X] a measured 120, four sectors, or a wrong coefficient would *falsify*  $(E_8)_1$ ; the data give  $248/1/\{1, 248, 4124\}$  (passed). SEAM.EQUIV.01 stays [O]. Wolfram-mirrored.
- **v458\_seam\_mmst\_citation\_audit.py** (SEAM.EQUIV.MMST.AUDIT.01). The exact hypothesis-by-hypothesis citation audit of MMST (arXiv:2107.13834) against the collar (G3, the “exakter Zitat-Abgleich”). [C] H1 CAR class, H2 massless limit, [E] H3 per-copy  $c=\frac{1}{2}$  (Thm 4.11), [C] H4 decoupled additivity to  $c=8$ , H5 the WZW range  $8 \leq 8 \leq 16$  with the 120 bilinear currents all match; [O] the single residual H6 is the 128-spinor extension  $SO(16)_1 \rightarrow (E_8)_1$  ( $\det K 4 \rightarrow 1$ ), outside MMST’s bilinear method. SEAM.EQUIV.01 stays [O], its open part narrowed to this one holomorphic extension.
- **v459\_seam\_lattice\_voa\_route.py** (SEAM.EQUIV.LATTICEVOA.01). The second, independent route to  $(E_8)_1$  supplying exactly the residual of v458 (G5). [E]  $E_8$  even unimodular ( $\det=1$ ); [E] the weight-1 space  $248=8+240$ ; [E] the 240 roots split 112 integer + 128 half-integer (the spinor,  $248=120+128$ ). [C] the lattice net  $A_{E_8}$  (Adamo–Giorgetti–Tanimoto, arXiv:2506.01008; OS arXiv:2407.18222) constructs the holomorphic extension; [O] the two routes leave one shared realisation input (collar =  $A_{E_8}$ , tied to  $P_1$  by v456). SEAM.EQUIV.01 stays [O]. Wolfram-mirrored.
- **Lean** (FORM.SEAM.RESIDUAL.01, `TfptCarrier/SeamResidualAxiom.lean`). Kernel-checks the whole G-block arithmetic (the one-sided  $8=|\mathbb{Z}_2| \int K/\pi$ , the reflection  $C \rightarrow -C$ ,  $c_-=8$ , the MMST range  $8 \leq 8 \leq 16$ , decoupled additivity,  $248=8+240=120+128$ ,  $240=112+128$ ,  $\det K 1$  vs 4) by `decide`, reducing the SEAM.EQUIV.01 residual to ONE named TFPT-internal realisation axiom (`CollarRealizedAsLatticePhase`) plus two cited external theorems (MMST v458, AGT v459); lake build OK.
- **Wolfram + integration**. Six exact mirrors added to `tfpt_readouts_extension.wl` (Sugawara  $c=8$  for 120/15 and 248/31; the one-sided  $8=2 \cdot 4$  and reflection  $C=-C \Rightarrow 0$ ; the  $(E_8)_1$  tower  $\{1, 248, 4124, 34752\}$  with  $248=120+128$  and  $|\det \text{Cartan}| 1$  vs 4; the root split  $112+128$  and  $248=8+240$ ) — extension total now 360/360. Registry, ledger, master index, website mirror and manifests updated in the same change.

**2026-06-30 (v449–v453 + Lean + Wolfram — the edge CFT named end-to-end: uniform-in- $N$  convergence, three independent reads of  $c_-=8$ , the Ising operator tower, the  $(E_8)_1$  modular data, and  $\mu_4$ -symmetry derived from the four marks)**

- **v449\_seam\_mmst\_uniform.py** (SEAM.EQUIV.MMST.UNIFORM.01). Strengthens the cited MMST hypothesis from convergence at a single cross-ratio (v444/v448) to the *uniformity* the theorem actually asserts, on the same v367/v368 collar. [E] cross-ratio A  $Q(1, 2, 4, 7)/12 \rightarrow 2+\sqrt{3}$ , the deviation shrinking monotonically; [E] an *independent* cross-ratio B  $Q(1, 3, 5, 9)/12 \rightarrow$  its own value 2, so convergence is not a tuned point; [E] a uniform  $1/N$  rate  $|Q(N)-Q_\infty| \cdot N \leq 4$  for *both* observables at every  $N_x \in \{36, 48, 60\}$  (one  $N$ -independent constant); [E] the bulk control is off-scale. [C] the MMST uniform-convergence hypothesis is empirically satisfied; SEAM.EQUIV.01 stays [O].
- **v450\_seam\_edge\_entanglement.py** (SEAM.EQUIV.EDGE.ENTANGLEMENT.01). A *third* independent reading of the edge central charge (after the correlator v444 and the topology v447). [E] the Calabrese–Cardy entanglement law gives  $c_{\text{Dirac}}=1.000$  for the critical free-fermion chain; [E] so  $c$  per Majorana =  $1/2$ ; [E] a gapped control saturates ( $c=0$ ). [C] bulk–edge:  $c_-=N_{\text{Maj}} \cdot \frac{1}{2}=16 \cdot \frac{1}{2}=8=g_{\text{car}}+N_{\text{fam}}$ . SEAM.EQUIV.01 stays [O].

- **v451\_seam\_edge\_cardy\_tower.py** (SEAM.EQUIV.EDGE.CARDY.01). Names the edge CFT by its *operator content*. [E] the Cardy finite-size spectrum gives the spin dimension  $h_\sigma=1/16$  (the periodic–antiperiodic ground-energy split) and [E] the energy dimension  $h_\epsilon=1/2$  (the lowest NS pair excitation), both [E]  $L$ -independent (conformal  $1/L$  scaling). [C]  $\{c, h_\sigma, h_\epsilon\}=\{\frac{1}{2}, \frac{1}{16}, \frac{1}{2}\}$  is the chiral Ising (free-Majorana) minimal model  $M(3, 4)$ ; 16 copies give  $c_-=8$ . SEAM.EQUIV.01 stays [O].
- **v452\_seam\_e8\_modular.py** (SEAM.EQUIV.E8.MODULAR.01). The  $(E_8)_1$  torus modular signature, extending v156’s  $q$ -expansion to the transformation data. [E] the character  $\chi=E_4/\eta^8$  is  $S$ -invariant  $\chi(-1/\tau)=\chi(\tau)$  (one primary,  $S=[1]$ ); [E]  $T$ -phase  $\chi(\tau+1)/\chi(\tau)=e^{-2\pi i/3}=e^{-2\pi i c/24}$  ( $c=8$ ); [E] leading power  $\chi q^{1/3} \rightarrow 1$  so  $q^{-c/24}=q^{-1/3} \Rightarrow c=8$ ; [E] holomorphic  $c=8 \equiv 0 \pmod{8}$  with  $T$ -phase order  $24/\gcd(8, 24)=3$ . [C] the  $(E_8)_1$  identity is now pinned on the torus *and* the edge. SEAM.EQUIV.01 stays [O].
- **v453\_seam\_mu4\_from\_marks.py** (SEAM.RIGIDITY.MU4FROMMARKS.01). Folds the last structural premise of the rigidity chain (v446’s residual QGEO.SYM.01) into the existing realisation premise. [E] the four Gauss–Bonnet marks *are*  $\mu_4$ = roots of  $z^4-1$ ; [E] the order-4 clock  $z \rightarrow iz$  is a 4-cycle permuting them (the v445 clock); [E] it preserves the cross-ratio  $\lambda=2$  (in the  $D_4$  stabiliser, v168); [E] the form basis  $\omega_k=z^{k-1}dz/(z^4-1) \rightarrow i^k\omega_k$  is  $\mu_4$ -graded (weights  $(1, 2, 3)=A_3$  exponents). [C] so QGEO.SYM.01 follows from marks= $\mu_4$  + the existing QGEO.REALIZE.01 — no new premise; the chain (v453→v446→v445) closes modulo realisation. SEAM.EQUIV.01 stays [O].
- **Lean: TfptCarrier/SeamEdgeChern.lean** (FORM.SEAM.EDGE.CHERN.01). Kernel-hardens the integer backbone the four edge reads converge on: the FHS Chern integers  $(+1/0/-1)$ , the bulk–edge channel count  $|C|=1$ ,  $N_{\text{Maj}}=2^{g_{\text{car}}-1}=16$ , the assembly  $2c_-=N_{\text{Maj}} \cdot |C|$  (so  $c_-=8$ ),  $c_-=8=g_{\text{car}}+N_{\text{fam}}=\text{rank } E_8$ , holomorphic  $c \equiv 0 \pmod{8}$  and the order-3 modular  $T$ -phase — all by kernel *decide* (edge\_kernel\_facts, axioms propext only). lake build OK, #print axioms clean.
- **Wolfram mirror.** tfpt\_readouts\_extension.wl gains nine exact checks (v452 modular data  $\times 4$ , v450 entanglement assembly, v451 Ising Kac weights, v453 marks= $\mu_4 \times 3$ ): 345  $\rightarrow$  354 checks, all passing (GATE.WOLFRAM.02).  $E_4$  is evaluated via its  $q$ -series since EisensteinE stays symbolic for complex  $\tau$ .
- **Papers.** The adversarial audit (tfpt\_5\_redteam, Target A) and the research-contracts note record the edge CFT as named end-to-end — central charge  $c_-=8$  from three independent observables (correlator v444, topology v447, entanglement v450), the Ising operator tower (v451), the  $(E_8)_1$  torus modular data (v452), and the  $\mu_4$  premise derived from the marks (v453) — while the only residual that stays [O] remains the cited continuum-existence theorem (v336).

**2026-06-30 (v446/v447/v448 + Lean + Wolfram — deriving clock-invariance from  $\mu_4$ -symmetry, an independent topological reading of the edge  $c_-=8$ , and pinning the one external MMST fact at four-point order)**

- **v446\_seam\_clock\_invariance.py** (SEAM.RIGIDITY.CLOCKINV.01). Closes v445’s single residual one notch further: the operator condition  $[\rho, \Lambda_\Sigma]=0$  (clock-invariance) is shown to be a *consequence* of a structural symmetry, not an independent assumption. [E] the OS canonical transfer  $\Lambda=C(1)C(0)^{-1}=e^{-h}$  (fixed by RP+reflection, v54) of a  $\mu_4$ -invariant RP covariance satisfies  $[\rho, \Lambda]=0$  (swept  $N=4..32$ ); [E] a  $\mu_4$ -breaking covariance gives  $[\rho, \Lambda] \neq 0$  (iff-sharp neg control); [E]  $\Lambda$  is a positive contraction (spectrum in  $(0, 1]$ ); [E] the free modular Hamiltonian  $K=\log((1-C)C^{-1})$  (v258) of the invariant state has  $[\rho, K]=0$  (Tomita–Takesaki), broken control  $\neq 0$ . [C] so with v445 (commute  $\Leftrightarrow$  block-diagonal), RP-definability *plus* the four marks *force* the block-diagonal  $\Lambda_\Sigma$ ; the residual reduces from an operator commutation to “the seam RP data is  $\mu_4$ -symmetric” (QGEO.SYM.01). SEAM.EQUIV.01 stays [O].

- **v447\_seam\_edge\_chern.py** (SEAM.EQUIV.EDGE.CHERN.01). An *independent* (topological) reading of the edge central charge, corroborating v444's correlator route from a different observable. [E] the  $M=1$  collar has integer Chern number  $C=+1$  (Fukui–Hatsugai–Suzuki link-variable method, no fit); [E]  $C=0$  in the trivial phase and  $C=-1$  at  $M=-1$  (opposite chirality); [E] the open strip carries exactly  $|C|=1$  chiral edge branch per boundary; [E] the Li–Haldane entanglement spectrum has in-gap modes at  $1/2$  only in the topological phase. [C] bulk–edge:  $c_- = N_{\text{Maj}} \cdot \frac{1}{2} \cdot |C| = 16 \cdot \frac{1}{2} = 8 = g_{\text{car}} + N_{\text{fam}}$  — the same  $c_- = 8$  as v444. SEAM.EQUIV.01 stays [O].
- **v448\_seam\_edge\_fourpoint.py** (SEAM.EQUIV.EDGE.FOURPOINT.01). Pins the one external MMST fact (continuum existence of the chiral scaling limit) empirically at the next order. [E] the ground-state correlation matrix is an exact projector  $G^2=G$  ( $\sim 10^{-13}$ ), so the state is quasi-free (the MMST CAR hypothesis holds); [E] the edge four-point cross-ratio  $Q=[G_{01}G_{23}]/[G_{02}G_{13}]$  at commensurate positions is Cauchy-convergent under refinement; [E]  $Q$  converges to the free-fermion conformal cross-ratio  $\sin(5\pi/12)/\sin(\pi/12)=2+\sqrt{3} \approx 3.732$ ; [E] the bulk control is off-scale (exponential). [C] so the one external fact is supported at *both* two-point (v444) and four-point order. SEAM.EQUIV.01 stays [O].
- **Lean: TftptCarrier/SeamRigidityForcing.lean** (FORM.SEAM.RIGIDITY.FORCING.01). Kernel-hardens v445 (and v442's uniformity): the entrywise forcing iff  $\text{clock4 } a = \text{clock4 } b \Leftrightarrow a=b$  (i.e.  $i^a=i^b \Leftrightarrow a \equiv b \pmod{4}$ , residue-only hence uniform in  $N$ ), the four distinct eigenphases, the exact commutant dimension  $\sum n_s^2=64<256$  and the order discriminator ( $128>64$ ) are all proved by kernel **decide** (**forcing\_kernel\_facts**, no domain axioms); only the operator-level lift to the full  $L^2$  is the named premise. **lake build OK, #print axioms clean**.
- **Wolfram mirror. tftpt\_readouts\_extension.wl** gains three exact v445 checks (entrywise forcing for all  $a, b \in 0..31$ ; commutant dimension  $64<256$ ; order-2 commutant  $128>64$  swept  $N=4..64$ ):  $342 \rightarrow 345$  checks, all passing (GATE.WOLFRAM.02).
- **Papers.** The adversarial audit (**tftpt\_5\_redteam**, Target A) records the five-direction narrowing of the SEAM.EQUIV.01 keystone (edge scaling v444, forcing converse v445 + Lean, derived clock-invariance v446, independent topological  $c_- = 8$  v447, four-point pinning v448) explicitly as “not promoted”: the only residual that stays [O] is still the cited continuum-existence theorem (v336). The research-contracts note lists the same as “eleven convergent attacks”.

**2026-06-30 (v445 — the rigidity FORCING converse: commuting with the order-4 clock is *exactly*  $\mu_4$ -block-diagonality, upgrading the rigidity step from “permit” to “force”)**

- **v445\_seam\_rigidity\_forcing.py** (SEAM.RIGIDITY.FORCING.01). The rigidity route (v398/v442) showed a  $\mu_4$ -block-diagonal  $K$  *commutes* with the order-4 clock  $\rho = \text{diag}(i^n)$ . This module proves the *converse* and the exact structure, isolating the single remaining physical premise. [E] FORCING (entrywise iff):  $[\rho, K]=0$  iff  $K_{ij}=0$  whenever  $i \not\equiv j \pmod{4}$ , verified entrywise and swept  $N=4..128$  — commuting *forces*  $\mu_4$ -block-diagonality, it does not merely permit it. [E] EXACT COMMUTANT DIMENSION:  $\dim\{K : [\rho, K]=0\} = \sum_s n_s^2$  (the  $\text{ad}_\rho$  nullspace), a proper subspace of the full  $N^2$  algebra ( $N=16$ :  $64<256$ ) — a genuine quantitative restriction. [E] OFF-BLOCK  $\Leftrightarrow$  NON-COMMUTING: every off-block matrix unit has  $\|[\rho, E_{ij}]\| \geq \sqrt{2} > 0$  (zero slack). [E] THE ORDER IS THE DISCRIMINATOR: the order-4 clock gives 4 sectors; the order-2 clock leaves a strictly larger commutant ( $N=16$ :  $64$  vs  $128$ ) — the four Gauss–Bonnet marks (order  $4=|\mu_4|=h(A_3)$ ) force strictly more structure than two, and only the index-4 extension is  $(E_8)_1$  ( $\det=1$ ; v281/v154/v351). [E] NONDEGENERACY: exactly 4 distinct eigenvalues  $\{1, i, -1, -i\}$  for all  $N \geq 4$ . [C] SYNTHESIS: commuting with the order-4  $\mu_4$  clock is *equivalent* to  $\mu_4$ -block-diagonality, so RP-definability (v54/v398)



+ clock-invariance *force* the block-diagonal  $\Lambda_\Sigma$  — v442’s “permit” upgraded to “force”. [O] the single residual is now precisely that the physical transfer lies in that commutant (clock-invariance = QGEO.SYM.01, v323/v335); SEAM.RIGIDITY.01/SEAM.EQUIV.01 stays [O]. Python-only (numpy; det facts Wolfram-mirrored via v89/v281/v422). Registered in `run_all.py`, `script_registry.csv`, the ledger, and cited in `tfpt_research_contracts` (route (xii) of the convergent attacks, and the sharpened residual in the Seam State Rigidity contract).

**2026-06-30 (Lean kernel-hardening of two of the seven SEAM.EQUIV.01 routes — the Borchers/Wiesbrock standard-pair relations and the applicability-ledger count, machine-verified with clean axioms; rigor without new mathematics)**

- **Context.** Two of the seven convergent attacks on the keystone residual are turned from numerical witnesses into kernel-checked facts, in the audit-contract style of `SeamScalingLimit.lean` / `BWKeystone.lean`: the load-bearing algebra/bookkeeping is proved by the Lean kernel, while the cited theorems and the one open continuum input remain named axioms. `lake build OK`; `#print axioms clean`. Tracked as two new FORM.\* ledger rows; `\veri`-context in `tfpt_research_contracts`; Lean manifest refreshed.
- **FORM.SEAM.BW.HSMI.01 (TfptCarrier/SeamStandardPair.lean) — standard-pair relations, kernel-checked (hardens v438).** The four Borchers/Wiesbrock relations on the centred 5-mode ladder over  $\mathbb{Z}$  are proved by kernel `decide` (axioms `propext` / `Classical.choice` / `Quot.sound` only, no domain axiom):  $[K, P]=P$ ,  $J^2=1$ ,  $JKJ=-K$ ,  $JPJ=P_+$  (and  $\text{tr } K=0$ ). The cited Borchers/Wiesbrock theorem and the one continuum input (the continuum standard pair) are named axioms; the composition pins face (ii) of SEAM.EQUIV.01 to exactly those two — intrinsic Bisognano–Wichmann, no rotation-covariance presupposed.
- **FORM.SEAM.APPLICABILITY.01 (TfptCarrier/SeamApplicabilityLedger.lean) — the audit count, kernel-checked (hardens v441).** The headline bookkeeping is itself machine-verified: of the 12 cited-theorem hypotheses (MMST/NPW26/Adamo, mirrored in order) exactly 11 are internal and 1 external (`total_count=12`, `internal_count=11`, `external_count=1`), plus `mmst_range` ( $8 \leq 8 \leq 16$ ) and `npw_outside_bucket` ( $\det K=1 \neq 4$ ) — ALL with NO axioms (pure kernel `decide`). The single external fact (continuum scaling-limit existence) is the named axiom; the keystone reduction pins its external dependence to exactly that one fact.
- **Net.** Both modules are imported by the root `TfptCarrier.lean`, axiom-audited in `AxiomCheck.lean`, and signature-locked in `AuditContract.lean`; full `lake build` succeeds. No new mathematics — the same facts v438/v441 verify numerically are now kernel-proved; SEAM.EQUIV.01 stays [O].

**2026-06-30 (the chiral-edge scaling motor v444 — the edge half of the one SEAM.EQUIV.01 continuum residual: the lattice edge already carries the conformal data of a chiral  $c=1/2$  Majorana CFT and converges under refinement, grounding the EDGE the way v439 grounded the BULK; the keystone stays [O])**

- **Context.** v439 grounded the *bulk* half of the cited MMST existence (uniform gap, exponential clustering, finite light cone) and explicitly narrowed the one open analytic step to the chiral massless *edge* scaling limit. v444 attacks exactly that residual on the SAME explicit v367/v368  $p+ip$  collar. vN-registered, runs in `run_all.py` (suite ends ALL CHECKS PASSED), `\veri`-cited in the body of `tfpt_research_contracts` (the  $G_{\text{net}}$  contract). Python-only (numpy; no new exact identity to Wolfram-mirror).
- **v444\_seam\_edge\_scaling.py (SEAM.EQUIV.CONTINUUM.EDGE.01) — chiral-edge motor.** The lattice edge carries the kinematic and conformal data of a chiral  $c=1/2$  Majorana CFT:



[E] a linear, chiral in-gap dispersion (velocity  $v \approx 0.98$ ,  $R^2 = 0.9999$ , with *opposite* per-edge velocities — one chiral Majorana per edge); [E] an *algebraic* equal-time edge two-point  $\sim r^{-\eta}$  with  $\eta \approx 1$ , versus the *exponential* bulk row and the exponential trivial-phase ( $M=3$ , no edge) control; [E] a Cauchy-convergent scaling-collapse under refinement ( $\eta = 1.001 \pm 0.007$ , successive profile spread  $0.028 \rightarrow 0.013$  across  $N_x = 36..60$ ) — the numerical hallmark of an existing scaling limit. [C]  $\eta = 1 \Leftrightarrow$  one chiral Majorana ( $c = 1/2$ ) per edge, so the 16-copy carrier has a chiral  $c_- = 8$   $(E_8)_1$  edge CFT ( $c_- = g_{\text{car}} + N_{\text{fam}}$ ). [O] the rigorous uniform-convergence theorem (cited MMST, v336) is not supplied.

- **Net.** With v439 (bulk) and v444 (edge), *both halves* of the one continuum residual are numerically grounded on our explicit model and pinned to a single cited existence theorem; SEAM.EQUIV.01 stays [O]. Ledger row, `script_registry.csv` index and the generated surfaces (`verification.tex`, `ScriptIndex.tsx`) updated in the same change; manifest refreshed last.

**2026-06-30 (six convergent attacks on the one SEAM.EQUIV.01 continuum residual — intrinsic Bisognano–Wichmann, a Lieb–Robinson continuum motor, the  $\beta$ -homotopy, an applicability audit, uniform rigidity, and falsifiable kill tests; none closes the keystone, together they reduce it to a single analytic fact and make the claim falsifiable)**

- **Context.** The keystone SEAM.EQUIV.01 is “closed modulo a cited theorem”: its one open input is the continuum existence of the seam’s chiral massless scaling limit. Six new modules attack that residual from independent directions; each is honest about *not* closing it. All six are vN-registered, run in `run_all.py` (suite ends ALL CHECKS PASSED), and are \veri-cited in the body of `tfpt_research_contracts` (the  $G_{\text{net}}$  contract). Python-only (no new exact identity to Wolfram-mirror).
- **v438\_seam\_hsmi\_borchers.py (SEAM.EQUIV.BW.HSMI.01) — intrinsic BW.** The seam supplies a Longo *standard pair* / +half-sided modular inclusion (positive lightlike translation  $P \geq 0$ , boost  $K$  with  $[K, P] = P$ , reflection  $J$  with  $JKJ = -K$ ,  $JPJ = P_-$ ), so by Borchers’ theorem the modular flow is geometric *intrinsically* — *without* presupposing rotation-covariance of the vacuum (complementary to Adamo–Giorgetti–Tanimoto). [E] ladder algebra exact,  $sl(2, \mathbb{R})$  positivity ( $\min \text{eig} > 0$ ); [C] synthesis; [O] the continuum standard pair.
- **v439\_seam\_lieb\_robinson.py (SEAM.EQUIV.CONTINUUM.LR.01) — continuum motor.** The explicit gapped  $p+ip$  collar (v367) has a uniform-in- $N$  gap, exponential clustering with  $N$ -independent  $\xi$ , and a finite Lieb–Robinson light cone, so by Nachtergaele–Sims its *bulk* thermodynamic limit exists — grounding the bulk half of the cited MMST existence in our model and narrowing the open step to the chiral *edge* scaling limit. [E]/[C], [O] the edge CFT.
- **v440\_seam\_lto\_rp\_beta.py (SEAM.EQUIV.LTORP.BETA.01) —  $\beta$ -homotopy.** The (LTO-RP)/BW data ( $\Theta K \Theta = -K$ ,  $[\rho, K] = 0$ ) are  $\beta$ -independent, and along the whole path  $C_\beta = 1/(1+e^{\beta K})$  from the NPW26-proven *trace* state ( $\beta=0$ ) to the seam KMS state ( $\beta=1$ ) every covariance is reflection-related,  $\mu_4$ -invariant and valid — a norm-continuous, obstruction-free homotopy, upgrading the single point of v426 to a path argument for sub-step (ii) of v424.
- **v441\_seam\_applicability\_ledger.py (SEAM.EQUIV.APPLICABILITY.01) — audit.** A hypothesis-by-hypothesis ledger of the three cited theorems (MMST / NPW26 / Adamo–Giorgetti–Tanimoto) finds 11 of 12 hypotheses are TFPT-internal computed facts and exactly one is external — the continuum existence of the chiral scaling limit; covariance is an Adamo consequence (defuses the v215 circularity), not an input. The keystone’s external dependence is pinned to that one analytic fact.

- **v442\_seam\_rigidity\_uniform.py** (SEAM.RIGIDITY.UNIFORM.01) — **uniform rigidity.** The band-limited  $\mu_4$ -character rigidity of v398 is uniform in  $N$ : exact 4-sector commutation to  $N=256$  (not merely the  $N \leq 64$  sweep), a uniform gap, and an  $N$ -independent clock-breaking lower bound  $\|[\rho, V]\|=2$  — so it is not a small- $N$  artifact (the full- $L^2$  forcing stays [O]).
- **v443\_seam\_e8\_discriminator.py** (SEAM.E8.DISCIMINATOR.01) — **kill tests.** Mode D of the closure-mode classification (v347) as concrete *falsifiable* discriminators against the nearest  $c=8$  rival  $SO(16)_1$ : torus ground-state degeneracy 1 vs 4, level-1 currents 248 vs 120, lattice theta 240 vs 112 norm-2 roots;  $c=8$  alone cannot decide (v277), only  $\det K=1$  (the genus-1 statement, v90) selects  $E_8$ . A seam with  $GSD>1$  would falsify  $(E_8)_1$ . [X]/[C].
- **Status discipline.** SEAM.EQUIV.01 stays [O] throughout; the six modules reduce, audit and falsify the residual but do not supply the one cited continuum-existence theorem. Ledger rows, the `script_registry.csv` index, and the generated surfaces (`verification.tex`, `ScriptIndex.tsx`) are updated in the same change; manifest refreshed last.

**2026-06-30 (shadow-export coherence — the tfpt5-shadow mirror is now reproducible *as shipped*: the audit is shadow-aware, the content maps are non-destructive on the website-free subset, and an export gate blocks any incoherent mirror)**

- **Context (reviewer finding, Alessandro).** On the exported `tfpt5_shadow.zip` the independent surfaces reproduced (`build.sh` notes  $\rightarrow$  11 ok; Wolfram base/extension; Red Team; Lean transcript), but the *package* layer did not: `make_manifest.py -check` reported a stale `verification/website_map.csv` and `build.sh` audit crashed because it unconditionally required `website/lib/release.ts`, which the shadow subset deliberately excludes.
- **make\_docs\_map.py is now shadow-aware.** `website_map.csv` mirrors the `website/` sources; on a tree without `website/` the generator now *skips* it instead of regenerating it down to a single `README.md` row. This removes the silent corruption that made `build.sh` notes/gen leave `website_map.csv` stale against the manifest on the export (it now matches the established skips in `make_script_index.py` and `make_changelog_web.py`).
- **audit\_sync.py is now shadow-aware.** When `website/` is absent it skips the website mirror (section D), the website version stamps (E) and the website Wolfram-count card (F), and the website generated-surface freshness in (A), printing exactly which sections were skipped. `build.sh` audit therefore runs to a clean AUDIT OK on the shadow subset instead of raising `FileNotFoundError`.
- **Export coherence gate.** `scripts/export-shadow.sh` now re-runs `make_manifest.py -check` and the shadow-aware `audit_sync.py` on the *filtered tree* before the mirror is published. The manifest check catches the earlier v235–v298 symptom at the source: a module referenced by `run_all.py/script_registry.csv/manifest.sha256` but not committed (and thus dropped by `git ls-files`) now fails the export loudly instead of shipping a package that says ALL CHECKS PASSED while `run_all.py` stops at the first missing script.
- **Manifest refreshed.** `manifest.sha256` was regenerated as the last release step so the shipped digests match the committed `changelog.tex` and `website_map.csv`; the full Python suite reproduces end-to-end on the regenerated export. No theory content, status marker or vN module changed — this is an export/infrastructure fix only.

**2026-06-29 (discoverability — an SEO/GEO pass on the public website: a generated /11ms.txt, explicit AI-crawler robots.txt, query-targeted metadata, and the first real backlinks from the README and the Zenodo deposit)**

- **New /11ms.txt context file** for AI / answer engines (ChatGPT, Perplexity, Claude, Gemini), generated at build time from the same data layer as the site (`papers.ts`, `predictions.ts`, `version.ts`) so it never drifts. It carries the one-line summary, the “claims / does not claim” split, the document set with PDF links, the headline predictions, and an explicit *disambiguation* from the unrelated Brouwer–Lefschetz “topological fixed point theory” of mathematics.
- **robots.txt** now lists the major search and AI crawlers (Googlebot, Google-Extended, Bingbot, GPTBot, ChatGPT-User, OAI-SearchBot, PerplexityBot, ClaudeBot, anthropic-ai, ...) explicitly as allowed; the site-wide meta description and keywords are reworded toward natural search queries (“derivation of the fine-structure constant”, “Standard Model from first principles”, “where does 137 come from”) and carry the same physics/mathematics disambiguation.
- **Backlinks (the missing piece).** The root README.md and the Zenodo deposit description (`zenodo_description.html`) previously contained *no* outbound links; both now link to `fixpoint-theory.com` and the source repository, and the README gains a citation block with the archived deposit DOI 10.5281/zenodo.20846087.
- Website / documentation infrastructure only — no theory, suite, ledger or status change.
- **New module v437 (E8.DEGREE.JOINT.01) — a consolidation of v6/v66/v355/v431 that derives more from the  $E_8$  Casimir degrees *without* crossing the v354/v355 anti-numerology line.** It adds *no* new per-degree coincidence — only one joint lock and a reconstruction.
  - **[E] all of  $E_8$  is fixed by its degree multiset.** From  $\deg(E_8) = \{2, 8, 12, 14, 18, 20, 24, 30\}$  alone:  $\text{rank} = \# \deg = 8$ ,  $h = \max \deg = 30$ ,  $\# \text{positive roots} = \sum \exp = 120$ ,  $\# \text{roots} = 240$ ,  $\dim E_8 = 248$ ,  $|W(E_8)| = \prod \deg = 696,729,600$ ,  $\sum \deg = 128 = \dim S^+$ .
  - **[E] the joint quadratic (the new lock).** The two TFPT structural integers ( $g_{\text{car}}, N_{\text{fam}}$ ) are the *unique* root pair of  $x^2 - (\text{rank } E_8)x + (h/2) = x^2 - 8x + 15 = (x-3)(x-5)$ : sum of roots =  $\text{rank } E_8 = 8$  (the rank-fill, v6), product =  $h/2 = 15 = g_{\text{car}}N_{\text{fam}}$  ( $2g_{\text{car}}N_{\text{fam}} = h$ , v431);  $\{3, 5\}$  is the unique positive-integer factor pair of 15 summing to 8. So  $g_{\text{car}} = 5$  and  $N_{\text{fam}} = 3$  are forced *together* by two  $E_8$  degree-invariants, not chosen one at a time. The three primary-readout degrees  $\{2, 8, 30\}$  are exactly  $\{\min, \text{rank}, \max\}$ .
  - **[E] discipline upheld:** no new per-degree mining beyond the joint quadratic; the five non-primary degrees keep their v431 family homes; the v354/v355 line is not crossed.
- **Surfaces updated.** Registered in `run_all.py`, `script_registry.csv` and `status_ledger.csv` (E8.DEGREE.JOINT.01); cited in the `tfpt_5` degree-audit winbox beside v431; mirrored in `wolfram/tfpt_readouts_extension.wl` (extension 339  $\rightarrow$  342). Exact integer/lattice; no new physical number, no status change.

**2026-06-29 (v436 — hardening the unconditional “is it numerology?” floor to an assumption-minimal counting floor; plus a literature note grounding the  $\alpha^3$  Chern level)**

- **New module v436 (OVERDET.FLOOR.02) — it *hardens* OVERDET.FLOOR.01 (v432) so the numerology verdict no longer rests on any subjective chance assignment.** v432’s nominal  $\sim 10^{-10}$  multiplies the counting census by hand-chosen plausibilities (input-“8” independence 0.1<sup>3</sup>, seed 0.012, kaon 0.15) a skeptic can contest; v436 shows the conclusion survives without them.

- **[E] assumption-minimal floor.** The  $\alpha$  cubic-class census (v100 layer D) is *pure counting*: of  $N = 94500$  complexity-matched  $F_{U(1)}$  variants only *one* lands in the  $4 \times 10^{-8}$  CODATA window — the TFPT instance, reproducing  $\alpha^{-1} = 137.0359992$  — so  $p_\alpha \leq 1/94500 = 1.06 \times 10^{-5} \sim 4.40\sigma$  with *no* subjective probability, prior or multi-observable grammar.
- **[E] monotone concession ladder.**  $-\log_{10} p$  from generous to adversarial: full scorecard+ $\alpha \sim 30.7 \rightarrow$  scorecard-only  $\sim 25.8 \rightarrow \alpha$ -census-only  $\sim 4.98$ ; each rung uses strictly less evidence and the “not chance ( $> 4\sigma$ )” verdict holds at the *most adversarial* rung, so it never rests on one contestable piece. Stripping *all* of v432’s subjective factors ( $\sim 1.8 \times 10^{-6}$ ) leaves the floor at  $4.40\sigma$  — they only sharpen it.
- **[C] honest  $5\sigma$  gap.** The conventional  $5\sigma$  line ( $p = 5.7 \times 10^{-7}$ ) is crossed unconditionally by the census times exactly *one* further independent factor  $\leq 0.054$  (the input-“8” four-fold over-determination, or one foreign witness) — stated, not hidden. **[C]** Firewall unchanged: bounds only “is it numerology?”, not the physics bridge (v187).
- **Literature note (research log, not a status change).** A `next.txt` entry grounds the  $\alpha^3$  Chern-level reading of v435 in named theorems — Dai–Freed ( $\eta$ -invariants and determinant lines, the exponentiated  $\eta$ -invariant as a section of the inverse Quillen determinant line), Bismut–Freed (determinant-line curvature = the  $\alpha^2$  Calderón side) and Redlich/Stora–Zumino (the cubic  $\text{Tr } F^3$  anomaly descends to a Chern–Simons level, quantised to  $\mathbb{Z}$ ). It records that the EM1 proof obligation reduces to constructing the seam Dirac family — i.e. it is a *facet* of SEAM.EQUIV.01 — and remains **[O]** external math (no fabricated closure).
- **Surfaces updated.** Registered in `run_all.py`, `script_registry.csv` and `status_ledger.csv` (OVERDET.FLOOR.02); cited in the `tfpt_safeguards` “unconditional floor” keybox beside v432; mirrored on the website. Python-only; no new physical number, no status change.

**2026-06-29 (v435 — a fourth honest step on the  $\alpha^{-1}$  Quillen target: a  $\pi$ -power test isolates the cubic  $\alpha^3$  as the unique metric-independent (topological) rung, and types its coefficient as a conditional integer level)**

- **New module v435 (ALPHA.QUILLEN.PROGRESS.04) — it attacks the *single* remaining **[O]** left after v434 (residual (2), the cubic  $\alpha^3$  Chern/Maxwell moment) and *sharpens* it; it does *not* close it.** The target ALPHA.QUILLEN.EXACT.01 (v382) stays **[O]**;  $\alpha^{-1} = 137.0359992$  stays **[E]**.
  - **[E]  $\pi$ -power / metric-independence test.** The three EM-closure coefficients carry  $\pi$ -powers exactly  $\{0, 3, 6\}$ , equal to their  $c_3$ -orders: Maxwell  $k_0 = 1 \rightarrow \pi^0$ , Calderón  $-2c_3^3 = -1/(256\pi^3) \rightarrow \pi^3$ , transport  $-8b_1c_3^6 = -41/(327680\pi^6) \rightarrow \pi^6$ . So the Maxwell  $a^3$  is the *unique*  $\pi^0/c_3^0$  (metric-independent) term. A heat-kernel boundary coefficient always carries the  $(4\pi)^{-d/2}$  measure; a metric-independent term carries none — the signature of a *topological* term. The test thus partitions the closure into *one* topological level ( $a^3$ ) + *two* heat-kernel boundary terms, confirming v342/v434’s “ $a^3$  is a level, not a Seeley–DeWitt curvature integral” from a *test* rather than a claim.
  - **[C] conditional level quantisation.** If the  $a^3$  term is the seam  $U(1)$  Chern–Simons/ $\eta$  boundary level (the EM1 reading, v48), large-gauge invariance quantises the level to  $\mathbb{Z}$ , so  $k_0 = 1$  is the *unit* (minimal) level — an integer, not a tunable real. This converts “the coefficient happens to be 1” into “the coefficient is an integer level and 1 is the unit”.
  - **[O] not closed:** the EM1 lemma itself — the from-first-principles Chern–Simons boundary proof that  $\partial_\tau \log Z_{\text{Maxwell}} = a^3$  — stays external math (type A, v384); the seam  $F$ -normalisation stays the one **[C]** (residuals 1 & 3, v434);  $\alpha^{-1}$  stays **[E]**.
- **Surfaces updated.** Registered in `run_all.py`, `script_registry.csv` and `status_ledger.csv`

(ALPHA.QUILLEN.PROGRESS.04); cited in the `tfpt_1` EM-closure subsection beside `v433/v434`; mirrored on the website residual-gap lab. Python-only; no new physical number.

**2026-06-28 (v434 — a third honest step on the  $\alpha^{-1}$  Quillen target:  $b_1$  is the  $a_4$  heat-kernel coefficient, and the three residuals collapse to one [C] + one [O])**

- **New module v434 (ALPHA.QUILLEN.PROGRESS.03)** — it settles the status of the three residuals named after **v433** and shows they are *not* three independent open problems. The target ALPHA.QUILLEN.EXACT.01 (v382) stays [O];  $\alpha^{-1} = 137.0359992$  stays [E].
  - [E]  $b_1$  *is* the  $U(1)$   $a_4$  **heat-kernel coefficient (the  $\beta = a_4$  theorem)**. The one-loop  $\beta$  function of a gauge coupling *is* the  $a_4$  (Seeley–DeWitt  $a_2$  in  $d=4$ ) heat-kernel coefficient of the charged-field operator (Vassilevich). Computed from the carrier hypercharge content with the standard spin weights ( $\frac{2}{3}$  per Weyl fermion,  $\frac{1}{3}$  per complex scalar),  $\sum \frac{2}{3}Y_f^2 + \frac{1}{3}Y_s^2 = 41/6$  (SM) and  $(3/5) \cdot \text{that} = 41/10 = b_1$  (GUT). So the  $b_1$  in the counterterm  $8b_1c_3^6$  is the  $a_4$  coefficient of the carrier  $U(1)_Y$  content (the same number as the  $\beta$ , v159/v246), the *same kind* of heat-kernel object as the  $\alpha^2$  Calderón  $a_4$  (v342) — not a fitted multiplicity.
  - [E] **the geometry factors off**:  $8b_1c_3^6 = b_1 \cdot (\text{rank } E_8) \cdot c_3^6$ , with  $\text{rank } E_8 = 8$  the  $\chi=2$  seam boundary count (v216) and  $c_3^6$  the six boundary insertions (the  $\pi$ -power test, v342).
  - [C] **residual (1)  $\equiv$  residual (3)**. With  $b_1$  and the  $c_3$ -powers heat-kernel-forced, the only freedom left in the exact seam  $a_4$  value is how the gauge curvature couples to the boundary measure — the seam  $F$ -normalisation. So “the actual seam  $a_4 = 8b_1c_3^6$ ” and “the seam  $F$ -normalisation” are *one* [C] obligation, not two.
  - [E] **residual (2) located (not closed)**: the determinant-line variation factors  $a^3 - 2c_3^3a^2 = a^2(a - 2c_3^3)$ ;  $a^2$  is the Gilkey gauge-curvature (Quillen/Chern) density (v342), so the leading  $a^3 = a \cdot a^2$  is the *first moment* of the Chern density — a topological level at boundary order  $c_3^0$ , not a Seeley–DeWitt curvature integral. Its from-first-principles (Chern–Simons-type) origin stays [O].
  - [E] **net reduction**: the EM-Ward residual is *one* [C] (residuals 1 & 3) + *one* [O] (residual 2), not three independent problems — a genuine narrowing of the open surface. [O] ALPHA.QUILLEN.EXACT.01 is *not* closed (the  $\alpha^3$  Chern level and the from-first-principles proof stay external, type A, v384).
- **Surfaces updated**. Registered in `run_all.py`, `script_registry.csv` and `status_ledger.csv` (ALPHA.QUILLEN.PROGRESS.03); cited in the `tfpt_1` EM-closure subsection beside `v433`; mirrored on the website residual-gap lab. Python-only; no new physical number.

**2026-06-27 (v433 — a second honest step on the  $\alpha^{-1}$  Quillen target: a solvable 4D model reaches the  $a_4$  order, connecting v391 and v342)**

- **New module v433 (ALPHA.QUILLEN.PROGRESS.02)** — a second honest step on the external target ALPHA.QUILLEN.EXACT.01 (**v382**); it *grounds and connects*, it does *not* close it. The target stays [O] (external math, type A, v384); the value  $\alpha^{-1} = 137.0359992$  stays [E]. The earlier honest attempt v391 gave a solvable model on a 1D circle whose  $\zeta$ -determinant variation reached only the low Seeley–DeWitt orders  $a_0, a_1$ ; separately, v342 grounded the quadratic Calderón term in the Gilkey gauge-curvature coefficient  $a_4$  ( $30/360 = \frac{1}{12}, \Omega^2/12$ ). This module supplies the missing link between the two.
  - [E] **a solvable 4D model reaches the  $a_4$  order**: the flat operator  $\Delta = -\partial^2 + m^2$  on a 4-torus of volume  $V$  has heat trace  $\text{Tr } e^{-t\Delta} = V/(4\pi)^2 t^{-2} e^{-tm^2}$ , with Gilkey coefficients  $a_0 = V/(4\pi)^2$ ,  $a_2 = -Vm^2/(4\pi)^2$  and  $a_4 = Vm^4/(2(4\pi)^2)$ ; the  $t^0$  (conformal-anomaly) order  $a_4$  is *non-zero* — exactly the order v391’s 1D circle toy could not reach.



- **[E] it is the order v342 uses:** so v391 (“a  $\zeta$ -determinant variation *is* closed heat-kernel data”) and v342 (“the seam term *is* the Gilkey  $a_4$ ”) are unified — the variation is heat-kernel data *at the  $a_4$  order*, not only the low orders v391 reached.
- **[E] the measure is  $c_3$ -built:** the universal 4D heat-kernel measure  $(4\pi)^{-2} = 1/(16\pi^2) = (2c_3)^2 = 4c_3^2$  in the seam’s one-sided 2D-boundary normalisation  $c_3 = \frac{1}{2}(4\pi)^{-1}$ ; and the v382 split is re-verified at  $\alpha^{-1} = 137.0359992$ .
- **[O] not closed:** computing the *actual* seam  $U(1)$   $\zeta$ -determinant  $a_4$  coefficient ( $= 8b_1c_3^6$ ) on the  $\mu_4$  seam moduli, and the origin of the cubic  $\alpha^3$  Maxwell/Chern moment, stay open (external math, type A, v384); the exact seam  $F$ -normalisation fixing the coefficient *value* stays **[C]** (v342). A second honest step that grounds and connects, it does not close.
- **Surfaces updated.** Registered in `run_all.py`, `script_registry.csv` and `status_ledger.csv` (ALPHA.QUILLEN.PROGRESS.02); cited in the `tfpt_1` EM-closure subsection beside v382/v391; mirrored on the website (the residual-gap lab). Python-only (like v391); no new physical number.

## 2026-06-27 (v431, v432 — the “unmapped” $E_8$ degrees are forced spine+flavor structure; the unconditional over-determination floor)

- **New module v431 (E8.DEGREE.LADDER.01) — the reverse-audit “unmapped”  $E_8$  Casimir degrees are NOT diffuse overhead but a forced two-family decomposition.** The structural complement of the reverse audit (v354/v355) and its sheet/deck complement (v430):  $\deg(E_8) = 6 \cdot \text{spine}\{2, 3, 4, 5\} \sqcup (\{2\} \cup \text{det-ladder}\{8, 14, 20\})$ . **[E]** the two-family split is *forced* by  $h(E_8) = 30 = 2 \cdot 3 \cdot 5$  — the exponents are the  $\varphi(30) = 8$  totatives of 30, hence coprime to 6, hence  $\equiv \pm 1 \pmod 6$ , so the degrees occupy only the residue classes  $\{0, 2\} \pmod 6$ . **[E]** the  $6k$  family  $\{12, 18, 24, 30\}/6 = \{2, 3, 4, 5\}$  is the v91 spine; the  $6k+2$  family  $\{8, 14, 20\} = (\det R, \det C, \det L)$  is the winding line  $\det M(s, 0) = 6s + 8$  of the flavor diamond (v135). **[E]** “ $18 = 6N_{\text{fam}}$ ” is not a holdout (spine family), and the clean split is special to  $E_8$  among the simply-laced exceptionals ( $E_6/E_7/D_5$  fail). **[C] honest v355 line:** the arithmetic decomposition is exact, but the *functorial* claim “ $6k+2$  Casimir stratum = flavor sheet operators” needs a representation-theoretic map and stays **[C]** (12, 24 admit two canonical readings each) — a structure theorem, not a forced functorial flavor map. It reclassifies the v354 overhead to zero orphan degrees and closes no gate.
- **New module v432 (OVERDET.FLOOR.01) — the unconditional improbability floor and the  $L_1 \leftrightarrow L_2$  bridge.** The complement of the conditional null model (v100) and the multiply-vs-compress framework (v427/v428): the defensible over-determination number that survives even if a skeptic *rejects* the declared formula grammar. **[E]** compression removed (v428), the genuinely multiplicative pieces are the input “8” forced four independent ways plus the foreign witness  $\alpha^{-1} \sim 137$ . **[C]** multiplying only across disjoint pieces (input over-determination  $\times$  the  $\alpha$  census  $1/94500 \times$  the  $\varphi_0^{\text{ret}}$ -seed cluster counted once  $\times$  one downstream witness) gives a floor  $\leq 10^{-6}$  across a grid of generous assignments, nominal  $\sim 10^{-10}$ . **[E]** that floor sits  $\sim 20$  orders *above* the v100 conditional  $10^{-30.7}$ ; the gap is exactly the evidential value of proving the forms forced, so the  $\alpha$  form-derivation (ALPHA.QUILLEN.EXACT.01, v382) and this floor are the *same* lever. **[C] honest scope:** bounds only the “is it numerology?” question, not the physics bridge (firewall v187); a conservative floor, not a posterior.
- *Surfaces synced in the same change:* v431/v432 added to `run_all.py`, `script_registry.csv` (master index + website `ScriptIndex`), `status_ledger.csv`, `tfpt_5_redteam` (reverse-audit winbox + Casimir-degree figure) and `tfpt_safeguards` (the over-determination layer); v431 mirrored in `wolfram/tfpt_readouts_extension.wl` (v432 is a conservative bound, Python-only).

## 2026-06-27 (v430 — the seam’s “other side” is forced-disjoint from $E_8$ ’s unmapped Casimir degrees)

- **New module v430 (E8.OTHERSIDE.AUDIT.01) — the sheet/deck complement of the reverse audit (v354/v355).** The reverse audit found five  $E_8$  Casimir degrees  $\{12, 14, 18, 20, 24\}$  with no primary readout (“hull overhead”); the double cover has an *other side* (the one-sided  $\mathbb{Z}_2$  collar / conjugate half-spinor  $S^-$ ), so the sharp adversarial follow-up is “*is the leftover where the second sheet lives?*” The answer is a disciplined **negative**. [E] the deck is the degree-2 invariant: the one-sided cover contributes  $|\mathbb{Z}_2| = 2$  (the  $\frac{1}{2}$  in  $c_3 = 1/(|\mathbb{Z}_2| 4\pi) = 1/(8\pi)$ , v58/v216), and  $2 = \min \deg(E_8)$  is the quadratic/metric — one of the *matched* primary readouts  $\{2, 8, 30\}$ , never one of the unmapped degrees. [E] the two sheets are the 128-spinor:  $\dim S^+ = \dim S^- = 16$ , the spinor block  $128 = \text{rank } E_8 \cdot \dim S^+ = 2^{\text{rank}-1} = \sum \deg$ , the spinor half of  $248 = 120 + 128$ ;  $S^-$  enters 248 as the conjugate  $(\overline{16}, \overline{4})$  ( $128 = 2 \cdot 64$ ), not a spare singlet. [E] forced-disjoint: the sheet/deck set  $\{2, 16, 32, 128\}$  has empty intersection with  $\{12, 14, 18, 20, 24\}$ , its only degree contact the matched 2. [O] each unmapped degree, “explained” from sheet atoms, needs an unforced coefficient and admits  $\geq 2$  readings (the v355 discriminator), so there is no forced sheet→unmapped-degree identity.
- **A structural negative, not a status change.** It consolidates the  $S^-$ -dark-matter downgrade (v227) and the no-spare-singlet WIMP no-go of the frontier  $E_8$  scan (tfpt\_4\_frontier): the other side is matched conjugate matter, the five unmapped degrees are the hull overhead, and the two do not meet. The only live “other side” question — whether the 128 phase/glue channel hosts a dark sector (v227) — concerns *matched* structure, not the unmapped five. Closes no gate, upgrades nothing.
- **Surfaces synced.** run\_all.py, script\_registry.csv, status\_ledger.csv; the reverse-audit Outcome winbox + degree-audit table caption and the Target D winbox of tfpt\_5\_redteam, the magnitude/phase keybox of tfpt\_3\_e8\_audit\_bootstrap, the WIMP-no-go passage of tfpt\_4\_frontier; the E8 node of the website verification DAG; the Wolfram extension/README/ledger counts and the website verification count; manifests refreshed.

## 2026-06-26 (v429 — the ‘unmapped’ golden $E_8$ structure is the geometry of the one external input $\theta_i$ )

- **New module v429 (DM.AXION.PENTAGON.01) — a geometric re-reading answering “why does so much idle  $E_8$  structure exist?” for the one case where the idle structure has a job.** The reverse audit (v354) had argued the golden ratio  $\varphi$  is *unmapped* (it appears only internally — the affine- $E_8$  attractor angle v312, the icosian lattice v348 — and in no physical readout), and used exactly that to conclude  $\varphi$  cannot be numerology. [E] this sharpens that statement: because  $N_{\text{fam}} = g_{\text{car}} - 2$ , the axion misalignment *spine* angle (v211, DM.AXION.SPINE.01) is  $\theta_i = \pi N_{\text{fam}} / g_{\text{car}} = (g_{\text{car}} - 2)\pi / g_{\text{car}}$ , which is *exactly* the interior angle of the regular  $g_{\text{car}}$ -gon — for  $g_{\text{car}} = 5$  the pentagon,  $\theta_i = 3\pi/5 = 108^\circ$ . [E] its cosine is golden,  $\cos(3\pi/5) = (1 - \sqrt{5})/4 = -1/(2\varphi)$  (partner  $2\cos(2\pi/5) = 1/\varphi = \varphi - 1$ ,  $\varphi = 2\cos(\pi/5)$ ). [E] the golden character is *unique* to  $g_{\text{car}} = 5$ : among the small regular  $n$ -gons only  $n = 5$  has an irrational interior-angle cosine ( $n = 3, 4, 6$  give  $1/2, 0, -1/2$ ), so  $\theta_i$  is golden *because* the carrier is 5 (P2).
- **An honest [C] bridge, not a status change.** It connects two standing facts —  $\varphi$  is the unmapped  $g_{\text{car}} = 5$  signature (v354/v313) and  $\theta_i = 3\pi/5$  is the one external cosmological input (v211) — and *refines* v354 from “ $\varphi$  in no physical readout” (it is the unmapped  $g_{\text{car}} = 5$  signature, v313) to “ $\varphi$  touches *only* the [C] spine misalignment angle” (conditional, branch-level, never a frozen prediction). It is a geometric motive for the spine over the hilltop  $\theta_i \approx 170^\circ$  (whose cosine is not a clean pentagon value), consistent with — not replacing — the

finite- $T$  solver that decides the branch (v185/v211). It does *not* derive  $\theta_i$ , does *not* upgrade DM.AXION.SPINE.01 (stays [C]) and closes no gate.

- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, the dark-matter section of `tfpt_4_frontier`; the Wolfram extension/README/ledger counts (GATE.WOLFRAM.02) and the website verification count; manifests refreshed.

## 2026-06-25 (External-works references appendix — web-verified)

- **New shared appendix `tex-artefacts/tfpt_references.tex`** (“External works built upon”), \input into the six math-heavy documents (`tfpt_1`, `tfpt_2`, `tfpt_4`, `tfpt_horizon_readouts`, `origin_theory`, `tfpt_research_contracts`). The active papers named the foundational results they build on by author only, with no reference list; the appendix supplies web-verified locators (journal, volume, year / arXiv) grouped by topic: modular theory & AQFT (Tomita–Takesaki LNM 128 1970; Bisognano–Wichmann J. Math. Phys. 16/17; Osterwalder–Schrader CMP 31/42; Doplicher–Haag–Roberts CMP 23/35; Haag–Ruelle; Longo CMP 126/130; Kawahigashi–Longo–Müger CMP 219), conformal nets/VOAs/lattices (Frenkel–Kac Invent. Math. 62; Segal CMP 80; MMST; Adamo–Moriwaki–Tanimoto; Adamo–Giorgetti–Tanimoto 2506.01008/2508.07109; Naaijken–Penneys–Wallick), lattices/quadratic forms (Conway–Sloane; Siegel Ann. Math. 36 1935), singularities (Brieskorn Invent. Math. 2 1966; Milnor AMS 61 1968) and spectral geometry/gravity/thermal time (Chamseddine–Connes CMP 186; Connes–Rovelli CQG 11; Wald PRD 48; Casini–Huerta–Myers JHEP 05 2011). Added to the `make_manifest.py` TEX list. No claim or status change — a provenance/citation completeness pass.

## 2026-06-25 (External AQFT citations + the four-fold $\mu_4$ identity sharpened)

- **Two recent Adamo–Giorgetti–Tanimoto theorems cited where the keystone builds on them.** The open SEAM.EQUIV.01 continuum/realisation legs (v336) now cite, alongside MMST and Adamo–Moriwaki–Tanimoto (arXiv:2407.18222, already in use): **(a)** the construction and classification of two-dimensional conformal nets from even lattices (arXiv:2506.01008, 2025) — “the lattice is a conformal net” is a theorem of the same operator-algebraic lineage, strengthening the lattice-realisation leg; and **(b)** automatic positive-energy/diffeomorphism covariance of conformal-net representations (arXiv:2508.07109, 2025), which defuses the “Bisognano–Wichmann presupposes the covariance it should produce” circularity (v215) on the intrinsic-BW residual (v424). Added to v336 docstring/registry, the SEAM.EQUIV.CONTINUUM.01 ledger `external_data`, and `tfpt_research_contracts`.
- **The four-fold  $\mu_4$  identity made explicit (v422).** v422 already proved that the seam deck/divisor  $\mu_4$ , the carrier Galois group  $\text{Gal}(\mathbb{Q}(\zeta_5))$ , the discriminant  $\text{disc}(D_5) = \text{disc}(A_3) = \mathbb{Z}/4$  and the  $(E_8)_1$  simple-current glue are ONE cyclic  $\mathbb{Z}/4$  (with the cyclic-vs-Klein negative control). It is now named the *four-fold  $\mu_4$  identity* and typed, per v428, as a *compression* (one object read four ways), NOT four independent witnesses – in v422, the registry, the SEAM.GALOIS.NET.01 ledger row and `origin_theory`. (No new module: a separate v429 would have duplicated v422 and violated the very compression discipline.)

## 2026-06-25 (v428 — honest self-correction of the over-determination map)

- **v428 (OVERDET.WITNESS.RECLASS.01) — applying v427’s own multiply-vs-compress test to v427 itself.** A pattern search and the right objection exposed that calling the seven arithmetic witnesses “disjoint grammars that *multiply*” overstates independence. [E] by the Brieskorn classification (v236) the  $(2, 3, 5)$  singularity is the ONE generator behind the order-30 clock (Milnor  $\mu = (2-1)(3-1)(5-1) = 8 = \text{rank } E_8$ ,  $30 = 2 \cdot 3 \cdot 5 = h(E_8)$ ,  $\varphi(30) = 8$ ), so each witness is a facet of that same object (primes 2, 3, 5 of 30, the clock 30, or  $E_8$ ): all

seven collapse to one generator  $\Rightarrow$  *compression*, not seven multiplications. [E] what genuinely multiplies is the input forced four independent ways — the “8” in  $c_3=1/(8\pi)$  from rank  $E_8$ ,  $h(D_5)=2(5-1)$ ,  $\varphi(30)$  and the Milnor number — plus the foreign witness  $\alpha^{-1}\approx 137$ . [C] refines and partly walks back v427, consistent with v236/v305/v313.

- **Surfaces corrected (same change).** v427’s prose, the ledger row (OVERDET.WITNESS.MAP.01, now noting the refinement), tfpt\_safeguards Layer 3 (the seven compress one object; a second keybox states what multiplies), the witness-map figure (redrawn), the website (papers.ts, Safeguards.tsx, release.ts, the intro-video description) and the Zenodo description. The over-determination claim is now the narrower, harder-to-attack “one richly over-determined input + a foreign readout”.

## 2026-06-25 (Safeguards visuals + paper expansion + the website safeguards band)

- **Five safeguards figures** (make\_figures.py, in manifest.sha256): the layered defense, the  $\alpha^{-1}$  uniqueness scan (one  $F_{U(1)}$  variant of 94,500 in the CODATA window), the reverse audit (3/8 vs 5/8), the witness-independence map (seven disjoint grammars), and the look-elsewhere null model (13/13 vs  $\leq 5/13$ ).
- **tfpt\_safeguards expanded.** Layer 5 now states the null model explicitly ( $\prod p_i=10^{-25.8}$ ,  $2\times 200k$  pseudo-theories  $\leq 5/13$ ,  $\varphi_0^{\text{ret}} \pm 10\% \rightarrow 6/13$ , shuffle 1.2) and the  $\alpha$  uniqueness ( $\leq 10^{-30.7}$ ,  $\sim 102$  bits, v100), with the data watchdog (v307) and kill board (v321); the five figures are embedded.
- **Website: a prominent animated Safeguards band** (components/Safeguards.tsx) in the home flow (after the trust beat) — headline statistics, the seven layers, the figures, links to the paper and the suite — to address the coincidence/numerology suspicion proactively.
- **Koide flow-time investigation (honest negative).** Post-v425: the flow-time is *not* native and does *not* reduce to one scale — v99 already rejects a scale reading as look-elsewhere; the only theory-side content is the conditional integer reading  $t=N_{\text{fam}}=3 \Rightarrow m_\tau=1776.94$  MeV, a Belle II kill test. No new module.

## 2026-06-25 (Three closure-strategy modules v425–v427 + the Safeguards paper)

- **v425 (DYN.TRANSFER.UNIVERSAL.01) — the single-flow reduction of  $F_{\text{transfer}}$ .** The four frontier transfers do not each need a separate external solver: their *dynamics* is the ONE native flow (the seam modular/recovery semigroup, v238/v240, gap  $6\ln(3/2)$ ) restricted to each sector. [E] the v99 Koide generator linearised at  $q=2$  has rate  $-\Delta$ , time-1 contraction  $(2/3)^6$ ; Boltzmann washout is that contraction (CP phase  $4\pi/3$  native, v320); QCD rate  $b_3=-7$ . [C] only the anchors ( $v_{\text{geo}}$ ,  $C_p$ ,  $M_R$ ) stay external;  $F_{\text{relic}}$ ’s cosmological IC  $\theta_i$  is the one wall. A reduction (extends v383 static $\rightarrow$ dynamic), not internalisation; the firewall holds.
- **v426 (SEAM.EQUIV.LTORP.KMS.01) — the invertible  $\beta=1$ -KMS variant of (LTO-RP).** The corner NPW26 leave open (they prove it for topologically-ordered models; the seam is invertible + KMS, not a trace). [E] a gapped reflection-symmetric  $\mu_4$ -even collar at  $\beta=1$  KMS satisfies  $\Theta K \Theta^{-1} = -K$  ( $u_\Theta = J$ ) and  $[\rho, K]=0$ , so (LTO-RP) holds by direct Tomita–Takesaki;  $|\det \text{Cartan}(E_8)|=1$  (no anyons), state not a trace; neg controls have teeth. Advances v424 sub-step (ii); SEAM.EQUIV.01 stays [O] (continuum sub-step (i) remains).
- **v427 (OVERDET.WITNESS.MAP.01) — the honest over-determination accounting.** A reviewer correction made machine-explicit: witnesses from *disjoint* grammars multiply, projections of the *same* generator compress. [E] seven disjoint-grammar witnesses each reproduce a carrier-skeleton integer from their own structure — Gauss  $N(3+2i)=13$ , Eisenstein  $N(3+2\omega)=7$ , cyclotomy  $N(3+2\zeta_5)=55$ , Galois  $|(\mathbb{Z}/5)^\times|=4$ ,  $|\det \text{Cartan}(E_8)|=1$ , Pascal  $\binom{4}{0}+\binom{4}{1}+\binom{4}{2}=11$  ( $2^4=16$ ), Coxeter  $h(E_8)=30=2\cdot 3\cdot 5$  with  $\varphi(30)=8$  — and the anchor  $a=(1,1,2)$  is the compression generator. Extends v305.



- **New paper `tfpt_safeguards` (companion S).** A dedicated methodology paper collecting, in one language, every mechanism by which TFPT defends a claim against coincidence and numerology: the status calculus + ledger + audit, no-free-pattern + reverse audit, the over-determination map (`v427`), the firewall + No-Unit theorem, frozen predictions + null model, the independent Wolfram and Lean paths, and the red team — with a final kill-test summary. Registered in `build.sh`, `make_docs_map.py`, `make_manifest.py` and the document-set box; the document set is now **eleven** deliverables (five numbered papers + four companions + introduction + changelog).
- **Surfaces synced.** `run_all.py` (421 modules), the registry, the ledger, `tfpt_research_contracts` (the  $F_{\text{transfer}}$  and keystone sections), `tfpt_safeguards`; manifests refreshed.

## 2026-06-25 (The (LTO-RP) reduction of the keystone `SEAM.EQUIV.01: v424`)

- **New module `v424` (`SEAM.EQUIV.LTORP.01`).** An honest MMST-style reduction of the one open keystone — `SEAM.EQUIV.01`'s intrinsic Bisognano–Wichmann lemma (the `v308/v309` residual) — to the recent commuting-projector theorem of Naaijken–Penneys–Wallick (`arXiv:2605.10693`, [NPW26]), with the central mapping *corrected*. [E] the  $\mu_4$  clock is a modular *symmetry* ( $[\rho, K]=0$ , `v309/v199`), not the flow —  $\sigma^\psi=\rho$  is type-wrong (a one-parameter group versus a finite order-4 element). [E] NPW26's axiom (LTO-RP) means  $u_\Theta=J$  (reflection = modular conjugation) = the intrinsic BW condition  $\theta K \theta = -K$ , *distinct* from  $\omega \circ \rho = \omega$ : a positive character-block-diagonal  $K$  has  $[\rho, K]=0$  yet never  $\theta K \theta = -K$ , so clock-invariance does *not* imply (LTO-RP) (refuting that identification). [O] make-or-break: NPW26 prove (LTO-RP) for *topologically ordered* models (Toric Code  $|\det K|=4$  via a trace; Levin–Wen anyonic), while the seam is invertible ( $|\det \text{Cartan}(E_8)|=1$ , no anyons) and a  $\beta=1$  KMS state (not a trace) — in neither proved bucket. [C] verdict: the BW residual reduces to NPW26 Theorem A modulo two open sub-steps (realise the seam as a  $\mathbb{Z}_2$ -reflection commuting-projector LTO; extend (LTO-RP) to the invertible KMS case) — a reduction, *not* a closure; `SEAM.EQUIV.01` stays [O]. Cited in `tfpt_research_contracts`; Python-only (numpy; the det discriminators mirrored via `v89/v281`).
- **Validation provenance.** The three pivotal references of the closure-strategy note were independently verified real and correctly described: NPW26 (`arXiv:2605.10693`), the RP–Yang–Mills complete-monotonicity reconstruction (Int. J. Geom. Meth. Mod. Phys. 2026), and Marolf (`arXiv:2203.07421`).
- **Surfaces synced.** `run_all.py`, the script registry, the status ledger and the `tfpt_research_contracts` keystone section; manifests refreshed.

## 2026-06-25 (No-Ambient-Dependency: no frozen readout is computed from `QG.AMB.01: v423`)

- **New module `v423` (`QG.AMB.NODEP.01`).** Sharpens the `v369/v384` prose redundancy (“`QG.AMB.01` is a [C] redundancy, not dynamics”) into a machine-checked DAG+script fact, told honestly. [E] gap-decoupling margin  $\Delta_{\text{eff}} = 6 \ln \frac{3}{2} - 31/(4\pi^2) \approx 1.648 > 0$  ( $31 = 2^{g_{\text{car}}} - 1$ ), susceptibility  $\chi = 729/665$  finite (negative control gap  $\rightarrow 0 \Rightarrow \chi \rightarrow \infty$ ); [E] `QG.AMB.01` is a certification *sink* (every claim that lists it is a gravity-gate/QG row, never a frozen prediction) and *not a value source* (no producing script, while  $\alpha^{-1}$ , the mixing angles and cosmology are each *computed* by a compiler script, `v3/v9/v268/v7`); [E] the one transitive link is architectural — of the frozen readouts only  $\theta_{13}$  references it, via the 4d-QFT contract spine (`PS.DIRAC`→`CONTRACT.QFT4D`→`QG.G2`→`GATE.METRIC`), a coherence edge, not a numerical input. [C] verdict: `QG.AMB.01` is gap-decoupled certification — `v369` as a DAG/script fact; honest scope — this does *not* discharge `SEAM.EQUIV.01`, on which the readouts still depend. Cited in `tfpt_4`; Python-only (gap algebra mirrored via `v337`).



- **Paper sharpening (seam selector, up front).** The `tfpt_4` quantum-gravity keybox now states explicitly that the open `SEAM.EQUIV.01` premise is the *holomorphic*  $|\det K|=1$  point that selects  $(E_8)_1$  — the central charge  $c=8$  alone does *not*, being shared by the non-holomorphic  $SO(16)_1$  counterexample (v277/v281).
- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, `tfpt_4`; manifests refreshed.

## 2026-06-25 (The Galois $\leftrightarrow$ Net bridge: v422)

- **New module v422 (SEAM.GALOIS.NET.01).** The seam  $\mu_4 = \text{Gal}(\mathbb{Q}(\zeta_5))$  is the SAME cyclic  $\mathbb{Z}/4$  as the  $(E_8)_1$  simple-current glue — not a mere order-4 coincidence — bridging the Galois gearbox of v419 to  $G_{\text{net}}/\text{SEAM.EQUIV.01}$ . [E]  $\text{disc}(A_3) = \text{disc}(D_5) = \mathbb{Z}/4$  (one Smith invariant factor  $4 = \text{cyclic, det } 4$ );  $D_n$  disc is  $\mathbb{Z}/4$  for  $n$  odd,  $\mathbb{Z}_2 \times \mathbb{Z}_2$  for  $n$  even, so the carrier  $D_5$  (rank  $5 = g_{\text{car}}$ , odd) is cyclic while  $D_8$  (rank 8, even) is Klein; [E]  $(\mathbb{Z}/5)^\times = \langle 2 \rangle$  is cyclic order 4 (only the carrier prime 5 gives order 4); [E] in  $\mathbb{Z}_4 \times \mathbb{Z}_4$  ( $16 = \dim S^+$ ) the glue  $(1, 1)$  is order 4 with Lagrangian quotient  $16/4^2 = 1 = (E_8)_1$  (v89/v154); [E] negative control: the Klein / order-2  $\langle (2, 2) \rangle$  gives  $16/2^2 = 4 = |\text{disc}(D_8)| = SO(16) (\det 4)$ , NOT  $E_8 (\det 1)$  — cyclic  $\mathbb{Z}/4 \rightarrow E_8$ , Klein  $\rightarrow D_8$ . [C] the bridge: one cyclic  $\mathbb{Z}/4$  threads the seam deck  $\mu_4$  (v419),  $\text{Gal}(\mathbb{Q}(\zeta_5))$  and the  $(E_8)_1$  glue — the Galois gearbox  $(\mathbb{Z}/30)^\times$  IS the holomorphic net's glue, cyclicity forced by 5, Klein excluded. Cited in `origin_theory`; Wolfram-mirrored ( $324 \rightarrow 327$ ).
- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, `origin_theory`; the Wolfram extension/README/ledger counts ( $327/327$ ) and the website verification count; manifests refreshed.

## 2026-06-25 (F\_pole external: the Koide source $\rightarrow$ pole transfer is one-loop EM: v420)

- **New module v420 (FR.POLE.QED.01).** The honest external-physics follow-up to v93 and the clock No-Go: the Koide source $\rightarrow$ pole transfer is a one-loop *electromagnetic* effect, not an algebraic step. [E] Koide  $Q$  is rescaling-invariant ( $Q(cm) = Q(m)$ ), so flavor-universal running leaves it fixed and the source $\rightarrow$ pole shift is purely flavor-split; [C] the tree deficit  $\frac{2}{3} - Q_{\text{src}} = 0.00220$  is one-loop-EM-sized ( $= \varphi_0/24$  to 0.6%,  $= \alpha/\pi$  to  $\sim 5\%$ ), the PDG pole sits at  $2/3$  (the IR attractor); [C] a 1-loop QED pole $\leftrightarrow \overline{\text{MS}}$  per-lepton-log estimate shifts  $Q$  by EM-loop order toward  $2/3$  (the v82 direction); [O] the exact flow time  $t \approx 2.84$  (v93) is scheme/scale-dependent external QED, only sized. [C] verdict:  $F_{\text{pole}}$  is a one-loop EM effect — the lens forces the rate  $(2/3)^6$  (v327), external EM supplies the flow; `F_pole` stays [C], firewalled (v187), so  $F_{\text{transfer}}$  is confirmed genuinely external — the open theme sharpened (QED), not closed. Cited in `tfpt_4`; numerical (mpmath), Python-only (no Wolfram check, by the suite convention like v93/v371).
- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, the Koide section of `tfpt_4`; manifests refreshed.

## 2026-06-25 (The seam $\mu_4$ is the carrier's Galois group: v419 + figure)

- **New module v419 (SEAM.GALOIS.01).** A positive resolution of the cyclic/Galois asymmetry (v409, `RES.COXETER.SYMMETRY.01`): why the seam  $\mu_4$  sits on the Galois side, not on the cyclic  $\zeta_{30}$  dial. [E]  $h(E_8) = 30$  is squarefree, so  $\mathbb{Z}/30$  has element orders  $\{1, 2, 3, 5, 6, 10, 15, 30\}$  — *no* order 4, so the seam  $\mu_4$  cannot be a rotation hand and is forced onto the Galois side; [E] by CRT  $(\mathbb{Z}/30)^\times = (\mathbb{Z}/2)^\times \times (\mathbb{Z}/3)^\times \times (\mathbb{Z}/5)^\times = \mu_4 \times \mathbb{Z}_2$  (order  $\varphi(30) = 8 = \text{rank } E_8$ ), and the  $\mu_4 (\mathbb{Z}/4)$  factor IS  $(\mathbb{Z}/5)^\times$  (the carrier prime 5), the  $\mathbb{Z}_2$  IS  $(\mathbb{Z}/3)^\times$  (family 3 = CP conjugation, v316); [E] so the seam  $\mu_4 = \text{Gal}(\mathbb{Q}(\zeta_5))$ , realised on the carrier clock  $C_5$

(v418) by the Frobenius  $\zeta_5 \mapsto \zeta_5^2$  — the explicit order-4 operator  $G$  with  $GC_5G^{-1} = C_5^2$ ,  $G^4 = I$ ,  $G^2C_5G^{-2} = C_5^{-1}$ . [C] this is the positive content of v409: the  $2^2=4$  is the order of  $(\mathbb{Z}/5)^\times$ , and “no contraction rate” is because the seam is an *automorphism*, not a hand; so  $\mathbb{Q}(\zeta_{30})$  carries both  $E_8$  invariants (cyclic  $h=30$ , Galois rank=8). Cited in `origin_theory`; Wolfram-mirrored (321  $\rightarrow$  324).

- **New figure `coxeter_galois.pdf`.** The order-30 cyclic dial with the four flavor hands (sheet/family/carrier/CP) beside the Galois gears ( $\mu_4 = \text{Aut}(\zeta_5)$  the seam,  $\mathbb{Z}_2 = \text{Aut}(\zeta_3)$  the CP conjugation); added to `make_figures.py`, the manifest figure list and `origin_theory`.
- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, the Wolfram extension/README/ledger counts (324/324) and the website verification count; manifests refreshed.

## 2026-06-25 (The cyclotomic norm triple + the carrier-5 clock: v418)

- **New module v418 (ARITH.NORMTRIPLE.01).** The carrier split (3,2) read in the three atom-rings gives (7, 13, 55)=(scalaron,  $\Delta_Q$ , quark numerator), and the carrier-5 clock missing from v417 is found as the  $4 \times 4$   $\Phi_5$ -companion  $C_5$ . [E]  $C_5^5 = I$ ,  $\text{tr } C_5 = -1$ ,  $\det C_5 = 1$  (the four primitive 5th roots), with the golden ratio as its real-part data ( $2 \cos 72^\circ = 1/\varphi$ ,  $2 \cos 144^\circ = -\varphi$ ; output-side, v313/v349; dim 4, not 3 — this refines v417’s gap from “no operator” to “no  $3 \times 3$  operator”); [E]  $N_{\mathbb{Z}[\zeta_5]}(3+2\zeta_5) = \det(3I+2C_5) = 55 = 5 \cdot 11 = g_{\text{car}} \| \text{Pl}(K) \|_1$ , the  $c_u/c_d$  numerator (v410/v411); [E] the triple  $\det(3I+2 \text{Comp}(\Phi_n)) = (7, 13, 55)$  for  $n=3, 4, 5$  over  $\mathbb{Z}[\omega]$ ,  $\mathbb{Z}[i]$ ,  $\mathbb{Z}[\zeta_5]$  (atoms 3, 2, 5, v416); [E] NEG control: the anchor (5, 4)  $\rightarrow$  (21, 41, 461) reaches named values only in the  $\omega$ - and  $i$ -rings ( $21=N_\omega(5, 4)$ ,  $41=10b_1$ , v222) but 461 (prime) in the carrier ring, so the carrier ring separates (3, 2)  $\rightarrow$  55 from (5, 4)  $\rightarrow$  461. [C] honest: (3, 2) is the *forced* carrier split (v14), not the unique clean one — a scan finds  $\{(2, 1) \rightarrow (3, 5, 11), (3, 2) \rightarrow (7, 13, 55)\}$ , a small ladder. The sharper discriminator is *cross-sector*: only (3, 2) spans three physics sectors (gravity 7, flavor 13, masses 55), the unique cross-sector seed, while (2, 1) is all compiler-core. Cited in `origin_theory`; Wolfram-mirrored (318  $\rightarrow$  321).
- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, a triple paragraph in `origin_theory`, the Wolfram extension/README/ledger counts (321/321) and the website verification count; manifests refreshed.

## 2026-06-25 (The Eisenstein/CP operator + the order-30 clock: v417)

- **New module v417 (DIAMOND.EISEN.01).** The hexagonal dual of v415: the  $\tau=\omega$  flavor side put on operators and unified with the seam into one order-30 Coxeter clock. [E] the family rotation  $P = (1\ 2\ 3)$  is the order-3 Eisenstein deck ( $P^2+P+I = \text{ONES}$ , so  $\omega^2+\omega+1=0$  on the plane  $\perp (1, 1, 1)$ ), the order-3 dual of  $J^2=-I$ ; [E] it realises the hex CM norm as  $(3I+2P)(3I+2P^2) = 7I+6 \text{ONES}$ ,  $\text{Spec}\{7, 7, 25\}$  ( $7=N_{\mathbb{Z}[\omega]}(3+2\omega) = \text{scalaron}$ ,  $25=g_{\text{car}}^2$ ) — the dual of v415’s {13, 13, 9}; [E] the CP clock  $W=-P^2$  ( $\rho$  of v233) has  $W^2=P$ ,  $W^3=-I$ ,  $\text{Spec}\{-1, \zeta_6, \zeta_6^{-1}\}$ , giving  $\delta_{\text{CKM,lead}} = \arg \zeta_6 = \pi/3$  and  $\delta_{\text{PMNS}} = \arg \zeta_6^4 = 4\pi/3$  (v316/v320); [E] all flavor clocks are powers of  $\zeta_{30}$  ( $\zeta_{30}^{15}=-1$ ,  $\zeta_{30}^{10}=\omega$ ,  $\zeta_{30}^6=\zeta_5$ ,  $\zeta_{30}^5=\zeta_6$ ) while the seam  $\mu_4=i$  is the Galois side  $(\mathbb{Z}/30)^\times$  of order  $\varphi(30)=8=\text{rank } E_8$  — so  $\mathbb{Q}(\zeta_{30})$  carries both  $E_8$  invariants ( $h=30$  and  $\text{rank}=8$ ). [C] honest gap: the carrier  $\mu_5$  has no  $3 \times 3$  operator (the golden  $\varphi$  is an output-side trace, v313/v349); negatives: no  $U, V$ -diamond operator carries  $\zeta_6$ ,  $[D, P] \neq 0$  (no  $\mu_{12}$ ),  $N(3+2\zeta_6)=19 \neq 7$ . Cited in `tfpt_2`; Wolfram-mirrored (315  $\rightarrow$  318).
- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, the Sheet-Diamond keybox of `tfpt_2`, the Wolfram extension/README/ledger counts (318/318) and the website verification count; manifests refreshed.

## 2026-06-25 (The Gaussian operator + the atom trichotomy: v415, v416)

- **New module v415 (DIAMOND.GAUSS.01).** The square CM-norm dictionary of v222/v230 is *operator-realised*, and the  $\mu_4$  deck becomes an intrinsic diamond object. [E] the integer commutator  $J := \frac{1}{3}[U, V] = \begin{pmatrix} -1 & 1 & 0 \\ -2 & 1 & 0 \\ -3 & 1 & 0 \end{pmatrix}$  has  $\text{Spec}(J) = \{i, -i, 0\}$ ,  $J^2 = -I$  on the rank-2 image and  $J^3 + J = 0$  — a  $\mu_4$  quarter-turn born from the binary  $V$  and ternary  $U$  compilers (v410/v95); [E]  $3I + 2J$  and  $5I + 4J$  have spectra  $\{3+2i, 3-2i, 3\}$ ,  $\{5+4i, 5-4i, 5\}$ , so the Gaussian integers  $3+2i$  (norm  $13=\Delta_Q$ ) and  $5+4i$  (norm  $41=10b_1$ ) are *literal eigenvalues*; [E]  $(aI+bJ)(aI-bJ)$  reads  $(a^2+b^2)$  on the  $\mu_4$ -plane and the real part<sup>2</sup> on the kernel ( $\{13, 13, 9\}$ ,  $\{41, 41, 25\}$ ); [E] the intrinsic order-4 deck  $D = -I + J - J^2$  ( $D^4 = I$ ,  $\chi_D = (x+1)(x^2+1)$ ) has  $\chi_2$ -line  $\ker[U, V] = a - 1$ . [C] the geometric-deck identification sharpens (does not close) the v140  $\mathbb{Z}_3$  residue. Cited in `tfpt_2`; Wolfram-mirrored.
- **New module v416 (ARITH.TRICHOTOMY.01).** The atoms  $\{2, 3, 5\}$  are ramified/inert/split in the two exceptional CM rings. [E] in  $\mathbb{Z}[i]$  (square/seam)  $2=|\mathbb{Z}_2|$  ramifies,  $3=N_{\text{fam}}$  is inert,  $5=g_{\text{car}}$  splits as  $(2+i)(2-i)$ ; [E] in  $\mathbb{Z}[\omega]$  (hex/flavor)  $3=N_{\text{fam}}$  ramifies,  $2, 5$  are inert; [E] the ramified prime of each ring is its own atom; [E] the three quadratic facets  $\mathbb{Q}(i), \mathbb{Q}(\sqrt{-3}), \mathbb{Q}(\sqrt{5})$  of  $K = \mathbb{Q}(i, \sqrt{-3}, \sqrt{5})$  (v403) ramify over exactly one atom each, product  $2 \cdot 3 \cdot 5 = 30 = h(E_8)$ , two imaginary (CM/dynamic) + one real (RM/static). [C] the seam/flavor/golden reading. Cited in `origin_theory`; Wolfram-mirrored.
- **Wolfram extension 308  $\rightarrow$  315 + a pre-existing fix.** Added the seven exact checks for v415/v416 and corrected a one-character sign bug in the v412 `CharacteristicPolynomial` check (Wolfram returns  $-\chi$  for odd rank), so `tfpt_readouts_extension.wl` now runs genuinely clean at 315/315 (GATE.WOLFRAM.02); README and ledger counts updated.
- **Surfaces synced.** `run_all.py`, `script_registry.csv` and `status_ledger.csv` updated; a new keybox paragraph in the Sheet-Diamond section of `tfpt_2` and a trichotomy sentence in `origin_theory`; manifests refreshed.

## 2026-06-24 (The sheet generator $V$ as a binary internal compiler: v410–v414)

- **New module v410 (SHEET.GEN.BINARY.01).** The rank-2 sheet axis  $V = Q \text{diag}(0, 1, 1)$  of the centered cross (v95/v218) is a binary internal compiler: its powers print the carrier spine. [E] the closed form  $V^n = \begin{pmatrix} 0 & 2^{n-1} & 0 \\ 0 & 2^n & 0 \\ 0 & 2^{n+1}-2 & 1 \end{pmatrix}$  (proved by the inductive step  $F(n)V=F(n+1)$ ), so  $V^n \mathbf{1} = (2^{n-1}, 2^n, 2^{n+1}-1) = (1, 2, 3), (2, 4, 7), (4, 8, 15), (8, 16, 31)$ ; [E] the four bilinear families  $\mathbf{1}^T V^n \mathbf{1} = 7 \cdot 2^{n-1} - 1$ ,  $\mathbf{1}^T V^n a = 7 \cdot 2^{n-1}$ ,  $a^T V^n a = 11 \cdot 2^{n-1}$ ,  $a^T V^n \mathbf{1} = 11 \cdot 2^{n-1} - 2$  give  $(6, 7, 9, 11)$  at  $n=1$  and  $\mathbf{1}^T V^2 \mathbf{1}=13=\Delta_Q$ ,  $\mathbf{1}^T V^3 \mathbf{1}=27=\mathbf{1}^T R a$ ,  $\mathbf{1}^T V^4 \mathbf{1}=55$ ,  $\mathbf{1}^T V^4 a=56=\dim \mathbf{56}_{E_7}$ ; [C] the binary spine is FORCED by  $\text{Spec}(V)=\{0, 1, 2\}$ , so the carrier/Lie matches (rank  $E_8=8$ ,  $\dim S^+=16$ ,  $248/8=31$ ) and the theta cross-link ( $\sigma_3(2)=9=a^T V \mathbf{1}$ ,  $\sigma_3(3)=28=\mathbf{1}^T V^3 a=\det(I+R)$ ,  $\sigma_3(5)=126$ ) are audit-typed (the v95  $G_2$ -center typing).
- **New module v411 (FLAV.UD.VPOWER.01).** The quark ratio is a pure  $V$ -power readout,  $c_u/c_d = (\mathbf{1}^T V^4 \mathbf{1})/((a^T V \mathbf{1})(\mathbf{1}^T V^2 \mathbf{1})) = 55/(9 \cdot 13) = 55/117$ , identical to the established  $g_{\text{car}} \|\text{Pl}(K)\|_1/(N_{\text{fam}}^2 \Delta_Q) = 5 \cdot 11/(9 \cdot 13)$  (v94). [E] the identity; [C] an exact *re-encoding* (no new physical input) — the value and the  $u/d$  identification stay conditional, coupled to the ( $U_{\text{wall}}$ ) readout rigidity.
- **New module v412 (DIAMOND.SHEET.SOURCE.J.01).** Names the operator on the  $\mathbb{Z}_2$  wall ( $t=-2$ ,  $\det=2$ , v135),  $J=M(1, -2)=C-2V$ , completing the sheet axis  $J, K, C, F=M(1, -2..1)$ . [E]  $\chi_J = \lambda^3 - 6\lambda^2 + 3\lambda - 2 \Rightarrow (\text{tr}, e_2, \det)=(6, 3, 2)$ ;  $a^T J a=30=h(E_8)$ ,  $\det(I+J)=12=\dim \mathbf{g}_{\text{SM}}$  and  $\det(2I+J)=40=|R(D_5)|$ ; [C] the Lie readings audit-typed like the  $G_2/F_4$  shadows.
- **New module v413 (DIAMOND.SHEET.CALCULUS.01).** On  $M_t=C+tV$  the elementary-symmetric coefficients encode the atoms as difference orders:  $e_1=3t+12$ ,  $e_2=2t^2+15t+25$ ,

$e_3=4t^2+14t+14$  give  $\Delta e_1=3=N_{\text{fam}}$ ,  $\Delta^2 e_2=4=|\mu_4|$ ,  $\Delta^2 e_3=8=\text{rank } E_8$  (the  $e_3$  curvature is the  $\sqrt{218}/\sqrt{224}$  sheet quadratic; the full triple is new); [E] the anchor energy  $a^T M_t a = 52 + 11t$  ( $30 \rightarrow 41 \rightarrow 52 \rightarrow 63$ , step  $11 = a^T V a$ ); [C] atom/Lie readings audit-typed.

- **New module v414 (DIAMOND.CENTER.RESOLVENT.01).** The center  $C=M(1,0)$  is a resolvent portal: [E]  $\det C=14=\dim G_2$ ,  $\det(I+C)=52=\dim F_4$ ,  $\det(2I+C)=120=|R^+(E_8)|$ , the exceptional shell ladder  $G_2 \rightarrow F_4 \rightarrow E_8^+$  as  $\det(C+kI)$ ,  $k=0,1,2$  (with  $\text{tr } C=12=\chi_{E_8}(i)$ ,  $\sum C=31=2^{g_{\text{car}}}-1$ , v95); [C] the portal reading.
- **Surfaces synced.** Integrated into the Sheet-Diamond section of `tfpt_2` (new keybox, five \veri citations) and the  $E_8$  theta-raster of `tfpt_3`; `run_all.py`, `script_registry.csv` and the ledger updated; Wolfram extension mirror  $303 \rightarrow 308$ .

## 2026-06-24 (Two new tracked targets + axion-branch consistency: v408, v409)

- **New module v408 (HYP.PHI0.PUNCTURE.01).** Names the  $\varphi_0$  puncture term  $48c_3^4$  as the order- $|\mu_4|=4$  local boundary heat-kernel (Seeley–DeWitt) contact term of the  $\mu_4$  puncture divisor, summed over  $\Omega_{\text{adm}}=N_{\text{fam}} \dim S^+=48$  admissible states — the analytic complement of the icosahedral tree term (v396). [E] the exact split  $\varphi_0=1/(6\pi)+48c_3^4$ , the puncture count, the closed form  $3/(256\pi^4)$  and the hypergraph negative (reused from v396/ARCH.QUAD.01); [O] the from-first-principles heat-kernel proof; [C] a FACE of the seam boundary analysis (the  $c_3$  Gauss–Bonnet datum), not a new gate. Cited in `origin_theory`; no new number.
- **New module v409 (RES.COXETER.SYMMETRY.01 closed as a structural lemma).** prime-2 (the seam sheet involution) is *necessary* but *not autonomous*:  $\sigma=\text{diag}(1,-1,-1)$  has spectrum  $\{\pm 1\}$  and projector  $(I+\sigma)/2$  spectrum  $\{0,1\}$ , none in  $(0,1)$ , so prime-2 carries no contraction rate, whereas the genuine rates  $2/3$  and  $(1/3)^6$  lie in  $(0,1)$  and carry prime-2 only as a factor. [E] no prime-2-only attractor exists; [C] dynamics begins only after coupling to family-3 or carrier-5 ( $|\mu_4|=2^2$  the Galois bridge). Resolves the v394 open question’s falsifiable corollary; cited in `tfpt_research_contracts`; the ledger row RES.COXETER.SYMMETRY.01 retyped [O]  $\rightarrow$  [E]/[C].
- **Axion branch consistency (tfpt\_4\_frontier, website predictions.ts, v185).** The dark-matter section’s opening + keybox are realigned with the finite- $T$  result already stated later in the same section: the robust dominant-DM branch is the *spine* angle  $\theta_i=\pi N_{\text{fam}}/g_{\text{car}}=3\pi/5=108^\circ$  (untuned, in band), while the seam hilltop  $\theta_i \approx 170^\circ$  over-produces ( $\sim 5.4\times$ ) and is disfavoured; the misalignment angle is retyped from a closed [E] input to a [C] branch decision. The website prediction card now leads with the spine angle.
- **Lean closure of the prime-2 corollary (FORM.COXETER.PRIME2.01).** The v409 result is now also machine-proved in Lean 4 / Mathlib (new module `TfptCarrier/CoxeterPrime2.lean`): an involution eigenvalue ( $r^2=1$ ) is  $\pm 1$ , a projector eigenvalue ( $r^2=r$ ) is 0 or 1, neither lies in the open interval  $(0,1)$ , while  $2/3$  does — so *no prime-2-only attractor exists*. Built clean (lake build); the new theorems are axiom-pure (`#print axioms = propext`, `Classical.choice`, `Quot.sound` only) and pass the `AuditCheck/AuditContract` signature locks. This lifts RES.COXETER.SYMMETRY.01 to the machine-proof tier.
- **Verified, no change (Plücker 11).** The quark constant  $11=\|Pl(K)\|_1=\det(1|a|\sigma)$  is already typed honestly as an [E] identity with “why 11” the lone [O] residue (v407, `tfpt_1`); the repo carried no overstatement, so no edit was required.

## 2026-06-24 (Cross-surface consistency sweep: papers, website, gates, figures realigned)

- **Ledger hygiene.** Resolved a duplicate active `claim_id`: GATE.UWALL.06-09 was carried by two distinct chains. It is an ID collision (not a supersede): the monodromy/residue chain



v114–v117 keeps .06–.09 (as asserted by v123\_inventory\_update.py and run\_all.py); the older exterior-leg chain v42–v45 is renumbered to GATE.UWALL.15–.18 (internal depends\_on carried along). No claim content changed.

- **Website.** papers.ts document count  $8 \rightarrow 9$  (the Red-Team paper was missing from the tally); Red-Team subtitle/highlight *Targets A–E*  $\rightarrow$  *A–F* with a new Target F section + contribution bullet (the abstract already carried F); StatusMatrix.tsx Doc 7  $\rightarrow$  Doc 8 for the research contracts and the residual interfaces re-typed  $v_{\text{geo}}$  [O],  $G_{\text{net}}$  [C],  $F_{\text{transfer}}$  [C]; objections Paper 1  $\rightarrow$  Doc 1.
- **Papers (marker/wording alignment, no canonical status change).** tfpt\_1 Strong CP [E]  $\rightarrow$  [E]/[C] (given RP) and  $\theta_{12}$  “genuinely open”  $\rightarrow$  “conditionally derived”; tfpt\_2 Starobinsky  $A_s$  [E]  $\rightarrow$  [E]/[C] ( $M$  exact,  $A_s$  a band over  $N_\star \in [50, 60]$ ) and the Part-II abstract “opens the one remaining gap”  $\rightarrow$  “closes the last structural gap”; tfpt\_research\_contracts the QFT4D “nonperturbative measure stays open” line and the “only  $U_{\text{point}}$  stays open” headline reconciled with the [C] redundancy /  $v_{\text{geo}}$  reduction stated alongside; origin\_theory class C re-typed to the ambient-measure [C] redundancy; introduction ( $U_{\text{wall}}$ )  $\rightarrow$   $v_{\text{geo}}$ . Forbidden old-paper attributions (“Papers 1–7”, “Papers 2/6”, “Paper-7”, “Paper-1”) replaced by current document / neutral wording (the tfpt\_3 cascade bridge stays exempt).
- **Suite README.** The residual-gates table (Alessandro-5.0 vintage) brought to the current model ( $v_{\text{geo}}$  [O];  $G_{\text{net}}$ /SEAM.EQUIV.01,  $F_{\text{transfer}}$ ,  $N_\star$ , QG.AMB.01 [C]; Wolfram 116/116 + 303/303).
- **Figures.** Regenerated status\_heatmap (ledger-driven) and extended the narrative figures script\_timeline/residual\_chain/qft\_skeleton through the v303–v407 arc (typed solvers, parameter-free local Einstein equation v359, QG.AMB.01 [C] redundancy, seam closed modulo cited theorems); the qft\_skeleton “Stelle ghost” open box is retyped (resolved truncation artefact). Added qft\_skeleton/qft\_unification to the manifest figure list.

## 2026-06-24 (The $\tau=\omega$ dual keystone + the flavor residuals it homes: v405–v407)

- [E]/[C]/[O] v405\_seam\_equiv\_omega.py (SEAM.EQUIV.02): the *dual keystone* at  $\tau=\omega$  – the family/flavor sector as the order-3 Eisenstein/ $A_2$  face of  $E_8$ , dual to the seam  $= (E_8)_1$  at  $\tau=i$  (v404). A named contract (genre of v286/v398), NOT a closure. [E]  $\chi_{E_8}(\omega) = 0$  (the family CM point degenerates  $E_8$ ) vs  $\chi_{E_8}(i) = 12$ ; [E]  $A_2$  is the  $\tau=\omega$  Eisenstein lattice (Cartan  $\det = 3 = N_{\text{fam}}$ ,  $h = 3 = N_{\text{fam}}$ ); [E]  $R$  lives in  $E_6 \times A_2$  ( $248 = 78 + 8 + 2 \cdot 27 \cdot 3$ ,  $\det R = 8 = \dim A_2 = \text{rank } E_8$ ); [E]  $\delta_{\text{PMNS}} = 4\pi/3$  is the  $\tau=\omega$  CM phase; [E] the cusp weights  $\{0, 1/3, 2/3\}$  are the order-3 deck  $= N_{\text{fam}} = \det Q$ . [C] the family sector is the order-3 Eisenstein/ $A_2$  deck; the two keystones sit at the only two elliptic points of  $\text{PSL}(2, \mathbb{Z})$ . [O] residual = the canonical  $\mathbb{Z}_3$  deck map (v140) + the  $v_{\text{geo}}$ -class scale. Cited in origin\_theory; Python-only.
- [E]/[C]/[O] v406\_pmns\_phase\_origin.py (FLAV.PMNS.CLOSE.01): the PMNS dynamics closed *modulo* the  $\tau=\omega$  keystone – the CP-phase ORIGIN is the family CM point and the absolute scale is one  $v_{\text{geo}}$ -class unit, so PMNS has *zero free flavor dials beyond*  $v_{\text{geo}}$ . [E] the three angles (v9/v268)  $\rightarrow$  unitary  $U_{\text{PMNS}}$  (v270); [C]  $\delta_{\text{PMNS}} = 4\pi/3$  is the  $\tau=\omega$  hexagonal CM phase (v405), upgrading v270’s [O] “assembled input” to a keystone consequence; [C]  $J_{\text{PMNS}} = -0.0297$  derived; [E] the absolute scale = one seesaw ratio =  $v_{\text{geo}}$ -class (v272/v153). [O] residual = the seesaw operator realisation ( $M_R$ , v184) + the  $\Sigma m_\nu$  floor kill test. Cited in tfpt\_2\_standard\_model; Python-only.
- [E]/[C]/[O] v407\_dn\_pairings\_omega.py (FLAV.SELECTOR.CLOSE.01): the  $R4'$  selector residue folds into the  $\tau=\omega$  keystone – the three  $(d, n)$  pairings ARE the  $\tau=\omega$  family-slice atoms. [E] the frame  $(\mathbf{1}, a, \sigma)$  has  $\det = 11 = \|Pl(K)\|_1$  and  $n = (5, -9, 6)$  is the unique covector with  $n \cdot \mathbf{1} = 2$ ,  $n \cdot a = 8$ ,  $n \cdot \sigma = 121 = 11^2$  ( $\Rightarrow (d, n) \Rightarrow R$ , v136/v139);



[E]  $n \cdot a = 8 = \text{rank } E_8 = \dim A_2 = \det R$  (the  $A_2$  block reading  $R$  in  $E_6 \times A_2$ , v405); [E] the pairings  $\{2, 8, 121\} = \{|Z_2|, \dim A_2 = \det R, \|Pl(K)\|_1^2\}$ . [C] deriving  $(d, n)$  folds into SEAM.EQUIV.02 ( $d$  anchor-derived, v139); [O] the lone residue “why  $11 = \|Pl(K)\|_1$ ”. Cited in `tfpt_1_architecture_e8`; Python-only.

## 2026-06-24 (The arithmetic hull + the $E_8$ -character CM duality: v403, v404)

- [E]/[C] `v403_facet_compositum.py` (ARITH.HULL.01): the three single-prime number-field facets of the order-30 clock (v390/v394, used so far only one at a time) *compose* to one field  $K = \mathbb{Q}(i, \sqrt{-3}, \sqrt{5})$ . [E] the squarefree parts  $-1, -3, 5$  are independent in  $\mathbb{Q}^\times/(\mathbb{Q}^\times)^2$ , so  $[K : \mathbb{Q}] = 2^3 = 8 = \text{rank } E_8 = g_{\text{car}} + N_{\text{fam}}$  with Galois group  $(\mathbb{Z}/2)^3$  (order 8); [E] exactly 7 quadratic subfields  $\{-1, -3, 5, 3, -5, -15, 15\}$ , splitting 4 imaginary  $= |\mu_4|$  (CM/dynamic) + 3 real  $= N_{\text{fam}}$  (RM/static); [E] ramified *only* over  $\{2, 3, 5\}$  = the atoms = the primes of  $h(E_8) = 30$ ; [E]  $2^{\omega(h)} = \text{rank}$  is  $E_8$ -special (the  $E_7$  control  $h = 18$  gives  $2^2 = 4 \neq 7$ ). [C]  $K$  is the abelian  $(\mathbb{Z}/2)^3$  *arithmetic* hull dual to the  $E_8$  *lattice* hull; the 7 quadratic subfields are a candidate for the 7  $E_8$  audit slices. A synthesis composing the established facets, no new number; cited in `origin_theory`; Python-only by the synthesis-module convention (cf. v390/v394).
- [E]/[C] `v404_e8_char_cm_duality.py` (E8.CMDUAL.01): the  $(E_8)_1$  vacuum character  $\chi_{E_8} = E_4/\eta^8 = j^{1/3}$  read at *both* CM points, extending v282 ( $\chi_{E_8}(i) = 12$ ) to the second elliptic point. [E]  $\chi_{E_8}^3 = j$  (q-series); [E] SEAM  $\tau=i$  ( $\mathbb{Q}(i)$ , order 4): cross-ratio  $2 \Rightarrow j = 1728 \Rightarrow \chi_{E_8}(i) = 1728^{1/3} = 12$  ( $E_8$  present, v214/v282); [E] FAMILY  $\tau=\omega$  ( $\mathbb{Q}(\sqrt{-3})$ , order 6): the equianharmonic  $\lambda$  ( $\lambda^2 - \lambda + 1 = 0$ )  $\Rightarrow j = 0 \Rightarrow \chi_{E_8}(\omega) = 0$  ( $E_4$  vanishes, v220/v267); [E]  $\tau=i, \tau=\omega$  are the *only* elliptic points of  $\text{PSL}(2, \mathbb{Z})$ , so the values are forced. [C] the seam/flavor rigidity duality: the seam is rigid where  $E_8$  is present ( $\chi = 12$ , the keystone SEAM.EQUIV.01 closes) while the flavor/CP sector is the soft residual where  $E_8$  degenerates ( $\chi = 0$ :  $U_{\text{wall}}, v_{\text{geo}}$ , the CP texture), a candidate dual keystone at  $\tau=\omega$ . Cited in `origin_theory`; core values  $j = 1728/j = 0$  already mirrored (v214/v220/v267/v282), Python-only.

## 2026-06-24 (The five TOE/QFT closure contracts v398–v402 + a ledger rescope)

- [E]/[O] `v398_seam_state_rigidity.py` (SEAM.RIGIDITY.01, Paper A): the *Seam State Rigidity Theorem* as ONE named target, consolidating the scattered state-invariance reductions (v177/v199/v308/v309/v335) and *hardening* the finite-block evidence to a multi-mode sweep. [E] RP-definability hinge (OS canonical transfer  $T=e^{-H}$  from RP + reflection alone, v54, quasi-free v155, no  $\mu_4$  normal form); [E] band-limited character-block-diagonal core  $[\rho, |k|]=0$  swept  $N=4.64$  (beyond the 3-dim  $H^1$ ); [E] holomorphy  $\Rightarrow (E_8)_1$  ( $c=8, |\det \text{Cartan}|=1$  vs 4, index  $4=|\mu_4|$ ). [O] the one residual:  $[\rho, \Lambda_\Sigma]=0$  on the full  $L^2$  (intrinsic Bisognano–Wichmann). NOT a closure of SEAM.EQUIV.01 (continuum = cited MMST v336).
- [E]/[C]/[O] `v399_gravity_complete.py` (GRAVITY.COMPLETE.01, Paper B): the explicit verdict *Gravity-complete* = *Boundary-complete* + *Ambient-redundancy*, combining the parameter-free local equation (v358/v359) with the ambient redundancy (v369). [E] the five gravitational readout classes (Newton  $8\pi$ ,  $\Lambda$ , scalaron  $c_3^7$ , horizon  $1/|\mu_4|$ , recovery  $(2/3)^6$ ) are all boundary/seam/gap, so  $\mathcal{O}_{\text{phys}} \subset \mathcal{A}_\Sigma$ ; [O] residual = SEAM.EQUIV.01 + intrinsic BW (no new gate).
- [E]/[O] `v400_grav_nonlocal_action.py` (GRAV.NONLOCAL.01, Paper C): the nonlocal spectral-gravity action with the local  $R+R^2$  as its IR projection (consolidating v304/v370/v380/v386, adding the IR-matching check). [E] entire  $a=e^u \Rightarrow$  one pole (ghost-free), every truncation carries a Stelle zero (monotone, v380), and the IR order reproduces  $R+R^2$  with the atom-fixed scale  $M_{\text{scal}}^2/\bar{M}_{\text{Pl}}^2=c_3^7$  (v36/v253). [O] the entire- $a$  assumption + perturbative-only.

- **[E]/[O] v401\_metrology\_closure.py** (METROLOGY.CLOSURE.01, Paper D): the final input set  $\{a, \pi, v_{\text{geo}}\}$  certified — 0 dimensionless dials + 1 unit (consolidating v153/v274/v364/v384). **[E]** dimensional bookkeeping  $\text{mass}/1G/\Lambda/U_{\text{point}}/(m/\mu) = \text{number} \times v_{\text{geo}}^{1,2,4,1,0}$ ; **[E]** the  $G$ -vs- $\Lambda$  over-determination 0.11% (v274); **[O]**  $v_{\text{geo}}$  theorem-forbidden ( $\text{TOE}_{\text{strict}} = \text{TOE}_{\text{dimensionless}} + [v_{\text{geo}}]$ ).
- **[E]/[C] v402\_ftransfer\_suite.py** (FTRANSFER.SUITE.01, Paper E): the four frontier sectors as ONE functor  $F_{\text{transfer}}$ , the composition  $F_{\text{obs}} \circ F_{\text{thr}} \circ F_{\text{RG}}$  (consolidating the FR.\*.SOLVE solvers), each with an exact atom source, a committed kill test, and the firewall (physical prediction stays **[C]/[O]**, only the algebraic source **[E]**; never re-pressed into the compiler, v187).
- **Ledger rescope (repo hygiene)**: GATE.METRIC.01 no longer reads “full covariant metric-sector field equation open” — the *local* covariant equation is closed parameter-free (GRAV.NONLINEAR.01/v359); the gate is rescoped so the open item is *only* the global nonperturbative ambient measure (QG.AMB.01, a **[C]** redundancy v369), removing the stale contradiction with the status card.

## 2026-06-24 ( $\varphi_0$ leading term from icosahedral combinatorics: v396)

- **[E]/[C] v396\_phi0\_icosahedral.py** (HYP.PHI0.01): the tree-level seed  $\varphi_0^{\text{tree}} = 1/(6\pi)$  is exactly  $F/(4h\pi) = F/(|\text{Aut}|\pi) = (F/(g_{\text{car}}N_{\text{fam}}))c_3$  on the icosahedral hypergraph ( $F = 20$  triangular hyperedges,  $h = 30$ ,  $|\text{Aut}| = 120$ );  $F/h = |Z_2|/N_{\text{fam}} = 2/3$  (the same survival ratio as v327). **[E]** Gauss–Bonnet consistent with  $c_3 = 1/(|Z_2|4\pi)$  (v216/v342). **[E]** honest negative: the puncture  $48c_3^4$  is still analytic, not a hypergraph fraction. **[C]**  $\varphi_0^{\text{tree}}$  is a combinatorial reinterpretation, not a new independent injection; the puncture stays **[O]**.

## 2026-06-24 (Coupled hypergraph rewrite: v395 unifies carrier $\times$ family in one local rule)

- **[E] v395\_hypergraph\_coupled.py** (HYP.COUPLED.01): one coupled local rewrite on a  $9 \times 3$  labelled grid (affine- $E_8$  nodes  $\times$  family channels) unifies v299 (carrier growth), v327 (branching rule  $M$ ) and v324 (fiber product). Micro-step = network lazy diffusion on each cusp column +  $M$  on each node’s family vector ( $T_{\text{micro}} = T_{\text{net}} \otimes M$ ); joint attractor marks  $\otimes e_0$ ; one clock hand (6 family steps) gives recovery  $(2/3)^6$ ; v324’s  $T_{\text{net}} \otimes T_{\text{cusp}}$  emerges as the cusp-readout basis. **[O]** the 3-arm seed ( $P_2$ ) and analytic  $\varphi_0$  remain irreducible (v312).

## 2026-06-24 (A falsifiable clock probe of the frontier: v397 how external is the “external physics”?)

- **v397 (FR.CLOCK.PROBE.01)** — **do the “external” frontier scheme-numbers land on the  $\{2, 3, 5\}$  clock?** A falsifiable, anti-numerology probe (strict No-Free-Pattern, v100: “on the clock” means an *exact* match, not a wide-band fit). **[E]** the  $\eta_B$  decuple  $A_\Lambda = 10 = |Z_2|g_{\text{car}}$  is ON the clock (v212); **[E]** the Koide rate  $(2/3)^6 = (|Z_2|/N_{\text{fam}})^6$  is ON the clock (v303); **[E]** the axion spine angle  $\theta_i = \pi N_{\text{fam}}/g_{\text{car}} = 3\pi/5$  is ON the clock (v211/v373); **[E]** the proton  $C_p \sim 2.83 \pm 0.15$  is *not* a discriminating clock value — within its  $\pm 5.3\%$  band several simple values fit  $(2^{3/2}, 17/6, 14/5)$ , so by No-Free-Pattern it is declared EXTERNAL (the firewall v187/v374 is correct there). **[C]** verdict: 3 of the 4 frontier scheme-numbers land *exactly* on the clock, so the irreducible “external physics” residue shrinks to essentially the proton’s non-perturbative QCD number  $C_p$ . Cited in `tfpt_4_frontier`; a falsifiable probe whose answer could have been “none on the clock”; Python-only.

## 2026-06-24 (Clockwork coherence + an open question: v394 primes = atoms = facets, and RES.COXETER.SYMMETRY.01)

- **v394 (COXETER.ATOMS.01) — the three Coxeter primes ARE the three anchor atoms ARE the three number-field facets.** A bird's-eye [E] re-reading tying v390 (the prime facets) to the anchor  $a = (1, 1, 2)$  and v319 (the  $5 \times 6$  clock). [E]  $a = (1, 1, 2)$  has  $e_3=2=|Z_2|$ ,  $p_0=3=N_{\text{fam}}$ ,  $e_2=5=g_{\text{car}}$ , product  $2 \cdot 3 \cdot 5 = 30 = h(E_8)$  — the clock's primes *are* the anchor's atoms. [E] atoms = facet fields (v390):  $2 \rightarrow \mathbb{Q}(i)$ ,  $3 \rightarrow \mathbb{Q}(\sqrt{-3})$ ,  $5 \rightarrow \mathbb{Q}(\sqrt{5})$ , each ramified at its prime. [E] the QG susceptibility is pure  $\{2, 3\}$ :  $\chi = 1/(1-(2/3)^6) = 729/665 = 3^6/(3^6-2^6)$ ,  $(2/3)^6 = 2^6/3^6$ , exponent  $6 = 2N_{\text{fam}}$ . [E] both dynamic rates carry prime-2:  $2/3 = |Z_2|/N_{\text{fam}}$  and  $(\varphi+2)/4 = (\varphi+2)/|\mu_4|$  with  $|\mu_4|=4=2^2$ . [E]  $30 = 5 \times 6 = g_{\text{car}}(2N_{\text{fam}})$  (v319). [C] the prime-2 seam is the universal dynamical engine; prime-3 (family) and prime-5 (carrier golden) the two attractors.
- **RES.COXETER.SYMMETRY.01 — a new registered open research question.** Is the prime-2-as-common-factor reading *forced* (the seam involution provably in every sector's gap, no independent prime-2 attractor), or is there a residual asymmetry? Resolving it would either tighten v383's "every sector is the same gapped operator" to "the same *seam-driven* operator" (with the falsifiable corollary that no prime-2-only attractor exists), or reveal a missing dynamical slot. Tracked in `tfpt_research_contracts`; a question, not a claim. Cited in `origin_theory`; Python-only.

## 2026-06-24 (Correction error-bars: v393 the typed budget as actual first-correction magnitudes)

- **v393 (CORRECTIONS.NUMERIC.01) — the typed correction budget (v388) as actual numbers.** The short-term, testable harvest of v388: instead of merely typing each prediction's correction, this gives the magnitude, so a parameter-free central value can be quoted as "value  $\pm$  computed first correction". [E] FIXED-POINT band (the  $\varphi_0$ -derived  $\theta_{12}, \theta_{13}, \lambda_C, \beta_{\text{rad}}$ ) = the explicit  $\varphi_0$  puncture term  $36c_3^4/c_3 = 9/(128\pi^3) \approx 0.227\%$  (exact;  $= \theta_{12}$ 's  $\varepsilon$  first-correction,  $\varepsilon = (3/4)\varphi_0 = c_3 + 36c_3^4$ ); [E] SEAM-GAPPED band (Koide  $F_{\text{pole}}$ , recovery)  $= (2/3)^6 \approx 8.78\%$ , the QG-decoupling bound capped by  $\chi - 1 = 64/665 \approx 9.62\%$  (v337); [E] EXACT-IDENTITY band = 0 structurally ( $\det R = 8$ ,  $N_{\Phi} = 1$ ,  $\theta_{\text{eff}} = 0$ ); [E] EXTERNAL-RATE band = external physics, quoted  $(m_p/m_e \pm 7.5\%$  v374;  $n_s/r$  the  $N_{\star} \in [50, 60]$  band;  $\eta_B 1.07 \times$  v212), fenced by v187. [C] the assembled budget gives each prediction a computed theoretical-uncertainty band. Cited in `origin_theory`, mirrored on the website prediction surface; a harvest of v388, no new number; Python-only.

## 2026-06-24 (Attacking the last open lemma: v392 the scaling-limit central charge converges)

- **v392 (SEAM.S3.SCALINGLIMIT.01) — an honest attack on SEAM.EQUIV.01's one open lemma (continuum existence, v336).** It does *not* close v336 (the abstract existence is the cited MMST theorem); it strengthens the case with the existence-relevant observable v376 did not test: that the finite-size data *converge*. [E] the Calabrese–Cardy entanglement slope  $c_1(L)$  at  $L = 66, 130, 258, 386, 514$  is a monotone Cauchy-like sequence approaching 1 ( $|c_1(L) - 1|$  strictly decreasing,  $4.6 \times 10^{-4} \rightarrow 7 \times 10^{-6}$ ); [E] a  $1/L$  Richardson fit extrapolates to  $c_1(\infty) \approx 1.000$ , so the 16-Majorana collar  $c = 8c_1 \approx 8.00 = g_{\text{car}} + N_{\text{fam}}$  is reached *in* the limit; [E] the successive differences are themselves strictly decreasing (a numerical Cauchy sequence). [O] convergent data are evidence, not a proof of existence (the cited MMST theorem, v336), and do not pin  $(E_8)_1$  over the same- $c$   $SO(16)_1$  (the holomorphy discriminator, v377/v378). Strengthens the residual, does *not* close it. Cited in `tfpt_research_contracts`; Python-only.

## 2026-06-24 (Lean 4: FORM.SPECTRALGAP.01 extended to the finite-dimensional Perron–Frobenius case)

- **FORM.SPECTRALGAP.01** — the multi-dimensional case, reduced to the scalar core. The Lean module `TfptCarrier/SpectralGapAttractor.lean` now also proves the *finite-dimensional* gapped-operator statement (lake build OK, no sorry): [E] `component_tendsto` (a non-dominant eigencomponent  $|r| < 1$  decays,  $r^n c \rightarrow 0$ ); [E] `deviation_tendsto` (with the dominant eigenvalue normalised to 1, the deviation vector — all non-dominant components — tends to 0, so the iterate converges to the dominant eigendirection, the unique attractor, for any start); [E] `deviation_unique` (two starts give deviation vectors both  $\rightarrow 0$ , so the selected eigendirection is independent of the start — parameter-freeness in finite dimensions); [E] `flavor_deviation_tendsto` (a concrete instance on the cusp-transfer spectrum  $\{1, (2/3)^6, (1/3)^6\}$ , v56/v82/v383). Honest scope: this is the finite-dimensional *diagonalisable* case (the genuine multi-dimensional content, reduced to the scalar core); the general non-diagonalisable / infinite-dimensional Perron–Frobenius theorem stays the cited statement. Referenced in `origin_theory`.

## 2026-06-24 (Website: an animated “Universal Gap Lab” visualising v383–v391)

- **Website** — the **UniversalGapLab** animation on /verification. A new client component (Framer Motion, no new dependency) with three linked, module-grounded views: (A) every sector as the gap contraction  $\text{iter}_n = x_\star + r^n(x_0 - x_\star)$  racing to its attractor at  $r = (2/3)^6$  (seam-gapped, v303/v387/v388) or the golden  $(\varphi + 2)/4$  (v312), with a gapless  $r = 1$  track that never forgets its start (the Lean theorem FORM.SPECTRALGAP.01) — the rate *is* the first-correction size (v388); (B) the entire-form-factor graviton ratio  $e^{-u}$  falling faster than any power, ghost-free and UV-soft (v386/v389); (C) the residual matrix as certification, not construction — every open item external, zero open internal mechanisms (v384). Mirror-only: every number is grounded in a module; no theory change.

## 2026-06-24 (Honest attempt at an external target: v391 reduces it, not a closure)

- **v391** (ALPHA.QUILLEN.PROGRESS.01) — an honest attempt at the external target ALPHA.QUILLEN.EXACT.01 (v382, door 5). It REDUCES the open step to a sharper named sub-target via a solvable model but does *not* close it: ALPHA.QUILLEN.EXACT.01 stays [O] (external math, type A, v384). [E] a solvable model  $\det_\zeta(-\partial_\tau^2 + m^2)$  on a circle of circumference  $\beta$  equals  $(2 \sinh(\beta m/2))^2$ , whose  $m$ -variation  $\beta \coth(\beta m/2) = 2/m + (\beta^2/6)m - (\beta^4/360)m^3 + \dots$  is a closed heat-kernel/Laurent series (the  $m^3$  coefficient  $-1/360$  verified). [E] so a  $\zeta$ -determinant’s coupling-variation is structured heat-kernel data, and v382’s counterterm  $8b_1c_3^6$  (a Gauss–Bonnet heat-kernel power) is the right *kind* of object; the v382 split is re-verified at  $\alpha^{-1} = 137.0359992$ . [E] the reduction: the open step now reduces to the named sub-target “the seam  $U(1)$  operator’s Seeley–DeWitt coefficient  $= 8b_1c_3^6$ ”. [O] *not closed*: the actual seam-operator  $\zeta$ -determinant proof stays open, and SEAM.EQUIV.01 continuum existence (MMST, v336) stays the other external [O]; the value  $\alpha^{-1}$  stays [E]. An attempt that reduces, does not close. Cited in `tfpt_1_architecture_e8`; Python-only.

## 2026-06-24 (Lean 4: FORM.SPECTRALGAP.01 formalises the scalar core of the universal spectral-gap principle)

- **FORM.SPECTRALGAP.01** — a Lean 4 proof of the gap  $\Rightarrow$  unique-attractor  $\Rightarrow$  parameter-free core (door 4). New module `TfptCarrier/SpectralGapAttractor.lean` (lake build OK, no sorry, only the three standard kernel axioms), formalising the meta-theorem of v303/v383/v387 at the one-dimensional reduction — the affine gap contraction  $x \mapsto x_\star + r(x - x_\star)$  with multiplier  $r = \lambda_2/\lambda_1$ . [E] `iter_sub`: the closed form  $\text{iter}_n - x_\star = r^n(x_0 - x_\star)$  (general  $n$ ); [E] `iter_tendsto`:  $0 \leq r < 1$  (the gap)  $\Rightarrow \text{iter}_n \rightarrow x_\star$  (the unique attractor); [E]



**starts\_agree**: two starts differ by  $r^n(a-b) \rightarrow 0$ , so the attractor is independent of the initial state (parameter-freeness); **[E] no\_gap\_no\_forget**:  $r = 1$  (no gap) leaves  $\text{iter}_n - x_\star = x_0 - x_\star$  constant — the initial condition is then a free parameter, so the gap is exactly what removes it; **[E]** the seam rate  $(2/3)^6$  and the golden  $(\varphi + 2)/4 = (5 + \sqrt{5})/8$  both lie in  $[0, 1)$ . Honest scope: the scalar reduction (the contraction core **v303/v383** iterate numerically); the full multi-dimensional Perron–Frobenius statement stays the cited theorem. Referenced in **origin\_theory**; ledger row **FORM.SPECTRALGAP.01**.

## 2026-06-24 (Completing the Coxeter clock: **v390** the prime-2 (Gaussian / $\mathbb{Z}/4$ -seam) facet of $30 = 2 \cdot 3 \cdot 5$ )

- **v390 (COXETER.PRIME2.01)** — the prime-2 facet of the order-30 Coxeter clock. **v383/v387** named two number-field facets of the clock  $30 = 2 \cdot 3 \cdot 5 = h(E_8)$  — the golden  $\varphi \in \mathbb{Q}(\sqrt{5})$  (prime-5, carrier) and  $(2/3)^6 \in \mathbb{Q}$  (prime-3, family). This names the *prime-2* facet and identifies it with structure already in the theory: the Gaussian field  $\mathbb{Q}(i)$ , i.e. the CM point  $\tau = i$  and the order-4  $\mu_4 = \mathbb{Z}/4$  square seam deck (**v282/v288/v333**). **[E]**  $h(E_8) = 30$  factors into exactly three primes; one quadratic facet field per prime, each ramified at its prime:  $\text{disc } \mathbb{Q}(i) = -4$  (2 ramifies),  $\text{disc } \mathbb{Q}(\sqrt{-3}) = -3$  (3 ramifies),  $\text{disc } \mathbb{Q}(\sqrt{5}) = 5$  (5 ramifies); the ramified primes are  $\{2, 3, 5\}$  with product  $30 = h(E_8)$ . **[E]** the prime-2 (Gaussian) facet is the  $\mathbb{Z}/4$  seam deck: min. poly  $x^2 + 1$ ,  $j(i) = 1728 = 12^3$ ,  $\tau = i$  fixed by the modular  $S$  of order  $4 = 2^2$ ,  $|\mu_4| = 4 = 2^2$  — the static, even, seam-orientation facet. **[C]** so the three primes of the clock now each carry a facet (square/hexagon/pentagon = seam/CP/carrier); the prime set  $\{2, 3, 5\}$  is forced by  $h(E_8) = 30$ , not chosen. Cited in **origin\_theory**; a synthesis, no new number; Python-only.

## 2026-06-24 (Graviton loop finiteness: **v389** the entire form factor is UV-finite at the loop level by power counting)

- **v389 (GRAV.LOOP.FINITE.01)** — the entire-form-factor graviton is UV-finite at the *loop* level. The honest next step past **v386** (which established the tree amplitude), extending **v304/v370/v380/v386** from the propagator to the superficial degree of divergence of loop graphs. This is the standard infinite-derivative / nonlocal-gravity finiteness statement (Tomboulis; Biswas–Mazumdar–Siegel; Modesto), *not* a full proof of perturbative quantum gravity. **[E]** super-polynomial falloff:  $p^k e^{-p^2/M^2} \rightarrow 0$  for every degree  $k$ . **[E]** one dressed line turns a quadratic divergence into the finite  $(M^2/2)e^{-p_0^2/M^2}$ , while the GR  $\int p^3/p^2 dp$  diverges quadratically. **[E]** the superficial degree  $\rightarrow -\infty$  at every loop order  $L$  (the form factor multiplies the integrand by  $e^{-(\#\text{internal})p^2/M^2}$ ; dressed cutoff-ratio  $\rightarrow 1$  vs GR  $\rightarrow 4$ ). **[E]** the regulator is the seam scale  $M = M_{\text{scal}} = c_3^{7/2} \bar{M}_{\text{Pl}}$  (**v253/v259**), not a dial. **[C][O]** loop *unitarity* of the nonlocal form factor (Lorentzian continuation / macro-causality, Pius–Sen / Briscese–Modesto) stays the honest open point — tree unitarity is **v386**; not claimed solved. Cited in **tfpt\_4\_frontier**; Python-only.

## 2026-06-24 (Typed correction budget: **v388** the gap correction is keyed to the mechanism class, not a uniform band)

- **v388 (CORRECTIONS.BUDGET.01)** — the gap-driven correction (**v387**) is a *typed* budget. A machine-checked answer to “does the  $(\lambda_2/\lambda_1)^n$  correction apply to *all* predictions?” — the answer is *no*, and for one whole class a band would be *wrong*. Because the size is the distance from a gapped operator’s leading eigenvector, it is defined only where a subleading  $\lambda_2$  exists, so the prediction surface splits into four classes. **[E]** *seam-gapped* (Koide  $F_{\text{pole}}$  **v303**, recovery **v221**, the QG bound capped by  $\chi = 729/665$  **v337**): rate  $(2/3)^6$ ; the discrete compiler: golden  $(\varphi + 2)/4$  (**v312**). **[E]** *exact identity* ( $\det R = 8$ ,  $N_\Phi = 1$ ,  $\theta_{\text{eff}} = 0$ ,  $\sin^2 \theta_{13} = \varphi_0 e^{-5/6}$ ): no  $\lambda_2$ , so the correction is 0 structurally and a band would contradict the **[E]** status. **[E]** *external-rate*



( $\eta_B$  washout, axion freeze,  $m_p/m_e$  RG): the gapped *shape* with an external rate — only  $F_{\text{pole}}$  has the seam rate (v303), firewalled (v187). [E] *fixed-point*: the texture is already explicit (e.g.  $\sin^2 \theta_{12}$ ,  $\varepsilon = (3/4)\varphi_0 = c_3 + 36c_3^4$ ). [E] the Koide first correction (door 2):  $F_{\text{pole}}$  is the one seam-gapped prediction, rate  $(2/3)^6 \approx 0.0878$  (v371), the source quotient 0.33% below  $2/3$  — the rate is the suppression *scale*, not the residual. [C] so a uniform band on all predictions is wrong; the size is keyed to the mechanism class. A typing of v387 (as v384 is of v383); cited in `origin_theory`; Python-only.

## 2026-06-24 (Harvesting the QFT/gravity machinery: v385 the UV-branch kill-test surface, v386 the graviton-exchange amplitude, v387 the gap-driven correction calculus)

- **v385 (UVBRANCH.KILLTEST.01) — the carrier-Pati–Salam UV branch on the all-order footing.** A consolidation that ties the all-order EG/BRST closure (v381) to the optional UV branch and computes the one genuinely new thing: the *proton-decay safety hierarchy*. [E] the carrier content is the SM with no new states, so the SM-non-unification (v246) is now all-order-robust; [X] the plain-SM 4D-GUT stays killed ( $\sin^2 \theta_W = 3/8$  vs 0.231); [C] the optional carrier-PS branch sits at  $M_{PS} = \kappa M_{\text{scal}} \approx 3 \times 10^{13}$  GeV ( $\kappa \in [1.0, 1.2]$ , v253); [E] minimal PS gauge bosons are  $SU(4)$  leptoquarks that mediate rare LFV ( $K_L \rightarrow \mu e$ ), *not*  $p \rightarrow e^+ \pi^0$ , so the branch clears the binding bound by  $M_{PS}/M_{\text{bound}} \sim 10^7$  — proton-safe, with *no* fake  $\tau_p$  window (v265). Cited in `tfpt_5_redteam`; Python-only.
- **v386 (GRAV.AMPLITUDE.01) — the graviton-exchange amplitude is finite and UV-soft.** Extends v304/v370/v380 from the propagator to the actual tree amplitude. [E] the dressed spin-2 propagator  $e^{-p^2/M^2}/p^2$  has its only pole at  $p^2 = 0$  (residue  $1 > 0$ , ghost-free), is exponentially softened in the UV (the ratio  $e^{-u}$  falls faster than any power) and reduces to GR + a calculable  $-p^2/M^2$  correction at low energy ( $M = M_{\text{scal}}$ , v253); [C] a single positive-residue pole gives tree unitarity, so perturbative graviton scattering  $A \sim (T \cdot P_2 \cdot T') e^{-p^2/M^2}/p^2$  is an explicit finite computation (perturbative-only; QG.AMB.01 stays a [C] redundancy, v369). Cited in `tfpt_4_frontier`; Python-only.
- **v387 (CORRECTIONS.GAP.01) — the gap-driven correction calculus.** The practical harvest of v383: the same spectral gap that makes a sector parameter-free also sets the *size* of its first correction ( $\text{correction}_n \sim (\lambda_2/\lambda_1)^n$ ). [E] flavor, horizon recovery and the QG bound all share the  $(2/3)^6$  rate (the family-3 facet; QG susceptibility  $\chi = 729/665$ ), while the discrete compiler transient decays at the golden  $(\varphi + 2)/4$  (the carrier-5 facet) — the two number-field facets of v314/v383. [C] the gap does double duty (why no free parameter *and* how big the first correction), so sub-leading magnitudes are organised, not free. Cited in `origin_theory`; Python-only.

## 2026-06-23 (Bird’s-eye synthesis: two new structural patterns — v383 the universal spectral-gap principle and v384 the residual is certification, not construction)

- **v383 (DYNAMICS.UNIVERSAL.01) — the universal spectral-gap / Perron–Frobenius principle.** A fresh bird’s-eye pass found that *every* TFPT sector is the *same* structural object: a gapped operator with a *unique* leading attractor (the physics) and a spectral gap (the reason there is no free parameter). v303 named this only for  $F_{\text{transfer}}$ ; this module extends it to the sectors made parameter-free *after* it. [E] three gaps are computed — the flavor transfer  $\{1, (2/3)^6, (1/3)^6\}$  with gap  $6 \ln(3/2)$  (v56/v82); the affine- $E_8$  adjacency with Perron eigenvalue 2 (Kac marks) and subleading  $2 \cos(\pi/5) = \varphi$ , gap  $2 - \varphi = 1/\varphi^2$  (v312/v313); the decoupling margin  $\Delta_{\text{eff}} \approx 1.648$  (v76/v337). [E] the two number-field facets (v314): golden  $\varphi \in \mathbb{Q}(\sqrt{5})$  (carrier-5, static) and  $(2/3)^6 \in \mathbb{Q}$  (family-3, dynamic) are the prime-5 and prime-3 facets of the order-30 Coxeter clock  $30 = 2 \cdot 3 \cdot 5 = h(E_8)$ . [C] the other sectors are the same structure — gravity ( $\delta S=0$  equilibrium, v358/v359), the QG measure (Gaussian saddle, v365),

the boundary QFT (holomorphic  $c=8$  RG fixed point, v157/v158), the all-order S-matrix (adiabatic limit, v381), recovery (Petz fixed point, v221),  $\tau=i$  (v333). [C] the meta-theorem: gapped  $\Rightarrow$  unique attractor  $\Rightarrow$  parameter-free, theory-wide. Cited in `origin_theory`; a synthesis (no new number), Python-only.

- **v384 (RESIDUAL.CERTIFICATION.01) — the residual is certification, not construction.** After v369/v381/v382 the *kind* of every open item has shifted: there is *no* open TFPT physics mechanism left. A ledger-CI audit classifies every residual into exactly three external kinds: [E] (A) external math proof (SEAM.EQUIV.01 continuum existence = the cited MMST/Adamo theorem v336; ALPHA.QUILLEN.EXACT.01 v382; QG.AMB.01 the constructive measure, a [C] redundancy v369), [E] (B) theorem-forbidden ( $v_{\text{geo}}$ , the No-Unit theorem v153/v364), [E] (C) external physics ( $F_{\text{transfer}}$ , v371–v374, firewalled v187). The two load-bearing facts are verified (No-Unit dimensionlessness; the gap-decoupling margin  $1.648 > 0$ ), and the count of open *internal* TFPT mechanisms is **zero**. [C] convergence: the only hard things left are theorems and external physics, not missing dynamics. Cited in `tfpt_research_contracts`; an audit (no new number), Python-only.

**2026-06-23 (External-review closure: the two genuine residual targets named — v381 the all-order Epstein–Glaser/BRST contract for  $S_{\text{pert}}$ , and v382 the Quillen determinant-line variation as a tracked target)**

- **v381 (QFT4D.EG.ALLORDER.01) — the all-order 4D perturbative-QFT closing statement.** A mathematical-physics review of the QFT/TOE layers identified exactly one genuinely new, buildable item: the *all-order* Epstein–Glaser/BRST contract for the perturbative 4D S-matrix  $S_{\text{pert}}$  (the previous documents had the  $S_{\text{pert}}$  skeleton and one-loop, v269/v271/v273/v278, but no named all-order claim). This module types the standard causal-perturbation-theory closure for the TFPT finite interaction class — it does *not* build a path integral. [E] power-counting: every gauge+matter vertex (Yang–Mills + Dirac–Yukawa + Higgs + ghost) is a mass-dimension-4 marginal operator, so the EG singular order  $\omega \leq 4$  and the counterterm space is finite at every order; [E] BRST nilpotency  $s^2=0$  verified exactly via the Jacobi identity for  $su(2)$  and  $su(3)$  (the carrier weak+colour content); [E] the seam gap  $\Delta = 6 \log \frac{3}{2} > 0$  (v302) gives the IR-safe adiabatic limit. [C] the two heavy legs are imported rigorous theorems — all-order  $T_n$  existence by causal factorisation (Epstein–Glaser 1973) and all-order BRST invariance (Piguet–Sorella / Kugo–Ojima) — applied to the TFPT class. Matter+gauge *only*: the  $R^2/\text{Weyl}^2$  gravity sub-sector is the entire-form-factor resummation (v304/v370/v380), and  $S_{\text{pert}}$  is not the ambient measure QG.AMB.01 (a [C] redundancy, v369). NET [C], prerequisites [E]; not [E] and not a path integral. Cited in `tfpt_5_redteam` and `tfpt_2`; Python-only.
- **v382 (ALPHA.QUILLEN.EXACT.01) — the Quillen determinant-line variation, named as a target.** Per the same review, the “why *this*  $F_{U(1)}$  functional” obligation (previously tracked only as EM.WARD.01’s residual, v341/v342) is elevated to its own citable target:  $\delta_\tau(\log \det_\zeta \Delta_{U(1)} + 8b_1c_3^6 \log \varphi_{\text{seam}}) = 0$  on the determinant line over the  $\mu_4$  seam moduli. [E] the Quillen split holds at the  $\alpha^{-1}=137.0359992$  root with every coefficient a named atom (index  $8b_1$ , heat-kernel  $c_3^{3,6}$ , discriminant  $q(D_5)+q(A_3)=2$ ); [E] the value stays closed (v3); [O] the from-first-principles exactness proof is the target. [C] it is a *face* of SEAM.EQUIV.01 (v336) — the seam  $U(1)$  determinant line lives on the same holomorphic net — so it adds *no* second independent gate; it makes the existing residual trackable. No new number (reuses the Wolfram-mirrored v341 identities). Cited in `tfpt_1_architecture_e8`; Python-only.

## 2026-06-23 (Track 2 of the forward plan v2 — the AMBIENT REDUNDANCY statement (v369): the holographic route to Gravity-complete that sidesteps building the general Euclidean QG measure)

- **v369 (QG.AMB.REDUNDANCY.01) — Track 2 (forward plan v2).** Instead of constructing the non-perturbative ambient measure (the conformal-factor swamp, v332/v334), assemble the established discriminators that prove the ambient sector carries *no* physical degrees of freedom relative to the holomorphic boundary algebra, so the QG measure (gate C7/QG.AMB.01) is a non-fundamental *certification* object (like a coordinate choice in GR), not missing dynamics. [E] holomorphy  $\Rightarrow$  DHR( $(E_8)_1$ ) = Vec: #primaries =  $|\det \text{Cartan}| = 1$  for  $E_8$  vs 4 for  $SO(16)$   $\Rightarrow$  no nontrivial superselection charge a bulk could carry invisibly; [E] no torus ground-state degeneracy ( $|\det K| = 1$ ); [E] finite Petz recovery rate  $(2/3)^6$  (v221); [E] the gap-decoupling margin  $\Delta_{\text{eff}} = 6 \log \frac{3}{2} - \frac{31}{4\pi^2} \approx 1.648 > 0$  (v76/v337). [C] with the Bisognano–Wichmann reconstruction map (v258/v309/v323) the redundancy statement  $\mathcal{O}_{\text{phys}}^{\text{bulk}}/\text{Diff} \cong \mathcal{A}_{\Sigma}^{\mu_4}$  is well-posed. [O] residual: the two consumed premises — SEAM.EQUIV.01 (the S3 lattice input, v366) and the BW intrinsicality of the raw collar (v329); it does *not* construct QG.AMB.01 but provides the *alternative* (holographic redundancy) route: Gravity-complete = Boundary-complete + Ambient-redundancy. Cited in `tfpt_research_contracts`; Python-only.

## 2026-06-23 (Status propagation: the v365–v380 + FORM.SEAM.MMST.01 status carried into the remaining papers, the README and the Zenodo description)

- **Deep-sync of the post-S3 status.** The canonical surfaces (`tfpt_status`, introduction) were already current; this pass propagated the same honest wording into `origin_theory`, `tfpt_1–tfpt_5`, `tfpt_research_contracts`, the root README.md, the Zenodo description and `website/lib/papers.ts`: SEAM.EQUIV.01 is *closed modulo a cited theorem* (lattice v367/v368 + the  $S_3$  stack v376–v379 + Lean FORM.SEAM.MMST.01; residual [O] = the cited MMST/OS existence only); QG.AMB.01 is a [C] *redundancy* (v369/v379), a certification object not missing dynamics; perturbative graviton unitarity is established (the Stelle ghost is a Seeley–DeWitt truncation artefact, v304/v370/v380); and the frontier transfers are typed runnable solvers with kill tests (v371–v374) + the observatory CI (v375). No new claim — a wording propagation so every surface matches the ledger; AUDIT OK.

## 2026-06-23 (v380 — the KMS Entire Hessian: the Stelle ghost is EXACTLY the Seeley–DeWitt truncation, and resummation pushes it to infinity)

- **v380 (GRAV.KMS.HESSIAN.01).** Upgrades v304/v370 from the *assumption* “the resummed graviton form factor is entire” to a derived statement. [E] the seam KMS cutoff  $f(u)=e^{-u}$  (v259) gives the dressed spin-2 propagator  $e^{-p^2/M^2}/p^2 = 1/(p^2 a)$  with  $a(u)=e^u$  entire and nowhere zero (only the  $p^2=0$  pole). [E] each finite Seeley–DeWitt truncation is a Taylor partial sum of  $a$  and carries a Stelle-ghost zero ( $T_1=1+u \rightarrow u=-1$ ;  $T_2 \rightarrow -1 \pm i$ ); the entire  $a$  has none. [E] the modulus of the nearest truncation-zero grows monotonically with the order  $(1, 1.41, \dots, 3.07 \text{ for } n=1..8) \rightarrow \infty$  — the ghost decouples, “entire” = “do not truncate”. [C]  $\Rightarrow$  perturbative graviton unitarity; [O] the exact off-shell curved-space Hessian identification is the cited heat-kernel resummation, perturbative-only. Cited in `tfpt_5_redteam` (Target F); Python.

## 2026-06-23 (Lean: the S3 continuum leg formalised as “closed modulo a cited theorem” — FORM.SEAM.MMST.01)

- **FORM.SEAM.MMST.01 — Lean 4 formalisation.** The S3 continuum leg is now machine-formalised in a new module (`SeamScalingLimit.lean`), lake build OK and `#print axioms` clean, in the audit-contract style of `SeamEquivChain.lean`. The MMST applicability hypotheses for the seam collar are *provable* by kernel `decide` (no axioms):  $D = 16$ ,  $\text{rank} = c = 8$ ,

the range  $8 \leq c \leq 16$ ,  $c_- = 8 \neq 0$ , and  $\det K = 1$  vs  $SO(16)$ 's 4. The MMST scaling-limit theorem (CMP 2022) and the Adamo–Moriwaki–Tanimoto OS reconstruction (2024) are *named cited axioms*; the conclusion  $(E_8)_1$  follows by a well-typed composition whose residual (`#print axioms`) is exactly those two published theorems plus the carrier gap — no hidden assumption, no sorry. Mirrors the S3 closure stack `v376–v379`; so `SEAM.EQUIV.01` is now “closed modulo a cited theorem”, the honest formalisation ceiling. Cited in `tfpt_research_contracts`; `lean_manifest` refreshed.

**2026-06-23 (S3 closure stack — the target of the seam scaling limit pinned at EVERY computable level: central charge  $c=8$  from the lattice (`v376`), the  $(E_8)_1$  character (`v377`), the genus-1 torus count = 1 (`v378`), and reflection positivity (`v379`); only the abstract continuum existence (cited MMST) stays external)**

- **`v376 (SEAM.S3.CENTRALCHARGE.01)`**. The Calabrese–Cardy finite-size entanglement-entropy scaling of the critical free-fermion content gives  $c = 1.0000$  per complex mode (Peschel correlation-matrix EE; the  $L=4m+2$  sizes avoid the half-filling zero-mode), so the 16-Majorana (= 8 complex) collar has  $c = 8 = g_{\text{car}} + N_{\text{fam}}$ , chiral ( $c_- = 8$ , `v367/v368`). Numerical  $c=8$  from the lattice. Python (numpy).
- **`v377 (SEAM.S3.E8CHARACTER.01)`**. The exact affine character  $\chi = E_4/\eta^8 = q^{-1/3}(1 + 248q + 4124q^2 + \dots)$  gives  $c = 8$  and 248 level-1 currents (one primary); the  $\mu_4/\text{GSO}$  promotion  $248 = 240 + 8 = 120 + 128$  ( $240 = 16 \cdot 5 \cdot 3$ ) pins  $(E_8)_1$  over the same- $c$  rival  $SO(16)_1$  (120 currents, 4 primaries). Python (sympy q-series).
- **`v378 (SEAM.S3.MODULAR.01)`**. The genus-1 discriminator: KLM condensation  $\mu(\text{carrier})=16 \rightarrow \mu(E_8)=16/4^2=1$ ,  $|\det \text{Cartan}(E_8)|=1$  vs  $D_8=4$ , and the single character's modular  $T$ -eigenvalue  $e^{-2\pi i/3}$  give torus ground-state degeneracy 1 (holomorphic =  $(E_8)_1$ ) — the “hard part” `v344` isolated. Python (sympy).
- **`v379 (SEAM.S3.RP.01)`**. Reflection positivity (the OS axiom) of the explicit gapped collar: the positive spectral measure makes the Hankel/reflection Gram matrix PSD (a ghost mode breaks it). With  $c=8 + (E_8)_1$  in hand, the OS inputs are complete and `C7/QG.AMB.01` is then discharged by the redundancy theorem `v369`. Python (numpy).
- **Net:** S3 is the master key — on the explicit lattice model the target net  $(E_8)_1$  is pinned exactly (central charge, character, genus-1 count, RP); the only remaining residual is the abstract *continuum existence* of the scaling limit (the cited MMST theorem `v336`), which lives outside the computational suite. Cited in `tfpt_research_contracts`.

**2026-06-23 (Track 4 — the prediction OBSERVATORY: a status-typed CI over the frozen prediction registry that makes the falsifiability surface machine-checkable, with a live data scorecard (`v375`))**

- **`v375 (OBSERVATORY.REGISTRY.01)`**. A CI over `freeze_file.csv`: **[E]** every prediction carries a kill criterion (falsifiability complete, no orphan claims); **[E]** the headline values re-derive from  $\{\varphi_0^{\text{ret}}\}$  ( $\sin^2 \theta_{12} = 1/3 - \varphi_0^{\text{ret}}/2$ ,  $\sin^2 \theta_{13} = \varphi_0^{\text{ret}} e^{-5/6}$ ,  $\beta_{\text{rad}} = \varphi_0^{\text{ret}}/(4\pi)$ ); **[C]** a live scorecard records  $\theta_{12}$  vs the JUNO first run ( $0.28\sigma$ , compatible),  $\theta_{13}$  vs NuFIT 6.0 NO ( $2.0\sigma$ , the flagged most-tensioned core prediction),  $\beta_{\text{rad}}$  vs ACT DR6 ( $0.37\sigma$ , a hint),  $r < 0.036$  (compatible). Tooling that makes the closed pieces usable and the open ones honestly fenced; no new physics. Cited in `tfpt_5_redteam`; Python (stdlib).



**2026-06-23 (Track 3 — the  $F_{\text{transfer}}$  functor promoted from contracts to TYPED runnable solvers with kill tests: Koide source→pole (v371),  $\eta_B$  Boltzmann (v372), axion finite- $T$  relic (v373),  $m_p/m_e$  uncertainty budget (v374))**

- **v371 (FR.POLE.SOLVE.01).** A clean *negative*: standard QED/EW running cannot be the Koide source→pole transfer (it pushes  $Q$  above  $2/3$ ; the required  $\epsilon$  is negative while QED's is positive), so the transfer is the non-QED  $53/54$  operator /  $(2/3)^6$  Moebius generator (v183/v82). [C], plain running excluded as the kill. Python (numpy+scipy).
- **v372 (FR.BOLTZMANN.SOLVE.01).** The integrated Buchmueller–Di Bari–Pluemacher Boltzmann ODE at the TFPT-frozen  $M_1=M_{\text{scal}}(\varphi_0^{\text{ret}})^2/A_\Lambda$  (v212) gives  $\eta_B$  within a factor  $\sim 1.1$  of  $6.1 \times 10^{-10}$  with no  $M_R$  dial; replaces the v326 toy washout. [C]/[X].
- **v373 (FR.RELIC.SOLVE.01).** The finite- $T$  axion misalignment ODE decides the branch: the frozen spine angle  $3\pi/5$  gives  $\Omega_a h^2 \sim 0.12$  (in  $[0.08, 0.16]$ , lands on  $\Omega_{\text{DM}}$  untuned) while the hilltop  $170^\circ$  over-produces ( $\sim 5.4\times$ ); replaces the v326 toy. [C]/[X] (lattice  $\chi(T)$  input).
- **v374 (FR.QCD.BUDGET.01).** The  $m_p/m_e$  uncertainty budget: with  $b_3 = -7$  [E] and the two declared externals ( $\alpha_s(M_Z)$ ,  $C_p$ ,  $\sim 5.3\%$  each), the central  $\sim 1824$  (v262) carries a propagated  $\pm 7.5\%$  band [ $\sim 1690, \sim 1960$ ] CONTAINING 1836.15; lattice tightening is the kill. The falsifiability companion to v262; [C].

**2026-06-23 (Track 1 — the S3 input made CONCRETE: an explicit gapped chiral-Majorana ( $p+ip$ ) lattice model with numerical Chern  $|C|=1$  realises the  $c_-=8$  edge (v367), and a strip shows the chiral edge exists by anomaly inflow (v368))**

- **v367 (SEAM.S3.LATTICE.01) — Track 1.** The abstract S3 residual of v356/v366 (“the collar realised as a genuine lattice chiral free-fermion invertible phase”) is turned into a RUNNABLE model. [E] the  $p+ip$  Bloch model is gapped at  $M=1$  with a numerically computed Fukui–Hatsugai–Suzuki Chern number  $|C|=1$  ( $C=0$  trivially at  $M=3$ ); [E]  $N_{\text{Maj}} = \dim S^+ = 16$  copies give  $c_- = 16/2 = 8 = g_{\text{car}} + N_{\text{fam}}$ , with the edge K-matrix  $SO(16)_1$  ( $\det 4$ )  $\rightarrow (E_8)_1$  ( $\det 1$ ) by the order-4  $\mu_4$  clock (v351/v369). [C] the  $\mu_4$  condensation (v154/v351); [O] the continuum scaling-limit existence (MMST, v336) stays open. Cited in `tfpt_research_contracts`; Python (numpy + sympy).
- **v368 (SEAM.S3.INFLOW.01) — Track 1.** The bulk-edge / anomaly-inflow leg (S4) made concrete on the v367 model: [E] a finite strip has an edge-localised in-gap chiral branch ( $\min_{k_x} |E| \sim 0$ ) in the topological phase and a clean gap in the trivial phase (neg control); [E] bulk-edge correspondence gives one chiral Majorana per edge ( $c_-=8$ ). [C] so by anomaly inflow the chiral edge EXISTS as a consequence of the invertible bulk, discharging the v336/v351 “does the edge exist?” residual on the lattice. [O] the continuum scaling limit stays open. Cited in `tfpt_research_contracts`; Python (numpy).

**2026-06-23 (Track 2 — perturbative graviton unitarity sector-by-sector: the Barnes–Rivers spin decomposition localises the Stelle ghost in the spin-2 sector, cured by the entire KMS form factor; the spin-0 mode by the GHP contour (v370))**

- **v370 (GRAV.SPIN2.UNITARITY.01) — Track 2.** Extends v304 (the scalar  $1/(p^2(p^2+M^2))$  pole algebra) to the actual tensor spin structure. [E] the Barnes–Rivers projectors are complete ( $\text{tr } P_2 = (d-2)(d+1)/2$ ,  $\text{tr } P_1 = d-1$ ,  $\text{tr } P_{0s} = \text{tr } P_{0w} = 1$  sum to  $d(d+1)/2$ ;  $d=4$  split  $5+3+1+1=10$ ), the EH propagator is  $P_2/p^2 - \frac{1}{d-2}P_{0s}/p^2$ , and the Stelle ghost is a *spin-2* state. [E] the entire KMS form factor  $e^{p^2/M^2}$  (v304/v259) makes the spin-2 sector ghost-free; [C] the wrong-sign spin-0 conformal mode (v332) is cured separately by the GHP contour



(v334). [C] so the full *perturbative* graviton is ghost-free sector-by-sector (conditional on v304’s entire-analyticity assumption). [O] residual: perturbative-only + the open analyticity assumption; not the non-perturbative measure. [O] DECLINE:  $\text{tr } P_2=5$  is dimension counting, not  $g_{\text{car}}$  (v354/v355). Cited in `tfpt_5_redteam` (Target F); Python-only.

**2026-06-23 (Directions 3 & 7 — the two hard functional-analysis residuals ADVANCED: the QG measure as one-loop fluctuations around the parameter-free saddle (v365), and the seam collar verified IN MMST’s free-fermion class so the scaling limit is  $(E_8)_1$  (v366))**

- **v365 (QGAMB.SADDLE.01) — Direction 3.** With the parameter-free saddle (v358/v359) the QG *measure* (gate C7/QG.AMB.01) organises as a one-loop Gaussian fluctuation determinant. [E] the fluctuation stiffness is  $1/c_3 = 8\pi$  (no free dial, so the one-loop problem is now well-posed); [E] the fluctuation operator is gapped  $M \geq \Delta_{\text{eff}} = 6 \log \frac{3}{2} - \frac{31}{4\pi^2} \approx 1.648 > 0$  (v76/v337)  $\Rightarrow$  mode-by-mode convergence, finite model  $\log\text{-det } \text{Tr}' \log M = 6 \log \frac{9}{2}$ ; [C] the conformal mode converges on the GHP contour (v334); [E] the admissible projective limit exists ( $\chi = 729/665$ , v330) with 0 new dials (v364). [O] residual: the full non-perturbative projective limit on the ambient sector. Sharpens C7 from “diffuse full QG” to a one-loop Gaussian problem around the parameter-free saddle; does not close it. Cited in `tfpt_research_contracts`; Python-only.
- **v366 (SEAM.MMST.INCLASS.01) — Direction 7.** The seam collar is verified IN MMST’s free-lattice-fermion scaling-limit class hypothesis-by-hypothesis: [E]  $D = \dim S^+ = 16$  Majoranas,  $c = g_{\text{car}} + N_{\text{fam}} = 8$ ,  $\text{rank } E_8 = 8$  so  $\text{rank} \leq c \leq D$  is  $8 \leq 8 \leq 16$  (in range); [E] gapped ( $\Delta = 6 \log \frac{3}{2} > 0$ , derived v302); [C] quasi-free CAR class (v155/v160); [E] chiral ( $c_- = D/2 = 8 \neq 0$ ). [C] the MMST scaling-limit theorem then applies, and [E] the order-4  $\mu_4$  clock pins the target to  $(E_8)_1$  ( $\det K=1$  vs  $SO(16)$   $\det K=4$ , v351). [O] residual: the single S3 input (the collar as a genuine lattice chiral invertible phase, the seam one-sidedness, v297/v356). Reduces the continuum side of SEAM.EQUIV.01 to one named input; ADVANCES, does not close. Cited in `tfpt_research_contracts`; Python-only.

**2026-06-23 (Consistency pass: the parameter-free *full covariant* Einstein equation (v358/v359) propagated across ALL stale paper and website surfaces; new v359 keybox; homepage gravity animation)**

- **Deep-sync of the parameter-free gravity status.** A cross-surface audit found several documents and website components still framing gravity as “exact only in the  $R + R^2$  regime” or the metric-sector field equation as “open”. These were reconciled with the canonical post-v358/v359 status: the *local* covariant field equation  $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$  is PARAMETER-FREE (both coefficients fixed); what remains open is the Jacobson equation-of-state status, the global ambient measure (QG.AMB.01/C7), and the single unit  $v_{\text{geo}}$ . Touched: `introduction`, `tfpt_2_standard_model`, `tfpt_3_e8_audit_bootstrap`, `tfpt_4_frontier` (summary table), `tfpt_research_contracts`, `origin_theory`.
- **New v359 keybox** added to `tfpt_research_contracts` (full covariant equation: fixed-volume  $\Rightarrow$  Einstein tensor, Lovelock  $\Rightarrow$  conservation an output); the v358 residual corrected from “linear $\rightarrow$ nonlinear” (stale — v359 closed it) to the equation-of-state + C7 +  $v_{\text{geo}}$  residual, including the generated script index (`script_registry.csv`).
- **Website.** The GravityEmergence animation (three origins of  $c_3$  converging to  $1/(8\pi)$ ; the parameter-free Einstein equation) was updated to the full covariant result and surfaced as a dedicated homepage section (`/#gravity`); gravity-status wording aligned across `ReconstructionChain`, `VerificationDag`, `StatusMatrix`, `WhatTfptClaims`, `ThreeDecoderMap`, `StatusPyramid`, `papers.ts`, `glossary.ts` and the review page.

**2026-06-23 (Direction 6,  $v_{\text{geo}}$  sharpened: after the parameter-free gravity, the gravity sector adds NO new dimensionful input either, so  $v_{\text{geo}}$  is the SINGLE dimensionful input of the complete theory — final tally  $\{v_{\text{geo}}, \pi\}$ , ZERO dimensionless dials: v364)**

- **v364 (VGE0.SHARPEN.01) — Direction 6.** The No-Unit Theorem (v153) already made  $v_{\text{geo}}$  a forced metrology primitive. [E] The SHARPENING after the parameter-free gravity (v358/v359/v361): the Einstein coefficient  $1/c_3 = 8\pi$  and the  $\Lambda$  prefactor  $(8\pi)^2 \cdot 48c_3^4 = 3/(4\pi^2)$  are dimensionless, so  $1/G \sim v_{\text{geo}}^2$  and  $\Lambda \sim (\text{dimensionless}) v_{\text{geo}}^4$  reduce to  $v_{\text{geo}}$  times atoms — the gravity sector adds no new scale. [E] FINAL TALLY: every dimensionful quantity = (a dimensionless ratio)  $\times$  (a power of  $v_{\text{geo}}$ ); TFPT's free content is  $\{v_{\text{geo}}, \pi\}$  with ZERO dimensionless dials, a  $\sim 26 \rightarrow 1$  reduction vs the SM. [O]  $v_{\text{geo}}$  stays irreducibly primitive (No-Unit;  $1/G$  UV-sensitive, Sakharov v68). Direction 6 is clarity — the last dimensionful anchor named and shown to be the only one. Cited in `tfpt_research_contracts`; Python-only.

**2026-06-23 (Direction 5, the matter–gravity backreaction: the carrier's gravitational anomaly is forced  $c_- = 8$ , and the matter vacuum-energy backreaction is finite ( $\rightarrow$  the forced  $\Lambda$  from  $\alpha$ , not  $M_{\text{Pl}}^4$ ) — the loop closes; no new SM-quantum-number read-out (declined): v361)**

- **v361 (GRAV.BACKREACT.01) — Direction 5.** With the parameter-free equation (v359) and the explicit J3 flux (v358), the backreaction of TFPT's carrier matter is computed. [E] the carrier's gravitational anomaly is forced:  $c_- = 16/2 = 8 = g_{\text{car}} + N_{\text{fam}}$  (the chiral central charge of the 16-Majorana content). [E] the backreaction is FINITE  $\rightarrow \Lambda$  from  $\alpha$ : the matter vacuum energy through  $G_{ab} = c_3^{-1} T_{ab}$  is gap-suppressed (Decoupling Thm v337) to  $\rho_{\Lambda}/\bar{M}_{\text{Pl}}^4 = (3/(4\pi^2))e^{-2\alpha^{-1}}$  (v60), the 123 orders =  $119.03 + 3.92$  — *not* the  $M_{\text{Pl}}^4$  catastrophe; the loop (v359+v60 + the carrier vacuum) closes. [E] J3 explicit:  $\delta\langle K_B \rangle = (8\pi^2 R^4/15)\delta\langle T_{00} \rangle$ , carrier  $T_{ab}$  = the SM's (v159). [C] the SM trace anomaly is reproduced (content = SM). [O] DECLINE: no new atom-forced read-out of individual SM quantum numbers beyond  $c_- = 8$  and  $\Lambda$  — per v354/v355 declined. The backreaction is finite and consistent, adding no new dial. Cited in `tfpt_4`; Python-only.

**2026-06-23 (Direction 2, the disciplined result: the gap-induced GR correction is the known  $R^2$  scalaron — already falsifiable via  $n_s/r$ ; a *new* forced near-term deviation does NOT survive the discriminator and is declined: v360)**

- **v360 (GRAV.GAPCORR.01) — Direction 2, run honestly.** Does the seam transfer gap  $\{1, (2/3)^6, (1/3)^6\}$  yield a NEW, calculable, FORCED deviation from GR (a near-term kill test) beyond the known  $R^2$  scalaron? With the v354/v355 discriminator the answer is *no*, and this is recorded rather than manufacturing a coefficient. [E] the leading higher-curvature term IS the  $R^2$  scalaron (spectral-action  $a_4 = R^2/72$ , v36;  $M_{\text{scal}}^2/\bar{M}_{\text{Pl}}^2 = c_3^7$ , exponent  $7 = g_{\text{car}} + N_{\text{fam}} - 1$ ) — the calculable beyond-Einstein correction, already in TFPT. [C] it is the entanglement-equilibrium NEXT order (the Wald first law  $\rightarrow f(R) = R + R^2$ , v28/v359). [E] it is ALREADY falsifiable via  $n_s = 1 - 2/N_*$ ,  $r = 12/N_*^2$  (live CMB-S4 kill test). [E] the gap  $\Delta = 6 \ln \frac{3}{2}$  is a dimensionless rate, so a distinct gap-induced curvature<sup>2</sup> term sits at  $\xi v_{\text{geo}} \sim$  the seam/Planck scale (not near-term observable). [O] DECLINE: a new dimensionless coefficient is not atom-forced (the scalaron exponent  $7 = g + N - 1$  and the gap exponent  $6 = 2N_{\text{fam}}$  are different;  $c_3^7$  and  $(2/3)^6$  unrelated) — per v354/v355 declined. Direction 2 yields no new near-term forced prediction; the answer is the existing scalaron. Cited in `tfpt_4`; Python-only.

**2026-06-23 (the FULL covariant Einstein equation, parameter-free: the fixed-volume entanglement equilibrium gives  $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$  with BOTH coefficients TFPT-fixed ( $8\pi = 1/c_3$  from v358,  $\Lambda$  from  $\alpha$  via v60); B6 closed at the local level: v359)**

- **v359 (GRAV.NONLINEAR.01) — Direction 1, the full covariant equation.** Extends v358 from the linearised (fixed-radius, Ricci-level) result to the FULL covariant equation. [C][E] Jacobson 2015: demanding stationarity of the small-ball entanglement entropy at fixed *volume* brings in the EINSTEIN TENSOR  $G_{ab}$  (not just  $R_{ab}$ ) — trace  $g^{ab}G_{ab} = (1 - d/2)R = -R$  in  $d=4$ , the structure that carries  $\Lambda$ . [E] Lovelock:  $G_{ab}$  is the unique symmetric divergence-free second-order metric tensor, so  $\nabla^a(G_{ab} + \Lambda g_{ab}) = 0$  forces  $\nabla^a T_{ab} = 0$  — conservation is an output. [E] coefficient 1:  $8\pi G \rightarrow 8\pi = 1/c_3$  fixed (v358). [E] coefficient 2: Jacobson leaves  $\Lambda$  free, TFPT fixes it —  $\rho_\Lambda / \bar{M}_{\text{Pl}}^4 = (3/(4\pi^2))e^{-2\alpha^{-1}}$ , prefactor  $(8\pi)^2 \cdot 48c_3^4 = 3/(4\pi^2)$  (v60); so BOTH coefficients are TFPT-fixed, not free integration constants. [C] all timelike directions  $\rightarrow$  the covariant tensor equation; the matter side is the assembled CHM ball modular flux (v358/v323). NET: the original B6 “full covariant Einstein-side field equation” is now PARAMETER-FREE at the local level. [O] residual: the Jacobson equation-of-state status (not a from-action quantisation) + the global ambient measure (C7/QG.AMB.01, separate, gap-decoupled);  $v_{\text{geo}}$  the one unit. Cited in `tfpt_research_contracts`; mirrored in Wolfram (303/303). (`next-plan.md` Direction 4 folded in: the  $\Lambda$  here matches the  $\alpha$ -readout.)

**2026-06-23 (the classical gravity field equation, parameter-free: the entanglement first law  $\delta S = \delta \langle K \rangle$  with TFPT’s atoms gives the LINEARISED Einstein equation with  $G$  from  $c_3$ ; the thermodynamic and geometric origins of  $c_3$  coincide via  $|\mu_4| = |\mathbb{Z}_2|\chi = 4$ ; the two formerly-open pieces (matter flux J3, entropy-density coefficient) closed/atom-fixed: v358)**

- **v358 (GRAV.ENTROPY.EQUILIBRIUM.01) — the bridge to the dynamic world, instantiated on the gravity side.** Carrying out the Jacobson-2015 / Faulkner-et-al. “first law of entanglement  $\Rightarrow$  Einstein equation” with TFPT’s atoms. [E] PARAMETER-FREE COEFFICIENT:  $\delta S = \delta \langle K \rangle$  gives  $G_{ab} = c_3^{-1} T_{ab}$  with  $c_3^{-1} = 8\pi$  fixed; all four  $c_3$ -roles consistent (Unruh  $1/(2\pi) = 4c_3$ , EH  $1/(16\pi) = c_3/2$ , Einstein  $8\pi = 1/c_3$ , Bekenstein  $1/4 = 1/|\mu_4|$ ) — no free dimensionless Newton dial. [E] THERMO = GEO COINCIDENCE (new): the first law’s coefficient  $2\pi/\eta$  equals the geometric  $|\mathbb{Z}_2|2\pi\chi$  iff  $|\mu_4| = |\mathbb{Z}_2|\chi = 4$  (v216) — the thermodynamic and geometric origins of  $c_3$  are ONE. [C] J3 SOLVED: the CHM ball modular Hamiltonian (boost via BW, v323) gives the matter flux  $\delta \langle K_B \rangle = \frac{8\pi^2 R^4}{15} \delta \langle T_{00} \rangle$ . [E] ENTROPY DENSITY ATOM-FIXED:  $1/4 = 1/|\mu_4|$  (v57) + central charge  $c = g_{\text{car}} + N_{\text{fam}} = 8$  (form v59). [O] residual: linear  $\rightarrow$  full non-linear (the original B6) + the  $v_{\text{geo}}$  unit. Closes the LINEARISED Einstein equation parameter-free; cited in `tfpt_research_contracts`; mirrored in Wolfram (302/302).

**2026-06-23 (the two frontier walls attacked directly: Direction A assembles the continuum chain of SEAM.EQUIV.01 theorem-by-theorem and ties its residual to the seam one-sidedness; Direction B was found ALREADY closed by v40 (harmonic metric = finite linear algebra, not a PDE): v356)**

- **v356 (SEAM.EQUIV.CONTINUUM.03) — Direction A, the continuum chain assembled.** After the reverse-audit/bandwidth work showed no hidden *discrete* lever remains, the only fundamental progress is to assemble the *continuum* chain. Link-by-link: **S1** the gapped ( $\Delta = 6 \ln(3/2) > 0$ , v302) quasi-free 16-Majorana collar (v155/v160)  $\rightarrow$  **S2** invertible (gapped free fermions have no intrinsic topological order, Kitaev, v301)  $\rightarrow$  **S3** the *nontrivial* invertible phase ( $c_- = 8 \neq 0$ )  $\rightarrow$  **S4** a *non-gappable* anomalous chiral edge (anomaly inflow / bulk-edge), so edge *existence* is a CONSEQUENCE of S2+S3, not a separate assumption — this directly

reduces the v336/v351 “does the collar have a chiral edge?” residual  $\rightarrow$  **S5** its chiral-CFT scaling limit (MMST, v336; range rank  $\leq c \leq D$ ,  $8 \leq 8 \leq 16$ )  $\rightarrow$  **S6** the order-4  $\mu_4$  clock + the AQFT stack pin  $(E_8)_1$  (v351/v297). [E] the only non-theorem link is S3, and [C] S3 is the seam being *one-sided* (chiral,  $c_- \neq 0$ ) — the *same* one-sidedness that makes  $c_3=1/(8\pi)$  a one-sided Gauss–Bonnet  $(|\mathbb{Z}_2| \nmid K, \text{v58})$ . [O] the one remaining input is “the abstract collar is realized as a genuine lattice chiral free-fermion invertible phase” (the v297 Flat-Away). NET: SEAM.EQUIV.01 reduced to ONE realization input tied to the constant that *defines* the theory; it ADVANCES, it does not close.

- **Direction B — found ALREADY closed by v40.** The U-wall “transcendental Hitchin solve” (v31) is the worst case for a *generic* Higgs bundle; but at the *polystable* point TFPT selects, Mehta–Seshadri makes the bundle *unitary*, so the harmonic metric is the constant invariant Hermitian form (*finite linear algebra*, Higgs  $\Phi=0$ ) and  $\det R=8$  is read off — not a PDE. Only the absolute amplitude  $U_{\text{point}}$  (an anchor) remains. A redundant new B module was built, found to duplicate/contradict v40, and *deleted* (no padding). Honest outcome: B needs no new work.

**2026-06-22 (bandwidth search of the unmapped  $E_8$  region, with a strict forced-identity discriminator: the genuine overlooked structure is COLLECTIVE ( $\sum \deg = 128 = \dim S^+$ ,  $\prod \deg = |W(E_8)|$ , exponents = totatives of 30); the per-degree atom matches are unforced and DECLINED; no new physical hit – reconciling v66 and v354: v355)**

- **v355 (E8.UNMAPPED.BANDWIDTH.01) — the next step after v354, run honestly.** The obvious follow-up (“search those five degrees for hits we overlooked”) is itself the numerology temptation v354 warns about: a rich object like  $E_8$  returns a numerical match for almost anything. So the scan uses a STRICT discriminator – a candidate counts only if it is a FORCED identity (a theorem about  $E_8$ ), never a coincidence. [E] FORCED, SET-LEVEL (the genuinely overlooked structure, collective not per-number):  $\sum \deg = 128 = \dim S^+$ , the spinor half of  $248 = 120 + 128$  (theorem:  $\sum \deg = \# \text{pos roots} + \text{rank} = 120 + 8$ ) – the invariant budget of  $E_8$  equals the fermion content the carrier reads;  $\sum \exp = 120 = \# \text{pos roots}$ ;  $\prod \deg = |W(E_8)| = 696,729,600$ ; exponents = the  $\varphi(30)=8$  totatives of 30 (v223), so the unmapped degrees’ exponents are part of the forced set;  $\max \deg = 30 = h(E_8)$ . [C] SUGGESTIVE, not a theorem:  $12=h(E_6)$ ,  $18=h(E_7)$ ,  $30=h(E_8)$  are all degrees of  $E_8$  (the  $2T/2O/2I$  McKay ladder, v219), but “ $\deg(E_8)$  contains its subalgebras’ Coxeter numbers” is not a general theorem – flagged, not claimed. [O] DECLINED (the numerology temptation): the per-degree atom matches  $14=\dim G_2$ ,  $20=\det L$ ,  $24=|W(A_3)|$ ,  $12=|R(A_3)|$ ,  $18=p_4(a)$  (v66 already flagged 12, 18 as fittable) each admit  $\geq 2$  readings, and the lone extra prime in  $|W(E_8)| = 2^{14}3^55^27$  “=” the scalaron exponent is one of many readings of 7 – all declined. [E] RECONCILES v66 (the 8 degrees coincide with the load-bearing integers) and v354 (only 2, 8, 30 are primary readouts): the search finds NO new forced physical hit, and the disciplined decline IS the anti-numerology result. (This also sharpens v354: the five are “no PRIMARY readout / hull overhead”, not literally “unmapped” – they coincide with structural atoms, but unforced.) Cited in `tfpt_5_redteam` (with a full audited degree table); mirrored in Wolfram (301/301).

**2026-06-22 (the REVERSE numerology audit, and what the golden ratio means: of  $E_8$ ’s 8 Casimir invariants only 3 feed a primary readout and 5 are hull overhead; the golden ratio is UNMAPPED structure, so it cannot be numerology: v354)**

- **v354 (E8.REVERSE.AUDIT.01) — the reverse complement of v305.** TFPT’s forward discipline (v305) is one-directional: every physical READOUT must map onto an  $E_8$  structure (anti-fitting). The sharp adversarial question runs the OTHER way — how much  $E_8$  structure



maps onto NOTHING, and is the recurring golden ratio numerology? [E] THE REVERSE AUDIT: of  $E_8$ 's eight Casimir invariant degrees  $\{2, 8, 12, 14, 18, 20, 24, 30\}$  (exponents  $+1$ ), exactly THREE feed a readout — 2 (metric), 8 (rank  $\rightarrow c_3=1/(8\pi)$ ), 30 (Coxeter  $\rightarrow g_{\text{car}}=5$  and the order-30 cycle) — and FIVE carry no primary readout  $\{12, 14, 18, 20, 24\}$  (they coincide with structural atoms per v66, but unforced – reconciled in v355); so 3/8 are primary readouts, 5/8 hull overhead. [E] ECONOMY, NOT PROMISCUITY: a small fixed set (rank 8,  $h=30$ ,  $\det K=1$ ,  $D_5 \oplus A_3$ , the 16 half-spinor) generates the ENTIRE readout set, and the higher Casimirs are never reached into to fit — a numerological use would be PROMISCUOUS (mine the rich structure), TFPT is ECONOMICAL (few invariants, many readouts) and leaves most of  $E_8$  untouched, the anti-numerology signature being precisely the large unused remainder. [E] THE GOLDEN RATIO IS UNMAPPED  $\Rightarrow$  NOT NUMEROLOGY:  $\varphi = 2 \cos(\pi/5)$  appears only in internal structure (the affine- $E_8$  attractor eigenvalue v312, the icosian lattice v348) and in NO physical readout; numerology means a number FITTED to an observation,  $\varphi$  is fitted to nothing — it is the algebraic signature of the 5-fold ( $h=30=2 \cdot 3 \cdot 5$ ), the carrier announcing it is icosahedral. [O] the 5/8 unmapped is the HULL OVERHEAD ( $E_8$  is the minimal even-unimodular hull, not the world group); where a future higher-invariant readout could live, or genuine over-capacity — stated plainly, not hidden. Cited in `tfpt_5_redteam`; mirrored in Wolfram (300/300).

**2026-06-22 (the bird's-eye rethink: TFPT is a unique self-consistent LOOP with ZERO free dials, not a linear axiom system;  $\pi$  is a fixed geometric constant, not a dial; the one residual is the continuum closure: v353)**

- **v353 (TFPT.SELFLOOP.01) — refining v352.** Stepping all the way back: TFPT is not a LINEAR theory (axioms  $\rightarrow$  theorems) but a CLOSED SELF-CONSISTENT LOOP whose outputs fix its own inputs. [E] it is a loop:  $g_{\text{car}}=5 = \max$  prime of the output  $h(E_8)=30$ , and the 8 in  $c_3 = \text{rank } E_8$  – the outputs fix the inputs (v6), with a unique fixed point (v56); “which is the axiom?” is the wrong question for a loop. [E] ZERO free adjustable dimensionless dials: every load-bearing number is forced/over-determined. [E]  $\pi$  is a FIXED geometric constant (the boundary 2-sphere,  $\oint K=4\pi$ ), not a dial. [C] this refines v352's “framework +  $\pi$ ”:  $\pi$  is not a free input and the framework is a SELECTION PRINCIPLE (the unique self-consistent loop), not a free meta-axiom – the honest characterisation is “a unique self-consistent loop with zero free dials”. [O] the one residual is the CONTINUUM closure of the loop (the chiral-edge existence, v336/v351); structural-realism (loop complete) vs constructivism (continuum open) is a philosophical, not numerical, fork – no free numbers either way. A capstone synthesis.

**2026-06-22 (two consequences: the  $c=8$  which-net ambiguity resolved by the order-4 clock, and BOTH axioms shown bootstrap-forced so the only irreducibles are the framework +  $\pi$ : v351, v352)**

- **v351 (SEAM.EQUIV.CONTINUUM.02) — the which-net ambiguity falls.** The long-standing “ $c=8$  ambiguity” ( $E_8 \det 1$  vs  $SO(16)=D_8 \det 4$ , both  $c=8$ , flagged open in v277/v344) is RESOLVED, non-circularly, by the seam's ORDER-4  $\mu_4$  clock: the index-4 simple-current extension of  $D_5 \oplus A_3$  is  $E_8$  (v154), the index-2 would be  $SO(16)$ , so the order-4 clock (the four Gauss–Bonnet marks, v216) selects  $E_8$  (full  $\mu_4$  condensation  $16 \rightarrow 4 \rightarrow 1$ ) not  $SO(16)$  (partial  $\mathbb{Z}_2$   $16 \rightarrow 4$ ). The bootstrap agrees ( $h(E_8)=30$  max prime  $5=g_{\text{car}}$  vs  $h(D_8)=14$  max prime 7). [E] so  $\det K=1$  is a consequence, not an open input. [O] the residual of SEAM.EQUIV.01 is then PURELY the chiral-edge / scaling-limit existence (v336), not which-net.
- **v352 (TFPT.IRREDUCIBLE.01) — the inputs are not axioms.** The deepest answer to “the inputs are back-determined”: BOTH P1 and P2 are bootstrap-forced fixed points. [E]  $g_{\text{car}}=5$  is forced three ways (v6); the 8 in  $c_3$  four ways (rank  $E_8 = h(D_5) = \phi(30) = \det R$ , v54); the  $E_8$  hull is forced (unique even unimodular rank-8, v83/v154). [E] so the ONLY irreducibles



are the FRAMEWORK (a discrete reflection-positive boundary with a finite carrier) and the transcendental  $\pi$ . [O] the framework is the META-AXIOM — not math-reducible, the foundational stance (like “spacetime is a manifold”), minimal, tested by its consequences not proved. TFPT’s true input is ONE framework +  $\pi$ . Wolfram mirror +2 (297 → 299).

**2026-06-22 (correction: the inputs are bootstrap-forced fixed points, not free axioms –  $g_{\text{car}}=5$  IS the icosahedral 5 of  $h(E_8)=30$ , and the golden ratio is emergent from the  $\mu_4$ -glue, not external: v350)**

- **v350 (SEAM.EQUIV.BOOTSTRAP.01) — correcting v349.** An honest correction prompted by the right objection that the inputs are not axioms but are back-determined by the theory. [E] the inputs ARE bootstrap-forced (v6):  $g_{\text{car}}=5$  is over-determined three ways – rank-fill ( $g_{\text{car}}+N_{\text{fam}}=8$ ), Coxeter-match ( $g_{\text{car}}=\max$  prime of  $h(E_8)=30$ ), Pascal ( $2^4=C(5,0)+C(5,1)+C(5,2)$ ). [E] so  $g_{\text{car}}=5$  IS the icosahedral 5-fold (the 5 in  $h(E_8)=30=2\cdot3\cdot5$ ), not the  $D_5$  rank; the atoms (2,3,5) ARE the primes of the icosahedral Coxeter number 30. [E] the golden ratio is EMERGENT from the glue:  $D_5$  ( $h=8$ )/ $A_3$  ( $h=4$ ) are non-golden alone, but  $D_5\oplus A_3 \rightarrow E_8$  ( $\mu_4$  index-4, v154) gives  $h(E_8)=30$  (golden) – the seam’s own  $\mu_4$  produces the 5-fold, so v349 checked the un-glued pieces, the wrong object, and its pentagon-vs-count dichotomy was false. [O] the residual relocates: the golden is bootstrap-forced, the only open thing is the physical continuum realisation (v336), NOT the golden ratio. Honest caveat: self-consistency within the framework, not from-nothing. Wolfram mirror +1 (296 → 297).

**2026-06-22 (the honest test – does the raw seam carry  $\varphi$ ? – answer NO: the bare carrier gives  $\sqrt{2}$ , golden is the icosahedral input; the keystone reduces to “is  $g_{\text{car}}=5$  a pentagon or a count?”: v349)**

- **v349 (SEAM.EQUIV.GOLDEN.01).** The honest, computed test of the v348 question, and a clarifying NEGATIVE: the raw seam does *not* carry the golden ratio. [E] the golden ratio  $2\cos(\pi/5)$  is the signature of a 5-fold rotation ( $5 \mid h$ ); the carrier  $D_5$  has  $h=8$  (eigenvalue  $\sqrt{2}$ ) and  $A_3$  has  $h=4$  – no 5-fold; the dynamical data (cusp  $\{0, 1/3, 2/3\}$ , recovery  $(2/3)^6$ , seed  $\varphi_0$ ) are rational/transcendental. [E]  $5 \mid h$  holds only for  $A_4=SU(5)$  ( $h=5$ ),  $H_3$  ( $h=10$ ),  $E_8$  ( $h=30$ ) – the output/icosahedral side, so  $\varphi$  is the icosahedral INPUT. [E]  $\varphi$  alone would not suffice ( $2I \neq C_5 \times \mu_4$ ,  $20 \neq 120$ ). [O] the resolution: the keystone reduces to one question – is  $g_{\text{car}}=5$  a PENTAGON (a 5-fold rotation  $\Rightarrow$  golden  $\Rightarrow$  icosahedral  $2I = E_8$ ) or just a COUNT (the rank of  $D_5 \Rightarrow \sqrt{2}$ )? The “absurdly simple solution” is real but is a *reading* of axiom P2 (the golden ratio is the pentagon content of  $g_{\text{car}}=5$ , not derivable from  $g_{\text{car}}=5$ -as-a-count); there is no hidden  $\varphi$ . L2 is foundational: P2-as-pentagon closes it (mode C), else a rigidity theorem must produce the 5-fold (mode B); not closed. Wolfram mirror +1 (295 → 296). A NEGATIVE/clarification; prevents a false closure.

**2026-06-22 (Route B carried out: rigidity + the icosian-ring identity reduce the whole keystone to ONE question – does the raw seam carry the golden ratio? – the simplest form of L2: v348)**

- **v348 (SEAM.EQUIV.RIGID.01).** Executes the rigidity route (mode B of v347) and lands on a strikingly simple statement. [E] rigidity closes the crystallographic part: the order-4 clock has the unique normal form  $z \mapsto iz$  (Nielsen/Kerckhoff, v180), Troyanov gives the unique flat pillowcase metric (v284). [E] the downstream is a LATTICE identity, not just a graph: by Conway–Sloane the ring of icosians (120 unit icosians =  $2I$ , golden-ratio quaternion norm) IS the rank-8  $E_8$  lattice. [E] the single bridge is the golden ratio: the crystallographic restriction allows only orders  $\{1, 2, 3, 4, 6\}$  (the 5-fold absent), so  $\varphi = 2\cos(\pi/5)$  is exactly what extends the crystallographic  $\mu_4$  to the non-crystallographic icosahedral  $2I$  (the quasicrystal). [O] so

the entire keystone reduces to ONE question: *does the raw seam carry the golden ratio?* – if yes,  $\mu_4 + \varphi \Rightarrow 2I \Rightarrow E_8$  and rigidity fixes the rest. [O] honest subtlety: TFPT’s  $\varphi$  is currently  $E_8$ -side (v312/v313); the raw-seam data so far are not manifestly golden, so showing the RAW seam carries  $\varphi$  non-circularly is the residual (= L2, sharpest form). Wolfram mirror +1 (294  $\rightarrow$  295). An analysis/reduction; closes nothing.

**2026-06-22 (can the one open arrow be *fully* solved? the closure-mode classification: the flat $\rightarrow$ spherical base locus, four modes, no free theorem – only rigidity or axiom  $P_3$ : v347)**

- **v347 (SEAM.EQUIV.CLOSURE.01).** A direct, honest answer to “is there any other way to fully solve the one open arrow L2?”. [E] the precise locus: the seam gives the *flat* pillowcase base  $S^2(2, 2, 2, 2)$  ( $\chi_{\text{orb}} = 2 - 4(1 - \frac{1}{2}) = 0$ , v216) while the  $2I/E_8$  structure is the Seifert base over the *spherical*  $S^2(2, 3, 5)$  ( $\chi_{\text{orb}} = 2 - (\frac{1}{2} + \frac{2}{3} + \frac{4}{5}) = \frac{1}{30}$ ), so L2 bridges two geometric types – why it needs the carrier  $(2, 3, 5)$  input. [E] as a “physics-realises-geometry” statement L2 has exactly four closure modes: A construct (the analytic wall), B rigidity (a unique-realisation theorem – the only theorem-route w/o new input, pursued v284/v300/v331), C axiom  $P_3$  (then TFPT is complete relative to 3 axioms), D empirical. [E] the “no new state” discipline (v246) blocks deriving L2 from extra physics, so no route makes it a free theorem. [E] verdict: full closure is either a rigidity theorem (B) or axiom status (C), validated either way by (D); only B closes it as a theorem. [O] NOT closed – classifies the modes, locates the difficulty. Wolfram mirror +1 (293  $\rightarrow$  294). An analysis/roadmap; closes nothing.

**2026-06-22 (the geometric bridge made explicit: the keystone is ONE arrow – raw-seam  $\Rightarrow \det K=1$  is a six-link chain, five theorems + one open orbifold realisation, the group forced by the  $(2, 3, 5)$  atoms: v346)**

- **v346 (SEAM.EQUIV.GEOM.01).** The honest capstone of the SEAM.EQUIV.01 investigation: it lays out the full path raw-seam  $\Rightarrow \det K=1$  as six links and shows exactly one is open. [E]/[C] L1 raw seam  $\rightarrow \mu_4$  + four marks (Gauss–Bonnet, v216); [O] L2  $\mu_4$  + the  $(2, 3, 5)$  atoms  $\rightarrow$  the  $2I$  orbifold  $\mathbb{C}^2/\Gamma$  (THE ONE OPEN ARROW = the geometric content of SEAM.EQUIV.01); [E] L3  $2I \rightarrow$  affine  $E_8$  McKay (v219); [E] L4  $2I \rightarrow E_8$  du Val (v232); [E] L5  $\mathbb{C}^2/2I \rightarrow$  Poincaré link  $H_1=0$  (v345); [E] L6  $H_1=0 \rightarrow \det K=1$ . [E] five of six links are theorems; [E] the group is FORCED by the seam’s own atoms (the exceptional  $SU(2)$  axis signatures  $(2, 3, 3)/(2, 3, 4)/(2, 3, 5) \rightarrow 2T/2O/2I$ ;  $(2, 3, 5)$  unique to  $2I$ , and  $(|\mathbb{Z}_2|, N_{\text{fam}}, g_{\text{car}})=(2, 3, 5)$  ARE those axes); [E] everything downstream of L2 is then forced. [O] the irreducible open content is L2 alone (“the raw seam normal bundle is an ADE orbifold point  $\mathbb{C}^2/\Gamma$ ”) = SEAM.EQUIV.01; not closed, but the bridge is now demonstrably ONE arrow with the group pinned and all downstream proven.

**2026-06-22 (executing route R3: the  $(2, 3, 5)$ -rewrite attractor link computed to be the Poincaré homology sphere ( $\det K=1$ ), the combinatorial core complete, only the geometric bridge open: v345)**

- **v345 (SEAM.DETK.02).** Carries out the combinatorial core of the hypergraph-homotopy route (R3, the most promising fresh angle from v344) as far as it computably goes. By plumbing topology (Brieskorn/Hirzebruch) the link of a tree plumbing with the ADE Cartan intersection form has  $H_1 = \text{coker}(\text{Cartan})$ . [E] the Smith normal form of the  $E_8$  Cartan matrix is the identity, so  $\text{coker}(E_8) = 0$  – the  $E_8$ -graph plumbing boundary is an INTEGRAL HOMOLOGY SPHERE (the genus-1 meaning of  $\det E_8=1$ ); among ADE only  $E_8$  does this ( $A_n \rightarrow \mathbb{Z}/(n+1)$ ,  $D_n \rightarrow \mathbb{Z}/4$ ,  $E_6 \rightarrow \mathbb{Z}/3$ ,  $E_7 \rightarrow \mathbb{Z}/2$ ). [E]  $\pi_1 = 2I = SL(2, 5)$  is PERFECT (commutator subgroup = the whole order-120 group, computed directly), so  $H_1 = 0$  and the link is THE Poincaré sphere,  $|2I| = 120 = |\mu_4| h(E_8)$ . So R3’s combinatorial half is

[E]-complete: the  $E_8$  attractor link is forced to be the Poincaré sphere ( $\det K=1$ ), uniquely among ADE. [O] what stays open is only the geometric realisation bridge (the raw rewrite seam attractor *is* this  $E_8$  du Val plumbing) = the SEAM.EQUIV.01 content; not closed, but the keystone is now reduced to that single geometric identification. Wolfram mirror +1 (292  $\rightarrow$  293).

**2026-06-22 (the one keystone bit  $\det K=1$  mapped: six equivalent faces, the genus-1 obstruction, three access routes, the hypergraph-homotopy route the most promising fresh angle: v344)**

- **v344 (SEAM.DETK.01).** A bird's-eye, full-spectrum map of SEAM.EQUIV.01's only open residual, the holomorphy bit  $\det K=1$ . [E] it is ONE statement with six equivalent faces (holomorphy / homology-sphere link  $H_1=0$  / one 1-dim irrep /  $\det \text{Cartan} = 1$  / full  $\mu_4$  condensation /  $\tau=i$  CM), unified by  $|\det \text{Cartan}| = |H_1(\text{link})| = \#(1\text{-dim irreps})$  across ADE ( $A_n \rightarrow n+1$ ,  $D_n \rightarrow 4$ ,  $E_6 \rightarrow 3$ ,  $E_7 \rightarrow 2$ ,  $E_8 \rightarrow 1$ ) – only  $E_8$  gives 1. [E] the GENUS-1 OBSTRUCTION explains why it is hard: every easy argument (plane vacuum, gap, free-fermion) is genus-0, necessary but not sufficient (v87);  $\det K$  is the torus degeneracy. [E] THREE genus-1 access routes: R1 continuum-modular (v336), R2 anyon-condensation (v281), R3 hypergraph-homotopy. [C] the HYPERGRAPH route is the most promising fresh angle: the (2,3,5) rewrite attractor is the affine- $E_8$  graph (v312) and the  $E_8$  du Val link is the Poincaré sphere (v232), so  $\det K=1$  becomes a finite combinatorial statement (no continuum limit). [O] none closes it; the gap is the combinatorial-to-geometric bridge = the SEAM.EQUIV.01 content. A synthesis/roadmap; closes nothing.

**2026-06-22 (the four black-hole-birth completion routes investigated: none closes all or is 100% provable alone; all four converge on SEAM.EQUIV.01 + the decoupled QG contour: v343)**

- **v343 (FOUR.ROUTES.01).** A direct, honest answer to “which black-hole-birth route closes ALL problems and is 100% provable”: *none individually* — but the four converge on the same two residuals as v335. [E] (A) the finite causal diamond reframes QG.AMB to a finite thermal trace (de Sitter  $S_{\text{dS}} = 32\pi^4 e^{2\alpha^{-1}}$  finite,  $\chi = 729/665$ ) but inherits the GHP contour (v334, [O]); does not touch the seam. [C] (B) Carlip–Cardy gives a second route to existence+ $c$  ( $c=8$  consistent, but the Carlip ambiguity and no holomorphy  $\det K=1$ ) — inherits SEAM.EQUIV.01. [E] (C) the modular/thermal flow (v239) is well-defined but its geometric content is QGEO.SYM.01, a corollary of SEAM.EQUIV.01 (v335) — a reduction. [E]/[O] (D) the unique attractor (v56) gives parameter-freeness but the cyclic dynamics is [C] interpretation. [E] **Verdict:** independent closures = 0; irreducible residuals after all four = 2 (unchanged from v335); the horizon premise makes both more natural but eliminates neither — focus = SEAM.EQUIV.01. An analysis module; closes nothing, fabricates nothing.

**2026-06-22 (the heat-kernel origin of the  $\alpha$  terms: the Calderón quadratic is the Gilkey gauge-curvature coefficient; the cubic Maxwell moment stays the EM.WARD.01 residual: v342)**

- **v342 (EM.WARD.02) — the heat-kernel sharpening of EM.WARD.01.** Derives the ORIGIN and STRUCTURE of the  $F_{U(1)}$  determinant-line terms from textbook Seeley–DeWitt / Gilkey coefficients, advancing the EM-Ward origin from [C] (conjectured) to [E] for the term structure; no new gate. [E]  $c_3 = 1/(8\pi)$  *is* the boundary Gauss–Bonnet coefficient ( $\oint_{S^2} K = 2\pi\chi = 4\pi$ ,  $\chi=2$ , v216). [E] the  $\alpha^2$  (Calderón) term *is* the Gilkey gauge-curvature coefficient ( $30\Omega^2/360 = \frac{1}{12}\Omega^2$ ,  $\Omega=i\alpha F \Rightarrow -(\alpha^2/12)F^2$ ), so it is forced, not an ansatz. [E] the  $c_3$ -orders  $\{0, 3, 6\}$  are an arithmetic ladder and  $2c_3^3 = 1/(256\pi^3)$  carries  $\pi^3$  (three boundary insertions). [C] the exact coefficient needs the seam  $F$ -normalisation. [O] the cubic  $\alpha^3$

Maxwell moment (a topological/Chern level) + the full “this is the seam  $U(1)$   $\zeta$ -determinant” identification stay the EM.WARD.01 residual, tied to SEAM.EQUIV.01.  $\alpha^{-1}$  stays [E] (v3). Wolfram mirror +1 (290  $\rightarrow$  291).

**2026-06-22 ( $\alpha$  as the stationarity of a  $U(1)$  determinant line: every ingredient an index / heat-kernel coefficient / discriminant form, the open EM-Ward obligation named: v341)**

- **v341 (ALPHA.QUILLEN.01).** Answers the external review’s sharpest valid point — “derive the form of  $\alpha$ , not more decimals”. The EM closure  $F_{U(1)} = \alpha^3 - 2c_3^3\alpha^2 - \frac{4}{5}c_3^6M \log(1/\varphi_{\text{seam}}) = 0$  is rewritten as a *stationarity* of the boundary  $U(1)$  determinant line:  $[\alpha^3 - 2c_3^3\alpha^2]_{\delta \log \det \Delta_{U(1)}} = [8b_1c_3^6]_{\text{counterterm}} \cdot [\log(1/\varphi_{\text{seam}})]_{\text{seam response}}$ . [E] every ingredient is a NAMED object, not a knob:  $M=41$  is a  $U(1)$  index ( $\frac{4}{5}M = 8b_1$ ,  $b_1=41/10$  the SM  $U(1)_Y$  beta, EM Ward v48);  $c_3=1/(8\pi)$  the Gauss–Bonnet boundary coefficient ( $8=\text{rank } E_8$ );  $-5/4 = -q(D_5)$  a discriminant form and  $\varphi_{\text{base}}=1/(6\pi)=\frac{4}{3}c_3$  carries the other,  $c_3/\varphi_{\text{base}}=3/4=q(A_3)$ ;  $\delta_{\text{top}}=\Omega_{\text{adm}}c_3^4$ . The Quillen split is verified at the  $\alpha^{-1}=137.0359992$  root. **No new gate:**  $\alpha^{-1}$  stays closed [E]; v341 adds no open item — it *sharpens* the pre-existing [O]/physical-origin obligation EM.WARD.01 (“why *this* function”, already conjectural). [O] the one still-open part *is* that residual: the from-first-principles AQFT proof that  $F_{U(1)}$  is the EXACT Quillen functional (the variational principle derived, not assembled), unchanged in scope. Wolfram mirror +1 (289  $\rightarrow$  290).

**2026-06-22 ( $A_s$  reconciled with the archived derivation: predicted & dimensionless (so it does not define a mass), the tension is the reheating channel: v340)**

- **v340 (AS.RECONCILE.01).** Answers “why don’t we have  $A_s$  / the old versions derived it” and “can a ratio define mass itself”. [E]  $A_s$  *is* predicted and *dimensionless*:  $A_s = N_\star^2(M_{\text{scal}}/\bar{M}_{\text{Pl}})^2/(24\pi^2) = N_\star^2c_3^7/(24\pi^2)$  with  $M_{\text{scal}}/\bar{M}_{\text{Pl}} = c_3^{7/2} = 1.26 \times 10^{-5}$  a pure ratio, so  $A_s$  pins the e-fold count  $N_\star$ , NOT  $\bar{M}_{\text{Pl}}$  — it cannot define an absolute mass (No-Unit, v153, stands). [C] the  $-11.4\sigma$  at the slow Higgs channel ( $N_\star=51.4$ ) is the slow-vs-fast reheating dichotomy; the  $A_s$ -preferred  $N_\star=56.2 \sim$  the instantaneous bound. [C] the archived TFPT v4.5/v4.6 reported a clean  $A_s = 2.124 \times 10^{-9}$  using the algebraic  $N_\star = 3/\varphi_0^{\text{ret}} - 1 = 55.42$ ; at that  $N_\star$  the current normalisation gives  $2.047 \times 10^{-9}$  ( $-1.9\sigma$ ) — the old clean  $A_s$  is near-instantaneous reheating, not the slow channel. [E] nothing lost: the current theory replaced the posited algebraic  $N_\star$  with one derived from reheating physics and exposed honestly that  $A_s$  requires fast (pre)heating. Record band  $N_\star \in [50, 60]$  (v84) untouched.

**2026-06-22 (the first-principles boundary, mapped: can  $M_{\text{Planck}}$  be computed? can  $F_{\text{transfer}}$  be first-principles? both NO, with the exact reason: v339)**

- **v339 (FIRSTPRINCIPLES.BOUNDARY.01).** An honest boundary audit answering two head-on questions and fabricating nothing. [E] *The Planck mass cannot be computed* —  $v_{\text{geo}}/M_{\text{Planck}}$  is forbidden by theorem (No-Unit, v153): a mass is mass-dimension 1 while every TFPT datum is dimension 0, so no dimensionless map outputs an absolute scale. It is not a gap but over-determined (gravity = dark energy to 0.11%, v274); every dimensionless *ratio* is derived, only the one unit is not. [E]  $F_{\text{transfer}}$  *cannot be made first-principles* either: for  $m_p/m_e$  the structure is in place (v262, carrier  $b_3=-7$  + closed electron) but two inputs stay external. [X]  $\alpha_s(M_Z)$  is external *because of a tension* — the SM (= TFPT content, no new states) does not unify (1-loop spread  $\sim 8.8$  at  $2 \times 10^{16}$  GeV, v246). The lattice  $C_p \sim 2.83$  is parameter-free but lattice-only (v35);  $\eta_B$ /axion ride on cosmological initial conditions; Koide  $Q=2/3$  is structural [E]. [E] The residuals classify into four honest kinds: forbidden-by-theorem, external-and-tensioned, external-but-parameter-free, cosmological. A classification, not a solution.



**2026-06-22 (next-steps round: the one open lemma literature-anchored, the Decoupling Theorem, the holomorphy discriminator hardened in Lean, and the  $\theta_{13}$  budget: v336, v337, v338, FORM.CARTAN.DET.01)**

- **v336 (SEAM.EQUIV.CONTINUUM.01) — the one open lemma, literature-anchored.** SEAM.EQUIV.01’s only open input (the continuum OS reconstruction of the gapped quasi-free collar) is split into a CITABLE part and a narrow open part. [C] continuum leg: Osborne–Stottmeister (*CFT from Lattice Fermions*, doi:10.1007/s00220-022-04521-8, CMP **398** (2023) 219) give, building on the MMST (Morinelli–Morsella–Stottmeister–Tanimoto) operator-algebraic (wavelet) renormalization framework (CMP 387 (2021) 299; PRL 127 (2021) 230601), the chiral-CFT scaling limit of free lattice fermions (Koo–Saleur  $\rightarrow$  Virasoro, correct  $c$ ); WZW range  $\text{rank} \leq c \leq D$ , and  $(E_8)_1$  ( $c=8$ , rank 8,  $D=16$  Majoranas,  $SO(16)_1 \rightarrow (E_8)_1$ ) is inside it. [C] OS leg: Adamo–Moriwaki–Tanimoto (arXiv:2407.18222, 2024) prove the conformal OS axioms for unitary lattice VOAs. [E] target pinned by  $c=g_{\text{car}}+N_{\text{fam}}=8$ ,  $\det \text{Cartan}(E_8)=1$ . [O] narrowed residual = “the raw collar’s massless scaling limit is the holomorphic  $(E_8)_1$  net” ( $\det K=1$ ), not a from-scratch construction.
- **v337 (DECOUPLING.THEOREM.01) — the Decoupling Theorem.** [E] every dimensionless readout factors through the gapped admissible transfer spectrum  $\{1, (2/3)^6, (1/3)^6\}$ ; finite susceptibility  $\chi = 1/(1 - (2/3)^6) = 729/665$  gap-suppresses the ambient contribution (margin  $6 \ln \frac{3}{2} - 31/(4\pi^2) \approx 1.648 > 0$ , with  $2 \dim(E_8)c_3^2 = 31/(4\pi^2)$  exactly), so no readout depends on the ambient measure  $\Rightarrow$  QG.AMB.01 decoupled; negative control (gap closed  $\Rightarrow \chi$  diverges) makes it non-vacuous. This is the theorem that makes the QG.AMB.01 reclassification (v335) honest; consolidates v76/v311/v330/v332.
- **v338 (THETA13.BUDGET.01) — the  $\theta_{13}$  budget, fenced.** The  $+2\sigma$  pressure point (v328) quantified against *both* NuFIT 6.0 NO variants:  $+2.0\sigma$  vs the no-SK central (0.02195) but  $+1.7\sigma$  vs the with-SK best fit ( $0.02215 \pm 0.00057$ ), a  $\sim 0.3\sigma$  systematic spread; [E] 0.023108 lies inside the with-SK  $3\sigma$  band (upper 0.02388), an upper-edge pressure point not a falsification; [X] JUNO’s sub-1% precision turns it into a  $\sim 6\sigma$  kill-or-confirm.
- **Lean (FORM.CARTAN.DET.01).** CartanDeterminants.lean extended with the explicit  $D_8 = SO(16)$  Cartan matrix and a fully-proven (kernel-only) holomorphy discriminator:  $D_8$  is even and symmetric just like  $E_8$  (same  $c=8$ ) but  $4 = \det \text{Cartan}(D_8)$  is genuinely not a unit while  $\det \text{Cartan}(E_8)$  is — so unimodularity (one anyon) is the load-bearing selector. lake build green.

**2026-06-22 (keystone unification + status reframe: ONE open theorem, ONE decoupled foreign problem: v335, FORM.QGEO.BW.01 qgeoSymIsCorollary)**

- **v335 (SEAM.EQUIV.UNIFY.01) — the keystone unification.** A bird’s-eye consolidation of the v308/v323/v325/v329/v333 arc that collapses the “two open structural items” narrative. [E] QGEO.SYM.01 is a COROLLARY of SEAM.EQUIV.01 (no extra premise): a chiral conformal net has, by axiom, a Möbius-covariant vacuum (the unique invariant positive-energy vector); rotations  $U(1)$  are the compact subgroup of the Möbius group, so the vacuum is rotation-invariant, and the  $\mu_4$  clock = rotation by  $\pi/2$  ( $(e^{i\pi/2})^4=1$ ) fixes it  $\Rightarrow \omega \circ \rho = \omega$ . So SEAM.EQUIV.01  $\Rightarrow$  QGEO.SYM.01 with NO extra premise (the v323 “rotation-invariant vacuum” is a net axiom) — the two bedrock items collapse to ONE. [E] **the one keystone theorem:** SEAM.EQUIV.01 = a construction theorem ( $c=g_{\text{car}}+N_{\text{fam}}=8$ ,  $\det \text{Cartan}(E_8)=1$ , gap  $6 \ln \frac{3}{2} > 0$  derived) with EXACTLY one open continuum lemma (the rigorous continuum OS reconstruction + invertibility of the raw collar). [E] QG.AMB.01 decoupled, not a gate: margin  $\Delta - 31/(4\pi^2) \approx 1.648 > 0$  (v76/v311/v330/v332)  $\Rightarrow$  no TFPT prediction depends on the ambient measure; it is the *general* Euclidean-QG conformal-factor problem (Gibbons–Hawking–Perry 1978), inherited not TFPT-specific — reclassified from “TFPT structural frontier” to “decoupled general QG problem”. [E] **collapsed status:** open structural items



$2 \rightarrow 1$ ; live residual =  $v_{\text{geo}} + G_{\text{net}} (= \text{SEAM.EQUIV.01}) + F_{\text{transfer}}$ . [O] the one residual = SEAM.EQUIV.01's continuum OS reconstruction.

- **Lean (FORM.QGEO.BW.01, qgeoSymIsCorollary).** BWKeystone.lean extended with the cited conformal-net vacuum axiom `conformal_net_vacuum_rotation_invariant`; with it, `StateInvariance (=QGEO.SYM.01)` follows from `SeamIsE8` alone, so QGEO.SYM.01 is a COROLLARY of SEAM.EQUIV.01 with no independent open premise (`#print axioms` confirms only the chain + cited BW/net axioms). lake build green (3362 jobs).
- **Status deep-sync.** The ledger (QG.AMB.01 reclassified to “decoupled general QG; non-TFPT”; QGEO.SYM.01 marked a corollary; new SEAM.EQUIV.UNIFY.01 row), the canonical status card (`tfpt_status.tex`), introduction (status core + gate table), `origin_theory`, `tfpt_4_frontier`, `tfpt_5_redteam`, `tfpt_research_contracts`, `README`, `next.txt` and the website (OpenGates, FAQ, `papers.ts`) all moved to “one solvable theorem (SEAM.EQUIV.01) + one decoupled foreign problem (QG.AMB.01)”. No falsifiable prediction changed; this is a status-framing sharpening backed by v335 + the Lean corollary.

**2026-06-22 (closing round: the CM-selection mechanism (the last keystone residual), the conformal-mode resolution, and the Conway–Sloane even-unimodular side in Lean: v333, v334, FORM.CARTAN.DET.01)**

- **v333\_cm\_selection.py (QGEO.CMSELECT.01, [E]/[O]).** The CM-selection mechanism for the last keystone residual:  $\tau=i$  (the square) is the *unique* order-4 modulus —  $S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$  has  $S^4=I$  and fixes  $i$ , and an order-4 deck selects  $j=1728$  (vs the order-3 hexagon  $j=0$ ) — AND the square is the Fekete attractor of symmetric mark-equilibration (four points minimizing the log-energy converge to equal  $\pi/2$  spacing). So given the  $\mu_4$  deck,  $\tau=i$  is forced; the residual sharpens from “why the square” to “the deck acts geometrically” (= QGEO.SYM.01). NOT a closure.
- **v334\_conformal\_resolution.py (QGAMB.CONFORMAL.01, [E]/[C]/[O]).** The conformal-mode resolution for the QG metric sector: the Gibbons–Hawking–Perry contour rotation  $\phi \rightarrow i\phi$  flips the wrong-sign kinetic term, making the divergent conformal Gaussian  $\int e^{+|c|\phi^2}$  into the convergent  $\int e^{-|c|\phi^2} = \sqrt{\pi/|c|}$ ; the entire IDG form factor  $e^{p^2/M^2}$  dresses the propagator pole-free (the polynomial  $R+R^2$  ghost as control). Together they give a *conditional* mode-by-mode construction of the metric-sector measure; the residual is only the contour’s nonperturbative validity (gap-decoupled, blocks no test). NOT a closure.
- **CartanDeterminants.lean extended (FORM.CARTAN.DET.01, [E]).** The Conway–Sloane lattice side: the  $E_8$  Cartan matrix is proved to be an EVEN (diagonal = 2), symmetric, unimodular ( $|\det|=1$ ) integer Gram matrix — the defining data of an even unimodular rank-8 lattice — all by kernel `decide` / the integer-inverse argument. The remaining content (Witt’s uniqueness:  $E_8$  is the only even unimodular lattice in  $\dim < 16$ ) is the cited classification residual.

**2026-06-22 (endgame round: the necessity of  $H$  reduced to the order-4 CM selection, the QG metric sector characterized as the conformal-mode problem,  $|\det E_8|=1$  and the  $\mu_4$  commutation grounded in Lean: v331, v332, FORM.MU4COMM.01)**

- **v331\_necessity\_of\_H.py (QGEO.NECESS.01, [E]/[O]).** The necessity companion to v329: a concrete Steklov (DtN) computation shows the seam  $\Lambda=|k|+M_f$  commutes with the  $\mu_4$  clock *iff* the four marks sit at the square ( $\tau=i$ ) — moving a mark off the square breaks  $\mathbb{Z}_4$  and  $[\rho, \Lambda] \neq 0$ . The square has cross-ratio 2 ( $j=1728$ ), the unique 4-config with an order-4 automorphism (v214/v267), so  $H \Leftrightarrow \text{square} \Leftrightarrow \mu_4$  and “necessity of  $H$ ” sharpens to: the raw

dynamics places the four Gauss–Bonnet marks (v216) at the order-4 CM point. Residual = that dynamical selection.

- **v332\_qgamb\_metric\_sector.py** (QGAMB.METRIC.01, [E]/[C]/[O]). The open metric sector of QG.AMB.01 is characterized precisely as the Gibbons–Hawking–Perry conformal-factor problem: the conformal mode has a negative kinetic coefficient  $-(D-1)(D-2)/4$  ( $= -\frac{3}{2}$  at  $D=4$ ), so the bare Euclidean measure diverges over it. The obstruction is in the conformal/gauge direction, gap-insulated from the admissible sector (v330/v76); the only route is the conditional IDG taming (v304), with the polynomial  $R+R^2$  ghost as the control. Open, but pinned to one named obstruction.
- **CartanDeterminants.lean extended** (FORM.CARTAN.DET.01, [E]). The rank-8 holomorphy discriminator  $|\det \text{Cartan}(E_8)|=1$  ( $E_8$  has one anyon, vs  $D_8$ ’s four) is now proved principally and kernel-only: `cartanE8 * cartanE8inv = 1` by `decide` on the  $8 \times 8$  PRODUCT gives `IsUnit (det E8)` via  $\det E_8 \cdot \det E_8^{-1} = \det 1 = 1$  — no 8! determinant, no `native_decide` (Mathlib itself leaves `E8_det = 1` a `proof_wanted`).
- **Mu4Commutation.lean** (FORM.MU4COMM.01, Lean 4, [E]). The v201/v323 mark-locality criterion as exact linear algebra over the Gaussian integers  $\mathbb{Z}[i]$ :  $i^4=1$ ,  $i^2=-1$ , and (on concrete  $6 \times 6$  matrices) an offset-4 coupling commutes with the  $\mu_4$  clock while an offset-2 coupling does not — all by kernel `decide`. Built lake build on Lean v4.29.1 + Mathlib.

**2026-06-22 (frontier round: the OS step discharged at the many-body level, the ambient measure built on the gap-decoupled sector, and the carrier Cartan determinants grounded in Lean: v329, v330, FORM.CARTAN.DET.01)**

- **v329\_os\_gap\_reduction.py** (QGE0.OSGAP.01, [E]/[O]). The one analytic input the v308 keystone chain still cited — “the OS-reconstructed bulk gap = the transfer gap” — is discharged at the many-body level: the OS/Euclidean Hamiltonian is the second quantization  $d\Gamma(-\log T)$ , whose spectrum is the set of sums of single-mode energies, so the bulk gap = the smallest positive single-mode energy  $= 6 \ln \frac{3}{2}$ , *independent of system size*. Together with  $H \Rightarrow \omega \circ \rho = \omega$  and  $H \Rightarrow \text{SRE} \Rightarrow (E_8)_1$ , this exhibits the rotation-invariant flat-pillowcase premise  $H$  as the *single sufficient premise* of both bedrock IDs; residual = the necessity of  $H$  on the raw collar.
- **v330\_qgamb\_admissible\_measure.py** (QGAMB.MEASURE.01, [E]/[O]). The constructive step v275’s roadmap calls for: on the gap-decoupled *admissible* sector the ambient measure is built as a bona fide OS object — reflection-positive (symmetric PSD transfer), exponentially clustering ( $C(n)/C(n-1) \rightarrow (2/3)^6$ ), with finite susceptibility  $\chi = 729/665$  so  $\text{Var}(S_\Lambda)/\Lambda \rightarrow \chi$  converges  $\Rightarrow$  uniform tightness (Prokhorov)  $\Rightarrow$  the projective limit exists; OS reconstruction gives  $H_{\text{OS}} \geq 0$  with gap  $6 \ln \frac{3}{2}$ . The full  $\text{QG}_{\text{amb}}$  (the metric sector) stays open, gap-decoupled — a genuine *partial* construction, not a closure.
- **CartanDeterminants.lean** (FORM.CARTAN.DET.01, Lean 4, [E]). The carrier piece of the holomorphy/anyon discriminator is grounded in real `Matrix.det` computations (kernel `decide`, no extra axiom):  $|\det \text{Cartan}(A_3)|=4=|\mu_4|$  ( $3 \times 3$ , `det_fin_three`) and  $|\det \text{Cartan}(D_5)|=4$  ( $5 \times 5$ , `decide`), giving the carrier  $|\det \text{Gram}|=4 \cdot 4=16$  (the v281 anyon count). The rank-8  $E_8/D_8$  case stays the Nat discriminator — Mathlib’s own `Data.Matrix.Cartan` leaves `E8_det = 1` as a `proof_wanted`. Built lake build on Lean v4.29.1 + Mathlib.

**2026-06-22 (next-steps round: the keystone collapsed to one shared premise (Lean + one citable theorem), the  $F_{\text{transfer}}$  solver suite, the cusp weight derived from a minimal rewrite, and the  $\theta_{13}$  pressure point: FORM.QGEO.BW.01, v325, v326, v327, v328)**

- **BWKeystone.lean** (FORM.QGEO.BW.01, Lean 4, [E]/[O]). The v323 Bisognano–Wichmann linkage formalised: the nodes `RotationInvariantVacuum`  $\rightarrow$  `GeometricModularFlow`  $\rightarrow$  `Mu4ModularSymmetry`  $\rightarrow$  `StateInvariance` (=QGEO.SYM.01) composed over the v308 chain. The theorem `bedrockReducesToRotationInvariance` shows the WHOLE bedrock QGEO.SYM.01 reduces to the SINGLE shared open premise `RotationInvariantVacuum` — the two open bedrock items collapse to one — with `#print axioms` confirming the residual = the cited chain steps + recovery gap + the named BW steps + that one premise. Plus a `decide`-proven discriminator  $(1 \mid 4) \wedge \neg(4 \mid 1)$  (BW geometric flow strictly stronger than v309 clock invariance). Built lake build on Lean v4.29.1 + Mathlib.
- **v325\_pillowcase\_keystone.py** (QGEO.KEYSTONE.01, [E]/[O]). The keystone as ONE citable theorem: IF the raw seam state is the flat  $\tau=i$  pillowcase (rotation-invariant) state THEN the whole bedrock closes (4 square marks,  $\mu_4$  deck,  $\omega \circ \rho = \omega$ , holomorphic  $c=8 \Rightarrow (E_8)_1$ , with the recovery gap as input); all pieces machine-checked (incl. the pillowcase  $\chi_{\text{orb}}=0$ ), the single open lemma shared by both bedrock IDs and Lean-pinned. An assembly certificate, not a closure.
- **v326\_ftransfer\_suite.py** (FR.SUITE.01, [E]/[C]). The four  $F_{\text{transfer}}$  readouts productised into one runnable harness, each a gapped relaxation to a unique attractor:  $F_{\text{pole}}$  (Koide, seam rate  $(2/3)^6$ ),  $F_{\text{Boltzmann}}$  ( $\eta_B$ , H-theorem),  $F_{\text{relic}}$  (axion adiabatic invariant),  $F_{\text{QCD}}$  ( $m_p/m_e$ , RG to the UV fixed point), plus the  $F_{\text{RareKaon}}$  bridge. Only  $F_{\text{pole}}$  carries the seam rate; the four external rates are honestly fenced (v187/v303).
- **v327\_hypergraph\_rewrite.py** (HYP.REWRITE.01, [E]/[O]). The cusp weight  $2/3$  is DERIVED from a minimal rewrite: a 3-channel/1-absorbing family rule has spectrum  $\{1, 2/3, 0\}$ , so  $2/3 = (N_{\text{fam}} - 1)/N_{\text{fam}} = |\mathbb{Z}_2|/N_{\text{fam}}$  emerges from the rule arity and  $(2/3)^6 = 64/729$  is the recovery gap; with a PROOF that  $2/3$  is not a root of the affine- $E_8$  charpoly ( $p(2/3) = -3520/19683 \neq 0$ , sharpening v312). v324's injected datum is reduced to the anchor arity  $\{2, 3\}$ .
- **v328\_theta13\_pressure.py** (FLAV.TH13.PRESSURE.01, [C]/[X]) + **v307 hardening**. The honest weak spot named and quantified:  $\sin^2 \theta_{13} = \varphi_0^{\text{ret}} e^{-5/6} = 0.023108$  is  $+2.0\sigma$  above NuFIT 6.0 NO, a SEPARATE carrier-trace channel (v268), not relieved by any freeze-respecting correction (RG < 1%, exponent  $5/6$  exact), pre-registered as the kill. The v307 watchdog gains a data provenance/date check and the rare-kaon  $F_{\text{transfer}}$  bridge row.
- **Wolfram mirror** 285  $\rightarrow$  288. v325 ( $\chi_{\text{orb}}=0$ ) and v327 (rewrite spectrum  $\{0, 2/3, 1\}$ ,  $(2/3)^6 = 64/729$ , and the non-graph-spectral proof) added to `tfpt_readouts_extension.wl`; GATE.WOLFRAM.02 and the Wolfram README updated to 288/288.

**2026-06-22 (consistency + de-accretion editorial pass: current-state alignment across papers, website and ledger; no new modules)**

- **Keystone naming reconciled to SEAM.EQUIV.01**. Following v323/QGEO.BW.01 (the deck premise is downstream of the seam being a chiral net), the single named open keystone is now SEAM.EQUIV.01 *everywhere*, with QGEO.SYM.01 stated as its *downstream* conformal-deck face. The ledger row QGEO.SYM.01 was updated (it no longer calls itself “the single structural residual of the whole theory”; it now depends on SEAM.EQUIV.01/QGEO.BW.01), and the superseded “GATE.QGEO open gate” / “two research gates ( $U_{\text{wall}}$ ) and  $G_{\text{metric}}$ ” framing was replaced by the current  $\text{Rest} = v_{\text{geo}} \oplus G_{\text{net}} \oplus F_{\text{transfer}}$  with the two structural

items SEAM.EQUIV.01 + QG.AMB.01 (intro status card, tfpt\_4\_frontier, origin\_theory, tfpt\_5\_redteam, tfpt\_research\_contracts, and the website).

- **De-accretion: the sprint-by-sprint reduction history was moved to this changelog, where it belongs.** Several reader-facing sections had grown by accretion (each round appended a clause). They were rewritten to state the *current* result once, with the round-by-round narrative left here: the introduction closure ledger (“honest bottom line”) and hard-reduction subsection; the `origin_theory` “residual inventory” (the five “Update/.../Terminal” passes collapsed to one current inventory); the `tfpt_5_redteam` Target-A Outcome; and on the website the FAQ “What is still open?”, the `OpenGates` closing-condition block, the `papers.ts` research-contracts/red-team entries, the verification-DAG `qft` node, the glossary `G_net` entry, the Rosetta lexicon, and the verification/review pages. No scientific content or `\veri` citation was dropped; only the embedded version-stamp logs were removed.
- **Stale facts corrected.** Version label 5.0→v5.2 (intro, single-sourced via `\TFPTversion`); Lean build 1886→3357 jobs (`FORM.SEAMEQUIV.01`); README Python suite v213/212→v324/322 modules; Wolfram extension 269/269 / 249/249→285/285 (README, website replication page, `zenodo_description.html`); residual “Paper-3” old-paper attributions in `tfpt_2` neutralised.

(Original 2026-06-22 entry follows.)

**2026-06-22 (external-review pass: a registry bug fix, the NA62 2026 rare-kaon update, CP-phase prose hygiene, and three firewall/visualisation enhancements: v202, v203, v305, v307, v321)**

- **Bug fix —  $\delta_{\text{CKM}}$  display value.** The predictions table (`tex-artefacts/predictions.tex`) showed  $\delta = \frac{\pi}{3} + 3\lambda^2 = 66.96^\circ$ , but that formula evaluates to  $68.65^\circ$  (the canonical frozen `DELTA_CKM_RAD` = 1.198232 rad, used by v88/v202/FLAV.CP.01 everywhere else). Corrected to  $68.65^\circ$ . An external review correctly flagged the inconsistency; it was an isolated display typo (the website was already clean).
- **v202\_rare\_kaon.py — NA62 2016–2024 update ([E]/[C]).** The charged-mode comparison is moved from the historical NA62 2016–2022 observation  $(13.0^{+3.3}_{-3.0}) \times 10^{-11}$  ( $\sim 1.2\sigma$  above) to the full 2016–2024 combination  $\text{BR}(K^+) = (9.6^{+1.9}_{-1.8}) \times 10^{-11}$  (La Thuile 2026), on which the prediction  $9.45 \times 10^{-11}$  lands at  $+0.08\sigma$ . Framed honestly: a strong *consistency* hit for the closed CKM point, but an  $F_{\text{transfer}}$  bridge — *not* a unique TFPT-vs-SM discriminator (the SM  $(8.6 \pm 0.4) \times 10^{-11}$  is also inside the NA62 error). Propagated to the ledger (`FR.RAREKAON.01`), `freeze_file` (new `rare_kaon` kill row), the watchdog (v307, a new  $F_{\text{transfer}}$  bridge row), the forward board (v321, rank 9), and `tfpt_2_standard_model` + the website.
- **CP-phase prose hygiene.** The Galois CP lock is tightened from “a relation to the *measured* quark phase” to “a relation to the *leading* ( $\pi/3$ ) component of the quark phase”: the lock is to the structural  $\pi/3$ , not the full measured  $\gamma = \delta_{\text{CKM}}^{\text{lead}} + 3\lambda_C^2 \approx 68.7^\circ$ , so  $\delta_{\text{PMNS}} = 240^\circ$  (not  $248.7^\circ$ ). Fixed in `tfpt_2_standard_model`, `origin_theory`, `papers.ts` and `predictions.ts`.
- **v305\_witness\_independence.py — the No-Golden-Dynamics rule ([E]).** The number-field split is made an explicit anti-numerology firewall rule:  $\phi = 2 \cos(\pi/5) \in \mathbb{Q}(\sqrt{5})$  is the *static* carrier ( $\mathbb{Z}/5$ ) field, every *dynamic* rate is rational ( $\mathbb{Q}$ , the  $\mathbb{Z}/6$  hand), so a dynamic rate explained by  $\phi$  is a cross-field false friend (v314/v315).
- **v203\_eht\_achromatic.py — the  $\mu_4$  Fourier fourth null ([X]).** The EHT residual pre-registration gains a fourth null: the azimuthal residual must carry power *only* at  $m \equiv 0 \pmod{4}$  (the  $\mu_4$  fingerprint of the four marks, v201); a  $\mu_4$ -orbit profile has  $\sim 0$  off-mode power while a  $\mathbb{Z}_2$  control leaks at  $m = 2$ .



- **Seed-hyperplane figure (Fig. ??).** A new figure visualises v306: every scorecard observable inverts *independently* to the one latent seed  $\varphi_0$ , the error-weighted mean matching the axiom to 0.04%, with  $\sin^2 \theta_{13}$  the  $\sim 2\sigma$  pressure point. Added to `make_figures.py`, `tfpt_5_redteam` and the manifest.

## 2026-06-22 (Wolfram mirror of the arithmetic arc + three sharpening steps: the CP-lock made quantitative, the Bisognano–Wichmann keystone, the minimal hypergraph fibre: v322, v323, v324)

- **Wolfram mirror of the cyclotomic/Galois arithmetic arc (the independent second path).** The exact algebraic results of the arc are now mirrored in `verification/wolfram/tfpt_readouts_extension.wl` (ten new `checkExact` blocks,  $275 \rightarrow 285$ , *all passing*): v313 (the affine- $E_8$  characteristic polynomial with the golden factor  $x^2-x-1$ ; the  $(2, 3, 5)$  atoms as  $2 \cos(\pi/k)$ ; the 31/30 selection), v314 (the rational kernel vs  $\mathbb{Q}(\sqrt{5})$  dynamic-rate split), v315 ( $30=5 \cdot 6$ ,  $\varphi(30)=8$ , the Galois group  $\mu_4 \times \mathbb{Z}_2$ , the  $\sqrt{5}$  Gauss sum), v316 ( $\rho=\zeta_6=\zeta_{30}^5$ ,  $\rho^4=-\rho$ ,  $\text{Gal}=\mathbb{Z}_2=\text{CP conjugation}$ ), v317 ( $N_{\text{fam}}=3$  as the  $\mu_3$  orbit, the  $1+2$  Galois split), v318 ( $\varphi_0^{\text{ret}}=(4/3)c_3+48c_3^4$  a pure function of  $\pi$ ) and v320 (the CP lock  $\delta_{\text{PMNS}}=240^\circ$ ). Ledger `GATE.WOLFRAM.02` and the Wolfram README updated to 285/285.
- **v322\_cp\_lock\_sharpened.py (GALOIS.CP.SHARPEN.01, [E]/[C]/[X]).** The v320 CP-lock sharpened into a quantitative, pre-registered kill test. [E] the leading phase sits on the six hexagonal nodes  $\arg \rho^k = k \cdot 60^\circ$ , and *which* one is selected is the deck order:  $\delta_{\text{PMNS}}^{\text{lead}} = |\mu_4| \delta_{\text{CKM}}^{\text{lead}} = 4(\pi/3)$  (node 4); [C] the sub-leading correction is bounded by the quark analogue  $3\lambda^2 \approx 8.7^\circ$ , so the prediction is the band  $240^\circ \pm \sim 9^\circ$ ; [C] against NuFIT 6.0 (NO,  $\delta_{\text{CP}}=212^{+26}_{-41}^\circ$ ) the leading value is  $+1.08\sigma$ , the band edge  $+0.74\sigma$  – compatible today; [X] the nearest wrong node is  $60^\circ$  away, far beyond the budget, so DUNE/Hyper-K ( $\sim 5\text{--}15^\circ$ ) discriminate cleanly. Cited in `origin_theory`; ledger `GALOIS.CP.SHARPEN.01`.
- **v323\_bw\_geometric\_modular.py (QGEO.BW.01, [E]/[C]/[O]).** The keystone pushed one honest step via Bisognano–Wichmann. [E] the  $\mu_4$  clock is a geometric rotation  $\rho = \text{diag}(i^n) = \exp(i\frac{\pi}{2}L)$ ,  $\mu_4 \subset U(1)_{\text{rot}}$ ; [E] for a rotation-covariant covariance  $C=f(L)$  the modular Hamiltonian  $K = \log((1-C)/C) = g(L)$ , so  $[K, L]=0$  (the modular flow is geometric) and the clock is a modular symmetry *for free* (the discriminator: a period-4 curvature preserves the clock but its flow is not geometric,  $[K, L] \neq 0$ ); [C] given the seam is the  $(E_8)_1$  chiral net (v308), BW/Hislop–Longo make the vacuum modular flow geometric, so `QGEO.SYM.01` ( $\omega \circ \rho = \omega$ ) is *downstream* of `SEAM.EQUIV.01` + a rotation-invariant vacuum, not an independent axiom; [O] residual: the raw collar vacuum is rotation-invariant (cleaner than v309, shared with v308). Cited in `origin_theory`; ledger `QGEO.BW.01`.
- **v324\_hypergraph\_fiber.py (HYP.FIBER.01, [E]/[C]).** The constructive v312 follow-up: the *minimal* substrate that carries both the  $E_8$  skeleton and the recovery gap is a *product*. [E] the carrier network  $T_{\text{net}} = (A+2I)/4$  (attractor = Kac marks = skeleton) fibred by the 3-node family cusp  $T_{\text{cusp}} = \text{diag}((1-w)^{2N_{\text{fam}}})$ ,  $w \in \{0, \frac{1}{3}, \frac{2}{3}\}$  (spectrum  $\{1, (2/3)^6, (1/3)^6\}$ ); the fibred  $T_{\text{net}} \otimes T_{\text{cusp}}$  has top eigenvalue 1 (eigenvector marks  $\otimes (w=0)$ ) and  $(2/3)^6$  as a genuine eigenvalue (marks  $\otimes (w=\frac{1}{3})$ ) – the  $N_{\text{fam}}=3$  cusp is the minimal fibre and the substrate = carrier  $\times$  family = the  $30 = g_{\text{car}}(2N_{\text{fam}}) = 5 \times 6$  of v315; [C] honestly (v312) the cusp weight is still *injected*, not derived from the graph – a consistent realisation, not a derivation of  $2/3$  from a pure rewrite. Cited in `origin_theory`; ledger `HYP.FIBER.01`.

## 2026-06-22 (the forward kill-test board: the decisive upcoming measurements ranked, each locked to a freeze row — incl. the new Galois-CP relation: v321)

- **v321\_killtest\_board.py (KILL.BOARD.01, [E]).** The forward companion to v307 (the current-data board): the decisive *upcoming* measurements, ranked and each locked to a `freeze_file.csv` kill row – #1 JUNO  $\sin^2 \theta_{12}=0.306747$  (fastest), #2 CMB-S4/LiteBIRD  $r$



(kill  $r > 0.01$ ), #3 DESI/Euclid  $w$  (kill  $w \neq -1$ ), #4 DUNE/Hyper-K  $\delta_{\text{PMNS}} = 240^\circ$  (the new v320 Galois-CP relation), #5 DUNE/LEGEND/nEXO ordering +  $m_{\beta\beta}$ , #6 PSI n2EDM, #7 JUNO  $\sin^2 \theta_{13}$  ( $\sim 2\sigma$  now), #8 LiteBIRD  $\beta$ . A forward decision ledger (publicly-stated experimental reach), complementing v307; no new physics. Cited in `origin_theory`; ledger row KILL.BOARD.01.

## 2026-06-22 (Lean: the SEAM.EQUIV.01 composition chain formalised — the keystone residual machine-pinned to the named cited steps: FORM.SEAMEQUIV.01)

- **SeamEquivChain.lean** (FORM.SEAMEQUIV.01, Lean 4, [E]/[O]). A Lean 4 mirror of the v308 closing chain: the four nodes  $\text{GapPositive} \rightarrow \text{InvertibleSRE} \rightarrow \text{HolomorphicC8} \rightarrow \text{SeamIsE8}$  assembled as a well-typed composition theorem `seamEquivChain`, with the three cited literature steps (Kitaev free-fermion invertibility; Mger / Kawahigashi–Longo–Mger holomorphy; Conway–Sloane  $E_8$  uniqueness) as named *axioms* and the carrier recovery gap (the OS step) as the single open input. [E] `#print axioms seamEquivChain` confirms the residual is exactly those named steps + `recovery_gap` (no hidden assumption); [E] a decide-proven discriminator `primariesE8 = 1  $\neq$  primariesD8 = 4` (the  $|\det \text{Cartan}|$  that separates the holomorphic  $E_8$  from the same- $c$  rival  $D_8 = SO(16)$ ). Built end-to-end (`lake build`, 3357 jobs) on Lean v4.29.1 + Mathlib. [O] The value is the machine-checked composition + axiom audit (pinning the keystone residual), *not* a from-scratch proof — it does NOT close SEAM.EQUIV.01. Ledger row FORM.SEAMEQUIV.01; mirrors v308.

## 2026-06-22 (a NEW falsifiable prediction from the Galois structure: the two CP phases are locked, $\delta_{\text{PMNS}} = \delta_{\text{CKM}}^{\text{lead}} + \pi$ : v320)

- **v320\_galois\_cp\_relation.py** (GALOIS.CP.PREDICT.01, [E]/[C]/[X]). The v316 follow-up turned into a testable cross-prediction. [E] both leading CP phases are powers of the *one* hexagonal unit  $\rho = \zeta_6$  of the family Galois factor:  $\delta_{\text{CKM}}^{\text{lead}} = \arg \rho = \pi/3$  ( $60^\circ$ ),  $\delta_{\text{PMNS}} = \arg \rho^4 = 4\pi/3$  ( $240^\circ$ ), and  $\rho^4 = -\rho$  locks them as  $\delta_{\text{PMNS}} = \delta_{\text{CKM}}^{\text{lead}} + \pi$  (the  $\mathbb{Z}_2$  sheet flip); [C] this upgrades the previously assigned  $\delta_{\text{PMNS}} = 240^\circ$  to a Galois-forced relation to the measured quark phase (the quark and lepton leading CP phases are not independent); [X] kill test: a  $\delta_{\text{PMNS}}$  robustly incompatible with  $240^\circ$  ( $> 3\sigma$  at DUNE/Hyper-K/JUNO) falsifies the Galois-CP organisation;  $240^\circ$  is currently within the broad NuFIT 6.0 range. A genuine new falsifiable consequence of the v313–v319 arithmetic arc (also a `freeze_file.csv` `cp_phase_relation` kill row). Cited in `origin_theory`; ledger row GALOIS.CP.PREDICT.01.

## 2026-06-22 (the translation clock: the discrete $\leftrightarrow$ dynamic bridge *is* the order-30 Coxeter element, a clock with two coprime hands $5 \times 6$ , read law-inclusive (0..5) or live-only (1..5): v319)

- **v319\_translation\_clock.py** (TRANSLATE.CLOCK.01, [E]/[C]). A synthesis of the v313–v318 arc with v124/v55, answering the question “is the static $\leftrightarrow$ dynamic translation a clock running 0, 1, 2, 3, 4, 5 or 1, 2, 3, 4, 5?”. [E] the only object holding both the static carrier ( $\mathbb{Q}(\sqrt{5})$ ) and the dynamic family (rational) data is the order-30 Coxeter element (v314);  $30 = h(E_8) = g_{\text{car}}(2N_{\text{fam}}) = 5 \cdot 6$  and  $\gcd(5, 6) = 1 \Rightarrow \mathbb{Z}/30 = \mathbb{Z}/5 \times \mathbb{Z}/6$  (two coprime hands, v315). [E] the *dynamic* hand is  $\mathbb{Z}/6 = 2N_{\text{fam}}$  (ticks 0, ..., 5): the recovery exponent is exactly  $6 = 2N_{\text{fam}}$ , rate  $(2/3)^6 = (|\mathbb{Z}_2|/N_{\text{fam}})^{2N_{\text{fam}}}$ , order-6 generator  $c^5$  ( $\zeta_6 = \zeta_{30}^5$ , v316). [E] position 0 is the *conserved law* — the resummed clock rate( $n$ ) =  $-p_2 \ln(1 - n/N_{\text{fam}})$  (v124) has rate(0) = 0 (the attractor, eigenvalue 1, v56), the live decaying modes from  $n=1$ , and the  $E_8$  live phases (totatives of 30) start at 1 ( $\{1, 7, \dots, 29\}$ ), the 0 phase not a totative — so “0, 1, 2, 3, 4, 5” is law-inclusive, “1, 2, 3, 4, 5” live-only. [E] the *static* hand is  $\mathbb{Z}/5 = g_{\text{car}}$  (ticks 1, ..., 5): the golden field  $\mathbb{Q}(\sqrt{5})$  ( $\varphi = 2 \cos(\pi/5)$ ), field-obstructed (no rational dynamic image, v314) — static-only;  $\zeta_5 = \zeta_{30}^6$  carries no phase. [E] one full turn

$= \text{lcm}(5, 6) = 30 = h(E_8)$ , rank  $E_8 = 8 = \varphi(30)$  live phases pairing into  $|\mu_4| = 4$  invariant planes (v55). [C] verdict: the discrete→dynamic translation *is* the order-30 clock = (static 5-ring)  $\times$  (dynamic 6-ring), the dynamic hand carrying the rate (exponent  $6 = 2N_{\text{fam}}$ ), the static hand the golden carrier (no rate), and “0 vs 1” = law-inclusive vs live-only; the arithmetic is [E], the “it is one clock” reading [C]. Python-only (sympy), like the rest of the arc. New figure [figures/translation\\_clock.pdf](#) (concentric 5- and 6-hands). Cited in [origin\\_theory](#); ledger row TRANSLATE.CLOCK.01.

**2026-06-21 (capstone: the SM structural sector is  $\mathbb{Q}(\zeta_{30})$ + Galois  $\mu_4 \times \mathbb{Z}_2$ , and the complete input is  $\{a, \pi, v_{\text{geo}}\}$  — zero dimensionless free parameters: v318)**

- **v318\_arithmetic\_capstone.py** (ARITH.CAPSTONE.01, [E]/[O]). The synthesis of the v313–v317 arc. [E] the SM *structural* sector (the 3 generations, the 2 CP phases, their orbit/hierarchy) is the cyclotomic field  $\mathbb{Q}(\zeta_{30})$  with Galois group  $\mu_4 \times \mathbb{Z}_2$  (degree  $\varphi(30) = 8 = \text{rank } E_8$ ,  $30 = |\mathbb{Z}_2|N_{\text{fam}}g_{\text{car}} = h(E_8)$ ), forced by the anchor atoms  $\{2, 3, 5\} = e(a)$ ; [E] the magnitude seed  $\varphi_0 = (|\mu_4|/N_{\text{fam}})c_3 + \Omega_{\text{adm}}c_3^4 = \frac{4}{3}c_3 + 48c_3^4$  ( $c_3 = 1/(8\pi)$ ) is a pure function of  $\pi$  (no free parameters, anchor-derived coefficients), so the magnitudes ( $\sin^2 \theta_{12} = \frac{1}{3} - \varphi_0/2$ ) are  $\{a, \pi\}$ -determined, not free; [E] free inventory: *zero* dimensionless free parameters,  $\pi$  the unique transcendental primitive (ARCH.CORE.01),  $v_{\text{geo}}$  the unique dimensionful input (No-Unit Theorem) — complete input  $\{a, \pi, v_{\text{geo}}\}$ . [O] honest residual: this rigidity holds *given* the axioms and does not realise the structure on the raw seam (QGEO.SYM.01/SEAM.EQUIV.01 stay [O]). Cited in [origin\\_theory](#); ledger row ARITH.CAPSTONE.01.

**2026-06-21 (are the 3 generations a  $\mu_3$ /Galois orbit? Yes ( $\mu_3$ ), Galois-refined 1+2 with the fixed generation = the recovery attractor: v317)**

- **v317\_galois\_family.py** (GALOIS.FAMILY.01, [E]/[C]). The v316 follow-up to the generation structure. [E] the three cusp weights  $\{0, \frac{1}{3}, \frac{2}{3}\}$  (with  $\text{Spec}(Q_+) = \{1, 2, 3\}$ , the  $A_3$  exponents) are the cube roots of unity  $\{1, \zeta_3, \zeta_3^2\}$  under  $w \mapsto e^{2\pi i w}$ , and the cyclic family symmetry  $w \mapsto w + \frac{1}{3}$  is a *transitive*  $\mu_3$  orbit (the family IS the cube-root orbit); [E] the transfer eigenvalues  $(1-w)^6 = \{1, (2/3)^6, (1/3)^6\}$  attach to the weights (attractor = eigenvalue 1 at  $w=0$ ); [E] the family Galois  $\mathbb{Z}_2$  ( $\zeta_3 \mapsto \zeta_3^2 = \bar{\zeta}_3$ , = the v316 CP conjugation) refines the orbit to 1+2 — fixing  $w=0$  (rational) and swapping  $w=\frac{1}{3} \leftrightarrow \frac{2}{3}$ ; [E] so the Galois-FIXED generation is the recovery ATTRACTOR (the law) and the Galois-conjugate pair the decaying hierarchy. [C] verdict: the family STRUCTURE is the same arithmetic as the CP phase (a  $\mu_3$  orbit, Galois-refined 1+2), the mass MAGNITUDES remaining the analytic seed. Cited in [origin\\_theory](#); ledger row GALOIS.FAMILY.01.

**2026-06-21 (does the Galois  $\mu_4 \times \mathbb{Z}_2$  organize the readouts? Yes — the CP phases are the family-factor cyclotomic data, Galois  $\mathbb{Z}_2 = \text{CP conjugation}$ : v316)**

- **v316\_galois\_readout.py** (GALOIS.READOUT.01, [E]/[C]). The v315 follow-up to phenomenology: does the Galois coupling  $\mu_4 \times \mathbb{Z}_2$  reach the physics? [E] it does, through the *family* factor — the hexagonal CP unit  $\rho = \zeta_6 = \zeta_{30}^5$  is exactly the order-6 =  $2N_{\text{fam}}$  factor  $c^5$  of v315 (the carrier being  $\zeta_5 = \zeta_{30}^6$ ); [C] the leading CP phases are its cyclotomic arguments  $\delta_{\text{CKM}}^{\text{lead}} = \pi/3 = \arg \zeta_6$  and  $\delta_{\text{PMNS}} = 4\pi/3 = \arg \zeta_6^4$  (power 4 =  $|\mu_4|$ ), with  $\zeta_6^4 = -\zeta_6$  the  $\mathbb{Z}_2$  sheet flip and  $\mu_6 = \mu_3 \times \mu_2$  (inherits v231/v233); [E] the family Galois  $\text{Gal}(\mathbb{Q}(\zeta_6)/\mathbb{Q}) = \mathbb{Z}_2$  ( $\zeta_6 \mapsto \zeta_6^5 = \bar{\zeta}_6$ ) *is* CP conjugation  $\delta \mapsto -\delta$ ; [E] the carrier factor  $\zeta_5$  (golden  $\sqrt{5}$ ) carries no phase ( $\mu_4$  static), and the magnitudes ( $\sin^2 \theta_{12} = \frac{1}{3} - \varphi_0/2$ , transcendental  $\varphi_0$ ) are not cyclotomic. [C] verdict: the Galois bridge reaches phenomenology in the phase sector predicted by the v314 field split. Cited in [origin\\_theory](#); ledger row GALOIS.READOUT.01.

## 2026-06-21 (couple or factorize? the order-30 Coxeter element couples the carrier and family sectors as a cyclotomic compositum, Galois = $\mu_4 \times \mathbb{Z}_2$ : v315)

- **v315\_coxeter\_coupling.py** (COX.COUPLE.01, [E]/[C]). Answers the v314 question — does  $30 = g_{\text{car}}(2N_{\text{fam}}) = 5 \cdot 6$  couple the carrier (5-fold, golden,  $\mathbb{Q}(\sqrt{5})$ ) and family ( $6 = 2 \cdot 3$ , rational) sectors, or just factorize? [E] the cyclic group *factorizes*:  $\gcd(5, 6) = 1 \Rightarrow \mathbb{Z}/30 = \mathbb{Z}/5 \times \mathbb{Z}/6$  (direct product, group-decoupled, in line with the v314 field obstruction); [E] the  $E_8$  exponents = totatives(30) split as totatives(5)  $\times$  totatives(6),  $\varphi(30) = 4 \cdot 2 = 8 = \text{rank } E_8$ ; [E] but the *field* couples:  $\mathbb{Q}(\zeta_{30})$  is the compositum of the carrier  $\mathbb{Q}(\zeta_5) \ni \sqrt{5}$  and the family  $\mathbb{Q}(\zeta_3)$ , degree  $8 = \text{rank } E_8$ ; [E] the Galois group is exactly  $(\mathbb{Z}/5)^\times \times (\mathbb{Z}/3)^\times = \mathbb{Z}/4 \times \mathbb{Z}/2 = \mu_4 \times \mathbb{Z}_2$  (order 8, max element order 4), so the  $\mu_4$  seam clock *is* the Galois group of the carrier's golden field and  $\mathbb{Z}_2$  that of the family field (refining v223); [E]  $\sqrt{5} = \phi + 1/\phi$  is the quadratic Gauss sum in  $\mathbb{Q}(\zeta_5)$ . [C] verdict:  $30 = 5 \cdot 6$  does not dynamically entangle the sectors — it couples them as the cyclotomic compositum, the static $\leftrightarrow$ dynamic bridge being *Galois*, not a dynamical map. Cited in *origin\_theory*; ledger row COX.COUPLE.01.

## 2026-06-21 (do the discrete-kernel and dynamic rates translate? An exact number-field split $\mathbb{Q}$ vs $\mathbb{Q}(\sqrt{5})$ : v314)

- **v314\_rate\_translation.py** (RATE.TRANSLATE.01, [E]/[C]). Answers, exactly, whether the discrete kernel (the affine- $E_8$  network, carrying the golden ratio) and the dynamic part (the seam transfer, carrying the recovery rate  $(2/3)^6$ ) translate. [E] FIELD SPLIT: the adjacency spectrum is the rational part  $\{0, \pm 1, \pm 2\} \subset \mathbb{Q}$  plus the golden part  $\{\pm \phi, \pm 1/\phi\} \subset \mathbb{Q}(\sqrt{5})$ , while the transfer spectrum  $\{1, (2/3)^6, (1/3)^6\}$  is *entirely* rational; [E] the family/sheet rates DO translate (the rational angles  $2 \cos(\pi/2)=0 = |\mathbb{Z}_2|$  and  $2 \cos(\pi/3)=1 = N_{\text{fam}}$  build  $(2/3)^6 = (|\mathbb{Z}_2|/N_{\text{fam}})^{2N_{\text{fam}}} \in \mathbb{Q}$ , same field, same atoms), but [E] the carrier 5-fold (golden,  $\mathbb{Q}(\sqrt{5})$ ) has NO rational dynamic image – a field obstruction ( $2/3$  is neither an adjacency nor a lazy-update eigenvalue; heat-kernel  $t = \Delta/(2 - \phi) \approx 6.37$  not clean), so the carrier is *static-only*; [E] the only object holding both is the order-30 Coxeter element  $30 = g_{\text{car}}(2N_{\text{fam}}) = 5 \cdot 6$ , and the  $\phi$ -quasicrystal  $E_8 = H_4 + \phi H_4$  ( $240 = 2 \cdot 120$ ) carries the carrier but not the (rational, 3-fold) cusp-weight family dynamics. Cited in *origin\_theory*; ledger row RATE.TRANSLATE.01.

## 2026-06-21 (the golden ratio made precise: the $(2, 3, 5)$ atoms are the affine- $E_8$ network spectral angles, $\phi = 2 \cos(\pi/5)$ the $g_{\text{car}}=5$ signature: v313)

- **v313\_golden\_atoms\_spectral.py** (GOLD.ATOMS.01, [E]/[C]). A follow-up to v312's observation, made exact. [E] the affine- $E_8$  adjacency characteristic polynomial factors as  $x(x^2-4)(x^2-1)(x^2-x-1)(x^2+x-1)$ , so the golden minimal polynomial  $x^2-x-1$  divides it and  $\phi = 2 \cos(\pi/5) = (1+\sqrt{5})/2$  is an *exact* eigenvalue; [E] the non-trivial eigenvalues are  $2 \cos(\pi/k)$  for exactly  $k \in \{2, 3, 5\} = \{|\mathbb{Z}_2|, N_{\text{fam}}, g_{\text{car}}\}$  ( $2 \cos(\pi/2)=0$ ,  $2 \cos(\pi/3)=1$ ,  $2 \cos(\pi/5)=\phi$ ,  $2 \cos(2\pi/5)=1/\phi$ ), so the golden ratio is *specifically* the  $g_{\text{car}}=5$  (5-fold) signature and  $g_{\text{car}}=5$  gains a new spectral witness distinct from Pascal/Riemann–Roch (raising the v305 multiplicity); [E] the  $(2, 3, 5)$  spherical excess  $1/2+1/3+1/5 = 31/30 > 1$  (the condition selecting the finite  $2I \Rightarrow E_8$ ) ties the v63/v76 gap-margin numerator  $31 = 2^{g_{\text{car}}} - 1 = 248/8 = 1+h(E_8)$  to the Coxeter  $30 = h(E_8)$  — a unification of already-anchored quantities, NOT new evidence (honest anti-numerology, v305); [E] honest null:  $\phi$  is a structural lattice/network quantity, not a phenomenological readout. [C] direction:  $\phi$  is the  $E_8$ -quasicrystal (Elser–Sloane cut-and-project) ratio, a candidate aperiodic substrate for the v312 question. Cited in *origin\_theory*; ledger row GOLD.ATOMS.01.

## 2026-06-21 (TOE-roadmap solver round: the SEAM.EQUIV.01 keystone certificate, the modular bedrock, the carrier→SM closure, the gap-decoupled measure, and the sharpened hypergraph substrate: v308–v312)

- **v308\_seam\_equiv\_chain.py** (SEAM.EQUIV.CHAIN.01, [E]/[O]). TOE attack point 1: the keystone “the raw seam-Calderón net is the holomorphic  $c=8$   $(E_8)_1$  net” assembled into ONE well-typed logical chain  $\text{gap}>0 \rightarrow \text{invertible/SRE (Kitaev)} \rightarrow \text{holomorphic } c=8 \text{ (Müger/KLM)} \rightarrow (E_8)_1 \text{ (Conway–Sloane)}$ , with the exact discriminators  $|\det \text{Cartan}(E_8)|=1$  vs  $|\det \text{Cartan}(D_8=SO(16))|=4$ , the carrier tower  $4 \cdot 4=16 \rightarrow 4 \rightarrow 1$  and  $c=g_{\text{car}}+N_{\text{fam}}=8$  all machine-checked; the input gap is the derived recovery gap  $6 \ln(3/2)>0$ . The only undischarged premise is the cited OS step — an assembly/reduction certificate (a Lean target), not an end-to-end closure. Cited in `tfpt_research_contracts`; ledger row SEAM.EQUIV.CHAIN.01.
- **v309\_modular\_bedrock.py** (QGEO.MODHAM.01, [E]/[O]). TOE attack point 2: the bedrock  $\omega \circ \rho = \omega$  via an explicit modular Hamiltonian, non-circular — the four marks come from Gauss–Bonnet (v216) and the order-4 clock from cross-ratio 2 (v214), so  $\mu_4$  is derived; the quasi-free modular Hamiltonian  $K = \log((1-C)/C)$  round-trips to the seam operator and  $[\rho, K]=0 \Leftrightarrow \omega \circ \rho = \omega$  (a  $\mathbb{Z}_2$  profile breaks it). Reduces the bedrock to the Bisognano–Wichmann intrinsicity of the raw collar; not a closure of QGEO.SYM.01. Cited in `tfpt_research_contracts`; ledger row QGEO.MODHAM.01.
- **v310\_carrier\_sm\_anomaly.py** (CAR.SM.ANOM.01, [E]). TOE attack point 3: the carrier half-spinor  $16 = \Lambda^{\text{even}}(\mathbb{C}^5) = 1+10+5$  is exactly one anomaly-free Standard-Model generation ( $SU(5) \subset SO(10)$ :  $10+\bar{5}+1$ ), with  $\sum Y = \sum Y^3 = [SU(2)]^2 U(1) = [SU(3)]^2 U(1) = 0$  (exact) and the one-loop  $(b_1, b_2, b_3) = (\frac{41}{10}, -\frac{19}{6}, -7)$ ; a fourth family shifts  $b_1$ , so  $N_{\text{fam}}=3$  pins the RG. Closes the spectrum/anomaly/ $\beta$  sub-question; the interacting Epstein–Glaser  $S$ -matrix and the Yukawa sector stay [C]/[O]. Cited in `tfpt_research_contracts`; ledger row CAR.SM.ANOM.01.
- **v311\_gap\_decoupled\_measure.py** (QGAMB.CLUSTER.01, [E]/[C]/[O]). TOE attack point 4: the physical QG measure as a gap-decoupled, exponentially-clustering object — the transfer spectrum  $\{1, (2/3)^6, (1/3)^6\}$  gives geometric clustering at rate  $(2/3)^6$  (correlation length  $1/(6 \ln(3/2))$ ), finite susceptibility  $729/665$ , and the v76 margin  $6 \ln(3/2) - 31/(4\pi^2) \approx 1.648 > 0$  decouples the physical sector from the un-built ambient. QG.AMB.01 stays [O] (gap-decoupled, inert). Cited in `tfpt_research_contracts`; ledger row QGAMB.CLUSTER.01.
- **v312\_hypergraph\_substrate.py** (HYP.INJECT.01, [E]/[O]). TOE attack point 5: the hypergraph substrate sharpened — the affine- $E_8$  Kac marks are the Perron eigenvector ( $A m=2m$ ) and the subleading eigenvalue is exactly  $2 \cos(\pi/5) =$  the golden ratio, but the recovery rate  $(2/3)^6$  is *not* any eigenvalue and  $\varphi_0$  is not a rational edge fraction; so a full-structure rewrite must inject exactly two non-graph-spectral data — the cusp weight  $2/3=|\mathbb{Z}_2|/N_{\text{fam}}$  and the analytic seed  $\varphi_0$ . The open question is precisely delimited, not closed. Cited in introduction; ledger row HYP.INJECT.01.

## 2026-06-21 (adversarial-review follow-ups: a structural anti-numerology firewall, an out-of-sample seed cross-validation, an operational data watchdog, and the Target-A holomorphy kill test: v305–v307)

- **v305\_witness\_independence.py** (WITNESS.ECON.01, [E]). The *structural* anti-numerology firewall complementing the v100 phenomenological null test, against the sharpest self-deception worry (“the same integer counted many times as if independent”). Every headline integer ( $|\mu_4|=4$ ,  $g_{\text{car}}=5$ ,  $|\mathbb{Z}_2|=2$ ,  $N_{\text{fam}}=3$ ,  $\dim S^+=16$ ,  $\text{rank } E_8=8$ , 240, 248,  $|2I|=120$ ,  $h(E_8)=30$ ,  $|W(D_5)|=1920$ ,  $\Omega_{\text{adm}}=48$ , the gap exponent 6) is a documented image of the single anchor  $a=(1, 1, 2)$  with  $\pi$  the only transcendental primitive (13 atoms from 2 free primitives,  $6.5 \times$



compression made explicit), so the recurring  $2/3=|\mathbb{Z}_2|/N_{\text{fam}}$  and  $(2/3)^6$  are *one* ratio; the  $E_8$  spine is overdetermined ( $\geq 3$  structurally independent witnesses) while the pure-seed readouts are single-witness (scored jointly by **v100**, never multiplied); the negative control shows  $\alpha^{-1}$ 's “137” is foreign to the anchor family (an independent witness via the  $F_{U(1)}$  cubic, **v3**). Cited in the **tfpt\_5\_redteam** body; ledger row **WITNESS.ECON.01**. Python-only.

- **v306\_seed\_crossval.py** (**SEED.CROSSVAL.01**, **[E]**). A leave-one-out cross-validation turning the “many observables, one seed  $\varphi_0$ ” worry into a falsifiable out-of-sample test: back-solving  $\varphi_0$  from each of the five seed-grammar observables ( $\sin^2 \theta_{12}$ ,  $\sin^2 \theta_{13}$ ,  $\beta$ ,  $\Omega_b$ ,  $\lambda_C$ ; NuFIT 6.0 / ACT DR6 / Planck / PDG) and error-weighted-averaging reproduces the axiom seed  $\varphi_0 = 1/(6\pi) + 48c_3^4 = 0.0531720$  to 0.04%; leave-one-out predicts every held-out observable within  $3\sigma$  out of sample (most-tensioned =  $\sin^2 \theta_{13}$ ,  $\sim 2\sigma$ ); a data $\leftrightarrow$ formula shuffle explodes the  $\chi^2$  by  $> 100\times$ . Cited in **tfpt\_5\_redteam**; ledger row **SEED.CROSSVAL.01**. Python-only.
- **v307\_data\_watchdog.py** (**DATA.WATCHDOG.01**, **[E]**). The operational decision pipeline over the frozen registry (**v84**): it classifies all twenty registry entries PASS/WATCH/TENSION/KILL by live  $n_\sigma$  against the repo-documented current bests (CODATA 2022, NuFIT 6.0, ACT DR6, Planck 2018, PDG 2024, BK18). No decidable prediction is in KILL; the watch items are  $\alpha^{-1}$  ( $1.9\sigma$ ),  $s_{23}$  ( $1.8\sigma$ ) and the most-tensioned core prediction  $\sin^2 \theta_{13}$  ( $2.0\sigma$ , the honest current pressure point);  $\sin^2 \theta_{12} = 0.306747$  is compatible ( $0.02\sigma$ , the JUNO falsifier), the  $r \in [0.0033, 0.0048]$  band is below the BK18 limit and the kill, and  $w=-1$  is flagged as the DESI dark-energy watchdog. Cited in **origin\_theory**; ledger row **DATA.WATCHDOG.01**. Python-only.
- **Target-A holomorphy kill test (freeze\_file.csv)**. A new **seam\_holomorphy** freeze row records the sharpened *physical* kill criterion for Target A: a raw quasi-free seam bulk showing topological ground-state degeneracy on the torus ( $|\det K| > 1$ , the same- $c$  rival  $SO(16)_1$ ) would break holomorphy and kill “seam net =  $(E_8)_1$ ” — since holomorphic  $c=8 \Leftrightarrow E_8$  is machine-proven (**v83/v143**) and  $\det K=1 \Leftrightarrow$  invertible/SRE (**v237/v301/v302**).

## 2026-06-19 (an infinite-derivative narrowing of the Stelle-ghost attack, the double-pushout figure, and the “constancy of $c$ ” framing of **SEAM.EQUIV.01**: **v304**)

- **v304\_idg\_nonlocal\_ghost.py** (**QGAMB.IDG.01**, **[C]/[O]**). A conditional, parameter-free narrowing of the red-team **rt\_F/v278** Stelle-ghost attack on the perturbative gravity sector. **[E]** the local  $R+R^2/\text{Weyl}^2$  four-derivative propagator  $1/(p^2(p^2+M^2))$  carries a negative-residue massive spin-2 mode (the Stelle ghost) only as a *truncation artefact*; **[E]** for an *entire* graviton form factor  $a(p^2) = e^{p^2/M^2}$  (nowhere zero) the dressed propagator  $1/(p^2 a)$  has no extra pole (Tomboulis 1997; Biswas–Mazumdar–Siegel 2006), whereas any *polynomial* truncation has a real zero = the ghost; **[C]** the spectral-action cutoff is the seam KMS weight  $f(u) = e^{-u}$  (**v259**), so the un-truncated trace points to a nonlocal form factor with *no new parameter*. **[O]** It does *not* close **QG.AMB.01**: the trace regulator is not yet shown to resum to an entire  $a(\square)$ , and a ghost-free *perturbative* graviton is not the nonperturbative ambient measure. Cited in the red-team Target F body; ledger row **QGAMB.IDG.01**.
- **Double-pushout figure (tfpt\_1\_architecture\_e8)**. A new shared TikZ macro (**TFPTdoublepushout** in **tex-artefacts/tfpt\_figures.tex**) draws the commuting square showing that the order- $|\mu_4|=4$  glue is *one* operation in two categories: the lattice pushout  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  (**v1**) and the conformal-net / anyon-condensation extension  $(D_5)_1 \otimes (A_3)_1 \rtimes \mu_4 \cong (E_8)_1$  (**v92/v281/v234/v277/v260**). Expository only; no new claim.
- **Framing (introduction, tfpt\_research\_contracts)**. **SEAM.EQUIV.01** is now stated explicitly as the *one* irreducible structural postulate of the theory — the role the constancy of  $c$  plays in relativity (cf. **v267**). Prose sharpening; no status change.



## 2026-06-19 (the discrete→dynamic lens on **F\_transfer**: one shape, one seam-rate, v303)

- **v303\_ftransfer\_dynamics.py** asks whether the main-branch mechanism — a gapped, positivity-preserving relaxation to a *unique* attractor (the local-averaging update → the  $E_8$  marks, gap  $6 \ln(3/2)$ ) — is also the shape of the four **F\_transfer** instances, or an unrelated bolt-on. It is the *same* shape: **[E]** **F\_pole** (Koide) *is* the seam dynamics (Möbius multiplier  $F'=(2/3)^6=\lambda_2$ , the seam-clock eigenvalue  $\Rightarrow$  a unique attractor at the seam rate, **v82/v54**); **[C]** **F\_Boltzmann** ( $\eta_B$ ) is a gapped positive contraction (washout  $\kappa_f \in (0,1)$ , H-theorem); **[C]** **F\_relic** (axion) freezes at an adiabatic fixed point; **[E]** **F\_QCD** ( $m_p/m_e$ ) is the 1-loop RG flow with the carrier  $b_3 = -7$  (asymptotic freedom, Gaussian UV attractor). **[O]** Verdict: one shape underlies all four; **F\_pole** has the seam rate  $(2/3)^6$  *exactly*, the other three share only the shape with external rates (thermal/cosmological/RG). So **F\_transfer** is the readout end of the one discrete→dynamic principle, *not* a bolt-on; the theory's simplicity is preserved because the external rates are honestly fenced (**v187**), not reduced to the seam. **[E]** suite v303 5/5, AUDIT OK.

## 2026-06-19 (the last lever: the seam mass gap *is* the derived Recovery gap $6 \ln(3/2)$ , v302)

- **v302\_seam\_gap.py** closes the chain on the single statement **v301** had left: “the quasi-free seam bulk is gapped”. That gap is *not* a free input — it is the established *Recovery gap* of the seam transfer operator, derived from the carrier. **[E]** the frozen transfer spectrum is  $\text{spec}(T) = \{1, (2/3)^6, (1/3)^6\}$  (**v160/v162**), a simple Perron eigenvalue 1 separated from the rest; **[E]** the gap is  $\Delta = -\ln((2/3)^6) = 6 \ln(3/2) = 2N_{\text{fam}} \ln(3/2) \approx 2.4328 > 0$  (multiplicative gap  $1 - (2/3)^6 = 665/729$ ;  $(2/3)^6$  the  $\mu_4$ -deck transfer eigenvalue forced by the carrier); **[E]** with the decoupling margin  $\Delta - 31/(4\pi^2) \approx 1.648 > 0$  (**v76**) it is robust, and Perron–Frobenius primitivity makes the leading mode simple; **[C]** by Osterwalder–Schrader / quasi-free clustering a transfer-operator gap  $\Delta > 0$  *is* a mass gap of the reconstructed Hamiltonian. **[O]** **Bottom line:** with **v300/v301**, **SEAM.EQUIV.01**'s entire residual is now a composition of *standard cited theorems* (OS/quasi-free clustering + Kitaev free-fermion invertibility + the **v297** AQFT stack) over *established* TFPT facts (16 quasi-free Majoranas; transfer gap  $6 \ln(3/2)$ ) — *no undischarged TFPT-internal assumption remains*. It stays **[O]** (not machine-proved end-to-end), but the open front is reduced to citable machinery over a spectral gap that is provably positive. **[E]** suite v302 5/5, AUDIT OK.

## 2026-06-19 (Route A's last hypothesis discharged: the gapped quasi-free bulk is invertible by the free-fermion classification, v301)

- **v301\_route\_a\_invertible.py** attacks the single remaining *unconditional* gap of **SEAM.EQUIV.01**. Since **v300** collapsed Route B into Route A's rationality, that gap is Route A's open hypothesis (**v297** LIT-A): “the raw quasi-free seam bulk is invertible (SRE)”. A *gapped* free-fermion (Gaussian) 2+1D bulk cannot carry intrinsic topological order — free fermions give the integer (IQHE/Chern) phases, all *invertible*; intrinsic topological order needs interactions (Kitaev's periodic table, class D). **[E]** the carrier is 16 Majoranas,  $c = g_{\text{car}} + N_{\text{fam}} = 8 = 16/2$  (**v148**); **[E]**  $SO(16)_1 \supset (E_8)_1$  ( $\dim \mathfrak{so}(16) = 120$  currents + 128 spinor = 248, **v154**); **[C]** so a gapped quasi-free bulk is invertible automatically; **[E]** the invariant  $|\det K| = \#\text{anyons}$  is 1 for  $E_8$  (invertible) vs 4 for the gauged  $D_8=SO(16)$  (anyons), which free fermions cannot produce. **[O]** LIT-A's ‘invertible bulk’ now *follows* from ‘gapped 16-Majorana quasi-free bulk’ (**v148/v160**+the mass gap), so the residual changes from a topological-order statement to a spectral one. **[O]** **Bottom line:** with **v300**, the whole open content of **SEAM.EQUIV.01** reduces to the *single spectral statement* “the quasi-free seam bulk is gapped” — the topological-order obstruction (hidden anyons) is removed; the gap is not

manufactured. [E] suite v301 6/6, AUDIT OK.

## 2026-06-19 (the Flat-Away attack: a hard discrete obstruction + the pin derived from $(E_8)_1$ , v300)

- **v300\_flataway\_rigid.py** attacks the single external fact the three Flat-Away routes (v291/v293/v295) left open, on two fronts. *Harden*: the flat seam was only a *soft* minimum (“the spectral mismatch grows with  $\varepsilon$ ”); here the obstruction becomes a *discrete* on/off invariant. [E] the flat fingerprint —  $\text{spec}(|D_\theta|)$  is the integer ladder with every level  $n \geq 1$  of multiplicity exactly 2; [E] the resonant mode  $4k$  splits level  $2k$  by gap  $= g_k$  (sympy  $2 \times 2$   $[[2k, g/2], [g/2, 2k]] \Rightarrow 2k \pm g/2$ , + an  $O(g^2)$  common shift), validated against the full Toeplitz operator; [E] any  $g_k \neq 0$  lifts the resonant degeneracy ( $2 \rightarrow 1+1$ ) and changes the spectral *multiset*, so preserving the flat ladder *with* its mult-2 degeneracies forces every  $g_k = 0 \Rightarrow f = 0$  — strictly stronger than the “deviation grows” scan; [E] with the v295 convexity, flat is isolated analytically *and* discretely. *Derive the pin*: [C] the  $(E_8)_1$  character  $E_4/\eta^8 = q^{-1/3}(1 + 248q + 4124q^2 + \dots)$  has integer  $q$ -coefficients (248 at level 1;  $E_8$  even unimodular  $\Rightarrow$  integer weights), so by 2d Steklov conformal rigidity the integer ladder  $\Leftrightarrow$  flat seam, i.e. the external pin *is* the  $(E_8)_1$  rationality (Route A). [O] verdict: Flat-Away carries *zero* new open content beyond Route A — the two SEAM.EQUIV.01 routes converge on one rationality fact. [E] suite v300 6/6, AUDIT OK.

## 2026-06-19 (the hypergraph/Wolfram bridge hardened: the $(2, 3, r)$ seed reaches $E_8$ as the last finite point, v299)

- **v299\_hypergraph\_0d\_autonomous.py** hardens the Wolfram/hypergraph reading into a precisely typed bridge (builds on v219/v298): [E] the icosahedron is a 3-uniform hypergraph ( $|\text{Aut}(\text{graph})| = 120 = |I_h| = |A_5 \times \mathbb{Z}_2|$ ; the McKay group is the  $SU(2)$  binary lift  $2I$ , a *different* order-120 group whose McKay graph is affine  $E_8$ ; 30 edges  $= h(E_8)$ ); [E] the  $1 \rightarrow 4$  subdivision rewrite is equivariant (all 120 symmetries at every scale); [E] the  $2I$  fusion tensor is a weighted 3-uniform hypergraph whose 2-dim slice is the affine  $E_8$  McKay graph; [E] Smith ( $\rho \leq 2 \Leftrightarrow$  affine ADE) + Collatz–Wielandt ( $\rho \leq 2 \Leftrightarrow$  a local witness  $Am \leq 2m$ ); [E]\* the autonomous rule (diffusion-generated witness + *local* gate  $Am \leq 2m$ , Perron only as audit) runs  $E_6 \rightarrow E_7 \rightarrow E_8 \rightarrow \hat{E}_8$ ; [E] the negative controls  $\rho(A_n) = 2 \cos \frac{\pi}{n+1}$  and  $\rho(D_n) = 2 \cos \frac{\pi}{2n-2}$  are  $< 2$  for *all*  $n$  (closed form), so neither family ever reaches  $E_8$ . Honest type: [O] the  $(2, 3, \cdot)$  seed is the single irreducible input = P2 — *the rule reaches  $E_8$  as the last finite point on this seed; it is not a derivation of P2*. [E] build green, AUDIT OK.

## 2026-06-18 ( $E_8$ as a network attractor — the Wolfram-program reading, v298)

- **v298\_e8\_network\_attractor.py** is an honest audit of the “TFPT as a hypergraph/rewrite system” idea, on the v219 McKay bedrock. [E] the affine- $E_8$  Kac marks  $(1, 2, 3, 4, 5, 6, 4, 2, 3)$  are the *unique dynamical attractor* of a minimal local-averaging update  $m \leftarrow (A+2I)/4m$  on the  $(2, 3, 5)$  McKay graph (converges from any positive start;  $Am = 2m$ , top eigenvalue 2); the spherical tower terminates at  $E_8$  and the cascade is nested-subgraph coarse-graining. [C] the Wolfram reading “ $E_8$  is the universal attractor of a minimal  $(2, 3, 5)$  network” holds at the Coxeter-skeleton level. [O] a minimal hypergraph *rewrite* reproducing the *full* structure stays open; honest negatives:  $\varphi_0^{\text{ret}}$  is the analytic seed  $\frac{4}{3}c_3 = 1/(6\pi)$ , not an edge fraction, and recovery is a 1-D Möbius fixed point, not a rewrite. [E] build green, AUDIT OK.

## 2026-06-18 ( $a_2$ closed form + Route A literature stack, v296–v297)

- **v296\_flataway\_a2\_closed.py** (FLATAWAY.A2.CLOSED.01) evaluates the exact  $a_2$  coefficient in *closed form*: the operator tails telescope geometrically ( $C_k + D_k e^{-4kt} = 4kt(1 - e^{-4kt})$ ) to  $W_k^{\text{tail}} = t(1 - e^{-4kt})/(8k(1 - e^{-t}))$ , leaving a finite  $(4k-1)$ -term middle; e.g.  $W_1(t) =$

$t(2te^t + e^{3t} + 3e^{2t} + e^t - 1)e^{-3t}/8$ . Manifestly positive, small- $t$  law  $W_k = t/2 + O(t^3)$  (the  $t^2$  term cancels). Reproduces v295 to machine precision.

- **v297\_route\_a\_literature\_stack.py** (SEAM.EQUIV.A.LIT.01) writes Route A’s one import as a citable stack: LIT-A (Kitaev; Freed–Hopkins)  $\rightarrow$  LIT-B (Müger; Kawahigashi–Longo–Müger)  $\rightarrow$  LIT-C (Conway–Sloane; Dong–Xu) compose into “invertible bulk  $\Rightarrow (E_8)_1$ ”, all legs established; the single open hypothesis (seam-bulk invertibility) coincides with Route B’s Flat-Away, and the stack is non-circular with Route B (the v286 firewall holds). [E] build green, AUDIT OK.

## 2026-06-18 (Flat-Away heat route made exact + Lean-formalised, v295 / FORM.FLATAWAY.01)

- **v295\_flataway\_a2\_exact.py** (FLATAWAY.A2.01) upgrades the v292 positive-definiteness from a numerical scan to a *proof*: the first variation of  $\text{Tr } e^{-t\Lambda}$  vanishes (zero-mean off-mark curvature, Gauss–Bonnet), and since  $\Lambda \mapsto \text{Tr } e^{-t\Lambda}$  is convex (Klein/Peierls),  $\varepsilon=0$  is the *global* minimum — so  $\Delta \text{Tr} \geq 0$  for all deformations,  $= 0$  iff  $f = 0$ . The exact second-order coefficient is the divided-difference kernel of  $e^{-tx}$  (Duhamel), validated to rel. err  $< 10^{-6}$ .
- **Lean FORM.FLATAWAY.01** (SeamDeckClosure.lean Part 5): the formal  $\Delta=0 \Leftrightarrow$  flat step — a positive-weighted sum of squares is positive-definite (heatDeviation\_eq\_zero\_iff); built and axiom-checked (only propext, Classical.choice, Quot.sound). So the heat route’s positive-definiteness is now both analytically proved and machine-formalised; only the single external fact (the seam fixes  $a_2$ ) remains. [E] build green, AUDIT OK.

## 2026-06-18 (Flat-Away attacked on all three routes, v292–v294)

- **v292\_flataway\_heat\_reduction.py** (FLATAWAY.HEAT.01): the heat route, made precise. The heat-trace deviation  $\Delta \text{Tr}(e^{-t\Lambda}) = c(t)\varepsilon^2 + O(\varepsilon^4)$  is a *positive-definite* quadratic form in the off-mark curvature ( $c_k(t) > 0$  for every  $\mathbb{Z}_4$  mode, positive on all combinations), leading invariant  $\|f\|_{L^2}$  — so Flat-Away reduces to the single fact “the seam fixes the  $a_2$  heat coefficient”.
- **v293\_flataway\_spectral\_hessian.py** (FLATAWAY.SPEC.01): the spectral route, upgraded from one direction to the *full* smooth- $\mathbb{Z}_4$  space. Mode  $4k$  splits the degenerate Steklov level  $|2k|$  at first order; the Hessian of the spectral mismatch over modes  $\{4, 8, 12\}$  is positive-definite (eigenvalues  $\{0.497, 0.500, 0.503\}$ ), so the flat seam is a strict isolated minimum — the “only one direction” objection is answered.
- **v294\_flataway\_rp\_energy.py** (FLATAWAY.ENERGY.01): the variational route. With the rigid mark divisor (4 cone angles  $\pi$ , Gauss–Bonnet  $4\pi$ ), Troyanov uniqueness gives a unique flat cone metric, and the curvature energy  $\int f^2$  is strictly convex (Hessian  $2\pi I$ ) with a unique minimiser — Flat-Away reduces to “RP+gap realises the energy-minimiser”. The three routes converge on one selection principle for the flat metric; closing any one closes Route B. [E] build green, AUDIT OK.

## 2026-06-18 (Route-B red-team: the smooth $\mathbb{Z}_4$ adversary + Flat-Away as a mini-theorem, v290–v291)

- **v290\_z4\_smooth\_curvature\_adversary.py** (SEAM.ADVERSARY.01) is the honest red-team that  $\mathbb{Z}_4$  block-diagonality is *not* mark-locality: a smooth  $\mathbb{Z}_4$  off-mark curvature  $f = \varepsilon \cos(4\theta)$  [E] passes the v288 commutator ( $\|[\rho, \Lambda]\| = 0$ , same as mark-local) yet [E] shifts the DtN spectrum ( $\max |\Delta\lambda| = 0.155$ ) and the heat trace ( $\Delta = 0.019$ ). So the commutator is necessary, not sufficient; the modulus is excluded by the spectral data, not  $[\rho, \Lambda]=0$  — which is exactly why Flat-Away cannot be read off the commutator.

- **v291\_flataway\_contract.py** (FLATAWAY.RP.01) states Flat-Away as its own named mini-theorem and types three independent proof routes: [E] spectral-rigidity and [E] heat-kernel (the flat seam is the *unique* spectral/heat match of the smooth  $\mathbb{Z}_4$  family — deviation zero only at  $\varepsilon=0$ ) and [C] RP-energy/Trojanov (constant-curvature minimiser). Closing any one closes Route B. [E] build green, AUDIT OK.

## 2026-06-18 (Route B sharpened: raw mark-locality decomposed + reviewer firewalls, v289 / v287–v288)

- **v289\_marklocal\_raw.py** (MARKLOCAL.RAW.01) decomposes the one remaining Route-B residual (‘why is the raw seam sub-principal term mark-local?’) into a 5-lemma chain and isolates the *single* open analytic lemma: [C] L1 conic parametrix (b-calculus/Melrose, imported); [O] L2 **Flat-Away** — RP+gap+4 marks  $\Rightarrow$  curvature vanishes away from the marks; [E] L3 orbit-source (v195/v216), L4 Fourier-support on  $4\mathbb{Z}$  (v264/v288), L5 full-operator  $[\rho, \Lambda]=0$  (v288). 4/5 discharged — Route B reduces to exactly L2.
- **v288 reviewer firewall**: a SCOPE FIREWALL check makes the exact-label split explicit (Fourier lemma Formal/Exact; numerical test Numerical Exact; transfer to the raw conic DtN Conditional, v264; mark-locality from the RP seam Open Selection, v289) — so it can never be read as ‘full- $L^2$  QGEO proved’.
- **v287 literature matrix**: L3 (‘invertible bulk  $\Rightarrow$  single-sector’) is typed as a composition of citable results (LIT-A Kitaev/Freed–Hopkins, LIT-B Muger/Kawahigashi–Longo–Muger, LIT-C Kawahigashi–Longo) conditional on one open hypothesis — seam-bulk invertibility, which coincides with Route B. [E] build green, AUDIT OK.

## 2026-06-18 (closing sprint: the Seam Equivalence Theorem named + both routes attacked, v286–v288)

- **v286\_seam\_equivalence\_contract.py** (SEAM.EQUIV.01) names the one open theorem — *the raw RP seam state is the holomorphic  $(E_8)_1$  net at  $\tau=i$*  — and concentrates the whole open structure on it: four objects, six arrows (2 closed, 1 conditional v282, 1 open Gral), an **import firewall** proving the two routes (v276/v280/v284 vs v277/v281/v285) never import each other (no ‘ $E_8$  into the geometry and back’ circularity), and an exact-label taxonomy. Writes `seam_equivalence.json`.
- **v287\_free\_rp\_bulk\_to\_holomorphic\_boundary.py** (SEAM.EQUIV.A01): Route A (AQFT) dependency checker — the 5-lemma chain RP+gap+chirality+Gaussian+invertible  $\Rightarrow \mu=1 \Rightarrow$  holomorphic  $\Rightarrow (E_8)_1$  is logically complete *modulo one* standard theorem: *invertible (SRE) Gaussian bulk  $\Rightarrow$  single-sector boundary* (the AQFT form of  $\det K=1$ ). 4/5 discharged.
- **v288\_full\_l2\_subprincipal\_z4.py** (SEAM.EQUIV.B01): Route B (DtN) — *proves* the full- $L^2$  lift of the sub-principal  $\mathbb{Z}_4$  block-diagonality (a mark-sourced curvature has Fourier support only on  $m \equiv 0 \pmod{4}$ , so  $\Lambda = |n| + M_f$  commutes with  $\rho = \text{diag}(i^n)$  on the full operator,  $\|[\rho, \Lambda]\| \sim 2 \times 10^{-15}$ ; generic curvature  $\rightarrow 4.44$ ), lifting v201/v284 to  $L^2$ . The residual shrinks to one sharper question: why is the raw seam sub-principal term mark-local? [E] build green, AUDIT OK.

## 2026-06-18 (the two proof routes decomposed: Route (i) and Route (ii) into lemma chains, v284/v285)

- **v284\_route\_i\_rp\_uniqueness.py** (QGEO.ROUTEI.01) turns Route (i) (RP-state uniqueness) into a 6-lemma chain: [E] 5/6 lemmas are already discharged — L1 RP $\rightarrow$ DtN (v194), L2 marks $\rightarrow$ pillowcase (v195/v216/v214), L3 flat Steklov =  $\sqrt{-\Delta_{\text{flat}}}$  (verified here), L4 covariance from DtN (v258), L5  $\mu_4$ -invariance (v276/Lean/v280); [C] Trojanov ( $\chi_{\text{orb}}=0 \Rightarrow$  unique flat

$\tau=i$ ) reduces the residual to one lemma. [O] the one open lemma: *the raw seam is the constant-curvature geometric state*.

- **v285\_route\_ii\_seam\_condensation.py** (QGAMB.ROUTEII.01) turns Route (ii) (seam-net condensation) into a 4-lemma chain: [E]  $3/4$  discharged (no abelian sector  $\Leftrightarrow \det K=1 \Leftrightarrow$  holomorphic  $\Leftrightarrow E_8$ ; the tower  $16 \rightarrow 4 \rightarrow 1$ ), and **its open lemma coincides with Route (i)’s** — both are “the raw seam is the  $(E_8)_1$ -at- $\tau=i$  object” (v282), so proving *either* route closes *both*. [O] the unified open lemma: a canonical equivalence between the raw seam KMS/DtN state and the holomorphic  $(E_8)_1$  net at  $\tau=i$ . Build green, AUDIT OK.

## 2026-06-18 (self-investigation round: pillowcase DtN, modular data, the E8-at- $\tau=i$ unification, live scorecard, v280–v283)

- **v280\_pillowcase\_steklov.py** (QGEO.STEKLOV.01) runs the QGEO chain on the actual flat  $\tau=i$  pillowcase orbifold: the order-4 deck permutes the 4 cone points (2 fixed + 1 swap) and the geometric chain  $[\rho, \Delta]=0 \Rightarrow [\rho, H]=0 \Rightarrow [\rho, C]=0 \Rightarrow \omega \circ \rho = \omega$  is numerically exact — route-(i) evidence for QGEO.OBLIG.01; the premise stays [O].
- **v281\_holomorphy\_modular\_data.py** (QGAMB.MODULAR.01):  $\# \text{anyons} = |\det \text{Gram}|$  ( $E_8 \rightarrow 1$ ,  $SO(16) \rightarrow 4$ ), the anyon-condensation tower  $16 \rightarrow 4 \rightarrow 1$ , Gauss–Milgram  $\rightarrow c=8$ ; holomorphy ( $\det K=1$ ) is the single Tier-B discriminator.
- **v282\_e8\_tau\_i\_unification.py** (QGAMB.UNIFY.01): the candidate *simplification* —  $\chi_{E_8}=j^{1/3}$  gives  $\chi_{E_8}(i)=1728^{1/3}=12$ , so  $\tau=i$  is both the order-4 CM point (QGEO) and the  $(E_8)_1$  modulus (QG.AMB): the two obligations are two faces of one object (the seam is  $(E_8)_1$  at  $\tau=i$ ), **obligation count**  $2 \rightarrow 1$ .
- **v283\_live\_killtest\_scorecard.py** (PRED.LIVE.01): the recent round’s computable predictions vs current data — all consistent  $< 2\sigma$ , with two sharp near-term kill tests ( $\Sigma m_\nu$  floor 0.0586 eV; proton decay  $1.55 \times 10^{35}$  yr).
- Wolfram 275/275 (v281/v282 exact-mirrored); build green, AUDIT OK.

## 2026-06-18 (the Gräl: QGEO.SYM.01 as one precise lemma + the all-orders step Lean-formalised, v279/FORM.QGEO.03)

- **v279\_qgeo\_obligation\_lemma.py** (QGEO.OBLIG.01) writes the bedrock as **ONE precise constructive-QFT lemma**. Given the premise *the raw seam carries the flat  $\tau=i$  pillowcase metric* (Trojanov-unique,  $\Lambda_\Sigma = \sqrt{-\Delta_{\text{flat}}}$ ), then  $\omega_\Sigma \circ \rho = \omega_\Sigma \rightarrow$  mark-locality  $\rightarrow E_8$ . [E] the reduction DAG from ‘free RP seam’ has exactly *one* open leaf — ‘the raw seam state is the flat state’; [E] the geometric side is fixed (flat DtN diagonal,  $\rho$  commutes to all orders), so the obligation is a state-identification, with [C] two proof routes (RP-state uniqueness, or Trojanov + OS).
- **FORM.QGEO.03: the all-orders step is now machine-proved in Lean**. `SeamDeckClosure.lean` gains `diagonal_commutatesClock` and `flat_all_orders_clock`: any spectral function  $g(\Delta_{\text{flat}})$  of the Fourier-diagonal flat Laplacian commutes with the carrier clock  $\rho = \text{diag}(i^n)$  — so the v276 all-orders closure is a Lean theorem (only `propext`, `Classical.choice`, `Quot.sound`; no `sorry`; full lake build green, 3356 jobs). The premise (the seam *is* flat) stays the [O] QGEO.SYM.01 input. Build green, AUDIT OK.

## 2026-06-18 (4D QFT: the $S_{\text{pert}} \rightarrow S_{\text{phys}}$ LSZ bridge + one-loop unitarity, v278)

- **v278\_lszy\_bridge\_unitarity.py** (QFT4D.SPERT.04) connects the perturbative S-matrix to the physical asymptotic one and verifies one-loop unitarity. [E] LSZ bridge: the EG  $T$ -products of  $S_{\text{pert}}$  (v269) are the OS Wick-rotated correlators



(v240), so amputate+on-shell-residue gives  $S_{\text{pert}} \xrightarrow{\text{LSZ}} S_{\text{phys}}$ ; [C] the gap  $\Delta=6\log(3/2)>0$  gives Haag–Ruelle asymptotic states. [E] *one-loop optical theorem*: the bubble discontinuity  $\text{Im } B(s)/\pi = x_+ - x_- = \sqrt{1-4m^2/s}$  *exactly* = the two-body phase space, so  $2\text{Im } M = \sum \int d\Pi |M|^2$  (cutting rule) holds — the matter+gauge sector is unitary, with the cut turning on at the physical two-particle threshold. [O] the gravity sector’s Stelle ghost (v269/rt\_F) is non-unitary  $\rightarrow$  QG.AMB.01. Exact core mirrored in Wolfram (273/273). Build green, AUDIT OK.

## 2026-06-18 (QG.AMB.01 Tier-B made concrete: the seam-Calderón $\rightarrow (E_8)_1$ match, v277)

- **v277\_seam\_calderon\_e8\_match.py** (QGAMB.TIERB.01) **reduces Tier-B to ONE bit**. [E] the full  $(E_8)_1$  matching target is computed and matches:  $c=8$ ,  $\det \text{Cartan}(E_8)=1$ , the holomorphic character  $E_4/\eta^8=j^{1/3}=q^{-1/3}(1+248q+\dots)$  with a single primary and exactly  $248=\dim E_8$  weight-1 currents (240 roots). [E] the discriminator is holomorphy — the  $c=8$  counterexample  $(D_8)=SO(16)_1$  has  $\det \text{Cartan}=4$  (non-holomorphic), so the one condition is  $|\det K|=1$ . [C] the seam side matches via v276 (flat  $\tau=i$ ) + v260 (Kummer/K3). [O] residual = the single holomorphy bit; reduced, not closed. Exact core mirrored in Wolfram (272/272). Build green, AUDIT OK.

## 2026-06-18 (QGEO.SYM.01 narrowed: the flat pillowcase closes the commutator to all orders, v276)

- **v276\_qgeo\_flat\_closes\_commutator.py** (QGEO.SYM.03) **collapses the bedrock to ONE geometric postulate**. The state-level residual the whole bedrock rests on is the operator equation  $[\rho, H]=0$ . [E] if  $H=f(\Delta_{\text{flat}})$  and the order-4 deck  $\rho(z \mapsto iz)$  is a flat- $\tau=i$  isometry, then  $[\rho, \Delta]=0 \Rightarrow [\rho, H]=0$  *exactly*, to all orders (verified on the discretised square torus:  $\rho^4=I$ ,  $[\rho, \Delta]=0$ ,  $[\rho, \sqrt{-\Delta}]=0$ ; generic non-isometry fails) — upgrading the leading-order v198/v201 to the full  $H$ . [C]  $\tau=i$  is forced by order-4 ( $j=1728$ ; hexagonal  $j=0$  is order-6, excluded). [O] QGEO.SYM.01 therefore collapses to the single statement “the raw seam IS the flat  $\tau=i$  pillowcase quasi-free state” — one human constructive-QFT proof, or an honest axiom. Narrows, does not close. Build green, AUDIT OK.

## 2026-06-18 (Red Team Target F — adversarial audit of the v269–v275 round, rt\_F)

- **redteam/rt\_F\_qft4d.py** (REDTEAM.F.01) **stress-tests the just-published round** (the perturbative  $S_{\text{pert}}$  layer, the PMNS Jarlskog, the  $\nu$ -scale, the over-determination, the QG.AMB.01 roadmap). SURVIVES (NARROWED): two attacks land and are folded in. (1) the gravity  $R^2/\text{Weyl}^2$  sub-sector is power-counting renormalizable but *non-unitary* (the Stelle ghost: the four-derivative propagator  $1/(p^2(p^2+M^2))$  has a negative-residue massive spin-2 mode), so  $S_{\text{pert}}$  is unitary for the matter+gauge sector only — which is exactly why the nonperturbative QG.AMB.01 is the real frontier (caveat added to v269). (2) the 0.11% anchor over-determination is conditional on the  $\Lambda$ -branch readout (LAMBDA.BRANCH.01, [C]), the gravity route being unconditional (caveat added to v274). Added as Target F in `tfpt_5_redteam` + the red-team table; `run_redteam.py` green.

## 2026-06-18 (Zenodo release 5.2 (rev 191) — the perturbative 4D-QFT layer + scale closure, v269–v275)

- **Published to Zenodo**. Version 5.2 (rev 191) of the ten-PDF document set was published as a new version of the TFPT deposit — record 20743347, DOI 10.5281/zenodo.20743347. This release carries the perturbative 4D-QFT layer (the Epstein–Glaser  $S_{\text{pert}}$  skeleton v269, the

concrete one-loop quartic **v271** and gauge  $\beta$ -coefficients **v273**), the complete complex PMNS matrix and leptonic CP strength **v270**, the honest absolute  $\nu$ -mass-scale typing **v272**, the over-determination of the single mass anchor **v274**, and the **QG.AMB.01** obligation roadmap **v275**. Suite 273 modules (highest ID **v275**; **v186/v226** skipped), Wolfram 271/271, **run\_all.py** green, **AUDIT OK**, **npm** build clean. The Zenodo description (**zenodo\_description.html**) and the public website were synced in the same release.

## 2026-06-18 (scale + frontier round: $\nu$ -mass scale, EG gauge running, over-determination, QG roadmap, **v272–v275**)

- **v272\_nu\_mass\_scale.py** (**FLAV.NUSCALE.01**) **types the absolute neutrino-mass scale honestly** (no fabricated  $\Sigma m_\nu$  — would contradict the **v184** firewall). **[E]** the absolute scale reduces to *one* seesaw ratio  $m_3 = (y_\nu v_{EW})^2/M_R$  (one UV input, like  $v_{geo}$ , **v153**); **[C]** the observed  $m_3=0.050$  eV with the TFPT PS→GUT window  $M_R \in [4.2 \times 10^{13}, 2.4 \times 10^{15}]$  GeV (**v249**) needs  $y_\nu \in [0.26, 2.0]$  —  $O(1)$ , no tuning; **[O]**  $M_R$  is not a clean  $\bar{M}_{Pl}$  power, so the absolute value stays open; **[X]** the NO floor  $\Sigma m_\nu=0.0586$  eV is the cosmological kill test. Closes the PMNS bookkeeping (the last **[O]** is one seesaw ratio).
- **v273\_eg\_gauge\_running.py** (**QFT4D.SPERT.03**) **derives the SM one-loop  $b_i$  via the EG gauge self-energy**. **[E]** the gauge 2-point is the same scaling-degree-4 extension as the quartic (**v271**), giving  $(b_1, b_2, b_3) = (41/10, -19/6, -7)$  from the carrier/SM content (the same 41 as the carrier algebra, **v159**); **[O]** two-loop + all-order stay open. Exact  $b_i$  mirrored in Wolfram.
- **v274\_scale\_overdetermination.py** (**SCALE.OVERDET.01**) **shows the single mass anchor is over-determined**. **[E]/[C]** two independent routes to  $\bar{M}_{Pl}$  agree — gravity  $(8\pi G)^{-1/2} = 2.4353 \times 10^{18}$  GeV and cosmology (the compiler prediction  $\rho_\Lambda/\bar{M}_{Pl}^4 = (3/4\pi^2)e^{-2\alpha^{-1}}$  with  $\rho_\Lambda^{1/4} = 2.24$  meV) =  $2.438 \times 10^{18}$  GeV, to 0.11%; the seam being a conformal  $c=8$  CFT makes “no derivable absolute scale” a theorem, not a gap. Sharpens **v153/v78**.
- **v275\_qgamb\_roadmap.py** (**QGAMB.ROADMAP.01**) **gives the open gate QG.AMB.01 a module**. **[E]** Tier A gap decoupling ( $\Delta_{eff} = 1.648 > 0$ , **v36/v76**); **[E]/[C]** Tier B reduction to the holomorphic  $(E_8)_1$  boundary net (**v77**); **[C]** the perturbative layer in hand (**v269/v271/v273**); **[O]** the nonperturbative ambient measure stays the one structural frontier.
- Suite 271/271 Wolfram, **run\_all** green, **AUDIT OK**, build green.

## 2026-06-18 (4D QFT: a concrete Epstein–Glaser one-loop quartic renormalization, **v271**)

- **v271\_eg\_oneloop\_quartic.py** (**QFT4D.SPERT.02**) **takes the **v269**  $S_{pert}$  skeleton from “exists” to an actual EG number** (no path integral). **[E]** the first renormalization is the order-2 product  $T_2$ ; the massless propagator has scaling degree  $sd=2$  in  $d=4$ , the one-loop bubble has  $sd=4=d$ , so  $\omega=sd-d=0 \Rightarrow$  exactly one logarithmic local counterterm ( $C(\omega+d, d)=1$ , finite, no infinite tower); the loop factor  $\Omega_3/(2(2\pi)^4) = 1/(16\pi^2)$  exactly — the *same* factor that normalises the spectral-action  $a_4$  geometric quartic. **[C]** the  $\phi^4$  one-loop  $\beta_\lambda = 3\lambda^2/(16\pi^2)$  (symmetry  $3 = s, t, u$  channels), EG initial condition  $\lambda_\Phi(UV) = 1/(16\pi^2)$ . **[O]** order-2/one-loop only; the all-order EG construction and the nonperturbative measure **QG.AMB.01** stay open. Exact core mirrored in Wolfram (270/270). **[E]** build green, **AUDIT OK**.

## 2026-06-18 (PMNS: the complete complex matrix + the leptonic Jarlskog CP strength, **v270**)

- **v270\_pmns\_jarlskog\_assembly.py** (**FLAV.PMNS.03**) **assembles the four TFPT channels** ( $\sin^2 \theta_{12} = \frac{1}{3} - \frac{\varphi_0^{ret}}{2}$ ,  $\sin^2 \theta_{23} = \frac{1}{2}$ ,  $\sin^2 \theta_{13} = \varphi_0^{ret} e^{-5/6}$ ,  $\delta = 4\pi/3$ ) **into one exactly-unitary**

**complex PMNS matrix and derives the leptonic CP strength.** [E]  $U = R_{23}U_{13}(\delta)R_{12}$  is unitary; [C] the Jarlskog  $J_{\text{PMNS}} = \Im(U_{e1}U_{\mu 2}U_{e2}^*U_{\mu 1}^*) = -0.02965$  ( $J_{\text{max}} = 0.03424$ ) is a *derived* number, the leptonic analogue of the CKM  $J$  (v88); [C] data:  $J_{\text{max}}$  vs NuFIT 5.2 0.0332 ( $\sim 3\%$ ),  $J$  vs best fit  $-0.026$ , the negative CP-violating sign reproduced. [O]  $\delta$  is the  $\mu_6$ /triality phase (assembled, not from  $M_\nu/D_F$ ) and the absolute  $\nu$ -mass scale stays open — closes the CP-*strength* readout, not the phase origin nor the scale. [E] build green, AUDIT OK.

## 2026-06-18 (4D QFT: the perturbative S-matrix as an Epstein–Glaser / pAQFT skeleton, v269)

- **v269\_spert\_paqft\_skeleton.py** (QFT4D.SPERT.01) **builds the honest middle layer of the three-layer S-matrix** (the first concrete 4D-QFT step beyond the spectral-action contract), *not* a nonperturbative closure. [E] three distinct layers:  $S_{\text{top}}$  (2d DHR braiding  $|M|=1$ , statistics, v243),  $S_{\text{pert}}$  (4d Epstein–Glaser/BV S-matrix of the spectral action),  $S_{\text{phys}}$  (LSZ on the OS Wightman functions, v240). [E] the spectral-action  $a_4$  interaction is power-counting renormalizable (all operators  $\dim \leq 4$ ; 7 marginal + 3 relevant, a finite counterterm basis), so [C] by the Epstein–Glaser theorem the perturbative S-matrix exists order by order (causal factorisation + finite local extension; no path integral, no UV divergences). [C] the gap  $\Delta = 6 \log(3/2)$  gives an IR-safe adiabatic limit ( $S_{\text{pert}} \rightarrow S_{\text{phys}}$  via LSZ). [O] perturbative only — the ambient measure QG.AMB.01 stays open, the canonical 4d reading remains boundary-only (v265). [E] build green, AUDIT OK.

## 2026-06-18 (PMNS: the reactor-angle exponent closed as the carrier hypercharge trace, v268)

- **v268\_theta13\_carrier\_trace.py** (FLAV.TH13.01) **closes the  $\theta_{13}$  exponent origin.** The value  $\sin^2 \theta_{13} = \varphi_0^{\text{ret}} e^{-5/6} = 0.0231$  was already checked (v16); now the exponent is exact: [E]  $\gamma = 5/6 = \text{tr}_E Y^2$ , the carrier hypercharge-squared trace ( $Y = \text{diag}(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2})$ , roots of  $6Y^2 - Y - 1 = 0$ ; complement  $\frac{1}{6}$  the neutrino ratio). [E] own channel: the  $\mu\tau$ -breaking  $\varepsilon = \frac{3}{4}\varphi_0^{\text{ret}}$  induces only  $\sim 1.6 \times 10^{-3} \ll 0.023$ , so  $\theta_{13}$  is a separate carrier-trace channel — correcting the tempting (and wrong) “ $\theta_{13}$  from  $M_\nu$ ” route flagged by the new script atlas. [O] the CP magnitude and the absolute  $\nu$ -mass scale remain the open PMNS items. Exact core mirrored in Wolfram. [E] build green, AUDIT OK.

## 2026-06-18 (Infrastructure: a generated causal “script atlas” so the suite is navigable at a glance)

- **verification/make\_script\_atlas.py** generates **verification/script\_atlas.md** — a single grouped, dependency-aware view fusing the existing single sources (**status\_ledger.csv** for claim→status→dependencies→supersedes, **script\_registry.csv/script\_clusters.csv** for the theme grouping, **docs\_map.csv** for where each script is cited). It lists every registered script by cluster with its claim(s), four-class marker, one-liner, resolved dependencies and citing papers, plus a *supersede map* (which old claim each new one replaces) and a dependency/most-depended-on overview — the antidote to duplicating work or forgetting superseded results. No new data, a re-projection only; wired into **bash build.sh gen**. [E] build green, AUDIT OK.

## 2026-06-18 (The bedrock in its minimal-axiom form: a rigidity theorem for QGEO.SYM.01, v267)

- **v267\_qgeo\_rigidity\_minimal\_axiom.py** (QGEO.SYM.02) **sharpens the one bedrock postulate to its irreducible kernel** (it does *not* close it). [E] the *single* bare statement “the seam admits the carrier order-4 clock as a conformal symmetry permuting its four marks”

( $= \omega \circ \rho = \omega$ ) forces *everything*: an order-4 Möbius orbit  $\{a, ia, -a, -ia\}$  has cross-ratio 2 independent of  $a$ ; cross-ratio 2  $\Leftrightarrow j=1728$  (only  $\lambda \in \{-1, \frac{1}{2}, 2\}$ )  $\Leftrightarrow$  the order-4 CM automorphism on  $y^2=x^3-x$ ; and the unique flat pillowcase metric, mark-locality,  $[\rho, \Lambda]=0$  and  $\omega \circ \rho = \omega$  all follow (v214/v201/v264/v198). [E] neg controls: generic config  $j \neq 1728$  ( $\mathbb{Z}_2$  only), hexagonal  $j=0$  ( $\mathbb{Z}_6$ , order 6) — only the square (order-4) realises the  $\mu_4$  clock. [C] second, independent reason:  $\{1, i, -1, -i\}$  over  $\mathbb{Q}$  with  $j=1728$  (CM by  $\mathbb{Z}[i]$ ), Belyi/dessins Galois-rigidity. [O] the bare order-4 symmetry stays the one foundational postulate (like  $c=\text{const}$ ) — the *reduction* is closed (axiom  $\Rightarrow$  everything), the axiom is not. [E] build green, AUDIT OK.

## 2026-06-18 (Branch B decided: the explicit PS threshold model + proton-decay window — minimal content killed, $+(15, 1, 1)$ survives, Hyper-K decisive, v266)

- **v266\_ps\_threshold\_proton.py** (PS.PROTON.02/PS.THRESHOLD.02) decides branch B of the **v265 fork**. The explicit two-step Pati–Salam thresholds give  $d=6$   $\tau_p(p \rightarrow e^+ \pi^0)$  (writes `ps_threshold_proton.json`): [X] the minimal 16-Higgs content ( $M_{GUT} \sim 2.4 \times 10^{15}$ ) gives  $\tau_p \sim 4.4 \times 10^{33}$  yr  $<$  Super-K and is *killed* even at the high hadronic band; [C] the  $E_8$ -allowed  $(15, 1, 1)$  [the single 45] raises  $M_{GUT} \sim 5.9 \times 10^{15} \Rightarrow \tau_p \sim 1.6 \times 10^{35}$  yr, proton-safe *now*; [X] but that sits at the Hyper-K reach ( $\sim 1.4 \times 10^{35}$  yr) — a *dated* kill test decisive within the decade. So branch B is promoted from “well-mannered candidate” to *content-selected* (*must be*  $+(15, 1, 1)$ ), *Hyper-K-decisive UV branch*; [O] structurally marginal (only one 45 in the hull, v247) with an  $O(3)$   $\tau_p$  band. Branch A (boundary-only) is unchanged and canonical. [E] build green, AUDIT OK.

## 2026-06-18 (The 4D-QFT fork frozen as a branch policy: boundary-only canonical, Pati–Salam a falsifiable UV branch, SM-only GUT killed, v265)

- **v265\_qft4d\_fork\_freeze.py** (QFT4D.FORK.01) turns the 4D-QFT fork from an open ambiguity into a frozen decision tree (`qft4d_fork_freeze.json`), no new physics. **A (canonical)**: boundary-only is the default 4D reading — a 4D-GUT is *not claimed by default* ( $E_8$  is the audit hull). [X] SM-only 4D-GUT unification is killed (1-loop: at the  $\alpha_1=\alpha_2$  scale  $\alpha_3^{-1}$  off by  $\sim 5.6$ , spread  $\sim 8.8$  at  $2 \times 10^{16}$  GeV, re-deriving v246) — the canonical expectation, not a failure. **B (conditional)**: carrier-native Pati–Salam is a falsifiable UV branch only with the  $E_8$ -allowed content  $\{1, 10, 16, 45\}$  (no **126**, v247),  $\sin^2 \theta_W = 3/8$  (v248), the  $O(1)$   $\kappa = M_{PS}/M_s$  band (v249/v253), a threshold model and a proton-decay kill test ([X], no fake  $\tau_p$  window; carrier gauging stays the contract). [E] a prose guard (QFT4D.NO\_OVERCLAIM.01) forbids bare 4D over-claims and requires the scoped policy wording. New ledger rows QFT4D.BOUNDARY.DEFAULT.01, QFT4D.SMONLYGUT.01, PS.UVBRANCH.01, PS.THRESHOLD.01, PS.PROTON.KILL.01. A fork-policy box added to `tfpt_research_contracts`; [E] build green, AUDIT OK.

## 2026-06-18 (Three open-residual advances: the $F_{\text{QCD}}$ solver, $M_\nu$ from $D_F$ , and the raw-seam $\Rightarrow$ mark-locality reduction, v262–v264)

- **v262\_fqcd\_mp\_me.py** (FR.MPME.02) implements the one contract-only  $F_{\text{transfer}}$  pipeline. A real two-loop  $\alpha_s$  solver runs  $\alpha_s(M_Z)=0.1179$  with  $n_f$  thresholds (carrier  $b_3 = -7$ , [E]) to  $\alpha_s(2 \text{ GeV}) \sim 0.30$  and  $\Lambda_{\overline{\text{MS}}}^{(3)} \sim 0.33 \text{ GeV}$ ; with the lattice bridge  $C_p$  (external  $O(1)$ ) and the closed  $m_e$  it reproduces  $m_p/m_e \sim 1824$  vs the measured 1836.15 within the  $\sim 7\%$  external budget. A [C] transfer reproduction (firewall: no compiler integer for 1836), advancing FR.MPME.01 ([O]) to FR.MPME.02 ([C]).
- **v263\_mnu\_from\_df.py** (FLAV.PMNS.02) makes  $M_\nu$  a seesaw operator of  $D_F$ . [E]  $M_\nu = -m_D M_R^{-1} m_D^\top$  is symmetric, seesaw-suppressed and (for  $\mu\tau$ -symmetric inputs)  $\mu\tau$ -symmetric;



[C] it reproduces  $\theta_{23}=45^\circ$  and  $\sin^2 \theta_{12}=\frac{1}{3}-\frac{\varphi_0^{\text{ret}}}{2}$  (v9) under the seam misalignment; [O]  $\theta_{13}=0$  in the leading texture, the CP *magnitude* ( $240^\circ$ ) is the  $\mu_6$ /triality phase (v231/v233), and  $M_R$  is the free seesaw scale. An operator advance, not a closure of full PMNS dynamics.

- **v264\_raw\_seam\_marklocal.py (QGEO.ENERGY.03) sharpens the bedrock.** Composing v216+v201+v198 on the canonical pillowcase: the flat-pillowcase curvature lives only at the 4 marks, so the DtN sub-principal symbol is  $\mathbb{Z}_4$ -invariant (Fourier on  $4\mathbb{Z}$ , FFT-verified), block-diagonal, and  $[\rho, \Lambda]=0 \Rightarrow \omega \circ \rho=\omega$  on this metric. So the whole bedrock collapses to the single metric statement “the raw RP seam carries the uniformising flat pillowcase metric” (=QGEO.SYM.01). A reduction/sharpening, [O] — it does *not* prove the seam must carry that metric (negative control: curvature off the  $\mu_4$  orbit breaks the chain). Paper citations added in tfpt\_4\_frontier, tfpt\_2\_standard\_model and tfpt\_research\_contracts. [E] AUDIT OK, build green.

**2026-06-18 (Paper consistency pass: the closure ledger and the status/QFT sections updated for the Modular Spectral Closure, with an explicit “what is *not* closed” line)**

- **Five paper sections that predated the v238–v261 round are brought up to date** (additive, no status marker moved). The introduction closure-ledger table and the “five questions” item now carry a *boundary QFT (Modular Spectral Closure)* row — one relative object  $\text{TFPT}_{\text{QFT}} = (\mathcal{A}_\Sigma, \omega_\Sigma, \Delta_\Sigma, \rho, A_F, H_F, D_F, J, \gamma, S_{\text{rel}})$  closed modulo QGEO.SYM.01 (v258–v261) — plus an explicit list of what the closure does *not* reach: the full measured pole masses, the absolute QCD scales ( $m_p/m_e$ ), the complete PMNS dynamics, and the real 4D interacting QFT / ambient measure. The tfpt\_2 “ambient closure” section now states that the cutoff of its  $S_{\text{rel}}$  is the seam KMS weight ( $f_2/f_0=1$ , v259) and  $D_{\text{rel}}$  the covariance induction (v258); the tfpt\_2 “(4) Constructive QFT closure” section adds the relative-object pointer; tfpt\_4\_frontier’s full-quantum-gravity section records that the cutoff moment  $f_0$  is pinned by the seam state (v255/v259); and origin\_theory’s honest-status keybox notes that the whole boundary QFT collapses onto the same QGEO.SYM.01 postulate (no sixth structural class). An editorial consistency pass so every status surface agrees with the ledger. [E] build green (10 ok, 0 failed), AUDIT OK.

**2026-06-18 (Consistency pass: the Modular Spectral Closure surfaced structurally across the website and the intro status card)**

- **The v258–v261 closure is now reflected on every “what is open” surface, not just the research-contracts paper.** The homepage open-gates section and compiler-pipeline diagram, the hostile-referee FAQ, the glossary  $G_{\text{net}}$  entry, the reviewer page and the verification residual-chain caption now state that the *entire* emergent-QFT layer (Dirac, cutoff, gauging, glue, orientability) collapses onto the *same* premise QGEO.SYM.01 via the Modular Spectral Closure — the boundary QFT is one relative object that adds *no new open item*, ambient QG separate. The introduction status card (“the structural residual is one condition”) carries the same sentence. No status marker moved; this is an editorial consistency pass so the public surfaces agree with the ledger and the paper. [E] build green, AUDIT OK.

**2026-06-17 (The Modular Spectral Closure: Dirac as covariance induction, the modular cutoff, the Kummer/K3 unification, and the assembly certificate, v258–v261)**

- **v258\_dirac\_covariance\_induction.py (PS.DIRAC.03) closes the “posited Yukawa vs derived operator” gap of v250/v252.** The finite Dirac operator  $D_F$  is the modular/covariance *induction* of the boundary seam quasi-free (KMS,  $\beta=1$ ) state: a Gaussian state is fixed by its covariance  $C$  (a positive contraction) with one-particle modular Hamiltonian



$H = \log((1-C)C^{-1})$  (Araki; Casini–Huerta), inverse to the KMS occupation  $C = (1+e^H)^{-1}$ . [E] the inversion identity holds exactly; [E]  $C_F$  is a positive contraction; [E] the induced operator is chirality-odd; [E] it is unitarily equivalent to the v252  $D_F$  (same mass spectrum); [E] the Majorana/seesaw  $m_D^2/M_R$  is an off-diagonal covariance entry. [C]  $C_F = P_F C_\Sigma P_F$ , so  $D_F$  is the Kasparov shadow  $[D_\Sigma] \hat{\otimes}_{(E_8)_1} [\mathcal{K}_{\text{car}}]$  of the boundary geometry and the v18 Yukawas are the readout of  $C_\Sigma$ .

- **v259\_modular\_cutoff\_kappa.py (PS.SPECACT.02) fixes the last open spectral-action input.** The cutoff  $f$  is the seam’s own modular/KMS weight: by Tomita–Takesaki/Bisognano–Wichmann the seam is KMS at  $\beta=1$  (v239) and  $2\pi=1/(4c_3)$  (v58) fixes  $\beta$ , so the heat-kernel cutoff  $f(u) = e^{-u}$  is *derived*. [E] its moments give  $f_2/f_0 = f_4/f_2 = 1$  exactly; [E] a generic Gaussian cutoff gives  $\sqrt{\pi}/2 \neq 1$  (the choice is non-vacuous); [E] hence  $\kappa = \sqrt{(f_2/f_0) c_{PS}/c_{\text{grav}}} = \sqrt{c_{PS}/c_{\text{grav}}}$  loses its scheme factor. [C] the open [O] cutoff freedom of v253/v255 becomes a finite trace ratio over the 96-dim  $H_F$  (the RG  $\kappa \approx 1.3$  pins  $c_{PS}/c_{\text{grav}} \approx 1.7$ ).
- **v260\_k3\_kummer\_unification.py (ARCH.K3.01) unifies seam, carrier and lattice on one surface.** The pillowcase is the elliptic-involution quotient  $T_{\tau=i}^2/(z \mapsto -z)$  (v214); squaring it gives the Kummer/K3 surface, whose three facets are three TFPT objects. [E] the  $E_8$  Cartan is even/det 1/positive-definite; [E]  $H^2(K3, \mathbb{Z}) = U^3 \oplus E_8(-1)^2$  (rank 22, det  $-1$ , even, signature  $(3, 19)$ ); [E] the seam is  $T^2/(z \mapsto -z)$  with 4 marks (orbifold Euler char 0); [E] the carrier-16 are the  $|A[2]|=2^4=16$  Kummer nodes  $= \dim S^+$  (v197),  $16=4 \times 4$ ; [E] honest note — the rank-16 Kummer lattice is *not* the  $E_8(-1)^2$  summand. [C] so the v1 “glue  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ ” is the lattice shadow of a single object;  $E_8$  appears both locally (du Val  $\mathbb{C}^2/2I$ , v232/v236) and globally ( $H^2$  summand). A unification, not a closure.
- **v261\_modular\_spectral\_closure.py (QFT.MSC.01) is the assembly capstone.** The whole QFT round (v238–v260) is certified as *one* relative object  $\text{TFPT}_{\text{QFT}} = (\mathcal{A}_\Sigma, \omega_\Sigma, \Delta_\Sigma, \rho, A_F, H_F, D_F, J, \gamma, S_{\text{rel}})$  reduced to *one* open premise. [E] the cross-consistency invariants agree:  $4 = [B:A] = |\mu_4| = 2\chi = |(\mathbb{Z}/2)^2|$  (index = marks = fixed points = glue order), one carrier-16 (spinor = lattice = triple), one gap  $6 \log(3/2)$  (OS = recovery = Lindblad); [E] the three closures re-derived; [E] the ten-component manifest present. [O] the QFT layer reduces to QGEO.SYM.01 (seam realisation), the ambient QG measure QG.AMB.01 kept separate. [C] the *Modular Spectral Universality* master statement: the unique RP holomorphic index-4 KMS seam state induces the 96-dim KO-6 triple whose inner fluctuations give the gauged Pati–Salam spectral action — the honest sense in which the boundary QFT is complete.

**2026-06-17 (Infrastructure: the changelog is now a public website page, generated from changelog.tex and audit-gated)**

- **The canonical changelog now also drives a public /changelog website page.** A new generator `verification/make_changelog_web.py` parses this file (dated `\subsection*` entries, `itemize` items, and the inline markup `\textbf`/`\emph`/`\texttt`/`\vref`, the status markers [E]/[C]/[O]/[X], inline math and accents) into the typed data file `website/lib/changelog.ts`, rendered server-side (math via KaTeX) by `website/app/changelog/page.tsx` + `website/components/Changelog.tsx`. The generator is wired into `bash build.sh gen`; its freshness is enforced by `audit_sync.py` (section A.generated), so the page can never drift from this file – editing the changelog and re-running `gen` is the only supported path (the `.ts` mirror is never hand-edited). `/changelog` is linked from the navbar, footer and sitemap. The `tfpt-workflow`, `website-sync` and `sync-maps` cursor rules were updated to record the new single-source generation step. [E] `build (npm run build green, /changelog prerendered static)` and `bash build.sh audit (AUDIT OK)`.

## 2026-06-17 (Orientability + Poincaré duality: the honest, literature-confirmed status — strict fail, weakened hold, v257)

- **v257\_quasiorient\_modpd.py** (PS.NCG.ORIENT.02) **corrects the slightly too-generous v256 typing**. The published result (Cacic–Stephan arXiv:0902.2068; CCM07; SM-vacuum hep-th/0601192) is that the Chamseddine–Connes–Marcolli Standard-Model finite triple (KO-dimension 6, with three right-handed neutrinos — exactly the TFPT seesaw case) *genuinely fails* strict orientability and strict Poincaré duality, and instead satisfies the physically correct weakened versions. Verified on the explicit **v252** triple: **[E]** strict orientability fails ( $\gamma \notin \text{span}\{\lambda(a)\rho(b)\}$ ); **[E]** *quasi-orientability* holds (the left/right  $A$ -irrep boxes are disjoint by chirality:  $L = H$  weak-doublet,  $R = \mathbb{C}$  singlet, so  $L^{\text{LR}}(\mathcal{H}^{\text{even}}, \mathcal{H}^{\text{odd}}) = \{0\}$ ); **[E]** strict Poincaré duality fails (KO-6 skew intersection form, corank 1, the equal- $L/R$ -neutrino case); **[C]** modified Poincaré duality holds (CCM), Stephan’s alternative triple (St06) restoring strict PD with identical physics. A known property of the NCG-SM model class, not a TFPT defect — no faked closure.

## 2026-06-17 (Remaining NCG obligations discharged or honestly typed: inner fluctuations, spectral action, orientability/Poincaré, v254–v256)

- **v254\_inner\_fluctuations.py** (PS.NCG.FLUCT.01) **closes v250 #5 + the no-junk obligation**. The inner fluctuations  $\Omega_D^1(A_F)$  of the finite Dirac operator are computed explicitly: **[E]** they are purely chirality-odd (scalars), and for the SM algebra sit *exactly* in the one Higgs doublet  $(1, 2)_{1/2}$  — no junk, no triplet, no  $(10, 1, 3)$ /**126**. **[E]** the  $\sigma$  ( $\nu_R$  Majorana) direction is absent for the SM algebra but present for Pati–Salam (the CCvS beyond-first-order scalar), and **[E]**  $\sigma$  is an SM singlet  $(1, 1, 0)$  with  $B-L = 2$ . **[C]**  $\sigma$  = scalaron.
- **v255\_spectral\_action\_expansion.py** (PS.SPECACT.01) **closes the spectral-action expansion (v252 #11)**. The Seeley–DeWitt expansion  $S = 2f_4\Lambda^4 a_0 + 2f_2\Lambda^2 a_2 + f_0 a_4$ : **[E]** structure ( $a_0$  cosmological,  $a_2$  Einstein–Hilbert+Higgs-mass,  $a_4$  Yang–Mills+Weyl<sup>2</sup> +  $R^2$ +Higgs-quartic,  $N = 96$ ); **[E]**  $b/a^2 = 1/3$  (top-dominated); **[E]**  $\sin^2 \theta_W = 3/8$ ; **[E]** the  $R^2$  spin-0 mode = the Starobinsky scalaron; **[C]** Higgs quartic  $\rightarrow \sim 125$  GeV with  $\sigma$ ; **[C]**  $\kappa = \sqrt{(f_2/f_0)c_{PS}/c_{\text{grav}}}$  explicit; **[O]** exact decimal needs the cutoff moments.
- **v256\_orientability\_poincare.py** (PS.NCG.ORIENT.01) — **honest status, no faked proof**. KO-dimension 6 forces the intersection form  $\text{Tr}(\gamma \pi(p) J \pi(q) J^{-1})$  to be *antisymmetric*: **[E]** skew form (verified), **[E]** rank 2, corank 1, null  $\sim (1, 1, -1)$  on the SM K-theory  $\mathbb{Z}^3$ ; **[E]** the orientability local condition  $[\gamma_F, \pi(a)] = 0$ . **[C]** the canonical hypercharge-faithful SM/PS triple satisfies orientability and Poincaré duality (van Suijlekom; CCM); **[O]** a self-contained machine proof needs the complete canonical bimodule.

## 2026-06-17 (Full 96-dim finite spectral triple built + NCG axioms verified, and $\kappa$ pinned to $O(1)$ , v252–v253)

- **v252\_full\_finite\_triple.py** (PS.DIRAC.02) **advances the Dirac contract v250 from a 7-obligation contract to a concrete object discharging 5 to [E]**. The full 96-dim finite spectral triple  $(A_F, H_F, D_F, J, \gamma)$  —  $H_F$  the particle $\oplus$ antiparticle doubling of three  $SO(10)$  16-plets — is built explicitly and the NCG axioms are checked by direct computation: **[E]** reality ( $J^2 = +1$ ,  $JD = DJ$ ), grading ( $\gamma^2 = 1$ ,  $[\gamma, a] = 0$ ,  $J\gamma = -\gamma J$ ), KO-dimension 6, order-zero, the first-order condition (HOLDS for the SM algebra including the Majorana; VIOLATED for the Pati–Salam algebra exactly by the Majorana, localized to  $\{\nu_R, e_R\}$  — the CCvS beyond-first-order mechanism on the full triple), and the spectrum of  $D_F$  reproducing the fermion masses and the type-I seesaw  $m_{\text{light}} = m_D^2/M_R$ . **[C]** Poincaré duality; **[O]** orientability, no-junk, spectral-action expansion.

- **v253\_kappa\_scalaron\_ps.py** (PS.KAPPA.01) discharges residual 6(b) of **v249**.  $\kappa$  in  $M_{PS} = \kappa M_s$  ( $M_s = c_3^{7/2} \bar{M} \approx 3.06 \times 10^{13}$  GeV, exact) is pinned by an RG scan (PDG errors  $\times$  the two  $E_8$ -allowed contents) to a tight  $O(1)$  interval ( $\approx 1.3$ ). [C] heat-kernel origin:  $\kappa = \sqrt{(f_2/f_0) c_{PS}/c_{\text{grav}}}$  is a scale-free ratio of heat-kernel traces over the 96-dim  $H_F$  ( $\Lambda$  cancels). [X] the **126**-content drives  $\kappa$  out of the  $O(1)$  window (content-selective). [O] the exact decimal needs  $f_2/f_0$  fixed.

**2026-06-17 (Spectral triple, first step: the first-order condition is violated exactly by the Majorana/ $\sigma$  — the CCvS Pati–Salam mechanism, v251)**

- **v251\_first\_order\_sigma.py** (PS.DIRAC.01) advances the Dirac contract **v250** from a contract to an exhibited mechanism. On a faithful lepton-block model ( $H = H_F \oplus H_F^c$ ,  $H_F = \mathbb{C}^3$ ), the real even structure is checked ( $J^2=+1$ ,  $\gamma^2=1$ ,  $D\gamma=-\gamma D$ , KO-dim-6 signs), order-zero holds, the first-order condition HOLDS for the Yukawa block, and is VIOLATED *exactly* by the Majorana ( $\nu_R \nu_R^c = \sigma$ ) block — the Chamseddine–Connes–van Suijlekom “beyond first order” mechanism by which the inner fluctuations promote the SM to Pati–Salam and generate  $\sigma$ . Honest correction to the review:  $\sigma$  does *not* rescue the first-order condition; its controlled violation *is* the mechanism. [C]  $\sigma = \text{scalaron}$  (v249); [O] the full 96-dim triple + spectral action.

**2026-06-17 (Carrier-native Pati–Salam program:  $E_8$  forbids the 126, PS algebra, unification kill-test, Dirac contract, v247–v250)**

- **Four scripts turning the gauge-unification finding into typed claims with negative controls.** **v247\_e8\_branching\_no126.py** (PS.E8BRANCH.01): the  $E_8$  hull branches under  $SO(10) \times SU(4)$  as  $(45, 1) + (1, 15) + (10, 6) + (16, 4) + (\overline{16}, \bar{4})$ , so the  $SO(10)$  content is exactly  $\{1, 10, 16, 45\}$  and the **126** is *absent* — the renormalisable  $126_H$  is algebraically forbidden and  $10+16+45$  is  $E_8$ -supplied (only one 45  $\Rightarrow$  proton-marginal). **v248\_ps\_algebra.py** (PS.ALGEBRA.01):  $A_F = \mathbb{H}_L \oplus \mathbb{H}_R \oplus M_4(\mathbb{C}) \rightarrow SU(2)_L \times SU(2)_R \times SU(4)_c$ , exact hypercharges  $Y = T_{3R} + (B-L)/2$ , matching  $\alpha_1^{-1} = \frac{3}{5}\alpha_{2R}^{-1} + \frac{2}{5}\alpha_4^{-1}$ ,  $\sin^2 \theta_W = 3/8$  (colour  $SU(4)_c \subset D_5$ , distinct from the family  $A_3$ ). **v249\_ps\_unification.py** (PS.RGTEST.01): the two-step  $PS \rightarrow SO(10)$  unification kill-test with negative controls (SM-only fails; 126 breaks the scalaron match; proton decay selects content),  $M_{PS}/M_s = 1.0\text{--}1.7$ . **v250\_dirac\_triple\_contract.py** (CONTRACT.QFT4D.DIRAC.01): the spectral-triple contract ( $H_F = 3 \times 16$ , the seven NCG axioms,  $\sigma = \text{scalaron}$ ) that would turn “carrier is gauged” from a fork into a theorem. The root-level **qft.txt** carries the full analysis.

**2026-06-17 (Honest data cross-check + emergent-QFT figures: the unification tension, v246)**

- **v246\_unification\_data\_crosscheck.py** (QFT4D.RGTEST.01) — the first hard data confrontation of the spectral-action prediction. TFPT’s gauge-charged content *is* the Standard Model (v159) and the theory adds no new states; the  $E_8$  cascade is an arithmetic spine (v5), not a threshold tower. So the run *is* the SM run: at one loop the couplings spread  $\sim 10^{13}\text{--}10^{17}$  GeV (residual  $\Delta\alpha^{-1} \approx 9$  at  $2 \times 10^{16}$  GeV), and at two loops (RK4) the minimal spread is  $\Delta\alpha^{-1} \approx 3.2$  at  $1.7 \times 10^{14}$  GeV — the three never meet. Typed [X]/[O]: with “no new state” there is *no* admissible threshold source, so the spectral-action 4d-GUT route is in genuine tension (same status as NCG-SM), likely falsified unless extended — never a confirmation, and *not* rescued by a cascade. Two new figures (**figures/qft\_skeleton.pdf**, **figures/qft\_unification.pdf**) added via **make\_figures.py** (PDF + website PNG) and embedded in **tfpt\_research\_contracts.tex**; a root-level **qft.txt** summarises the emergent-QFT skeleton, what each script proves, the data status and the open questions.

## 2026-06-17 (Spectral-action matter content: the $SO(10)$ $\mathbf{16}$ = one anomaly-free SM generation, v245)

- **v245\_spectral\_matter\_content.py** (QFT4D.MATTER.01) discharges the computable core of the 4d contract **v244**. The carrier half-spinor (the  $SO(10)$   $\mathbf{16}$ ) is verified, in exact rational arithmetic, to be one Standard-Model generation:  $\mathbf{16} = SU(5)(\mathbf{10} + \bar{\mathbf{5}} + \mathbf{1})$ , the SM multiplet dimensions sum to  $16 = \dim S^+$  (15 Weyl fermions +  $\nu_R$ ), the electric charges  $Q = T_3 + Y$  are exactly the SM values, all four gauge anomalies cancel ( $\sum Y = \sum Y^3 = [SU(2)]^2 U(1) = [SU(3)]^2 U(1) = 0$ ), the spectral-action normalisation gives  $\sin^2 \theta_W = 3/8$  ( $g_3 = g_2 = \sqrt{5/3} g_1$ ), and the Higgs is the  $(1, 2)_{1/2}$  inner fluctuation. So the matter-content + tree-relation half of CONTRACT.QFT4D.01 is now **[E]**; only the seam→Dirac realisation and the run-down couplings stay **[O]**. Wired into `run_all.py`, the ledger, the registry, cited in `tfpt_research_contracts.tex`, mirrored to the website.

## 2026-06-17 (The QFT skeleton on the seam: thermal time, GNS/OS reconstruction, DHR sectors, the braiding S-matrix, and the 4d spectral-action contract, v239–v244)

- **Six emergent-QFT skeleton scripts (v238)**. `v239_kms_thermal_time.py` (DYN.KMS.01): the modular flow is KMS at  $\beta = 1$ , and  $2\pi = 1/(4c_3)$  fixes the modular temperature as  $T_H = c_3/M$  — thermal time = horizon time. `v240_gns_os_reconstruction.py` (DYN.GNS.01): GNS reconstructs the Hilbert space from  $(\mathcal{M}, \omega)$  and the OS Hamiltonian  $H_{OS} = -\log T = -L$  is positive with gap  $\Delta$  (the **v64** mass gap). `v241_dhr_sectors.py` (DHR.SECTORS.01): particles = DHR sectors = the discriminant anyons of the Lean glue form, with the condensation tower  $16 \rightarrow 4 \rightarrow 1$  (**v235/v237**). `v242_spin_statistics_modular.py` (DHR.MODULAR.01): spin-statistics, the modular  $(S, T)$  data, Verlinde fusion, and Gauss–Milgram  $c = 8$ . `v243_haag_ruelle_braiding.py` (DHR.SMATRIX.01): the 2-particle S-matrix = the braiding monodromy (factorised, crossing), with the **v238** mass gap supplying Haag–Ruelle asymptotic states. `v244_spectral_action_contract.py` (CONTRACT.QFT4D.01): the 4d Lagrangian via the Connes spectral action — the carrier half-spinor  $\mathbf{16}$  is one generation, the SM gauge group sits in the carrier, gravity is **v36**; the matter Lagrangian is the open **[O]** contract. Net: the residual to a full 4d QFT is *one premise* (QGEO.SYM.01) plus *one contract* (CONTRACT.QFT4D.01). All wired into `run_all.py`, the ledger and registry, cited in `tfpt_research_contracts.tex`, mirrored to the website.

## 2026-06-17 (The static→dynamic flip: modular flow + a recovery Lindblad generator, v238)

- **New v238\_modular\_lindblad\_dynamics.py** (DYN.SEMIGROUP.01, **cluster registry**). A companion to the modular round (**v196/v198**) and the recovery code (**v221/v64**) that reads the *same* seam objects *dynamically* rather than statically. **[E]** the modular flow  $\sigma_t = e^{it\Lambda_\Sigma}$  is a unitary one-parameter group (Tomita–Takesaki) whose conservation of the  $\mu_4$  clock,  $\rho \sigma_t \rho^{-1} = \sigma_t \Leftrightarrow [\rho, \Lambda_\Sigma] = 0$ , IS the **v198** state-invariance — so the static commutator is the conservation law of modular time. **[E]** the recovery channel’s generator  $L = \log T$  is a doubly-stochastic/CPTP-classical Markov generator ( $e^L = T$ ,  $e^{L\tau} \rightarrow$  the democratic projector = the arrow of time). **[E]** keystone: the dissipative gap  $= -\log((2/3)^6) = 6 \log(3/2) =$  the OS mass gap  $\Delta$  (**v64**) — the static recovery bound (**v221**) and the dynamical mass gap (**v64**) are one number, one exponentiated. **[E]** the GKSL split  $G = i\Lambda_\Sigma + L$  (imaginary + real  $\leq 0$  spectrum). **[O]** the residual is the *geometric* modular flow (Connes–Rovelli “thermal time = physical time”) on the full seam algebra = QGEO.SYM.01 (**v198**) — a target, not a closure. Wired into `run_all.py`, `status_ledger.csv`, `script_registry.csv` and cited in `tfpt_research_contracts.tex`; mirrored to `website/public/verification`.



## 2026-06-17 (Experiments: black-hole-cosmology signatures, the $\eta_B$ Boltzmann solve, the Petz/baby-universe realisation, and a real-data GW-echo pipeline)

- **Three new standalone black-hole-cosmology experiments (experiments/, all data\_limited, not load-bearing).** From the problem\_b black-hole-cosmology survey: (i) **ccbh-dark-energy** — the de Sitter seam interior ( $w_{\text{in}}=-1$ ) forces the Croker–Weiner cosmological-coupling index  $k = -3w_{\text{in}} = 3$ , hence a population  $w = -1$ ; vs. Farrah+2023  $k = 3.11 \pm 0.79$  ( $-0.14\sigma$ ), a *contested* downstream bridge and an *alternative reading* of the same  $w = -1$  the DESI watchdog tests (**alternative\_group w\_de\_eos**, never double-counted). (ii) **gravastar-compactness** — the Nariai  $Q_{\text{geom}}=3/8$  equals the Jampolski–Rezzolla (2026) maximum horizonless compactness  $\mathcal{C}=3/8$  (shared de Sitter endpoint  $1/2$ ); since  $1/3 < 3/8 < 4/9 < 1/2$  this is a light-trapping ECO, yielding an echo template (round-trip delay  $\sim 0.7$  ms at  $62 M_\odot$ , amplitude  $\leq (2/3)^6$ ) — a **[C]** structural echo, no  $\mathcal{C} \leftrightarrow Q_{\text{geom}}$  map proven. (iii) **cosmic-handedness** — a parity watchdog (Shamir JADES  $\sim 3.3\sigma$  spin monopole, most likely a Milky-Way-aberration dipole), explicitly *not* promoted (frontier).
- **$\eta_B$  — full BDP Boltzmann ODE solve confirms the route at the frozen scale** (**experiments/ftransfer/leptogenesis\_boltzmann/fboltzmann\_solve.py; tfpt\_4\_frontier, [C]**). Integrating the Buchmüller–Di Bari–Plümacher Boltzmann network (integrated efficiency  $\kappa_f = 0.092$ , validating the analytic fit) at the *frozen* heavy scale  $M_1 = M_{\text{scal}}(\varphi_0^{\text{ret}})^2/A_\Lambda \approx 8.65 \times 10^9$  GeV with  $\delta_{\text{CP}} = 4\pi/3$  gives  $\eta_B = 6.5 \times 10^{-10}$  — a factor 1.07 from the observed  $6.1 \times 10^{-10}$ , with *no* free  $M_R$  dial. This moves the leptogenesis route from “solve pending” to a consistent **[C]** readout (the full *flavored* density-matrix solve is the next refinement); **tfpt\_4\_frontier** Route C synced.
- **The Petz recovery map and the rank-one baby universe, realised numerically (experiments/recovery-channel; tfpt\_horizon\_readouts, [C]; companion to v221).** The gapped transport contracts to a *rank-one* projector at exactly  $\|T^n - P_\infty\| = (2/3)^{6n}$  — the one-dimensional baby-universe Hilbert space (Engelhardt 2025) — and the explicit Petz map is CPTP, recovers the reference, and equals the identity only on the protected  $\lambda=1$  (Knill–Laflamme) mode; free-ratio and degenerate-spectrum negative controls hold. So the Petz identification **v221** deferred is now verified at the channel level (**internal\_consistency**, no new datum); the full holographic reading stays gated on the Seam–Horizon theorem.
- **GW ringdown echo — Stage-1 matched filter validated, then run on real GWOSC strain (experiments/gw-ringdown-echo; tfpt\_horizon\_readouts search-targets).** The Stage-1 machinery (Kerr-ringdown subtraction  $\rightarrow$  matched filter on the residual, ratio  $(2/3)^6$  frozen, lag free, + free-ratio control) classifies 3/3 synthetic injections; run on *real* GWOSC strain (GW150914, GW190521; PSD-whitening + off-source background): **no kernel-ratio echo coincident in  $\geq 2$  detectors**, consistent with the  $(2/3)^6$  *upper* bound (a single loud H1 excess has  $\hat{q} \approx 2$ , far from  $(2/3)^6$ , so the free-ratio control flags it as residual ringdown, not an echo). First pass: dominant-QNM subtraction, incoherent combination — full multi-mode + coherent stacking is the next step; no detection claim.
- **Scorecard + website synced.** **experiments/evidence\_scorecard.json** now 51 rows (26 consistent, 7 null, 13 data-limited, 4 tension, 1 parked); the website **EXPERIMENTS\_AUDIT** mirror and the  $\eta_B$  prediction entry are updated to match. These are standalone **experiments/** surfaces (not in the verification suite or ledger); the **tfpt\_4\_frontier/tfpt\_horizon\_readouts** and **papers.ts/predictions.ts** edits are the in-same-change sync.



## 2026-06-17 (Red-team layer re-synced to the paper; shadow export mirrors figures/)

- **Red Team Target A** re-synced to `tfpt_5_redteam` (`rt_A_e8net.py`, `redteam_table.txt`, `redteam/README.md`, `infrastructure`). The Python red-team surface still emitted the historical *three-residual* “A reduced, not closed” form, lagging the reader-facing note. It now encodes the sharper narrowing already in the paper: the residual collapses to the *single* statement “the seam–Calderón boundary net is holomorphic with  $c = 8$ ” ( $\Leftrightarrow$  the index-4 simple-current extension), with the index-4  $(D5)_1 \times (A3)_1 \rightarrow (E8)_1$  glue realised at Lie level ([v143](#)) and its odd sectors twisted/Ramond ( $E_8 = 120_{\text{NS}} + 128_{\text{R}}$ , [v148](#)), while the  $q(A_3)$  normalisation collapses into the one declared anchor ([v152](#)). Two exact adversarial checks added ( $120+128 = 248$ ; glue index  $= |\mu_4| = N_{\text{fam}}+1 = 4$ ); status stays [\[C\]](#)/[\[O\]](#) (reduced, not closed).
- **Shadow export now mirrors figures/ but not website/** (`scripts/export-shadow.sh`, `verification/make_script_index.py`, `infrastructure`). `export-shadow.sh` ships `figures/*.pdf` (so `make_manifest.py --check` passes literally on the exported tree) while still excluding `website/` and the compiled paper PDFs; the `SHADOW_MIRROR.md` table is corrected accordingly. `make_script_index.py` now detects a missing `website/` and skips its `ScriptIndex.tsx` mirror, so `build.sh` notes runs on the subset without crashing.

## 2026-06-17 (CP phases as $\mu_6$ powers — a reduction of red-team Target D)

- **Both CP phases are  $\mu_6$  powers of one hexagonal CM unit, split by the sheet** ([v231\\_cp\\_mu6\\_phases](#), `tfpt_5_redteam`, [\[E\]](#)/[\[C\]](#)). Sharper than [v220/v225](#) (which only *located* the CP residual). With  $\rho = e^{i\pi/3}$  the primitive 6th root (the  $j=0$  Eisenstein CM unit):  $\delta_{\text{CKM}}$  leading  $= \arg(\rho^1) = \pi/3$  (quark) and  $\delta_{\text{PMNS}} = \arg(\rho^4) = 4\pi/3$  (lepton, the phase lattice), with  $\rho^4 = -\rho$ , so  $\delta_{\text{PMNS}} = \delta_{\text{CKM,lead}} + \pi = (\text{CM unit}) \times (\mathbb{Z}_2 \text{ sheet}, \rho^3 = -1)$ . The two CP phases are *one* hexagonal unit on the two sheets, not two independent inputs; the dual-frame Jarlskog orientations are sheet-flipped ( $\pm 21 \sin(\pi/3)$ ). A structural reduction of Target D (removes one of its two phase inputs); the quark seam correction  $3\lambda_C^2$  stays the [v88](#) misalignment, the deck stays  $\mathbb{Z}/4$ , and the physical CP derivation stays [\[C\]](#) (CP.MU6.01).
- **The seam as the  $E_8$  Kleinian singularity** ([v232\\_e8\\_kleinian\\_seam](#), `origin_theory`, [\[E\]](#)/[\[C\]](#)). The du Val side of the McKay correspondence ([v219](#)) gives the open seam-realisation premise `QGE0.REALIZE.01` a canonical algebraic-geometric model: dropping the trivial-rep node of the affine- $E_8$  McKay graph of  $2I$  leaves the finite  $E_8$  Dynkin = the dual intersection graph of the eight exceptional  $\mathbb{P}^1$ 's of the minimal resolution of  $\mathbb{C}^2/2I$ ; the eight curves = rank  $E_8 = g_{\text{car}} + N_{\text{fam}}$  (a fourth reading of the 8), their negated intersection form is the  $E_8$  Cartan (det 1, even,  $(-2)$ -curves), and the link is the Poincaré homology 3-sphere  $S^3/2I$  ( $2I$  perfect  $\Rightarrow H_1=0$ ,  $\pi_1=2I$  order 120). The raw seam ( $\mathbb{P}^1$  with marks) is canonically a du Val exceptional curve — a model for the bedrock, not a P2 proof (it presupposes  $E_8$ ; `TOP0.E8.01`, bedrock stays [\[O\]](#)).
- **CP is the universal family/triality phase, only sheet-split** ([v233\\_cp\\_triality\\_phase](#), `tfpt_5_redteam`, [\[E\]](#)/[\[C\]](#)). One step past [v231](#) in the  $\mu_6 = \mu_3(\text{family/triality}) \times \mu_2(\text{sheet})$  factorisation:  $\rho = \omega^2(-1)$  with  $\omega = e^{2\pi i/3}$  the  $\mathbb{Z}_3$  triality centre (the  $2/3$  cusp weight). Since  $\rho^4 = \rho^1 \rho^3$  with  $\rho^3 \in \mu_2$ , the quark ( $\rho^1$ ) and lepton ( $\rho^4$ ) CP phases share the *same* family/triality class and differ only by the sheet — so CP *is* the universal triality phase, sheet-split, not a free power choice (CP.TRIALITY.01).
- **The Seam-Holomorphy selection certificate: the structural residual is one gate** ([v234\\_seam\\_holomorphy\\_selection](#), `tfpt_research_contracts`, [\[E\]](#)/[\[O\]](#)). The whole structural residual is *one* condition — “the seam carries no nontrivial abelian sector” — with three provably-equivalent faces, all forcing  $E_8$ : (i) holomorphy ( $\mu$ -index 1; Target A), (ii) the

seam link is a homology 3-sphere ( $\Gamma$  perfect  $\Leftrightarrow 2I$ ; v232), (iii) exactly one 1-dim irrep (v219). All equal  $\#(\text{mark-1}) = |\Gamma^{\text{ab}}| = |H_1|$ , which is 1 only for  $E_8$ ; and holomorphic  $c=8=g_{\text{car}}+N_{\text{fam}}$  gives the unique even unimodular rank-8 lattice  $E_8$ . So P2,  $G_{\text{net}}$ , Target A and the Kleinian seam are one gate. The stated closing theorem ([O]):  $\text{RP}+\text{gap}+\text{chirality} \Rightarrow \text{quasi-free bulk}$  (v160)  $\Rightarrow$  holomorphic boundary  $\Rightarrow E_8$ ; the single residual analytic step is “free bulk  $\Rightarrow$  holomorphic boundary” (GATE.HOLO.01).

- **The closing step in Chern–Simons language: holomorphic  $\Leftrightarrow \det K=1$**  (v235\_seam\_chern\_simons, tfpt\_research\_contracts, [E]/[O]). A genuine reduction (not a closure) of GATE.HOLO.01 into standard anyon-condensation physics: a free gapped bosonic 2+1d bulk is an even integer  $K$ -matrix with  $\#\text{anyons} = |\det K|$ , edge  $c = \text{signature}(K)$ , and the edge net holomorphic  $\Leftrightarrow |\det K|=1$ . The TFPT extension tower (v92) is the condensation tower read by determinant: carrier  $D_5 \oplus A_3$  (det 16, 16 anyons)  $\rightarrow SO(16)_1 = D_8$  (det 4)  $\rightarrow (E_8)_1$  (det 1, holomorphic) — all rank 8,  $c=8=g_{\text{car}}+N_{\text{fam}}$  — so holomorphic =  $E_8$  = the Kitaev  $E_8$  quantum-Hall state. Holomorphy  $\Leftrightarrow$  condensing the order- $|\mu_4|$  Lagrangian glue (det 16  $\rightarrow$  1, vs. a partial order-2 isotropic  $\rightarrow 4=D_8$ ). The residual narrows to one integer condition “the free RP seam condenses the Lagrangian glue (det=1)” = the sheet/QGEO selection, now with holomorphy  $\Leftrightarrow \det 1$  a theorem (GATE.HOLO.02).
- **The (2,3,5) Brieskorn singularity is the one generator of the skeleton** (v236\_brieskorn\_capstone, origin\_theory, [E]). Capstone of the icosahedral round: the singularity  $x^2+y^3+z^5$ , whose exponents are the atoms ( $|\mathbb{Z}_2|, N_{\text{fam}}, g_{\text{car}})=(2,3,5)$ , has Milnor number  $\mu=(2-1)(3-1)(5-1)=8=\text{rank } E_8$  (a *fifth* origin of the “8”), Milnor monodromy = the order-30  $E_8$  Coxeter element (eigenvalues the primitive 30th roots = the  $E_8$  exponents, charpoly  $\Phi_{30}$ ), Milnor lattice  $E_8$ , link the Poincaré sphere; the two clocks are facets of this one monodromy — the  $\mu_3$  triality (CP, v233) is  $h^{10}$  in  $\langle h \rangle = \mathbb{Z}/30$ , the  $\mu_4$  clock (v223) is the order-4 Galois automorphism of  $\mathbb{Q}(\zeta_{30})$  (TOPO.BRIESKORN.01).
- **The closing step as a physical condition: no topological degeneracy  $\Leftrightarrow \det K=1 \Leftrightarrow$  the seam is SRE** (v237\_seam\_sre\_closure, tfpt\_research\_contracts, [E]/[O]). Sharpens GATE.HOLO.02’s residual from definitional to falsifiable: the  $K$ -matrix ground-state degeneracy on a genus- $g$  surface is  $|\det K|^g$  (torus: carrier 16,  $D_8$  4,  $E_8$  1), so “no topological degeneracy on any closed surface”  $\Leftrightarrow \det K=1$ ; among rank-8 even positive-definite  $K$  ( $c=8=g_{\text{car}}+N_{\text{fam}}$ ) only  $E_8$  is short-range-entangled (the Kitaev  $E_8$  phase). A unique vacuum on the plane (RP+gap+cluster) is necessary but not sufficient (the plane cannot see  $\det K$ ); so the one open step is the physical statement “the free RP seam is SRE (no topological order)” (GATE.HOLO.03).

## 2026-06-16 (the icosahedral bedrock, CM-norm duality, and a structural-finds round)

- **McKay bedrock: why the atoms are  $\{2,3,5\}$**  (v219\_icosahedral\_mckay, origin\_theory, [E]).  $E_8$  is the exceptional *top* of the McKay tower of finite  $SU(2)$  subgroups ( $2T \rightarrow \hat{E}_6$ ,  $2O \rightarrow \hat{E}_7$ ,  $2I \rightarrow \hat{E}_8$ ). Built *from the group*: the 120 unit-quaternion icosians close to the binary icosahedral group  $2I$  (element orders  $\{1,2,3,4,5,6,10\}$  — it contains the 2,3,5 rotation-axis orders), 9 conjugacy classes; the 9 irreducible character degrees (Dixon, from the class algebra) are  $\{1,2,2,3,3,4,4,5,6\}$ , sum  $30=h(E_8)$ , sum of squares  $120=|R^+(E_8)|$ ; the McKay adjacency from the natural 2-dim rep *is* the affine  $E_8$  graph with Kac marks = the degrees. A backward certificate of the closed  $E_8$ , not a P2 proof (MCKAY.E8.01).
- **CM-norm duality: 41 (square) and 7 (hex)** (v222\_cm\_norm\_duality, origin\_theory, [E]). The square seam (Gaussian  $\mathbb{Z}[i]$ ,  $j=1728$ ) gives  $N(g_{\text{car}}+i|\mu_4|)=5^2+4^2=41=10b_1$  (the EM index *is* the Gaussian norm of the carrier-glue vector) and  $N(3+2i)=13=\Delta_Q$ ; the hexagonal partner (Eisenstein  $\mathbb{Z}[\omega]$ ,  $j=0$ ) gives  $N(3+2\omega)=7=\text{scalaron}$ . The (3,2) split generates (5,6,7,13) by sum/product/Eisenstein/Gauss norm; rings rigid (CMNORM.DUAL.01).

- **The  $\mu_4$  clock is the order-4 character of the  $E_8$  Coxeter cycle (v223\_coxeter\_totative\_clock, origin\_theory, [E]).**  $(\mathbb{Z}/30)^\times$  are the  $E_8$  exponents; the order-4 generator 7 ( $\langle 7 \rangle = \{1, 7, 13, 19\}$ ) is a literal  $\mu_4$  clock permuting the four conjugate planes; only  $E_8$  has  $\varphi(h) = \text{rank}$  and  $h = 2 \cdot 3 \cdot 5$  (COX.CLOCK.01).
- **248=120+128 as a magnitude/phase channel typing (v227\_degree\_exponent\_channel\_split, tfpt\_5\_redteam, [E]).** Exponents sum  $120 = |R^+(E_8)|$  (magnitude channel), degrees sum  $128 = \text{rank} \cdot \dim S^+$  (phase/glue channel); the red-team Target D magnitude bijection lives in 120, the un-covered CP phases in 128; the dark-sector reading of 128 stays Frontier (replaces the over-strong  $S^-$  reading; E8.CHAN.01).
- **P2 as the mode space of a degree-4 seam divisor (v228\_rr\_index\_gate, origin\_theory, [E]/[C]/[O]).** A Riemann–Roch index gate:  $\deg D = 4 = |\mu_4|$  on  $\mathbb{P}^1$  forces  $h^0 = 5 = g_{\text{car}}$ ,  $\text{rank } H_1 = 3 = N_{\text{fam}}$ ,  $h^0 + H_1 = 8 = \text{rank } E_8$ ,  $\Lambda^{\text{even}}(\mathbb{C}^5) = 16 = \dim S^+$  — bedrock language for P2, conditional on the seam producing the degree-4 divisor (QGE0.RR.01).
- **The seam as a finite recoverability code (v221\_seam\_qecc, origin\_theory, [E]/[C]).** Code dimension  $\dim S^+ = 16$ ; the gapped transport is a CPTP doubly-stochastic contraction with spectrum  $\{1, (2/3)^6, (1/3)^6\}$ , recovery bound  $\|T^n \delta\| \leq (2/3)^{6n} \|\delta\|$  (Knill–Laflamme type); free ratio breaks it. Holographic/Petz reading [C], gated on the Seam–Horizon theorem (QEC.SEAM.01).
- **CP lives in the hexagonal phase fiber (v220\_cp\_hexagonal\_modulus, tfpt\_5\_redteam, [E]/[C]).** The seam deck is the square modulus  $j = 1728$  ( $\mathbb{Z}/4$ ); its partner is the hexagonal  $j = 0$  ( $\mathbb{Z}/6$ ,  $\arg \rho = \pi/3$ ). The frozen  $\delta = \pi/3 + 3\lambda_C^2 = 68.654^\circ$  has leading term  $\pi/3 = \arg \rho$ ; the deck stays  $\mathbb{Z}/4$ , CP is the  $\mathbb{Z}/6$  fiber (CP.MOD.01).
- **The dual normal frame  $(d, n)$  with CP as orientation (v225\_dual\_normal\_frame, tfpt\_5\_redteam, [E]/[C]).**  $d = a^\top R^{-1} = (-\frac{1}{2}, -\frac{1}{2}, 1) = -\frac{1}{2}$  Nariai;  $n = (5, -9, 6)$  reads  $\det$  on the first-generation column;  $\det(1, d, n) = 21 = 3 \cdot 7$ ;  $\text{Im } \det(1, d, \rho n) = 21 \sin(\pi/3)$  — CP as orientation (DUAL.FRAME.01).
- **$F_{\text{transfer}}$  as one path on the Sheet Diamond (v224\_diamond\_ftransfer\_path, tfpt\_2\_standard\_model, [E]/[C]).**  $M(s, t) = R + Q \text{diag}(s, t, t)$ :  $K, C, F$  on the sheet axis (curved, 2nd diff 8), the winding axis flat (slope 6), Plücker steps  $(1, 8, 10), (1, 8, 16)$ ; the Koide source  $\rightarrow$  pole is the 1-D projection (FTR.PATH.01).
- **The charged leptons as an étale Frobenius algebra (v229\_lepton\_frobenius\_algebra, tfpt\_2\_standard\_model, [E]/[C]).**  $(c_e, c_\mu, c_\tau) = (\frac{16}{7}, \frac{4}{3}, \frac{7}{6})$ , ring closure  $c_e c_\tau = \frac{8}{3} = |\mathbb{Z}_2| c_\mu$ , product  $\frac{32}{9}$ ;  $A = \mathbb{Q}[t]/(m)$  is étale (trace-form Gram nondegenerate), a commutative Frobenius algebra over the  $\mu_6$  resolvent (LEP.FROB.01).
- **The center budget  $(7, 11, 13)$  as three local norms (v230\_center\_budget\_norms, tfpt\_2\_standard\_model, [E]).**  $C = R + Q \text{diag}(1, 0, 0)$  row sums  $(7, 11, 13)$ :  $7 = N_{\mathbb{Z}[\omega]}(3 + 2\omega)$  (hex),  $13 = N_{\mathbb{Z}[i]}(3 + 2i)$  (square),  $11 = \binom{4}{0} + \binom{4}{1} + \binom{4}{2}$  (QBL Fock count); the two CM rings and the boundary count meet in one operator center (CENTER.NORM.01).

## 2026-06-15 (the diamond axis geometry — two axes + the transfer corner)

- **The Sheet Diamond is a discrete geometry with two axes (v218\_diamond\_axis\_geometry, tfpt\_2\_standard\_model, [E]).** Sharpens v94/v95 with three exact [E] lemmas (no new numbers). **(1) Axis curvature:** along the centered cross the determinant is linear on the winding axis  $\det(C + xU) = 14 + 6x$  (slope  $6 = |R^+(A_3)|$ ) and quadratic on the sheet axis  $\det(C + yV) = 14 + 14y + 4y^2$  (2nd difference  $8 = \text{rank } E_8$ ); the anchor block has  $\det B(C + xU) = 2 \det(C + xU)$  and sheet curvature  $6 = |R^+(A_3)|$  — two curvatures  $(8, 6) = (\text{rank } E_8, |R^+(A_3)|)$ . **(2) Plücker transfer ladder:**  $\text{Pl}(K) \rightarrow \text{Pl}(C) \rightarrow \text{Pl}(F)$  lifts in steps  $(1, 8, 10) = (N_\Phi, \text{rank } E_8, A_\Lambda)$  then  $(1, 8, 16) = (N_\Phi, \text{rank } E_8, \dim S^+)$  — the

decouple then the full generation (identity [E], source→pole reading [C]). **(3) Spectral ramification:** for  $Q, K, C, F$  the cubic discriminant factors as  $q(r)^2 \text{Disc}(q)$  with squares  $(1, 3, 4, 6)$  and kernels  $(13, 48, 65, 105)$ ,  $F$  carrying  $105 = 3 \cdot 5 \cdot 7$ . The anchor-defect spectrum and pair-sum/staircase blocks are honest *audit* fingerprints ( $28 = 2 \dim G_2$ ,  $52 = \dim F_4$  stay audit-only, not spine, as in v94/v95);  $F$  as the transfer-completion corner is a *heuristic*, not a hard filter. Ledger DIAMOND.AXIS.01, DIAMOND.PLUCKER.01, DIAMOND.SPECTRAL.01; Wolfram-mirrored (249/249).

## 2026-06-15 (QGEO: the four marks emerge from Gauss–Bonnet — K1 executed)

- **The four marks emerge from Gauss–Bonnet — count derived, not assumed** (v216\_marks\_gauss\_bonnet, tfpt\_research\_contracts, [E]; ledger QGEO.MARKS.03). The K1 lever of the v215 kill-test (“exactly four marks”) is *executed*: if the marks are  $\mathbb{Z}_2$  branch points (deficit  $\pi$ ) on the flat seam sphere ( $\chi = 2$ , the same  $\chi$  as the 8 in  $c_3$ ), Gauss–Bonnet forces  $n\pi = 2\pi\chi = 4\pi$ , so  $n = 2\chi = 4 = |\mu_4| = N_{\text{fam}} + 1$ . The closed Euclidean sphere 2-orbifolds are exactly  $(2, 3, 6)$ ,  $(2, 4, 4)$ ,  $(3, 3, 3)$ ,  $(2, 2, 2, 2)$ , and three independent criteria — all cone orders = 2 (the  $|\mathbb{Z}_2|$  sheet branch), rank  $H^1 = \# \text{marks} - 1 = 3 = N_{\text{fam}}$  (the three generations pick the 4-mark square over the 3-mark hexagonal), and a free order-4 deck — converge on the pillowcase  $(2, 2, 2, 2)$ . The numerical sibling v217\_marks\_emergence\_scan ([C], ledger QGEO.EMERGE.01) confirms it on the actual DtN/state (number  $n$  and positions free, the 4-square config is the unique joint minimiser of clock-invariance + 3-family cohomology + gap  $(2/3)^6$ ). The mark *count* and *equality* move from input to derived; only the square *modulus* (cross-ratio 2, the order-4 clock) stays the one carrier input, and the from-zero raw-seam emergence QGEO.REALIZE.01 stays [O]. v216 exact (Wolfram extension → 248/248); v217 numerical (Python-only). Suite → 217 modules.

## 2026-06-15 (QGEO: the bedrock recast as a falsifiable kill-test)

- **QGEO.SYM.01 as a kill-test, not a proof-promise** (v215\_seam\_deck\_killtest, tfpt\_research\_contracts, [E]/[O]; ledger QGEO.KILL.01). A foundational symmetry cannot be derived from nothing (Bisognano–Wichmann would presuppose the covariance it should produce), so the bedrock  $\omega \circ \rho = \omega$  is recast as four independently-falsifiable predictions about the raw seam DtN. The non-circular reduction (v198/v199/v201) gives  $\omega \circ \rho = \omega \Leftrightarrow [\rho, C] = 0$  with  $|k|$  commuting exactly, so the residual is the bounded sub-principal  $M_f$  (equivalently the raw Gaussian bulk form  $A$  is  $\mu_4$ -deck-invariant). Levers: **K1** exactly four marks ( $b_1 = \# \text{marks} - 1 = 3 = N_{\text{fam}} \Rightarrow \# \text{marks} = 4 = |\mu_4|$ ); **K2** the square config (cross-ratio 2  $\Rightarrow j = 1728$ , Aut =  $\mathbb{Z}/4$ ; order 4 selects  $\mathbb{Z}/4$  not the  $j = 0$   $\mathbb{Z}/6$ ; generic  $\Rightarrow \mathbb{Z}/2$ ); **K3** mod-4 sub-principal support ( $[\rho, M_f] = 0 \Leftrightarrow f$   $\mathbb{Z}_4$ -invariant); **K4** transfer gap  $(2/3)^6 = 64/729$ . Freeze-bound via freeze\_file.csv (seam\_deck\_symmetry), carrier-side [E] (regression guards), with the *decisive* kill deferred to QGEO.REALIZE.01 [O] (the raw collar DtN must *produce* four square marks without inputting them — an emergence test). A kill-test, not a closure. Python-only (falsification layer; the exact sub-parts  $j = 1728$  and  $(2/3)^6$  are Wolfram-mirrored via v214/v54/v56). Suite → 215 modules.

## 2026-06-15 (QGEO: the pillowcase reduction — two residuals become one canonical metric)

- **Pillowcase reduction of QGEO.SYM.01** (v214\_seam\_pillowcase, tfpt\_research\_contracts, [E]/[O]; ledger QGEO.PILLOW.01). The two separately-tracked open QGEO residuals — QGEO.ISO.01 (v180: the carrier clock is an order-4 *isometry* of the seam metric) and QGEO.SUBPRIN.01 (v201: the DtN sub-principal symbol is *mark-local*, the seam flat away from the  $\mu_4$  marks) — are shown to be *one* statement: the seam carries the uniformising flat orbifold (*pillowcase*) metric. **(1)** The four marks make the Euclidean orbifold  $S^2(2, 2, 2, 2)$



( $\chi_{\text{orb}} = 2 - 4(1 - \frac{1}{2}) = 0$ ), so by Troyanov uniformisation the metric is flat (unique up to scale) with curvature only at the cone points (Gauss–Bonnet  $4\pi = 2\pi\chi$ ) — this *is* the v201 “flat away from the marks”. **(2)** New link: cross-ratio( $\mu_4$ ) = 2 (v168)  $\Rightarrow j = 1728$  ( $j(\lambda) = 256(\lambda^2 - \lambda + 1)^3/(\lambda^2(\lambda - 1)^2$ ); all six harmonic cross-ratios give 1728)  $\Rightarrow$  the *square* modulus  $\tau = i$ , whose CM by  $\mathbb{Z}[i]$  *derives* the order-4 clock  $z \mapsto iz$  (explicit  $(x, y) \mapsto (-x, iy)$  on  $y^2 = x^3 - x$ ); generic cross-ratio gives  $j \notin \{0, 1728\}$  and only  $\mathbb{Z}/2$ . **(3)** Unification: “seam = uniformising flat pillowcase metric” implies *both* v201 and v180, so two residuals collapse to one canonical-metric premise. **[O]** It does *not* close QGEO.SYM.01: the choice “RP seam metric = the constant-curvature representative” stays open, milder and more canonical than the bare isometry premise. Wolfram-mirrored (the exact orbifold/ $j$ -invariant/CM arithmetic); Klein- $J$  values mpmath-numerical. Suite  $\rightarrow$  214 modules.

## 2026-06-15 (the F\_transfer transfer functor contract CONTRACT.F.01)

- $F_{\text{transfer}}$  formalised as a typed functor with four axioms (v213\_ftransfer\_functor, tfpt\_research\_contracts, **[C]**). The four frontier transfers are consolidated into ONE functor  $F_{\text{transfer}} = F_{\text{observable}} \circ F_{\text{threshold}} \circ F_{\text{RG}}$ , each instance a discrete compiler kernel **[E]** composed with an external continuous solver **[C]**: **(1)**  $\mu_4$ -deck equivariance (the Koide multiplier  $\lambda_2 = (2/3)^6 = 64/729$  is the deck transfer eigenvalue); **(2)** Plücker preservation ( $53 = a^T(R+Q)\mathbf{1}$ ,  $\text{Pl}(F) = (1, 22, 30)$ ,  $\|\text{Pl}(K)\|_1 = 11$ ); **(3)** positivity/stochasticity ( $\text{spec } T$  positive,  $\kappa_f \in (0, 1)$ ); **(4)** external modules explicit ( $b_3 = -7$  a carrier output). The four instances are  $F_{\text{pole}}$  (v183),  $F_{\text{Boltzmann}}$  (v212),  $F_{\text{relic}}$  (v211),  $F_{\text{QCD}}$  (v164). CONTRACT.F.01 is the third research contract alongside ( $U_{\text{wall}}$ ) and ( $G_{\text{metric}}$ ), the structural complement of the v187 typing guard — an architectural consolidation, *not* a closure of any frontier number. Ledger CONTRACT.F.01.

## 2026-06-15 (frontier: two sharper [C] scenario branches — axion angle + leptogenesis)

- **Axion DM spine-angle branch**  $\theta_i = \pi N_{\text{fam}}/g_{\text{car}} = 3\pi/5$  (v211\_axion\_spine\_angle, tfpt\_4\_frontier, **[C]**). A more *robust* misalignment angle than the determinant-line hilltop  $\theta_i \approx 170^\circ$  (v185, which over-produces  $\Omega_a h^2 \approx 0.66$ ): the central spine quotient gives  $\theta_i = 3\pi/5 = 108^\circ$  exactly (no fit), the finite- $T$  solver reaches  $\Omega_{\text{DM}}$  at  $\approx 106^\circ$ , and  $108^\circ$  sits in the *mild*-anharmonic regime ( $62^\circ$  below the hill-top) — not exponentially sensitive. An alternative ansatz to  $\theta_i = \pi(1 - \varphi_\Sigma)$  (mutually exclusive, the full solver decides, DM.AXION.SPINE.01); **[X]**  $\Omega_a h^2$  outside  $\sim [0.08, 0.16]$  demotes it. A sharper scenario, not a derivation.
- **Leptogenesis scalaron-decuple branch** (v212\_leptogenesis\_decuple, tfpt\_4\_frontier, **[C]**). Both Boltzmann inputs share the decuple  $A_\Lambda = 10 = |E(K_5)|$ :  $M_1 = M_{\text{scal}}(\varphi_0^{\text{ret}})^2/A_\Lambda \approx 8.65 \times 10^9$  GeV and  $\tilde{m}_1 = m_3/A_\Lambda \approx 5$  meV.  $M_1$  uses *only*  $\{M_{\text{scal}}, \varphi_0^{\text{ret}}, A_\Lambda\}$  — no hidden seesaw scale, cleaner than v184’s  $M_R \varphi_0^4$  — and lands in the thermal window where  $\eta_B \approx 1.2 \times 10^{-9}$  brackets the observed  $6.1 \times 10^{-10}$ . The coupling mechanism is posited not derived, so  $\eta_B$  stays **[C]**; **[X]** a precise Boltzmann/RGE solve outside the band falls the route, not the theory (FR.ETAB.04; Route A independent).

## 2026-06-15 (Lean: QGEO.COHO.M.01 character grading + MODULE parity machine-proved)

- **The cohomology character grading is now Lean-formalised** (CohomologyGrading.lean, ledger FORM.QGEO.03, **[O]**). A new module machine-proves the COHO.M node of the QGEO proof tree (v177; AUDIT: PASS, no sorry/admit, axioms propext, Classical.choice, Quot.sound only): the three  $H^1(\mathbb{P} \setminus \mu_4)$  eigenforms  $\omega_k = z^{k-1}/(z^4 - 1)$  ( $k=1, 2, 3$ ) have pullback character  $\rho^* \omega_k = i^k \omega_k$  under the clock  $\rho: z \mapsto iz$  (each an exact rational-function identity, the denominator clock-invariant since  $i^4=1$ ), so the grading is  $(i, -1, -i)$  with weights  $(1, 2, 3) =$  the  $A_3$  exponents  $= \text{Spec}(Q_+)$ , rank 3  $= N_{\text{fam}}$ . The MODULE *parity* is also



proved: the reflection  $\sigma : z \mapsto 1/z$  satisfies  $\sigma^*\omega_1 = \omega_3$ ,  $\sigma^*\omega_2 = \omega_2$ ,  $\sigma^*\omega_3 = \omega_1$  (swap  $\omega_1 \leftrightarrow \omega_3$ ,  $\omega_2$  fixed). Signature-locked in `AuditContract.lean`. The geometric COHOM/MODULE-parity nodes are now [E]; the MARKS/KERNEL obligations and the MODULE *uniqueness* stay [O].

## 2026-06-15 (Lean: QGEO.UNIFORM.01 geometric normal form machine-proved)

- **The Möbius uniformisation normal form is now Lean-formalised** (`MöbiusUniformisation.lean`, ledger `FORM.QGEO.02`, [O]). A new module machine-proves the constructive UNIFORM node of the QGEO proof tree (v177; AUDIT: PASS, no `sorry/admit`, axioms `propext`, `Classical.choice`, `Quot.sound` only): the clock  $\rho : z \mapsto iz$  has  $\rho^4 = \text{id}$  and  $\rho^2 \neq \text{id}$  (order exactly 4); the reflection  $\sigma : z \mapsto 1/z$  is an involution with  $\sigma\rho\sigma = \rho^{-1}$  (so  $\langle \rho, \sigma \rangle$  is the dihedral  $D_4$  of order 8); a non-fixed orbit  $\{a, ia, -a, -ia\}$  scales to  $\mu_4 = \{1, i, -1, -i\}$ ;  $\sigma$  permutes  $\mu_4$  (fixes 1,  $-1$ , swaps  $i, -i$ ); and the multiplier classification “ $z \mapsto \zeta z$  has order exactly 4  $\Leftrightarrow \zeta = \pm i$ ”. Signature-locked in `AuditContract.lean`. This formalises the geometric half of the seam realisation *given* the four marks and the clock; the raw-seam marking obligation `QGEO.MARKS.01` stays [O], *not* closed.

## 2026-06-15 (Lean: the QGEO.SYM.01 conditional theorem machine-proved)

- **The mark-local  $\Rightarrow \omega \circ \rho = \omega$  implication is now Lean-formalised** (`SeamDeckClosure.lean`, ledger `FORM.QGEO.01`, [O]). A new module in the carrier-rigidity Lean project machine-proves the algebraic core of v201/v210 (AUDIT: PASS, no `sorry/admit`, no domain axioms — `#print axioms = propext`, `Classical.choice`, `Quot.sound` only): (i) the 4th-root character orthogonality  $\sum_{j<4} \zeta^j = (\text{if } \zeta=1 \text{ then } 4 \text{ else } 0)$  for  $\zeta^4=1$  (so a  $\mu_4$ -mark sum is supported only on modes  $\equiv 0 \pmod{4}$ ); (ii) the clock generator  $-i$  has order 4; (iii) `markLocal_blockDiagonal`: a mark-local Toeplitz symbol connects only equal clock-characters,  $[\rho, M_f] = 0$ ; (iv) the premise is a typed structure `SeamDeckPremise` (the physical seam DtN *is* mark-local) — consumed, *not* proved, exactly as `CalderonProjector` encodes the Paper-1 analytic input; (v) `SeamDeckPremise.clock_invariant`: *given* the premise, the clock commutes with the DtN, whence  $\omega \circ \rho = \omega$ . So the *implication* is now [E] (formally proved); the *premise* (the physical seam being mark-local) stays [O] — the one fundamental seam-identification postulate, *not* closed.

## 2026-06-15 (QGEO sharpening: mark-local DtN certified on realistic profiles)

- **Mark-local DtN  $\Rightarrow \omega \circ \rho = \omega$ , certified on realistic Steklov profiles and at the state level** (`v210_mark_local_dtn`, `tfpt_research_contracts`, [O]). A numerical sharpening of `QGEO.SUBPRIN.01/v201`: a real von-Mises curvature bump summed over the four  $\mu_4$  marks,  $f(\theta) = \sum_{j=0}^3 g(\theta - j\pi/2)$ , builds the Steklov DtN  $\Lambda = |D_\theta| + M_f$  whose off-(mod 4) Fourier support vanishes to  $< 10^{-16}$ ,  $\|[\rho, \Lambda]\| < 10^{-15}$ , and — the genuinely new step — the quasi-free covariance  $C = \frac{1}{2}(1 + \text{sgn } H_1)$  is clock-invariant ( $\rho C \rho^\dagger = C$  to  $< 10^{-13}$ ), i.e. the actual  $\omega \circ \rho = \omega$ . Negative controls (single off-centre bump,  $\mathbb{Z}_3$  source, four generic marks) break both by  $O(1)$ ; convergence stable for  $N \in \{16, 32, 64\}$ . This certifies the *implication*; the premise that the physical seam *is* mark-local stays [O] (it does *not* close `QGEO.SYM.01`; net existence and full-cone RP remain the [E] of v175). Ledger `QGEO.MARKLOCAL.01`. Python-only (the exact mark-sum identity is Wolfram-mirrored via v201).

## 2026-06-15 (archive integration #5: six dormant results revived from the old papers)

- **Muon  $g-2$  as a seam vertex** (`v204_muon_seam_g2`, `tfpt_4_frontier`, [C]). The carrier’s second-order topological defect  $\delta_2 = \frac{5}{4}\delta_{\text{top}}^2$  projected through the seam-loop phase gives  $a_\mu^{\text{seam}} = \delta_2/(2\pi) = 45/(524288\pi^9) \approx 2.879 \times 10^{-9}$  ( $0.81\sigma$  vs the dispersive  $\Delta a_\mu$ ;  $\sim 1.5\sigma$  vs

lattice/CMD-3). The value is exact [E]; the vertex identification is the [C] bridge. Ledger FR.MUONG2.01; Wolfram-mirrored.

- **An independent gravitational 3/4 (v205\_xi\_threequarter, tfpt\_4\_frontier, [E]/[C]).**  $\xi = c_3/\varphi_{\text{tree}} = 3/4$  exactly (Einstein limit of  $\kappa^2 = \xi\varphi_0^{\text{ret}}/c_3^2$ ) — the same  $3/4 = q(A_3) = \ln(m/\mu)$  as the seam-determinant replica (v152), an over-determination. Ledger GRAV.XI.01; Wolfram-mirrored.
- **Quantitative  $H_0$  from the  $\Lambda$  branch (v206\_h0\_lambda\_branch, tfpt\_4\_frontier, [C]).**  $\delta_\Sigma = 48 e^{-2\pi\alpha^{-1}/3} \approx 1.085 \times 10^{-123}$ , the  $(8\pi)^2$  Planck-convention split, and  $H_0 = 66.5\text{--}67.1$  km/s/Mpc (below Planck, far below SH0ES — the Hubble tension is *not* relieved). Ledger GRAV.H0.01.
- **Asymptotic Safety as a second QG route (v207\_asymptotic\_safety, tfpt\_4\_frontier, [O]).** A complementary continuum-FRG argument (a non-Gaussian UV fixed point from the same axioms;  $y^2 = 16c_3^2 = 1/(4\pi^2)$ ) — a plausibility cross-check, *not* load-bearing, does *not* close  $G_{\text{metric}}$ . Ledger GRAV.ASYMP.01.
- **Black-hole thermodynamics from the seam (v208\_bh\_thermodynamics, tfpt\_horizon\_readouts, [E]/[C]).** Modular  $T_H = \kappa/(2\pi)$  (the  $2\pi = 1/(4c_3)$  seam unit, ties to v198/v199) and the scalaron-corrected Wald entropy  $S_W = (A/4G)(1 + R_h/3M_s^2)$  (exact from v28). Ledger HOR.BHTHERMO.01; Wolfram-mirrored.
- **The black hole as a seam fixed point (v209\_bh\_regular\_core, tfpt\_horizon\_readouts, [C]).** No singularity ( $\varphi \rightarrow \varphi_*$  at the gapped attractor  $(2/3)^6$ ); the maximal BH = the anchor (Nariai roots  $(1, 1, -2)$ ); information via Page recovery; a Planck-scale holographic floor. The superseded RN/torsion metric is *not* resurrected. Ledger HOR.BHCORE.01.

## 2026-06-15 (archive integration #4: Möbius $\cong$ Lorentz foundation citation)

- **The Möbius seam is the boundary Lorentz structure (tfpt\_3, [O]).** Added a paragraph to the “Möbius origin of  $\pi$ ” section:  $\text{Möb}(\hat{\mathbb{C}}) \cong \text{PSL}(2, \mathbb{C}) \cong \text{SO}^+(3, 1)$  (Penrose & Rindler, *Spinors and Space-time*, Vol. 1, §1.2–1.4), the celestial sphere of null directions via stereographic projection. This gives the self-consistency loop a spacetime meaning — the carrier clock  $z \mapsto iz$  is an elliptic  $\text{PSL}(2, \mathbb{C})$  rotation (v180) and the RP→OS causal cone is the same Lorentz structure — so “Möbius origin of  $\pi$ ” and “one Lorentzian cone” are two faces of one boundary  $\text{PSL}(2, \mathbb{C})$ . A foundation citation, not a new claim ( $\pi$  stays primitive).

## 2026-06-15 (archive integration #3: axion coupling $g_{a\gamma\gamma} = -4c_3 +$ retired small-angle reading)

- **Haloscope coupling tied to  $c_3$  (tfpt\_4\_frontier, [C]).** Made explicit: in the determinant-line normalization the axion–photon anomaly coefficient is  $g_{a\gamma\gamma} = -4c_3 = -1/(2\pi)$  ( $y^2 = 16c_3^2 = 1/(4\pi^2) \approx 0.0253$ ), so the haloscope  $F\tilde{F}$  coupling is set by the *same*  $c_3$  that fixes  $\alpha$  and  $\beta_{\text{rad}} = \varphi_0/4\pi$  — a [C] structural relation (the coefficient, not a parameter-free physical  $g_{a\gamma\gamma}$ ). The earlier small-angle  $\theta_i = \varphi_0$  reading (old determinant-line note, a much higher axion mass) is *superseded* by the closed near-hilltop  $\theta_i = \pi(1 - \varphi_{\text{seam}}) = 170.4^\circ$ , recorded as a retired reading and guarded by v188 (a new sentinel forbids that stale axion mass from re-entering the active docs).

## 2026-06-15 (archive integration #2: the EHT achromatic polarization intercept — v203)

- **HOR.EHT.01 (v203, [E]/[C]/[X]).** The surviving core of the old UFE/black-hole notes (*not* the RN-type metric — superseded by the Nariai/seam=horizon readouts v101–v104/v190; only the polarization signature survives): a local achromatic polarization-rotation intercept around a

horizon-scale black hole,  $\beta_{\text{BH}}(r) = 16c_3^4 Q_e^{\text{eff}} Q_m^{\text{eff}} / r^2$ . The coupling  $16c_3^4 = 1/(256\pi^4) = \delta_{\text{top}}/3$  is *exact* ([E], Wolfram-mirrored) — the *same* top-form coefficient  $\delta_{\text{top}} = 48c_3^4$  that fixes the  $\alpha$ -kernel precision-zone correction, one row from the cosmic-birefringence seed  $\beta_{\text{rad}} = \varphi_0/4\pi$ . TFPT fixes the  $1/r^2$  shape, achromaticity and sign-flip ([C]); the amplitude  $Q_e Q_m$  is an MHD/GR weight, never a pixel prediction. **Dated kill test [X]**: the residual intercept must pass three nulls (frequency, spatial  $1/r^2$ , sign-flip) after honest GRMHD subtraction. Integrated into `tfpt_horizon_readouts` (new EHT section), the ledger, registry, Wolfram (243/243), and the website (`predictions.ts`). Suite 202 modules.

## 2026-06-15 (archive integration #1: the rare kaon sector $K \rightarrow \pi \nu \bar{\nu}$ — v202)

- **FR.RAREKAON.01 (v202, [C])**. The closed TFPT CKM point ( $s_{12}=\lambda_C$ ,  $s_{23}=\varphi_0/(1+\lambda_C)=|V_{cb}|$ ,  $s_{13}=\lambda_C^3/3$ ,  $\delta=\pi/3+3\lambda_C^2=1.19823$  rad; v18/v88, no flavor fit) feeds the cleanest FCNC probe. With standard external short-distance input (Brod–Gorbahn–Stamou  $X_t, P_c, \kappa_{\pm}$ ) it gives  $\text{BR}(K^+ \rightarrow \pi^+ \nu \bar{\nu}) = 9.45 \times 10^{-11}$  and  $\text{BR}(K_L \rightarrow \pi^0 \nu \bar{\nu}) = 3.33 \times 10^{-11}$  — a [C] downstream flavor readout, *not* a compiler power (the SD functions are external). Recomputed from the *current* CKM point (an earlier archive scaffold quoted  $\text{BR}(K_L) = 3.47 \times 10^{-11}$  from an  $\Im(\lambda_t)/\lambda^5$  slip, corrected here). Grossman–Nir ratio  $0.352 \ll 4.3$ ; NA62  $(13.0^{+3.3}_{-3.0}) \times 10^{-11}$  is  $\sim 1.2\sigma$  above. **Dated kill test [X]**: a stable NA62  $\text{BR}(K^+)$  outside  $[7, 12] \times 10^{-11}$ , or a KOTO-II  $\text{BR}(K_L)$  off the GN point, breaks the flavor bridge (core untouched). Integrated into `tfpt_2_standard_model` (§rare kaon), the ledger, v187 guard (new FR.RAREKAON. family), registry, and the website (`predictions.ts`). Python-only. Suite 201 modules.

## 2026-06-15 (bedrock: the $\omega \circ \rho = \omega$ residual reduced once more — v201)

- **QGE0.SUBPRIN.01 (v201, [E]/[O])**. The v199 bedrock residual — “the bounded sub-principal symbol of the raw seam DtN has no off-character (mod 4) matrix elements” — is reduced one genuine step further: block-diagonality is *not* an independent postulate. Writing the DtN as  $\Lambda = |k| + M_f$  with  $M_f$  the bounded sub-principal piece = multiplication by a curvature  $f(\theta)$  (standard Steklov structure),  $M_f$  is  $\mu_4$ -character-block-diagonal iff  $f$  is  $\mathbb{Z}_4$ -invariant ([E]; principal  $|k|$  already commutes, v198); and a  $\mu_4$ -mark sum  $f = \sum_{j=0}^3 g(\theta - 2\pi j/4)$  has  $f_m = 4g_m[m \equiv 0 \pmod{4}]$  for any profile  $g$ , hence is automatically  $\mathbb{Z}_4$ -invariant ([E]). So the  $\mu_4$ -mark orbit (forced by v195) *forces* the sub-principal symbol block-diagonal; 3 marks ( $\mathbb{Z}_3$ ) and 4 generic marks break it (negative controls). The residual collapses to “the DtN sub-principal symbol is mark-local (seam flat away from the  $\mu_4$  marks = conformal deck, v177)”, structurally definitional — the narrowest form yet of QGE0.SYM.01, [O]. Registered in `run_all.py`, the ledger, `tfpt_research_contracts`, `origin_theory`; Wolfram-mirrored (the mark-sum Fourier identity is exact; extension now 242/242). Suite 200 modules.

## 2026-06-15 (F\_transfer workshop: the axion relic solver — an honest over-production tension)

- **Axion relic solver (experiments/ftransfer/axion\_relic/relic\_solver.py, [C])**. The first real  $F_{\text{transfer}}$  pipeline computation (work package C / review §6.2): a standard misalignment relic with a numerically-computed anharmonic factor  $F(\theta_i) \approx 4$  at the hilltop. Honest finding: at the *predicted*  $\theta_i \approx 170^\circ$  the estimate *over*-produces dark matter,  $\Omega_a h^2 \sim 0.6\text{--}2.3$  (robustly  $> 0.12$ ), because the large  $\theta_i^2 \approx 8.8$  times  $F$  overshoots. So the determinant-line axion as the *dominant* DM is in *mild tension* (it would prefer  $\theta_i \sim 120\text{--}130^\circ$  or extra dilution) — this *refines* v185’s optimistic “can be all-DM” toward an over-production tension. The near-hilltop regime is  $O(1)$ -uncertain, so a full finite- $T$  solve decides; over-production kills only the axion-as-dominant-DM *branch*, not the compiler. Reflected in `tfpt_4_frontier` (dark-matter section) and the v185 docstring; experiments-only, not a suite/ledger claim. No refitted exponents;  $\theta_i = \pi(1 - \varphi_{\text{seam}})$  only.

- **Full finite- $T$  solve confirms it (axion\_relic/full\_finiteT\_solve.py).** A converged nonlinear misalignment solve  $(\theta'' + (3 + d \ln H/dN)\theta' + (m_a(T)/H)^2 \sin \theta = 0$ , lattice  $\chi(T) \propto T^{-8.16}$ , realistic  $g_*(T)$ ; normalised so  $\theta_i=1 \rightarrow \Omega_a h^2 \approx 0.03$ , the standard value) gives  $\Omega_a h^2 \approx 0.66$  at the predicted  $\theta_i \approx 170^\circ$  —  $\sim 5.5\times$  above  $\Omega_{\text{DM}} = 0.12$ , reached only at  $\theta_i \approx 106^\circ$ . The over-production is thus *confirmed* (not the optimistic “all-DM”); `tfpt_4_frontier`, v185 docstring and the website updated accordingly. Stays [C]; kills only the axion-as-dominant-DM branch.

## 2026-06-14 (work packages A+B: attacking $\omega \circ \rho = \omega$ + the variational scan — v199, v200)

- **v199\_seam\_state\_invariance.py [E]/[O] (claim QGEO.STATE.01, work package A).** Attacking  $\omega \circ \rho = \omega$  directly on the *raw* seam. (1) [E] the setup is non-circular:  $\rho$  is the Coxeter element of  $W(A_3) = S_4$  ( $\rho^4 = 1$ , v117), an automorphism of the raw CAR algebra (16 Majoranas) — both carrier-side, no seam geometry. (2) [E] for a quasi-free RP state  $C_\Sigma = \frac{1}{2}(1 + \text{sgn } H_1)$ , so  $\omega \circ \rho = \omega \Leftrightarrow [\rho, H_1] = 0$ . (3) [E]  $[\rho, H_1] = 0 \Leftrightarrow H_1$  is block-diagonal in the four  $\mu_4$ -character classes  $\{n=r \bmod 4\}$  (verified). (4) [O] the residual is sharpened to “the bounded sub-principal symbol has no off-character matrix elements” — a bounded, finite-character condition, not a diffuse geometry; not a closure. Cited in `tfpt_research_contracts`; Wolfram-mirrored.
- **v200\_seam\_variational\_scan.py [C]/[O] (claim QGEO.VARI.02, work package B).** The discretised variational test: with the clock angle free,  $E_{\text{fail}}(\theta)$  has its unique minimum at  $\theta = \pi/2$  (the  $\mu_4$  clock  $z \mapsto iz$ ) — order-4 *and* the three distinct characters  $i, -1, -i =$  the  $(1, 2, 3)$  grading; decoys  $(0, \pi, \pi/3, 2\pi/3)$  all lose. A numerical sign that the variational principle selects  $\mu_4$ ; the full operator-valued minimisation stays the v199 residual [O]. Cited in `tfpt_research_contracts`.
- *Also (this pass, experiments/ only, not the ledger): the  $F_{\text{transfer}}$  four-pipeline scaffold (experiments/ftransfer/) with one CONTRACT per interface, and confirmation that the FRB preregistration (frb\_tfpt\_v1.yaml) and its status semantics already exist. Suite now 199 modules, highest ID v200; Wolfram extension 241/241.*

## 2026-06-14 (cracking the bedrock to a state-invariance: principal symbol + Tomita–Takesaki — v198)

- **v198\_modular\_commutator\_reduction.py [O] (claim QGEO.MODULAR.01).** The decisive crack on the last residual. (1) [E] the DtN principal symbol  $|k| = \sqrt{-\partial_\theta^2} = \text{diag}(|n|)$  and the clock  $\rho: z \mapsto iz = \text{diag}(i^n)$  are both diagonal in the Fourier basis, so  $[\rho, |k|] = 0$  *exactly* on all of  $L^2$  (verified on 33 modes) — the leading-order commutation is free on the whole boundary, never the open part. (2) [E] by Tomita–Takesaki the modular operator is intrinsic to  $(M, \Omega)$ , so a state-preserving symmetry ( $\rho\Omega = \Omega$ ) commutes with  $\log \Delta \sim \Lambda_\Sigma$ : hence  $[\rho, \Lambda_\Sigma] = 0$  *follows from*  $\omega \circ \rho = \omega$ , using only the state + reflection (RP) and *no* conformal covariance — which **removes the v194 Bisognano–Wichmann circularity worry**. (3) [E] the finite  $H^1$  block is already  $\mu_4$ -invariant (v177). So the entire residual collapses to the single *state-invariance* “the raw quasi-free seam state is  $\mu_4$ -invariant” ( $\omega \circ \rho = \omega \Leftarrow$  the Gaussian bulk measure is deck-invariant) — the maximally-operational, non-circular form of QGEO.SYM.01, [O]. *Not* a full closure (a foundational symmetry cannot be derived from nothing, like  $c = \text{const}$ ), but the sharpest falsifiable form, with the circularity gone. Cited in `tfpt_research_contracts`; OpenGates updated. Suite now 197 modules, highest ID v198; Wolfram extension 240/240.



## 2026-06-14 (three review levers: Marks-via-Lefschetz, E\_fail functional, RR→D5 — v195–v197)

- **v195\_marks\_lefschetz\_character.py** [E]/[O] (claim QGEO.MARKS.02). The four seam marks are *forced* (not posited):  $\rho: z \mapsto iz$  fixes only  $\{0, \infty\}$ ,  $H^1 = \chi_1 \oplus \chi_2 \oplus \chi_3$  (the  $A_3$  exponents,  $\text{Tr}(\rho|H^1) = i + i^2 + i^3 = -1$ ) has no trivial component  $\Rightarrow$  no puncture at a fixed point, and  $\text{rank } H^1 = n-1 = 3 \Rightarrow n = 4 \Rightarrow$  a single free  $\mu_4$  orbit (cross-ratio 2). So MARKS reduces from the geometric posit “ $D_4$  marks” to the milder premise “ $H^1$  carries the  $A_3$ -exponent rep” — less circular, still [O]. Cited in `tfpt_research_contracts`.
- **v196\_seam\_energy\_functional.py** [E]/[O] (claim QGEO.VARI.01). The failure functional  $E_{\text{fail}} = \|\rho, \Lambda_\Sigma\|_{HS}^2 + \|\rho^4 - 1\|^2 + \|\Theta\rho\Theta - \rho^{-1}\|^2$ , zero locus = the QGEO.SYM.01 conditions; on the finite  $H^1$  block it vanishes at the  $\mu_4$  config [E] and is  $> 0$  for any  $\mu_4$ -breaking perturbation, so  $\mu_4$  is a true minimiser. The full-operator minimisation is the v194 Bisognano–Wichmann residual — a concrete discretisable *target* on the open bedrock, [O]. Cited in `tfpt_research_contracts`.
- **v197\_rr\_carrier\_clifford\_d5.py** [E]/[C] (claim ARCH.RRCAR.02). Hardens the Riemann–Roch carrier from  $g_{\text{car}}$  to  $D_5$  itself:  $\Lambda^{\text{even}}(C_{\text{car}}) = \binom{5}{0} + \binom{5}{2} + \binom{5}{4} = 16 = 2^{g_{\text{car}}-1} = \dim S^+$ , the  $\mathfrak{so}(10) = D_5$  half-spinor — so  $D_5$  is the Clifford spinor of the 5-dim mode space, closing  $\mu_4 \rightarrow g_{\text{car}} \rightarrow D_5$ . [E] arithmetic, [C] identification (as v189). Cited in `origin_theory`.
- All three are validated external-review levers (the axion-hilltop “ $\pi - 3\phi_0$ ” proposal was *rejected* as a  $\pi \approx 3$  coincidence, not built). Suite now 196 modules, highest ID v197; Wolfram extension 239/239.

## 2026-06-14 (subclaim 2 of ENERGY.02 tackled: raw-seam-DtN RP-definability — v194)

- **v194\_raw\_seam\_dtn\_rp.py** [O] (claim QGEO.DTN.01). The next hard lever — “is the raw RP-seam DtN definable from RP data alone?” (sub-claim 2 of QGEO.ENERGY.02) — was tackled, and the honest result is a *reduction*, not a closure. (1) [E] the commutator  $[\rho, \Lambda_\Sigma] = 0$  closes on the finite cohomology  $H^1$  ( $\rho^* w_k = i^k w_k$ , distinct  $\mu_4$  characters; v177). (2) [E] the *hinge* is met: the DtN/transfer operator is RP-canonical (Osterwalder–Schrader, v54) on a quasi-free seam state (v155), so  $\Lambda_\Sigma$  is RP-definable without naming  $\mathbb{P}^1 \setminus \mu_4$ . (3) [O] the residual relocates: the full- $L^2$  commutation  $\Leftrightarrow$  Bisognano–Wichmann geometric modular flow of the quasi-free seam state (which must be intrinsic, else it re-circulates). So QGEO.SYM.01 reduces from a definitional postulate to “intrinsic BW geometric modular covariance” — one notch sharper, still [O]; the last structural fog point sharpens, it does *not* close to [E]. Cited in `tfpt_research_contracts`; OpenGates updated. Suite now 193 modules, highest ID v194.

## 2026-06-14 (QGEO.ENERGY.02 energy-commutator + QFT-wording hardening — v193)

- **v193\_qgeo\_energy\_commutator.py** [O] (claim QGEO.ENERGY.02). The non-circular sharpening of QGEO.ENERGY.01: the energy commutator  $[\rho, \Lambda_\Sigma] = 0$  on the rank-8 Calderón polarisation, with three sub-claims — (1) [E]  $\rho$  is carrier-defined (Coxeter of  $W(A_3)=S_4$ , v117), not imported from the seam; (2) [O] the non-circularity hinge:  $\Lambda_\Sigma$  must be the *raw* RP-seam DtN, not the  $\mu_4$  normal form; (3) [E] the commutator forces the  $(1, 2, 3)$  grading on  $H^1$  (QGEO.COHO.M.01). Exact [E] rider (Wolfram-mirrored): the induced EH coefficient  $k = (\ln m)/(12\pi)$  (v150) reproduces  $c_3/2$  iff  $\ln m = q(A_3) = \frac{3}{4}$  (*family*), not  $q(D_5) = \frac{5}{4}$  (carrier, off by  $5/3$ ) — gravity is family-geometry-induced. A *proof target*; if sub-claim (2) holds, QGEO.SYM.01 becomes an energy-rigidity theorem. Cited in `tfpt_research_contracts`. Suite now 192 modules, highest ID v193; Wolfram extension 236/236.



- **QFT-wording hardening (review point 7).** `tfpt_2_standard_model` §(4) was retitled to “Constructive QFT closure of the admissible physical sector”, and its standalone opening over-claim replaced by an explicitly scoped sentence (the unprojected ambient metric measure is not constructed;  $G_{\text{metric}}$  frontier). The two superseded standalone phrasings were added to the v188 prose guard so a frozen release cannot re-introduce them.

## 2026-06-14 (geometry round: Riemann-Roch carrier, Nariai bound, branch line, energy clock — v189–v192)

- **v189\_riemann\_roch\_carrier.py** [E]/[C] (claim ARCH.RRCAR.01). The pair  $(g_{\text{car}}, N_{\text{fam}}) = (5, 3)$  are the two canonical invariants of *one* object, the four-point seam divisor  $D = \mu_4$  on  $\mathbb{P}^1$ : the cycle side rank  $H_1(\mathbb{P}^1 \setminus \mu_4) = \deg -1 = 3 = N_{\text{fam}}$  (already QGEO.COHO.01) and the function side  $h^0(\mathbb{P}^1, \mathcal{O}(\mu_4)) = \deg +1 = 5 = g_{\text{car}}$ , matched by  $\text{rank}(D_5) = 5$ . The carrier-as-mode-space reading is [C] (geometric, does not displace the over-determined  $g_{\text{car}} = 5$ ). Cited in `origin_theory`; Wolfram-mirrored.
- **v190\_nariai\_entropy\_bound.py** [E] (claim HOR.NARIAI.02). A variation-free one-line proof of the SdS entropy floor:  $3(x^2+1) - 2\Phi_3(x) = (x-1)^2 \geq 0$ , so  $S_{\text{tot}} \geq \frac{2}{3}S_{dS}$  with equality iff  $x = 1$  (the Nariai merge). Cited in `tfpt_horizon_readouts`; Wolfram-mirrored.
- **v191\_universal\_branch\_line.py** [C] (claim HOR.BRANCHLINE.01). The affine map  $q = \frac{7}{2} + \frac{N_{\text{fam}}^2}{2}m$  carries the SdS branch points  $\pm 1/N_{\text{fam}}$  onto the flavor labels  $\{2, 5\}$ . *Honestly typed* [C] — an exact *relabeling*, not a theorem (a decoy pair  $\{3, 4\}$  admits an equally exact map; the shared content is the known  $2/3$  ramification). Cited in `tfpt_horizon_readouts`; Wolfram-mirrored.
- **v192\_qgeo\_energy\_clock.py** [O] (claim QGEO.ENERGY.01). An operator-checkable restatement of the bedrock:  $\rho^* \Lambda_{\Sigma} \rho = \Lambda_{\Sigma}$  (the clock preserves the DtN energy form). *Honestly typed*: it *relocates* QGEO.SYM.01 to a more falsifiable surface but does *not* eliminate it (plausibly equivalent unless  $\rho$  is independently defined and the invariance proven). Cited in `tfpt_research_contracts`. Suite now 191 modules, highest ID v192 (v186 skipped).
- *All four are validated external-review proposals (problem\_b); each is built with the honest typing determined on review — the branch line is recorded as an alignment, not a theorem, and the energy clock as a relocation, not a closure.*

## 2026-06-14 (Zenodo release 5.1 (rev 120) — F\_transfer firewall, v184–v188)

- **Published to Zenodo.** Version 5.1 (rev 120) of the ten-PDF document set was published as a new version of the TFPT deposit — record 20687564, DOI 10.5281/zenodo.20687564. This version carries the F\_transfer-firewall round (v184–v188): the honest  $\eta_B$  anchored-Boltzmann test (sharper scenario, not a closure), the axion hilltop-relic typing, the ledger v187 and prose v188 guards, and the Route B/Koide/dark-matter wording fixes plus the P1/P2-vs-QGEO.SYM.01 two-layer clarification. The uploader’s DEFAULT\_RECORD\_ID was advanced to the new record for the next release. Suite 187 modules (highest ID v188; v186 skipped). `run_all.py` green, audit clean, website built.

## 2026-06-14 (F\_transfer round: eta\_B honest test v184, axion relic v185, discipline guard v187)

- **v184\_etaB\_anchored\_boltzmann.py** [C] (claim FR.ETAB.03). An honest test of the external-review  $\eta_B$  proposal ( $M_1 = M_R \varphi_0^4$ ,  $\tilde{m}_1 = m_3/A_{\Lambda}$ ). Firewall verdict: *sharper scenario, not a closure*. The washout anchors plausibly ( $\tilde{m}_1 = m_3/10 \approx 5$  meV,  $A_{\Lambda} = 10$  atom, in the strong-washout window) [C]; but  $M_1$  does *not* —  $M_R$  is the seesaw scale  $\sim v^2/m_3$  and the nearest clean TFPT powers of  $\bar{M}_{\text{P1}}$  both miss it ( $c_3/\varphi_0$ -powers off  $\sim 75\%/ \sim 40\%$ ), so

$M_1 = M_R \varphi_0^4$  merely relocates the free input.  $\eta_B$  stays [C], the route viable but with one (not zero) free input. Cited in `tfpt_4_frontier`.

- **v185\_axion\_relic\_solver.py** [C] (claim FR.DM.03). Honest misalignment estimate: the determinant-line axion  $f_a = M_{\text{scal}}/128 \approx 2.39 \times 10^{11}$  GeV ( $m_a \approx 23.8 \mu\text{eV}$ ) CAN be all dark matter, but the harmonic misalignment under-produces ( $\sim 21\%$  at  $\theta \sim O(1)$ ), so the  $\theta_i \approx 170^\circ$  hilltop anharmonic enhancement is *required* — where the relic is exponentially sensitive. Hence a [C] scenario, not a sharp prediction; the relic contract is a kill test for the BRANCH, not the theory. Cited in `tfpt_4_frontier`.
- **v187\_ftransfer\_laws.py** [E] (claim FR.TRANSFER.01). A CI-style firewall guard: the four continuous-transfer frontier observables (Koide,  $\eta_B$ , axion,  $m_p/m_e$ ) are read from the ledger and required to carry a [C]/[O] marker — never a primitive [E] compiler prediction (exact sub-parts like the Koide 53/54 factor or  $b_3 = -7$  may be [E], the physical prediction never is). Records the four interfaces  $F_{\text{pole}}/F_{\text{Boltzmann}}/F_{\text{relic}}/F_{\text{QCD}}$ ; prevents any future edit from quietly promoting a frontier number to a closed output. Cited in `tfpt_4_frontier`.
- **v188\_frontier\_wording\_guard.py** [E] (claim AUDIT.WORDING.01). A prose sentinel (no new physics): v187 protects the ledger *typing*, this guard protects the *prose*. It scans the 10 active documents and forbids five stale phrasings whose semantics earlier rounds superseded (the old [E]-typed Route B leptogenesis claim and its heavy-scale wording  $\rightarrow$  v184 [C]; the old “open relic scale” dark-matter line  $\rightarrow$  v185; the old “near, not exact” Koide line  $\rightarrow$  v183; and the module-count/ID conflation), so a frozen Zenodo release can never re-introduce wording that was true yesterday and is half-true today. The same pass updated the `tfpt_4_frontier` Route B (to [C]), the Koide and dark-matter summary lines, and added the P1/P2-vs-QGEO.SYM.01 two-layer clarification to the introduction. Suite now has **187 modules**, highest verification ID v188 (v186 was intentionally skipped as redundant —  $m_p/m_e$  is already typed [O] as a QCD matching contract, QCD.LAMBDA.01).
- *Review framing points (P1/P2 as projections of the seam-deck postulate,  $v_{\text{geo}}$  as unit gauge,  $G_{\text{net}}$  a corollary of QGEO.SYM.01, grammar tuning = data) were validated as already implemented in the prior rounds;  $m_p/m_e$  is already typed [O] as a QCD matching contract (QCD.LAMBDA.01), so no redundant module was added.*

## 2026-06-14 (operator origin of the Koide 53/54 factor: the missing Sheet-Diamond corner, v183)

- **v183\_koide\_f\_corner\_transfer.py** [E]/[C] (claim FR.KOIDE.07). An external-review proposal, validated exactly: the Koide source $\rightarrow$ pole factor  $\frac{53}{54}$  is not a bare arithmetic guess but an *operator* identity on the carrier,  $u_{\text{pole}}/u_{\text{source}} = a^T(R+Q)\mathbf{1}/(2\mathbf{1}^T Ra) = 53/(2\cdot 27) = \frac{53}{54}$ . Here  $\mathbf{1}^T Ra = 27$  is the  $E_6 \times A_2$  cubic family block ( $248 = \dim E_6 + \det R + 2(\mathbf{1}^T Ra)N_{\text{fam}} = 78 + 8 + 2\cdot 27\cdot 3$ ) and  $a^T(R+Q)\mathbf{1} = 53 = 52 + 1$ , where  $F = R+Q$  is the *missing* Sheet-Diamond corner with  $\det B_F = 52 = \dim F_4$  (v80). Hard negative control:  $a^T M \mathbf{1}$  for  $M \in \{R, Q, K, L\}$  is  $\{32, 21, 35, 56\}$  — only the absent corner  $F$  gives 53. This upgrades FR.KOIDE.03 from “near miss + conjecture” to “operator-sourced source $\rightarrow$ pole transfer”: the *factor* is [E] (exact, mirrored in Wolfram, 232/232); the *mechanism* (the leptonic source $\rightarrow$ pole transfer reads the  $F$  corner) stays [C], an  $F_{\text{transfer}}$  special case. The prediction itself stays [C]. Cited in `tfpt_4_frontier` (Koide section). Suite now 183 modules.
- **Review point 2 noted as already-present:** the claim that flavor and Nariai share one double-cover ClockFunctor (flavor rate  $(2/3)^{2N_{\text{fam}}}$  vs gravity curvature  $(2/3)^{N_{\text{fam}}}$ , exponent ratio  $|\mathbb{Z}_2|$ , “one clock, two geometries”) is exactly HOR.TRISECT.01/v103 — a rediscovery of an existing result, no new module.

## 2026-06-14 (review recommendations 1/3/4 + the four concerns A–D mapped to the named residuals)

- **v182\_reviewer\_residual\_map.py** [E]/[C] (claim REVIEW.MAP.01). A careful external review raised four concerns (A mapping  $2D \rightarrow 4D$ , B UV-gravity abstractness, C grammar-tuning/meta-look-elsewhere, D Koide Möbius flow). This module shows three of them *reduce*, by the same discipline as v175–v181, onto the *two already-named* residuals  $\{\text{QGEO.SYM.01}, F_{\text{transfer}}\}$ : A (Plücker-11 = QBL boundary count v174 + holographic reduction), B ( $\equiv \text{QGEO.SYM.01}$ ), D (Möbius forced by  $\text{Aut}(\mathbb{P}) = \text{PSL}(2, \mathbb{C})$  on the deck; the missing generator =  $F_{\text{transfer}}$ ). Only C is genuinely *experiment-only* (frozen + minimal + over-determined grammar; residual falls to JUNO/CMB-S4). A unification of the concern *structure*, not a closure. Cited in `tfpt_5_redteam`. Suite now 182 modules.
- **Review recommendation #3 — a “Rosetta” translation lexicon** (website RosettaLexicon): TFPT’s compiler-speak (seam, deck, readout, microcode, audit hull, gap, anchor,  $F_{\text{transfer}}$ ,  $v_{\text{geo}}$ , QGEO.SYM.01) mapped onto standard QFT / mathematical-physics language (Wilsonian boundary CFT, orbifold deck, index/intersection number, RG irrelevance rate, renormalization condition, fundamental postulate) — a reading aid for mainstream physicists, not a new claim.
- **Review recommendation #1 — kill-tests up front** (website TrustContract): the frozen pre-data predictions ( $\sin^2\theta_{12} = 0.3067$ , JUNO;  $r \approx 0.004$ , CMB-S4) and the null model ( $P \leq 10^{-30.7}$ ) are now a prominent strip directly under the hero.
- **Review recommendation #4 — the bedrock as a postulate**. QGEO.SYM.01 is now framed confidently as TFPT’s *one fundamental postulate* (the role “ $c = \text{const}$ ” plays in relativity), not a gap — in the `tfpt_research_contracts` box and the website “what is open” card; the achievement is that the whole theory is compressed onto exactly one sharply-located, falsifiable premise.

## 2026-06-14 (external-review response: CSV fix, G\_net 3-level, QGEO.SYM.01 page, claim detox)

- **Ledger CSV bug fixed + guarded**. Three rows (AX.P1.01, AX.P2.01, QGEO.ISO.01) had *unquoted commas* in a status/external field (e.g. “a=(1,1,2)”, “PSL(2,C)”); so a strict CSV parser read 11–12 columns instead of 10 — a real machine-readability defect in the single source of truth. All offending fields are now quoted (every row parses to exactly 10 columns), and `audit_sync.py` gained a `check_csv_wellformed` guard (ledger, registry, clusters, freeze must all parse to constant width) so it cannot regress.
- **$G_{\text{net}}$  typed in three explicit levels** (website “What is open” card): (1) *algebra* [E]  $(D_5)_1 \otimes (A_3)_1 \rtimes \langle (1,1) \rangle \cong (E_8)_1$ ; (2) *AQFT machinery* [E] (net existence + full-cone RP, CAR functor, v175); (3) *seam instantiation* [O] (QGEO.SYM.01). The target object’s mathematics is closed; only the physical coupling of the raw seam is open.
- **QGEO.SYM.01 now has its own one-page box** in `tfpt_research_contracts`: what is identified, why it is weaker than CONF/ISO, what follows automatically (the conditional theorem), and *what would falsify it* (genus  $> 0$  double; non-deck transport;  $N_{\text{fam}} \neq 3$ ); with the explicit conditional framing “*given* QGEO.SYM.01, the closure follows” rather than “the theory derives the seam geometry”.
- **Claim-language detox**. “no free fundamental number” softened to “no free *dimensionless* compiler dial beyond the anchor and  $\pi$  (dimensionful scale + transfer physics stay typed)” (`origin_theory`, `introduction`, website); the site title softened from “the Standard Model Derived” to “the Standard-Model *Skeleton* Derived”.

- **External-reviewer map** added to the website verification page: four boxes (exact kernel [E], conditional physics [C], open premises [O], kill tests [X]) so a reviewer sees the attack surface without reading 181 scripts. Suite unchanged at 181.

## 2026-06-14 (Zenodo release 5.1 (rev 114) — v175–v181 bedrock + figures + P1/P2)

- **Published to Zenodo.** Version 5.1 (rev 114) of the ten-PDF document set was published as a new version of the TFPT deposit — record 20686284, DOI 10.5281/zenodo.20686284. This version carries the AQFT-closure-to-bedrock chain (v175–v181, the structural residual reduced to the single definitional premise QGEO.SYM.01), the two overview figures (residual\_chain, script\_timeline), and the P1/P2 anchor-reduction provenance on the axiom surfaces. The README and the Zenodo description were refreshed (181 modules), and the uploader’s DEFAULT\_RECORD\_ID was advanced to the new record for the next release.

## 2026-06-14 (overview figures + P1/P2 reduction reflected on the axiom surfaces)

- **Two overview figures** (make\_figures.py; PDF + website PNG). (i) `residual_chain` — the structural-residual reduction chain v175–v181 as a descending staircase from “build a quantum-gravity measure” to the single definitional bedrock QGEO.SYM.01; embedded in `tfpt_research_contracts`. (ii) `script_timeline` — the eight-phase journey of the ~181 scripts (foundations → SM readouts → seam= horizon → horizon/flux geometry → R1–R5/premise-(A) → PyR@TE cross-checks → AQFT bridges → AQFT closure), what each did mathematically and physically; embedded in `introduction` ahead of the master script index, and both shown on the website verification page.
- **P1/P2 reduction now reflected on the axiom surfaces.** The reduction of the two axioms to the single parabolic anchor  $a = (1, 1, 2)$  plus  $\pi$  (with  $g_{\text{car}} = 5$  an over-determined bootstrap fixed point — forced three ways, v6, Pascal-unique and Lean-formalised, FORM.LADDER.01) was already established (ANCHOR.GEN.01/v23, ARCH.CORE.01/v53) and stated in the paper-s/glossary, but the *axiom surfaces* still read as bare “declared inputs”. Fixed: the ledger rows AX.P1.01/AX.P2.01 now carry the anchor/bootstrap reduction and point to it, and the website dependency-graph nodes P1/P2 now show the anchor input ( $c_3 = 1/(2e_1(a)\pi)$ ;  $g_{\text{car}} = e_2(a)$ , bootstrap-forced) rather than “declared axiom”. No marker change (still declared inputs), only honest provenance.
- Manifests refreshed (two new figures added to the FIG list). Suite unchanged at 181.

## 2026-06-14 (the final reduction to bedrock v181 + full non-changelog status sweep)

- **v181\_clock\_is\_conformal\_symmetry.py** [E]/[O] (claim QGEO.SYM.01). The isometry premise QGEO.ISO.01 (v180) is weakened one more genuine step — to a conformal-*symmetry* premise — via *equivariant uniformisation* (Nielsen realisation; for cyclic groups Kerékjártó + uniformisation): a *conformal* automorphism (strictly weaker than an isometry — every isometry is conformal, conformal allows any positive scale) already suffices to put the order-4 clock in the form  $z \mapsto iz$ . Since the clock is the Coxeter element of  $W(A_3) = S_4$  (v117), the  $\mu_4$  deck, and a deck transformation preserves the pulled-back conformal structure by definition, the residual collapses to the *definitional* identification QGEO.SYM.01: “the seam’s conformal structure *is* the  $\mu_4$ -deck structure of the carrier clock”. This is bedrock — a physical/definitional statement tying P2 to the seam’s conformal geometry, *not* a finite computation and not reducible further without relabeling. The chain `REALIZE(v175)→⋯→SYM.01(v181)` ends here; we stop honestly. Python-only. Suite now 181 modules.

- **Full non-changelog status sweep.** The current bedrock framing was propagated to every live status surface (not just the changelog): the introduction “latest reductions” prose, `tfpt_5_redteam` Target A, the `tfpt_research_contracts` central-theorem keybox, `origin_theory`; and on the website `OpenGates`, `VerificationDag`, `papers.ts` (Target A  $\times 2$ ,  $G_{\text{net}}$ ), the `glossary`  $G_{\text{net}}$  entry and the `faq` live-residual answer — all now end at `QGEO.SYM.01`.

## 2026-06-14 (the conformal premise reduces to a milder isometry premise v180)

- `v180_clock_is_mobius.py` [E]/[O] (claim `QGEO.ISO.01`). Cracks `QGEO.CONF.01` one level further: the conformal-realisation premise is *replaced*, via three classical theorems, by a strictly *milder, non-circular* premise. (1) *Uniformisation* (Poincaré–Koebe): a genus-0 Riemann surface is conformally  $\mathbb{P}$ , so the genus-0 seam double (P1, v179) with its RP-metric conformal structure *is*  $\mathbb{P}$  — the target is *produced*, not assumed; (2) *Kerékjártó* (1919; Constantin–Kolev 2003): a finite-order orientation-preserving homeomorphism of  $S^2$  is conjugate to a rotation, so the order-4 clock ( $h(A_3)=4$ ) is topologically  $z \mapsto iz$ ; (3) *isometry*  $\Rightarrow$  *conformal*  $\Rightarrow$  *Möbius*: an orientation-preserving isometry of a Riemannian surface is holomorphic,  $\text{Aut}(\mathbb{P}) = \text{PSL}(2, \mathbb{C})$ ; (4) an order-4 Möbius map is  $z \mapsto iz$  (verified: projective order 4, fixed  $\{0, \infty\}$ , multiplier  $i$ , free orbit  $\mu_4$ ). Hence `QGEO.CONF.01` is *implied* by the milder premise `QGEO.ISO.01`: “the carrier clock is an order-4 orientation-preserving *isometry* of the RP seam metric” — which does not name  $\mathbb{P}/\mu_4$  (uniformisation produces them) and is the natural statement that the seam transport is metric-compatible (a holonomy preserving the RP inner product). That milder premise is the single, now milder, structural residual of the whole theory; its surface realisation from the raw seam stays honestly [O]. Cited in `tfpt_research_contracts`; Python-only. Suite now 180 modules.

## 2026-06-14 (the two obligations collapse to ONE geometric premise v179)

- `v179_conformal_realisation.py` [E]/[O] (claim `QGEO.CONF.01`). Investigating the two residual premises left by v178 shows they are *not* independent — they collapse onto a single geometric premise (honest unification; the premise itself stays [O]). MARKS’s residual is not three independent assumptions: (1) genus-0 *is* P1 — the one-sided Gauss–Bonnet  $c_3 = 1/(|\mathbb{Z}_2| 2\pi \chi(S^2)) = 1/(8\pi)$  uses  $\chi(S^2) = 2$  and  $\chi = 2 - 2g \Rightarrow g = 0$ , so “genus-0” is the same  $\chi = 2$  that gives the “8” in  $c_3$  (v58/v73); (2) the order-4 clock *is* the carrier — the Coxeter element of  $W(A_3) = S_4$  (v117), order  $h(A_3) = 4 = |\mu_4|$ ; (3) marks = one orbit is *forced* (the parabolic marks are not the two elliptic fixed points, a free order-4 orbit has exactly  $4 = e_1(a)$  points). KERNEL then *follows* from the same geometry (DtN symbol  $|k|$ , Lee–Uhlmann v156;  $c = 8$  rigidity v158; the cusped-surface residual  $= H^1 = \mathbb{C}^3 = N_{\text{fam}}$ ). So the *entire* structural residual of the theory is the *single* geometric premise `QGEO.CONF.01`: “the raw RP seam normal double is conformally  $(\mathbb{P}, \mu_4)$  with the carrier clock as the order-4 Möbius automorphism”. The two doors of v177/v178 are one geometric realisation, left honestly [O]. Cited in `tfpt_research_contracts`; Python-only. Suite now 179 modules.

## 2026-06-14 (an honest attempt at the two obligations v178 — deeper reduction, not closure)

- `v178_marks_kernel_attempt.py` [E]/[O] (claim `QGEO.REDUCE.01`). An honest attempt at `QGEO.MARKS.01` and `QGEO.KERNEL.01`. These are genuine constructive-geometry/AQFT statements, *not* finite computations — neither is closed. What the module does is push each as far as computation allows: it closes the finite [E] core of each and reduces each to *one sharper* premise. *Marks core* [E]: given a genus-0 double with an order-4 clock  $\rho$  (fixed points  $\{0, \infty\}$ ) and reflection  $\tau$ , the marks are forced to be a generic  $\rho$ -orbit  $\{1, i, -1, -i\} = \mu_4$  (*not* the fixed points), distinct iff  $b_1 = 4 - 1 = 3 = N_{\text{fam}}$  (any collision drops  $b_1$ ), with  $\tau\rho\tau = \rho^{-1}$



giving a faithful  $D_4$  of order 8 — so MARKS reduces to the single topological/clock premise. *Kernel core* [E]: on the 3-dim multiplicity-free  $H^1$  the  $\mu_4$  characters  $(1, 2, 3)$  are distinct, so by Schur a  $\mu_4$ -equivariant operator is forced *diagonal* and completely determined by its eigenvalue per character; two such with the forced transfer spectrum  $\{1, (2/3)^6, (1/3)^6\}$  (v162) and the unique residue-normalised basis (v177) are *equal as operators*, not merely isospectral — the spectrum-vs-operator worry *vanishes* on the finite block. So KERNEL reduces to the single full-operator-reduction premise (principal symbol  $|k|$ , Lee-Uhlmann/v156; no drift v158). Net: neither obligation closed; both sharpened to milder premises with their finite cores [E]. Cited in `tfpt_research_contracts`; Python-only (Möbius/Schur symbolic + numeric). Suite now 178 modules.

## 2026-06-14 (QGEO proof tree v177 — the residual sharpened into two named obligations)

- `v177_seam_marking_kernel.py` [E]/[O] (claims QGEO.TREE.01, QGEO.MARKS.01, QGEO.KERNEL.01). A second external residual-class audit pointed out that the v176 statement takes the marks/ $D_4$  as a *hypothesis* — a near-circular trap. This module factors the central theorem honestly as `REALIZE = MARKS · UNIFORM · COHOM · MODULE · KERNEL · NET` and closes *four* nodes *constructively* (all validated symbolically before adoption): `UNIFORM` [E] — an order-4 Möbius map conjugates to  $z \mapsto iz$ , a non-fixed orbit scales to  $\mu_4$ , and  $\sigma\rho\sigma = \rho^{-1}$  for  $\sigma : z \mapsto 1/z$  gives a faithful  $D_4$  (order 8), so a genus-0 four-marked faithful- $D_4$  boundary *is*  $(\mathbb{P}, \mu_4)$  (strengthening v168's cross-ratio to the full uniformisation); `COHOM` [E] —  $\rho^*\omega_k = i^k\omega_k$ , grading  $(1, 2, 3) = A_3$  exponents; `MODULE` [E] — a *unique*  $\mu_4$ -equivariant iso (multiplicity-free + residue normalisation; reflection  $\omega_1 \leftrightarrow \omega_3$ ,  $\omega_2$  fixed); `NET` [E] (v154/v175). The structural residual is thereby sharpened from one diffuse premise into *exactly two* named open obligations, neither a finite computation: QGEO.MARKS.01 (the raw RP seam collar canonically produces the genus-0 four-marked  $D_4$  boundary) and QGEO.KERNEL.01 (the raw seam Calderón operator *is* the  $\mu_4$ -equivariant free gapped contraction, as an operator). Cited in `tfpt_research_contracts` (the central-theorem keybox); Python-only (Möbius/cohomology symbolic; numbers mirrored via v168/v137/v162/v154/v175). Suite now 177 modules.

## 2026-06-13 (Seam Collar Realisation Theorem v176 — the one central proof obligation)

- `v176_seam_collar_realisation.py` [E]/[O] (claim QGEO.REALIZE.02). Following an external residual-class audit, the single remaining structural item is formulated as one central theorem and certified as a *reduction* (not a fabricated proof). *Theorem*: given an oriented one-sided RP seam collar with a sheet-odd Calderón involution, faithful  $D_4$  marks, four parabolic marks with  $\mu_4$  decomposition and admissible  $c=8$  data, its conformal boundary is Möbius-equivalent to  $\mathbb{P} \setminus \mu_4$  with  $H^1$  grading  $(1, 2, 3)$ , the Calderón contraction is the rank-8  $K_\Sigma$  polarisation and the index-4 extension is  $(E_8)_1$ . The proof decomposes into six steps, *five already* [E]: geometry/uniformisation (v168), Calderón DtN symbol  $|k|$  (v156),  $H^1$  character = generation grading  $(1, 2, 3) = A_3$  exponents (v137/v168),  $\mu_4$ -equivariant contraction with gap  $(2/3)^6$  (v162), AQFT closure  $(D_5)_1 \otimes (A_3)_1 \rtimes \mu_4 = (E_8)_1 + \text{full-cone RP all } m$  (v154/v175). The whole structural residual thus reduces to the *one* open premise QGEO.REALIZE.01: that the physical seam collar's boundary *is* this  $\mathbb{P} \setminus \mu_4$  — a constructive-geometry statement, left honestly [O]. Cited in `tfpt_research_contracts` (the central theorem keybox); assembly/reduction certificate, Python-only (its exact identities are mirrored via v168/v156/v162/v154/v175). Suite now 176 modules. *Audit note*: the metrology residual the same review flags is already discharged — v153 (No-Unit Theorem) is the metrology-line typing ( $v_{\text{geo}} \mapsto \lambda^{-1}v_{\text{geo}}$ ,  $1/G \mapsto v_{\text{geo}}^2$ , dimensionless data invariant), and the Koide  $u \rightarrow \frac{53}{54}u$  source→pole transfer is already typed [C] (FR.KOIDE.03); neither is re-derived.

## 2026-06-13 (Zenodo release 5.1 (rev 105) — v174 + v175 + doc refresh)

- **Published to Zenodo.** Version 5.1 (rev 105) of the ten-PDF document set (introduction, origin\_theory, tfpt\_1–5, horizon, research\_contracts, changelog) was published as a new version of the TFPT deposit — record 20681451, DOI 10.5281/zenodo.20681451. This version carries the AQFT closure round (v170–v175): the seam Fock-space readings and the heavy-artillery discharge of net existence + full-cone reflection positivity to [E]. The README and the Zenodo description were refreshed to the current state (175 modules, Wolfram 116/116 + 231/231), and the uploader’s DEFAULT\_RECORD\_ID was advanced to the new record for the next release.

## 2026-06-13 (the heavy artillery v175: net existence + full-cone RP discharged to [E])

- **v175\_net\_existence\_full\_cone.py [E]/[O] (claim QFT.NETRP.01).** The two structural residuals that the premise-(A) chain had relocated to *already-open* items —  $A_2$  net existence and full-cone reflection positivity — are now *discharged to a theorem*, leaving the seam realisation as the single irreducible open premise (nothing fabricated). (1) *Full-cone RP for all  $m$* : the CAR second-quantisation functor  $\Gamma(t) = \bigoplus_m \Lambda^m(t)$  lifts the one-particle seam contraction  $t$  to the complete Fock space  $\Lambda^\bullet(\mathbb{R}^{16})$  ( $\dim 2^{16} = 65536$ ); by the exterior-power spectrum theorem every eigenvalue is a subset product of  $\text{spec}(t)$ , so all  $2^{16}$  products lie in  $[0, 1]$  (a contraction = full-cone RP), with eigenvalue-1 multiplicity  $2^8 = 256$  (wedges of the rank-8  $K_\Sigma$  polarisation) and sub-leading = the one-particle gap  $(2/3)^6$  — so full-cone RP reduces *for every  $m$*  (not only  $m \leq 6$ , v161) to the one-particle contraction (Shale–Stinespring), discharging the v160 scope premise. (2)  $A_2$  *assembled + verified*: the  $(E_8)_1$  character  $E_4/\eta^8 = 1 + 248q + 4124q^2 + 34752q^3 + 213126q^4 + 1057504q^5$  (level-1 = 248, extends the order-4 v156 check by one order), the embedding  $248 = 120 + 128$ , and the  $E_8$  Cartan matrix even with  $\det 1$  (the unique rank-8 even unimodular lattice, v83); the net *exists* by cited theorems (free-fermion net, lattice VOA, index-4 Q-system, v125). (3) Both then need the *same* input — the seam realisation QGEO.REALIZE.01 (finite half proven, v168) — which is a constructive-geometry/AQFT statement, *not* closeable by a finite computation, and is left honestly [O]. **Net:** net existence and full-cone RP become [E]; the one irreducible structural premise is the seam realisation. Cited in `tfpt_research_contracts` (premise (A) closure) and `tfpt_5_redteam` (Target A); the exact parts (Cartan unimodular, character order 5, functoriality integers) mirrored in the Wolfram extension (231/231). Suite now 175 modules.

## 2026-06-13 (seam Fock-space readings v174: CAR cone, Witten index, g-theorem)

- **v174\_seam\_fock\_readings.py [E] (claim QFT.FOCK.01).** Three more exact audit bridges from the same source note. (1) *CAR cone compression*: the even CAR algebra of the 16 seam Majoranas has  $\dim \Lambda^{\text{even}}(\mathbb{R}^{16}) = 2^{15} = 32768$  directions, but a quasi-free Bogoliubov transport reduces the whole cone to a 16-dim one-particle problem ( $2^{15}/16 = 2^{11}$ ) — the explicit count behind v161. (2) *Witten index = the Plücker norm 11*: the four  $\mu_4$ -corner fermions span a  $2^4 = 16 = \dim S^+$  Fock space; QBL ( $\deg \leq 2$ ) keeps  $\sum_{k \leq 2} \binom{4}{k} = 11 = \dim S^+ - g_{\text{car}}$ , so  $c_u/c_d = 55/117$  reads a topological state-space dimension. (3) *Affleck–Ludwig g-theorem*:  $c_{UV} = 5 + 3 = 8 = c_{IR}((E_8)_1)$ , so  $\Delta c = 0$  forces an exact isometry — the boundary-entropy form of the v157/v158 rigid fixed point. Cited in `tfpt_2` (Exterior Leg Lemma); mirrored in the Wolfram extension (230/230). Suite now 174 modules. (The categorical reframings in the source note remain framing, not asserted claims.)

## 2026-06-13 (CFT/QFT bridges: slice compression v170, OS moment v171, trace-anomaly seed v172, Pfaffian CP v173)

- **v170\_diamond\_slice\_compression.py** [E] (claim E8.SLICE.01). E8 Slice Compression Lemma: the seven  $E_8$  audit slices are *one* projection of two alphabets — anchor power sums  $P = (3, 4, 6, 10)$  ( $p_n = 2 + 2^n$ ) and Sheet-Diamond determinants  $(3, 4, 8, 14, 20, 32)$  summing to  $81 = N_{\text{fam}}^4$ . All seven 248-slices, the  $K_4$  edge graph, and the row-budget cross  $(7, 11, 13) + s(3, 3, 3) + t(1, 2, 3)$  follow;  $78 = \dim E_6 = p_2(p_0 + p_3)$ . Cited in `tfpt_3` (slice atlas). Exact; reading is audit (anti-numerology).
- **v171\_os\_moment\_cluster.py** [E]/[C] (claim QFT.OSMOMENT.01). Atomic OS Moment Lemma:  $\text{spec}(T) = \{1, 64/729, 1/729\}$  makes every correlator a positive moment sequence, so the OS Hankel matrices factorise as Vandermonde Grams (a clean RP certificate); cluster  $729/665$ ,  $\xi = 0.4111$ . Sugawara Gap Safety:  $31 = 1 + h^\vee(E_8)$ ,  $c((E_8)_1) = 248/31 = 8$ ,  $\Delta_{\text{eff}} = 6 \log \frac{3}{2} - 31/(4\pi^2) > 0$ . Cited in `tfpt_research_contracts` (G5).
- **v172\_trace\_anomaly\_seed.py** [E] (claim SEED.ANOMALY.01). The seed prefactor  $\frac{4}{3}$  is the halved  $(E_8)_1$  Weyl anomaly: over the seam  $S^2$  ( $\chi = 2$ ) the  $c = 8$  trace anomaly is  $c\chi/6 = \frac{8}{3}$ , halved by the one-sided  $|\mathbb{Z}_2|$  to  $\frac{4}{3} = 16/12$ . Plus QCD  $b_3 = 7 = \text{scalaron exponent } 48 - 41$  (confinement  $\leftrightarrow$  inflation), and the Volkov–Wipf  $-98/45 = -2 \cdot 7^2 / \dim(D_5)$  factorisation. Cited in `tfpt_2` (seed).
- **v173\_pfaffian\_cp\_car.py** [E] (claim FLAV.STRONGCP.02). Strong CP  $\theta_{\text{eff}} = 0$  as Pfaffian reality in the self-dual 16-Majorana CAR model:  $\{D, \Sigma\} = 0$  pairs the spectrum,  $\gamma_5$ -Hermiticity makes the Pfaffian real, and RP picks the positive branch. Cited in `tfpt_2` (Strong CP section).
- All four validated computationally before adoption; the exact identities (v170/v171/v172) are mirrored in the Wolfram extension (229/229). Suite now 173 modules. (The categorical reframings in the source note — compiler-as-single-functor,  $E_8 = K_0$ , GIT/monadic/Langlands — are recorded as framing, not asserted as machine-checked claims.)

## 2026-06-13 (Zenodo release 5.1 (rev 97))

- **Published to Zenodo.** Version 5.1 (rev 97) of the ten-PDF document set (introduction, `origin_theory`, `tfpt_1–5`, `horizon`, `research_contracts`, `changelog`) was published as a new version of the TFPT deposit — record 20678568, DOI 10.5281/zenodo.20678568. The uploader’s `DEFAULT_RECORD_ID` was advanced to the new record for the next release. Also fixed `build.sh zenodo` on macOS bash 3.2 (empty-array expansion under `set -u`).

## 2026-06-13 (QGEO rigidity theorem v168 + $\eta_B$ leptogenesis interface v169)

- **v168\_qgeo\_rigidity.py** [E]/[C] (claims QGEO.RIGID.01, QGEO.REALIZE.01). Hardens the *finite* half of the one remaining Tier-1 hinge GATE.QGEO (the seam = flavour = horizon identification). Exact, machine-checked:  $\mu_4 = \{1, i, -1, -i\}$  has cross-ratio 2 and a faithful Möbius stabiliser of order  $8 = |D_4|$ ;  $H^1(\mathbb{P}^1 \setminus \mu_4)$  has rank  $4 - 1 = N_{\text{fam}} = 3$ ; and the forms  $\omega_k = z^{k-1} dz / (z^4 - 1)$  ( $k = 1, 2, 3$ ) are  $\mu_4$ -eigenforms with characters  $\chi_1, \chi_2, \chi_3$  of weights  $(1, 2, 3) = A_3$  exponents =  $\text{Spec}(Q_+)$ . So a genus-0 boundary with faithful  $D_4$ , four parabolic marks and this  $\mu_4$ -decomposition is Möbius-equivalent to  $\mathbb{P}^1 \setminus \mu_4$  — the QGEO Rigidity Theorem (QGEO.RIGID.01, [E]). The seam-collar realisation stays the one premise (QGEO.REALIZE.01, [C]). Mirrored in the Wolfram extension (226/226). Cited in `tfpt_research_contracts` (U0). *Not* a new structural class — it hardens the existing hinge (consistent with v167’s completeness claim).
- **v169\_etaB\_boltzmann\_interface.py** [C] (claim FR.ETAB.02). Operationalises the Tier-3  $F_{\text{transfer}}$  leptogenesis path. Type-I thermal leptogenesis  $\eta_B \sim 0.96 \times 10^{-2} \varepsilon_1 \kappa_f$  (sphaleron

28/79), fed by TFPT’s normal-ordered neutrino spectrum and the predicted  $\delta_{CP} = 240^\circ$ . Over natural  $M_1 \in [3 \times 10^9, 3 \times 10^{10}]$  GeV and washout,  $\eta_B$  brackets the observed  $6.1 \times 10^{-10}$  (a canonical point gives  $6.0 \times 10^{-10}$ , untuned) — the route is *viable*. But  $M_1$ /washout are scenario inputs, so  $\eta_B$  stays [C]; if a precise solve excluded the window the leptogenesis *route* falls, not the theory (the downstream  $\Omega_b$  readout is independent). Cited in `tfpt_4_frontier` (Route B). Python-only.

## 2026-06-13 (Higgs quartic from the free seam: SM near-criticality explained, v166)

- `v166_higgs_free_seam.py` [E]/[C] (claim `HIGGS.FREESEAM.01`). A new prediction tying the Higgs mass to premise (A). The seam UV is the free chiral  $c=8$  fixed point (v158, Gaussian, no relevant/marginal interactions), so the one marginal SM scalar coupling vanishes there:  $\lambda(M_{\text{seam}}) = 0$  and  $\beta_\lambda(M_{\text{seam}}) = 0$  — the Shaposhnikov–Wetterich double-criticality, here *derived* from the free seam, not assumed. Integrating the PyR@TE-confirmed *two-loop* SM RGEs from  $M_Z$  up with the measured  $(m_H, m_t)$  gives  $\lambda(\bar{M}_{\text{Pl}}) \approx 0.002$  and  $\beta_\lambda(\bar{M}_{\text{Pl}}) \approx 0$ : the famous SM near-criticality is a *consequence* of the free seam (2-loop sharpens  $\lambda(\bar{M}_{\text{Pl}})$  from  $\sim 0.022$  at one loop to  $\sim 0.002$ ). The double condition predicts  $m_H \sim 129\text{--}134$  GeV; the same condition at the scalaron scale gives  $\sim 107$  GeV (too low), so the boundary condition lives at the reduced Planck scale (seam = horizon = Planck). Fills the Higgs-sector gap (was only  $N_\Phi = 1$ ); cited in `tfpt_4_frontier`.
- PyR@TE-sourced: the reproducer’s SM run supplies the two-loop  $\beta_\lambda$ , reduced to top-only (`verification/pyrate/sm_higgs_betas.txt`). Suite now 166 modules. Python-only (numerical 2-loop ODE).

## 2026-06-13 (PyR@TE $F_{\text{transfer}}$ follow-ups: flavor scale-tagging v163, QCD scale v164)

- `v163_rg_stability_flavor.py` [E]/[C] (claim `FLAV.RGSTAB.01`). Scale-tagging of the flavor sector. The lepton-mixing seeds ( $\sin^2 \theta_{12} = \frac{1}{3} - \frac{\varphi_0}{2} = 0.3067$ ,  $\sin^2 \theta_{13} = \varphi_0 e^{-5/6} = 0.0231$ ) are *RG-stable*: the PMNS running is  $y_\tau^2$ -suppressed (the only mixing-rotating Yukawa term is  $\frac{3}{2} Y_e Y_e^\dagger Y_e$ , PyR@TE SM RGEs), so  $\kappa_\tau = y_\tau^2 / (16\pi^2) \ln(M_{\text{scal}}/M_Z) \sim 1.8 \times 10^{-5}$  and  $|\Delta \sin^2 \theta_{ij}| \lesssim 10^{-4}$  — two to three orders below precision. Seam value = measured value; the  $y_t^2$ -driven quark mass ratios ( $m_t/m_b \sim 41$ ) by contrast carry a scale tag. Cited in `tfpt_2` (solar/reactor angle box).
- `v164_qcd_scale.py` [E]/[O] (claim `QCD.LAMBDA.01`). The QCD confinement scale is parameter-free from the carrier  $b_3 = -7$  (v159): running  $\alpha_s(M_Z) = 0.1179$  down (two-loop,  $n_f$  thresholds) reproduces  $\alpha_s(m_b) \simeq 0.22$ ,  $\alpha_s(m_c) \simeq 0.38$  and confinement at  $\sim 0.5$  GeV by dimensional transmutation — so the QCD-scale *numerator* of  $m_p/m_e$  is independently sourced. The  $O(1)$  proton/ $\Lambda$  factor is lattice, so  $m_p/m_e = 1836$  stays the honest “[O] — not forced” frontier ratio. Cited in `tfpt_4_frontier`.
- Both are PyR@TE-backed (the reproducer now also extracts the SM Yukawa  $\beta$ ’s, `verification/pyrate/sm_yukawa_betas.txt`). Suite now 164 modules. Python-only (no Wolfram mirror: magnitude bound / numerical running).

## 2026-06-13 (PyR@TE external cross-check of the $F_{\text{transfer}}$ gauge inputs — v159)

- **New module** `v159_pyrate_gauge_crosscheck.py` [E]/[C] (claim `EM.BUDGET.02`). The one-loop gauge coefficients used by the transfer functor are confirmed by an *independent* third-party RGE generator. Feeding the carrier / Standard-Model field content into the textbook one-loop formula (GUT normalisation) gives  $(b_1, b_2, b_3) = (\frac{41}{10}, -\frac{19}{6}, -7)$ , and **PyR@TE 3** (`arXiv:2007.12700`, `github.com/LSartore/pyrate`) writes  $\beta_{g_1} = (\frac{41}{10})g_1^3$ ,  $\beta_{g_2} = -\frac{19}{6}g_2^3$ ,



$\beta_{g_3} = -7g_3^3$  verbatim (plus the full standard 2-loop matrix). So  $b_1 = 41/10$  is pinned *three* independent ways — carrier algebra  $10b_1 = g_{\text{car}} 2^{g_{\text{car}}-2} + 1 = 41$ , the  $U(1)$  hypercharge index  $\sum \frac{3}{5} Y^2$ , and PyR@TE — and the split  $10b_1 = 40_{(\text{ferm.})} + 1_{(\text{Higgs})} = 41$  reproduces the EM-budget Gate 1 of `tfpt_2`. Cited there with `\veri`; the gauge sector of  $F_{\text{transfer}}$  is thus not a free knob.

- **PyR@TE reproducer (not vendored).** `verification/pyrate/reproduce.sh` clones PyR@TE on demand into `experiments/pyrate` (git-ignored), neutralises its interactive `pdflatex` end-command, runs the SM at one and two loops, and re-extracts the gauge  $\beta$  functions; the committed evidence is `verification/pyrate/sm_gauge_betas.txt`. PyR@TE is the right external tool for the transfer/RGE gates (gauge running, and — with a seesaw model — the neutrino-Yukawa running behind  $\eta_B$ ); it is *not* a tool for premise (A) (a 2d boundary-CFT question). Mirrored in the Wolfram extension; suite now 159 modules.

## 2026-06-13 (terminal residual inventory: (A) no longer a structural class; one hinge + $F_{\text{transfer}}$ + irreducible core, **v167**)

- **New module `v167_inventory_post_closure.py` [E]/[C].** The terminal residual inventory (line `v105`→`v123`→`v167`), re-pinned after the (A)-chain. With (A) discharged (`GATE.METRIC.16-19`) the `v123` *five* structural classes collapse: the seam clock (R1), the index-4 net (R2) and the  $Q$ -realisation (R5) all merge into *one* Tier-1 hinge — the seam=flavour=horizon identification =  $A_2$  net existence (assembly of established net constructions) + `GATE.QGEO` (seam boundary =  $\mathbb{P} \setminus \mu_4$ , cohomology  $b_1 = N_{\text{fam}} = 3$ ); the EH step (R3) folds into  $v_{\text{geo}}$  (**v152**) and the ambient measure into  $G_{\text{net}}$  (**v76**). **Terminal structure:** Tier 0 = the irreducible core  $\{\pi, v_{\text{geo}}\}$  (a theorem, No-Unit **v153**); Tier 1 = one hinge; Tier 2 = folded; Tier 3 =  $F_{\text{transfer}}$ , one functor with four typed conditional interfaces (Koide,  $\eta_B$ , axion,  $m_p/m_e$ ),  $\eta_B$  carrying the most open computational room. Completeness: a sixth independent structural class — or a new irreducible — fails the script (ledger `SYNTH.INVENTORY.03`). A unification + re-typing, NOT an unconditional proof. Python-only (reads the ledger).
- **Bookkeeping.** Suite at 167 modules, all green; Wolfram extension unchanged at 224/224; `origin_theory` (residual-inventory keybox) and `next.txt` moved together.

## 2026-06-13 (premise A maximal honest closure: $A_2$ by assembly, $R_1 = \text{GATE.QGEO}$ , zero independent open gates, **v165**)

- **New module `v165_premise_a_closure.py` [C]/[O].** Pins the two residual gates of the `v160`–`v162` (A)-chain to their floor.  $A_2$  (**net existence**) **closes by assembly** of established mathematics [C]: the 16 free Majoranas give a free-fermion conformal net (Buchholz–Mack–Todorov), the holomorphic  $c=8$  lattice net is  $(E_8)_1$  (Frenkel–Kac–Segal / Dong–Xu / Kawahigashi–Longo), the  $\mu_4$  extension is a Longo  $Q$ -system of index 4 (**v125**),  $\mu(B)=16/16=1$ ,  $248=120+128$ , character  $E_4/\eta^8=j^{1/3}$  (**v156**); the one TFPT-specific input (seam Calderón kernel = free-fermion two-point function) is done via  $\text{DtN} = |k|$  (**v156/v157**).  $R_1$  (**seam clock**) **reduces to `GATE.QGEO`** [C]: a  $\mu_4$ -equivariant flat gapped generator is *unique* ( $= A_0^*$ , **v115/v116**), so the gap  $(2/3)^6$  is forced and the residual is the one geometric identification “seam boundary =  $\mathbb{P} \setminus \mu_4$ ” = `GATE.QGEO` (cohomology established, **v137**,  $b_1=3=N_{\text{fam}}$ ). **Final accounting:** by the No-Unit Theorem (**v153**) the irreducibles stay  $\{\pi, v_{\text{geo}}\}$ ; premise (A) = ( $A_2$  assembly) + ( $R_1=\text{GATE.QGEO}$ ) + the  $\{\pi, v_{\text{geo}}\}$  core, so it carries *zero independent open gates* and ceases to be its own gate — a unification and re-typing, *not* an unconditional proof (ledger `GATE.METRIC.19`). Python-only (numpy/sympy + stdlib).
- **Bookkeeping.** Suite at 165 modules, all green; Wolfram extension unchanged at 224/224; `tfpt_research_contracts` (premise-A keybox) and `next.txt` moved together.



## 2026-06-13 (the final identification: premise A factors into the already-open A2 + R1, no new open content, **v162**)

- **New module `v162_seam_transport_identification.py`** [E]/[O]. Pushes the **v161** residual as far as it rigorously goes and shows it carries *no new open content*. The one-particle symbol splits into a gap value ( $\beta$ ) and a fixed space ( $\alpha$ ). ( $\beta$ ) **is forced**: a  $\mu_4$ -equivariant generator is diagonal in the cusp basis (the deck average is the diagonal projection,  $\sum_k U^k X U^{-k} = |\mu_4| \text{diag } X$  for  $U = \text{diag}(1, i, -i)$ ; the commutant of  $U$  is exactly the diagonal algebra), so its spectrum is the cusp weights  $\text{spec}(A_0^*) = \{0, \frac{1}{3}, \frac{2}{3}\}$  (**v115**) and the transfer eigenvalue  $(1-\alpha)^{p_2}$  with  $p_2=6=|R_+(A_3)|$  gives  $\{1, (2/3)^6, (1/3)^6\}$ , gap  $6 \log \frac{3}{2}$  — all carrier data; the lift to the 16 Majoranas is the  $A_3$  subnet grading (**v137**). ( $\alpha$ ) is the rank-8 Calderón polarisation of  $\omega_k = |k|$  (**v113/v156**). **Closure**: with **v160+v161**, (A) closes given only  $\mu_4$ -equivariance + admissibility, whose two non-derived inputs are the already-open A2 (net existence, Kawahigashi–Longo) and R1 (seam clock, **HOR.CLOCK**). So premise (A) factors exactly into  $A_2 + R_1$  — it adds *no* independent open item and is closed *conditionally* on them (a unification, not an unconditional proof; ledger **GATE.METRIC.18**). Python-only (numpy/sympy + stdlib).
- **Bookkeeping**. Suite at 162 modules, all green; Wolfram extension unchanged at 224/224; **tftpt\_research\_contracts** (premise-A keybox) and **next.txt** moved together.

## 2026-06-13 (the cone reduces to one particle: premise A's scope premise discharged to a 16-dim statement, **v161**)

- **New module `v161_one_particle_reduction.py`** [E]/[O]. *Discharges* the scope premise of **v160** (“the transport is gapped on the *full* Schwinger cone”) via Bogoliubov second quantisation. A quasi-free transport is  $T_\Sigma = \Gamma(t)$ , the second quantisation of a one-particle contraction  $t$  on the 16-dim Majorana space;  $\Gamma(t)$  acts on the degree- $m$  correlator sector as the exterior power  $\Lambda^m(t)$ . Machine-checked: (1)  $\text{spec}(\Gamma(t)|_{\deg m}) =$  products of  $m$  one-particle eigenvalues (compound-matrix theorem, all  $m=0..6$ ); (2) **the cone gap equals the one-particle gap** — the sub-leading eigenvalue over the whole cone is  $(2/3)^6$  exactly, so premise 5 on the cumulant space *is* premise 5 on the 16-dim space; (3) the eigenvalue-1 sector is  $\Lambda^*$  of the fixed subspace (disconnected Wick powers), with uniqueness closed by **v113** (one polarisation  $\Leftrightarrow$  one vacuum); (4) quasi-freeness is *forced* — the  $120 = \dim \mathfrak{so}(16)$  chiral currents are the Bogoliubov generators and **v158** makes every non-Gaussian deformation irrelevant. **Consequence**: **v160**’s three full-cone premises collapse to ONE finite (16-dim) one-particle statement (seam symbol = gapped Calderón Bogoliubov map, fixed = rank-8  $K_\Sigma$ , gap  $(2/3)^6$ ); all numbers supplied, only the physical identification (Kawahigashi–Longo class = the A2 net residual) stays open — the infinite-dimensional cone is eliminated (ledger **GATE.METRIC.17**). Python-only (numpy + stdlib).
- **Bookkeeping**. Suite at 161 modules, all green; Wolfram extension unchanged at 224/224; **tftpt\_research\_contracts** (premise-A keybox) and **next.txt** moved together.

## 2026-06-13 (premise A as a conditional fixed-point theorem; the load relocates to one scope premise, **v160**)

- **New module `v160_seam_gaussianity_from_pf.py`** [E]/[O]. Formalises the proposed closure of premise (A) (“the seam bulk is a reflection-positive free field”) as a fixed-point theorem and types it honestly. The *exact* parts: (1) a quasi-free state has *all* higher connected cumulants  $\kappa_{2n}=0$  — verified by the *signed* set-partition/Pfaffian moment  $\leftrightarrow$  cumulant inversion on a concrete pure Majorana covariance (the **v113** “one kernel = net” property at the cumulant level); (2) a kernel-preserving transport fixes the whole quasi-free state by Wick functoriality; (3) the Perron–Frobenius dichotomy (a *fixed*  $\kappa_{2n}$  is a second fixed ray  $\Rightarrow$  breaks uniqueness; a *non-fixed* one decays by the gap) forces Gaussianity; (4) the qualitative claim

needs only  $\text{gap} > 0$ , the value  $(2/3)^6$  only sets the rate. **The honest residual: v56's** gapped PF uniqueness lives on the *three-dimensional* scalar transfer spectrum  $\{1, (2/3)^6, (1/3)^6\}$ , while the 16-Majorana seam already carries  $\binom{16}{4}=1820$  and  $\binom{16}{6}=8008$  higher-cumulant directions — so (A) reduces to the single internal, exact-testable statement “the admissible TFPT seam transport is primitive + spectrally gapped on the *full* Schwinger cone, not only the readout spectrum of v56”, which the suite does *not* yet establish (ledger GATE.METRIC.16). Python-only (numpy + stdlib); no new exact-identity content for the Wolfram mirror.

- **Bookkeeping.** Suite at 160 modules, all green; Wolfram extension unchanged at 224/224 (v160 is numerical, Python-only); `tfpt_research_contracts` (the premise-A keybox) and `next.txt` moved together.

## 2026-06-13 (changelog date-order audit + reorder pass)

- **Changelog sort enforced.** `audit_sync.py` now checks that every dated `\subsection × {YYYY-MM-DD...}` block in this file is non-increasing (newest first; range titles such as 2026-06-07 / 2026-06-08 sort by the later date). A manual prepend had left five 2026-06-12 entries above seven 2026-06-13 blocks — reordered; future mis-sorts fail `build.sh audit` as `[B.changelog]`.

## 2026-06-13 (premise A: the destination fixed point is provably stable, v158)

- **The free chiral  $c=8$  fixed point is provably stable (v158, exact dimension count).** The finite core of (A)'s “reaches-the-fixed-point” residual: the chiral Majorana has  $h=\frac{1}{2}$ , the bilinear current  $h=1$ , the quartic  $h=2$ . The relevant window  $0 < h < 1$  for bosonic deformations is *empty*, the  $h=1$  currents are chiral (v157, not  $(1,1)$ -marginal), and the lowest interaction (quartic,  $h=2$ ) is irrelevant — so the fixed point is isolated and stable against all interactions. Premise (A) reduces to a basin-of-attraction statement with a *proven* stable target (ledger GATE.METRIC.15). *Tooling note:* PyR@TE (4d gauge-Yukawa RGEs) is the wrong tool for this 2d boundary-CFT residual, but the right tool for the  $F_{\text{transfer}}$  interfaces ( $\eta_B$  leptogenesis RGEs, gauge-coupling running, the  $b_1=41/10$  cross-check).
- **Bookkeeping.** Suite at 158 modules, all green; Wolfram extension 224/224; contracts / `origin_theory` / `next.txt` / website moved together.

## 2026-06-13 (premise A reframed: freeness is a rigid boundary fixed point, v157)

- **The free-bulk premise becomes a fixed-point property (v157, simplicity-first).** Instead of postulating Gaussianity, three facts click: (i) the DtN/Calderón symbol is *universally*  $|k|$  (Lee-Uhlmann — order-1  $\Psi$ DO, principal symbol  $|\xi|_g$ ; interactions live in the irrelevant lower-order symbol), so the free dispersion is bulk-detail-independent; (ii) a holomorphic  $c=8$  theory is *rigid* — every field has  $\bar{h}=0$ , so no marginal  $(1,1)$  field exists and the fixed point is isolated, with no knob for an interaction; (iii) the same  $|\mathbb{Z}_2|$  that halves  $8\pi$  in  $c_3 = \frac{1}{8\pi}$  is the Ramond/GSO projection giving  $\mu=1$  ( $248=120_{\text{NS}}+128_{\text{R}}$ ). With  $(E_8)_1$  unique (v83), premise (A) shrinks from “prove the seam is Gaussian” to “the gapped ( $\Delta=6\log\frac{3}{2}$ ) boundary flow reaches the holomorphic fixed point” — freeness then forced by rigidity, not fine-tuned (ledger GATE.METRIC.14).
- **Bookkeeping.** Suite at 157 modules, all green; Wolfram extension 223/223; contracts / `origin_theory` / `next.txt` / website moved together.

## 2026-06-13 (Route 3: the seam net constructed and $(E_8)_1$ verified, v156)

- **The constructive route carried to its exact endpoint (v156).** “ $B \cong (E_8)_1$ ” is no longer an assertion: the chiral seam net is built explicitly and the identification *checked*. (a) The

Dirichlet-to-Neumann (Calderón) operator of the 2d Laplacian is the free chiral dispersion  $\omega_k = |k|$  (harmonic extension  $e^{ikx-|k|y}$ ); (b) 16 free Majoranas give  $c = 8 = 5+3$ ; (c) the weight-1 currents are  $248 = 120_{\text{NS}} + 128_{\text{R}}$  ( $SO(16)$  bilinears + Ramond spinor, Frenkel–Kac); (d) the holomorphic character  $E_4/\eta^8 = q^{-1/3}(1 + 248q + 4124q^2 + 34752q^3 + 213126q^4) = j^{1/3}$  (verified by  $q$ -series) — the 248 at level 1 is  $\dim E_8$ , so the net *is*  $(E_8)_1$ . *Residual*: exact given (i) the free-bulk premise ( $\text{DtN} = |k|$ , **v155**) and (ii) the operator-algebraic existence of the net from the seam kernel (a Kawahigashi–Longo-type theorem, not a finite computation) — where TFPT meets constructive QFT, now sharply located (ledger **GATE.METRIC.13**).

- **Bookkeeping.** Suite at 156 modules, all green; Wolfram extension 222/222 (the  $\text{DtN} = |k|$  identity, the  $(E_8)_1$  character  $E_4/\eta^8 = j^{1/3}$  and the  $248=120+128$  split mirrored); contracts / redteam (Target A) / **origin\_theory** / **next.txt** / website moved together.

## 2026-06-13 (the seam-side identification reduced to one shared premise, **v155**)

- $G_{\text{net}}$ ’s last premise reduces to a single shared physical premise (**v155**, Python-only). The seam-side identification of the Simple-Current Extension Theorem (**v154**) splits (**v110**) into (a) “the Calderón involution is sheet-odd” (the double-cover deck, structural) and (b) “the seam 2-point kernel is quasi-free”. Part (b) is *not* independent: the boundary marginal of a reflection-positive *Gaussian* bulk is the compression  $PTP$  of the covariance, and the compression of a contraction is a contraction — so the seam state is automatically quasi-free (Wick inherited). Hence (b) *follows* from “the seam bulk is a reflection-positive free field”, the *same* premise already load-bearing in the area law (**v59**) and the EH mechanism (**v150/v151**). **The entire seam-side residual is therefore one physical premise**, shared by  $G_{\text{net}}$ , the area law and R3 — not reducible by a finite computation, but no longer three separate items (ledger **GATE.METRIC.12**).
- **Bookkeeping.** Suite at 155 modules, all green; Wolfram unchanged at 221/221 (**v155** is numerical/numpy, Python-only); contracts / **origin\_theory** / **next.txt** / website moved together.

## 2026-06-13 (external review formalised: No-Unit Theorem + Simple-Current Extension Theorem, **v153–v154**)

- $v_{\text{geo}}$  **re-typed: the No-Unit Theorem (**v153**)**. Every load-bearing compiler datum is mass-dimension 0 and invariant under a unit rescaling  $L \mapsto \lambda L$ , whereas a mass /  $1/G$  is not — so a dimensionless boundary compiler *provably cannot* select an absolute scale.  $v_{\text{geo}}$  is therefore an *irreducible metrology primitive*, not an open gap, and the three apparent residuals collapse onto it ( $U_{\text{point}} \sim v_{\text{geo}}$ ,  $1/G \sim v_{\text{geo}}^2$ ,  $m/\mu = e^{3/4}$  a pure ratio); the irreducibles stay {one unit,  $\pi$ } (ledger **ANCHOR.VGEO.02**).
- $G_{\text{net}}$  **stated as the central closing theorem (**v154**)**. The Simple-Current Extension Theorem:  $A=(D_5)_1 \otimes (A_3)_1$  extended by the isotropic glue  $L=\langle(1,1)\rangle$  has  $[B:A]=|L|=4$ ,  $c=5+3=8$ ,  $\mu(B)=\mu(A)/|L|^2=16/16=1 \Rightarrow$  holomorphic  $\Rightarrow B \cong (E_8)_1$  ( $SO(16)_1$  excluded). Every algebraic ingredient is machine-checked (**v89/v92/v125/v143/v148**); the only residue is the seam-side identification. Pulled to the front of **tfpt\_research\_contracts**, **tfpt\_5\_redteam** (Target A final form) and **origin\_theory** (ledger **GATE.METRIC.11**).
- **Status surfaces re-typed** (review): the canonical status card now reads  $v_{\text{geo}}$  as a metrology *primitive*,  $G_{\text{net}}$  as the simple-current extension  $\cong (E_8)_1$  (**[E]/[C]**), and  $F_{\text{transfer}}$  as one functor with four typed interfaces — “not structural gaps”. The final one-line framing (dimensionless boundary compiler; irreducibles  $\{\pi, v_{\text{geo}}\}$ ; one seam-net theorem; frontier =  $F_{\text{transfer}}$ ) is in **origin\_theory**, the introduction, the README and **next.txt**.
- **Bookkeeping.** Suite at 154 modules, all green; Wolfram extension 221/221; intro/README/**origin\_theory**/contracts/redteam/status-card/website moved together.

## 2026-06-13 (currency sweep: remaining old-series “Paper N” cross-references removed)

- **No old-version framing left in the active docs.** A consistency sweep found a residual layer of old-series “Paper N” attributions and removed them: four literal “Paper 6” references (`tfpt_4_frontier` leptogenesis/axion, `tfpt_2` seam determinant), the old “Paper 4” (OS/admissibility) and “Paper 5” (APS/Hodge) references in `tfpt_2`, the old “Paper 3” transport-kernel attributions in `tfpt_2`, and the internal “Paper 1/2/4” cross-references in `tfpt_3`, `tfpt_4` and `tfpt_horizon_readouts` — all rewritten as content-based phrasing (the flavor sector, the architecture document, the  $E_8$  decoder, the anchor characteristic polynomial, etc.). The active documents now attribute results to the current documents / axioms / scripts only, as the workflow rule requires; the machine-checked sync (scripts  $\leftrightarrow$  papers  $\leftrightarrow$  ledger  $\leftrightarrow$  website) was already current (152 scripts, AUDIT OK).

## 2026-06-13 (fifth round: the R3 normalisation is the dimensionful anchor, **v152**)

- **R3: the  $q(A_3)$  normalisation merges into the one anchor (**v152**).** The replica EH coefficient is a *log-ratio*  $k = \ln(m/\mu)/(12\pi)$  ( $\ln m$  alone is scale-ambiguous, only  $m/\mu$  is physical); pinning  $k = c_3/2$  fixes the dimensionless ratio  $m/\mu = e^{3/4}$ , and in  $d=2$  that ratio *is* the induced Newton constant  $1/G$  (**v68**). So  $v_{\text{geo}}$  (**v78**),  $1/G$  (**v68**) and  $m/\mu$  are *one* dimensionful anchor in three readings — the irreducibles stay {one scale,  $\pi$ }, and SEAM.THEOREM.01’s residual reduces to the kernel-identification premise alone. *Audit (not promoted)*: the log-ratio is numerically  $\frac{3}{4} = q(A_3)$ , the target value the anchor takes in seam units — not a derivation (ledger SEAM.EHMODEL.03).
- **Bookkeeping.** Suite at 152 modules, all green; Wolfram extension 219/219; intro/README/origin\_theory/next.txt/website moved together.

## 2026-06-12 (Zenodo version label: semantic + git rev)

- **Release version string.** `build.sh` now writes `tex-artefacts/release-version.txt` as `{TFPTversion}` (`rev {git count}`) (matches `\TFPTdatestamp` without the date). `scripts/zenodo_upload.py` and `build.sh zenodo/zenodo-publish` use this label on Zenodo; cursor rules extended for when `zenodo_description.html` must move with deposit-facing changes.

## 2026-06-12 (Zenodo API upload workflow)

- **Automated Zenodo versioning.** Added `scripts/zenodo_upload.py` and `.github/workflows/zenodo-release.yml`: creates a new draft version of record 20579275 (DOI 10.5281/zenodo.20579275), uploads the ten active paper PDFs, refreshes `zenodo_description.html`, bumps the version from `tex-artefacts/version.tex`, and optionally publishes via GitHub Actions (ZENODO\_TOKEN secret).

## 2026-06-12 (Overleaf shadow mirror: tfpt5-shadow)

- **One-way GitHub mirror for Overleaf sync.** Added `scripts/export-shadow.sh` and `.github/workflows/shadow-sync.yml`: on every push to main, a filtered copy is pushed to `sthamann/tfpt5-shadow` (papers, `tex-artefacts/`, `figures/`, `verification/`, `experiments/`; excludes `_archive/`, `website/`, all `*.pdf`, `.cursor/`).  $\sim 270$  files /  $\sim 3$  MB — within Overleaf GitHub-sync limits.

## 2026-06-12 (origin\_theory style alignment + box reduction)

- **Fewer boxes, embedded in text.** `origin_theory` from 22 to 16 boxes: the narrative/interpretation callouts (“Why the interpretation is internally consistent”, “Dependency chain of the birefringence test”, “The precise reformulation”, “The three theses honestly graded”, “The whole story”, “One-line summary”) become run-in bold prose; the boxed reformulation becomes a **keyeq**. The exact result keyboxes (**v53–v56**, **v101**, **v105**, **v113**, ...) and the master-principle / Seam–Horizon target boxes are kept.
- **Colour-marked module references + marker hygiene.** `\vref` applied to the inline `vN` references; stale literal `[A]`/`[I]`/`[L]` markers in running text replaced by the 4-class **[O]**/**[E]**.

## 2026-06-12 (tfpt\_5 / horizon / research-contracts pass + build hardening)

- **Forked document-set box removed (horizon).** `tfpt_horizon_readouts` carried a hand-written, *stale* copy of the document-set card (“five short documents”, missing `tfpt_5`) and the canonical `\input{tex-artefacts/tfpt_docset}`; the fork is deleted so the single-source card (all five papers + three companions) appears once.
- **Fewer boxes, embedded in text.** Narrative/result callouts converted to run-in bold prose (with the `\veri{}` citation moved out of the bold title into the body): `tfpt_horizon_readouts` from 22 to 7 boxes (the ten-box “clock” wall becomes prose); `tfpt_research_contracts` from 19 to 10 (the nine progress/result winboxes become prose, the named-theorem keyboxes kept); `tfpt_5_redteam` from 10 to 9 (the outcome summary; the per-target verdict boxes are kept as the structured red-team output).
- **Colour-marked module references + marker hygiene.** `\vref` applied to the inline `vN` references in all three documents; the horizon “Scope/typing” marker line and a stale literal `[A]` in the research contracts updated to the 4-class markers.
- **Build hardening.** `build.sh` now removes a document’s `.aux/.toc/.out` *on failure*, so a half-written cross-reference file from a crashed compile cannot poison the next build; artefacts are still kept on success.

## 2026-06-12 (fourth round: the Calderón transfer answered — the BFK split, **v151**)

- **R3: the Calderón/DtN kernel is conically clean (**v151**).** Reflection doubling derives that the Kac corner constant is boundary-condition independent ( $C_N = C_D$ ), so by the Burghilea–Friedlander–Kappeler gluing ledger the conical deficit of the two-sided Calderón jump determinant *vanishes identically* ( $C_{\text{cone}}(\gamma) - 2C_D(\gamma/2) = 0$ ): the seam-reduced effective action inherits the **v150** EH form *verbatim*, with all of it localised in the *local* half-space determinants — the “Calderón version” demanded by **v150** needs no separate derivation. `SEAM.THEOREM.01` narrows to the  $q(A_3)$  normalisation plus the standing kernel premise; the gate stays **[O]** (ledger `SEAM.EHMODEL.02`).
- **Bookkeeping.** Suite at 151 modules, all green; Wolfram extension 218/218; intro/README/origin\_theory/next.txt moved together.

## 2026-06-12 (third round: both normals anchor data; the R3 mechanism exists, **v149–v150**)

- **R4': both dual normals are anchor data (**v149**).** In the cusp frame the torsion normal pairs to  $(6, 3, 5) = (p_2, p_0, e_2)(a)$  — one anchor invariant per  $Q_+$  line (the cusp matrix is unimodular, so this determines  $n$  uniquely); with  $d = \frac{3}{2}a - 2 \cdot \mathbf{1}$  the dual normal pair is anchor data through and through. Equivalences to the  $(2, 8, 121)$  frame data and the  $w_0$ -lift exact.  $R4'$  residue = *one* discrete assignment (ledger `GATE.UWALL.14`).



- **R3: the mechanism exists at the gapped-model level (v150) — the first movement on SEAM.THEOREM.01.** The conical heat-trace deficit is exact and  $t$ -independent (Cheeger image sum  $(N^2-1)/(12N)$ ); for a *gapped* operator the  $\zeta$ -regularised replica variation is *finite and cutoff-independent*,  $\Delta \log \det' = 2C(\gamma) \ln m$ , and its linearisation is the Einstein–Hilbert form  $(\ln m/12\pi) \int \sqrt{g} R$ . The v73 normalisation becomes one exact target equation:  $k = c_3/2 \Leftrightarrow \ln m = \frac{3}{4} = q(A_3)$  (audit, not asserted). Remaining: the Calderón (boundary) version and that normalisation — the gate narrows but stays **[O]** (ledger SEAM.EHMODEL.01).
- **Bookkeeping.** Suite at 150 modules, all green; Wolfram extension 217/217; status surfaces (intro reductions + card, `origin_theory`, contracts, `next.txt`, website) moved together.

## 2026-06-12 (R1–R5 second round: every class at its floor, v145–v148)

- **R4': the pairing values are atom identities (v145).** Reading the v139 lift structurally ( $n = w_0(\sigma) + |\mu_4|e_3$ ,  $w_0 =$  the  $A_3$  exponent duality), all three pairings become atom arithmetic — including the *new* closure  $|\mu_4| + g_{\text{car}} = N_{\text{fam}}^2 (4+5=9, \sigma \perp w_0(a))$  and  $\|\sigma\|^2 = 110 = |\mathbb{Z}_2|g_{\text{car}}\|\text{Pl}(K)\|_1$ . R4' = derive *one* map (ledger GATE.UWALL.13).
- **R5: the Möbius  $D_4$  realisation (v146).** The *full* dihedral symmetry  $\langle z \mapsto iz, z \mapsto 1/z \rangle$  of the seam curve acts on  $H^1$  exactly as the integer model:  $\iota^*$  is the exponent duality with parity +1 on the self-conjugate line ( $= T_A$  in the cusp basis, glue-swap parity), and  $\Sigma$  is the  $\delta\iota$  class — the premise reduces to the module identification itself (ledger FLAV.QGEO.07).
- **R1: the clock is a Gaussian zero-mode integral, and the bend is  $\det'$ -clean (v147).** Variance ratio  $(1 - \alpha)$  (forced by norm = area) gives the Born-squared ratio  $(1 - \alpha)^{p^2}$ , the  $\ln$  series *is* the v127 ring sum; the two quantum weights share *one* Nariai geometry, so the bend  $\log_{3/2} 3$  carries no determinant correction — the v144 obstruction is localised to the  $\alpha = 0$  reference (ledger HOR.CLOCK.13).
- **R2: the NS/R sector census (v148).** The untwisted (Neveu–Schwarz) module carries only the *even* discriminant classes (integer weights), so the odd  $\mu_4$  simple currents have zero NS support — they are Ramond modules; the R zero-mode module (256) splits 128+128 into the odd sectors of the *two* Lagrangian glues: choosing the glue *is* choosing the R-projection ( $248 = 120_{\text{NS}} + 128_{\text{R}}$ ); the one remaining net statement is irreducibly twisted-sector (ledger GATE.METRIC.10). *Correction note (same day):* the first version of v148 graded the Ramond labels by the sum  $q_D + q_A$  (not the glue label) and called that module untwisted; the census was rewritten as above — conclusion unchanged, support corrected, the R-sector sheet split is new.
- **Bookkeeping.** Suite at 148 modules, all green; Wolfram extension 215/215 (README + GATE.WOLFRAM.02 + website counter in lock-step); status surfaces (intro card, `tfpt_1`, `origin_theory` third update, `next.txt`) moved together.

## 2026-06-12 (external review integrated: $\Lambda$ typing, marker migration, website parity)

- **$\Lambda$  normalisation made unambiguous.** The external review flagged the adjacent reduced/unreduced displays in `origin_theory` as a type trap (the formulas were individually correct —  $\bar{M}_{\text{P1}}$  vs  $M_{\text{P1}}$  — but easy to misread). Both normalisations are now shown explicitly side by side with their order counts ( $3/(4\pi^2) \Rightarrow 120.147$  reduced;  $3/(256\pi^4) \Rightarrow 122.948 = 119.028 + 3.920$  unreduced) plus a typing note.
- **Stale references cleaned.** “How each of the seven papers simplifies”  $\rightarrow$  “How the legacy layers simplify (bridge map)” with the lead-in “(Paper 6)” reference replaced by the scalaron-reheating chain (v86); “(Paper~6/7)” in the `tfpt_3` bridge table reworded; the `tfpt_1` “single remaining geometric input (H2)” passage and the `tfpt_2` “single remaining input (U)” passage

carry dated status updates pointing at the current decomposition (v118–v122, v139, v141, v142, v75).

- **Marker migration completed in the shared figures.** The claim stack, anchor cockpit,  $E_8$  glue,  $2/3$  motif and  $K_5$  figures (`tex-artefacts/tfpt_figures.tex`) now show the four public classes `[E]/[C]/[O]` instead of the legacy `[A]/[I]/[L]/[B]/[P]/[R]` badges.
- **Red-team front table pulled to the final state.** The Target-A residual column now states the one remaining statement (boundary-net holomorphy +  $c=8 \Leftrightarrow$  index-4 inclusion; v83/v87/v89, Lie-level realisation v143); the historical three-residual form is kept below, dated.
- **Website parity.** The Red Team document is now a first-class entry (`papers.ts` number 5, slug `redteam`) and the changelog PDF a download row; Appendix H / Origin Theory / Research Contracts are labelled *companions* (never “Paper 5–7”); the marker system on every website surface migrated to `[E]/[C]/[O]/[X]` (fine types named in prose; the script index keeps quoting the scripts’ internal fine-grained typing); the verification page counters are live (144 checks generated from the registry via `lib/suite.ts`, Wolfram 116/116 + 211/211 — now guarded by the sync audit) and the reproduce block matches the current pipeline. The `papers.ts` section bodies of the affected papers carry the v141–v144 outcomes (sheet question: deck derived;  $G_{net}$ : Lie-level realisation; resummed clock: in-family det-ratio cancellation; selector triangle: two line pairings).
- **Not adopted** (recorded for transparency): the review’s narrative reframings (anchor-algebra trilogy, “one seam clock” framing, one-page structure) and the front-matter/status-card layout changes are editorial decisions left to the authors; the claimed  $\Lambda$  error was in fact a correct-but-treacherous notation switch (see above).

## 2026-06-12 (R1–R5 attack round: four residual classes sharpened, v141–v144)

- **R5 closed at the discrete level (v141, deck selection theorem).** The v140  $\mathbb{Z}_3$  choice is *derived*: the established integer deck  $G = T_A \Sigma$  (v97/v98) pairs the  $Q_+ = 1$  cusp line with the self-conjugate character 2, so only the sheet-twisted assignment  $(2, 3, 1) = \text{chars}(-U)$  survives; the cusp exponential  $i^{Q_+}$  is excluded (same spectrum, wrong grading pairing). Explicit conjugacy constructed; under  $k \rightarrow -k$  only the established sheet  $\mathbb{Z}_2$  flips. Consequence: Euler grading =  $Q_+ \circ \rho$ . GATE.QGEO keeps only its realisation premise — no discrete freedom left (ledger FLAV.QGEO.06).
- **R4’ narrowed: three pairings  $\rightarrow$  two (v142, frame integrality).** Integer covectors in the  $(1, a, \sigma)$  dual frame form an index-11 sublattice ( $x - 3y + z \equiv 0 \pmod{11}$ ; the index is  $\|\text{Pl}(K)\|_1$ ); with  $(x, y) = (|\mathbb{Z}_2|, \text{rank } E_8)$  the  $\sigma$ -pairing is forced  $\equiv 0 \pmod{11}$ , and primitivity forces the line parameter even. Cramer reductions for both line pairings recorded as restructuring (ledger GATE.UWALL.12).
- **R2 realised at the finite Lie level (v143, graded Frobenius).** The  $\mathbb{Z}_4$ -graded hull carries exact coset duality ( $-C_1 = C_3$ ; Killing pairing nondegenerate per  $g_k \times g_{-k}$ ), the glue average is exactly the carrier projector (index 4), and each glue sector is one Weyl orbit (simple currents, fusion =  $\mathbb{C}[\mathbb{Z}_4]$ ) — the v125 Q-system realised by the hull; residue = one conformal-net statement (ledger GATE.METRIC.09).
- **R1’s det-ratio step derived in-family (v144).**  $e_2$ -rigidity of the traceless SdS cubic gives  $r_b r_c = 1 - \Delta^2/3$  exactly; with the v132 anomaly the non-zero-mode determinant ratio is  $(1 - \Delta^2/3)^{4/3}$  — no first-order term in the horizon split (the v131 cancellation derived, not assumed); deficit coefficient  $\frac{32}{27} = |\mu_4|(\frac{2}{3})^3$  (audit). Finite-weight absorption stays `[C]` with the  $6 \rightarrow \frac{14}{3}$  obstruction stated sharply (ledger HOR.CLOCK.12).
- **R3 attempted, honestly unchanged.** No defensible computation landed for the seam-determinant  $\rightarrow$  EH step (SEAM.THEOREM.01); the gate keeps its `[O]` typing untouched.

- **Wolfram mirror extended to v141–v144:** extension file now 211/211 (GATE.WOLFRAM.02 and the wolfram README updated in lock-step); Python suite at 144 modules, all green.

## 2026-06-12 (sync infrastructure: single-source script index, content maps, one audit)

- **Single-source script index.** The master script→check table (`tex-artefacts/verification.tex`) and the website `ScriptIndex.tsx` are now *generated* from one registry (`script_registry.csv` + `script_clusters.csv` in `verification/`) by `make_script_index.py`; both targets carry a generated-file header and are never edited by hand.
- **Content maps without splitting the papers.** `verification/make_docs_map.py` generates `docs_map.csv` (every paper section with line range, the vN scripts cited there, a content hash and a last-changed date) and `website_map.csv` (which website file mentions which scripts/documents) — the machine-readable enumeration of all sync surfaces.
- **One sync audit.** `verification/audit_sync.py` bundles and extends the previous loose shell checks, now in *both* directions: `suite` ↔ `run_all.py` ↔ `registry` ↔ `ledger`; every script cited in a paper *body* (the master index alone no longer counts); no stale `\veri/\vref` targets; every new module mentioned in this changelog; website mirror byte-identical (PDFs, scripts, `release.ts` hashes, version stamps); wolfram counts; overfull boxes. Frozen exceptions live in `audit_baseline.json` (remove-only).
- **Build pipeline.** `build.sh` gains `gen` / `website` / `audit` / `release` steps; `notes` now regenerates the single-source surfaces first; `website` copies all active PDFs (now incl. `tfpt_5_redteam.pdf` and `changelog.pdf`, with new `release.ts` entries) and the vN scripts, and stamps version, date and git revision into `website/lib/version.ts` (shown in the site header) and `release.ts`.
- **Citation gaps closed.** The audit immediately found three scripts cited only in the master index: `v17` and `v85` are now cited in their `tfpt_2` sections (hexagonal resolvent; master cover), and two short `\veri{v19}` citations were expanded to the full script name. `v91` (spine tetrahedron) had no paper section at all — it is now integrated: a keybox in `tfpt_3` (“Three views of one spine”: anchor closure, edge/face products, volume 120,  $K_6$  negative control, the tautology/sub-grammar caveats) and a closing note in the `tfpt_1` Spine Quotient Lemma; the ARCH.SPINE.01 ledger location is corrected to `tfpt_1;tfpt_3` and the baseline exception list is empty again.
- **Rules.** `.cursor/rules` updated (`tfpt-workflow`, `website-sync`) and a new `sync-maps` rule added describing the agentic procedure: *regenerate* the maps first, enumerate the affected surfaces from them, audit as the exit gate. The mirror map (`website_map.csv`) also covers the root `README.md` and `next.txt` (bare vN ids resolved to full script names), and both files are named explicitly as status surfaces that move with every finding.

## 2026-06-12 (tfpt\_3 / tfpt\_4 style alignment + box reduction)

- **Shared header/footer on tfpt\_3.** `tfpt_3_e8_audit_bootstrap` dropped its custom `\footmarkers` footer (the post-collapse four-[E] artefact) and inherits the shared `tfpt_style` header/footer with the version stamp; the box-orphan `needspace` guards are kept.
- **Fewer boxes, embedded in text.** Narrative/commentary callouts converted to run-in bold paragraphs (boxed one-line formulas to `keyeq`): `tfpt_3` from 39 to 27 boxes (e.g. “Purpose and discipline”, the five “(i)–(v)” audit-lemma boxes, “Net”, “The connection in one sentence”, “The unification”, “The question and the honest answer”, “Net: two axioms”); `tfpt_4` from 18 to 7 (the seven-box Koide stack and the two dark-matter conjecture boxes become prose). The per-item “Honest status of ...” keyboxes are kept — `tfpt_4` is the status authority for the frontier.

- **Colour-marked module references.** `\vref` applied to the inline `vN` references in `tfpt_3` and `tfpt_4` (TikZ node names excluded).
- **Marker hygiene.** A stale literal `[P]` in `tfpt_4` (Koide flow-time) replaced by the 4-class `[C]`; status tables checked against the ledger.

## 2026-06-12 (tfpt\_1 / tfpt\_2 style alignment + box reduction)

- **Shared header/footer everywhere.** `tfpt_1_architecture_e8` dropped its custom `\footmarkers` footer (which after the marker collapse showed four redundant `[E]`) and now inherits the shared `tfpt_style` header/footer with the version stamp, identical to every other document.
- **Fewer boxes, embedded in text.** The narrative/commentary callouts of both documents are converted to run-in bold paragraphs (and the boxed one-line formulas to the colour-marked `keyeq`): `tfpt_1` from 21 to 14 loud boxes (+3 `keyeq`); `tfpt_2` from 52 to 43 (e.g. “In one sentence”, “The one-paragraph version”, the four “Sharpening” notes, “What the simplification means net”, “The disciplined main statement”, “What improves...”, “Scope and honesty”). Only genuine results / theorems / gates keep a box.
- **Colour-marked module references.** `\vref` now also colours the inline `vN` references throughout `tfpt_1` and `tfpt_2` (TikZ node names accidentally matching the pattern were excluded).
- **Status currency.** `tfpt_2`’s inflation-pivot table gains the `v86` scalaron-reheating point ( $N_\star=51.4 \Rightarrow n_s=0.9611, r=0.0045, A_s$ -disfavoured) and states the band-of-record explicitly, matching `COSMO.NSTAR.01` and the introduction; the box is re-typed `[C]`. The other status tables were checked against the ledger and are current under the 4-class markers.

## 2026-06-12 (marker simplification to 4 classes + status reconciliation)

- **Four-class display markers.** The reader-facing marker system is collapsed from eleven symbols to four: `[E]` exact / machine-proven (identity, Lie/lattice, formalised, numerical FP), `[C]` conditional (physical / bridge / readout under named hypotheses), `[O]` open / axiom, and `[X]` kill test. `tex-artefacts/tfpt_style.tex` defines `\mE/\mC/\mO/\mX` and keeps the legacy `\tI ... \tX` names as aliases that render their class. All marker call-sites across the nine documents and the shared fragments were rewritten, collapsing same-class combinations (e.g. `[I]/[L] \rightarrow [E]`, `[N]/[P] \rightarrow [E]/[C]`). The fine-grained per-claim type (Axiom / Formal / Lattice / Numerical / Identity / Physical) is unchanged in `status_ledger.csv` — the single source of truth — so no fidelity is lost. The marker key, badge bar, README marker table and the workflow rule were updated to match.
- **Status tables reconciled against the ledger.** The introduction’s inflation/ $N_\star$  rows are made consistent with `v86/COSMO.NSTAR.01`:  $r$  and  $n_s$  are shown as the frozen *bands* over  $N_\star \in [50, 60]$  with the scalaron-reheating conditional point  $N_\star=51.4$  ( $n_s=0.9611, r=0.0045$ ), replacing the stale point values  $N_\star=55$  (predictions table) and  $N_\star \simeq 57$  (residual card); the closure-ledger inflation row is re-typed `[E] \rightarrow [C]` (conditional on  $M_{\text{Star}}=M_{\text{scal}} + \text{reheating}$ ), matching the other two tables and the ledger `[P]` typing.

## 2026-06-12 (version system, predictions/K5, readability pass B3)

- **Authors, auto-date and auto-version on every paper.** New single source `tex-artefacts/version.tex` defines `\TFPTauthors` (Stefan Hamann, Alessandro Rizzo), a curated `\TFPTversion`, and a title/footer date stamp (`\today + version + revision`). `build.sh` writes the auto build revision (git commit count) into the gitignored `tex-artefacts/version-auto.tex`, so every document’s title page and page footer carry

author, date and v5.4 (rev N) automatically. All nine active documents and `changelog.tex` now set `\author{\TFPTauthors}`.

- **Builder keeps build artefacts.** `build.sh` no longer deletes the `.aux/.log/.out/.toc` after a successful build (the `.log` is needed for the overfull-box audit and the `.aux/.toc` for correct cross-references on the next incremental build).
- **Predictions table outsourced.** The running/upcoming-experiment predictions table is moved from `introduction.tex` into the shared fragment `tex-artefacts/predictions.tex` (`\input` in the predictions section).
- **K5 figure fixed.** The `\TFPTactionkfive` mnemonic ( $K_5$  on the five carrier slots) is redrawn so the complete graph and the Pascal-row reading list sit side by side instead of overlapping.
- **Colour-marked script references.** New `\vref` macro colours inline module references (e.g. `v108–v113`, `v49/v71`) in the introduction, predictions and verification fragments, matching the blue of the `\veri` citation, so machine-checked references are visible at a glance.
- **Lighter, non-breaking, embedded boxes.** The `tcolorbox` families are restyled to a light tint + thin frame + light title band with dark coloured title (no heavy white-on-saturated bars); a neutral `navbox` carries the document map and a colour-marked `keyeq` sets off load-bearing display formulas. The introduction’s callouts use the non-breaking variants (`keyboxnb/winboxnb/gapboxnb`) so no box splits across a page, and the shared status card is non-breaking.
- **Readability / sectioning.** The dense run-in prose of the introduction is broken into `\subsection*` headings (In plain words; one anchor and  $E_8$  as a compiler hull; the companion documents; the compact status formula; the hard reduction; the Pascal action grammar; the closure ledger) with added whitespace.
- **Orphan results linked to scripts.** The Glue theorem, the “Reduction in one line” and the fermion-spectrum callouts now carry inline `\veri{}` citations (`v1`, `v6/v14/v23`, `v18/v20/v46`).

## 2026-06-12 (tex-artefacts refactor + introduction visual pass B2)

- **Shared imports moved to tex-artefacts/.** The shared, imported-only TeX fragments (`tfpt_style.tex`, `tfpt_figures.tex`, `tfpt_docset.tex`, `tfpt_status.tex`) now live in a dedicated `tex-artefacts/` folder; all nine active documents `\input` them via `tex-artefacts/<name>` (and `tfpt_style` loads `tex-artefacts/tfpt_figures`). `changelog.tex` stays at the repo root because it is also a standalone `build.sh` notes deliverable. The `make_manifest.py` TEX list and the workflow rule were updated to the new paths.
- **Master script index outsourced (tex-artefacts/verification.tex).** The long `vN→check` table is moved out of `introduction.tex` into the shared fragment `tex-artefacts/verification.tex` (`\input` in the verification appendix); the paper-consistency audit now also reads that file, so every registered script stays cited exactly once.
- **Introduction box reduction (B2).** The introduction’s eleven `keyboxes` are reduced to the single headline “In one sentence” card; the explanatory `keyboxes` (*In plain words*, *Sharper still*, *compact status formula*, *hard reduction*, *Cosmology Pascal*,  $E_6 \times A_2$ , *closure ledger*, *Plausibility balance*) are converted to inline run-in paragraphs, the *Glue theorem* is re-typed as a `winbox` (a result, not a claim), and the explanatory *plausible/follows next* `winboxes` become prose. The loud boxes now carry only genuinely load-bearing callouts (one `keybox`, five `winboxes`, one `gapbox`) plus the two shared orientation cards.



- **Document-count consistency.** The introduction’s document map gained the missing `tfpt_5_redteam` row; “seven/eight companion documents” and the ledger “six documents” wording are corrected to the true count (nine active documents); `README.md` (“10 ok”, 140 scripts) and the workflow rule (“9 docs”) are reconciled.

## 2026-06-12 (visual pass: box reduction B1)

- **Box reduction (B1).** The 26 non-load-bearing aside boxes (audit / recorded / interpretation: `auditbox/audbox/intbox/resbox/roadbox`) across `tfpt_1`, `tfpt_2`, `tfpt_3` and `origin_theory` are converted to inline bold run-in paragraphs — the reviewer’s prescription that an audit fingerprint must not carry the same visual volume as a theorem. The loud box families are now only `keybox` (claim), `winbox` (result) and `gapbox` (open/caveat).

## 2026-06-12 (visual pass: shared figures, badges, marker typology)

- **Shared figure library `tfpt_figures.tex`** (`\input` by `tfpt_style`): reusable TikZ macros for the architecture *claim stack* (`\TFPTclaimstack[zone]`, B3), the *anchor cockpit*  $a = (1, 1, 2)$  (B4), the  $E_8$  *glue* diagram (B5), the  $2/3$  *motif map* (B6) and the  $K_5$  *action lattice* (B7). The claim stack and a marker *badge bar* are now shown near the front of *every* document, each highlighting that document’s layer; the topical figures are placed in their natural homes (cockpit +  $K_5$  in the introduction, cockpit + glue in `tfpt_1`, motif in the horizon note).
- **Marker typology (A2).** `tfpt_style` adds the sharper sub-types `[E]` (exact algebra), `[E]` (exact number), `[C]` (readout / standard physics in TFPT units) to the existing `[C]` (bridge) and `[X]` (kill test); the marker key and the new `\TFPTbadges` bar show the full typology. (Inline body markers retain `[E]` where they are genuine exact identities.)

## 2026-06-12 (review pass on list.txt)

- **Claim hygiene (review section A).** The introduction’s one-sentence claim is softened from “derives all constants” to “constructs a discrete compiler; physical readouts run through named, status-typed bridges”; the ledger zombie in `FLAV.QRATIO.01` is removed (the statement no longer says “stays [P]” — ratios are `[E]` closed on the derived stratum, only the absolute scale stays `[O]`, with an explicit forbidden-wording note); the seam misalignment is written explicitly as  $\varepsilon = \frac{3}{4}\varphi_0^{\text{ret}} = c_3 + 36c_3^4$  (leading  $c_3$ , fourth-order puncture correction exact);  $\theta_{12}$  has a single prediction of record  $0.306747$  with the seam/non-linear values typed as scheme diagnostics; JUNE’s first 59.1-day result ( $0.3092 \pm 0.0087$ , compatible, precision pending) is recorded and NuFIT 6.0 is marked the pre-JUNE baseline; the ACT DR6 birefringence hint carries a systematics caveat (not a validation target); CODATA-2022 is dated; the null-model figure is moved out of the main text and typed as a grammar-internal bound. “Complete solutions of the five open questions” becomes “Closure status of the five former open questions”.
- **Interface language unified (review sections A5/C2/C3/C6/C7/C8).** The live residual is stated everywhere as  $v_{\text{geo}} \oplus G_{\text{net}} \oplus F_{\text{transfer}}$  (historical  $U_{\text{wall}}/G_{\text{metric}}/F_{\text{frontier}}$  kept only for ledger continuity); `tfpt_research_contracts` is retitled and its recommended order rewritten (selector-triangle pairings  $\rightarrow v_{\text{geo}} \rightarrow G_{\text{net}}$ );  $F_{\text{transfer}}$  is named as one missing functor with  $\text{Koide}/\eta_B/\text{axion}/m_p/m_e$  as its four instances; full QG closure is framed as a certification layer, not a prerequisite. Document numbering aligned with file names (the Red Team paper is now in the canonical set; document  $N \equiv \text{tfpt}_N$ ). Website mirrored throughout (`papers.ts`, `predictions.ts`, `faq.ts`, `glossary.ts`, `OpenGates`, `Hero`, orientation components).

## 2026-06-12 (later)

- **Shared style and central status artifact.** Two new single-source fragments: `tftpt_style.tex` (the canonical palette, status markers, marker key, `\veri` reference, the four `tcolorbox` families with back-compat aliases, and one running header/footer) is now `\input` by all eight documents — the per-paper duplicated preambles are gone; and `tftpt_status.tex` (“TFPT status at a glance”) gives one canonical overall-status card imported near the front of every document, replacing the divergent per-paper status statements. Both registered in `build.sh` and the manifest TEX list.

## 2026-06-12

- **Margin theorem (v122).** The established frozen selectors ( $D_4$  annihilator  $n = (5, -9, 6)$ ,  $\det R = 8$ ,  $\det K = 4$ ) pin  $R$  uniquely among hexagon matrices ( $64 \rightarrow 12 \rightarrow 1$ ); the atom margins become theorems — H2 introduces no new residual class.
- **Inventory update (v123).** Residual table re-pinned post v110–v122: exactly five structural classes (R1 clock, R2 index-4 net, R3 EH step, R4' selector establishment — replacing the discharged H2 dictionary — and R5  $Q$  realisation); R2 carries three loads (metric + carrier + QBL: one theorem, three doors).
- **Resummed clock + glue  $Q$ -system (v124/v125).** R1's bend in closed form:  $\text{rate}(n) = -p_2 \ln(1 - n/N_{\text{fam}})$  (pole at  $N_{\text{fam}}$  forces the three-level spectrum); the index-4  $\mu_4$  extension exists explicitly as a Longo  $Q$ -system on  $\mathbb{C}[\mathbb{Z}_4]$  (all Frobenius axioms exact) — R2 sharpens from “find” to “identify”.
- **Clock–wall bridge (v126) and ring resummation (v127).** The resummed-clock weights are the Mehta–Seshadri parabolic weights ( $\text{spec } A_0^*$ );  $\det(\mathbf{1} - A_0^*) = \frac{2}{9}$  = the entropy curvature; the clock is a log-determinant (RPA rings, one tower per hexagon site, coupling = parabolic weight).
- **Graded hull (v128).**  $E_8$  is a  $\mathbb{Z}_4$ -graded Lie algebra over the carrier (grading = glue  $Q$ -system; exact on all 6720 root-pair sums); ring multiplicity 6 = sheet-doubled Ginsparg–Perry zero modes. *Notation fix in the same round:* the clock factor 6 is  $p_2(a)$ , not  $2p_2$  — labels corrected across modules, ledger, papers, website (numbers unchanged).
- **Entropy power law  $\rightarrow$  zeta budget (v129–v133).** The clock is  $\Gamma_n \propto (S_n/S_{dS})^{p_2}$  (Gibbons–Hawking form);  $p_2 = 2h$  with  $h = N_{\text{fam}}$  from mode counting + the Born rule (v130); the zero-mode norm is the area,  $\|Y_{1m}\|^2 = A/(4\pi)$  exactly (v131); the det-ratio anomaly is the Koide constant,  $\zeta(0)|_{\det'} = -\frac{2}{3}$  (v132); both  $\zeta(0)$  budget routes computed exactly — the reduced seam route carries pure anchor ratios ( $-\frac{4}{3}$  = minus the seed gain), the naive 4d route ( $-\frac{109}{45}$ ) carries no atom (v133).
- **Dual anchor + determinant surface (v134/v135).**  $d := a^\top R^{-1} = a^\top L^{-1} = (-\frac{1}{2}, -\frac{1}{2}, 1)$  with  $(1, 1, -2) = -2d$  — the inverse flavor response is the Nariai root, invariance exact via Sherman–Morrison (third algebraic flavor–horizon bridge leg);  $\det M(s, t) = 3s(t+1)(t+2) + t^2 + 5t + 8$  with iso-volume walls at the  $\mathbb{Z}_2/\mu_4$  atoms, winding line  $(8, 14, 20)$ ,  $\det F = 32$ , row-budget axes with  $\text{rowsum}(V) = \text{Spec}(Q_+)$ ; micro-identities  $41 = 16 + 25$ ,  $52 = 16 + 36$ .
- **Dual-normal selector,  $Q_+$  cohomology, Volkov–Wipf firewall (v136–v138).**  $(d, n)$  pins  $R$  *columnwise* (each column the unique lattice point of the address box; kernel  $(6, 8, 7)$ ); the  $A_0^*$  zero mode carries the spectral normal  $\sigma = (2, -9, 5)$  with a  $W(A_3)$  orbit control excluding the naive lift;  $H^1(\mathbb{P}^1 \setminus \mu_4) = \chi_1 \oplus \chi_2 \oplus \chi_3$  with degrees  $(1, 2, 3) = \text{Spec}(Q_+) =$  the  $A_3$  exponents and cusp weights =  $\text{spec } A_0^*$ ; the external full-4d graviton/ghost value  $-\frac{98}{45}$  is pinned (arXiv:2506.02142 = Volkov–Wipf hep-th/0003081) and its *edge* part is exactly two copies of the reduced seam budget — R1 typed closed as an edge-sector object.

- **Selector triangle + canonical map (v139/v140).**  $d = \frac{3}{2}a - 2 \cdot \mathbf{1}$  is pure anchor data (first selector *derived*); the frame  $(\mathbf{1}, a, \sigma)$  has volume 11 and  $n$  is the unique covector with atom pairings  $(2, 8, 121)$  — the  $R4'$  residue is three pairings; the  $H^1 \rightarrow$  generation equivariant map exists and is Schur-rigid (only sign-twisted  $\mu_4$  actions match), so the `GATE.QGEO` residue collapses to one  $\mathbb{Z}_3$  choice.
- **Editorial and consistency passes.** Content-currency sweep (superseded “single remaining input” claims retired across contracts, paper 3, intro, website); introduction: self-explanatory title (spells out Topological Fixed-Point Theory), the old-version comparison removed throughout, redundant diagrams dropped, the unreadable status heatmap removed, the script-index master table converted to a page-breaking `xltable` (it had silently truncated  $\sim 110$  rows), status card extended with the v134–v140 reductions, predictions table completed (CP phase, cosmic birefringence); the canonical document-set box extracted into the shared `tfpt_docset.tex`; paper 1 cleaned of all 35 old-series references, every section now opens with its machine-check script references; smaller box fonts across all eight documents; this `changelog.tex` introduced as a dual-mode (standalone + imported) document.

## 2026-06-11

- **External-review integration (v83) and release hygiene.** Manifest `-check` mode and the release rule; holomorphy pins  $(E_8)_1$  (the unique even unimodular rank-8 lattice theory).
- **Sheet diamond and centered form (v94/v95).** All four flavor operators are points of one two-parameter surface  $M(s, t)$ ; centered cross  $R, L = C \mp U$ ,  $K, F = C \mp V$  with the cofactor seam normal  $n = (5, -9, 6)$ ; documentation pass (“The Sheet Diamond and the Winding Line” in paper 3).
- **Horizon/anchor round (v101–v103).** The maximal (Nariai) black hole is the anchor: roots  $(1, 1, -2)$ , entropy bound  $\frac{2}{3} = |\mathbb{Z}_2|/N_{\text{fam}}$ ; one repeller/attractor orientation in both sectors; trisection normal form (the canonical cosine coordinate).
- **Clock/ladder round (v104–v108) + Lean hardening.** The classical Nariai clock speaks anchor ( $\chi_{\text{clock}} = (\lambda - 1)(\lambda + 2)$ ); machine-pinned residual inventory; first review validation; quantum-clock target made quantitative; Pascal Ladder Theorem ( $g = 2K + 1 \Leftrightarrow$  the closure).
- **QBL theorem chain (v109–v113).** Sheet-Pairing Lemma; Calderón-sheet selection (a scalar seam datum exists iff the involution is sheet-odd); Quadratic-Transport Theorem (degree exactly 2); Self-Counting Channel (the Pascal closure as an identity); quasi-free kernel (one 2-point kernel determines the whole net, rank =  $c$ ) — the QBL input merges with the  $R2$ /holomorphy premise; communicated across all status surfaces.
- **$U_{\text{wall}}$  made exact (v114–v118).** Torsion normal form and  $\delta = \frac{1}{2}$  theorem; exact anchor residue  $A_0^*$  (diag =  $a/|\mu_4|$ , off-weights  $(8, 0, 5)/144$ ); resonance theorem ( $M_\infty = \mathbf{1} \iff a_{13} = 0$ , one gauge orbit); the holonomy is the exact 24-element  $W(A_3) = S_4$ ; the hypercharge hexagon is the sign-twisted family spectrum with the lepton coefficients as resolvent determinants.
- **Second review validation + address pinning (v119–v121).** Anchor ratio triad  $(\frac{2}{3}, \frac{4}{3}, \frac{5}{3})$ ; the 121 audit lemma; the lepton words are the compiler atoms  $(8, 5, 3)$  with divmod-hexagon addresses;  $R$  is the unique hexagon matrix with atom margins and  $\det 8$  — the address table carries zero residual information.

## 2026-06-10

- **B/C/D round closed.** Koide attractor toy model (v93), glue uniqueness (v92), the Wolfram extension as the independent second verification path, Lean hardening; papers and website synced with the round.

- **Content rounds v96–v100.** Sheet conjugation bridge, discriminant dictionary, Koide flow time, and the look-elsewhere-corrected numerology null test (v100: joint null probability  $\leq 10^{-30.7}$ , conditional on the declared grammar).

## 2026-06-09

- **Blind registry frozen (v84).** Machine-enforced prediction registry (`predictions_frozen.json`): exactly one  $\theta_{12}$  number of record before JUNO; conditional entries carry their hypotheses by name. Research-contracts document added to the set.

## 2026-06-07 / 2026-06-08

- **Compiler-closure consolidation.** The TFPT development is consolidated into the current document set with the verification suite (modules v1–v82:  $E_8$  glue, carrier Pascal, EM fixed point, flavor matrix, cascade, bootstrap, gravity/cosmology, transport, masses, registry and gate audits) as the single proof layer; manifests (`manifest.sha256`, `lean_manifest.sha256`) as content identity; website mirrors the suite (interactive DAG, in-browser script reproducer, script index).

## 2026-04-27

- **Initial commit.** TFPT papers and website baseline: electromagnetic closure and flavor transport (UFE bridge, birefringence seed, dyonic intercept), PDF-download tracking, accessibility and metadata passes.